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OF

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ADVERTISEMENT

THE present volume contains 69 papers numbered 417 to 485, published for the most part in the years 1866 to 1872; they include a series of astronomical papers published in the Memoirs and Monthly Notices of the Royal Astronomical Society.

The Portrait in Volume VI. is from the Painting in Oil by Mr Lowes Dickinson in the year 1874, presented by the Subscribers to Trinity College, Cambridge, and now in the Hall of the College: the portrait in the present volume is a photograph of a pencil sketch by Mr Lowes Dickinson in the year 1893.

The Table for the seven volumes is

Vol. I.	Numbers 1 to 100.
Vol. II.	Numbers 101 to 158.
Vol. III.	Numbers 159 to 222.
Vol. IV.	Numbers 223 to 299.
Vol. V.	Numbers 300 to 383.
Vol. VI.	Numbers 384 to 416.
Vol. VII.	Numbers 417 to 485.

CONTENTS

[417]

On the Locus of the Foci of the Conics which pass through Four Given Points.

[From the Philosophical Magazine, vol. XXXII. (1866), pp. 362–365.]

THE curve which is the locus of the foci of the conics which pass through four given points is, as appears from a general theorem of M. Chasles, a sextic curve having a double point at each of the circular points at infinity; and Professor Sylvester, in his "Supplemental Note on the Analogues in Space to the Cartesian Ovals *in plano*" (Phil. Mag., May 1866), has further remarked that the lines (eight in all) joining the circular points at infinity with any one of the four points are all of them double tangents of the curve; whence each of these points is a focus (more accurately a quadruple focus) of the curve. It is to be added that, besides the circular points at infinity, the curve has 6 double points (3 of these are the centres of the quadrangles formed by the 4 points), in all 8 double points; the class is therefore = 14. Hence also the number of tangents to the curve from a circular point at infinity is = 10; viz. these are the 4 double tangents each reckoned twice, and 2 single tangents; and the theoretical number of foci is = 100; viz. we have

16 quadruple foci, or intersections of a double tangent by a double tangent	$16 \times 4 = 64$
16 double foci, or intersections of a double tangent by a single tangent	$16 \times 2 = 32$
4 single foci, or intersections of a single tangent by a single tangent	4
	100

To verify the foregoing results, consider any two given points I, J, and the series of conics which pass through four given points A, B, C, D; we have thus a curve the locus of the intersections of the tangents from I and the tangents from J to any conic of the series; which curve, if I, J are the circular points at infinity, is the required curve of foci. Taking $U + \lambda V = 0$ for the equation of a conic of the series, the pair of tangents from I is given by an equation of the form

$$(\lambda,\, 1)^2 (x, y, z)^2 = 0,$$

and the pair of tangents from J by an equation of the like form

$$(\lambda,\, 1)^2 (x, y, z)^2 = 0;$$

and by eliminating λ from these equations, we obtain the equation of the required curve. This in the first instance presents itself as an equation of the eighth order; but it is to be observed that in the series of conics there are two conics each of them touching the line IJ, and that, considering the tangents drawn to either of these conics, the line IJ presents itself as part of the locus; that is, the line IJ twice repeated is part of the locus; and the residual curve is thus of the order $8 - 2, = 6$; that is, the required curve is of the order 6. The consideration of the same two conics shows that each of the points I, J is a double point on the curve. Moreover, by taking for the conic any one of the line-pairs through the four points, it appears that each of the points $(AB.CD)$, $(AC.BD)$, $(AD.BC)$ is a double point on the curve: this establishes the existence of five double points. The two conics of the series which touch the line IA are a single conic taken twice, and the consideration of this conic shows that the line IA is a double tangent to the curve; similarly each of the eight lines $I(A, B, C, D)$ and $J(A, B, C, D)$ is a double tangent to the curve.

Instead of seeking to establish directly the existence of the remaining three double points, the easier course is to show that, besides the four double tangents from I, the number of tangents from I to the curve is $= 2$; for, this being so, the total number of tangents from I to the curve will be $(2 \times 4 + 2 =)\,10$; that is, I being a double point, the class of the curve is $= 14$; and assuming that the depression $(6 \times 5 - 14 =)\,16$ in the class of the curve is caused by double points, the number of double points will be $= 8$. But observing that in the series of conics there is one conic which passes through J, so that the tangents from J to this conic are the tangent at J twice repeated, then it is easy to see that the tangents from I to this conic, at the points where they meet the tangent at J, touch the required curve, and that these two tangents are in fact (besides the double tangents) the only tangents from I to the curve; that is, the number of tangents from I to the curve is $= 2$.

Considering I, J as the circular points at infinity, and writing A, B, C, D to denote the squared distances of a point P from the four points A, B, C, D respectively, then, as remarked by Professor Sylvester, the equation

$$\lambda\sqrt{A} + \mu\sqrt{B} + \nu\sqrt{C} + \pi\sqrt{D} = 0$$

(where λ, μ, ν, π are constants) is in general a curve of the order 8; but the ratios $\lambda : \mu : \nu : \pi$ may be so determined that the order of the curve in question shall be $= 6$; the resulting curve of the order 6 is (not one of a group of curves, but the very curve) the locus of the foci of the conics through the four points. And the determination of the ratios $\lambda : \mu : \nu : \pi$ is in fact quite simple; for writing

$$\begin{aligned} A &= (x-a)^2 + (y-\alpha)^2 \\ &= \rho^2 - 2(ax + \alpha y) + \&\text{c.}, \quad (\text{if } \rho^2 = x^2 + y^2), \end{aligned}$$

and therefore

$$\sqrt{A} = \rho\sqrt{1 - \cdots},$$

with similar values for $\sqrt{B}$, $\sqrt{C}$, $\sqrt{D}$, it is easy to see that, considering λ, μ, ν, π as standing for $\pm\sqrt{A}$, $\pm\sqrt{B}$, $\pm\sqrt{C}$, $\pm\sqrt{D}$ respectively, the conditions for the reduction to the order 6 are

$$\begin{aligned} \lambda + \mu + \nu + \pi &= 0, \\ \lambda a + \mu b + \nu c + \pi d &= 0, \\ \lambda \alpha_1 + \mu b_1 + \nu c_1 + \pi d_1 &= 0, \end{aligned}$$

and hence that the required equation of the curve of foci is

$$\Sigma \begin{vmatrix} 1, & 1, & 1 \\ b, & c, & d \\ b_1, & c_1, & d_1 \end{vmatrix} \sqrt{(x-a)^2 + (y-\alpha_1)^2} = 0,$$

or, as this may also be written,

$$\Sigma \pm (B,\, C,\, D)\sqrt{A} = 0,$$

where (B, C, D), &c. are the areas of the triangles B, C, D, &c.

I remark, in conclusion, that the number of conditions to be satisfied in order that a curve may have for double points two given points I, J, may have besides six double points, and may have for double tangents eight given lines, is $(3 + 3 + 6 + 16 =)\,28$; the number of constants contained in the general equation of the order 6 is $= 27$. The conditions that a curve of the order 6 shall have for double points two given points I, J, shall besides have six double points, and shall have for double tangents four given lines through I and four given lines through J, are more than sufficient for the determination of the sextic curve; and the existence of a sextic curve satisfying these conditions is therefore a theorem.

In the case where the points I, J lie on a conic of the series, the consideration of this conic shows that the curve has a ninth double point, the pole of the line IJ in regard to the conic in question: in this case the sextic curve, as is known, breaks up into two cubic curves. [It need not do so, for a proper sextic curve may have nine (or indeed ten) double points.]

P.S. In general the curve $\lambda\sqrt{A} + \mu\sqrt{B} + \nu\sqrt{C} + \pi\sqrt{D} = 0$ has (exclusively of multiple points at infinity) six double points; viz. these are situate at the intersections of the pairs of circles,

$$\begin{gathered}(\lambda\sqrt{A} + \mu\sqrt{B} = 0,\;\; \nu\sqrt{C} + \pi\sqrt{D} = 0),\\ (\lambda\sqrt{A} + \nu\sqrt{C} = 0,\;\; \mu\sqrt{B} + \pi\sqrt{D} = 0),\\ (\lambda\sqrt{A} + \pi\sqrt{D} = 0,\;\; \mu\sqrt{B} + \nu\sqrt{C} = 0).\end{gathered}$$

In the case of the curve of foci, the first, second, and third pairs of circles intersect respectively in the points $(AB.CD)$, $(AC.BD)$, $(AD.BC)$, which, as mentioned above, are double points on the curve; and they besides intersect in three other points, which are the other three double points mentioned above.

Professor Sylvester reminds me that he mentioned to me in conversation that he had himself obtained the foregoing equation $\Sigma \pm (B, C, D)\sqrt{A} = 0$, for the locus of the foci of the conics which pass through the four points A, B, C, D.

Cambridge, October 10, 1866.

[418]

A Remark on Differential Equations.

[From the Philosophical Magazine, vol. XXXII. *(1866), pp. 379–381.]*

CONSIDER a differential equation $f(x, y, p) = 0$, of the first order, but of the degree n, where f is a rational and integral function of (x, y, p) not rationally decomposable into factors: the integral equation contains an arbitrary constant c, and represents therefore a system of curves, for any one of which curves the differential equation is satisfied: the differential equation is assumed to be such that the curves are algebraical curves. The curves in question may be considered as undecomposable curves; in fact, if the curve $U^\alpha V^\beta W^\gamma \ldots = 0$ (composed of the undecomposable curves $U = 0$, $V = 0$, $W = 0$,...) satisfies the differential equation, then either the curves $U = 0$, $V = 0$, $W = 0$,... each satisfy the differential equation, and instead of the curve $U^\alpha V^\beta W^\gamma \ldots = 0$ we have thus the undecomposable curves $U = 0$, $V = 0$, $W = 0$,... each satisfying the differential equation; or if any of these curves, for instance $W = 0$, &c., do not satisfy the differential equation, then W^γ, &c. are mere extraneous factors which may and ought to be rejected, and instead of the original curve $U^\alpha V^\beta W^\gamma \ldots = 0$, we have the undecomposable curves $U = 0$, $V = 0$ satisfying the differential equation. Assuming, as above, the existence of an algebraical solution, this may always be expressed in the form $\phi(x, y, c) = 0$, where ϕ is a rational and integral function of (x, y, c), of the degree n as regards the arbitrary constant c: this appears by the consideration that for given values (x_0, y_0) of (x, y) the differential equation and the integral equation must each of them give the same number of values of p. It is to be observed that ϕ regarded as a function of (x, y, c) cannot be rationally decomposable into factors; for if the equation were $\phi = \Phi\Psi \ldots = 0$, Φ, Ψ, &c. being each of them rational and integral functions of (x, y, c), then the differential equation would be satisfied by at least one of the equations $\Phi = 0$, $\Psi = 0$, ... that is, by an equation of a degree less than n in the arbitrary constant c.

But the equation $\phi(x, y, c) = 0$ is not of necessity the equation of an undecomposable curve, and the undecomposable curve which constitutes the proper solution of the differential equation cannot always be represented by an equation of the form in question. For although ϕ regarded as a function of (x, y, c) is not rationally decomposable into factors, yet it may very well happen that ϕ regarded as a function of (x, y) is rationally decomposable into factors (geometrically the sections by the planes $z = c$ of the undecomposable surface $\phi(x, y, z) = 0$ may each of them be composed of two or more distinct curves); and assuming that the function ϕ is thus decomposed into its prime factors, then each factor equated to zero gives an undecomposable curve satisfying the differential equation, and constituting the proper solution thereof.

It may be observed that, by the foregoing process of decomposition, we sometimes reduce the original equation $\phi(x, y, c) = 0$ into a like equation $\phi_1(x, y, c_1) = 0$ of a more simple form. Thus, for instance, if we have $\phi(x, y, c) = U^2 - c = 0$, U being a rational and integral function of (x, y), then instead of $U^2 - c = 0$ we have the equations $U + \sqrt{c} = 0$, $U - \sqrt{c} = 0$, each of which is an equation of the form $U - c_1 = 0$, or we pass from the original equation $\phi(x, y, c) = U^2 - c = 0$ to the simplified equation

$$\phi_1(x, y, c_1) = U - c_1 = 0.$$

Again, to take a somewhat more complicated instance, if the given integral equation be

$$\phi(x, y, c) = U^4 + c^2V^4 + (c+1)^2W^4 - 2c\,U^2V^2 - 2(c+1)\,U^2W^2 - 2c(c+1)\,V^2W^2 = 0,$$

then the equation $U^2 + V^2\sqrt{c} + W^2\sqrt{c+1} = 0$, writing therein $\sqrt{c} = \dfrac{c_1^2+1}{c_1^2-1}$, and therefore $\sqrt{c+1} = \dfrac{2c_1}{c_1^2-1}$, becomes

$$U(c_1^2 - 1) + V \cdot 2c_1 + W(c_1^2 + 1) = 0;$$

so that we pass from the original equation $\phi(x, y, c)$ to the simplified equation

$$\phi_1(x, y, c_1) = U(c_1^2 - 1) + V \cdot 2c_1 + W(c_1^2 + 1) = 0.$$

But observe that the possibility of the rationalization depends on the form of the radicals $\sqrt{c}$ and $\sqrt{c+1}$; if we had had $\sqrt{c}$ and $\sqrt{c^2+1}$ (or c and $\sqrt{c^4+1}$), the rationalization could not have been effected.

Returning to the case of an integral equation $\phi(x, y, c) = 0$, where ϕ regarded as a function of (x, y) is decomposable into factors, then equating to zero any one of the prime factors of ϕ, we obtain an integral equation $\psi(x, y, c_1, c_2, \ldots c_k) = 0$, where $c_1, c_2, \ldots c_k$ are irrational functions (not of necessity representable by radicals, and without any superior limit to the number of these functions) of c: here ψ regarded as a function of (x, y) is of course undecomposable, and the equation $\psi(x, y, c_1, c_2, \ldots c_k) = 0$ belongs to the undecomposable curve which is the proper solution of the differential equation. The result may be stated under a quasi-geometrical form; viz. regarding $c_1, c_2, \ldots c_k$ as the coordinates of a point in k-dimensional space, then as these are functions of the single parameter c, the point to which they belong is an arbitrary point on a certain curve or $(k - 1)$fold locus C in the k-dimensional space. And this curve must be such that to given values of (x, y) there shall correspond n points on the curve; that is, treating (x, y) as constants, the surface or onefold locus $\psi(x, y, c_1, c_2, \ldots c_k) = 0$, and the curve or $(k - 1)$fold locus C, shall meet in n points. The conclusion stated in the foregoing quasi-geometrical form is, that the solution of the differential equation may be exhibited in the form $\psi(x, y, c_1, c_2, \ldots c_k) = 0$; viz. ψ is a rational and integral function of $(x, y, c_1, c_2, \ldots c_k)$, where $(c_1, c_2, \ldots c_k)$ are the coordinates of an arbitrary or variable point on a curve or $(k-1)$fold locus C in a k-dimensional space, which curve meets the surface or onefold locus $\psi(x, y, c_1, c_2, \ldots c_k)$ in n points, and where ψ regarded as a function of (x, y) is not rationally decomposable into factors.

Cambridge, October 13, 1866.

[419]

A Theorem on Differential Operators.

[From a paper by Prof. Sylvester, "Note on the Test Operators which occur in the Calculus of Invariants, &c.," Philosophical Magazine, vol. XXXII. (1866), pp. 461–472, see p. 471.]

THE paper concludes with an Observation from Professor Cayley as follows:

"In the case of two variables, if

$$P_1 = (ax+by)\frac{d}{dx} + (cx+dy)\frac{d}{dy},$$

then in the notation of matrices,

$$P_1 = \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} (x,\, y)\left(\frac{d}{dx},\, \frac{d}{dy}\right),$$

$$P_2 = \tfrac{1}{2}\begin{pmatrix} a, & b \\ c, & d \end{pmatrix}^2 (x,\, y)\left(\frac{d}{dx},\, \frac{d}{dy}\right),$$

$$P_3 = \tfrac{1}{6}\begin{pmatrix} a, & b \\ c, & d \end{pmatrix}^3 (x,\, y)\left(\frac{d}{dx},\, \frac{d}{dy}\right);$$

whence also

$$P\cdot P_1 = P_1\cdot P + P_2 = \tfrac{1}{2}\begin{pmatrix} a, & b \\ c, & d \end{pmatrix}^2 (x,\, y)\left(\frac{d}{dx},\, \frac{d}{dy}\right) = 3P_2,$$

which accords with your theorem,

$$E_1 * E_2 * = E_2 * E_1 * = E_3 E_0 * + 3E_2 * ."$$

I have taken the liberty of writing in the above $\frac{d}{dx}, \frac{d}{dy}$ for δ_x, δ_y, and P for δ in the original. It will be useful to bear in mind that in any operator such as E_1* or E_2*, the asterisk forms an integral part of the symbol. Thus $E_1 * E_2*$, if we choose, may be written under the form of E_1* multiplied by E_2*, i.e. $(E_1*)\times(E_2*)$, where the cross is the sign of ordinary algebraical multiplication.

[420]

On Riccati's Equation.

[From the Philosophical Magazine, vol. XXXVI. (1868), pp. 348–351.]

THE following is, it appears to me, the proper form in which to present the solution of Riccati's equation.

The equation may be written

$$\frac{dy}{dx} + y^2 = x^{2q-2},$$

which is integrable by algebraic and exponential functions if $(2i+1)q = \pm 1$, i being zero, or a positive integer. To effect the integration, writing $y = \frac{1}{u}\frac{du}{dx}$, we have

$$\frac{d^2u}{dx^2} = x^{2q-2}\,u.$$

The peculiar advantage of this well-known transformation has not (so far as I am aware) been explicitly stated; it puts in evidence the form under which the sought-for function y contains the constant of integration. In fact if $u = P$, $u = Q$ be two particular solutions of the equation in u, then the general solution is $u = CP + DQ$; and denoting by P', Q' the derived functions, the value of y is

$$y = \frac{CP' + DQ'}{CP + DQ},$$

showing the form under which the constant of integration $C \div D$ is contained in y.

To complete the solution, assume

$$u = ze^{\frac{1}{q}x^q};$$

we find

$$\frac{d^2z}{dx^2} + 2x^{q-1}\frac{dz}{dx} + (q-1)\,x^{q-2}\,z = 0;$$

considering first the particular integral of the form

$$z = A + Bx^q + Cx^{2q} + Dx^{3q} + \&\text{c.},$$

we find that the equation will be satisfied if

$$\begin{aligned}
(\ q-1)\,A + \ \ q\,(\ q-1)\,B &= 0,\\
(3q-1)\,B + 2q\,(2q-1)\,C &= 0,\\
(5q-1)\,C + 3q\,(3q-1)\,D &= 0,\\
(7q-1)\,D + 4q\,(4q-1)\,E &= 0,\\
&\&\text{c.}
\end{aligned}$$

If $A = 1$, this is

$$\begin{aligned}
A &= \ \ 1,\\
B &= -\frac{q-1}{q(q-1)},\\
C &= +\frac{(q-1)(3q-1)}{q(q-1)\,2q(2q-1)},\\
D &= -\frac{(q-1)(3q-1)(5q-1)}{q(q-1)\,2q(2q-1)\,3q(3q-1)},\\
&\&\text{c.},
\end{aligned}$$

where it is to be noticed that the series may be considered to stop so soon as there is in the numerator a factor $= 0$. For instance, if $5q - 1 = 0$, then if the particular integral had been assumed to be $z = A + Bx^q + Cx^{2q}$, the only conditions to be satisfied by the coefficients are the first and second equations giving the foregoing values of A, B, C. It is immaterial that the analytical expressions of F and the subsequent coefficients contain in the denominators the evanescent factor $5q - 1$; the coefficients after C do not ever come into consideration.

Thus if $(2i+1)q = +1$, the series terminates, and we have for u the finite particular solution

$$u = P = \left(1 - \frac{q-1}{q(q-1)}\,x^q + \frac{(q-1)(3q-1)}{q(q-1)\,2q(2q-1)}\,x^{2q} - \&\text{c.}\right) e^{\frac{1}{q}x^q},$$

and it is easy to see that we may herein change the sign of x^q, thereby obtaining another finite particular solution,

$$u = Q = \left(1 + \frac{q-1}{q(q-1)}\,x^q + \frac{(q-1)(3q-1)}{q(q-1)\,2q(2q-1)}\,x^{2q} + \&\text{c.}\right) e^{-\frac{1}{q}x^q}.$$

Reverting to the equation in z, we have next a particular solution of the form

$$z = Ax + Bx^{q+1} + Cx^{2q+1} + Dx^{3q+1} + \&\text{c.},$$

giving between the coefficients the relation

$$\begin{aligned}
(\ \ q+1)\,A + (\ \ q+1)\,q\,B &= 0,\\
(3q+1)\,B + (2q+1)\,2q\,C &= 0,\\
(5q+1)\,C + (3q+1)\,3q\,D &= 0,\\
(7q+1)\,D + (4q+1)\,4q\,E &= 0,\\
&\&\text{c.}
\end{aligned}$$

If $A = 1$, we have

$$\begin{aligned}
A &= \ \ 1,\\
B &= -\frac{q+1}{(q+1)\,q},\\
C &= +\frac{(q+1)(3q+1)}{(q+1)\,q\,(2q+1)\,2q},\\
D &= -\frac{(q+1)(3q+1)(5q+1)}{(q+1)\,q\,(2q+1)\,2q\,(3q+1)\,3q},\\
&\&\text{c.},
\end{aligned}$$

where, as in the former case, the series is considered to terminate as soon as there is an evanescent factor in the numerator, without any regard to the subsequent coefficients which contain in the denominators the same evanescent factor. [In particular, $q = -1$, we have the solution $z = x$.]

Hence if we have $(2i+1)q = -1$, the series terminates, and we have for u the finite particular solution,

$$u = P = x\left(1 - \frac{q+1}{(q+1)\,q}\,x^q + \frac{(q+1)(3q+1)}{(q+1)\,q\,(2q+1)\,2q}\,x^{2q} - \&\text{c.}\right) e^{\frac{1}{q}x^q},$$

from which, changing the sign of x^q, we deduce the other finite particular solution,

$$u = Q = x\left(1 + \frac{q+1}{(q+1)\,q}\,x^q + \frac{(q+1)(3q+1)}{(q+1)\,q\,(2q+1)\,2q}\,x^{2q} + \&\text{c.}\right) e^{-\frac{1}{q}x^q}.$$

Hence, in the equation

$$\frac{dy}{dx} + y^2 = x^{2q-2},$$

where $q(2i+1) = \pm 1$, we have (writing $D = 1$)

$$y = \frac{CP' + Q'}{CP + Q},$$

where C is the constant of integration, P, Q are finite series as above, and P', Q' are the derived functions of P and Q. Writing successively $i = 0$, $i = 1$, $i = 2$, &c., we may tabulate the solutions:

$$\begin{array}{llll}
q + 2i = 1, & \frac{dy}{dx} + y^2 = 1, & P = e^{x}, & Q = e^{-x}, \\
 & \frac{dy}{dx} + y^2 = x^{-4}, & P = xe^{-\frac{1}{x}}, & Q = xe^{\frac{1}{x}}, \\
 & \frac{dy}{dx} + y^2 = x^{-\frac{8}{3}}, & P = (1 - 3x^{\frac{1}{3}})\, e^{3x^{\frac{1}{3}}}, & Q = (1 + 3x^{\frac{1}{3}})\, e^{-3x^{\frac{1}{3}}}, \\
 & \frac{dy}{dx} + y^2 = x^{-\frac{4}{3}}, & P = (1 - 5x^{\frac{1}{5}} + \frac{25}{3}x^{\frac{2}{5}})\, e^{5x^{\frac{1}{5}}}, & Q = (1 + 5x^{\frac{1}{5}} + \frac{25}{3}x^{\frac{2}{5}})\, e^{-5x^{\frac{1}{5}}}, \\
 & & & \text{\&c.}
\end{array}$$

It is hardly necessary to make the final step of calculating P' and Q' and substituting in y; but, as an example, take the above equation $\frac{dy}{dx} + y^2 = x^{-\frac{8}{3}}$: we have

$$y = \frac{-3x^{-\frac{1}{3}}\,(C\,e^{3x^{\frac{1}{3}}} + e^{-3x^{\frac{1}{3}}})}{C\,(1 - 3x^{\frac{1}{3}})\,e^{3x^{\frac{1}{3}}} + (1 + 3x^{\frac{1}{3}})\,e^{-3x^{\frac{1}{3}}}},$$

which is readily identified with the solution, p. 98 of Boole's *Differential Equations* (Cambridge, 1859).

Cambridge, September 29, 1868.

[421]

Note on the Solvibility of Equations by means of Radicals.

[From the Philosophical Magazine, vol. XXXVI. (1868), pp. 386, 387.]

In regard to the theorem that the general quintic equation of the nth order is not solvible by radicals, I believe that the proofs which have been given depend, or at any rate that a proof may be given that shall depend, on the following two lemmas:

I. A one-valued (or symmetrical) function of n letters is a perfect kth power, only when the kth root is a one-valued function of the n letters.

There is an exception in the case $k = 2$, whatever be the value of n: viz. the product of the squares of the differences is a one-valued function, a perfect square; but its square root, or the product of the simple differences, is a two-valued function. It is in virtue of this exception that a quadric equation is solvible by radicals; we have the one-valued function $(x_1 - x_2)^2$, the square of a two-valued function $x_1 - x_2$, and thence the two roots are each expressible in the form

$$\tfrac{1}{2}\{x_1 + x_2 \pm \sqrt{(x_1 - x_2)^2}\}.$$

II. A two-valued function of n letters is a perfect kth power, only when the kth root is a two-valued function of the n letters.

There is an exception in the case $k = 3$, when $n = 3$ or 4: viz. for $n = 3$ we have $(x_1 + \omega x_2 + \omega^2 x_3)^3$ (ω an imaginary cube root of unity) a two-valued function, and a perfect cube; whereas its cube root is the six-valued function $x_1 + \omega x_2 + \omega^2 x_3$. And similarly for $n = 4$ we have, for instance,

$$\{x_1x_2 + x_3x_4 + \omega(x_1x_3 + x_2x_4) + \omega^2(x_1x_4 + x_2x_3)\}^3$$

a two-valued function, and a perfect cube, whereas its cube root is a six-valued function. And it is in virtue of this exception that a cubic or a quartic equation is solvible by radicals. But I assume that for $n > 4$ the lemma is true without exception.

The course of demonstration would be something as follows: Imagine, if possible, the root of an equation expressed, by means of radicals, in terms of the coefficients; the expression cannot contain any radical such as $\sqrt[p]{X}$, $p > 2$, where X is a one-valued (or rational) function of the coefficients, not a perfect pth power, for the reason that, expressing the coefficients in terms of the roots, such function $\sqrt[p]{X}$ is not a rational function of the roots; if it were so, by lemma I. it would be a one-valued (that is, a symmetrical) function of the roots; consequently a rational function of the coefficients, or X expressed in terms of the coefficients, would be a perfect pth power.

The expression may however contain a radical $\sqrt{X}$, X a one-valued (or rational) function of the coefficients, not a perfect square: viz. X may be any square function multiplied into that function of the coefficients which is equal to the product of the squared differences of the roots, or, say, multiplied into the discriminant; that is, we may have $X = Q^2\nabla$, or $\sqrt{X} = Q\sqrt{\nabla}$.

We have next to consider whether the expression can contain any radical $\sqrt[p]{X}$, where X, not being a rational function of the coefficients, is a function expressible by radicals. But the foregoing reasoning shows that if this be so, X cannot contain any radical other than the radical $\sqrt{Q^2\nabla}$ or $Q\sqrt{\nabla}$, as above; that is, X must be $P + Q\sqrt{\nabla}$, where P and Q are rational functions of the coefficients, and where we may assume that $P + Q\sqrt{\nabla}$ is not a perfect pth power of a function of the like form $P' + Q'\sqrt{\nabla}$. But then, expressing the coefficients in terms of the roots, we have $P + Q\sqrt{\nabla}$, a (rational) two-valued function of the roots; and there is no radical $\sqrt[p]{P + Q\sqrt{\nabla}}$, which is a rational function of the roots; for by lemma II., if such radical existed we should have $\sqrt[p]{P + Q\sqrt{\nabla}}$ a (rational) two-valued function of the roots; that is, it would be

$= P' + Q'\sqrt{\nabla}$, P' and Q' one-valued (symmetrical) functions of the roots, consequently rational functions of the coefficients; or $P + Q\sqrt{\nabla}$ would be a perfect pth power $(P' + Q'\sqrt{\nabla})^p$.

The conclusion is that for $n > 4$ there is not (besides the function $P + Q\sqrt{\nabla}$) any function of the coefficients, expressible by means of radicals, which, when the coefficients are expressed in terms of the roots, will be a rational function of the roots, and consequently there is no possibility of expressing the roots in terms of the coefficients by means of radicals.

Cambridge, October 1, 1868.

[422]

On the Geodesic Lines on an Oblate Spheroid.

[From the Philosophical Magazine, vol. XL. *(1870), pp. 329–340.]*

THE theory of the geodesic lines on an oblate spheroid of any excentricity whatever was investigated by Legendre[1]; and the general course of them is well known, viz. each geodesic line undulates between two parallels equidistant from the equator (being thus either a closed curve, or a curve of indefinite length, according to the distance between the two parallels): at a point of contact with the parallel the curve is, of course, at right angles to the meridian; say this is V, a vertex of the geodesic line, and let the meridian through V meet the equator in A; the geodesic line proceeds from V to meet the equator in a point N, the node, where AN is at most $= 90°$; and the undulations are obtained by the repetition of this portion VN of the geodesic line alternately on each side of the equator and of the meridian.

[Figure 1: A diagram of the oblate spheroid showing the vertex V, the meridian VA through V, the equator, and the node N at which the geodesic line meets the equator.]

I consider in the present paper the series of geodesic lines which cut at right angles a given meridian AC, or, say, a series of geodesic normals. It may be remarked that as V passes from the position A on the equator to the pole C, the angular distance AN increases from a certain determinate value (equal, as will appear, to $\frac{C}{A} \cdot 90°$, if C, A are the polar and equatorial axes respectively) up to the value 90°; and it thus appears that, attending only to their course after they first meet the equator, the geodesic normals have an envelope resembling in its general appearance the evolute of an ellipse (see fig. 1 and also fig. 2), the centre hereof being the point B at the distance $BA = 90°$, and the axes coinciding in direction with the equator BA and meridian BC: this is in fact a real geodesic evolute of the meridian CA. The point α is, it is clear, the intersection of the equator by the geodesic line for which V is consecutive to the point A (so that $\angle BOA = \left(1 - \frac{C}{A}\right) 90°$); and the point γ is the intersection of the meridian CB by the geodesic line for which V is consecutive to the point C; and its position will be in this way presently determined. I was anxious, with a view to the construction of a drawing and a model, to obtain some numerical results in relation to a spheroid of considerable excentricity, and I selected that for which

$$\frac{C}{A} = \tfrac{1}{2} \quad \text{(polar axis} = \tfrac{1}{2} \text{ equatorial).}$$

[Figure 2: A diagram showing the geodesic evolute of the meridian, with the point B, the vertex region, the points α on the equator and γ on the meridian CB.]

Before proceeding further, I remark that Legendre's expression "reduced latitude" is used in what is not, I think, the ordinary sense; and I propose to substitute the term "parametric latitude": viz., in fig. 3, referring the point P on the ellipse by means of the ordinate MPQ to a point Q on the circle, radius OK ($= OA$, fig. 1), and drawing the normal PT, then we have for the point P the three latitudes,

$$\begin{aligned} \lambda &= \angle PTK, && \text{normal latitude,} \\ \lambda'' &= \angle POK, && \text{central latitude,} \\ \lambda' &= \angle QOK, && \text{parametric latitude;} \end{aligned}$$

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[1] *Mém. de l'Inst.* 1806; see also the *Exer. de Calcul Intégral*, t. I. (1811), p. 178, and the *Traité des Fonctions Elliptiques*, t. I. (1825), p. 360.

viz. λ' is the parameter most convenient for the expression of the values of the coordinates x, y ($x = A\cos\lambda'$, $y = C\sin\lambda'$) of a point P on the ellipse. The relations between the three latitudes are

$$\tan\lambda'' = \frac{C}{A}\tan\lambda' = \frac{C^2}{A^2}\tan\lambda,$$

so that λ'', λ', λ are in the order of increasing magnitude. I use in like manner l, l', l'' in regard to the vertex V. The course of a geodesic line is determined by the equation

$$\cos\lambda'\sin\alpha = \text{const.},$$

where λ' is the reduced latitude of any point P on the geodesic line, and α is at this point the azimuth of the geodesic line, or its inclination to the meridian. Hence, if l' be the parametric latitude of the vertex V, the equation is

$$\cos\lambda'\sin\alpha = \cos l'$$

(whence also, when $\lambda' = 0$, $\alpha = 90^\circ - l'$; that is, the geodesic line cuts the equator at an angle $= 90^\circ - l'$, the parametric latitude of the vertex). The equation in question, $\cos\lambda'\sin\alpha = \cos l'$, leads at once to Legendre's other equations: viz. taking, as above, A, C for the equatorial and polar semiaxes respectively, and δ for the excentricity, $\delta = \sqrt{1 - \frac{C^2}{A^2}}$; and to determine the position of P on the meridian, using (instead of the parametric latitude λ') the angle ϕ determined by the equation

$$\cos\phi = \frac{\sin\lambda'}{\sin l'},$$

and writing, moreover, s to denote the geodesic distance VP, and Λ to denote the longitude of P measured from the meridian CA which passes through the vertex V, these are

$$ds = d\phi\sqrt{C^2 + A^2\delta^2\sin^2 l'\cos^2\phi},$$
$$d\Lambda = \frac{\cos l'}{A}\cdot\frac{d\phi\sqrt{C^2 + A^2\delta^2\sin^2 l'\cos^2\phi}}{1 - \sin^2 l'\cos^2\phi},$$

which differential expressions are to be integrated from $\phi = 0$; and the equations then determine λ', s, and Λ, all in terms of the angle ϕ,—that is, virtually s and Λ, the length and longitude, in terms of the parametric latitude λ'.

Writing, with Legendre,

$$c^2 = \frac{A^2\delta^2\sin^2 l'}{C^2 + A^2\delta^2\sin^2 l'}, \quad = \delta^2\sin^2 l,$$
$$b^2 = 1 - c^2 = \frac{C^2}{C^2 + A^2\delta^2\sin^2 l'}, \quad = 1 - \delta^2\sin^2 l;$$

also

$$n = \tan^2 l', \qquad M = \frac{C}{bA\cos l'} = \frac{C}{A\cos l},$$

then the formulæ become

$$ds = \frac{C}{b}\,d\phi\sqrt{1 - c^2\sin^2\phi},$$
$$d\Lambda = M\,\frac{d\phi\sqrt{1 - c^2\sin^2\phi}}{1 + n\sin^2\phi}.$$

Hence integrating from $\phi = 0$, and using the notations F, E, Π of elliptic functions, we have

$$\begin{aligned} s &= \frac{C}{b} E(c,\, \phi), \\ \Lambda &= \frac{M}{n} \big\{ (n + c^2)\, \Pi\, (n,\, c,\, \phi) - c^2 F(c,\, \phi) \big\}; \end{aligned}$$

viz. these belong to any point P whatever on the geodesic line, parametric latitude of vertex $= l'$; and if we write herein $\phi = 90^\circ$, then they will refer to the node N, or point of intersection with the equator.

The position of the point α is at once obtained by writing $l' = 0$: viz. this gives $c = 0$, $b = 1$, $M = \dfrac{C}{A}$, $n = 0$: the differential expressions are $ds = C\, d\phi$, $d\Lambda = \dfrac{C}{A}\, d\phi$. Or integrating from $\phi = 0$ to $\phi = \frac{1}{2}\pi$, we have $s = A \cdot \dfrac{C}{A} \cdot \frac{1}{2}\pi$, $\Lambda = \dfrac{C}{A} \cdot \frac{1}{2}\pi$, agreeing with each other, and giving longitude of $\alpha = \dfrac{C}{A} \cdot \frac{1}{2}\pi$; or, what is the same thing, $\angle \alpha OB = \frac{1}{2}\pi \left(1 - \dfrac{C}{A}\right)$.

Writing in the formulæ $l' = 90^\circ$, we have $c = \delta$, $b = \dfrac{C}{A}$, $\dfrac{M}{n} = 0$; whence $d\Lambda = 0$, or $\Lambda = \text{const.} = \frac{1}{2}\pi$, since the geodesic line here coincides with the meridian CB; and moreover $s = A\, E(\delta,\, \phi)$; viz. this is merely the expression of the distance from C of a point P on the meridian CB. But we do not thus obtain the position of the point γ.

To find it we must consider a position of V consecutive to C, say, $l' = \frac{1}{2}\pi - \varepsilon$, where ε is indefinitely small; n is thus indefinitely large, and the integral $\Pi\, (n,\, c,\, \phi)$ is not conveniently dealt with. But it may be replaced by an expression depending on $\Pi\!\left(\dfrac{c^2}{n},\, c,\, \phi\right)$, where $\dfrac{c^2}{n}$ is indefinitely small; viz. (Legendre, *Fonct. Ellipt.* vol. I. p. 69) we have

$$\Pi\, (n,\, c,\, \phi) = F(c,\, \phi) + \frac{1}{\sqrt{a}} \tan^{-1} \frac{\sqrt{a}\, \tan\phi}{\sqrt{1 - c^2 \sin^2\phi}} - \Pi\!\left(\frac{c^2}{n},\, c,\, \phi\right),$$

where

$$a = (1 + n)\left(1 + \frac{c^2}{n}\right).$$

We thus have

$$\Lambda = \frac{M}{n} \left\{ nF(c,\, \phi) + \frac{\sqrt{n}\,\sqrt{c^2 + n}}{\sqrt{1 + n}} \tan^{-1} \frac{\sqrt{a}\, \tan\phi}{\sqrt{1 - c^2 \sin^2\phi}} - (c^2 + n)\, \Pi\!\left(\frac{c^2}{n},\, c,\, \phi\right) \right\},$$

where, $\dfrac{c^2}{n}$ being small,

$$\begin{aligned} \Pi\!\left(\frac{c^2}{n},\, c,\, \phi\right) &= \int \frac{d\phi}{\left(1 + \dfrac{c^2}{n} \sin^2\phi\right) \sqrt{1 - c^2 \sin^2\phi}} \\ &= \int \frac{\left(1 - \dfrac{c^2}{n} \sin^2\phi\right) d\phi}{\sqrt{1 - c^2 \sin^2\phi}} = \left(1 - \frac{1}{n}\right) F(c,\, \phi) + \frac{1}{n} E(c,\, \phi). \end{aligned}$$

And expanding also the $\tan^{-1}$ term, we thus have

$$\begin{aligned}\Lambda &= \frac{M}{n}\left\{nF(c,\,\phi) + \frac{\sqrt{n}\,\sqrt{c^2+n}}{\sqrt{1+n}}\left[\tfrac{1}{2}\pi - \frac{\sqrt{1-c^2\sin^2\phi}}{\tan\phi}\cdot\frac{\sqrt{n}}{\sqrt{(1+n)(c^2+n)}}\right]\right.\\ &\qquad\left. - (c^2+n)\left[\left(1-\frac{1}{n}\right)F(c,\,\phi) + \frac{1}{n}\,E(c,\,\phi)\right]\right\}\\ &= \frac{M}{n}\left\{\left(b^2+\frac{c^2}{n}\right)F(c,\,\phi) - \left(1+\frac{c^2}{n}\right)E(c,\,\phi) + \frac{\sqrt{n}\,\sqrt{c^2+n}}{\sqrt{1+n}}\cdot\tfrac{1}{2}\pi\right.\\ &\qquad\left. - \frac{n}{n+1}\cot\phi\,\sqrt{1-c^2\sin^2\phi}\right\},\end{aligned}$$

which, in the term in $\{\}$ neglecting negative powers of n, becomes

$$\Lambda = \frac{M}{n}\big\{\sqrt{n}\cdot\tfrac{1}{2}\pi + b^2F(c,\,\phi) - E(c,\,\phi) - \cot\phi\,\sqrt{1-c^2\sin^2\phi}\big\}.$$

We may moreover write $c=\delta$, $b=\dfrac{C}{A}$, $\phi = 90^\circ - \lambda'$, $n = \dfrac{1}{\varepsilon^2}$, $M=\varepsilon$, and therefore $\dfrac{M}{n}=\varepsilon$, so that the formula is

$$\begin{aligned}\Lambda &= \varepsilon\left\{\frac{1}{\varepsilon}\cdot\tfrac{1}{2}\pi + b^2F(c,\,90^\circ-\lambda') - E(c,\,90^\circ-\lambda') - \tan\lambda'\,\sqrt{1-c^2\cos^2\lambda'}\right\}\\ &= \tfrac{1}{2}\pi - \varepsilon\big\{\tan\lambda'\,\sqrt{1-c^2\cos^2\lambda'} + E(c,\,90^\circ-\lambda') - b^2F(c,\,90^\circ-\lambda')\big\},\end{aligned}$$

where I retain c, b as standing for $\sqrt{1-\dfrac{C^2}{A^2}}$, $\dfrac{C}{A}$ respectively.

Writing herein $\lambda'=0$, we have

$$\Lambda = \tfrac{1}{2}\pi - \varepsilon\,(E,\,c - b^2F,\,c),$$

where the coefficient $E,\,c - b^2F,\,c$ is

$$= \int_0^{\frac{1}{2}\pi} d\theta\left(\sqrt{1-c^2\sin^2\theta} - \frac{1-c^2}{\sqrt{1-c^2\sin^2\theta}}\right) = c^2\int_0^{\frac{1}{2}\pi}\frac{\cos^2\theta\,d\theta}{\sqrt{1-c^2\sin^2\theta}},$$

consequently positive; that is, Λ, the longitude of the node, is less than 90°, as it should be. Hence in order that Λ may be $=90^\circ$, we must have λ' negative, say, $\lambda' = -\mu'$, where μ' is positive; and, observing that we may under the signs E, F write $90^\circ-\mu'$ instead of $90^\circ+\mu'$, we thus have

$$\tfrac{1}{2}\pi = \tfrac{1}{2}\pi + \varepsilon\big\{\sqrt{1-c^2\cos^2\mu'}\,\tan\mu' - E(c,\,90^\circ-\mu') + b^2F(c,\,90^\circ-\mu')\big\};$$

that is, we must have

$$\tan\mu'\,\sqrt{1-c^2\cos^2\mu'} = E(c,\,90^\circ-\mu') - b^2F(c,\,90^\circ-\mu');$$

viz. μ' is here the parametric latitude (south) of the intersection of the meridian CB with the consecutive geodesic line—that is, of the point γ. As μ' increases from 0 to 90°, the left-hand side increases from 0 to ∞; and the right-hand side, beginning from a positive value and either attaining a maximum or not, ultimately decreases to 0; there is consequently a real root, which is easily found by trial.

Thus $\dfrac{C}{A}=\tfrac{1}{2}$, $C=\tfrac{1}{2}\sqrt{3}$ (angle of modulus $=60^\circ$), $b=\tfrac{1}{2}$; or the equation is

$$\tan\mu'\,\sqrt{1-\tfrac{3}{4}\cos^2\mu'} = E\,(90^\circ-\mu') - \tfrac{1}{4}F\,(90^\circ-\mu').$$

Using Legendre's Table IX., we have

μ'	$90^\circ - \mu'$	E	F	$E - \frac{1}{4}F$	$\tan\mu'\sqrt{1 - \frac{3}{4}\cos^2\mu'}$
0°	90°	1.21105	2.15651	0.6719	0.0
10	80	1.12248	1.81252	0.6693	
20	70	1.02663	1.49441	0.6530	
30	60	0.91839	1.21253	0.6153	0.3819
40	50	0.79538	0.96465	0.5542	0.6278

so that we see the required value is between 30° and 40°; and a rough interpolation gives the value $\mu' = 37^\circ\,40'$. But repeating the calculation with the values 37° and 38°, we have

μ'	$90^\circ - \mu'$	E	F	$E - \frac{1}{4}F$	$\tan\mu'\sqrt{1 - \frac{3}{4}\cos^2\mu'}$
37°	53°	0.833879	1.035870	0.57419	0.54425
38	52	0.821197	1.011849	0.56823	0.57108

whence, interpolating, $\mu' = 37^\circ\,55'$.

The semiaxes of the geodesic evolute, measured according to their longitude and parametric latitude respectively, are thus $B\alpha$, long. of $\alpha = 45^\circ$; $B\gamma$, param. lat. $= 37^\circ\,55'$. But measuring them according to their geodesic distance, the equatorial radius A being taken $= 1$, we have

$$\begin{aligned} B\alpha &= \tfrac{1}{4}\pi = 0.78540, \\ B\gamma &= \left(\frac{C}{b} - 1\right)\{E,\,c - E(52^\circ\,5')\} = 1.21106 - 0.82225 = 0.38881. \end{aligned}$$

Reverting to the general formulæ for s, Λ, but writing therein $A = 1$, and therefore $C = \sqrt{1-\delta^2}$; writing also $\phi = 90^\circ$ (that is, making the formulæ to refer to the node N of the geodesic line), we have

$$\begin{aligned} s &= \frac{\sqrt{1-\delta^2}}{\sqrt{1-\delta^2\sin^2 l}}\,E,\,c, \\ \Lambda &= \frac{\sqrt{1-\delta^2}}{n\cos l}\big\{(n+c^2)\,\Pi\,(n,\,c) - c^2F,\,c\big\}; \end{aligned}$$

but for the calculation of the second of these formulæ by means of Legendre's Tables it is necessary to express $\Pi\,(n,\,c)$ in terms of the functions E, F.

The proper formula is given in *Fonct. Ellipt.* vol. I. p. 137; viz. this is

$$\frac{\Delta\,(b,\,\theta)}{\sin\theta\cos\theta}\,\Pi\,(n,\,c) = \tfrac{1}{2}\pi + \frac{\sin\theta}{\cos\theta}\,\Delta\,(b,\,\theta)\,F,\,c + F,\,c\,F\,(b,\,\theta) - F,\,c\,E\,(b,\,\theta) - E,\,c\,F\,(b,\,\theta),$$

where $\Delta\,(b,\,\theta) = \sqrt{1 - b^2\sin^2\theta}$. θ is an angle given by the equation $\cot\theta = \sqrt{n}$; we have $n = \tan^2 l'$; consequently $\theta = 90^\circ - l'$. Substituting this value, except that for shortness I retain $E\,(b,\,\theta)$, $F\,(b,\,\theta)$ in place of $E\,(b,\,90^\circ - l')$, $F\,(b,\,90^\circ - l')$, we have

$$\begin{aligned} \Delta\,(b,\,\theta) &= \sqrt{1 - b^2\cos^2 l'} \\ &= \sqrt{1 - (1 - \delta^2\sin^2 l)\cos^2 l'} = \sin l, \end{aligned}$$

and thence

$$\tan\theta\,\Delta\,(b,\,\theta) = \cot l'\,\sin l = \frac{\cos l}{\sqrt{1-\delta^2}};$$

whence

$$\Pi\,(n,\,c) = \frac{\sin l'\cos l'}{\sin l}\left\{\tfrac{1}{2}\pi + F,\,c\left[\frac{\cos l}{\sqrt{1-\delta^2}} + F\,(b,\,\theta) - E\,(b,\,\theta)\right] - E,\,c\,F\,(b,\,\theta)\right\}.$$

But

$$n + c^2 = \tan^2 l' + \delta^2 \sin^2 l = \sin^2 l \sec^2 l'.$$

Hence

$$(n + c^2)\,\Pi\,(n,\, c) - c^2 F,\, c = \sin^2 l \bigl\{\sec^2 l'\,\Pi\,(n,\, c) - \delta^2 F,\, c\bigr\},$$

and multiplying this by $\dfrac{\sqrt{1-\delta^2}}{n\cos l}$, $= \dfrac{\sqrt{1-\delta^2}\cos l}{\tan^2 l' \cos^2 l'}$, the exterior factor is

$$\frac{\sqrt{1-\delta^2}\cos l \tan^2 l'}{\tan^2 l' \cdot \sqrt{1-\delta^2}} = \frac{\cos l}{\sqrt{1-\delta^2}},$$

and we have

$$\Lambda = \frac{\cos l}{\sqrt{1-\delta^2}}\bigl\{\sec^2 l'\,\Pi\,(n,\, c) - \delta^2 F,\, c\bigr\},$$

which is the formula I used in the calculations. It would, however, have been better to reduce a step further; viz. we have

$$\begin{aligned}\sec^2 l'\,\Pi\,(n,\, c) &= \frac{\sqrt{1-\delta^2}}{\cos l}\left\{\tfrac{1}{2}\pi + F,\, c\left[\frac{\cos l}{\sqrt{1-\delta^2}} + F\,(b,\, \theta) - E\,(b,\, \theta)\right] - E,\, c\,F\,(b,\, \theta)\right\} \\ &= \frac{\sqrt{1-\delta^2}}{\cos l}\,\bigl\{\tfrac{1}{2}\pi + F,\, c\bigl[F\,(b,\, \theta) - E\,(b,\, \theta)\bigr] - E,\, c\,F\,(b,\, \theta)\bigr\} + F,\, c,\end{aligned}$$

and thence

$$\sec^2 l'\,\Pi\,(n,\, c) - \delta^2 F,\, c = \frac{\sqrt{1-\delta^2}}{\cos l}\bigl\{\tfrac{1}{2}\pi + F,\, c\bigl[F\,(b,\, \theta) - E\,(b,\, \theta)\bigr] - E,\, c\,F\,(b,\, \theta)\bigr\} + \sqrt{1-\delta^2}\,F,\, c;$$

or, finally,

$$\Lambda = \tfrac{1}{2}\pi + F,\, c\,F\,(b,\, \theta) - F,\, c\,E\,(b,\, \theta) - E,\, c\,F\,(b,\, \theta) + \sqrt{1-\delta^2}\cos l\,F,\, c.$$

It is easy with this expression of Λ to obtain the results already found for the extreme values $l' = 0°$, $l' = 90°$.

As Legendre's Tables have for argument, not the modulus c, but the angle of the modulus, say χ (that is, $\sin\chi = c = \delta \sin l$), it is convenient to replace $\sqrt{1-\delta^2\sin^2 l}$ by its value $\cos\chi$; and the formulæ thus are

$$\begin{aligned}s &= \frac{\sqrt{1-\delta^2}}{\cos\chi}\,E,\, c, \\ \Lambda &= \tfrac{1}{2}\pi + F,\, c\Bigl[\sqrt{1-\delta^2}\cos l + F\,(b,\, \theta) - E\,(b,\, \theta)\Bigr] - E,\, c\,F\,(b,\, \theta),\end{aligned}$$

where

$$c = \sin\chi = \delta\sin l, \qquad \tan l' = \sqrt{1-\delta^2}\,\tan l, \qquad \theta = 90° - l';$$

and in the case intended to be numerically discussed, $\delta = \tfrac{1}{2}\sqrt{3}$, $\sqrt{1-\delta^2} = \tfrac{1}{2}$. I take l' as the argument, giving it the values $0°$, $10°$, …, $90°$, and perform the calculation as shown in the Table.

[Table: The Table of numerical values of l, χ, $90° - \chi$, $\tfrac{1}{2}\cos l$, $F,\, c$, $E,\, c$, $F\,(b,\, \theta)$, $E\,(b,\, \theta)$, their combinations, Λ in radians and degrees, and the length s, for $l' = 0°$, $10°$, …, $90°$, as displayed on page 23 of the original. Columns marked with an asterisk show respectively the longitude of the node and the length (or distance of node from vertex) for the geodesic lines belonging to the different values of the argument l'.]

The remarks which follow have reference to the stereographic projection of the figure on the plane of the equator, the centre of projection being the pole (say the South Pole) of the spheroid.

It is to be remarked that if a point P of the spheroid is projected as above, by means of an ordinate into the point Q of the sphere radius $OK\ (= OA)$, then projecting stereographically as to the spheroid and the sphere from the south poles thereof respectively, the points P and Q have the same projection. And it is hence easy to show that an azimuth α at a point of the meridian (parametric latitude λ', normal latitude λ, and therefore $\tan\lambda' = \frac{C}{A}\tan\lambda$) is projected into an angle (α) such that

$$\tan(\alpha) = \frac{\sin\lambda'}{\sin\lambda}\tan\alpha.$$

In fact in fig. 3, if we take therein OK, OC for the axes of x, z respectively, and the axis of y at right angles to the plane of the paper, and if we have at P on the surface of the spheroid an element of length PR at the inclination α to the meridian PK, then if x, y, z are the coordinates of P, and $x+\delta x$, $y+\delta y$, $z+\delta z$ those of R, we have

$$\begin{aligned}\delta x &= \rho\cos\alpha\sin\lambda,\\ \delta z &= -\rho\cos\alpha\cos\lambda,\\ \delta y &= \rho\sin\alpha,\end{aligned}$$

and thence

$$\tan\alpha = \frac{\delta y}{\sqrt{\delta x^2+\delta z^2}}.$$

Now, if the meridian and the points P, R are referred by lines parallel to Oz to the surface of the sphere radius OA, the only difference is that the ordinates z are increased in the ratio $C : A$; so that if the projected angle be (α), we have

$$\tan(\alpha) = \frac{\delta y}{\sqrt{\delta x^2 + \frac{A^2}{C^2}\delta z^2}};$$

and then projecting the sphere stereographically from its south pole, the angle in the projection is $= (\alpha)$. And according to the foregoing remark, the angle (α) thus obtained is also the projection of α from the south pole of the spheroid. We have thus

$$\frac{\tan(\alpha)}{\tan\alpha} = \frac{\sqrt{\delta x^2+\delta z^2}}{\sqrt{\delta x^2+\frac{A^2}{C^2}\delta z^2}} = \frac{\sqrt{\sin^2\lambda+\cos^2\lambda}}{\sqrt{\sin^2\lambda+\frac{A^2}{C^2}\cos^2\lambda}} = \sqrt{\frac{1+\cot^2\lambda}{1+\cot^2\lambda'}} = \frac{\sin\lambda'}{\sin\lambda},$$

which is the required relation.

The foregoing equations,

$$\cos\lambda'\sin\alpha = \cos l', \qquad \tan\lambda' = \frac{C}{A}\tan\lambda, \qquad \tan(\alpha) = \frac{\sin\lambda'}{\sin\lambda}\tan\alpha,$$

determine in the stereographic projection the inclination (α) to the radius, or projection of the meridian, of the geodesic line (parametric latitude of vertex $= l'$) at the point the parametric latitude of which is $= \lambda'$; viz. they enable the construction (in the projection) of the direction of the successive elements of the geodesic line. There would be no difficulty in performing the construction geometrically; but it would, I think, be more convenient to calculate (α) numerically for a given value of l' and for the successive values of λ'. Observe that for $\lambda' = 0$ we have (as above) $90^\circ - \alpha = l'$, and then $\frac{\sin\lambda'}{\sin\lambda} = \frac{\tan\lambda'}{\tan\lambda} = \frac{C}{A}$, consequently $\tan(\alpha) = \frac{C}{A}\cot l'$: but we have also $\cot l' = \frac{A}{C}\cot l$, so that this equation becomes $\tan(\alpha) = \cot l$, or we have $90^\circ - (\alpha) = l$; viz. in the projection, the geodesic line cuts the equator at an angle $l =$ the normal latitude of the vertex of the geodesic line.

The preceding formulæ and results have enabled me to construct a drawing, on a large scale, of the stereographic projection of the geodesic lines for the spheroid, polar axis $= \frac{1}{2}$ equatorial axis.

[End of pages 1–25. Paper 422 continues to pages 26 ff. in subsequent files.]

[423]

On the Plane Representation of a Solid Figure.

[From the Philosophical Magazine, vol. XLI. (1871), pp. 286–290.]

WE represent *in plano* the position of a point P whose coordinates in space are (x, y, z) by drawing these coordinates, on the same scale or on different scales, and in given directions from a fixed origin in the plane; $OM = x$, $MP' = y$, $P'P'' = z$. But observe that the point P'' *alone* does not completely represent the point P; in fact P'' represents a whole series of points lying in a line; any one such point is the point whose coordinates are Om, mp', $p'P''$. For the complete representation of P we require the *two points* P', P'': these might be distinguished as the projection P'', and the foot-point P'. The two points P', P'' are obviously such that the line joining them is in a given direction.

The preceding is, of course, the ordinary method of orthogonal projection, or geometrical delineation of a solid figure: it may be used under various forms; for example, the coordinates x, y, z may be taken on the same scale and in directions inclined to each other at angles of 120° (isometrical projection); or the coordinates x, y may be drawn on the same scale and at their actual inclination, 90°, to each other; and the coordinate z on the same or an altered scale in any given direction; the points P' then give a true ground-plan of the solid figure, and the lengths of the lines $P'P''$ give the altitudes of the several points P: this is also a method in ordinary use.

But it is to be observed that the points P', P'' are both of them *projections*, and that the general theory is as follows: we represent the position of the point P by means of its projections P', P'', from two fixed points Ω', Ω'' respectively; the line joining these points passes, it is clear, through a fixed point Ω which is the intersection of the plane of projection by the line which joins the two points Ω', Ω''.

Hence we say that a point P in space is represented *in plano* by any two points P', P'' which are such that the line joining them passes through a fixed point Ω. And we have thus a *system of constructive geometry* which is the more simple on account of the generality of its basis, and which is at once applicable to any of the special projections above referred to. I establish the fundamental notions of such a geometry, and by way of illustration apply it to the solution of the well-known problem of finding the lines which meet four given lines in space.

A point P (as already mentioned) is given by its projections P', P'', which are points such that the line joining them passes through the fixed point Ω.

A line L is given by its projections L', L'', which are any two lines in the plane. We speak of the point (P', P''), meaning the point P whose projections are P', P''; and similarly of the line (L', L''), meaning the line whose projections are L', L''.

If P', P'' coincide, then the point P is in the plane of projection; and so if L', L'' coincide, then the line L is in the plane of projection.

If through Ω we draw a line meeting L', L'' in the points P', P'' respectively, these are the projections of a point P on the line L. In particular the intersection of L', L'' (considered as two coincident points) represents the intersection of the line L with the plane of projection.

The line through the points (P', P'') and (Q', Q'') has for its projections the lines $P'Q'$ and $P''Q''$.

Two lines (L', L'') and (M', M'') intersect each other if only the intersections $L'M'$ and $L''M''$ are the projections of a point P—that is, if the line through the points $L'M'$ and $L''M''$ passes through Ω. And then clearly P is the intersection of the two lines.

A plane Π is conveniently given by means of its trace Θ on the plane of projection, and of the projections (P', P'') of a point on the plane; or, say, by means of the trace Θ, and of a point P on the plane.

Suppose, however, that a plane is given by means of a line L and a point P on the plane. The trace Θ passes through the point of intersection of the line L with the plane of projection—that is, through the point of intersection of the projections L', L''. To find another point on the trace, we have only to imagine on the line L a point Q, and, joining this with P, to suppose the line PQ produced to meet the plane of projection. The construction is obvious; but by way of illustration I give it in full. Through Ω draw a line meeting L', L'' in Q', Q'' respectively (then these are the projections of a point Q on the line L); the lines $P'Q'$ and $P''Q''$ are the projections of the line PQ, and the intersection of $P'Q'$ and $P''Q''$ is therefore the required point on the trace Θ.

The line of intersection of two planes passes through the point of intersection of their traces Θ_1, Θ_2; whence, if the planes have in common a point P, the line of intersection is the line joining P with the intersection of the traces Θ_1, Θ_2.

In what precedes we have the solution of the following problem:—"Given a point P, and two lines L_1, L_2, to find a line through P meeting the two lines L_1, L_2." The required line is in fact the line of intersection of the planes (P, L_1) and (P, L_2); we have seen how to construct the traces Θ_1 and Θ_2 of these planes respectively; and the required line is the line joining P with the intersection of Θ_1 and Θ_2.

I proceed now to the problem to find the two lines, each of them meeting four given lines, L_1, L_2, L_3, L_4 (these being, of course, given by means of their projections (L_1', L_1'') &c.). The question is in effect to find on the line L_1 a point P such that, drawing from it a line to meet L_2, L_3, and also a line to meet L_3, L_4, these shall be one and the same line.

Now, considering in the first instance P as an arbitrary point on the line L_1, the line from P to meet L_2, L_3 is any line whatever meeting the lines L_1, L_2, L_3: say it is a generating line of the hyperboloid whose directrices are L_1, L_2, L_3, or of the hyperboloid $L_1L_2L_3$. Hence projecting from any point Ω' whatever, the generating lines and directrices are projected into tangents of one and the same conic. We know the projections L_1', L_2', L_3' of the directrices; to find two other tangents of the conic, we take two arbitrary positions of P on the line L_1, and construct as above the projections M', N' of the lines from these to meet the lines L_2, L_3. The conic is then given as the conic touching the five lines L_1', L_2', L_3', M', N': say this is the conic Σ'. Similarly, instead of Ω', considering the point Ω'', we have the lines L_1'', L_2'', L_3'', and the lines M'', N'', which are the other projections of the lines through the two positions of P; and touching these five lines we have a conic Σ''. Each tangent T' of Σ', combined with the corresponding tangent T'' of Σ'', represents a line T meeting L_1, L_2, L_3; to establish the correspondence, observe that, inasmuch as the line T meets L_1, the intersections of T', L_1' and of T'', L_1'' must lie in a line with Ω; if T' be given, the point (T'', L_1'') is thus uniquely determined, and therefore also T'' (since L_1'' is a tangent of Σ''); and similarly if T'' be given, T' is uniquely determined; the correspondence T', T'' is thus, as it should be, a (1, 1) correspondence.

Considering in like manner the lines which meet L_1, L_3, L_4, we have touching L_1', L_3', L_4', M', N' a conic Σ_1'; and touching L_1'', L_3'', L_4'', M'', N'' a conic Σ_1''; each tangent T' of Σ_1', combined with the corresponding tangent T'' of Σ_1'', represents a line meeting L_1, L_3, L_4, the correspondence being a (1, 1) correspondence such as in the former case.

The conics Σ', Σ_1' both touch L_1', L_3'; hence they have in common two tangents. Say one of these is $T' = T_1'$; the corresponding tangents T'' and T_1'' will coincide with each other and be a common tangent of Σ'', Σ_1'' (these conics both touch L_1'', L_3'', and have thus in common two tangents). We have thus $T' = T_1'$, and $T'' = T_1''$, as the projections of a line meeting L_1, L_2, L_3, L_4; and taking the other common tangents of Σ', Σ_1' and of Σ'', Σ_1'', we have the projections of the other line meeting L_1, L_2, L_3, L_4.

The whole process is:—Construct M', M'' and N', N'' each of them the projections of a line through a point P of L_1, which meets L_2, L_3; and M_1', M_1'' and N_1', N_1'' each of them the projections of a line through a point P of L_1, which meets L_3, L_4; we have then the conics

$$\Sigma',\ \Sigma''\quad\text{touching}\quad L_1',\ L_2',\ L_3',\ M',\ N' \text{ and } L_1'',\ L_2'',\ L_3'',\ M'',\ N'' \text{ respectively,}$$
$$\Sigma_1',\ \Sigma_1''\quad\text{touching}\quad L_1',\ L_3',\ L_4',\ M_1',\ N_1' \text{ and } L_1'',\ L_3'',\ L_4'',\ M_1'',\ N_1'' \text{ respectively;}$$

and then the projections of each of the required lines are $T' = T_1'$, a common tangent of Σ', Σ_1', and $T'' = T_1''$, the corresponding common tangent of Σ'', Σ_1''.

It is material to remark how the construction is simplified when there is given one of the lines, say M, which meets L_1, L_2, L_3, L_4. Here if M is a common directrix of the two hyperboloids, we may for the hyperboloids Σ' and Σ'' consider, instead of L_1, L_2, L_3 and two new generating lines, the lines L_1, L_2, L_3, M, and a single new generating line N; and similarly for the hyperboloids Σ_1', Σ_1'' the lines L_1, L_3, L_4, M, and a single new generating line N_1. Σ', Σ_1' have thus in common the three tangents L_1', L_3', M', and therefore only a single other common tangent, $T' = T_1'$; and similarly Σ'', Σ_1'' have in common the three tangents L_1'', L_3'', M'', and therefore only a single other common tangent, $T'' = T_1''$; and we have thus the other line cutting the four given lines.

I take the opportunity of mentioning the following theorem:

"If in a given triangle we inscribe a variable triangle of given form, the envelope of each side of the variable triangle is a conic touching the two sides (of the given triangle) which contain the extremities of the variable side in question."

We have thence a solution of the problem (*Principia*, Book I. Sect. V. Lemma XXVII.), in a given quadrilateral to inscribe a quadrangle of given form. The question in effect is: in the triangle ABC to inscribe a triangle $\alpha\beta\gamma$ of given form; and in the triangle ADE a triangle $\alpha'\beta'\gamma'$ of given form, in such wise that the sides $\alpha\gamma$, $\alpha'\gamma'$ may be coincident. The envelope of $\alpha\gamma$ is a conic touching AD, AE, and the envelope of $\alpha'\gamma'$ a conic also touching AD, AE: there are thus two other common tangents, either of which may be taken for the position of the side $\alpha\gamma = \alpha'\gamma'$; and the problem admits accordingly of two solutions.

[424]

On the Attraction of a Terminated Straight Line.

[From the Philosophical Magazine, vol. XLI. *(1871), pp. 358–360.]*

WRITE for shortness $(a,\, b,\, c\,;\, \varepsilon)$ to denote the shell included between the ellipsoids

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1 \quad\text{and}\quad \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=(1+\varepsilon)^2,$$

(where ε is indefinitely small); then, if the ellipsoids

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1 \quad\text{and}\quad \frac{x^2}{a'^2}+\frac{y^2}{b'^2}+\frac{z^2}{c'^2}=1$$

are confocal, the attractions of the shells $(a,\, b,\, c\,;\, \varepsilon)$ and $(a',\, b',\, c'\,;\, \varepsilon)$ upon any exterior point P are proportional to their masses. Hence, considering a prolate spheroid of revolution, $c=b$, the attractions of the shell $(a,\, b,\, b\,;\, \varepsilon)$ will be proportional to those of the shell $\left(\sqrt{a^2-h},\ \sqrt{b^2-h},\ \sqrt{b^2-h}\,;\, \varepsilon\right)$; or if, as usual, $b^2=a^2(1-e^2)$, then, if h increases and becomes ultimately equal to b^2, to those of the shell $(ae,\, 0,\, 0\,;\, \varepsilon)$; viz. this last is the portion of the axis of x included between the limits $x=-ae$, $x=+ae$; or say it is the terminated line $x=\pm\, ae$; and I say that the mass is distributed over this line uniformly.

To see that this is so, observe in general that, in the spheroid $\dfrac{x^2}{a^2}+\dfrac{y^2+z^2}{b^2}=1$, the volume included between the planes $x=\alpha$, $x=\alpha+d\alpha$, is $=\pi(y^2+z^2)\,d\alpha\ =\pi b^2\left(1-\dfrac{\alpha^2}{a^2}\right)d\alpha$; and thence, writing $a(1+\varepsilon)$, $b(1+\varepsilon)$ for a, b, in the shell $(a,\, b,\, b\,;\, \varepsilon)$ the volume included between the planes $x=\alpha$, $x=\alpha+d\alpha$ is $=\pi b^2\cdot 2\varepsilon\, d\alpha$; viz. this is independent of α, and simply proportional to $d\alpha$. Hence, writing $b=0$, when the shell shrinks up into a line, the mass must be distributed uniformly over the line. It follows that for a line of uniform density the equipotential surfaces are each of them a prolate spheroid of revolution having the extremities of the line for its foci, and that, if we have a shell bounded by any such surface and the consecutive similar surface, with its mass equal to that of the line, then such shell and the line will exert the same attractions upon any point P exterior to the shell. The attractions of the line are most easily obtained by means of its potential V; viz. taking S, H for the extremities of the line, and, as above, the origin at the middle point, and the axis of x in the direction of the line, and writing $2ae$ for the length of the line, x, y, z for the coordinates of P, and r, s for the values of HP, SP $\big(thatis,\text{r} = \sqrt{(x-ae)^2+y^2+z^2},\ s = \sqrt{(x+ae)^2+y^2+z^2}\big), thenthepotentialisatoncefoundtobeV = \log\frac{x+ae+s}{x-ae+r}$, and we can hereby verify that the equipotential surface is in fact a spheroid of revolution having the foci S, H; for, taking the equation of such a spheroid to be

$$\frac{x^2}{\alpha^2} + \frac{y^2+z^2}{\alpha^2 - a^2e^2} = 1$$

(α is an arbitrary parameter, since only the value of ae has been defined), we have

$$s = \alpha + ex, \qquad r = \alpha - ex,$$

and thence

$$\begin{aligned} x + ae + s &= (1+e)(x+\alpha), \\ x - ae + r &= (1-e)(x+\alpha), \end{aligned}$$

and the quotient is $= \dfrac{1+e}{1-e}$, a constant value, as it should be. The equation $V =$ const. may in fact be written

$$\frac{1+e}{1-e} = \frac{x+ae+s}{x-ae+r},$$

viz. this equation, apparently of the fourth order, breaks up into the twofold plane $\dfrac{x^2}{a^2} + \dfrac{y^2+z^2}{a^2(1-e^2)} = 1$, and the spheroid $y^2 = 0$.

The foregoing results in regard to the attraction of a line are not new. See Green's *Essay on Electricity*, 1828, and *Collected Works*, Cambridge, 1871, p. 68; also

Joachimsthal, "On the Attraction of a Straight Line," with Sir W. Thomson's Note, *Camb. and Dubl. Math. Journ.*, vol. III. (1848), p. 93; but it does not appear to have been noticed that they are, in fact, included in the theory of the attraction of ellipsoids.

The like considerations show that the attractions of the ellipsoidal shell $(a, b, c; \varepsilon)$ upon an exterior point are equal to those of an elliptic disk $z = 0$, $\frac{x^2}{a^2 - c^2} + \frac{y^2}{b^2 - c^2} = 1$, the mass of which is equal to that of the shell, and which has the density at the point (x, y) proportional to $\left(1 - \frac{x^2}{a^2 - c^2} - \frac{y^2}{b^2 - c^2}\right)^{-\frac{1}{2}}$.

Sir W. Thomson informs me that the foregoing results have long been familiar to him.

[425]

Note on the Geodesic Lines on an Ellipsoid.

[From the Philosophical Magazine, vol. XLI. (1871), pp. 534, 535.]

THE general configuration of the geodesic lines on an ellipsoid is established by means of the known theorem (an immediate consequence of Jacobi's fundamental formulæ, but which was first given by Mr Michael Roberts, *Comptes Rendus*, vol. XXI. p. 1470, Dec. 1845) that every geodesic line touches a curve of curvature; that is, attending to the two opposite ovals which constitute the curve of curvature, the geodesic line is in general an infinite curve undulating between these opposite ovals, and so touching each of them an infinite number of times (but possibly in particular cases it is a re-entrant curve touching each oval a finite number of times). The geodesic lines thus divide themselves into two kinds, according as they touch a curve of curvature of the one or the other kind; and there is besides a third limiting kind, the lines which pass through an umbilicus: any such geodesic line passes through the opposite umbilicus, and is in general an infinite curve passing an infinite number of times alternately through the two umbilici; but possibly it is in particular cases a re-entrant curve passing a finite number of times through the two umbilici. I annex a figure giving a general idea of the configuration of the geodesic lines drawn in different directions from a given point P on the surface of the ellipsoid: this is drawn (as it were) on the plane of the greatest and least axes; but it is not a perspective or geometrical representation of any kind, but a mere diagram for the purpose in question. We have A, A_1, B, C, C_1 the extremities of the axes; U_1, U_2, U_3, U_4 the umbilici; P the point on the surface; $1P2$ and $1P4$ the curves of curvature through P, viz. these are ovals containing the umbilici U_1, U_2 and U_1, U_4 respectively. Then U_1PU_3 and U_4PU_2 are the limiting geodesics passing through the umbilici; the line TPT' represents a geodesic line of the one kind, viz. this at T touches an oval (curve of curvature) U_1U_4, and at T' the conjugate oval U_2U_3. Similarly SPS' is a geodesic line of the other kind, viz. this at S touches an oval (curve of curvature) U_1U_2, and at S' the conjugate oval U_3U_4; the dotted figure-of-eight curves are the loci of the points of contact T, T', S, S'.

[426]

On a supposed New Integration of Differential Equations of the Second Order.

[From the Philosophical Magazine, vol. XLII. (1871), pp. 197–199.]

THIS refers to a paper, Challis, "On the Application of a new Integration of Differential Equations of the Second Order to some unsolved Problems in the Calculus of Variations," *Phil. Mag.* same volume, pp. 28–40.

[427]

On Gauss's Pentagramma Mirificum.

[From the Philosophical Magazine, vol. XLII. *(1871), pp. 311, 312.]*

TAKE on a sphere (in the northern hemisphere) two points, A, B, whose longitudes differ by 90°, and refer them to the equator by the meridians AE and BG respectively; join A, B by an arc of great circle, and take in the southern hemisphere the pole D of this circle; and join D with E and C respectively by arcs of great circle. We have a spherical pentagon $ABCDE$, which is in fact the "Pentagramma mirificum," considered by Gauss, as appearing vol. III. pp. 481–490 of the *Collected Works*. Among its properties we have

$$\left.\begin{array}{l}\text{the distance of any two non-adjacent summits}\\ \text{the inclination of any two non-adjacent sides}\end{array}\right\} = 90^\circ;$$

so that each summit is the pole of the opposite side, or the pentagon is its own reciprocal.

Each angle is the supplement of the opposite side.

If the squared tangents of the sides (or angles) taken in order are α, β, γ, δ, ε, then

$$1+\alpha=\gamma\delta,\quad 1+\beta=\delta\varepsilon,\quad 1+\gamma=\varepsilon\alpha,\quad 1+\delta=\alpha\beta,\quad 1+\varepsilon=\beta\gamma,$$

equivalent to three independent equations, so that any three of the quantities may be expressed in terms of the remaining two. (This agrees with the foregoing construction, where the arbitrary quantities are the latitudes of A, B respectively.)

Projecting from the centre of the sphere upon any plane, we have a plane pentagon which is such that the perpendiculars let fall from the summits upon the opposite sides respectively meet in a point. This (as easily seen) implies that the two portions into which each perpendicular is divided by the point in question have the same product.

Conversely, starting from the plane pentagon, and erecting from the point of intersection a perpendicular to the plane, the length of this perpendicular being equal to the square root of the product in question, we have the centre of a sphere such that the projection upon it of the plane polygon is the pentagramma mirificum.

I remark as to the analytical theory, that, taking the origin at the intersection of the perpendiculars, and for the coordinates of the summits (α_1, β_1), …, (α_5, β_5) respectively, then we have

$$\alpha_1\alpha_3 + \beta_1\beta_3 = \alpha_2\alpha_4 + \beta_2\beta_4 = \alpha_3\alpha_5 + \beta_3\beta_5 = \alpha_4\alpha_1 + \beta_4\beta_1 = \alpha_5\alpha_2 + \beta_5\beta_2 = -\gamma^2,$$

where γ^2 is the above-mentioned product, or γ is the radius of the sphere.

Cambridge, September 14, 1871.

[428]

Note sur la Correspondance de Deux Points sur une Courbe.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXII. *(Janvier–Juin, 1866), pp. 586–590.]*

THIS is to the same effect with the paper 385, "On the Correspondence of two Points on a Curve"; four examples of the theory are given, the first, second, and third of them the same as in this paper—the fourth example is as follows:

4°. *Recherche du nombre des points sextactiques*, c'est-à-dire des points qui sont tels que par chacun passe une conique qui a dans ce point un contact du cinquième ordre avec la courbe. Il faut prendre pour les points P les intersections avec la courbe de la conique qui a au point P' un contact du quatrième ordre: les points unis seront ceux dont il s'agit. La courbe $\theta = 0$ est la conique qui a au point P' un contact du quatrième ordre. On a ainsi parmi les intersections le point P' 5 fois; donc $k = 5$. À chaque point P' correspondent $2m - 5$ points P; à chaque point P, $10m^2 - 20m - 5 - 20\delta$ points P' (j'emprunte le terme -20δ d'une formule que vient de donner M. Zeuthen); donc la formule donne pour le nombre des points unis

$$10m^2 - 18m - 10 - 20\delta + 10\delta,$$

c'est-à-dire

$$10m^2 - 18m - 10 - 20\delta.$$

Mais cette expression comprend le nombre $3m(m - 2) - 6\delta$ des inflexions; en effet pour un point d'inflexion la conique avec contact du quatrième ordre se réduit à la tangente prise deux fois, ce qui est une conique avec contact du cinquième ordre. Donc enfin le nombre des points sextactiques sera

$$m(12m - 27) - 24\delta,$$

ou, pour une courbe sans points doubles,

$$m(12m - 27),$$

ce qui s'accorde avec la valeur que j'ai trouvée par d'autres moyens.

[429]

Sur les Coniques déterminées par cinq Conditions de Contact avec une Courbe donnée.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXIII.
(Juillet–Décembre, 1866), pp. 9–12.]

THIS paper (dated Cambridge, 26 June 1866), contains the expressions for the numbers (5), (4, 1), (3, 2), (3, 1, 1), (2, 2, 1), (2, 1, 1, 1) and (1, 1, 1, 1, 1), of the conics which satisfy five conditions of contact with a given curve, as obtained in the paper 406 "On the Curves which satisfy given conditions," see p. 214, and which expressions were found by the same process, viz. by consideration of functional equations obtained by supposing the given curve to break up into two curves of the orders m and m' respectively; there was in the expression for (1, 1, 1, 1, 1) a numerical error as mentioned in the footnote of the same page. The paper contains also the formula $\mu'' + \iota'' - \tau'' + \rho'' - \sigma'' = 0$, and the expression for (2ν, 3ρ) given, pp. 203, 204.

[430]

Note sur quelques Formules de M. E. de Jonquières, relatives aux Courbes qui satisfont à des Conditions données.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXIII.
(Juillet–Décembre, 1866), pp. 666–670.]

LES formules dont il s'agit sont publiées dans les *Comptes Rendus*, séances du 3 et du 17 septembre 1866. En faisant une simple transformation algébrique pour y introduire la classe $M\,(= m^2 - m)$ de la courbe donnée U^m, et en changeant un peu la forme, les théorèmes de M. de Jonquières peuvent s'énoncer comme il suit:

1°. Le nombre des contacts des courbes C^r qui ont un contact de l'ordre n avec une courbe fixe U^m, et qui passent en outre par $\frac{1}{2}r(r+3) - n$ points donnés, est

$$= \tfrac{1}{2}(n+1)\bigl[nM + (2r - 2n)m\bigr].$$

Observation. Énoncé de cette manière, le théorème s'applique même au cas $n = 0$. En effet, pour $n = 0$, le nombre donné par le théorème est $= mr$, qui est le nombre des contacts de l'ordre 0 (intersections simples) de la courbe donnée U^m avec une courbe déterminée de l'ordre r.

2°. Le nombre des contacts de l'ordre n' ($=$ ou $< n$) des courbes C^r qui ont deux contacts des ordres n et n' respectivement avec une courbe fixe U^m, et qui passent en outre par $\frac{1}{2}r(r+3)-n-n'$ points donnés est

$$\begin{aligned} = \tfrac{1}{4}(n+1)(n'+1)\Bigl\{&[nM + (2r-2n)m]\,[n'M + (2r-2n')m] \\ &- 2(n^2 + nn' + n'^2 + n + n')\,M \\ &+ [-4r(n+n'+1) + 4(n^2 + nn' + n'^2 + n + n')]\,m\Bigr\}. \end{aligned}$$

Observation. Énoncé de cette manière, le théorème s'applique même aux cas $n' = 0$, et $n' = n$. En effet, pour $n' = 0$, le nombre donné par le théorème est $= (rm - n - 1) \cdot \frac{1}{2}(n + 1)\,[nM + (2r - 2n)m]$, ce qui est égal au nombre des courbes qui ont avec la courbe donnée U^m un contact de l'ordre n, multiplié par $rm - n - 1$, nombre des contacts de l'ordre 0 (intersections simples) de chacune de ces courbes avec la courbe U^m. Et pour $n' = n$, le nombre des contacts est le double du nombre des courbes C^r.

Je remarque que les deux théorèmes peuvent se démontrer de la manière dont je me suis servi en cherchant le nombre des coniques qui satisfont à cinq conditions données; car, en remplaçant la courbe U^m par l'ensemble de deux courbes m et m', on trouve que pour le théorème 1° le nombre cherché est

$$\tfrac{1}{2}(n+1)\,[nM + (2r-2n)m] + \alpha M + \beta m,$$

où les coefficients $(\alpha,\, \beta)$ ne dépendent que de $(r,\, n)$; et puis, en supposant que ce théorème soit connu, on trouve que pour le théorème 2° le nombre cherché est

$$\tfrac{1}{4}(n+1)(n'+1)\,[nM + (2r-2n)m]\,[n'M + (2r-2n')m] + \alpha M + \beta m,$$

où de même les coefficients $(\alpha,\, \beta)$ ne dépendent que de $(r,\, n)$.

Or voici comment on peut déterminer les coefficients dans les deux théorèmes:

Pour le théorème 1°, on démontre que pour U^m une droite, le nombre cherché est $= (n + 1)(r - n)$; et que pour U^m une conique, le nombre cherché se déduit de là en écrivant $2r$ au lieu de r; c'est-à-dire, que pour la conique, le nombre est $= (n + 1)(2r - n)$. On a donc

$$\begin{aligned} \alpha + \beta &= (n+1)(r-n), \\ 2\alpha + 2\beta &= (n+1)(2r-n), \end{aligned}$$

et de là

$$\alpha = \tfrac{1}{2}(n+1)n, \quad \beta = \tfrac{1}{2}(n+1)(2r-2n);$$

ce qui achève la démonstration.

Pour le théorème 2°, on démontre que pour U^m une droite, le nombre cherché est $= (n + 1)(n' + 1)(r - n - n')(r - n - n' - 1)$, et que pour U^m une conique, le nombre cherché se déduit de là en écrivant $2r$ au lieu de r; c'est-à-dire, pour la conique, le nombre est

$$= (n+1)(n'+1)(2r-n-n')(2r-n-n'-1).$$

On a donc

$$\begin{aligned} (n+1)(n'+1)(r-n-n')(r-n-n'-1) &= (n+1)(n'+1)(r-n)(r-n') + \alpha + \beta, \\ (n+1)(n'+1)(2r-n-n')(2r-n-n'-1) &= (n+1)(n'+1)(2r-n)(2r-n') + \alpha + 2\beta, \end{aligned}$$

cela donne pour α et β les valeurs

$$\begin{aligned} \alpha &= \tfrac{1}{4}(n+1)(n'+1)\,[-2(n^2 + nn' + n'^2 + n + n')], \\ \beta &= \tfrac{1}{4}(n+1)(n'+1)\,[-4r(n+n'+1) + 4(n^2 + nn' + n'^2 + n + n')], \end{aligned}$$

et la démonstration est ainsi achevée.

Je remarque que sous les formes ici données les deux théorèmes s'appliquent à une courbe U^m avec des points doubles, mais sans point de rebroussement.

Le théorème dont je me suis servi pour la détermination des coefficients peut s'énoncer sous la forme plus générale que voici, savoir:

En dénotant par $\phi(r, n, n', \ldots)$ le nombre des courbes C^r qui ont avec une droite donnée des contacts des ordres $n, n', \ldots$ et qui passent en outre par $\frac{1}{2}r(r+3)-n-n'\ldots$ points donnés, alors si, au lieu de la droite donnée, on a une conique donnée, le nombre des courbes C^r sera $=\phi(2r, n, n', \ldots)$.

En effet, l'équation de la courbe cherchée C^r contient des coefficients indéterminés, lesquels, par les conditions de passer par les points donnés, se réduisent linéairement à $n+n'+\ldots+1$ coefficients; en dénotant par $(A, B, \ldots)$ ces coefficients, l'équation de la courbe contiendra linéairement $(A, B, \ldots)$ et sera ainsi de la forme $(A, B, \ldots \between x, y, z)^r = 0$. L'équation de la droite donnée est satisfaite en prenant pour (x, y, z) des fonctions linéaires déterminées d'un paramètre variable θ; donc, en coupant la courbe C^r par la droite donnée, on obtient une équation $(A, B, \ldots \between \theta, 1)^r = 0$, et en exprimant que cette équation ait n racines égales, n' racines égales, etc., on obtient entre $(A, B, C, \ldots)$ des équations, lesquelles, en éliminant tous les coefficients, exceptés deux quelconques (A, B), conduisent à une équation finale $(A, B)^p = 0$, et le degré p de cette équation est ce qu'il s'agissait de trouver, le nombre des courbes C^r. Si au lieu d'une droite donnée on a une conique donnée, il n'y a rien à changer, sinon que les coordonnées (x, y, z) doivent être remplacées par des fonctions quadratiques de θ; on a ainsi une équation $(A, B, \ldots \between \theta, 1)^{2r} = 0$, qui conduit à une équation finale $(A, B)^{p'} = 0$, où p' est la même fonction de $(2r, n, n', \ldots)$ qu'est p de $(r, n, n', \ldots)$; et le nombre des courbes C^r est $= p'$. Le théorème est donc démontré. Et, précisément de la même manière, on démontre le théorème encore plus général:

En dénotant par $\phi(r, n, n', \ldots)$ le nombre des courbes C^r qui ont avec une droite donnée des contacts des ordres $n, n', \ldots$, et qui passent en outre par $\frac{1}{2}r(r+3)-n-n'\ldots$ points donnés, alors si, au lieu de la droite donnée, on a une courbe unicursale donnée de l'ordre m, le nombre des courbes C^r est $=\phi(mr, n, n', \ldots)$.

On aurait pu se servir directement de cela pour démontrer les théorèmes 1° et 2°. Par exemple, pour le théorème 1°, la considération de la courbe unicursale U^m donne

$$\alpha M + \beta m = \alpha(2m-2) + \beta m = (n+1)(mr-n),$$

c'est-à-dire

$$\alpha = \tfrac{1}{2}(n+1)n, \quad \beta = \tfrac{1}{2}(n+1)(2r-2n),$$

comme auparavant.

[431]

Sur la Transformation Cubique d'une Fonction Elliptique.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXIV. (Janvier–Juin, 1867), pp. 560–563.]

SOIT $U = (a,\, b,\, c,\, d,\, e \between x,\, 1)^4$ une fonction quartique quelconque de x: I, J les deux invariants:

$$(I = ae - 4bd + 3c^2, \quad J = ace - ad^2 - b^2e + 2bcd - c^3),$$

et prenons $\Omega = \dfrac{I^3 - 27J^2}{I^3}$ pour l'invariant absolu de U. Soient de même $U' = (a',\, b',\, \ldots \between x,\, 1)^4$ et $\Omega' = \dfrac{I'^3 - 27J'^2}{I'^3}$ l'invariant absolu de U'. En supposant que $\sqrt[4]{U}$, $\sqrt[4]{U'}$ soient les radicaux des deux fonctions elliptiques liées par la transformation du troisième ordre ou cubique, on peut se proposer la question quelle est la relation entre les deux invariants absolus Ω, Ω'? J'ai trouvé cette relation d'abord par des considérations géométriques qui me furent suggérées par une lettre de M. Sylvestre; puis je l'ai déduite des formules pour la transformation cubique données par M. Hermite, *Crelle*, t. LX., 1862, p. 304), et enfin, à l'aide d'une considération tirée de ces formules, j'ai réussi à l'obtenir au moyen des formules des *Fundamenta Nova*. Je vais donner ici cette dernière investigation de la relation dont il s'agit.

En supposant que les fonctions U, U' soient transformées linéairement en $(1 - x^2)(1 - k^2x^2)$, $(1 - y^2)(1 - \lambda^2y^2)$ respectivement, pour exprimer la liaison entre les modules k, λ, au lieu de l'équation explicite entre $\sqrt[4]{k}$, $\sqrt[4]{\lambda}$ (*Fund. Nova*, p. 23), je me sers des formules, p. 25, lesquelles en y écrivant

$$-\beta = \frac{\alpha + 2}{2\alpha + 1},$$

c'est-à-dire

$$2\alpha\beta + \alpha + \beta = -2,$$

deviennent

$$k^2 = -\alpha^2\beta, \quad \lambda^2 = \alpha\beta^2.$$

Les transformations linéaires donnent sans peine

$$\Omega = \frac{108k^2(k^2-1)^2}{(k^4+14k^2+1)^3}, \quad \Omega' = \frac{108\lambda^2(\lambda^2-1)^2}{(\lambda^4+14\lambda^2+1)^3},$$

et il s'agit entre ces équations d'éliminer α, β, k, λ de manière à obtenir une équation entre Ω, Ω'.

J'écris

$$\Omega = \frac{\frac{1}{4}(2\alpha+1)(\alpha+2)(\alpha-1)^4}{(\alpha^2+4\alpha+1)^3}, \quad \Omega' = \frac{\frac{1}{4}(2\beta+1)(\beta+2)(\beta-1)^4}{(\beta^2+4\beta+1)^3}.$$

L'équation entre α, β donne

$$2\beta+1 = \frac{-3(\alpha+1)}{2\alpha+1}, \quad \beta+2 = \frac{3(\alpha^2+4\alpha+1)}{2\alpha+1}, \quad \beta-1 = \frac{-3(\alpha+1)^2}{2\alpha+1}, \quad \beta^2+4\beta+1 = \frac{3(\alpha^2+4\alpha+1)}{(2\alpha+1)^2};$$

et on a de là

$$\beta' = \frac{\beta^2(\beta-1)^2(\beta+1)^2}{\beta^2(\beta+2)^2}$$

puis, en faisant attention à l'identité

$$(2\alpha+1)(\alpha+2)(\alpha-1)^4 + 27\alpha(\alpha+1)^4 = 2(\alpha^2+4\alpha+1)^3,$$

on obtient entre α', β', la relation très simple $\alpha' + \beta' = 1$.

L'expression de k^2 donne

$$\begin{aligned}
k^2 &= \frac{\alpha^2(\alpha+2)}{2\alpha+1}, \\
k^2 - 1 &= \frac{(\alpha-1)(\alpha+1)^2}{2\alpha+1}, \\
k^4 + 14k^2 + 1 &= \frac{1}{(2\alpha+1)^2}\big\{\alpha^4(\alpha+2)^2 + 14\alpha^2(\alpha+2)(2\alpha+1) + (2\alpha+1)^2\big\}, \\
&= \frac{1}{(2\alpha+1)^2}\big\{(\alpha^2+4\alpha+1)(\alpha^4+3\alpha^3+16\alpha^2+3\alpha+1)\big\},
\end{aligned}$$

et on a de là

$$\Omega = \frac{108\alpha^4(2\alpha+1)(\alpha+2)(\alpha-1)^2(\alpha+1)^4}{(\alpha^2+4\alpha+1)^3\cdot(\alpha^4+3\alpha^3+16\alpha^2+3\alpha+1)^3}.$$

Or on a

$$\begin{aligned}\alpha' - 1 &= \frac{\frac{1}{4}(2\alpha+1)(\alpha+2)(\alpha-1)^4 - (\alpha^2+4\alpha+1)^3}{(\alpha^2+4\alpha+1)^3} = \frac{-\frac{27}{4}\alpha(\alpha+1)^4}{(\alpha^2+4\alpha+1)^3},\\ 8\alpha' + 1 &= \frac{4(2\alpha+1)(\alpha+2)(\alpha-1)^4 + (\alpha^2+4\alpha+1)^3}{(\alpha^2+4\alpha+1)^3} = \frac{9(\alpha^4+3\alpha^3+16\alpha^2+3\alpha+1)^2}{(\alpha^2+4\alpha+1)^3},\end{aligned}$$

et de là, en formant l'expression de la fonction $\dfrac{-64\alpha'(\alpha'-1)^2}{(8\alpha'+1)^3}$, on la trouve égale à la valeur qui vient d'être donnée pour Ω en termes de α: on a donc

$$\Omega = \frac{-64\alpha'(\alpha'-1)^2}{(8\alpha'+1)^3}$$

et de même

$$\Omega' = \frac{-64\beta'(\beta'-1)^2}{(8\beta'+1)^3}.$$

Avec la relation $\alpha' + \beta' = 1$, l'élimination de α', β' entre ces équations ne présente pas de difficulté.

[432]

Théorème relatif à la Théorie des Substitutions.

Extrait d'une lettre adressée à M. J. A. Serret.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXVII. *(Juillet–Décembre, 1868), pp. 784, 785.]*

ON peut énoncer par rapport aux substitutions un théorème qui comprend les trois théorèmes III., IV., V., t. II. pp. 260–263 de votre *Cours d'Algèbre Supérieure.*

Pour un nombre quelconque μ on peut former avec les $\phi(\mu)$ nombres inférieurs et premiers à μ plusieurs systèmes de nombres lesquels sont chacun un système conjugué (mod. μ); c'est-à-dire que le produit de deux nombres quelconques d'un tel système est congru suivant le module μ, à un nombre du système. Comme cas extrêmes, l'unité est un tel système, et les $\phi(\mu)$ nombres forment aussi un système conjugué.

Pour μ premier, en dénotant par a une racine primitive de μ et par f un diviseur quelconque de $\mu-1$, les nombres $\equiv a^{fx}$ (mod. μ), x étant un entier quelconque, forment un système conjugué. Et généralement pour un nombre μ quelconque ce nombre a des racines quasi-primitives α, β, γ, …, aux exposants A, B, C, …, tels que $\alpha^A \equiv 1$ (mod. μ), $\beta^B \equiv 1$ (mod. μ), … et $ABC \cdots = \phi(\mu)$. En choisissant une combinaison quelconque, par exemple α, β de ces racines, soient f, g des diviseurs quelconques de A, B respectivement, les nombres $\equiv \alpha^{fx}\beta^{gy}$ (mod. μ) forment un système conjugué, l'ordre du système ou nombre des termes étant $= \dfrac{AB}{fg}$.

Cela étant, on a ce théorème:

Une substitution T quelconque de l'ordre μ étant formée avec n lettres, l'on forme toutes les substitutions S telles que le produit STS^{-1} se réduise à une puissance de T dont l'exposant soit un nombre quelconque appartenant à un système conjugué (mod. μ)*, les substitutions S constitueront un système conjugué de l'ordre θM, où θ dénote l'ordre du système conjugué* (mod. μ) *et M le nombre des substitutions échangeables avec T.*

La démonstration est tout à fait la même que celle que vous donnez p. 62 pour votre théorème IV, en y ajoutant seulement que les nombres i, j qui appartiennent au système conjugué (mod. μ) auront leur produit ij congru à un nombre de ce même système conjugué.

[433]

Sur les Surfaces Tétraédrales.

[Notes to the work De la Gournerie, "Recherches sur les surfaces réglées tétraédrales symétriques," 8vo. Paris, 1867.]

Premier Mémoire. Notes pp. 190–193.

Équations de certaines

1°. L'équation

$$\begin{aligned}
0 &= a^4f^4v^4 + b^4g^4u^4 + c^4h^4w^4 \\
&+ 2c^2ag\,(af - bg)\,y^2z^2 - 2c^2bf\,(af - bg)\,x^2z^2 \\
&- 2b^2ah\,(ch - af)\,z^2x^2 + 2b^2ch\,(ch - af)\,y^2z^2 \\
&+ 2a^2gh\,(bg - ch)\,w^2x^2 + 2a^2bh\,(bg - ch)\,z^2y^2 \\
&+ 2g^2hf\,(ch - af)\,w^2y^2 \\
&+ a^2\,(b^2y^2 + c^2h^2 - 4bg\,ch)\,y^2z^2 + b^2\,(c^2h^2 + a^2f^2 - 4ch\,af)\,z^2x^2 \\
&+ g^2\,(c^2h^2 + a^2f^2 - 4ch\,af)\,w^2y^2 \\
&+ f^2\,(b^2y^2 + c^2h^2 - 4bg\,ch)\,w^2x^2 \\
&+ 2bf\,(afbg + c^2h^2 - 2ch\chi)\,x^2z^2w^2 \\
&- 2ag\,(afbg + c^2h^2 - 2ch\chi)\,y^2w^2z^2 \\
&+ 2ah\,(chaf + b^2y^2 - 2bg\chi)\,z^2hu^2 - 2bh\,(bgch + a^2f^2 - 2af\chi)\,z^2\chi w^2 \\
&- 2gh\,(bcgh + af^2 - 2af\chi)\,w^2y^2z^2 - 2hf\,(cahf + by^2 - 2bg\chi)\,w^2z^2u^2 \\
&+ 2\Omega hf^2z^2w^2.
\end{aligned}$$

Surfaces du Huitième Ordre.

$$\begin{aligned}
&+ 2b^2cf\,(ch - af)\,x^2z^2 - 2bcf^2\,(bg - ch)\,x^2w^2 \\
&- 2a^2cg\,(bg - ch)\,y^2z^2 - 2cag^2\,(ch - af)\,y^2w^2 \\
&- 2abh^2\,(af - bg)\,z^2w^2 + 2h^2fg\,(af - bg)\,w^2z^2 \\
&- c^2\,(ch - af)\,z^2x^2 + c^2\,(a^2f^2 + b^2g^2 - 4afbg)\,x^4y^4 \\
&- 4\,(ch - af)\,w^2y^2 + h^2\,(a^2f^2 + b^2g^2 - 4afbg)\,w^2z^2 \\
&- 2cf\,(chaf + b^2f^2 - 2bg\chi)\,x^2w^2y^2 + 2bc\,(bcgh + a^2f^2 - 2af\chi)\,x^2y^2z^2 \\
&+ 2cg\,(bgch + a^2f^2 - 2af\chi)\,y^2x^2w^2 + 2ca\,(cahf + by^2 - 2bg\chi)\,y^2z^2x^2 \\
&+ 2ab\,(abfg + c^2h^2 - 2ch\chi)\,z^2x^2y^2 \\
&- 2\,(abfg + c^2h^2 - 2ch\chi)\,w^2x^2y^2.
\end{aligned}$$

(où, pour abréger, on a écrit $\chi = af + bg + ch$, et Ω est une quantité quelconque) est celle d'une surface du huitième ordre qui a pour courbes doubles les quatre coniques

$$\begin{array}{ll} x = 0, & + cy^2 - bz^2 + fw^2 = 0; \\ y = 0, & -cx^2 \phantom{{}+cy^2} + az^2 + gw^2 = 0; \\ z = 0, & + bx^2 - ay^2 \phantom{{}-bz^2} + hw^2 = 0; \\ w = 0, & -fx^2 - gy^2 - hz^2 = 0. \end{array}$$

En déterminant Ω, savoir, en écrivant $\lambda + \mu + \nu = 0$, $af\lambda^2 + bg\mu^2 + ch\nu^2 = 0$ (ce qui donne deux systèmes de valeurs de $\lambda : \mu : \nu$), et puis

$$\Omega = 4 \sum \frac{\mu^2\nu^2}{\lambda^2} (af + bg)^2\, ch - 2 \sum \frac{\mu^2\nu^2}{\lambda^2} (af + bg)^2 - 4\,(af - bg)\,(bg - ch)\,(ch - af),$$

la surface devient une surface réglée, savoir, la quadrispinale de M. de la Gournerie; et, en particulier, en supposant $\dfrac{1}{af} + \dfrac{1}{bg} + \dfrac{1}{ch} = 0$, on obtient

$$\Omega = (-2 - 4) = -6\,(af - bg)\,(bg - ch)\,(ch - af),$$

et la surface sera développable.

2°. L'équation

$$\begin{aligned} 0 = {} & a^2y^2z^4 + b^2z^2x^4 + c^2x^2y^4 + f^2z^4w^4 + g^2x^4w^4 + h^2y^4w^4 \\ & + 2bfx^2z^2w^2 - 2cfzx^2y^2w^2 + 2bczx^2y^2z^2 \\ & - 2agy^2z^2w^2 + \ldots + 2cgx^2y^2w^2 + 2cax^2y^2z^2 \\ & + 2ahz^2y^2w^2 - 2bhz^2x^2w^2 + \ldots + 2abx^2y^2z^2 \\ & - 2ghw^2y^2z^2 - 2hfw^2z^2x^2 - 2fgw^2x^2y^2 \\ & + 2\Omega\, x^2y^2z^2w^2 \end{aligned}$$

(où Ω est une quantité quelconque) est celle d'une surface du huitième ordre ayant pour courbes doubles les quatre courbes du quatrième ordre

$$\begin{array}{ll} x = 0, & -hz^2w^2 + gw^2y^2 - ay^2z^2 = 0; \\ y = 0, & -hz^2w^2 + fw^2x^2 + bz^2x^2 = 0; \\ z = 0, & + gw^2x^2 - fw^2y^2 + cx^2y^2 = 0; \\ w = 0, & - ay^2z^2 - bz^2x^2 - cx^2y^2 = 0. \end{array}$$

En écrivant

$$\lambda + \mu + \nu = 0, \quad af\lambda^2 + bg\mu^2 + ch\nu^2 = 0,$$

. . .

(ce qui donne quatre systèmes de valeurs de $\lambda : \mu : \nu$), et puis

$$\Omega = af\,\frac{\nu-\mu}{\lambda} + bg\,\frac{\lambda-\nu}{\mu} + ch\,\frac{\mu-\lambda}{\nu}\ :$$

la surface devient une surface réglée, savoir, la *quadricuspidale* de M. de la Gournerie ; et, en supposant

$$(af)^{\frac{1}{3}} + (bg)^{\frac{1}{3}} + (ch)^{\frac{1}{3}} = 0,$$

et

$$\lambda : \mu : \nu \ = \ (af)^{\frac{1}{3}} \ : \ (bg)^{\frac{1}{3}} \ : \ (ch)^{\frac{1}{3}},$$

ce qui donne

$$\Omega \ = \ \left[(af)^{\frac{1}{3}} - (bg)^{\frac{1}{3}}\right] \left[(bg)^{\frac{1}{3}} - (ch)^{\frac{1}{3}}\right] \left[(ch)^{\frac{1}{3}} - (af)^{\frac{1}{3}}\right] :$$

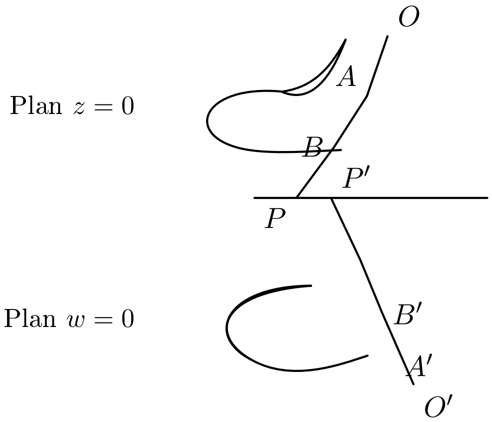
la surface devient développable.

Cambridge, 18 Octobre 1866.

Deuxième Mémoire. Notes pp. 279–283.

Note I. Sur la décomposition du lieu des génératrices en surfaces tétraédrales distinctes.

Il me semble qu'une de vos conclusions a besoin d'être modifiée. Ainsi la surface tétraédrale dérivée de deux courbes triangulaires à exposant $\frac{1}{m}$ (m étant un entier positif), laquelle, selon un de vos théorèmes, serait de l'ordre $2m^2$, paraît se décomposer en m surfaces chacune de l'ordre $2m$. Il y a pour cela une raison à *priori* ; en effet,



pour deux triangulaires de la forme en question, en employant votre construction, on peut établir une correspondance non-seulement entre les deux systèmes de points $A, B, \ldots$, et $A', B', \ldots$, mais aussi entre chaque point A et un seul point correspondant A', car l'équation de la première courbe étant de la forme

$$fx^{\frac{1}{m}} + gy^{\frac{1}{m}} + hz^{\frac{1}{m}} = 0,$$

on satisfait à cette équation en écrivant

$$x : y : z \;=\; a\,(\theta + \alpha)^m \;:\; b\,(\theta + \beta)^m \;:\; c\,(\theta + \gamma)^m,$$

où θ est un paramètre variable. De même, l'équation de la seconde courbe étant

$$f' x^{\frac{1}{m}} + g' y^{\frac{1}{m}} + k' w^{\frac{1}{m}} = 0,$$

on satisfait à cette équation en écrivant

$$x : y : w \;=\; a'\,(\theta' + \alpha')^m \;:\; b'\,(\theta' + \beta')^m \;:\; d'\,(\theta' + \delta')^m,$$

où θ' est aussi un paramètre variable.

Pour la droite OP, on a

$$\frac{x}{y} \;=\; \frac{a\,(\theta + \alpha)^m}{b\,(\theta + \beta)^m},$$

et pour la droite $O'P'$

$$\frac{x}{y} \;=\; \frac{a'\,(\theta' + \alpha')^m}{b'\,(\theta' + \beta')^m};$$

donc, la condition pour la correspondance des droites est

$$\frac{a\,(\theta + \alpha)^m}{b\,(\theta + \beta)^m} \;=\; \frac{a'\,(\theta' + \alpha')^m}{b'\,(\theta' + \beta')^m},$$

ce qui donne m valeurs différentes pour θ' en termes de θ. Mais chacune de ces valeurs est de la forme

$$\theta' \;=\; \frac{A\theta + B}{C\theta + D},$$

et, en ne faisant attention qu'à une seule valeur de θ', on a le point

$$x : y : z \;=\; a\,(\theta + \alpha)^m \;:\; b\,(\theta + \beta)^m \;:\; c\,(\theta + \gamma)^m,$$

qui correspond à un point unique

$$x : y : w \;=\; a'\,(\theta' + \alpha')^m \;:\; b'\,(\theta' + \beta')^m \;:\; d'\,(\theta' + \delta')^m.$$

Pour le cas de l'exposant $\frac{1}{3}$, on a, de cette manière, une surface de l'ordre 6. J'ai vérifié cela dans le cas particulier de la surface développable. Il est très-singulier (c'est M. Salmon qui m'a fait cette remarque) qu'en écrivant dans cette équation (x^2, y^2, z^2, w^2) au lieu de (x, y, z, w), on obtient l'équation d'une surface du douzième ordre, lieu des centres de courbure d'un ellipsoïde.

Cambridge, 15 Février 1866.

[433] p. 53

NOTE II. À L'OCCASION DE L'ORDRE DES SURFACES TÉTRAÉDRALES.

Je crois que j'ai négligé de vous faire connaître un théorème assez général au sujet de l'ordre de ces surfaces. En considérant dans l'espace deux courbes (planes ou à double courbure) des ordres m et m' respectivement, et en supposant qu'il y ait entre les points de ces deux courbes une correspondance (a, a'), (c'est-à-dire qu'à un point donné de la courbe m correspondent a' points sur la courbe m', et à un point donné de la courbe m' correspondent a points sur la courbe m), alors la surface réglée que l'on obtient en unissant par des droites les points correspondants des courbes m et m' sera de l'ordre $ma' + m'a$.

Cambridge, 18 Octobre 1866.

NOTE III. SUR LA SURFACE COMPLÉMENTAIRE.

Je puis reconnaître, par mes propres formules, que, des pq^2 surfaces de l'ordre $2p^2q$, il n'y en a que pq qui passent par la troisième directrice. En effet, le rapport anharmonique k est donné en termes des paramètres de la troisième directrice, au moyen d'une équation qui contient la quantité irrationnelle $k^{\frac{p}{q}}$. En rationalisant cette équation, on obtient pour k une équation de l'ordre pq ; à chaque racine k_1 correspondent q surfaces, savoir celles qui appartiennent aux q valeurs de $k_1^{\frac{p}{q}}$; mais l'équation irrationnelle n'est satisfaite que par une seule valeur de $k_1^{\frac{p}{q}}$, à savoir la valeur de $k_1^{\frac{p}{q}}$ donnée par l'équation irrationnelle, en y substituant pour k, en tant que k y entre rationnellement, la valeur $k = k_1$. Donc, à chaque racine k_1 correspond une seule surface qui passe par la troisième directrice. La question à laquelle donne lieu cette circonstance paraît très-intéressante. La surface déterminée par les trois directrices est composée de pq surfaces chacune de l'ordre $2p^2q$, et d'une surface résiduale de l'ordre $2p^2(q^2 - q)$. Quelles sont la nature et les propriétés de cette surface résiduale ? Je serais bien aise de savoir si vous avez fait des recherches à ce sujet.

Cambridge, 29 Mars 1866.

[434]

On certain Skew Surfaces, otherwise Scrolls.

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THE investigations contained in the present Memoir were suggested to me by the Memoirs of M. De la Gournerie, presented by him to the Academy of Sciences in 1865 and 1866, published in extract in the *Comptes Rendus*, and reproduced in his work "Recherches sur les surfaces réglées tétraédrales symétriques, par Jules De la Gournerie, avec des notes par Arthur Cayley," 8vo. Paris, 1867. Although the results, or the greater part of them, agree with those in the work just referred to, the mode of treatment is different, and more general, the orders, &c. of the different scrolls being obtained by considerations founded on the theory of Correspondence; and I have thought it not improper to submit to geometers in this altered form the theory of the very interesting class of Scrolls for which they are indebted to M. De la Gournerie's researches.

Article Nos. 1 to 10. Geometrical Construction of a Class of Scrolls.

1. Consider any two curves (plane or of double curvature) U, U', of the orders m, m' respectively, and let the points of U have with those of U' an (α, α') correspondence; viz. let the points of the two curves be so related that to each point of U correspond α' points of U', and to each point of U' correspond α points of U; then the lines joining the corresponding points of U, U' form a scroll the order of which is $= m\alpha' + m'\alpha$.

2. In particular let U, U' be plane curves in the planes Π, Π' respectively; and let the correspondence between the points of the two curves be established as follows; viz. consider in the plane Π a curve Ω of the class μ, and in the plane Π' a curve Ω' of the class μ'; and (to avoid useless generality) let the tangents of these two curves Ω, Ω' have to each other a $(1, 1)$ correspondence; that is, to each tangent of Ω there corresponds a single tangent of Ω', and to each tangent of Ω' a single tangent of Ω (this assumes that the curves Ω, Ω' are rational transformations one of the other, and that they have consequently the same Deficiency). This being so, let the points of U which lie on any tangent of Ω and the points of U' which lie on the corresponding tangent of Ω' be taken to be corresponding points of U, U'. The correspondence is then $(\mu m', \mu' m)$: in fact, through a given point of U there pass μ tangents of Ω, and the corresponding μ tangents of Ω' meet U' in $\mu m'$ points, that is, to a given point of U correspond $\mu m'$ points of U'; and similarly to a given point of U' correspond $\mu' m$ points of U. And hence the order of the scroll formed by the lines joining the corresponding points of U, U' is $= (\mu + \mu')\, mm'$.

3. This conclusion may be otherwise established as follows; let K, K' be any two corresponding points of U, U', so that the scroll we are concerned with is that generated by the series of lines KK'; and let I denote the line of intersection of the planes Π, Π'. The line I meets the curve U in m points, and taking one of these points for a point K we may from this point draw μ tangents to the curve Ω, that is, the point in question is a point K in respect of μ different tangents of the curve Ω; to each of these tangents there corresponds a single tangent of Ω', and such tangent of Ω' meets the curve U' in m' points, that is, to the point K in question there correspond $\mu m'$ points K' and consequently $\mu m'$ lines KK' in the plane Π'; hence to each of the m points K on the line I there correspond $\mu m'$ lines KK' in the plane Π', and we have thus $\mu mm'$ generating lines in the plane Π'; there are in like manner $\mu' mm'$ generating lines in the plane Π.

Take K an arbitrary point on the curve U; there are $\mu m'$ corresponding points K', and consequently $\mu m'$ generating lines through K, that is, through each point of the curve U; or the curve U (which is of the order m) is a $\mu m'$-tuple line on the scroll; similarly the curve U' (which is of the order m') is a $\mu' m$-tuple line on the scroll.

The complete section of the scroll by the plane Π' consists of the curve U taken $\mu m'$ times (order $\mu mm'$) and of the $\mu' mm'$ generating lines in the plane Π'; that is, the order of the section is $= (\mu + \mu')\, mm'$; and we thus see that the order of the scroll is $= (\mu + \mu')\, mm'$. Of course in like manner the complete section of the scroll by the plane Π' consists of the curve U' taken $\mu' m$ times (order $\mu' mm'$) and of the $\mu mm'$ generating lines in the plane Π', the order of the section being thus $= (\mu + \mu')\, mm'$.

4. There are on the scroll certain singular tangent planes; viz. if we have two corresponding tangents of Ω, Ω' meeting the line I in the same point, then we have m points K and m' points K' all lying in the plane of the two tangents; and of course the mm' lines KK' will all lie in the plane of the two tangents, that is, the intersection of the scroll by the plane in question will be made up of the mm' lines, and of a curve of the order $(\mu + \mu' - 1)\, mm'$; and the plane in question is thus a singular tangent plane.

5. The number of these singular tangent planes is $= \mu + \mu'$; in fact, considering as corresponding points on the line I, the intersection of this line by any tangent of Ω and the intersection by the corresponding tangent of Ω', the correspondence is obviously (μ, μ'); viz. through a given point P considered as belonging to the first system there pass μ tangents of Ω, and corresponding thereto we have μ tangents of Ω' each intersecting I in a single point P'; that is, to a given point P correspond μ points P'; and similarly to a given point P' correspond μ' points P. And this being so, the number of united points, that is, points of I through which pass corresponding tangents of Ω, Ω', is $= \mu + \mu'$.

6. In particular the curves Ω, Ω' may reduce themselves each to a point: the tangents to the two curves are here the lines passing through the points Ω, Ω' respectively: and the condition for the $(1, 1)$ correspondence of the two tangents is that the pencils of lines shall be homographically related; or, what is the same thing, that these two pencils shall determine on the line I two ranges which are homographically related; the entire construction is then as follows:

Given in the plane Π a curve U and a point Ω, and in the plane Π' a curve U' and a point Ω'; and taking in the plane Π a pencil of lines through Ω, and in the plane Π' a pencil of lines through Ω', in such wise that the two pencils correspond homographically; then if a line of the first pencil meets the curve U in the m points K, and the corresponding line of the second pencil meets the curve U' in the m' points K', the scroll in question is that generated by the mm' lines KK'.

7. By what precedes, the scroll is of the order $2mm'$; the curve U is a m'-tuple line, and the complete section by the plane Π is made up of this curve taken m' times and of mm' generating lines; similarly the curve U' is a m-tuple line, and the complete section by the plane Π' is made up of this curve taken m times and of mm' generating lines; there are two singular tangent planes such that the section by each of them is made up of mm' generating lines and of a curve of the order mm'; the planes in question are obviously those through the line $\Omega\Omega'$ and the coincident points of the two ranges on the line I, say the points A, B respectively.

8. The foregoing results will be modified in special cases. Suppose, for instance, that the curve U passes ω times, α times, and β times through the points Ω, A, B respectively, and that the curve U' passes ω' times, α' times, and β' times through the points Ω', A, B respectively. Then to each point on the curve U there correspond the $m' - \omega'$ intersections (other than the point Ω') on a line through Ω', so that U' is a $(m' - \omega')$-tuple line on the surface. The curve U' meets the line I in m' points; and corresponding to each of them we have a line through Ω meeting the curve U in $(m - \omega)$ points, exclusive of the point Ω; this would give $m'(m - \omega)$ generating lines in the plane Π; but among the m' points are included the point A α' times, and the point B β' times; the $(m - \omega)$ points corresponding to A include the point A α times, and we have thus the point A corresponding to itself $\alpha\alpha'$ times, and giving a reduction $= \alpha\alpha'$ in the number $m'(m - \omega)$ of generating lines; similarly the $m - \omega$ points corresponding to B

include the point B β times, and we have thus the point B corresponding to itself $\beta\beta'$ times and giving a reduction $= \beta\beta'$ in the number $m'(m-\omega)$ of generating lines; the number of generating lines in the plane Π is thus $= m'(m-\omega) - \alpha\alpha' - \beta\beta'$. The complete section by the plane Π is made up of the curve U $(m-\omega')$ times (order $m(m'-\omega')$) and of the $m'(m-\omega) - \alpha\alpha' - \beta\beta'$ generating lines; the order of the section, and consequently also the order of the scroll, is thus $= 2mm' - m\omega' - m'\omega - \alpha\alpha' - \beta\beta'$. It is clear that in like manner the curve U' is a $(m-\omega)$-tuple line on the surface, and that the complete section by the plane Π' is made up of this curve taken $(m-\omega)$ times, order $m'(m-\omega)$, and of $m(m'-\omega') - \alpha\alpha' - \beta\beta'$ generating lines.

9. The section by the tangent plane through A is made up of $(m-\omega)(m'-\omega') - \alpha\alpha'$ generating lines (viz. these are, the line $\Omega' A$ $\alpha'(m-\omega-\alpha)$ times, the line ΩA $\alpha(m'-\omega'-\alpha')$ times, and $(m-\omega-\alpha)(m'-\omega'-\alpha')$ other generating lines) and of a curve of the order $mm' - \omega\omega' - \beta\beta'$; similarly the section by the tangent plane through B is made up of $(m-\omega)(m'-\omega') - \beta\beta'$ generating lines (viz. these are, the line ΩB $\beta'(m-\omega-\beta)$ times, the line $\Omega' B$ $\beta(m'-\omega'-\beta')$ times, and $(m-\omega-\beta)(m'-\omega'-\beta')$ other generating lines), and of a curve of the order $mm' - \omega\omega' - \alpha\alpha'$.

10. A very interesting case is when (m, m' being each even) we have

$$\omega = \alpha = \beta = \tfrac{1}{2}m, \qquad \omega' = \alpha' = \beta' = \tfrac{1}{2}m'.$$

Here the curve U is a $\frac{1}{2}m'$-tuple line on the scroll, and the complete section by the plane Π is this curve taken $\frac{1}{2}m'$ times; the order of the section, and therefore of the scroll, is thus $= \frac{1}{2}mm'$: of course in like manner the curve U' is a $\frac{1}{2}m$-tuple line on the scroll, and the complete section by this plane is the curve U' taken $\frac{1}{2}m$ times; the section by each of the planes $\Omega\Omega' A$, $\Omega\Omega' B$ is a curve of the order $\frac{1}{2}mm'$, the planes in question being in the present case no longer singular tangent planes, or even tangent planes at all, of the scroll.

Article Nos. 11 to 14. Analytical Theory.

11. It will be convenient to denote by D, C respectively the points heretofore called Ω, Ω' respectively: this being so, we have a tetrahedron $ABCD$, of which the faces ABD, ABC are the planes heretofore called Π, Π' respectively, and the other two faces CDA, CDB are the singular tangent planes $\Omega\Omega' A$, $\Omega\Omega' B$ respectively. And then, taking

$$x = 0, \quad y = 0, \quad z = 0, \quad w = 0$$

for the equations of the faces BCD, CDA, DAB, ABC of the tetrahedron, we may write for the equations of the curve U, $z = 0$, $f_1(x, y, w) = 0$, for those of the curve U', $w = 0$, $f_2(x, y, z) = 0$; and take the homographic ranges on the line I ($z = 0$, $w = 0$) to be given as the intersections of this line with the pencils of planes $x - \theta y = 0$, $x - k\theta y = 0$ respectively (θ a variable parameter, k a constant). The points K are therefore given by

$$x - \theta y = 0, \qquad z = 0, \qquad f_1(x, y, w) = 0,$$

the points K' by

$$x - k\theta y = 0, \qquad w = 0, \qquad f_2(x, y, z) = 0;$$

and then the lines KK' belonging to the different values of the parameter θ generate the scroll.

12. Or, what is the same thing, taking (X, Y, Z, W) as the coordinates of K, (X', Y', Z', W') as the coordinates of K', we have

$$\begin{aligned} X - \theta Y &= 0, & Z &= 0, & f_1(X, Y, W) &= 0, \\ X' - k\theta Y' &= 0, & W' &= 0, & f_2(X', Y', Z') &= 0, \end{aligned}$$

and then the equations of the line KK' are

$$\begin{vmatrix} x & y & z & w \\ X & Y & 0 & W \\ X' & Y' & Z' & 0 \end{vmatrix} = 0;$$

or, as these may be written,

$$\begin{aligned} -WZ'y + W'Y'z + YZ'w &= 0, \\ WZ'x \quad + W'X'z - XZ'w &= 0, \\ -W'Yx + WX'y + (XY' - X'Y)\,w &= 0, \\ -YZ'x + XZ'y - (XY' - X'Y)\,z &= 0, \end{aligned}$$

equivalent of course to two equations. The elimination of X, Y, W, X', Y', Z', θ from all the equations gives the equation of the scroll.

13. Substituting the values $X = \theta Y$, $X' = k\theta Y'$, we have

$$f_1(\theta Y, Y, W) = 0, \qquad f_2(k\theta Y', Y', Z') = 0,$$

and

$$\begin{aligned} -WZ'y + WY'z + YZ'w &= 0, \\ WZ'x - k\theta WY'z - \theta YZ'w &= 0, \\ -WY'x + k\theta WY'y + \theta(1-k)\,YY'w &= 0, \\ -YZ'x + \theta YZ'y - \theta(1-k)\,YY'z &= 0; \end{aligned}$$

or, what is the same thing, writing $\dfrac{W}{Y} = \omega$ and $\dfrac{Z'}{Y'} = \zeta$, we have

$$f_1(\theta, 1, \omega) = 0, \qquad f_2(k\theta, 1, \zeta) = 0,$$

$$\begin{aligned} \omega\zeta\, x - \omega\zeta\, y \quad + \zeta\, w &= 0, \\ \omega\zeta\, x - k\theta\, \omega\zeta\, z - \theta\zeta\, w &= 0, \\ -\omega\, x + k\theta\, \omega\, y \quad + \theta(1-k)\, w &= 0, \\ -\zeta\, x + \theta\zeta\, y - \theta(1-k)\, z &= 0. \end{aligned}$$

Recollecting that the last four equations are equivalent to two equations only, and substituting for ω, ζ their values in terms of θ, we have in effect two equations, which by the elimination of θ lead to a relation in (x, y, z, w), the equation of the scroll.

14. We may find the sections of the scroll by the planes $x = 0$, $y = 0$ respectively. Writing first $x = 0$, we have

$$\omega y = \frac{k-1}{k}\, w, \qquad \zeta y = -(k-1)\, z.$$

Hence taking the other two equations in the form

$$f_1\Big(\theta y, y, \frac{k-1}{k}\, w\Big) = 0, \qquad f_2\Big(u, \frac{y}{k}, \frac{-(k-1)z}{k}\Big) = 0,$$

and putting $\theta y = u$, we have

$$f_1\Big(u, y, \frac{k-1}{k}\, w\Big) = 0, \qquad f_2\Big(u, \frac{y}{k}, \frac{-(k-1)z}{k}\Big) = 0,$$

from which eliminating u we obtain an equation $f_3(y,z,w)=0$, the equation of the section by the plane $x=0$.

Similarly, writing $y=0$, we have

$$\frac{\omega x}{\theta}=-(k-1)\,w,\qquad \frac{\zeta x}{\theta}=(k-1)\,z,$$

whence taking the equations in the form

$$f_1\Big(x,\frac{x}{\theta},\frac{-(k-1)w}{\theta}\Big)=0,\qquad f_2\Big(kx,v,\frac{(k-1)z}{\theta}\Big)=0,$$

and writing $\dfrac{x}{\theta}=v$, we have

$$f_1(x,v,-(k-1)\,w)=0,\qquad f_2(kx,v,(k-1)\,z)=0,$$

from which, eliminating v, we obtain an equation $f_4(x,z,w)=0$, the equation of the section by the plane $y=0$.

Article Nos. 15 to 29. The Curves U, U' are henceforward "triangular" curves.

15. Let $r=\pm\dfrac{p}{q}$, where p, q are positive integers prime to each other, and let the given sections be

$$\begin{aligned}z=0,&\quad Ax^r+By^r+Dw^r=0,\\ w=0,&\quad A'x^r+B'y^r+C'z^r=0,\end{aligned}$$

where it is to be observed that r being $=+\dfrac{p}{q}$, the two given sections are of the order pq, the order of the scroll is $=2p^2q^2$; each of the given sections is a pq-tuple line on the scroll, and the plane thereof meets the scroll in the section taken pq times, and in the pq generating lines: but r being $=-\dfrac{p}{q}$, the two given sections are each of the order $2pq$, with three pq-tuple points ($\omega=\alpha=\beta=pq$, $\omega'=\alpha'=\beta'=pq$), and thence the order of the scroll is $\frac{1}{2}(2pq)^2=2p^2q^2$; each of the sections is a pq-tuple line on the scroll, and the plane meets the scroll only in the section taken pq times. But in either case, if q be >1, that is, if r be fractional, it will presently appear that the scroll of the order $2p^2q^2$ breaks up into q scrolls each of the order $2p^2q$.

16. To find the section by the plane $x=0$, we have

$$Au^r+By^r\qquad+D\Big(\frac{k-1}{k}\,w\Big)^r=0,$$

$$A'(ku)^r+B'y^r+C'\left(\frac{-(k-1)z}{k}\right)^r=0,$$

and eliminating u we obtain

$$(AB'-A'Bk^r)\,y^r+AC'\left(\frac{-(k-1)}{k}\right)^r z^r-A'D\Big(\frac{k-1}{k}\Big)^r w^r=0;$$

writing $\left(\dfrac{k-1}{k}\right)^r=(-)^r\left(\dfrac{1-k}{k}\right)^r$, this is

$$\frac{AB'-A'Bk^r}{(1-k)^r}\,y^r-(-)^r\,AC'\,z^r+A'D\,w^r=0;$$

or, what is the same thing, it is

$$\frac{AB' - A'B\,k^r}{(1-k)^r}\,y^r - (-)^r\,AC'\,z^r + A'D\,w^r = 0.$$

And in regard to this and the other equations which contain $(-)^r$, it is to be observed that r being integral we have $(-)^r = (-)^r$, and that r being fractional, every value of $(-)^r$ is also a value of $(-)^r$; so that we may in every case write $(-)^r$ in place of $(-)^{-r}$.

Similarly for the section by the plane $y = 0$, we have

$$\begin{aligned} Ax^r + Bv^r \qquad\qquad\quad &+ D\,(-(k-1)w)^r = 0, \\ A'\,(kx)^r + B'v^r + C'\,((k-1)z)^r \qquad &\qquad\qquad\quad\;\; = 0, \end{aligned}$$

and eliminating v, we have

$$(AB' - A'Bk^r)\,x^r - BC'\,(k-1)^r\,z^r + BD\,(-(k-1))^r\,w^r = 0;$$

or, what is the same thing,

$$\frac{AB' - A'Bk^r}{(1-k)^r}\,x^r - (-)^r\,BC'\,z^r + B'D\,w^r = 0.$$

17. The four sections thus are

$$\begin{array}{ll} x = 0, & \dfrac{AB' - A'Bk^r}{(1-k)^r}\,y^r - (-)^r\,AC'\,z^r + A'D\,w^r = 0, \\ y = 0, & \dfrac{AB' - A'Bk^r}{(1-k)^r}\,x^r - (-)^r\,BC'\,z^r + B'D\,w^r = 0, \\ z = 0, & \qquad\qquad Ax^r + By^r \qquad\quad + Dw^r = 0, \\ w = 0, & \qquad\qquad A'x^r + B'y^r + C'z^r \qquad\;\; = 0. \end{array}$$

It will be convenient to speak of these four curves as directrices of the scroll.

18. Suppose for a moment that r is integral; as either of the given equations may be multiplied by a constant, we may assume that $D = -(-)^r\,C'$; substituting this value and dividing the first and second equations each by C', we have

$$\begin{array}{ll} x = 0, & \dfrac{AB' - A'Bk^r}{(1-k)^r\,C'}\,y^r - (-)^r\,Az^r - A'w^r = 0, \\ y = 0, & \dfrac{AB' - A'Bk^r}{(1-k)^r\,C'}\,x^r - (-)^r\,Bz^r - B'w^r = 0, \\ z = 0, & \qquad Ax^r + By^r \qquad\quad - (-)^r\,C'w^r = 0, \\ w = 0, & \qquad\qquad A'x^r + B'y^r + C'z^r \qquad\; = 0, \end{array}$$

so that the diagonally opposite coefficients differ only by the factor $-(-)^r$; viz. the matrix is symmetrical or skew symmetrical according as r is odd or even.

19. If r be fractional, it is to be observed that, although the three symbols $(-)^r$ and the two symbols $(1-k)^r$ which enter into the first and second equations of No. 17, do not in the first instance represent of necessity the same values of $(-)^r$ and $(1-k)^r$ respectively, yet there is no loss of generality in assuming that they do so — the irrational equations are mere symbols for the rational equations to which they respectively give rise — and the irrationalities $(-)^r$ and $(1-k)^r$ will on the rationalisation of the equations disappear along with the irrationalities of x^r, y^r, z^r, w^r to which they are attached. But the case is otherwise with the irrationality k^r

involved in the expression $AB' - A'Bk^r$; writing as before $r = \pm\frac{p}{q}$ (p and q positive integers prime to each other), the symbol k^r has q different values; and there is not in the first instance any relation between the k^r of the first equation and the k^r of the second equation: for each of these equations the rationalised equation (that is, the equation rationalised in regard to the coordinates) will contain the irrationality k^r, and will thus for each of the q values of k^r represent a distinct curve. The given equations (viz. the first and second equations) represent each of them a single curve of the order pq or $2pq$, according as r is positive or negative; the first and second equations represent each of them q such curves.

20. Hence, starting from the two given curves in the planes $z = 0$ and $w = 0$, respectively, and with a given value of k, the section of the scroll by the plane $y = 0$ is made up of q curves, viz. the curves obtained from the second equation of No. 17, by assigning to the radical k^r each of its q different values; the scroll consequently breaks up into q different scrolls, viz. the lines passing through the two given curves, and any one of the q curves in the plane $y = 0$, constitute a distinct scroll. The lines in question meet the plane $x = 0$, not indifferently in any one of the q curves in that plane, but in a certain one of these curves, viz. in that curve for which the radical k^r has the same value as for the curve in question in the plane $y = 0$. Hence we may in the first and second equations regard the radicals k^r as having the same meaning, and the system of four equations in effect breaks up into q systems, viz. the systems obtained by giving to the radical k^r its q different values; each of these systems gives a scroll, and the scroll derived from the two given curves with a given value of k is made up of these q scrolls. And hence, attaching a unique value to each of the symbols $(-)^r$, $(1-k)^r$ and k^r, we may, as before, write $D = -(-)^r\, C'$, and so reduce the original equations as in the case r integral, to the form No. 18, in which the diagonally opposite coefficients differ only by the factor $-(-)^r$.

21. Let the two given equations be taken to be

$$\begin{array}{lllll} z = 0, & bx^r + yay^r & & + hw^r & = 0, \\ w = 0, & -(-)^r fx^r - (-)^r gy^r + (-)^r cz^r & & & = 0; \end{array}$$

we have then

$$\frac{AB' - A'Bk^r}{(1-k)^r\, C'} = \frac{(-)^r\, bg + afk^r}{(-)^r\, c\,(1-k)^r},$$

or, putting this $= -(-)^r\, c$, that is, writing

$$af\, k^r + bg\,(-)^r + ch\,(1-k)^r = 0,$$

the four equations become

$$\begin{array}{ll} x = 0, & + cy^r - (-)^r\, bz^r + fw^r = 0, \\ y = 0, & -(-)^r\, cx^r \qquad + az^r + gw^r = 0, \\ z = 0, & bx^r - (-)^r\, ay^r \qquad + hw^r = 0, \\ w = 0, & -(-)^r\, fx^r - (-)^r\, gy^r - (-)^r\, hz^r = 0, \end{array}$$

where c being considered as given, k is determined as mentioned above; or, what is the same thing, $k : -1 : 1-k = \lambda : \mu : \nu$, we have $\lambda : \mu : \nu$, and thence k, determined by the equations

$$\lambda + \mu + \nu = 0, \qquad af\,\lambda^r + bg\,\mu^r + ch\,\nu^r = 0.$$

22. Consider for a moment λ, μ, ν, as the coordinates of a point in a plane, then $\bigl(r = \pm\frac{p}{q}$ as before$\bigr)$, the equation $af\,\lambda^r + bg\,\mu^r + ch\,\nu^r = 0$, is that of a curve of the order pq or $2pq$, according as r is positive or negative: and this curve is met by the line $\lambda + \mu + \nu = 0$, in pq or $2pq$ points, that is, k has this number pq or $2pq$, of values: but to each of these values of k there corresponds (not q values but) only a single value of k^r, viz. that value for which $af\,k^r + bg\,(-)^r + ch\,(1-k)^r = 0$; that is, starting from the two directrices in the planes $z=0$, $w=0$, respectively, and a given third directrix in the plane $y=0$ (or in the plane $x=0$), we may by means of each of the pq or $2pq$ values of k construct a scroll passing through the three directrices, and which will also pass through the fourth directrix in the plane $x=0$ (or in the plane $y=0$), but such scroll is only one (not each) of the q scrolls which can be constructed from the two given sections in the planes $z=0$, $w=0$, respectively, and from the assumed value of k. It has been mentioned that whether r is $=+\frac{p}{q}$, or $=-\frac{p}{q}$, the total scroll constructed from the two given directrices in the planes $z=0$, $w=0$, and from a given value of k is of the order $2p^2q^2$, and that such scroll breaks up into q distinct scrolls, hence the order of each of the distinct scrolls is $=2p^2q$. Whence, starting with the given directrices in the planes $z=0$, $w=0$, and a given third directrix in the plane $y=0$ (or in the plane $x=0$), we have pq or $2pq$ scrolls each of the order $2p^2q$, and passing through these three directrices, and through the given fourth directrix in the plane $x=0$ (or in the plane $y=0$).

23. It is to be observed that when q is >1, then considering the three directrices as given, the pq or $2pq$ scrolls each of the order $2p^2q$, do not make up the total scroll generated by the lines which pass through the three given directrices. I call to mind that for three given directrices the orders of which are m, n, p, respectively, and which meet, the second and third, the third and first, and the first and second, in α points, β points, and γ points respectively, the order of the scroll generated by the lines which meet the three directrices is $=2mnp-\alpha m-\beta n-\gamma p$. Suppose first, that $r=+\frac{p}{q}$, then the directrices are each of the order pq, and they do not any two of them meet; the order of the scroll is $=2p^3q^3$. Suppose secondly, $r=-\frac{p}{q}$, then the directrices are each of the order $2pq$, but each two of them have in common two pq-tuple points counting as $2p^2q^2$ intersections; the order of the scroll is thus $(16-2\cdot 4)\,p^3q^3 = 4p^3q^3$. In the first case the lines which meet the three directrices generate a residuary scroll of the order $2p^3(q^3-q^2)$, and the pq scrolls each of the order $2p^2q$; in the second case they generate a residuary scroll of the order $4p^3(q^3-q^2)$, and the $2pq$ scrolls each of the order $2p^2q$.

24. In the case $r=+\frac{p}{q}$, by way of illustration of the origin of the pq scrolls each of the order $2p^2q$, I consider the particular case $p=1$, that is, $r=\frac{1}{q}$, the reciprocal of a positive integer q, and where it is to be shown that we have q scrolls each of the order $2q$. The given directrices are here

$$\begin{array}{llllll} z=0, & Ax^{\frac{1}{q}} & +\,By^{\frac{1}{q}} & & +\,Dw^{\frac{1}{q}} & =0, \\ w=0, & A'x^{\frac{1}{q}} & +\,B'y^{\frac{1}{q}} & +\,C'z^{\frac{1}{q}} & & =0, \end{array}$$

each of them a unicursal curve; we may in fact satisfy the two equations respectively, by writing in the first of them

$$x:y:w \;=\; a\,(\phi+\alpha)^q \;:\; b\,(\phi+\beta)^q \;:\; d\,(\phi+\delta)^q;$$

and in the second

$$x:y:z \;=\; a'\,(\phi'+\alpha')^q \;:\; b'\,(\phi'+\beta')^q \;:\; c'\,(\phi'+\gamma')^q,$$

where a, b, d, α, β, δ, a', b', c', α', β', γ' are properly determined constants, ϕ, ϕ' are variable parameters. It follows that, considering the points K, K' which are the intersections of the first curve by the line $x-\theta y=0$, and of the second curve by the corresponding line $x-k\theta y=0$, we have not only a correspondence of q points K with q points K', but we may establish, and that in q different manners, a correspondence between single points K and K'. For, substituting the foregoing values of $x : y$ in the equations $x - \theta y = 0$ and $x - k\theta y = 0$ respectively, we have

$$\theta = \frac{a\,(\phi+\alpha)^q}{b\,(\phi+\beta)^q}, \qquad k\theta = \frac{a'\,(\phi'+\alpha')^q}{b'\,(\phi'+\beta')^q},$$

and thence

$$\frac{a\,(\phi+\alpha)^q}{b\,(\phi+\beta)^q} = \frac{1}{k}\cdot\frac{a'\,(\phi'+\alpha')^q}{b'\,(\phi'+\beta')^q};$$

so that, extracting the qth root of each side, we have, in q different ways corresponding to the q values of the radical $\left(\frac{1}{k}\cdot\frac{ba'}{ab'}\right)^{\frac{1}{q}}$, a relation of the form $\phi' = \dfrac{\Lambda\phi + M}{N\phi + R}$; and considering ϕ' as having this value, the points K, K' as given by the equations

$$\begin{aligned} z = 0, \quad x : y : w &= a\,(\phi+\alpha)^q : b\,(\phi+\beta)^q : d\,(\phi+\delta)^q, \\ w = 0, \quad x : y : z &= a'\,(\phi'+\alpha')^q : b'\,(\phi'+\beta')^q : c'\,(\phi'+\gamma')^q, \end{aligned}$$

respectively, correspond as single points to each other. We have thus in q different ways a series of corresponding points K, K', and consequently q series of lines KK' each of them generating a scroll which (as the order of the scroll generated by all the q series is $= 2q^2$), must be each of them of the order $2q$; and the decomposition in question is thus explained.

25. In the scroll of the order $2q^2$, each directrix is a q-tuple line, and the complete section by the plane of the directrix is made up of the directrix q times (order q^2), and of q q-fold generating lines, in fact, of q q-fold generating lines: to show that this is so, consider the directrix in the plane $z = 0$, viz. the equation of this is $Ax^{\frac{1}{q}} + By^{\frac{1}{q}} + Dw^{\frac{1}{q}} = 0$. Writing herein $w = 0$, we have $Ax^{\frac{1}{q}} + By^{\frac{1}{q}} = 0$, that is, $A^q x - (-)^q B^q y = 0$; it is clear that the rationalised equation must reduce itself to $[A^q x - (-)^q B^q y]^q = 0$, and that the line $w = 0$, is thus a tangent of q-pointic intersection at the point $w = 0$, $A^q x - (-)^q B^q y = 0$. Taking K at this point we have, in each of the scrolls of the order $2q$, q coincident positions of K', that is, a q-fold line KK' in the plane $w = 0$; and the like for the plane $z = 0$, so that the total section by the plane $z = 0$ is made up of the directrix q times and of q q-fold generating lines; and it follows that for each of the scrolls of the order $2q$, the section by the plane $z = 0$ is made up of the directrix once, and of a q-fold generating line.

26. It is easy to see that in the general case $r = +\frac{p}{q}$, the like conclusion holds for the scroll of the order $2p^2q^2$, the section by the plane of the directrix consists of the directrix pq times (order p^2q^2), and of p^2q q-fold generating lines; whence for each of the q component scrolls of the order $2p^2q$, the section is made up of the directrix p times (that is, the directrix is a p-tuple line on the scroll) and of p^2 q-fold generating lines.

27. In the case $r = -\frac{p}{q}$, where the order of the directrix is $= 2pq$, then in the scroll of the order $2p^2q^2$, the directrix is a pq-tuple line on the scroll, and taken pq times it constitutes the complete section by the plane of the directrix; whence in each of the q component scrolls of the order $2p^2q$, the directrix is a p-tuple line; and taken p times it constitutes the complete section by the plane of the directrix.

28. It is convenient to exhibit the foregoing results in a tabular form as follows:

$r = +\frac{p}{q}$.	$r = -\frac{p}{q}$.
Each directrix is of order pq.	Each directrix is of order $2pq$, with three pq-tuple points.
Scroll belonging to two directrices, and a given value of k, is of the order	
$2p^2q^2$, breaking up into q scrolls each of order $2p^2q$, each which scroll of the order $2p^2q$ has each directrix for a p-tuple line and has besides p^2 q-fold generating lines in the plane of the directrix.	$2p^2q^2$, breaking up into q scrolls each of the order $2p^2q$, each which scroll of the order $2p^2q$ has each directrix for a p-tuple line, and consequently no generating line in the plane of the directrix.
Considering two directrices and a given third directrix,	
k has pq values.	k has $2pq$ values.
Total scroll for the three directrices is made up of	
pq scrolls each of order $2p^2q$ (viz. one for each value of k), and residuary scroll of order $2p^3(q^3 - q^2)$.	$2pq$ scrolls each of order $2p^2q$ (viz. one for each value of k), and residuary scroll of order $4p^3(q^3 - q^2)$.

29. The following are noticeable cases; $r = 1$ gives the hyperboloid as derived from three directrix lines; $r = -1$ the hyperboloid as derived from three plane sections thereof; $r = 2$, an octic surface, M. De la Gournerie's Quadrispinal; $r = -2$, an octic surface, his Quadricuspidal; $r = \frac{1}{3}$, a sextic surface which (as remarked by Dr Salmon), on writing therein (x^2, y^2, z^2, w^2), in place of (x, y, z, w), is converted into a surface of the twelfth order, locus of the centres of curvature of an ellipsoid.

[435]

On the Six Coordinates of a Line.

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THE notion of the six coordinates of a line was, so far as I am aware, first established in my paper "On a new analytical representation of Curves in Space," *Quart. Math. Jour.* t. III. (1860), pp. 225–236, [**284**]; see p. 226, where writing p, q, r, s, t, u for the six determinants of the matrix $\left(\begin{smallmatrix} x_1 & y_1 & z_1 & w_1 \\ x_2 & y_2 & z_2 & w_2 \end{smallmatrix}\right)$, I remark that these values give identically $ps + qt + ru = 0$; and I consider a cone as represented by a homogeneous equation $V = 0$ between the six coordinates (p, q, r, s, t, u); and many of the investigations of the present memoir, in which these coordinates are employed, have been in my possession for some years past. But these coordinates presented themselves independently to Prof. Plücker, and the theory of them is set forth in his most interesting and valuable memoir, "On a new Geometry of Space," *Phil. Trans.* t. CLV. (1865), pp. 725–791; the course of development there given to the theory is however altogether different from that in the present memoir. They have also more recently been made use of in a paper by Herr Lüroth, "Zur Theorie der windschiefen Flächen," *Crelle*, t. LXII. (1867), pp. 130–152.

I have in the present memoir applied these coordinates to the question of the Involution of six lines; the notion of this relation of six lines is due to Prof. Sylvester, to whom it presented itself in the year 1861, in connexion with a theorem in the *Lehrbuch der Statik*, by Möbius (Leipzig, 1837), that if four forces acting on a solid body are in equilibrium the lines along which the forces act are the generating lines of a hyperboloid. Prof. Sylvester was thereby led to consider six lines such that (regarding them as lines in a solid body) there exist along them forces which are in equilibrium; and he thence obtained, by the statical considerations reproduced in the present memoir, the construction (when five of the lines are given) of a sixth line to pass through a given point or to be situate in a given plane.

Article, Nos. 1 to 8. The Six Coordinates of a Line; definition and general notions.

1. Using any quadriplanar coordinates (x, y, z, w) whatever, consider a line; on the line two points the coordinates of which are $(\alpha, \beta, \gamma, \delta)$ and $(\alpha', \beta', \gamma', \delta')$ respectively; and through the line two planes, the equations whereof are $(A, B, C, D \between x, y, z, w) = 0$, and $(A', B', C', D' \between x, y, z, w) = 0$ respectively; we have

$$\begin{aligned}(A, B, C, D \between \alpha, \beta, \gamma, \delta) &= 0,\\ (A, B, C, D \between \alpha', \beta', \gamma', \delta') &= 0,\\ (A', B', C', D' \between \alpha, \beta, \gamma, \delta) &= 0,\\ (A', B', C', D' \between \alpha', \beta', \gamma', \delta') &= 0.\end{aligned}$$

2. From the first and second equations, eliminating successively A, B, C, D, we find

$$\begin{vmatrix} 0 & \alpha\beta' - \alpha'\beta & -(\gamma\alpha' - \gamma'\alpha) & \alpha\delta' - \alpha'\delta \\ -(\alpha\beta' - \alpha'\beta) & 0 & \beta\gamma' - \beta'\gamma & \beta\delta' - \beta'\delta \\ \gamma\alpha' - \gamma'\alpha & -(\beta\gamma' - \beta'\gamma) & 0 & \gamma\delta' - \gamma'\delta \\ -(\alpha\delta' - \alpha'\delta) & -(\beta\delta' - \beta'\delta) & -(\gamma\delta' - \gamma'\delta) & 0 \end{vmatrix} (A, B, C, D) = 0,$$

and from the third and fourth equations we find the like system with (A', B', C', D') in place of (A, B, C, D). Comparing the corresponding equations of the two systems, we find an equality of ratios, as will presently be mentioned.

3. From the first and third equations, eliminating successively α, β, γ, δ, we find

$$\begin{vmatrix} 0 & AB'-A'B & -(CA'-C'A) & AD'-A'D \\ -(AB'-A'B) & 0 & BC'-B'C & BD'-B'D \\ CA'-C'A & -(BC'-B'C) & 0 & CD'-C'D \\ -(AD'-A'D) & -(BD'-B'D) & -(CD'-C'D) & 0 \end{vmatrix}(\alpha,\beta,\gamma,\delta)=0,$$

and from the second and fourth equations we find the like system with $(\alpha',\beta',\gamma',\delta')$ in place of $(\alpha,\beta,\gamma,\delta)$; comparing the corresponding equations of the two systems, we find the same equality of ratios as before, viz.

4. This is

$$\begin{aligned}&\beta\gamma'-\beta'\gamma:\gamma\alpha'-\gamma'\alpha:\alpha\beta'-\alpha'\beta:\alpha\delta'-\alpha'\delta:\beta\delta'-\beta'\delta:\gamma\delta'-\gamma'\delta\\ &\qquad=AD'-A'D:BD'-B'D:CD'-C'D:BC'-B'C:CA'-C'A:AB'-A'B,\end{aligned}$$

and putting each of these two equal sets of ratios

$$=a:b:c:f:g:h,$$

then the quantities (a,b,c,f,g,h), which it is easy to see satisfy the condition

$$af+bg+ch=0,$$

are said to be the 'six coordinates' of the line: as only the ratios of the six quantities are material, and as the last-mentioned equation establishes a single relation between these ratios, the system of the six coordinates contain four arbitrary ratios or parameters, for the determination of the particular line.

5. A line is thus determined by its six coordinates (a,b,c,f,g,h), which are such that $af+bg+ch=0$; and conversely any six quantities (a,b,c,f,g,h) satisfying this relation may be taken to be the six coordinates of a line.

6. It is proper to show that the ratios $a:b:c:f:g:h$ are independent of the particular two points on the line, or two planes through the line, used for their determination. In fact, if instead of the points

$$\alpha,\beta,\gamma,\delta,\qquad\alpha',\beta',\gamma',\delta',$$

we have any other two points on the line, say the points

$$\lambda\alpha+\mu\alpha',\ \ \lambda\beta+\mu\beta',\ \ \lambda\gamma+\mu\gamma',\ \ \lambda\delta+\mu\delta',$$

$$\nu\alpha+\rho\alpha',\ \ \nu\beta+\rho\beta',\ \ \nu\gamma+\rho\gamma',\ \ \nu\delta+\rho\delta',$$

then the six determinants have their original values each multiplied by $\lambda\rho-\mu\nu$; and the ratios are unaltered.

And the like is the case, if instead of the planes

$$A,B,C,D,\qquad A',B',C',D',$$

we have any other two planes through the line, say the planes

$$\lambda A+\mu A',\ \ \lambda B+\mu B',\ \ \lambda C+\mu C',\ \ \lambda D+\mu D',$$

$$\nu A+\rho A',\ \ \nu B+\rho B',\ \ \nu C+\rho C',\ \ \nu D+\rho D';$$

the determinants have their original values each multiplied by $\lambda\rho-\mu\nu$; and the ratios are unaltered.

7. It may be remarked, that the theory of the six coordinates considered as derived from the two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, and as derived from the two planes (A, B, C, D), (A', B', C', D') is precisely the same in each case; and we may confine ourselves to the first point of view, regarding therefore the six coordinates as derived from the two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$. I further remark, that I do not at present in anywise fix the absolute magnitudes of the coordinates (a, b, c, f, g, h): it is only the ratios that we are concerned with.

8. The values of the ratios $a : b : c : f : g : h$ of the six coordinates do however depend on the particular coordinate planes $x = 0$, $y = 0$, $z = 0$, $w = 0$, made use of for their determination; and in the sequel it will be necessary to investigate the formulae of transformation to a new set of coordinate planes $x_0 = 0$, $y_0 = 0$, $z_0 = 0$, $w_0 = 0$. And I shall also show in what manner the absolute magnitudes of the coordinates may be fixed. But deferring the consideration of these questions, I consider the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$ as given planes, and take the six coordinates (a, b, c, f, g, h) of a line to be determined as above in reference to these given planes, the absolute values of these coordinates remaining indeterminate, and their ratios only being attended to. And I proceed to consider the various questions which present themselves in the geometry of the line, considered as thus determined by means of its six coordinates (a, b, c, f, g, h).

Article, Nos. 9 to 18. (Various Sub-headings.) Elementary Theorems.

Condition that a line may be in a given plane.

9. Taking the line to be (a, b, c, f, g, h), the equation of the given plane to be

$$(A, B, C, D \between x, y, z, w) = 0;$$

then if $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$ are the coordinates of any two points on the line, we have the system of equations *ante*, No. 2, and substituting therein for $\beta\gamma' - \beta'\gamma$, &c. the values (a, b, c, f, g, h), we find

$$\left|\begin{array}{cccc} 0 & c & -b & f \\ -c & 0 & a & g \\ b & -a & 0 & h \\ -f & -g & -h & 0 \end{array}\right| (A, B, C, D) = 0;$$

which equations, equivalent to a twofold relation, are the required condition. It may be remarked that, treating (A, B, C, D) as current plane coordinates, each equation of the system is that of a point lying in the line.

Condition that a line may pass through a given point.

10. The coordinates of the given point are taken to be $(\alpha, \beta, \gamma, \delta)$. If

$$(A, B, C, D \between x, y, z, w) = 0, \qquad (A', B', C', D' \between x, y, z, w) = 0,$$

are the equations of any two planes through the line, then we have the system of equations *ante* No. 3, and substituting therein for $AB' - A'B$, &c. their values in terms of the coordinates (a, b, c, f, g, h) of the line, we have

$$\left|\begin{array}{cccc} 0 & h & -g & a \\ -h & 0 & f & b \\ g & -f & 0 & c \\ -a & -b & -c & 0 \end{array}\right| (\alpha, \beta, \gamma, \delta) = 0;$$

which equations, equivalent to a twofold relation, are the required condition. It is obvious that, treating $(\alpha, \beta, \gamma, \delta)$ as current point coordinates, each equation of the system is the equation of a plane through the given line.

Condition for the intersection of two lines.

11. The coordinates of the lines are taken to be (a, b, c, f, g, h), and $(a_1, b_1, c_1, f_1, g_1, h_1)$, respectively. If $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, are the coordinates of any two points in the first line, and $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, $(\alpha'_1, \beta'_1, \gamma'_1, \delta'_1)$, are the coordinates of any two points on the second line, then the four points are in a plane, that is, we have

$$\begin{vmatrix} \alpha & \beta & \gamma & \delta \\ \alpha' & \beta' & \gamma' & \delta' \\ \alpha_1 & \beta_1 & \gamma_1 & \delta_1 \\ \alpha'_1 & \beta'_1 & \gamma'_1 & \delta'_1 \end{vmatrix} = 0,$$

that is, expanding the determinant and substituting for $\beta\gamma' - \beta'\gamma$, &c. and $\beta_1\gamma'_1 - \beta'_1\gamma_1$, &c. their values in terms of the coordinates of the two lines respectively, we have

$$af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1 = 0,$$

or, as this may also be written,

$$(f_1, g_1, h_1, a_1, b_1, c_1 \between a, b, c, f, g, h) = 0,$$

for the condition that the two lines may intersect.

12. The same result will be obtained if we take

$$(A, B, C, D \between x, y, z, w) = 0, \qquad (A', B', C', D' \between x, y, z, w) = 0,$$

for the equations of any two planes through the first line, and

$$(A_1, B_1, C_1, D_1 \between x, y, z, w) = 0, \qquad (A'_1, B'_1, C'_1, D'_1 \between x, y, z, w) = 0,$$

for the equations of any two planes through the second line. The four planes will meet in a point, that is, we have

$$\begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A_1 & B_1 & C_1 & D_1 \\ A'_1 & B'_1 & C'_1 & D'_1 \end{vmatrix} = 0,$$

or, expanding and substituting, we have the same condition as before.

13. In the case of any two lines (a, b, c, f, g, h), and $(a_1, b_1, c_1, f_1, g_1, h_1)$, we may define the 'moment' of the two lines to be the function

$$af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1,$$

it being understood that we have not as yet any complete quantitative definition of the moment; this being so, we have, in what precedes, the theorem that the moment of two intersecting lines is $= 0$.

Plane through two intersecting lines.

14. Let $(A, B, C, D \between x, y, z, w) = 0$ be the equation of the plane through the two intersecting lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$. We have two systems of equations, as in No. 9, and comparing the corresponding equations of the two systems, we find in the first instance

$$\begin{aligned} A : B : C : D = \lambda : bf_1 - b_1f : cf_1 - c_1f &: -(bc_1 - b_1c) \\ = ag_1 - a_1g : \mu : cg_1 - c_1g &: -(ca_1 - c_1a) \\ = ah_1 - a_1h : bh_1 - b_1h : \nu &: -(ab_1 - a_1b) \\ = gh_1 - g_1h : hf_1 - h_1f : fg_1 - f_1g &: \rho, \end{aligned}$$

where λ, μ, ν, ρ, are in the first instance unknown; the different sets of ratios are of course identical in virtue of the relation

$$(f_1, g_1, h_1, a_1, b_1, c_1 \backslash\!\backslash\, a, b, c, f, g, h) = 0,$$

and comparing them we have equations which lead to the values of λ, μ, ν, ρ; and we thus obtain more completely.

$$\begin{aligned} A : B : C : D &= f_1a + b_1g + c_1h : bf_1 - b_1f : cf_1 - c_1f : -(bc_1 - b_1c) \\ &= ag_1 - a_1g : a_1f + g_1b + c_1h : cg_1 - c_1g : -(ca_1 - c_1a) \\ &= ah_1 - a_1h : bh_1 - b_1h : a_1f + b_1g + h_1c : -(ab_1 - a_1b) \\ &= gh_1 - g_1h : hf_1 - h_1f : fg_1 - f_1g : f_1a + g_1b + c_1h. \end{aligned}$$

15. It is in these equations easy to verify the identity of the different sets of values: we ought, for instance, to have

$$\frac{a_1f + bg + h_1c}{fg_1 - f_1g} = \frac{ab_1 - a_1b}{a_1f + bg_1 + ch_1};$$

that is,

$$(h_1c + a_1f + b_1g)(h_1c + af_1 + bg_1) + (ab_1 - a_1b)(fg_1 - f_1g) = 0,$$

and, observing that

$$\begin{aligned} (a_1f + b_1g)(af_1 + bg_1) + (ab_1 - a_1b)(fg_1 - f_1g) &= (af + bg)(a_1f_1 + b_1g_1) \\ &= ch \cdot c_1h_1, \end{aligned}$$

the left-hand side is

$$\begin{aligned} &= ch_1\,(ch_1 + af_1 + bg_1 + a_1f + b_1g + c_1h), \\ &= ch_1\,(af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1) = 0. \end{aligned}$$

Point on two intersecting lines.

16. Let $(\alpha, \beta, \gamma, \delta)$ be the coordinates of the point of intersection of the two intersecting lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$. We have two systems of equations such as in No. 10, and comparing the corresponding equations of the two systems, we find

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= L : ag_1 - a_1g : ah_1 - a_1h : gh_1 - g_1h \\ &= bf_1 - b_1f : M : bh_1 - b_1h : hf_1 - h_1f \\ &= cf_1 - c_1f : cg_1 - c_1g : N : fg_1 - f_1g \\ &= -(bc_1 - b_1c) : -(ca_1 - c_1a) : -(ab_1 - a_1b) : P, \end{aligned}$$

where L, M, N, P, are in the first instance unknown; but, comparing the different sets of values, we have equations for finding the values of these quantities, and we thus obtain the more complete system

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= f_1a + b_1g + c_1h : ag_1 - a_1g : ah_1 - a_1h : gh_1 - g_1h \\ &= bf_1 - b_1f : a_1f + g_1b + c_1h : bh_1 - b_1h : hf_1 - h_1f \\ &= cf_1 - c_1f : cg_1 - c_1g : a_1f + b_1g + h_1c : fg_1 - f_1g \\ &= -(bc_1 - b_1c) : -(ca_1 - c_1a) : -(ab_1 - a_1b) : f_1a + g_1b + h_1c, \end{aligned}$$

where it is to be observed that the right-hand side considered as a matrix is the transposed matrix of that which occurs in No. 13, in the formula for $A : B : C : D$. The verification of the identity of the different sets of values can of course be effected as in No. 15.

Expression for an arbitrary plane through a line.

17. The condition in order that the plane $(A, B, C, D \between x, y, z, w) = 0$, may pass through the line (a, b, c, f, g, h), is the twofold relation given, No. 9; it is satisfied by any one of the four systems

$$\begin{aligned} A : B : C : D = {} & 0 : h : -g : a, \\ \text{or} = {} & -h : 0 : f : b, \\ \text{or} = {} & g : -f : 0 : c, \\ \text{or} = {} & -a : -b : -c : 0; \end{aligned}$$

and consequently also by

$$\begin{aligned} A : B : C : D = {} & (0, -h, g, -a \between \xi, \eta, \zeta, \omega) \\ & : (h, 0, -f, -b \between \xi, \eta, \zeta, \omega) \\ & : (-g, f, 0, -c \between \xi, \eta, \zeta, \omega) \\ & : (a, b, c, 0 \between \xi, \eta, \zeta, \omega); \end{aligned}$$

or, what is the same thing, by

$$\begin{aligned} A : B : C : D = {} & (0, h, -g, a \between \xi, \eta, \zeta, \omega) \\ & : (-h, 0, f, b \between \xi, \eta, \zeta, \omega) \\ & : (g, -f, 0, c \between \xi, \eta, \zeta, \omega) \\ & : (-a, -b, -c, 0 \between \xi, \eta, \zeta, \omega), \end{aligned}$$

where $(\xi, \eta, \zeta, \omega)$ are arbitrary: there is, however, no loss of generality in putting any two of these quantities $= 0$.

Expression for an arbitrary point in a line.

18. The condition in order that the point $(\alpha, \beta, \gamma, \delta)$, may lie in the line (a, b, c, f, g, h), is the twofold relation given, No. 10; it is satisfied by any one of the four systems

$$\begin{aligned} \alpha : \beta : \gamma : \delta = {} & 0 : c : -b : f, \\ \text{or} = {} & -c : 0 : a : g, \\ \text{or} = {} & b : -a : 0 : h, \\ \text{or} = {} & -f : -g : -h : 0; \end{aligned}$$

and consequently, also by

$$\begin{aligned} \alpha : \beta : \gamma : \delta = {} & (0, -c, b, -f \between x, y, z, w) \\ & : (c, 0, -a, -g \between x, y, z, w) \\ & : (-b, a, 0, -h \between x, y, z, w) \\ & : (f, g, h, 0 \between x, y, z, w); \end{aligned}$$

or, what is the same thing, by

$$\begin{aligned} \alpha : \beta : \gamma : \delta = {} & (0, c, -b, f \between x, y, z, w) \\ & : (-c, 0, a, g \between x, y, z, w) \\ & : (b, -a, 0, h \between x, y, z, w) \\ & : (-f, -g, -h, 0 \between x, y, z, w), \end{aligned}$$

where (x, y, z, w) are arbitrary: there is, however, no loss of generality in putting two of these quantities $= 0$.

Article Nos. 19 to 25. Geometrical considerations in regard to three, four, five, and six lines.

Before proceeding further, I will establish certain geometrical notions in regard to three, four, five, and six lines. I use the term 'tractor' to denote a line which meets any given lines.

19. Three given lines have an infinity of tractors; viz. these are the generating lines of a hyperboloid having the three given lines for directrices.

[435] ON THE SIX COORDINATES OF A LINE (CONTINUED).

29. If the two sets of coefficients are each of them the coordinates of a line, then the two equations express that the line (a, b, c, f, g, h) is any line whatever cutting each of the two given lines. And the general case is in fact reducible to this particular one; for suppose that neither set of coefficients belongs to a line, then we may from the two given linear relations form the relation

$$(\lambda F+\lambda_1 F_1,\ \lambda G+\lambda_1 G_1,\ \lambda H+\lambda_1 H_1,\ \lambda A+\lambda_1 A_1,\ \lambda B+\lambda_1 B_1,\ \lambda C+\lambda_1 C_1)\cdot(a,b,c,f,g,h)=0,$$

and if the ratio $\lambda : \lambda_1$ be properly determined, then $(\lambda A + \lambda_1 A_1, \ldots)$ will be the coordinates of a line. This will in fact be the case if

$$(\lambda A+\lambda_1 A_1)(\lambda F+\lambda_1 F_1)+(\lambda B+\lambda_1 B_1)(\lambda G+\lambda_1 G_1)+(\lambda C+\lambda_1 C_1)(\lambda H+\lambda_1 H_1)=0,$$

that is, if

$$(AF+BG+CH,\ AF_1+BG_1+CH_1+FA_1+GB_1+HC_1,\ A_1F_1+B_1G_1+C_1H_1)(\lambda,\lambda_1)^2=0,$$

a quadric equation giving two values of the ratio $\lambda : \lambda_1$, that is, two linear relations in each of which the coefficients are the coordinates of a line: we have thus two derived lines, and the line (a, b, c, f, g, h) meets each of these derived lines.

There is no real difference if one or the other of the given systems of coefficients, say the system (A, B, C, F, G, H), are the coordinates of a line. We have then $AF + BG + CH = 0$; the quadric equation in $\lambda : \lambda_1$ has a root $\lambda_1 : \lambda = 0$, and rejecting it, the other root is determined by a simple equation: this only means that the line (A, B, C, F, G, H) is itself one of the two derived lines.

But there is a real difference in the case where the equation in $\lambda : \lambda_1$ has equal roots; to explain this special case, observe that if in the general case we consider the two derived lines as a pair of tractors of any four lines, then the linear relations express that the line (a, b, c, f, g, h) has with these four lines a pair of tractors; and in the special case under consideration the linear relations express that the line (a, b, c, f, g, h) has with the four lines, or (what is the same thing) with any three of them, that is with some three lines, a twofold tractor. According to what precedes (No. 21), the construction of the line (a, b, c, f, g, h) is in fact as follows, viz. if on the twofold tractor considered as given, we take a series of points p, and through the tractor, homographic with the range, a pencil of planes P, then the sought-for line will be any line through a point p, in the corresponding plane P. But it is proper to give an analytical proof of the construction.

30. I observe that we may without loss of generality assume $A_1F_1 + B_1G_1 + C_1H_1 = 0$, and this being so, the condition for the equality of the roots of the quadric equation is

$$AF_1+BG_1+CH_1+FA_1+GB_1+HC_1=0,$$

that is, writing $(a_1, b_1, c_1, f_1, g_1, h_1)$ in place of $(A_1, B_1, C_1, F_1, G_1, H_1)$, the case in question may be taken to be that of

Two linear relations

$$(f_1,g_1,h_1,a_1,b_1,c_1)\,(a,b,c,f,g,h)=0,$$
$$(F,G,H,A,B,C)\,(a,b,c,f,g,h)=0,$$

where $(a_1, b_1, c_1, f_1, g_1, h_1)$ are, (A, B, C, F, G, H) are not, the coordinates of a line, and where

$$(f_1,g_1,h_1,a_1,b_1,c_1)\,(A,B,C,F,G,H)=0;$$

that is, where the twofold derived line is in fact the original line

$$(a_1,b_1,c_1,f_1,g_1,h_1).$$

31. To simplify, we may take $x = 0$, $y = 0$ for the equations of the line; the coordinates of the line then are $(a_1, b_1, c_1, f_1, g_1, h_1) = (0, 0, 0, 0, 0, 1)$. Taking moreover $x = 0$, $y = 0$, $\dfrac{z}{\gamma} = \dfrac{w}{\delta}$ for the coordinates of the point p, and $\dfrac{x}{\alpha} = \dfrac{y}{\beta}$ for the equation of the plane P, the homographic relation of the point and plane is given by an equation of the form

$$-F\beta\gamma + G\alpha\gamma - A\alpha\delta - B\beta\delta = 0,$$

or, as this may be written,

$$(F, G, H, A, B, C)\,(-\beta\gamma,\ \alpha\gamma,\ 0,\ -\alpha\delta,\ -\beta\delta,\ \omega) = 0,$$

where H and ω, being each multiplied by 0, do not really enter into the equation.

The equations of any line whatever through the point p and in the plane P may be written $\beta x - \alpha y = 0$, $A'x + B'y + \delta z - \gamma w = 0$, where A', B' are arbitrary: hence arranging the coefficients in the order

$$\begin{array}{cccc} \beta, & -\alpha, & 0, & 0, \\ A', & B', & \delta, & -\gamma, \end{array}$$

the coordinates (a, b, c, f, g, h) of the line in question are

$$(-\alpha\beta\gamma,\ \alpha\gamma\beta',\ 0,\ -\alpha\delta,\ -\beta\delta,\ A'\alpha + B'\beta);$$

correctly written:

$$(a, b, c, f, g, h) = (-\beta\gamma,\ \alpha\gamma,\ 0,\ -\alpha\delta,\ -\beta\delta,\ A'\alpha + B'\beta);$$

so that we have

$$(f_1, g_1, h_1, a_1, b_1, c_1)\,(a, b, c, f, g, h) = (0, 0, 1, 0, 0, 0)\,(a, b, c, f, g, h) = c = 0;$$

and moreover the homographic relation, replacing therein the arbitrary quantity ω by $A'\alpha + B'\beta$, becomes

$$(F, G, H, A, B, C)\,(a, b, c, f, g, h) = 0.$$

Hence the linear relations satisfied by the coordinates (a, b, c, f, g, h) of the line in question are

$$\begin{aligned} (f_1, g_1, h_1, a_1, b_1, c_1)\,(a, b, c, f, g, h) &= 0, \\ (F, G, H, A, B, C)\,(a, b, c, f, g, h) &= 0, \end{aligned}$$

with the coefficients

$$\begin{aligned} (f_1, g_1, h_1, a_1, b_1, c_1) &= (0, 0, 1, 0, 0, 0), \\ (A, B, C, F, G, H) &= (A, B, 0, F, G, H), \end{aligned}$$

values which satisfy the condition

$$(f_1, g_1, h_1, a_1, b_1, c_1)\,(A, B, C, F, G, H) = 0.$$

Hence the line (a, b, c, f, g, h) through the point p and in the plane P is a line the coordinates of which satisfy two linear relations as mentioned in the heading; and the theorem is thus proved. The demonstration would be simplified by taking, as is allowable, the homographic relation to be $\dfrac{\alpha}{\beta} = k\dfrac{\gamma}{\delta}$.

32. It appears from the foregoing examination of the case of two linear relations that in the following cases of three or more linear relations there is no real loss of generality in assuming that the coefficients of each set are the coordinates of a line; for if originally this be not so, we have only to replace the given relations by linear functions of these relations, and to assign such values to the multipliers λ, λ_1, $\lambda_2 \ldots$ as in each case to make the new coefficients to be the coordinates of a line; and as there are two or more arbitrary ratios $\lambda : \lambda_1 : \lambda_2 : \ldots$ to be assigned at pleasure and only a single condition to be satisfied, no cases of failure can arise. The remaining cases may consequently be stated in a more simple form.

Three linear relations, the coefficients of each set being the coordinates of a line.

33. The three relations express that the line (a, b, c, f, g, h) meets each of the three given lines; that is, that the line is any generating line of a hyperboloid having the three given lines for directrices.

Four linear relations, the coefficients of each set being the coordinates of a line.

34. The four relations express that the line (a, b, c, f, g, h) meets each of four given lines; or what is the same thing, that the line is a tractor of four given lines. It is to be noticed that the four linear relations serve to express the ratios $a : b : c : f : g : h$ linearly in terms of any one of these ratios, or what is the same thing, to express the several ratios in terms of an arbitrary ratio $u : v$. Substituting the resulting values in the equation

$$af + bg + ch = 0,$$

we have a quadric equation for the determination of the remaining ratio, or of the ratio $u : v$; and then each of the ratios of the coordinates can be expressed rationally in terms of either root of the quadric equation; we thus obtain the coordinates of each of the two tractors of the four given lines; or we have a complete analytical solution of the problem, to find the tractors of four given lines. The quadric equation may have equal roots; that is, the four given lines may have a twofold tractor, which is then determined linearly.

35. The theory of the linear relations of the coordinates (a, b, c, f, g, h) of a line may be considered in a different manner. It will be convenient to take the different cases in a reverse order, beginning with the extreme case (not before mentioned) of a fivefold relation and ascending to the case of a onefold or single relation.

Case of the fivefold relation.

36. The fivefold relation

$$\left\Vert \begin{matrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \end{matrix} \right\Vert = 0,$$

expresses that the quantities (a, b, c, f, g, h) are proportional to $(a_1, b_1, c_1, f_1, g_1, h_1)$. As the former set are by hypothesis the coordinates of a line, the given set $(a_1, b_1, c_1, f_1, g_1, h_1)$ must, it is clear, also be the coordinates of a line, and the relation then expresses that the line (a, b, c, f, g, h) coincides with the given line.

Case of the fourfold relation.

37. The fourfold relation is

$$\begin{Vmatrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \end{Vmatrix} = 0,$$

or what is the same thing, we have the six equations $\lambda a + \lambda_1 a_1 + \lambda_2 a_2 = 0$, &c., involving the indeterminate quantities λ, λ_1, λ_2. If the coefficients $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, b_2, c_2, f_2, g_2, h_2)$ are not either set the coordinates of a line; then substituting the foregoing values $-\lambda a = \lambda_1 a_1 + \lambda_2 a_2$, &c. in the equation $af + bg + ch = 0$, we have a quadric equation in $(\lambda_1 : \lambda_2)$: and for each root of this equation, the coefficients $\lambda_1 a_1 + \lambda_2 a_2$, &c. will be the coordinates of a line. There are thus in general two derived lines; and the fourfold relation expresses that the line (a, b, c, f, g, h) coincides with one or other of these derived lines. There is no real difference if one or the other of the two sets $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, b_2, c_2, f_2, g_2, h_2)$, or if each set, are the coordinates of a line; one of the derived lines or both of them will in these cases coincide with one or both of the given lines. And if the quadric equation has equal roots, then instead of two derived lines there is a twofold derived line, and the line (a, b, c, f, g, h) must coincide with this twofold line.

38. A case presenting peculiarity is however that in which the coefficients of the quadric equation vanish identically; this is only so when the coefficients $(a_1, b_1, c_1, f_1, g_1, h_1)$ and $(a_2, b_2, c_2, f_2, g_2, h_2)$ are the coordinates of two intersecting lines. The equations $-\lambda a = \lambda_1 a_1 + \lambda_2 a_2$, &c. here show that every line whatever which meets each of the two lines $(a_1, b_1, c_1, f_1, g_1, h_1)$ and $(a_2, b_2, c_2, f_2, g_2, h_2)$ meets also the line (a, b, c, f, g, h); that is, the line (a, b, c, f, g, h) is any line whatever in the plane and through the point of intersection of the two intersecting lines. We see moreover that not only

$$a_1 f_2 + b_1 g_2 + c_1 h_2 + a_2 f_1 + b_2 g_1 + c_2 h_1 = 0,$$

but also that $af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1 = 0$ and $af_2 + bg_2 + ch_2 + fa_2 + gb_2 + hc_2 = 0$; that is, the moment of each pair of lines is $= 0$. It may be remarked that the ratios $\lambda : \lambda_1 : \lambda_2$ may be determined from any two of the six equations

$$\lambda a + \lambda_1 a_1 + \lambda_2 a_2 = 0, \ \ldots \ \lambda h + \lambda_1 h_1 + \lambda_2 h_2 = 0;$$

but that in consequence of the moments being each $= 0$, there is not for the determination of these ratios any such set of equations as occur in the cases subsequently considered of a threefold relation, &c.

39. In what follows we have three or more sets $(a_1, b_1, c_1, f_1, g_1, h_1)$, &c.; and we may without loss of generality assume that each of these are the coordinates of a line: for replacing the several coefficients $a_1, \ldots$ by linear functions $\mu_1 a_1 + \mu_2 a_2 + \mu_3 a_3 +$ &c., &c., the multipliers may be determined so that these are the coordinates of a line: and since for each set there is only a single condition to be satisfied by the two or more ratios $\mu_1 : \mu_2 : \mu_3 \ldots$, it is easy to see that no cases of failure will arise.

Case of the threefold relation.

40. The threefold relation is

$$\begin{Vmatrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \end{Vmatrix} = 0,$$

where $(a_1,\ldots)$, $(a_2,\ldots)$, $(a_3,\ldots)$ are each the coordinates of a line. Here writing

$$\lambda a+\lambda_1a_1+\lambda_2a_2+\lambda_3a_3=0,\ \ldots$$

it is clear that every line which meets each of the lines $(a_1,\ldots)$, $(a_2,\ldots)$, $(a_3,\ldots)$ will also meet the line (a,b,c,f,g,h); the lines which meet the first-mentioned three lines are the generating lines of a hyperboloid having these three lines for directrices, and it hence appears that the line (a,b,c,f,g,h) is any directrix line whatever of the hyperboloid in question.

41. Using the notations 01, 02, 12, &c. to denote the moments of the several pairs of lines, viz.

$$\begin{aligned}01&=af_1+bg_1+ch_1+fa_1+gb_1+hc_1,\\12&=a_1f_2+b_1g_2+c_1h_2+f_1a_2+g_1b_2+h_1c_2,\\&\ \&\text{c.},\end{aligned}$$

then from the equations $\lambda a+\lambda_1a_1+\lambda_2a_2+\lambda_3a_3=0$, &c., we deduce

$$\begin{aligned}\lambda_1\,01+\lambda_2\,02+\lambda_3\,03&=0,\\\lambda\,10+\qquad+\lambda_2\,12+\lambda_3\,13&=0,\\\lambda\,20+\lambda_1\,21+\qquad+\lambda_3\,23&=0,\\\lambda\,30+\lambda_1\,31+\lambda_2\,32+\qquad&=0,\end{aligned}$$

and hence eliminating λ, λ_1, λ_2, λ_3, we find

$$\begin{vmatrix}\cdot&01&02&03\\10&\cdot&12&13\\20&21&\cdot&23\\30&31&32&\cdot\end{vmatrix}=0,$$

a relation between the moments satisfied in virtue of the given threefold relation; but which as a mere onefold relation is of course not equivalent to the threefold relation. It will subsequently appear that the equation expresses that any one of the four lines, say the line (a,b,c,f,g,h) touches the hyperboloid having the other three lines for generatrices; this condition is satisfied in virtue of the threefold relation which, as we have seen, expresses that the line (a,b,c,f,g,h) lies wholly in the hyperboloid in question.

42. The last mentioned determinant is the Norm of

$$\sqrt{01\cdot23}+\sqrt{02\cdot31}+\sqrt{03\cdot12},$$

so that the equation may be written

$$\sqrt{01\cdot23}+\sqrt{02\cdot31}+\sqrt{03\cdot12}=0,$$

or, what is the same thing,

$$\sqrt{01}\,\sqrt{23}+\sqrt{02}\,\sqrt{31}+\sqrt{03}\,\sqrt{12}=0,$$

it being of course understood that the signs of the radicals must be determined in accordance with this equation; we then find

$$\lambda:\lambda_1:\lambda_2:\lambda_3=\sqrt{23\cdot31\cdot12}:\sqrt{02\cdot03\cdot23}:\sqrt{03\cdot01\cdot31}:\sqrt{01\cdot02\cdot12},$$

or say

$$=\sqrt{23}\,\sqrt{31}\,\sqrt{12}:\sqrt{02}\,\sqrt{03}\,\sqrt{23}:\sqrt{03}\,\sqrt{01}\,\sqrt{31}:\sqrt{01}\,\sqrt{02}\,\sqrt{12};$$

in fact, substituting these last values in the linear equations for λ, λ_1, λ_2, λ_3, we find that the equations are all satisfied in virtue of the single equation

$$\sqrt{01}\,\sqrt{23}+\sqrt{02}\,\sqrt{31}+\sqrt{03}\,\sqrt{12}=0.$$

Case of the twofold relation.

43. We have here

$$\left\|\begin{matrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \\ a_4, & b_4, & c_4, & f_4, & g_4, & h_4 \end{matrix}\right\| = 0,$$

where $(a_1,\ldots)$, $(a_2,\ldots)$, $(a_3,\ldots)$, $(a_4,\ldots)$, are each the coordinates of a line. Here, writing

$$\lambda a+\lambda_1 a_1+\lambda_2 a_2+\lambda_3 a_3+\lambda_4 a_4=0,$$

it is clear that every line which meets each of the four given lines, will also meet the line (a,b,c,f,g,h); but the only lines meeting the four given lines are two determinate lines, the tractors of the four given lines; and the conclusion is, that the line (a,b,c,f,g,h) is any line whatever which meets the two tractors.

44. If, however, the four given lines have a twofold tractor, then the line (a,b,c,f,g,h) is still a line having two conditions imposed upon it; it is in fact a line determined as in No. 21, viz. if on the tractor we take a series of points p, and through the tractor a series of planes P, corresponding homographically to the points, then the line (a,b,c,f,g,h) is any line through a point p, in the corresponding plane P.

45. Using as before 01, 02, … 12, &c. to denote the moments of the several pairs of lines, we have

$$\begin{aligned} \lambda_1\,01+\lambda_2\,02+\lambda_3\,03+\lambda_4\,04&=0,\\ \lambda\,10+\quad+\lambda_2\,12+\lambda_3\,13+\lambda_4\,14&=0,\\ \lambda\,20+\lambda_1\,21+\quad+\lambda_3\,23+\lambda_4\,24&=0,\\ \lambda\,30+\lambda_1\,31+\lambda_2\,32+\quad+\lambda_4\,34&=0,\\ \lambda\,40+\lambda_1\,41+\lambda_2\,42+\lambda_3\,43+\quad&=0, \end{aligned}$$

and thence also

$$\begin{vmatrix} \cdot & 01 & 02 & 03 & 04 \\ 10 & \cdot & 12 & 13 & 14 \\ 20 & 21 & \cdot & 23 & 24 \\ 30 & 31 & 32 & \cdot & 34 \\ 40 & 41 & 42 & 43 & \cdot \end{vmatrix} = 0,$$

a relation between the moments satisfied in virtue of the original twofold relation; but which, as a single equation, is of course not equivalent to the twofold relation. It is in fact easy to see that this equation expresses that the five lines have a common tractor; this is true, since in virtue of the twofold relation there are really two common tractors.

I have not obtained from the linear equations any symmetrical expressions for the ratios $\lambda:\lambda_1:\lambda_2:\lambda_3:\lambda_4$.

Case of a onefold relation.

46. The onefold relation is

$$\left\|\begin{matrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \\ a_4, & b_4, & c_4, & f_4, & g_4, & h_4 \\ a_5, & b_5, & c_5, & f_5, & g_5, & h_5 \end{matrix}\right\| = 0,$$

where $(a_1, \ldots)$, $(a_2, \ldots)$, $(a_3, \ldots)$, $(a_4, \ldots)$, $(a_5, \ldots)$, are each the coordinates of points in a line. The preceding mode of dealing with the question is inapplicable, since there is not in general any line which meets the five given lines; in the particular case, however, where the five given lines are met by a single line, say when they have a common tractor, then the line (a, b, c, f, g, h) is any line meeting this common tractor. The general case is that of the involution of six lines, mentioned No. 25, and the consideration of which was deferred.

47. The onefold relation implies that we can find multipliers λ, μ, ν, ρ, σ, τ, such that

$$\begin{aligned} \lambda a + \mu b + \nu c + \rho f + \sigma g + \tau h &= 0, \\ \lambda a_1 + \mu b_1 + \nu c_1 + \rho f_1 + \sigma g_1 + \tau h_1 &= 0, \\ &\vdots \\ \lambda a_5 + \mu b_5 + \nu c_5 + \rho f_5 + \sigma g_5 + \tau h_5 &= 0; \end{aligned}$$

we may by means of the last five equations determine the ratios of $\lambda, \mu, \nu, \rho, \sigma, \tau$, viz. these quantities will be proportional to the determinants formed out of the matrix

$$\begin{pmatrix} a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \\ a_4, & b_4, & c_4, & f_4, & g_4, & h_4 \\ a_5, & b_5, & c_5, & f_5, & g_5, & h_5 \end{pmatrix}$$

and the first equation is then a linear relation in (a, b, c, f, g, h), expressing the relation that exists between these coordinates.

48. Consider an arbitrary point O on the line (a, b, c, f, g, h); taking this point as origin, the coordinates of O are $0, 0, 0, 1$; and if x, y, z, w, are the coordinates of any other point on the line, then writing

$$\begin{matrix} x, & y, & z, & w, \\ 0, & 0, & 0, & 1, \end{matrix}$$

we find $a : b : c : f : g : h = 0 : 0 : 0 : x : y : z$ and the equation

$$\lambda a + \mu b + \nu c + \rho f + \sigma g + \tau h = 0$$

becomes simply $\rho x + \sigma y + \tau z = 0$; viz. this equation expresses that the line (a, b, c, f, g, h), assumed to pass through a given point O, lies in a determinate plane Ω through this point.

49. To construct this plane Ω, I consider any four of the five given lines, say the lines 2, 3, 4, 5, and I endeavour to find the line OQ_1 through O, which has with these lines a pair of tractors. OQ_1 being a line through O, the coordinates of the line in question may be taken to be $0, 0, 0, F_1, G_1, H_1$, (where F_1, G_1, H_1, are in fact the coordinates x, y, z, of any point on the line OQ_1); and then the condition for the pair of tractors may be written

$$\begin{gathered} p_2a_2 + p_3a_3 + p_4a_4 + p_5a_5 = 0, \\ p_2b_2 + p_3b_3 + p_4b_4 + p_5b_5 = 0, \\ p_2c_2 + p_3c_3 + p_4c_4 + p_5c_5 = 0, \\ p_2f_2 + p_3f_3 + p_4f_4 + p_5f_5 = F_1, \\ p_2g_2 + p_3g_3 + p_4g_4 + p_5g_5 = G_1, \\ p_2h_2 + p_3h_3 + p_4h_4 + p_5h_5 = H_1, \end{gathered}$$

where $p_2, p_3, \ldots$ are arbitrary coefficients; and we hence deduce

$$\rho F_1 + \sigma G_1 + \tau H_1 = 0;$$

but in precisely the same way, if the line OQ_2 have with the lines $1, 3, 4, 5$, a pair of tractors, and if F_2, G_2, H_2, be the coordinates of a point on the line OQ_2, and similarly for the lines OQ_3, OQ_4, OQ_5, and the coordinates (F_3, G_3, H_3), (F_4, G_4, H_4), (F_5, G_5, H_5), we have

$$\begin{gathered} \rho F_2 + \sigma G_2 + \tau H_2 = 0, \\ \rho F_3 + \sigma G_3 + \tau H_3 = 0, \\ \rho F_4 + \sigma G_4 + \tau H_4 = 0, \\ \rho F_5 + \sigma G_5 + \tau H_5 = 0, \end{gathered}$$

and these equations show that the five lines $OQ_1, OQ_2, OQ_3, OQ_4, OQ_5$, lie in the plane

$$\rho x + \sigma y + \tau z = 0;$$

so that this plane is given as the plane through the lines $OQ_1, OQ_2, OQ_3, OQ_4, OQ_5$; and we have thus (given the lines $1, 2, 3, 4, 5$, and the arbitrary point O) the construction of the line (a, b, c, f, g, h) through O in involution with the given lines.

50. The original onefold relation may be replaced by the six equations

$$\begin{aligned} \lambda a + \lambda_1a_1 + \lambda_2a_2 + \lambda_3a_3 + \lambda_4a_4 + \lambda_5a_5 &= 0, \\ \lambda b + \lambda_1b_1 + \lambda_2b_2 + \lambda_3b_3 + \lambda_4b_4 + \lambda_5b_5 &= 0, \\ &\vdots \\ \lambda h + \lambda_1h_1 + \lambda_2h_2 + \lambda_3h_3 + \lambda_4h_4 + \lambda_5h_5 &= 0, \end{aligned}$$

and hence denoting as before the moments by $01, 02, 12$, &c. we have

$$\begin{aligned} \lambda_1\,01 + \lambda_2\,02 + \lambda_3\,03 + \lambda_4\,04 + \lambda_5\,05 &= 0, \\ \lambda\,10 + \quad + \lambda_2\,12 + \lambda_3\,13 + \lambda_4\,14 + \lambda_5\,15 &= 0, \\ \lambda\,20 + \lambda_1\,21 + \quad + \lambda_3\,23 + \lambda_4\,24 + \lambda_5\,25 &= 0, \\ \lambda\,30 + \lambda_1\,31 + \lambda_2\,32 + \quad + \lambda_4\,34 + \lambda_5\,35 &= 0, \\ \lambda\,40 + \lambda_1\,41 + \lambda_2\,42 + \lambda_3\,43 + \quad + \lambda_5\,45 &= 0, \\ \lambda\,50 + \lambda_1\,51 + \lambda_2\,52 + \lambda_3\,53 + \lambda_4\,54 + \quad &= 0, \end{aligned}$$

which lead to

$$\begin{vmatrix} \cdot & 01 & 02 & 03 & 04 & 05 \\ 10 & \cdot & 12 & 13 & 14 & 15 \\ 20 & 21 & \cdot & 23 & 24 & 25 \\ 30 & 31 & 32 & \cdot & 34 & 35 \\ 40 & 41 & 42 & 43 & \cdot & 45 \\ 50 & 51 & 52 & 53 & 54 & \cdot \end{vmatrix} = 0,$$

a relation between the moments equivalent to the original onefold relation, and consequently expressing that the six lines are in involution. I have not obtained a symmetrical system of values for the ratios $\lambda : \lambda_1 : \lambda_2 : \lambda_3 : \lambda_4 : \lambda_5$.

51. Reverting to the relation which exists between the point O and the plane Ω, it is proper to remark that, since to any given point O there corresponds a single plane Ω, and to any given plane Ω a single point O, it follows that the point O and plane Ω are reciprocal figures; viz. they are reciprocals of the particular kind treated of by Mobius, wherein the reciprocal of a point is a plane through the point, and the reciprocal of a plane a point in the plane; and of which the analytical character is that the reciprocal of the point $(\alpha, \beta, \gamma, \delta)$ is the plane

$$\begin{aligned}
&\big(\ \cdot + h\beta - g\gamma + l\delta\big)\, x\\
+&\big(-h\alpha + \cdot + f\gamma + m\delta\big)\, y\\
+&\big(\ g\alpha - f\beta + \cdot + n\delta\big)\, z\\
+&\big(-l\alpha - m\beta - n\gamma + \cdot\big)\, w = 0.
\end{aligned}$$

Article No. 52. A geometrical property of an involution of six lines.

52. The figure of six lines in involution is connected in various ways with the theory of cubic curves in space, for instance, considering a point A of the curve, this determines with any given line l a plane meeting the curve in two other points, and the line λ which joins these two points may be called the projection of the line l. This being so, if in any osculating plane of the cubic we have six lines $l, l_1, l_2, l_3, l_4, l_5$, tangents of a conic in that plane, the six projections $\lambda, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$ of these tangents will be a set of lines in involution. I do not stop to prove this theorem or to develope any of its consequences.

Article No. 53. To find the condition that four given lines may have a twofold tractor.

53. Taking the coordinates of the given lines to be

$$(a, b, c, f, g, h),\ (a_1, b_1, c_1, f_1, g_1, h_1),\ (a_2, b_2, c_2, f_2, g_2, h_2),\ (a_3, b_3, c_3, f_3, g_3, h_3),$$

then if (A, B, C, F, G, H) be the coordinates of a tractor of these lines, we have

$$\begin{aligned}
(F, G, H, A, B, C)\,(a, b, c, f, g, h) &= 0,\\
(F, G, H, A, B, C)\,(a_1, b_1, c_1, f_1, g_1, h_1) &= 0,\\
(F, G, H, A, B, C)\,(a_2, b_2, c_2, f_2, g_2, h_2) &= 0,\\
(F, G, H, A, B, C)\,(a_3, b_3, c_3, f_3, g_3, h_3) &= 0.
\end{aligned}$$

In virtue of these relations the ratios $A : B : C : F : G : H$ are given linear functions of any one of these ratios or of an arbitrary ratio $u : v$; and we then have $AF + BG + CH = 0$, a quadric equation for determining the unknown ratio. In the case of a twofold tractor, this equation must have equal roots; whence employing as usual the method of indeterminate multipliers, we find

$$\begin{aligned}
A + \lambda a + \lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 &= 0,\\
B + \lambda b + \lambda_1 b_1 + \lambda_2 b_2 + \lambda_3 b_3 &= 0,\\
C + \lambda c + \lambda_1 c_1 + \lambda_2 c_2 + \lambda_3 c_3 &= 0,\\
F + \lambda f + \lambda_1 f_1 + \lambda_2 f_2 + \lambda_3 f_3 &= 0,\\
G + \lambda g + \lambda_1 g_1 + \lambda_2 g_2 + \lambda_3 g_3 &= 0,\\
H + \lambda h + \lambda_1 h_1 + \lambda_2 h_2 + \lambda_3 h_3 &= 0.
\end{aligned}$$

Hence representing as before the moments of the pairs of lines by $01, 02$, &c., we deduce

$$\begin{aligned}
\lambda_1\, 01 + \lambda_2\, 02 + \lambda_3\, 03 &= 0,\\
\lambda\, 10 + \quad + \lambda_2\, 12 + \lambda_3\, 13 &= 0,\\
\lambda\, 20 + \lambda_1\, 21 + \quad + \lambda_3\, 23 &= 0,\\
\lambda\, 30 + \lambda_1\, 31 + \lambda_2\, 32 + \quad &= 0,
\end{aligned}$$

so that, as already mentioned, we have

$$\begin{vmatrix} \cdot & 01 & 02 & 03\\ 10 & \cdot & 12 & 13\\ 20 & 21 & \cdot & 23\\ 30 & 31 & 32 & \cdot \end{vmatrix} = 0,$$

as the condition that the four given lines may have a twofold tractor.

Article Nos. 54 to 56. Hyperboloid passing through three given lines.

54. The direct investigation is somewhat tedious; but I write down, and will afterwards verify, the equation of the hyperboloid passing through the three given lines

$$(a_1, b_1, c_1, f_1, g_1, h_1),\ (a_2, b_2, c_2, f_2, g_2, h_2),\ (a_3, b_3, c_3, f_3, g_3, h_3).$$

Writing for shortness (agh), &c. to denote the determinants

$$\begin{vmatrix} a_1, & g_1, & h_1\\ a_2, & g_2, & h_2\\ a_3, & g_3, & h_3 \end{vmatrix},\ \text{\&c.},$$

the equation of the hyperboloid is

$$\begin{aligned}
&(agh)\, x^2 + (bhf)\, y^2 + (cfg)\, z^2 + (abc)\, w^2\\
&\quad + \big[(afb) + (cah)\big]\, xw + \big[(bfg) + (chf)\big]\, yz\\
&\quad + \big[(bch) - (ahf)\big]\, yw + \big[(cgh) + (afg)\big]\, zx\\
&\quad + \big[(caf) - (bcg)\big]\, zw + \big[(ahf) + (bgh)\big]\, xy = 0.
\end{aligned}$$

In fact, we have

$$(agh) = a_1\,(g_2h_3 - g_3h_2) - g_1\,(h_2a_3 - h_3a_2) + h_1\,(a_2g_3 - a_3g_2) = a\cdot gh + g\cdot ha + h\cdot ag,$$

where a, &c. stand for a_1, &c. and gh, &c. for $g_2h_3 - g_3h_2$, &c. Hence the foregoing equation may be written

$$\begin{aligned}
&x^2\,(a\cdot gh + g\cdot ha + h\cdot ag)\\
&+ y^2\,(b\cdot hf + h\cdot fb + f\cdot bh)\\
&+ z^2\,(c\cdot fg + f\cdot gc + g\cdot cf)\\
&+ w^2\,(a\cdot bc + b\cdot ca + c\cdot ab)\\
&+ xw\begin{pmatrix} a\cdot fb + b\cdot ga + g\cdot ab\\ -\, c\cdot ah - a\cdot hc - h\cdot ca \end{pmatrix} + yz\begin{pmatrix} b\cdot fg + f\cdot gb + g\cdot bf\\ +\, c\cdot hf + h\cdot fc + f\cdot ch \end{pmatrix}\\
&+ yw\begin{pmatrix} b\cdot ch + c\cdot hb - h\cdot bc\\ -\, a\cdot bf - b\cdot fa - f\cdot ab \end{pmatrix} + zx\begin{pmatrix} c\cdot gh + g\cdot hc + h\cdot cg\\ +\, a\cdot fg + f\cdot ga + g\cdot af \end{pmatrix}\\
&+ zw\begin{pmatrix} c\cdot af + a\cdot fc + f\cdot ca\\ -\, b\cdot cg - c\cdot gb - g\cdot bc \end{pmatrix} + xy\begin{pmatrix} a\cdot hf + h\cdot fa + f\cdot ah\\ +\, b\cdot gh + g\cdot hb + h\cdot bg \end{pmatrix} = 0.
\end{aligned}$$

55. This is

$$\begin{aligned}
&bc\cdot w(\phantom{-hx+{}} hy-gz+aw)\\
&+ca\cdot w(-hx+fz+bw)\\
&+ab\cdot w(gx-fy+cw)\\
&+gh\cdot x(ax+by+cz)\\
&+hf\cdot y(ax+by+cz)\\
&+fg\cdot z(ax+by+cz)\\
&+af\bigl\{w(ax+by+cz)-x(hy-gz+aw)\bigr\}\\
&+bg\bigl\{w(ax+by+cz)-y(-hx+fz+bw)\bigr\}\\
&+ch\bigl\{w(ax+by+cz)-z(gx-fy+cw)\bigr\}\\
&-bf\cdot y(hy-gz+aw)\\
&-cf\cdot z(hy-gz+aw)\\
&-cg\cdot z(-hx+fz+bw)\\
&-ag\cdot x(-hx+fz+bw)\\
&-ah\cdot x(gx-fy+cw)\\
&-bh\cdot y(gx-fy+cw)=0.
\end{aligned}$$

56. Hence writing

$$(X,Y,Z,W)=\begin{pmatrix} \cdot & -h & g & a\\ h & \cdot & -f & b\\ -g & f & \cdot & c\\ -a & -b & -c & \cdot \end{pmatrix}(x,y,z,w),$$

the foregoing equation is

$$\begin{aligned}
&bc\cdot wX+ca\cdot wY+ab\cdot wZ-gh\cdot xW-hf\cdot yW-fg\cdot zW\\
&-af(wW+xX)-bg(wW+yY)-ch(wW+zZ)\\
&-bf\cdot yX-cg\cdot zY-ah\cdot xZ-cf\cdot zX-ag\cdot xY-bh\cdot yZ=0;
\end{aligned}$$

or, collecting and arranging, this is

$$\begin{aligned}
&X\bigl\{-af\cdot x-bf\cdot y-cf\cdot z+bc\cdot w\bigr\}\\
&+Y\bigl\{-ag\cdot x-bg\cdot y-cg\cdot z+ca\cdot w\bigr\}\\
&+Z\bigl\{-ah\cdot x-bh\cdot y-ch\cdot z+ab\cdot w\bigr\}\\
&+W\bigl\{-gh\cdot x-hf\cdot y-fg\cdot z+(af+bg+ch)w\bigr\}=0,
\end{aligned}$$

which is satisfied by $X=0$, $Y=0$, $Z=0$, $W=0$; that is, since (a,b,c,f,g,h) have been written in place of $(a_1,b_1,c_1,f_1,g_1,h_1)$, by $X_1=0$, $Y_1=0$, $Z_1=0$, $W_1=0$ (if we thus denote the corresponding functions of $(a_1,b_1,c_1,f_1,g_1,h_1)$), that is, the hyperboloid passes through the line $(a_1,b_1,c_1,f_1,g_1,h_1)$; and similarly it passes through the other two lines.

Article Nos. 57 and 58. The six coordinates defined as to their absolute magnitudes.

57. In all that precedes, the absolute magnitudes of the coordinates have been left indeterminate, only the ratios being attended to. But the magnitudes of the six coordinates may be fixed in a very simple manner as follows; viz. using ordinary rectangular coordinates, then for any line, if x_0, y_0, z_0 are the coordinates of a particular point on this line, and α, β, γ the inclinations of the line to the axes, the coordinates of another point on the line are

$$x_0 + r\cos\alpha,\ y_0 + r\cos\beta,\ z_0 + r\cos\gamma,$$

and hence writing

$$\begin{array}{cccc} x_0 + r\cos\alpha, & y_0 + r\cos\beta, & z_0 + r\cos\gamma, & 1, \\ x_0, & y_0, & z_0, & 1, \end{array}$$

we have

$$a : b : c : f : g : h = z_0\cos\beta - y_0\cos\gamma : x_0\cos\gamma - z_0\cos\alpha : y_0\cos\alpha - x_0\cos\beta : \cos\alpha : \cos\beta : \cos\gamma.$$

Or we may take

$$\begin{array}{ll} a = z_0\cos\beta - y_0\cos\gamma, & f = \cos\alpha, \\ b = x_0\cos\gamma - z_0\cos\alpha, & g = \cos\beta, \\ c = y_0\cos\alpha - x_0\cos\beta, & h = \cos\gamma, \end{array}$$

values which of course satisfy, as they should do, the relation $af + bg + ch = 0$. It is hardly necessary to remark, that the values of a, b, c are not altered on substituting for x_0, y_0, z_0 the coordinates $x_0 + s\cos\alpha$, $y_0 + s\cos\beta$, $z_0 + s\cos\gamma$ of any other point on the line.

58. Considering any two lines (a, b, c, f, g, h), $(a_1, b_1, c_1, f_1, g_1, h_1)$, if we define the moment of the two lines to be the product of the perpendicular distance into the sine of the inclination of the two lines, then we have, Moment

$$= af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1,$$

viz. we have now a quantitative definition of the function of the coordinates previously called the moment of the two lines.

For the demonstration of this formula it is to be remarked, that taking on the first line a segment of the length r, the coordinates of its extremities being (x_0, y_0, z_0) and $(x_0 + r\cos\alpha, y_0 + r\cos\beta, z_0 + r\cos\gamma)$, and on the second line a segment of the length r_1, the coordinates of its extremities being (x'_0, y'_0, z'_0) and $(x'_0 + r_1\cos\alpha_1, y'_0 + r_1\cos\beta_1, z'_0 + r_1\cos\gamma_1)$ and joining the extremities of these segments so as to form a tetrahedron, the volume of the tetrahedron is

$$\tfrac{1}{6} rr_1\,(af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1).$$

But the volume of the tetrahedron is also equal to $\frac{1}{6}$ of the product of the opposite edges into their perpendicular distance into the sine of the inclination of the two edges[1]; that is, it is $\frac{1}{6}rr_1$ into the moment of the two lines, and we have thus the formula in question.

Article Nos. 59 to 75. Statical and Kinematical Applications.

The coordinates (a, b, c, f, g, h), as last defined, are peculiarly convenient in kinematical and mechanical questions, as will appear from the following investigations.

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[1] I take the opportunity of mentioning a very simple demonstration of this formula: taking the opposite edges to be r, r_1, their inclination $= \theta$, and perpendicular distance $= h$; the section of the tetrahedron by a plane parallel to the two edges at the distances z, $h - z$ from the two edges respectively is a parallelogram, the sides of which are $\dfrac{rz}{h}$ and $\dfrac{r_1(h-z)}{h}$ respectively, and their inclination is $= \theta$; the area of the section is therefore $\dfrac{rr_1}{h^2}\sin\theta \cdot z(h-z)$, and the volume of the tetrahedron is $= \dfrac{rr_1}{h^2}\sin\theta\int_0^h z(h-z)\,dz = \frac{1}{6}rr_1\,h\sin\theta$. The same result is however obtained still more simply by drawing a plane through one of the two edges perpendicular to the other edge; the volume is then equal to the sum or the difference of the volumes of two tetrahedra standing on a common triangular base; and the required result at once follows.

59. Using the term rotation to denote an infinitesimal rotation, I say first that a rotation λ round the line (a, b, c, f, g, h) produces in the point (x, y, z) rigidly connected with this line the displacements

$$\begin{aligned} \delta x &= \lambda\,(\ \cdot - hy + gz - a), \\ \delta y &= \lambda\,(hx + \cdot - fz - b), \\ \delta z &= \lambda\,(-gx + fy + \cdot - c). \end{aligned}$$

In fact assuming for a moment that the axis of rotation passes through the origin, then for the point P coordinates x, y, z, the square of the perpendicular distance from the axis is

$$\begin{aligned} &(\cdot - y\cos\gamma + z\cos\beta)^2 \\ &+ (x\cos\gamma + \cdot - z\cos\alpha)^2 \\ &+ (-x\cos\beta + y\cos\alpha + \cdot)^2, \end{aligned}$$

and the expressions which enter into this formula denote as follows; viz. if through the point P at right angles to the plane through P and the axis of rotation we draw a line PQ, = perpendicular distance of P from the axis of rotation, then the coordinates of Q referred to P as origin are

$$\begin{aligned} \cdot\, &- y\cos\gamma + z\cos\beta, \\ x\cos\gamma &+ \cdot - z\cos\alpha, \\ -x\cos\beta &+ y\cos\alpha + \cdot, \end{aligned}$$

respectively. Hence the foregoing quantities each multiplied by λ are the displacements of the point P in the directions of the axes, produced by the rotation λ.

60. Suppose that the axis of rotation (instead of passing through the origin) pass through the point (x_0, y_0, z_0); the only difference is that we must in the formula write $(x - x_0, y - y_0, z - z_0)$ in place of (x, y, z): and attending to the significations of the six coordinates, it thus appears that the displacements produced by the rotation are equal to λ into the expressions

$$\begin{aligned} \cdot\, &- hy + gz - a, \\ hx &+ \cdot - fz - b, \\ -gx &+ fy + \cdot - c, \end{aligned}$$

respectively; which is the theorem in question.

61. I say secondly that considering in a solid body the point (x, y, z) situate in the line (a, b, c, f, g, h), and writing

$$a, b, c, f, g, h = z\cos\beta - y\cos\gamma,\ x\cos\gamma - z\cos\alpha,\ y\cos\alpha - x\cos\beta,\ \cos\alpha,\ \cos\beta,\ \cos\gamma,$$

then for any infinitesimal motion of the solid body the displacement of the point in the direction of the line is

$$= ap + bq + cr + fl + gm + hn,$$

where p, q, r, l, m, n are constants depending on the infinitesimal motion.

In fact for any infinitesimal motion of a solid body the displacements of the point (x, y, z) are

$$\begin{aligned} \delta x &= l + \cdot + ry - qz, \\ \delta y &= m - rx + \cdot + pz, \\ \delta z &= n + qx - py + \cdot, \end{aligned}$$

and hence the displacement in the direction of the line is

$$= \cos\alpha\,\delta x + \cos\beta\,\delta y + \cos\gamma\,\delta z,$$

which attending to the significations of (a, b, c, f, g, h) is

$$= ap + bq + cr + fl + gm + hn,$$

and we have thus the theorem in question.

62. It thus appears that for a system of rotations

$$\begin{array}{lll} \lambda_1 \text{ about the line} & & (a_1, b_1, c_1, f_1, g_1, h_1), \\ \lambda_2 & ,, & (a_2, b_2, c_2, f_2, g_2, h_2), \\ \&\text{c.} & ,, & \&\text{c.} \end{array}$$

the displacements of the point (x, y, z) rigidly connected with the several lines are

$$\begin{aligned} \delta x &= \qquad\quad \cdot - y\Sigma h\lambda + z\Sigma g\lambda - \Sigma a\lambda, \\ \delta y &= x\Sigma h\lambda + \cdot - z\Sigma f\lambda - \Sigma b\lambda, \\ \delta z &= -x\Sigma g\lambda + y\Sigma f\lambda + \cdot - \Sigma c\lambda, \end{aligned}$$

and when the rotations are in equilibrium then the displacements $(\delta x, \delta y, \delta z)$ of any point (x, y, z) whatever must each of them vanish; that is, we must have

$$\Sigma\lambda a = 0,\ \Sigma\lambda b = 0,\ \Sigma\lambda c = 0,\ \Sigma\lambda f = 0,\ \Sigma\lambda g = 0,\ \Sigma\lambda h = 0,$$

which are therefore the conditions for the equilibrium of the system of rotations.

63. And it further appears that for a system of forces acting on a rigid body,

$$\begin{array}{lll} \lambda_1 \text{ along the line} & & (a_1, b_1, c_1, f_1, g_1, h_1), \\ \lambda_2 & ,, & (a_2, b_2, c_2, f_2, g_2, h_2), \\ \&\text{c.} & & \end{array}$$

the conditions of equilibrium as given by the Principle of Virtual Velocities is

$$\Sigma\lambda\,(ap + bq + cr + fl + gm + hn) = 0,$$

or what is the same thing, that we have

$$\Sigma\lambda a = 0,\ \Sigma\lambda b = 0,\ \Sigma\lambda c = 0,\ \Sigma\lambda f = 0,\ \Sigma\lambda g = 0,\ \Sigma\lambda h = 0,$$

for the conditions of equilibrium of the system of forces λ_1, λ_2, &c. The conditions of equilibrium are thus precisely the same in the case of a system of rotations (infinitesimal rotations) and in that of a system of forces.

64. It now appears that the greater portion of the investigations in the first part of the present paper are applicable, and may be considered as relating, to the equilibrium of forces (or of rotations; but as the two theories are identical, it is sufficient to attend to one of them), and that we have in effect solved the following question, "Given any system of two, three, four, five or six lines considered as belonging to a solid body, to determine the relations between these lines in order that there may exist along them forces which are in equilibrium;" but for greater clearness I will consider the several cases in order; it is hardly necessary to remark that when the forces exist the equilibrium will depend on the ratios only, and that the absolute magnitude of any one of the forces may be assumed at pleasure.

65. The condition in the case of two lines is of course that these shall coincide together, or form one and the same line; and the forces are then equal and opposite forces.

66. In the case of three lines, these must meet in a point and lie in a plane; and the force along each line must then be as the sine of the angle between the other two lines.

67. Supposing that the forces are λ along the line (a, b, c, f, g, h), λ_1 along the line $(a_1, b_1, c_1, f_1, g_1, h_1)$, and λ_2 along the line $(a_2, b_2, c_2, f_2, g_2, h_2)$, the conditions of equilibrium are $\lambda a + \lambda_1 a_1 + \lambda_2 a_2 = 0, \ldots\ \lambda h + \lambda_1 h_1 + \lambda_2 h_2 = 0$, any two of which determine the ratios $\lambda : \lambda_1 : \lambda_2$; these ratios were not worked out *ante* No. 38 for the reason that with the coordinates there made use of, a symmetrical solution was not obtainable; but in the present case, selecting the last three equations, these are

$$\begin{aligned}
\lambda \cos\alpha + \lambda_1 \cos\alpha_1 + \lambda_2 \cos\alpha_2 &= 0,\\
\lambda \cos\beta + \lambda_1 \cos\beta_1 + \lambda_2 \cos\beta_2 &= 0,\\
\lambda \cos\gamma + \lambda_1 \cos\gamma_1 + \lambda_2 \cos\gamma_2 &= 0,
\end{aligned}$$

giving in the first instance an equation which expresses that the three lines (assumed to meet in a point) lie in the same plane: and then if 01, 02, 12 be the angles between the pairs of lines respectively, giving by an easy transformation

$$\begin{aligned}
\lambda + \lambda_1 \cos 01 + \lambda_2 \cos 02 &= 0,\\
\lambda \cos 10 + \lambda_1 + \lambda_2 \cos 12 &= 0,\\
\lambda \cos 20 + \lambda_1 \cos 21 + \lambda_2 &= 0.
\end{aligned}$$

68. Putting for shortness A, B, C in the place of $12, 20, 01$ respectively, we thence find

$$\begin{vmatrix} 1 & \cos C & \cos B \\ \cos C & 1 & \cos A \\ \cos B & \cos A & 1 \end{vmatrix} = 0,$$

that is

$$1 - \cos^2 A - \cos^2 B - \cos^2 C + 2\cos A\cos B\cos C = 0,$$

equivalent to $A + B + C = 2\pi$; and then from the first and second equations

$$\begin{aligned}
\lambda : \lambda_1 : \lambda_2 &= \cos A - \cos B\cos C : \cos B - \cos C\cos A : 1 - \cos^2 C,\\
&= \sin A\sin C : \sin B\sin C : \sin^2 C,\\
&= \sin A : \sin B : \sin C,
\end{aligned}$$

which is the required formula.

69. In the case of four given lines the condition (as noticed by Mobius) is that the four lines shall be generating lines of the same hyperboloid. In fact every line which meets three of the four lines must also meet the fourth line; for otherwise the moment of the system about such line would not be $= 0$. Calling the lines $0, 1, 2, 3$ and writing as before 01, 02, &c. for the moments of the several pairs of lines, then taking the moments of the system about the four lines respectively, we obtain directly the before-mentioned system of equations

$$\begin{aligned}
\lambda_1\, 01 + \lambda_2\, 02 + \lambda_3\, 03 &= 0,\\
\lambda\, 10 + \quad + \lambda_2\, 12 + \lambda_3\, 13 &= 0,\\
\lambda\, 20 + \lambda_1\, 21 + \quad + \lambda_3\, 23 &= 0,\\
\lambda\, 30 + \lambda_1\, 31 + \lambda_2\, 32 + \quad &= 0,
\end{aligned}$$

leading as before to the relation

$$\sqrt{01}\,\sqrt{23}+\sqrt{02}\,\sqrt{31}+\sqrt{03}\,\sqrt{12}=0,$$

and to the values

$$\lambda:\lambda_1:\lambda_2:\lambda_3=\sqrt{12}\,\sqrt{23}\,\sqrt{31}:\sqrt{23}\,\sqrt{30}\,\sqrt{02}:\sqrt{30}\,\sqrt{01}\,\sqrt{13}:\sqrt{01}\,\sqrt{12}\,\sqrt{20}$$

for the proportional magnitudes of the forces. These last equations give

$$\lambda\,\lambda_1\,01=\lambda_2\,\lambda_3\,23,$$

which, representing each force by a segment on the line along which the force acts, denotes that the tetrahedron of any two of the forces is equal to the tetrahedron of the other two forces; this is in fact equivalent to the theorem of M. Chasles, that if a system of forces be in any manner whatever reduced to two forces, the tetrahedron formed by these two forces has a constant volume.

70. In the case of five given lines, the lines must have a pair of tractors. Any four of the lines have in fact two tractors; and each of these tractors must meet the fifth line, for otherwise the moment of the system about the tractor would not be $= 0$. In the case where the four lines have a twofold tractor, the foregoing consideration shows only that the fifth line meets the twofold tractor, but it fails to show that the twofold tractor is a twofold tractor in regard to the fifth line.

71. I stop to consider this particular case under the present statical point of view. Taking the twofold tractor for the axis of z; let the line meet this line in the point $(0,0,c)$, the coordinates (a,b,c,f,g,h) of this line being consequently

$$(-c\cos\beta,\ c\cos\alpha,\ 0,\ \cos\alpha,\ \cos\beta,\ \cos\gamma)$$

and the like for the other four lines $1,2,3,4$. Using the sign Σ to refer to the last-mentioned four lines the equations of equilibrium become

$$\begin{aligned}-\lambda c\cos\beta+\Sigma\lambda_i c_i\cos\beta_i&=0,\\ \lambda c\cos\alpha+\Sigma\lambda_i c_i\cos\alpha_i&=0,\\ \lambda\cos\alpha+\Sigma\lambda_i\cos\alpha_i&=0,\\ \lambda\cos\beta+\Sigma\lambda_i\cos\beta_i&=0,\\ \lambda\cos\gamma+\Sigma\lambda_i\cos\gamma_i&=0.\end{aligned}$$

These equations give

$$\frac{\Sigma\lambda_i c_i\cos\beta_i}{\Sigma\lambda_i\cos\alpha_i}=\frac{c\cos\beta}{\cos\alpha},$$

we may without loss of generality take the homographic conditions which express that the axis of z is a twofold tractor of the four lines to be

$$\frac{c_1\cos\beta_1}{\cos\alpha_1}=\frac{c_2\cos\beta_2}{\cos\alpha_2}=\frac{c_3\cos\beta_3}{\cos\alpha_3}=\frac{c_4\cos\beta_4}{\cos\alpha_4},$$

and this being so, the last-mentioned equation becomes

$$\frac{c\cos\beta}{\cos\alpha},$$

and it thus appears that the axis of z is a twofold tractor in regard also to the line 0.

72. In the case of six lines such that there exist along them forces which are in equilibrium, taking this as a definition of the involution of six lines, we may very readily obtain from statical considerations the before-mentioned construction of the sixth line; viz. it may be shown that given any five of the lines, say the lines $1, 2, 3, 4, 5$, and a point O, we can through the point O determine a plane Ω, such that any line whatever through the point O and in the plane Ω is in involution with the five given lines. Consider the tractors of any four of the lines, say the lines $2, 3, 4, 5$; we may through the point O draw a line OA meeting the two tractors; that is, the lines $2, 3, 4, 5$ and the line OA will have a pair of common tractors. There consequently exist along these lines forces which are in equilibrium; and since only the ratios are material, the absolute magnitude of the force along the line OA may be anything whatever. Similarly, considering the tractors of the lines $1, 3, 4, 5$, and through O a line OB meeting these tractors, then there exist along the lines $1, 3, 4, 5$ and the line OB forces which are in equilibrium, and the absolute magnitude of the force along the line OB may be anything whatever. Hence, combining the two sets of forces, we have, along a line through O in the plane OA, OB, but otherwise indeterminate in its direction, a force in equilibrium with forces along the lines $1, 2, 3, 4, 5$; that is, the line found as above is a line in involution with the lines $1, 2, 3, 4, 5$.

73. It is to be added, that through O we cannot, out of the plane OA, OB, draw a line in involution with the lines $1, 2, 3, 4, 5$; for if any such line OK existed, then we should have along each of the lines OA, OB, OK forces in equilibrium with forces along the lines $1, 2, 3, 4, 5$; and the magnitudes of the three forces being each of them anything whatever, it would follow that along any line whatever through the point O there would exist a force in equilibrium with forces along the lines $1, 2, 3, 4, 5$; that is, any line whatever through the point O would be a line in involution with these lines.

74. It hence appears that drawing OA to meet the tractors of $2, 3, 4, 5$; OB to meet those of $3, 4, 5, 1$; OC to meet those of $4, 5, 1, 2$; OD to meet those of $5, 1, 2, 3$; and OE to meet those of $1, 2, 3, 4$; the lines OA, OB, OC, OD, OE will be in one plane, say the plane Ω: and that any line through O in the plane Ω will be a line in involution with the lines $1, 2, 3, 4, 5$.

75. There is another statical representation of the involution of six lines. If a system of forces act on a solid body, then taking six lines at random, the system will be in equilibrium if the sum of the moments be $= 0$ in regard to each of the six lines. But if the six lines be in involution; then, for the very reason that a rotation about one of these lines is resolvable into rotations about the other five lines, if the sum of the moments be $= 0$ for each of the five lines, it will also be $= 0$ for the sixth line: that is, it is not sufficient for the equilibrium of the forces that the sum of the moments shall be $= 0$ for each of the six lines. And we thus see that six lines in involution are lines such that the equilibrium of a system of forces about each of the six lines as axes does not insure the equilibrium of the system.

Article Nos. 76 and 77. Transformation of Coordinates.

76. Reverting to the general definition of the six coordinates (a, b, c, f, g, h) of a line by means of the points $(\alpha, \beta, \gamma, \delta)$ and $(\alpha', \beta', \gamma', \delta')$ on the line; suppose that instead of the original coordinate planes $x = 0$, $y = 0$, $z = 0$, $w = 0$ (forming a tetrahedron $ABCD$) we have new coordinate planes $x_0 = 0$, $y_0 = 0$, $z_0 = 0$, $w_0 = 0$ (forming a tetrahedron $A_0B_0C_0D_0$); and that the relations between the two sets of current coordinates are given by the equations

$$\begin{aligned}
x &= (\lambda_1, \mu_1, \nu_1, \rho_1)\,(x_0, y_0, z_0, w_0)\\
y &= (\lambda_2, \mu_2, \nu_2, \rho_2)\,(x_0, y_0, z_0, w_0)\\
z &= (\lambda_3, \mu_3, \nu_3, \rho_3)\,(x_0, y_0, z_0, w_0)\\
w &= (\lambda_4, \mu_4, \nu_4, \rho_4)\,(x_0, y_0, z_0, w_0),
\end{aligned}$$

with, of course, the like relations between the original coordinates $(\alpha, \beta, \gamma, \delta)$ and new coordinates $(\alpha_0, \beta_0, \gamma_0, \delta_0)$, and between the original coordinates $(\alpha', \beta', \gamma', \delta')$ and the new coordinates $(\alpha'_0, \beta'_0, \gamma'_0, \delta'_0)$, of the two points on the line (a, b, c, f, g, h); then taking $(a_0, b_0, c_0, f_0, g_0, h_0)$ as the new values of the six coordinates of the line, viz. writing

$$\begin{array}{rcccccc}
 & a_0, & b_0, & c_0, & f_0, & g_0, & h_0\\
= & \beta_0\gamma'_0 - \beta'_0\gamma_0, & \gamma_0\alpha'_0 - \gamma'_0\alpha_0, & \alpha_0\beta'_0 - \alpha'_0\beta_0, & \alpha_0\delta'_0 - \alpha'_0\delta_0, & \beta_0\delta'_0 - \beta'_0\delta_0, & \gamma_0\delta'_0 - \gamma'_0\delta_0,
\end{array}$$

we obtain a system of formulae which may be conveniently written as follows:

$$\begin{array}{l}
a : b : c : f : g : h =\\
\quad \dfrac{\mu\nu}{23}\,a_0 + \dfrac{\nu\lambda}{23}\,b_0 + \dfrac{\lambda\mu}{23}\,c_0 + \dfrac{\lambda\rho}{23}\,f_0 + \dfrac{\mu\rho}{23}\,g_0 + \dfrac{\nu\rho}{23}\,h_0\\
\quad (\text{suffix } 31)\\
\quad (\text{suffix } 12)\\
\quad (\text{suffix } 14)\\
\quad (\text{suffix } 24)\\
\quad (\text{suffix } 34)
\end{array}$$

viz. the top line stands for $(\mu_2\nu_3 - \mu_3\nu_2)\,a_0 + (\nu_2\lambda_3 - \nu_3\lambda_2)\,b_0 +$ &c., and the other lines are obtained from this by mere alterations of the suffixes.

77. As to the interpretation of these formulae, taking $ABCD$ as the fundamental tetrahedron for (x, y, z, w), then

$$\begin{array}{rlll}
(\lambda_1, \mu_1, \nu_1, \rho_1)\,(x_0, y_0, z_0, w_0) = 0 & \text{is the equation of plane} & BCD,\\
(\lambda_2, \mu_2, \nu_2, \rho_2)\,(\quad ,, \quad) = 0 & \quad ,, & CDA,\\
(\lambda_3, \mu_3, \nu_3, \rho_3)\,(\quad ,, \quad) = 0 & \quad ,, & DAB,
\end{array}$$

whence, observing that the second and third equations belong to two planes each passing through the line DA, it appears that the coefficients

$$\frac{\mu\nu}{23},\ \frac{\nu\lambda}{23},\ \frac{\lambda\mu}{23},\ \frac{\lambda\rho}{23},\ \frac{\mu\rho}{23},\ \frac{\nu\rho}{23}$$

are the six coordinates of the line DA, expressed in regard to the tetrahedron $(A_0B_0C_0D_0)$; and similarly that the coefficients in the six expressions of the transformation formula are the six coordinates of the lines AD, BD, CD, BC, CA, AB respectively in regard to the tetrahedron $(A_0B_0C_0D_0)$.

In the preceding formulae for the transformation of coordinates the ratios only have been attended to, no determinate absolute magnitudes have been assigned to the coordinates (a, b, c, f, g, h). But I will nevertheless show how we may attribute absolute magnitudes to these coordinates.

Article Nos. 78 to 80. New definition of the six coordinates as to their absolute magnitudes.

78. I assume (x, y, z, w) to be "volume" coordinates; viz. taking as before $ABCD$ for the fundamental tetrahedron, and denoting the point (x, y, z, w) by P, I assume that we have

$$x : y : z : w : 1 = PBCD : APCD : ABPD : ABCP : ABCD,$$

where $PBCD$, &c. denote the volumes of the several tetrahedra $PBCD$, &c. It is to be noticed that the volume is in every case taken with a determinate sign: analytically the sign may be fixed by taking (x_a, y_a, z_a), &c., as the Cartesian coordinates of the points A, &c. and writing

$$PBCD = \begin{vmatrix} x_p, & x_b, & x_c, & x_d \\ y_p, & y_b, & y_c, & y_d \\ z_p, & z_b, & z_c, & z_d \\ 1, & 1, & 1, & 1 \end{vmatrix}, \text{ \&c.}$$

(whence of course $PBCD = PCDA = -PCBD$, &c. according to the rule of signs): or we may in an equivalent manner, but less easily, determine the sign, by considering the sense of the rotation about CD (considered as an axis drawn from C to D) which would be produced by a force along PB (from P to B).

79. It is to be observed that the foregoing values give identically $x + y + z + w = 1$, so that the equation of the plane infinity is $x + y + z + w = 0$. The values of the coordinates (x, y, z, w) may be written

$$x : y : z : w : 1 = PBCD : PCAD : PABD : PCBA : ABCD;$$

or in the original form

$$x : y : z : w : 1 = PBCD : APCD : ABPD : ABCP : ABCD,$$

as may be most convenient.

80. Denoting the points $(\alpha, \beta, \gamma, \delta)$ and $(\alpha', \beta', \gamma', \delta')$ by Q, Q' respectively, we have

$$\alpha : \beta : \gamma : \delta : 1 = QBCD : AQCD : ABQD : ABCQ : ABCD,$$

and

$$\alpha' : \beta' : \gamma' : \delta' : 1 = Q'BCD : AQ'CD : ABQ'D : ABCQ' : ABCD,$$

and writing

$$(a, b, c, f, g, h) = (\beta\gamma' - \beta'\gamma,\ \gamma\alpha' - \gamma'\alpha,\ \alpha\beta' - \alpha'\beta,\ \alpha\delta' - \alpha'\delta,\ \beta\delta' - \beta'\delta,\ \gamma\delta' - \gamma'\delta),$$

viz. the two sets being taken to be equal, $a = \beta\gamma' - \beta'\gamma$, &c. instead of merely proportional, then it is easily seen that we obtain

$$a : b : c : f : g : h : 1 = AQQ'D : Q'BQD : QQ'CD : QBCQ' : AQCQ' : ABQQ' : ABCD,$$

that is, in order to form the first six combinations we successively replace

$$(B, C),\ (C, A),\ (A, B),\ (A, D),\ (B, D),\ (C, D)$$

in $ABCD$ by (Q, Q').

Article No. 81. Resulting formulae of Transformation.

81. For the transformation of coordinates if we assume

$$\begin{aligned} x &= (\lambda_1, \mu_1, \nu_1, \rho_1)\,(x_0, y_0, z_0, w_0), \\ y &= (\lambda_2, \mu_2, \nu_2, \rho_2)\,(\quad ,, \quad), \\ z &= (\lambda_3, \mu_3, \nu_3, \rho_3)\,(\quad ,, \quad), \\ w &= (\lambda_4, \mu_4, \nu_4, \rho_4)\,(\quad ,, \quad), \end{aligned}$$

and take also (a, b, c, f, g, h), $(a_0, b_0, c_0, f_0, g_0, h_0)$ respectively equal, instead of merely proportional, to the foregoing values, then, observing that for the point A_0 we have $(x_0, y_0, z_0, w_0) = (1, 0, 0, 0)$ we see that $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are the $A_0(BCD)$ coordinates of A_0; and the like as to the other sets of coefficients; viz. we have

$$\begin{aligned} \lambda_1 : \lambda_2 : \lambda_3 : \lambda_4 : 1 &= A_0BCD : AA_0CD : ABA_0D : ABCA_0 : ABCD, \\ \mu_1 : \mu_2 : \mu_3 : \mu_4 : 1 &= B_0\cdot,,\cdot : \cdot,,\cdot B_0\cdot,,\cdot : \cdot,,\cdot B_0\cdot,,\cdot : \cdot,,\cdot B_0 : \cdot,,\cdot, \\ \nu_1 : \nu_2 : \nu_3 : \nu_4 : 1 &= C_0\cdot,,\cdot : \cdot,,\cdot C_0\cdot,,\cdot : \cdot,,\cdot C_0\cdot,,\cdot : \cdot,,\cdot C_0 : \cdot,,\cdot, \\ \rho_1 : \rho_2 : \rho_3 : \rho_4 : 1 &= D_0\cdot,,\cdot : \cdot,,\cdot D_0\cdot,,\cdot : \cdot,,\cdot D_0\cdot,,\cdot : \cdot,,\cdot D_0 : \cdot,,\cdot, \end{aligned}$$

and we hence find

$$\begin{gathered} \frac{\mu\nu}{23} : \frac{\nu\lambda}{23} : \frac{\lambda\mu}{23} : \frac{\lambda\rho}{23} : \frac{\mu\rho}{23} : \frac{\nu\rho}{23} : 1 \\ = AB_0C_0D : AC_0A_0D : AA_0B_0D : AA_0D_0D : AB_0D_0D : AC_0D_0D : ABCD \end{gathered}$$

viz. multiplying the last-mentioned set of terms by $A_0B_0C_0D_0 \div ABCD$, in order to make the last term equal to unity, we see that the coefficients $\frac{\mu\nu}{23}$, &c. are equal to $\frac{A_0B_0C_0D_0}{ABCD}$ into the six $(A_0B_0C_0D_0)$-coordinates respectively of the line AD by means of the points A, D thereof. And similarly in the six expressions which enter into the formula of transformation, the coefficients are $\frac{A_0B_0C_0D_0}{ABCD}$ into the six $(A_0B_0C_0D_0)$-coordinates of the

line AD	in regard to points	A, D thereof
,, BD	,,	B, D
,, CD	,,	C, D
,, BC	,,	B, C
,, CA	,,	C, A
,, AB	,,	A, B

The foregoing theory of the transformation of coordinates seemed to me interesting for its own sake, and I have developed it in preference to the more simple theory which might easily be established of the case in which the coordinates are quantitatively defined as being equal to

$$(z_0 \cos\beta - y_0 \cos\gamma,\ x_0 \cos\gamma - z_0 \cos\alpha,\ y_0 \cos\alpha - x_0 \cos\beta,\ \cos\alpha,\ \cos\beta,\ \cos\gamma)$$

respectively.

[436] ON A CERTAIN SEXTIC TORSE.

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The torse (developable surface) intended to be considered is that which has for its edge of regression an excubo-quartic curve, or say a unicursal quartic curve. I call to mind that (excluding the plane quartic) a quartic curve is either a quadriquadric, viz. it is the complete intersection of two quadric surfaces; or else it is an excubo-quartic, viz. there is through the curve only one quadric surface, and the curve is the partial intersection of this quadric surface with a cubic surface through two generating lines (of the same kind) of the quadric surface. Returning to the quadriquadric curve, this may be general, nodal, or cuspidal; viz. if the two quadric surfaces have an ordinary contact, the curve of intersection is a nodal quadriquadric; if they have a stationary contact, the curve is a cuspidal quadriquadric.

The unicursal quartic is a curve such that the coordinates (x, y, z, w) of any point thereof are proportional to rational and integral quartic functions $(*\not\theta, 1)^4$ of a variable parameter θ; and the general unicursal quartic is in fact the excubo-quartic; but included as particular cases of the unicursal curve (although not as cases of the excubo-quartic as above defined) we have the nodal quadriquadric and the cuspidal quadriquadric. The torse having for its edge of regression a unicursal curve is a sextic torse; and this is in fact the order of the torse derived from the excubo-quartic, and from the nodal quadriquadric; but for the cuspidal quadriquadric, there is a depression of one, and the torse becomes a quintic torse. The equations have been obtained of (1) the sextic torse derived from the nodal quadriquadric, (2) the quintic torse derived from the cuspidal quadriquadric, (3) the sextic torse derived from a certain special excubo-quartic; but the equation of the torse derived from the general unicursal quartic has not yet been found. To show at the outset what the analytical problem is, I anticipate the remark that the coordinates (x, y, z, w) of a point on the curve may by an obvious reduction be rendered proportional to the fourth powers $(\theta+\alpha)^4$, $(\theta+\beta)^4$, $(\theta+\gamma)^4$, $(\theta+\delta)^4$ in the parameter θ; this leads to an equation

$$\frac{x}{(\theta+\alpha)^4} + \frac{y}{(\theta+\beta)^4} + \frac{z}{(\theta+\gamma)^4} + \frac{w}{(\theta+\delta)^4} = 0,$$

for the osculating plane at the point (x, y, z, w); or observing that this equation, when integralised, is of the form $(x, y, z, w)(\theta, 1)^{12} = 0$, we see that the equation is obtained by equating to zero the discriminant of a certain sextic function in θ; the discriminant is of the order 10 in the coordinates (x, y, z, w), but it obviously contains the factor $xyzw$, or throwing this out we have an equation of the order 6, so that the torse is (as above stated) a sextic torse.

Theorem relating to Four Binary Quartics.

1. Consider the four quartics:

$$\begin{gathered}(a_1, b_1, c_1, d_1, e_1)(x, y)^4,\\ (a_2, b_2, c_2, d_2, e_2)(x, y)^4,\\ (a_3, b_3, c_3, d_3, e_3)(x, y)^4,\\ (a_4, b_4, c_4, d_4, e_4)(x, y)^4,\end{gathered}$$

then if $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are any four quantities, these may be determined, and that in four different ways, so that

$$\lambda_1\,(a_1, \ldots)(x, y)^4 + \lambda_2\,(a_2, \ldots)(x, y)^4 + \lambda_3\,(a_3, \ldots)(x, y)^4 + \lambda_4\,(a_4, \ldots)(x, y)^4 = (\beta x + \alpha y)^4,$$

a perfect fourth power; in fact, equating the coefficients of the different powers of $(x, y)^4$, we have five equations, which determine the ratios of the unknown quantities $\lambda_1, \lambda_2, \lambda_3, \lambda_4$; α, β: eliminating $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, we find the equation

$$\begin{vmatrix} \beta^4 & 4\beta^3\alpha & 6\beta^2\alpha^2 & 4\beta\alpha^3 & \alpha^4 \\ a_1, & b_1, & c_1, & d_1, & e_1 \\ a_2, & b_2, & c_2, & d_2, & e_2 \\ a_3, & b_3, & c_3, & d_3, & e_3 \\ a_4, & b_4, & c_4, & d_4, & e_4 \end{vmatrix} = 0,$$

giving four different values of the ratio $\alpha : \beta$; or, assigning at pleasure a value to α or β (say $\beta = 1$), then to each of the four sets of values of (α, β) there correspond a determinate set of values of $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$; that is, we have as stated four sets of values of $\lambda_1, \lambda_2, \lambda_3, \lambda_4$; α, β.

[436] ON A CERTAIN SEXTIC TORSE (CONTINUED).

Standard Equation of the Unicursal Quartic.

2. The coordinates (x, y, z, w) being originally taken to be proportional to any four given quartic functions $(* \not\theta, 1)^4$ of the parameter θ, then forming a linear function of the coordinates, we have four sets of values of the multipliers, each reducing the function of θ to a perfect fourth power; that is, writing (X, Y, Z, W) for the linear functions of the original coordinates, and taking (X, Y, Z, W) as coordinates, it appears that the unicursal quartic may be represented by the equations

$$X : Y : Z : W = (\theta+\alpha)^4 : (\theta+\beta)^4 : (\theta+\gamma)^4 : (\theta+\delta)^4.$$

Tangent Line, and Osculating Plane of the Unicursal Quartic.

3. The equations of the tangent line at the point (θ) (that is, the point the coordinates whereof are as $(\theta+\alpha)^4 : (\theta+\beta)^4 : (\theta+\gamma)^4 : (\theta+\delta)^4$) are at once seen to be

$$\begin{vmatrix} X, & Y, & Z, & W \\ (\theta+\alpha)^4, & (\theta+\beta)^4, & (\theta+\gamma)^4, & (\theta+\delta)^4 \\ (\theta+\alpha)^3, & (\theta+\beta)^3, & (\theta+\gamma)^3, & (\theta+\delta)^3 \end{vmatrix} = 0,$$

and that of the osculating plane to be

$$\begin{vmatrix} X, & Y, & Z, & W \\ (\theta+\alpha)^4, & (\theta+\beta)^4, & (\theta+\gamma)^4, & (\theta+\delta)^4 \\ (\theta+\alpha)^3, & (\theta+\beta)^3, & (\theta+\gamma)^3, & (\theta+\delta)^3 \\ (\theta+\alpha)^2, & (\theta+\beta)^2, & (\theta+\gamma)^2, & (\theta+\delta)^2 \end{vmatrix} = 0.$$

Writing as in the sequel

$$\begin{aligned} a &= \beta-\gamma, \quad & f &= \alpha-\delta, \\ b &= \gamma-\alpha, \quad & g &= \beta-\delta, \\ c &= \alpha-\beta, \quad & h &= \gamma-\delta, \end{aligned}$$

the equations of the tangent line become

$$\frac{hY}{(\theta+\beta)^4} - \frac{gZ}{(\theta+\gamma)^4} + \frac{aW}{(\theta+\delta)^4} = 0,$$

$$-\frac{hX}{(\theta+\alpha)^4} + \frac{fZ}{(\theta+\gamma)^4} + \frac{bW}{(\theta+\delta)^4} = 0,$$

$$\frac{gX}{(\theta+\alpha)^4} - \frac{fY}{(\theta+\beta)^4} + \frac{cW}{(\theta+\delta)^4} = 0,$$

$$-\frac{aX}{(\theta+\alpha)^4} - \frac{bY}{(\theta+\beta)^4} - \frac{cZ}{(\theta+\gamma)^4} = 0$$

(equivalent of course to two equations), and the equation of the osculating plane becomes

$$\frac{ahgX}{(\theta+\alpha)^2} + \frac{hbfY}{(\theta+\beta)^2} + \frac{gfcZ}{(\theta+\gamma)^2} + \frac{abcW}{(\theta+\delta)^2} = 0.$$

Modification of the foregoing notation, and final form for the Unicursal Quartic.

4. If instead of the coordinates (X, Y, Z, W) we introduce the coordinates (x, y, z, w) connected therewith by the relations

$$ahgX : hbfY : gfcZ : abcW = x : y : z : w,$$

or, what is the same thing,

$$X : Y : Z : W = bcfx : cagy : abhz : fghw,$$

then the curve is given by the equations

$$x : y : z : w = ahg(\theta+\alpha)^4 : hbf(\theta+\beta)^4 : cfg(\theta+\gamma)^4 : abc(\theta+\delta)^4.$$

The equations of the tangent line are

$$\begin{aligned}
\frac{cy}{(\theta+\beta)^2} - \frac{bz}{(\theta+\gamma)^2} + \frac{fw}{(\theta+\delta)^2} &= 0,\\
-\frac{cx}{(\theta+\alpha)^2} + \frac{az}{(\theta+\gamma)^2} + \frac{gw}{(\theta+\delta)^2} &= 0,\\
\frac{bx}{(\theta+\alpha)^2} - \frac{ay}{(\theta+\beta)^2} + \frac{hw}{(\theta+\delta)^2} &= 0,\\
-\frac{fx}{(\theta+\alpha)^2} - \frac{gy}{(\theta+\beta)^2} - \frac{hz}{(\theta+\gamma)^2} &= 0,
\end{aligned}$$

and the equation of the osculating plane is

$$\frac{x}{(\theta+\alpha)^2} + \frac{y}{(\theta+\beta)^2} + \frac{z}{(\theta+\gamma)^2} + \frac{w}{(\theta+\delta)^2} = 0.$$

Determination of the Sextic Torse.

5. Starting from the equation of the osculating plane written under the form

$$\begin{aligned}
&x(\theta+\beta)^2(\theta+\gamma)^2(\theta+\delta)^2\\
&+y(\theta+\gamma)^2(\theta+\delta)^2(\theta+\alpha)^2\\
&+z(\theta+\delta)^2(\theta+\alpha)^2(\theta+\beta)^2\\
&+w(\theta+\alpha)^2(\theta+\beta)^2(\theta+\gamma)^2 = 0,
\end{aligned}$$

the equation of the torse is obtained by equating to zero the discriminant of the sextic function. Writing as before

$$\begin{aligned}
a &= \beta-\gamma, & f &= \alpha-\delta,\\
b &= \gamma-\alpha, & g &= \beta-\delta,\\
c &= \alpha-\beta, & h &= \gamma-\delta,
\end{aligned}$$

equations which give

$$\begin{array}{rrrrrrl}
 & & & h\,\cdot & -g\,\cdot & +a\,\cdot & = 0,\\
 & & -h\,\cdot & +f\,\cdot & +b\,\cdot & & = 0,\\
 & g\,\cdot & -f\,\cdot & +c\,\cdot & & & = 0,\\
-a\,\cdot & -b\,\cdot & -c\,\cdot & & & & = 0,
\end{array}$$

$$\begin{array}{rl} & h\beta - g\gamma + a\delta = 0, \\ -h\alpha & + f\gamma + b\delta = 0, \\ g\alpha - f\beta & + c\delta = 0, \\ -a\alpha - b\beta - c\gamma & = 0, \end{array}$$

and also

$$af + bg + ch = 0,$$

then the discriminant is a function of (x, y, z, w), (a, b, c, f, g, h) of the degree 10 in (x, y, z, w) and the degree 30 in (a, b, c, f, g, h). But the equation in θ has two equal roots, or the discriminant vanishes, if any one of the quantities (x, y, z, w) is $= 0$; and again, if any one of the differences $\alpha - \beta$, &c. (that is any one of the quantities a, b, c, f, g, h) is $= 0$: the discriminant thus contains the factors $xyzw$ and $(abcfgh)^2$, and throwing these out, we have an equation of the form

$$\Delta = (a, b, c, f, g, h)^{18}(x, y, z, w)^6 = 0,$$

which is the equation of the sextic torse.

Principal Sections of the Torse.

6. Consider for instance the section by the plane $w = 0$. Writing $w = 0$, the equation of the osculating plane is

$$(\theta + \delta)^2 \left[x(\theta + \beta)^2(\theta + \gamma)^2 + y(\theta + \gamma)^2(\theta + \alpha)^2 + z(\theta + \alpha)^2(\theta + \beta)^2\right] = 0.$$

The discriminant of the sextic function vanishes identically in virtue of the double factor $(\theta+\delta)^2$. But omitting this factor, the equation becomes

$$x(\theta + \beta)^2(\theta + \gamma)^2 + y(\theta + \gamma)^2(\theta + \alpha)^2 + z(\theta + \alpha)^2(\theta + \beta)^2 = 0.$$

The discriminant of this quartic function of θ is a function of x, y, z, a, b, c of the degree 6 in (x, y, z) and 12 in (a, b, c); it contains however the factors xyz, $a^2b^2c^2$, and the remaining factor is of the degree 3 in (x, y, z) and 6 in (a, b, c); this remaining factor is as will presently be seen

$$= (a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2\, xyz.$$

The last mentioned sextic equation in θ will have a triple root $\theta = -\delta$, if only the value $\theta = -\delta$ makes to vanish the factor in [ ], that is if we have

$$0 = g^2h^2x + h^2f^2y + f^2g^2z.$$

The foregoing results lead to the conclusion that for $w = 0$, we have

$$\Delta = (g^2h^2x + h^2f^2y + f^2g^2z)^3\left[(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2\, xyz\right];$$

but this will appear more distinctly as follows.

7. First, as to the factor $(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2\, xyz$: writing in the equation of the osculating plane $w = 0$, the equation becomes

$$\frac{x}{(\theta + \alpha)^2} + \frac{y}{(\theta + \beta)^2} + \frac{z}{(\theta + \gamma)^2} = 0,$$

which equation is therefore that of the trace of the osculating plane on the plane $w = 0$; the envelope of the trace in question is a part of the section of the torse by the plane $w = 0$. To find the equation of this envelope we must eliminate θ from the foregoing, and its derived equation

$$\frac{x}{(\theta + \alpha)^3} + \frac{y}{(\theta + \beta)^3} + \frac{z}{(\theta + \gamma)^3} = 0;$$

the two equations give

$$x : y : z = a(\theta+\alpha)^3 : b(\theta+\beta)^3 : c(\theta+\gamma)^3,$$

and thence

$$(a^2x)^{1/3} + (b^2y)^{1/3} + (c^2z)^{1/3} = a(\theta+\alpha) + b(\theta+\beta) + c(\theta+\gamma) = 0,$$

that is, we have

$$(a^2x)^{1/3} + (b^2y)^{1/3} + (c^2z)^{1/3} = 0,$$

or, what is the same thing,

$$(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2\,xyz = 0$$

for a part of the section in question.

8. I have said that the foregoing cubic is a part of the section; the equations

$$x : y : z : w = ahg(\theta+\alpha)^4 : bhf(\theta+\beta)^4 : cfg(\theta+\gamma)^4 : abc(\theta+\delta)^4,$$

which for $w = 0$ give $\theta = -\delta$, and thence $x : y : z = af^4 : bg^4 : ch^4$, show that the last-mentioned point is a four-pointic intersection of the curve with the plane $w = 0$. But the curve, having four consecutive points, will have three consecutive tangents in the plane $w = 0$; that is, the tangent at the point in question will present itself as a threefold factor in the equation of the torse. Writing in the equations of the tangent $w = 0$, $\theta = -\delta$, we find for the equation of the tangent in question

$$\frac{x}{f^2} : \frac{y}{g^2} : \frac{z}{h^2} = \frac{1}{a} : \frac{1}{b} : \frac{1}{c},$$

or, what is the same thing,

$$g^2h^2x + h^2f^2y + f^2g^2z = 0.$$

Hence the section by the plane $w = 0$ is made up of this line taken three times, and of the last mentioned cubic curve.

9. By symmetry, we conclude that the sections by the principal planes $x = 0$, $y = 0$, $z = 0$, $w = 0$, are each made up of a line taken three times, and of a cubic curve: viz. these are

$$\begin{array}{lllll}
x = 0, & & f^2h^2y + c^2f^2z + b^2c^2w = 0, & & (b^2y)^{1/3} + (g^2z)^{1/3} + (a^2w)^{1/3} = 0,\\
y = 0, & a^2f^2x & + c^2g^2z + c^2a^2w = 0, & & (h^2x)^{1/3} + (f^2z)^{1/3} + (b^2w)^{1/3} = 0,\\
z = 0, & a^2h^2x + b^2h^2y & + a^2b^2w = 0, & & (g^2x)^{1/3} + (f^2y)^{1/3} + (c^2w)^{1/3} = 0,\\
w = 0, & g^2h^2x + h^2f^2y + f^2g^2z & = 0, & & (a^2x)^{1/3} + (b^2y)^{1/3} + (c^2z)^{1/3} = 0,
\end{array}$$

where for shortness I have written the equations of the four cubics in their irrational forms respectively.

Partial Determination of the Equation.

10. As the value of Δ is known when any one of the coordinates x, y, z, w is put $= 0$, we in fact know all the terms of Δ, except those which contain the factor $xyzw$, which unknown terms, as Δ is of the degree 6, are of the form $(*)(x, y, z, w)^2$.

I remark that if $(xyzw)$ is any homogeneous function $(*)(x,y,z,w)^n$, and $(xyz), (xy), (x)$ are what $(xyzw)$ become on putting therein $(w=0)$, $(z=0, w=0)$, $(y=0, z=0, w=0)$ respectively, and the like for the other similar symbols, then that

$$\begin{aligned}(xyzw) = (x) + (y) + (z) + (w) \\ &- (xy) - (xz) - (xw) - (yz) - (yw) - (zw) \\ &+ (xyz) + (xyw) + (xzw) + (yzw) \\ &+ \text{terms multiplied by } xyzw\end{aligned}$$

in fact, omitting the last line, this equation on writing therein $x=0$ or $y=0$ or $z=0$ or $w=0$, becomes an identity, that is, the difference of the two sides vanishes when any one of these equations is satisfied, and such difference contains therefore the factor $xyzw$; which proves the theorem. It hence appears that the equation $\Delta = 0$ of the torse is

$$\begin{aligned}\Delta = {} & a^2g^2h^2x^6 + b^2h^2f^2y^6 + c^2f^2g^2z^6 + a^2b^2c^2w^6 \\ &- (g^2h^2x + h^2f^2y)^3(a^2x + b^2y)^3 \\ &- (h^2f^2y + f^2g^2z)^3(b^2y + c^2z)^3 \\ &- (g^2h^2x + f^2g^2z)^3(a^2x + c^2z)^3 \\ &- (a^2h^2x + a^2b^2w)^3(g^2x + b^2w)^3 \\ &- (b^2h^2y + a^2b^2w)^3(f^2y + c^2w)^3 \\ &- (c^2f^2z + c^2a^2w)^3(g^2z + b^2w)^3 \\ &+ (b^2h^2y + c^2f^2z + b^2c^2w)^3\left[(h^2y + g^2z + a^2w)^3 - 27a^2g^2h^2\, yzw\right] \\ &+ (a^2f^2x + c^2g^2z + c^2a^2w)^3\left[(h^2x + f^2z + b^2w)^3 - 27b^2h^2f^2\, zxw\right] \\ &+ (a^2h^2x + b^2h^2y + a^2b^2w)^3\left[(g^2x + f^2y + c^2w)^3 - 27c^2f^2g^2\, xyw\right] \\ &+ (g^2h^2x + h^2f^2y + f^2g^2z)^3\left[(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2\, xyz\right] \\ &+ xyzw \cdot (*)(x,y,z,w)^2,\end{aligned}$$

where the ten coefficients of $(*)(x,y,z,w)^2$ remain to be found.

Process for the Determination of the Unknown Coefficients.

11. At a point of the cubic curve in the plane $w=0$, we have

$$x : y : z = a(\theta+\alpha)^3 : b(\theta+\beta)^3 : c(\theta+\gamma)^3;$$

and the tangent plane at this point is the osculating plane of the curve; that is, it is the plane

$$\frac{x'}{(\theta+\alpha)^2} + \frac{y'}{(\theta+\beta)^2} + \frac{z'}{(\theta+\gamma)^2} + \frac{w'}{(\theta+\delta)^2} = 0,$$

if for a moment (x', y', z', w') are the current coordinates of a point in the tangent plane. But the equation of the tangent plane as deduced from the equation $\Delta = 0$ is

$$x'\frac{\mathrm{d}\Delta}{\mathrm{d}x} + y'\frac{\mathrm{d}\Delta}{\mathrm{d}y} + z'\frac{\mathrm{d}\Delta}{\mathrm{d}z} + w'\frac{\mathrm{d}\Delta}{\mathrm{d}w} = 0,$$

where in the differential coefficients of Δ, the coordinates (x,y,z,w) are considered as having the values

$$x : y : z : w = a(\theta+\alpha)^3 : b(\theta+\beta)^3 : c(\theta+\gamma)^3 : 0.$$

Hence, with these values of (x,y,z,w), we have

$$\frac{\mathrm{d}\Delta}{\mathrm{d}x} : \frac{\mathrm{d}\Delta}{\mathrm{d}y} : \frac{\mathrm{d}\Delta}{\mathrm{d}z} : \frac{\mathrm{d}\Delta}{\mathrm{d}w} = \frac{1}{(\theta+\alpha)^2} : \frac{1}{(\theta+\beta)^2} : \frac{1}{(\theta+\gamma)^2} : \frac{1}{(\theta+\delta)^2},$$

conditions which determine the values of certain of the coefficients of $(*)(x, y, z, w)^2$, viz. the six coefficients of the terms independent of w; and when these are known the values of the remaining four coefficients are at once obtained by symmetry.

12. To develope this process, disregarding the higher powers of w, we may write

$$\Delta = \Theta + 3\Phi w + xyzw(*)(x, y, z)^2,$$

where Θ denotes the terms independent of w, $3\Phi w$ the known terms which contain the factor w, and $xyzw(*)(x, y, z)^2$ the unknown terms which contain this same factor; the value of $(*)(x, y, z)^2$ being clearly $= (*)(x, y, z, 0)^2$.

We have, moreover,

$$\Theta = (g^2h^2x + h^2f^2y + f^2g^2z)^3\big[(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2\,xyz\big],$$

and

$$\begin{aligned}\Phi = {} & f^2(h^2y + g^2z)^2\big[a^2f^2(h^4y^2 - 7h^2g^2\,yz + g^4z^2)(b^2y + c^2z) + b^2c^2(h^2y + g^2z)^2\big]\\ & + g^4(a^2x + c^2z)^2\big[b^2g^2(h^4x^2 - 7h^2f^2\,xz + f^4z^2)(a^2x + c^2z) + c^2a^2(h^2x + f^2z)^2\big]\\ & + h^4(a^2x + b^2y)^2\big[c^2h^2(g^4x^2 - 7g^2f^2\,xy + f^4y^2)(a^2x + b^2y) + a^2b^2(g^2x + f^2y)^2\big]\\ & - a^2h^4g^4(b^2y + c^2z)^2x^2\\ & - b^2h^4f^4(c^2z + a^2x)^2y^2\\ & - c^2f^4g^4(a^2x + b^2y)^2z^2.\end{aligned}$$

13. The equations, putting after the differentiations $w = 0$, and writing for shortness $(*)$ in place of $(*)(x, y, z)^2$, become

$$\frac{\mathrm{d}\Theta}{\mathrm{d}x} : \frac{\mathrm{d}\Theta}{\mathrm{d}y} : \frac{\mathrm{d}\Theta}{\mathrm{d}z} : 3\Phi + xyz(*) = \frac{1}{(\theta+\alpha)^2} : \frac{1}{(\theta+\beta)^2} : \frac{1}{(\theta+\gamma)^2} : \frac{1}{(\theta+\delta)^2}.$$

Now, observing that the second factor of Θ vanishes for the values

$$a(\theta+\alpha)^3,\quad b(\theta+\beta)^3,\quad c(\theta+\gamma)^3 \text{ of } (x, y, z),$$

we have simply

$$\frac{\mathrm{d}\Theta}{\mathrm{d}x} = (g^2h^2x + h^2f^2y + f^2g^2z)^3 \cdot 3a^2\big[(a^2x + b^2y + c^2z)^2 - 9b^2c^2\,yz\big].$$

But

$$a^2x + b^2y + c^2z = a^2(\theta+\alpha)^3 + b^2(\theta+\beta)^3 + c^2(\theta+\gamma)^3,$$

in virtue of the relation $a(\theta+\alpha) + b(\theta+\beta) + c(\theta+\gamma) = 0$, and hence

$$\begin{aligned}\big[(a^2x + b^2y + c^2z)^2 - 9b^2c^2\,yz\big] &= 9b^2c^2(\theta+\beta)^2(\theta+\gamma)^2\big[a^2(\theta+\alpha)^2 - bc(\theta+\beta)(\theta+\gamma)\big],\\ &= 9b^2c^2(\theta+\beta)^2(\theta+\gamma)^2\,Q,\end{aligned}$$

where

$$\begin{aligned}Q &= a^2(\theta+\alpha)^2 - bc(\theta+\beta)(\theta+\gamma),\\ &= b^2(\theta+\beta)^2 - ca(\theta+\gamma)(\theta+\alpha),\\ &= c^2(\theta+\gamma)^2 - ab(\theta+\alpha)(\theta+\beta).\end{aligned}$$

Hence

$$\frac{\mathrm{d}\Theta}{\mathrm{d}x} = 27a^2b^2c^2\,(g^2h^2x + h^2f^2y + f^2g^2z)^3(\theta+\beta)^2(\theta+\gamma)^2\,Q,$$

and similarly

$$\frac{d\Theta}{dy} = 27a^2b^2c^2\,(g^2h^2x + h^2f^2y + f^2g^2z)^3(\theta+\gamma)^2(\theta+\alpha)^2\,Q,$$

$$\frac{d\Theta}{dz} = 27a^2b^2c^2\,(g^2h^2x + h^2f^2y + f^2g^2z)^3(\theta+\alpha)^2(\theta+\beta)^2\,Q;$$

whence the above-mentioned conditions reduce themselves to the single condition

$$(\theta+\delta)^2\big\{3\Phi + xyz(*)\big\} = 27a^2b^2c^2\,(g^2h^2x + h^2f^2y + f^2g^2z)^3(\theta+\alpha)^2(\theta+\beta)^2(\theta+\gamma)^2\,Q.$$

14. But we have

$$\begin{aligned} g^2h^2x + h^2f^2y + f^2g^2z &= g^2h^2a(\theta+\delta+f)^3 + h^2f^2b(\theta+\delta+g)^3 + f^2g^2c(\theta+\delta+h)^3, \\ &= (\theta+\delta)^3\big[(g^2h^2a + h^2f^2b + f^2g^2c)(\theta+\delta) + 3(gha + hfb + fgc)fgh\big], \\ &= -abc(\theta+\delta)^2\big[(gh + hf + fg)(\theta+\delta) + 3fgh\big], \\ &= -abc(\theta+\delta)^2\big[gh(\theta+\alpha) + hf(\theta+\beta) + fg(\theta+\gamma)\big], \\ &= -abc(\theta+\delta)^2\,P, \end{aligned}$$

if for shortness

$$P = gh(\theta+\alpha) + hf(\theta+\beta) + fg(\theta+\gamma).$$

Hence, substituting,

$$3\Phi + abc(\theta+\alpha)^3(\theta+\beta)^3(\theta+\gamma)^3\,(*) = -27(abc)^5(\theta+\delta)^4(\theta+\alpha)^2(\theta+\beta)^2(\theta+\gamma)^2\,P^3Q$$

which when the values $a(\theta+\alpha)^3$, $b(\theta+\beta)^3$, $c(\theta+\gamma)^3$ for (x, y, z) are substituted in the functions Φ and $(*)$, will be an identical equation in θ.

15. It is right to remark that what we require is the expression of $(*)$, $= (*)(x, y, z)^2$; the foregoing equation leads to the value of $(*)$ expressed in terms of θ; and it is necessary to show that this leads back to the expression for $(*)$ as a function of (x, y, z); in fact, that the function of θ is transformable in a definite manner into a function of (x, y, z). Suppose that the function of θ could be expressed in two different manners as a function of (x, y, z); then we should have two different functions $(x, y, z)^2$ each equivalent to the same function of θ, and the difference of these functions would be identically $= 0$; that is, we should have a function $(x, y, z)^2$ vanishing identically by the substitution

$$x : y : z = a(\theta+\alpha)^3 : b(\theta+\beta)^3 : c(\theta+\gamma)^3;$$

but these relations are equivalent to the single relation

$$(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2\,xyz = 0,$$

which, *quâ* cubic equation, is not equivalent to any equation whatever of the form

$$(x, y, z)^2 = 0;$$

that is, the function of θ is equivalent to a definite function $(x, y, z)^2$.

16. To proceed with the reduction, I remark that we have

$$\begin{aligned}\Phi = {} & (a^2x+b^2y+c^2z)\\ & \times \Big\{ f^4\big[a^2f^2(h^4y^2-7h^2g^2\,yz+g^4z^2)(b^2y+c^2z)+b^2c^2(h^2y+g^2z)^2\big]\\ & \quad + g^4\big[b^2g^2(h^4x^2-7h^2f^2\,xz+f^4z^2)(a^2x+c^2z)+c^2a^2(h^2x+f^2z)^2\big]\\ & \quad + h^4\big[c^2h^2(g^4x^2-7g^2f^2\,xy+f^4y^2)(a^2x+b^2y)+a^2b^2(g^2x+f^2y)^2\big]\\ & \quad - a^2h^4g^4(b^2y+c^2z)^2x^2\\ & \quad - b^2h^4f^4(c^2z+a^2x)^2y^2\\ & \quad - c^2f^4g^4(a^2x+b^2y)^2z^2\Big\}\\ & + xyz\,\Omega,\end{aligned}$$

where

$$\begin{aligned}-\Omega = {} & 2(b^2c^2Ax^2+c^2a^2By^2+a^2b^2Cz^2)\\ & + (My+Nz)(a^4x+2a^2b^2y+2a^2c^2z)\\ & + (Oz+Px)(2a^2b^2x+b^4y+2b^2c^2z)\\ & + (Qx+Ry)(2a^2c^2x+2b^2c^2y+c^4z),\end{aligned}$$

if for shortness

$$\begin{aligned}A &= a^2f^4h^4(b^2g^2+c^2h^2),\\ B &= b^2g^4f^4(c^2h^2+a^2f^2),\\ C &= c^2h^4g^4(a^2f^2+b^2g^2),\end{aligned}$$

$$\begin{aligned}M &= f^2h^2(a^2f^2b^2h^2-7a^2f^4g^2+3b^2g^4c^2h^2), & N &= f^2g^2(-7a^2f^2c^2h^2+a^2f^2b^2h^2+3b^2c^4g^2),\\ O &= g^2f^2(b^2g^2c^2f^2-7b^2c^2h^2g^2+3c^2h^4a^2f^2), & P &= g^2h^2(-7b^2c^2a^2f^2+b^2c^4h^2+3c^2h^4a^2f^2),\\ Q &= h^2g^2(c^2h^2b^2g^2-7c^2h^2a^2b^2+3a^2f^4b^2g^2), & R &= h^2f^2(-7c^2h^2b^2g^2+c^2h^2a^2f^2+3a^2f^4b^2g^2),\end{aligned}$$

and I represent the foregoing equation by

$$\Phi = (a^2x+b^2y+c^2z)U + xyz\,\Omega.$$

Hence, writing for x, y, z the foregoing values, we have

$$\Phi = 9a^2b^2c^2(\theta+\alpha)^2(\theta+\beta)^2(\theta+\gamma)^2\,U + abc(\theta+\alpha)^3(\theta+\beta)^3(\theta+\gamma)^3\,\Omega,$$

and thence

$$27\,U + \tfrac{\Omega}{(\theta+\alpha)(\theta+\beta)(\theta+\gamma)}\big(3\Omega+(*)\big) = -27(abc)^3(\theta+\delta)^4\,P^3Q,$$

that is

$$27\big[U+(abc)^3(\theta+\delta)^4\,P^3Q\big] + \tfrac{\Omega}{(\theta+\alpha)(\theta+\beta)(\theta+\gamma)}\big(3\Omega+(*)\big) = 0.$$

In order that this may be the case, it is clear that we must have

$$U+(abc)^3(\theta+\delta)^4\,P^3Q = (\theta+\alpha)(\theta+\beta)(\theta+\gamma)\,M,$$

viz. the left-hand side expressed as a function of θ must be divisible by the product $(\theta+\alpha)(\theta+\beta)(\theta+\gamma)$. Assuming for a moment that this is so, the quotient M will be a function $(\theta, 1)^6$ expressible in a unique manner in the form $(x, y, z)^2$, and assuming it to be so expressed, we have

$$27Mabc + 3\Omega + (*) = 0;$$

which equation, without any further substitution of the θ-values of (x, y, z), gives $(*)$ in its proper form as a function of (x, y, z).

Reduction of the Equation, $U + (abc)^3(\theta+\delta)^4P^3Q = (\theta+\alpha)(\theta+\beta)(\theta+\gamma)M$.

17. We have by an easy transformation

$$\begin{aligned}U = (a^2x+b^2y+c^2z)\Big\{ & a^2f^2(h^4y^2-7h^2g^2\,yz+g^4z^2)\\ & + b^2g^2(f^4z^2-7f^2h^2\,zx+h^4x^2)\\ & + c^2h^2(g^4x^2-7g^2f^2\,xy+f^4y^2)\Big\}\\ & + 7(a^4f^4+b^4g^4+c^4h^4)f^2g^2h^2\,xyz\\ & + U',\end{aligned}$$

if for shortness

$$\begin{aligned}U' = {} & y^2z\cdot a^2f^4h^4(3b^2g^2-c^2h^2)\\ & + yz^2\cdot b^2g^4f^4(3c^2h^2-b^2g^2)\\ & + z^2x\cdot c^2f^4g^4(3a^2h^2-a^2f^2)\\ & + zx^2\cdot c^2h^4g^4(3a^2f^2-b^2h^2)\\ & + x^2y\cdot b^2g^4h^4(3a^2f^2-b^2g^2)\\ & + xy^2\cdot a^2f^4h^4(3b^2g^2-a^2f^2).\end{aligned}$$

Substituting the θ-values, the terms of U, other than U', are at once seen to contain the factor $(\theta+\alpha)(\theta+\beta)(\theta+\gamma)$, and we have

$$\begin{aligned}M = 3abc\Big\{ & a^4(h^4y^2-7h^2g^2\,yz+g^4z^2)\\ & + b^4(f^4z^2-7f^2h^2\,zx+h^4x^2)\\ & + c^4(g^4x^2-7g^2f^2\,xy+f^4y^2)\Big\}\\ & + 7(a^4f^4+b^4g^4+c^4h^4)f^2g^2h^2\,abc\,(\theta+\alpha)^2(\theta+\beta)^2(\theta+\gamma)^2\\ & + M',\end{aligned}$$

where

$$U' + (abc)^3(\theta+\delta)^4\,P^3Q = (\theta+\alpha)(\theta+\beta)(\theta+\gamma)\,M'.$$

18. Write for shortness $p, q, r = (af, bg, ch)$; after a complicated reduction, I obtain

$$\begin{aligned}3abc\,M' = {} & a^2f^2h^2(r-p)(p-q)(-2p^4+5p^2qr-6q^2r^2)\,x^2\\ & + b^2h^2f^2(p-q)(q-r)(-2q^4+5q^2rp-6r^2p^2)\,y^2\\ & + c^2f^2g^2(q-r)(r-p)(-2r^4+5r^2pq-6p^2q^2)\,z^2\\ & + 2f^2g^2h^2b^2c^2(7p^4-20p^2qr+4q^2r^2)\,yz\\ & + 2f^2g^2h^2c^2a^2(7q^4-20pq^2r+4r^2p^2)\,zx\\ & + 2f^2g^2h^2a^2b^2(7r^4-20pqr^2+4p^2q^2)\,xy\\ & - 2f^2g^2h^2(p^4+q^4+r^4)(a^2x+b^2y+c^2z)^2.\end{aligned}$$

We then have

$$9abc\,M = \text{terms }(x,y,z)^2 + 9abc\,M',\quad \Omega = \text{terms }(x,y,z)^2$$

as above; and

$$27Mabc + 3\Omega + (*) = 0,$$

which gives $(*)$.

19. After all reductions we find:

$$\begin{aligned}
-\tfrac{1}{3}(*) = {} & a^2f^2h^2(28p^6 - 84p^4qr + 62p^2q^2r^2 - 28q^3r^3)\,x^2 \\
& + b^2h^2f^2(28q^6 - 84pq^4r + 62p^2q^2r^2 - 28r^3p^3)\,y^2 \\
& + c^2f^2g^2(28r^6 - 84pqr^4 + 62p^2q^2r^2 - 28p^3q^3)\,z^2 \\
& + f^2(-3p^6 + 14p^4qr - 130p^4q^2r^2 - 136p^2q^3r^3 + 42q^4r^4)\,yz \\
& + g^2(-3q^6 + 14pq^4r - 130p^2q^4r^2 - 136p^3q^2r^3 + 42r^4p^4)\,zx \\
& + h^2(-3r^6 + 14pqr^4 - 130p^2q^2r^4 - 136p^3q^3r^2 + 42p^4q^4)\,xy
\end{aligned}$$

or observing that the coefficients of $a^2f^2h^2x^2$, $b^2h^2f^2y^2$ and $c^2f^2g^2z^2$ are equal to each other and to

$$62p^2q^2r^2 - 28(q^3r^3 + r^3p^3 + p^3q^3),$$

the equation becomes

$$\begin{aligned}
(*) = {} & -3\big[62p^2q^2r^2 - 28(q^3r^3 + r^3p^3 + p^3q^3)\big](a^2f^2h^2\,x^2 + b^2h^2f^2\,y^2 + c^2f^2g^2\,z^2) \\
& + 3(3p^6 - 14p^4qr + 130p^4q^2r^2 + 136p^2q^3r^3 - 42q^4r^4)\,f^2yz \\
& + 3(3q^6 - 14q^4pr + 130q^4p^2r^2 + 136q^2p^3r^3 - 42r^4p^4)\,g^2zx \\
& + 3(3r^6 - 14r^4pq + 130r^4p^2q^2 + 136r^2p^3q^3 - 42p^4q^4)\,h^2xy;
\end{aligned}$$

and we thence obtain by symmetry the complete value of $(*)(x, y, z, w)^2$; viz. we have only to complete the literal parts of the foregoing expression into the forms

$$\begin{gathered}
f^2yz + a^2xw, \\
g^2zx + b^2yw, \\
h^2xy + c^2zw,
\end{gathered}$$

respectively.

20. The equation of the torse thus is

$$\begin{aligned}
\Delta = {} & a^2g^2h^2x^6 + b^2h^2f^2y^6 + c^2f^2g^2z^6 + a^2b^2c^2w^6 \\
& - g^6(f^2z + h^2x)^3(b^2z + a^2x)^3 \\
& - h^6(g^2x + f^2y)^3(c^2x + b^2y)^3 \\
& - a^6(g^2x + b^2w)^3(h^2x + b^2w)^3 \\
& - b^6(h^2y + a^2w)^3(f^2y + c^2w)^3 \\
& - c^6(f^2z + b^2y)^3(g^2z + a^2y)^3 \\
& + (a^2f^2x + c^2g^2z + c^2a^2w)^3\big[(h^2x + f^2z + b^2w)^3 - 27b^2h^2f^2\,zxw\big] \\
& + (a^2h^2x + b^2h^2y + a^2b^2w)^3\big[(g^2x + f^2y + c^2w)^3 - 27c^2f^2g^2\,xyw\big] \\
& + (g^2h^2x + h^2f^2y + f^2g^2z)^3\big[(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2\,xyz\big] \\
& + xyzw\,(*)(x, y, z, w)^2 = 0.
\end{aligned}$$

I recall that

$$\begin{aligned}
a &= \beta - \gamma, \ \ f = \alpha - \delta, \quad p = af = (\alpha - \delta)(\beta - \gamma), \\
b &= \gamma - \alpha, \ \ g = \beta - \delta, \quad q = bg = (\beta - \delta)(\gamma - \alpha), \\
c &= \alpha - \beta, \ \ h = \gamma - \delta, \quad r = ch = (\gamma - \delta)(\alpha - \beta).
\end{aligned}$$

Developing, we have finally the equation of the torse in the form following.

Equation of the Sextic Torse.

21. The equation is

$$
\begin{aligned}
0 = {} & (a^2g^2h^2,\ b^2h^2f^2,\ c^2f^2g^2,\ a^2b^2c^2)(x^6, y^6, z^6, w^6) \\
& + 3(q^2+r^2)(b^4h^4f^2,\ c^4g^4f^2,\ g^4h^4a^2,\ b^4c^4a^2)(y^4z^2, y^2z^4, x^4w^2, x^2w^4) \\
& + 3(r^2+p^2)(c^4f^4g^2,\ a^4h^4f^2,\ h^4f^4b^2,\ c^4a^4b^2)(z^4x^2, z^2x^4, y^4w^2, y^2w^4) \\
& + 3(p^2+q^2)(a^4g^4h^2,\ b^4f^4h^2,\ f^4g^4c^2,\ a^4b^4c^2)(x^4y^2, x^2y^4, z^4w^2, z^2w^4) \\
& + 3(q^4+3q^2r^2+r^4)(b^2h^2f^2,\ c^2g^2f^2,\ g^2h^2a^2,\ b^2c^2a^2)(y^4z^2, y^2z^4, x^4w^2, x^2w^4) \\
& + 3(r^4+3r^2p^2+p^4)(c^2f^2g^2,\ a^2h^2f^2,\ h^2f^2b^2,\ c^2a^2b^2)(z^4x^2, z^2x^4, y^4w^2, y^2w^4) \\
& + 3(p^4+3p^2q^2+q^4)(a^2g^2h^2,\ b^2f^2h^2,\ f^2g^2c^2,\ a^2b^2c^2)(x^4y^2, x^2y^4, z^4w^2, z^2w^4) \\
& + (6p^4+9p^2q^2+9p^2r^2-21q^2r^2)(a^2g^2h^2,\ a^2b^2c^2,\ f^2h^2b^2,\ f^2g^2c^2)(x^4yz, w^4yz, y^4wx, z^4wy) \\
& + (6q^4+9q^2r^2+9q^2p^2-21r^2p^2)(b^2h^2f^2,\ b^2c^2a^2,\ g^2f^2c^2,\ g^2h^2a^2)(y^4zx, w^4zx, z^4wy, x^4wz) \\
& + (6r^4+9r^2p^2+9r^2q^2-21p^2q^2)(c^2f^2g^2,\ c^2a^2b^2,\ h^2g^2a^2,\ h^2f^2b^2)(z^4xy, w^4xy, x^4wz, y^4wx) \\
& + (q^6+9q^4r^2+9q^2r^4+r^6)(f^6, a^6)(y^3z^3, x^3w^3) \\
& + (r^6+9r^4p^2+9r^2p^4+p^6)(g^6, b^6)(z^3x^3, y^3w^3) \\
& + (p^6+9p^4q^2+9p^2q^4+q^6)(h^6, c^6)(x^3y^3, z^3w^3) \\
& + 9(q^2r^4+q^4r^2+r^2p^4+r^4p^2+p^2q^4+p^4q^2-14p^2q^2r^2) \\
& \quad \times (f^2g^2h^2,\ f^2b^2c^2,\ g^2c^2a^2,\ h^2a^2b^2)(x^2y^2z^2, y^2z^2w^2, z^2x^2w^2, x^2y^2w^2) \\
& + 3\{p^6+3p^4(2q^2+r^2)+3p^2(q^4-7q^2r^2)+q^6\} \\
& \quad \times (g^4h^2,\ h^4g^2,\ g^4c^2,\ c^4b^2)(x^4y^2z, y^4wx^2, z^4w^2x, w^4yz^2) \\
& + 3\{q^6+3q^4(2r^2+p^2)+3q^2(r^4-7r^2p^2)+r^6\} \\
& \quad \times (h^4f^2,\ f^4c^2,\ h^4a^2,\ a^4c^2)(y^4z^2x, z^4wy^2, x^4w^2y, w^4zx^2) \\
& + 3\{r^6+3r^4(2p^2+q^2)+3r^2(p^4-7p^2q^2)+p^6\} \\
& \quad \times (f^4g^2,\ g^4a^2,\ f^4b^2,\ b^4a^2)(x^2yz^4, x^4wz^2, y^4w^2z, w^4xy^2) \\
& + 3\{p^6+3p^4(2r^2+q^2)+3p^2(r^4-7r^2q^2)+q^6\} \\
& \quad \times (g^4h^2,\ h^4b^2,\ g^4c^2,\ c^4b^2)(x^4yz^2, y^2w^2x, z^4wx^2, w^4y^2z) \\
& + 3\{q^6+3q^4(2p^2+r^2)+3q^2(p^4-7p^2r^2)+r^6\} \\
& \quad \times (h^4f^2,\ f^4a^2,\ h^4a^2,\ a^4c^2)(y^4zx^2, z^2wy^2, x^4wy^2, w^4z^2x) \\
& + 3\{r^6+3r^4(2q^2+p^2)+3r^2(q^4-7q^2p^2)+p^6\} \\
& \quad \times (f^4g^2,\ g^4c^2,\ f^4b^2,\ b^4a^2)(x^2yz^4, z^4w^2z, y^4wz^2, w^4xy^2) \\
& + xyzw\,\Big[\\
& \qquad - 3\{62p^2q^2r^2-28(q^3r^3+r^3p^3+p^3q^3)\}(a^2f^2h^2,\ b^2h^2f^2,\ c^2f^2g^2,\ a^2b^2c^2)(x^2, y^2, z^2, w^2) \\
& \qquad + 3(3p^6-14p^4qr+130p^4q^2r^2+136p^2q^3r^3-42q^4r^4)(f^2, a^2)(yz, xw) \\
& \qquad + 3(3q^6-14q^4rp+130q^4r^2p^2+136q^2r^3p^3-42r^4p^4)(g^2, b^2)(zx, yw) \\
& \qquad + 3(3r^6-14r^4pq+130r^4p^2q^2+136r^2p^3q^3-42p^4q^4)(h^2, c^2)(xy, zw) \\
& \Big].
\end{aligned}
$$

Comparison with the Equation of the Centro-surface of an Ellipsoid.

22. In the Equation

$$\frac{x}{(\theta+\alpha)^2} + \frac{y}{(\theta+\beta)^2} + \frac{z}{(\theta+\gamma)^2} + \frac{w}{(\theta+\delta)^2} = 0,$$

for x, y, z, w write $\xi^2, \eta^2, \zeta^2, \omega^2$, and then $\delta = \infty$, the equation is converted into

$$\frac{\xi^2}{(\theta+\alpha)^2} + \frac{\eta^2}{(\theta+\beta)^2} + \frac{\zeta^2}{(\theta+\gamma)^2} + \omega^2 = 0;$$

or writing a^2, b^2, c^2 for α, β, γ, and understanding $\xi^2, \eta^2, \zeta^2, \omega^2$ to mean $a^2x^2, b^2y^2, c^2z^2, -1$, this is

$$\frac{a^2x^2}{(\theta+a^2)^2} + \frac{b^2y^2}{(\theta+b^2)^2} + \frac{c^2z^2}{(\theta+c^2)^2} - 1 = 0.$$

This is an equation, the envelope of which in regard to the variable parameter θ, gives the surface which is the locus of the centres of curvature of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, or say the *Centro-surface of the Ellipsoid.* (Salmon's *Solid Geometry*, Ed. 2, p. 400, [Ed. 4, p. 465].)

Making the same substitution in the foregoing equation $(*)(x, y, z, w)^2 = 0$, the quantities f, g, h become equal to $-\delta$, and p, q, r to $-a\delta, -b\delta, -c\delta$ respectively, and the whole equation divides by δ^n; throwing out this factor, we have a result which is obtained more simply by changing

$$x, y, z, w,\ a, b, c, f, g, h,\ p, q, r,$$

into

$$\xi^2, \eta^2, \zeta^2, \omega^2,\ \alpha, \beta, \gamma, 1, 1, 1,\ \alpha, \beta, \gamma,$$

where α, β, γ now signify $b^2 - c^2, c^2 - a^2, a^2 - b^2$ respectively, and $\xi^2, \eta^2, \zeta^2, \omega^2$ are retained as standing for $a^2x^2, b^2y^2, c^2z^2, -1$ respectively; viz. the equation of the centro-surface is found to be

$$
\begin{aligned}
0 = {} & (\alpha^2, \beta^2, \gamma^2, \alpha^2\beta^2\gamma^2)(\xi^6, \eta^6, \zeta^6, \omega^6) \\
& + 3(\beta^2+\gamma^2)(\beta^2, \gamma^2, \alpha^2, \alpha^2\beta^2\gamma^2)(\eta^4\zeta^2, \eta^2\zeta^4, \xi^4\omega^2, \xi^2\omega^4) \\
& + 3(\gamma^2+\alpha^2)(\gamma^2, \alpha^2, \beta^2, \beta^2\alpha^2\gamma^2)(\zeta^4\xi^2, \zeta^2\xi^4, \eta^4\omega^2, \eta^2\omega^4) \\
& + 3(\alpha^2+\beta^2)(\alpha^2, \beta^2, \gamma^2, \gamma^2\alpha^2\beta^2)(\xi^4\eta^2, \xi^2\eta^4, \zeta^4\omega^2, \zeta^2\omega^4) \\
& + 3(\beta^4+3\beta^2\gamma^2+\gamma^4)(\beta^2, \gamma^2, \alpha^2, \alpha^2\beta^2\gamma^2)(\eta^4\zeta^2, \eta^2\zeta^4, \xi^4\omega^2, \xi^2\omega^4) \\
& + 3(\gamma^4+3\gamma^2\alpha^2+\alpha^4)(\gamma^2, \alpha^2, \beta^2, \beta^2\gamma^2\alpha^2)(\zeta^4\xi^2, \zeta^2\xi^4, \eta^4\omega^2, \eta^2\omega^4) \\
& + 3(\alpha^4+3\alpha^2\beta^2+\beta^4)(\alpha^2, \beta^2, \gamma^2, \gamma^2\alpha^2\beta^2)(\xi^4\eta^2, \xi^2\eta^4, \zeta^4\omega^2, \zeta^2\omega^4) \\
& + 3(2\alpha^4+3\alpha^2\beta^2+3\alpha^2\gamma^2-7\beta^2\gamma^2)(\alpha^2, \alpha^2\beta^2\gamma^2, \beta^2, \gamma^2)(\xi^4\eta\zeta, \omega^4\eta\zeta, \eta^4\omega\xi, \zeta^4\omega\xi) \\
& + 3(2\beta^4+3\beta^2\gamma^2+3\beta^2\alpha^2-7\gamma^2\alpha^2)(\beta^2, \beta^2\gamma^2\alpha^2, \gamma^2, \alpha^2)(\eta^4\zeta\xi, \omega^4\zeta\xi, \zeta^4\omega\eta, \xi^4\omega\zeta) \\
& + 3(2\gamma^4+3\gamma^2\alpha^2+3\gamma^2\beta^2-7\alpha^2\beta^2)(\gamma^2, \gamma^2\alpha^2\beta^2, \alpha^2, \beta^2)(\zeta^4\xi\eta, \omega^4\xi\eta, \xi^4\omega\zeta, \eta^4\omega\xi) \\
& + (\beta^6+9\beta^4\gamma^2+9\beta^2\gamma^4+\gamma^6)(1, \alpha^6)(\eta^3\zeta^3, \xi^3\omega^3) \\
& + (\gamma^6+9\gamma^4\alpha^2+9\gamma^2\alpha^4+\alpha^6)(1, \beta^6)(\zeta^3\xi^3, \eta^3\omega^3) \\
& + (\alpha^6+9\alpha^4\beta^2+9\alpha^2\beta^4+\beta^6)(1, \gamma^6)(\xi^3\eta^3, \zeta^3\omega^3) \\
& + 3\{\alpha^6+3\alpha^4(2\beta^2+\gamma^2)+3\alpha^2(\beta^4-7\beta^2\gamma^2)+\beta^6\}(1, \beta^2, \gamma^2, \beta^2\gamma^2)(\xi^4\eta^2\zeta, \eta^4\omega\xi^2, \zeta^4\omega^2\xi, \omega^4\eta\zeta^2) \\
& + 3\{\beta^6+3\beta^4(2\gamma^2+\alpha^2)+3\beta^2(\gamma^4-7\gamma^2\alpha^2)+\gamma^6\}(1, \gamma^2, \alpha^2, \gamma^2\alpha^2)(\eta^4\zeta^2\xi, \zeta^4\omega\eta^2, \xi^4\omega^2\eta, \omega^4\zeta\xi^2) \\
& + 3\{\gamma^6+3\gamma^4(2\alpha^2+\beta^2)+3\gamma^2(\alpha^4-7\alpha^2\beta^2)+\alpha^6\}(1, \alpha^2, \beta^2, \alpha^2\beta^2)(\zeta^4\xi^2\eta, \xi^4\omega\zeta^2, \eta^4\omega^2\zeta, \omega^4\xi\eta^2) \\
& + 3\{\alpha^6+3\alpha^4(2\gamma^2+\beta^2)+3\alpha^2(\gamma^4-7\gamma^2\beta^2)+\beta^6\}(1, \gamma^2, \beta^2, \beta^2\gamma^2)(\xi^4\eta\zeta^2, \zeta^2\omega^2\xi, \eta^4\omega\xi^2, \omega^4\eta^2\zeta) \\
& + 3\{\beta^6+3\beta^4(2\alpha^2+\gamma^2)+3\beta^2(\alpha^4-7\alpha^2\gamma^2)+\gamma^4\alpha^2\}(1, \alpha^2, \gamma^2, \gamma^2\alpha^2)(\eta^4\zeta\xi^2, \xi^2\omega^2\eta, \zeta^4\omega\eta^2, \omega^4\zeta^2\xi) \\
& + 3\{\gamma^6+3\gamma^4(2\beta^2+\alpha^2)+3\gamma^2(\beta^4-7\beta^2\alpha^2)+\alpha^4\beta^2\}(1, \beta^2, \alpha^2, \alpha^2\beta^2)(\zeta^4\xi\eta^2, \eta^2\omega^2\zeta, \xi^4\omega\zeta^2, \omega^4\xi^2\eta) \\
& + 9(\beta^2\gamma^4+\beta^4\gamma^2+\gamma^2\alpha^4+\gamma^4\alpha^2+\alpha^2\beta^4+\alpha^4\beta^2-14\alpha^2\beta^2\gamma^2)(1, \beta^2\gamma^2, \gamma^2\alpha^2, \alpha^2\beta^2)(\xi^2\eta^2\zeta^2, \eta^2\zeta^2\omega^2, \zeta^2\xi^2\omega^2, \\
& + \xi^2\eta^2\zeta^2\omega^2\,\big[\\
& \qquad - 3\{62\alpha^2\beta^2\gamma^2-28(\beta^3\gamma^3+\gamma^3\alpha^3+\alpha^3\beta^3)\}(\alpha^2, \beta^2, \gamma^2, \alpha^2\beta^2\gamma^2)(\xi^2, \eta^2, \zeta^2, \omega^2) \\
& \qquad + 3(3\alpha^6-14\alpha^4\beta\gamma+130\alpha^4\beta^2\gamma^2+136\alpha^2\beta^3\gamma^3-42\beta^4\gamma^4)(1, \alpha^2)(\eta^2\zeta^2, \xi^2\omega^2) \\
& \qquad + 3(3\beta^6-14\beta^4\gamma\alpha+130\beta^4\gamma^2\alpha^2+136\beta^2\gamma^3\alpha^3-42\gamma^4\alpha^4)(1, \beta^2)(\zeta^2\xi^2, \eta^2\omega^2) \\
& \qquad + 3(3\gamma^6-14\gamma^4\alpha\beta+130\gamma^4\alpha^2\beta^2+136\gamma^2\alpha^3\beta^3-42\alpha^4\beta^4)(1, \gamma^2)(\xi^2\eta^2, \zeta^2\omega^2) \\
& \big].
\end{aligned}
$$

This agrees with the result given in Salmon's *Solid Geometry*, Ed. 2, p. 151, [Ed. 4, p. 178], and *Quarterly Mathematical Journal*, vol. II. p. 220 (1858); in the latter place, however, the term

$$\beta^2\gamma^2\xi^4 + \gamma^2\alpha^2\eta^4 + \alpha^2\beta^2\omega^4 + \alpha^2\beta^2\gamma^2\xi^4\omega^4$$

is by mistake written

$$\beta^2\gamma^2\xi^4 + \gamma^2\alpha^2\eta^4 + \alpha^2\beta^2\omega^4 + \beta^2\gamma^2\xi^4\omega^4$$

viz. a factor α^2 is omitted in one of the coefficients.

Some of the coefficients are presented under slightly different forms; viz. instead of

$$62\alpha^2\beta^2\gamma^2 - 28(\beta^3\gamma^3 + \gamma^3\alpha^3 + \alpha^3\beta^3)$$

Salmon has

$$14(\beta^4\gamma^2 + \beta^2\gamma^4 + \gamma^4\alpha^2 + \gamma^2\alpha^4 + \alpha^4\beta^2 + \alpha^2\beta^4) + 20\alpha^2\beta^2\gamma^2;$$

and instead of

$$3\alpha^6 - 14\alpha^4\beta\gamma + 130\alpha^4\beta^2\gamma^2 + 136\alpha^2\beta^3\gamma^3 - 42\beta^4\gamma^4,$$

he has

$$-4\alpha^8 + 7\alpha^6(\beta^2+\gamma^2) + 196\alpha^4\beta^2\gamma^2 - 53\alpha^2\beta^2\gamma^2(\beta^2+\gamma^2) - 42\beta^4\gamma^4,$$

but these different forms are respectively equivalent in virtue of the relation

$$\alpha+\beta+\gamma=0.$$

[437] DÉMONSTRATION NOUVELLE DU THÉORÈME DE M. CASEY PAR RAPPORT AUX CERCLES QUI TOUCHENT À TROIS CERCLES DONNÉS.

[From the Annali di Matematica pura ed applicata, tom. I. (1867), pp. 132–134.]

This is in fact the investigation contained in the paper 414, "On Polyzomal Curves otherwise the curves $\sqrt{U} + \sqrt{V} + \&c. = 0$," Annex II. pp. 568–573, "On Casey's theorem for the circle which touches three given circles," viz. it is based on the identity of the two problems 1° to find a circle touching three given circles, 2° to find a cone-sphere (sphere of radius zero) passing through three given points in space.

[438] NOTE SUR QUELQUES TORSES SEXTIQUES.

[From the Annali di Matematica pura ed applicata, tom. II. *(1868), pp. 99, 100.]*

Je désire d'appeler attention aux surfaces développables, ou torses, données par l'équation

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0.$$

Dans cette équation (a, b, c, d, e) sont des fonctions linéaires quelconques des quatre coordonnées (x, y, z, t); ces quantités sont donc liées par une équation linéaire

$$Aa + 4Bb + 6Cc + 4Dd + Ee = 0,$$

et je remarque que la classification des torses comprises sous l'équation mentionnée dépend des propriétés invariantives de la fonction $(A, B, C, D, E)(t, 1)^4$.

En effet la torse a une courbe cuspidale, ou arête de rebroussement, donnée par les équations

$$ae - 4bd + 3c^2 = 0, \quad ace - ad^2 - b^2e + 2bcd - c^3 = 0,$$

et une courbe nodale donnée par les équations

$$\frac{ac - b^2}{a} = \frac{ad - bc}{2b} = \frac{ae + 2bd - 3c^2}{6c} = \frac{be - cd}{2d} = \frac{ce - d^2}{e},$$

ces deux courbes se rencontrent dans les points donnés par les équations

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e},$$

lesquels sont des points stationnaires de la courbe cuspidale. Pour trouver ces points, en écrivant $a : b : c : d : e = \tau^4 : \tau^3 : \tau^2 : \tau : 1$, on obtient pour le paramètre τ l'équation

$$(A, B, C, D, E)(\tau, 1)^4 = 0$$

et l'on voit ainsi qu'il y a un rapport entre la théorie de la surface et cette équation; à toute particularité invariantive de l'équation, il y correspondra quelque particularité de la torse.

Les cas à considérer sont:

1°. Racines inégales, sans aucune relation invariantive. C'est le cas général; je l'ai considéré dans le Mémoire, "On a certain Sextic Developable," *Quart. Math. Journ.*, t. IX. (1868), pp. 129–142. [398].

2°. Deux racines égales. Ce cas n'a pas été considéré; je remarque que la courbe cuspidale est du cinquième ordre. En effet on peut supposer que les racines égales soient $= 0$, ce qui revient à prendre $D = 0$, $E = 0$. On a donc $Aa + 4Bb + 6Cc = 0$, c'est-à-dire les équations $a = 0$, $b = 0$ impliquent l'équation $c = 0$; et on voit de là que les surfaces $ae - 4bd + 3c^2 = 0$, $ace - ad^2 - b^2e + 2bcd - c^3 = 0$ se coupent selon la droite $a = 0$, $b = 0$; il reste ainsi une courbe du cinquième ordre pour la courbe cuspidale.

3°. Trois racines égales. On peut supposer que ces racines sont $= 0$, ce qui donne $C = 0$, $D = 0$, $E = 0$; et l'on a ainsi $Aa + 4Bb = 0$, c'est-à-dire les plans $a = 0$, $b = 0$ sont ici un seul plan. L'équation de la torse contient le facteur a, et en l'écartant elle se réduit au cinquième ordre; on obtient ainsi la torse générale du cinquième ordre.

4°. Deux paires de racines égales. On peut supposer que ces racines sont $= \infty, 0$; cela donne $A = 0$, $B = 0$, $D = 0$, $E = 0$, et l'on a ainsi identiquement $c = 0$. L'équation de la torse est $(ae - 4bd)^3 - 27(-ad^2 - b^2e)^2 = 0$. J'ai considéré ce cas dans le Mémoire, "On a Special Sextic Developable," *Quart. Math. Journ.*, t. VII. (1866), pp. 105–113, [373]; la courbe cuspidale est du quatrième ordre, une courbe excubo-quartique d'une forme particulière.

5°. Quatre racines égales; en prenant ces racines $= 0$, on a $B = C = D = E = 0$, donc identiquement $a = 0$; l'équation de la torse contient le facteur b^2, et en l'écartant elle se réduit au quatrième ordre: on a dans ce cas la torse générale du quatrième ordre.

Il y a encore deux cas à considérer.

6°. L'invariant I de la fonction $(A, B, C, D, E)(t, 1)^4$ est $= 0$;

7°. L'invariant J de cette fonction est $= 0$;

Mais je n'ai pas encore examiné ce que cela veut dire[1]. Il n'y a pas le cas à considérer où l'on a à la fois $I = 0$, $J = 0$; car cela revient au cas 3° de trois racines égales.

Cambridge, le 16 mai 1868.

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[1] La courbe cuspidale étant du genre 0 (unicursale), on peut considérer la série des points de la courbe comme correspondant anharmoniquement aux points d'une droite. Si le système des quatre points stationnaires est harmonique on a $J = 0$; si ce système est équi-anharmonique, on a $I = 0$.

[439] ADDITION À LA NOTE SUR QUELQUES TORSES SEXTIQUES.

[From the Annali di Matematica pura ed applicata, tom. II. (1868), pp. 219–221.]

Je viens de trouver ce que signifie la condition $J = 0$. Considérons deux surfaces quadriques qui se touchent (d'un contact ordinaire). Les équations tangentielles peuvent s'écrire sous la forme

$$a\xi^2 + b\eta^2 + c\zeta^2 + 2n\zeta\omega = 0,$$
$$a'\xi^2 + b'\eta^2 + c'\zeta^2 + 2n'\zeta\omega = 0,$$

et l'on satisfait à ces équations par des valeurs de ξ, η, ζ, ω qui contiennent un paramètre arbitraire θ, en écrivant

$$\xi : \eta : \zeta : \omega = \alpha\left(\theta + \frac{1}{\theta}\right) : \beta\left(\theta - \frac{1}{\theta}\right) : \epsilon : \gamma\left(\theta^2 + \frac{1}{\theta^2}\right) + \delta;$$

en effet cela donne

$$a\alpha^2 + b\beta^2 + 2n\gamma\epsilon = 0, \quad 2a\alpha^2 - 2b\beta^2 + c\epsilon^2 + 2n\delta\epsilon = 0,$$
$$a'\alpha^2 + b'\beta^2 + 2n'\gamma\epsilon = 0, \quad 2a'\alpha^2 - 2b'\beta^2 + c'\epsilon^2 + 2n'\delta\epsilon = 0,$$

ce qui détermine les valeurs de $\alpha : \beta : \gamma : \delta : \epsilon$. L'équation du plan tangent commun sera donc

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

ou, en multipliant par $12\theta^2$, cette équation sera

$$(a, b, c, d, e)(\theta, 1)^4 = 0,$$

les valeurs des coefficients étant

$$a = 12\,\xi w,$$
$$b = 3(\alpha x + \beta y),$$
$$c = 2(\delta w + \epsilon z),$$
$$d = 3(\alpha x - \beta y),$$
$$e = 12\,\gamma w.$$

En représentant par $Aa + 4Bb + 6Cc + 4Dd + Ee = 0$ l'équation qui lie les fonctions linéaires a, b, c, d, e, cette équation sera $a - e = 0$; on a donc $A = -E = 1$, $B = C = D = 0$; l'invariant J de la fonction $(A, B, C, D, E)(\tau, 1)^4$ est donc $= 0$.

Nous arrivons ainsi à la conclusion que la torse sextique

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e - c^3 + 2bcd)^2 = 0,$$

où les fonctions linéaires a, b, c, d, e sont liées par une équation

$$Aa + 4Bb + 6Cc + 4Dd + Ee = 0,$$

telle que l'invariant

$$J = ACE - AD^2 - EB^2 - C^3 - 2BCD$$

de la fonction $(A, B, C, D, E)(t, 1)^4$ est $= 0$ (cas 7° de la Note), est la torse enveloppée par le plan tangent commun de deux surfaces quadriques qui se touchent d'un contact ordinaire. J'ai trouvé l'équation de cette torse dans le Mémoire, "On the Developable Surfaces which arise from two Surfaces of the Second Order," *Camb. and Dubl. Math. Jour.*, t. V. (1850), pp. 46–57, voir

p. 56, [85]; où, en supposant que $a, b, c, n, a', b', c', n'$ dénotent les mêmes coefficients comme à présent, et en écrivant

$$\begin{aligned} bc'-b'c&=f, & an'-a'n&=p,\\ ca'-c'a&=g, & bn'-b'n&=q,\\ ab'-a'b&=h, & cn'-c'n&=r, \end{aligned}$$

(et de là $pf+qg+rh=0$), l'équation trouvée est

$$f^2g^2h^2w^4+\ldots+4pqr(qx^2+py^2)^2=0.$$

Je vais vérifier ces termes. Partant de l'équation $(a, b, c, d, e)(\theta, 1)^4=1$, l'équation de la torse, en y introduisant pour commodité le facteur $-\frac{1}{27}pqr$, sera

$$\begin{aligned} &-\tfrac{1}{27}pqr\left\{[(\delta w+\epsilon z)^2+12\gamma^2-3(\alpha^2x^2-\beta^2y^2)]^3\right.\\ &\quad\left.-\left[2(\delta w+\epsilon z)^3-9(\delta w+\epsilon z)(\alpha^2x^2-\beta^2y^2)-27(\delta w+\epsilon z)\gamma w^2+54(\alpha^2x^2+\beta^2y^2)\gamma w\right]^2\right\}=0. \end{aligned}$$

En prenant $\gamma\epsilon=ab'-a'b=h$, on obtient pour $\alpha, \beta, \gamma, \delta, \epsilon$ les valeurs

$$\alpha^2=2q,\quad \beta^2=-2p,\quad \gamma\epsilon=h,\quad \epsilon^2=-\frac{9pq}{r},\quad \delta\epsilon=\frac{2(fp-gq)}{r},$$

et de là

$$\gamma^2=-\frac{h^2r}{9pq},\quad \delta^2=-\frac{(fp-gq)^2}{2pqr}=-\frac{h^2r+4fgpq}{2pqr},\quad \gamma^2\delta^2=\frac{h^2(h^2r+4fgpq)}{18p^2q^2r},\quad \delta^2-4\gamma^2=\frac{2fg}{r}.$$

Les termes en w^4 et $(x^2, y^2)^2$ sont

$$\begin{aligned} &-\tfrac{1}{27}pqr\left\{[4(\delta^2+12\gamma^2)^3-(2\delta^3-72\gamma^2\delta)^2]w^4-108(\alpha^2x^2-\beta^2y^2)^3\right\},\\ &=-\tfrac{1}{27}pqr\left\{432\gamma^2(\delta^2-4\gamma^2)^2\,w^4-108(\alpha^2x^2-\beta^2y^2)^3\right\}; \end{aligned}$$

ces termes sont donc

$$\begin{aligned} &=-\tfrac{1}{27}pqr\left\{-432\cdot\frac{h^2r}{9pq}\cdot\frac{4f^2g^2}{r^2}\,w^4+864(qx^2+py^2)^3\right\}\\ &=f^2g^2h^2w^4+4pqr(qx^2+py^2)^3, \end{aligned}$$

comme cela doit être.

Cambridge, le 22 septembre 1868.

[440] NOTE SUR UNE TRANSFORMATION GÉOMÉTRIQUE.

[From the Journal für die reine und angewandte Mathematik (Crelle), tom. LXVII. (1867), pp. 95, 96.]

La lecture de la Note de M. Hesse, "Ein Uebertragungsprincip" (t. LXVI. p. 15 de ce Journal) m'a suggéré les remarques suivantes:

Soient (a_1, b_1, c_1, d_1), (a_2, b_2, c_2, d_2), (a_3, b_3, c_3, d_3) des constantes données, on peut supposer que les coordonnées (x, y) d'un point quelconque dans un plan soient exprimées en fonctions des paramètres variables (u, v) par les équations

$$x = \frac{a_1 + b_1 u + c_1 v + d_1 uv}{a_3 + b_3 u + c_3 v + d_3 uv}, \quad y = \frac{a_2 + b_2 u + c_2 v + d_2 uv}{a_3 + b_3 u + c_3 v + d_3 uv}.$$

En introduisant une nouvelle indéterminée s, ces équations peuvent être écrites dans la forme

$$sx = a_1 + b_1 u + c_1 v + d_1 uv,$$

$$sy = a_2 + b_2 u + c_2 v + d_2 uv,$$

$$s = a_3 + b_3 u + c_3 v + d_3 uv;$$

pour des valeurs données des coordonnées (x, y) la quantité s est en général déterminée par une équation quadratique, et les paramètres u et v sont des fonctions linéaires données de s; il y a cependant deux cas particuliers qu'il convient de distinguer.

1°. L'équation quadratique en s peut avoir la racine $s = 0$, et débarrassée de ce facteur, se réduire par conséquent à une équation linéaire; ce cas particulier a lieu si la condition $(abc)(bcd) = (abd)(acd)$ est remplie, où la notation (abc) désigne le déterminant

$$\begin{vmatrix} a_1, & b_1, & c_1 \\ a_2, & b_2, & c_2 \\ a_3, & b_3, & c_3 \end{vmatrix}.$$

Dans ce cas u et v sont des fonctions rationnelles de (x, y) et la transformation a la signification géométrique suivante:

En considérant deux droites quelconques L, M dans l'espace et en menant par le point donné (x, y) la droite unique G qui rencontre ces deux droites, on peut supposer que u et v soient des paramètres qui déterminent les positions des points de rencontre sur les deux droites respectivement; c. à. d. que u soit la distance d'un point fixe sur la droite L au point de rencontre avec la droite G, et de même que v soit la distance d'un point fixe sur la droite M au point de rencontre avec la droite G.

2°. Supposons $b_1 : b_3 = c_1 : c_3$ ou ce qui est au fond la même chose $b_2 : b_3 = c_2 : c_3$, $b_1 - c_1 = 0$, $b_2 - c_2 = 0$, $b_3 - c_3 = 0$; alors s est déterminée par une équation simple, mais u et v ne sont plus des fonctions rationnelles de s; on voit que dans ce cas $u + v$ et uv sont des fonctions rationnelles de (x, y), et que par conséquent u et v sont les racines d'une équation quadratique qui contient (x, y) linéairement. On peut supposer que u et v soient les paramètres de deux points sur une droite donnée, c. à. d. que u et v soient les distances de ces deux points respectivement à un point fixe situé sur la droite donnée; on a ainsi la transformation de M. Hesse.

Je n'ai pas cherché la signification géométrique des formules générales.

Cambridge, 10 octobre 1866.

[441] NOTE SUR L'ALGORITHME DES TANGENTES DOUBLES D'UNE COURBE DU QUATRIÈME ORDRE.

[From the Journal für die reine und angewandte Mathematik (Crelle), tom. LXVIII. (1868), pp. 176–179.]

On n'a pas, je crois, assez fait attention à l'algorithme (tiré de la considération d'une figure dans l'espace) qu'a trouvé M. Hesse (dans le mémoire "Ueber die Doppeltangenten der Curven vierter Ordnung," t. XLIX de ce Journal, 1855) pour dénoter les tangentes doubles (ou bitangentes) d'une courbe du quatrième ordre. Voici en quoi cet algorithme consiste. En employant les huit symboles $1, 2, 3, \ldots 8$, les 28 bitangentes sont représentées par les combinaisons binaires $12, 13, 14, \ldots 78$. Cela posé, considérons une expression quelconque 12.13.14, ou 12.34,… ou disons un "terme" qui représente un système d'une seule ou de plusieurs des bitangentes. On peut opérer sur ce terme avec deux espèces de substitutions; la substitution ordinaire qui consiste à changer l'arrangement 12345678 des huit symboles en un autre arrangement quelconque; et la substitution "bifide" représentée par un symbole tel que 1234.5678, lequel dénote qu'il faut entrechanger les combinaisons 12 et 34, 13 et 24, 14 et 23, 56 et 78, 57 et 68, 58 et 67, en ne changeant pas les autres combinaisons. Par exemple en opérant avec 1234.5678 sur 34.45.56.17 on obtient 12.45.78.17. Le nombre de ces substitutions bifides est 35, ou en comptant la substitution unité, qui ne change aucune des combinaisons, ce nombre est 36.

Appelons "homotypiques" deux termes qui se dérivent l'un de l'autre par une substitution ordinaire; "syntypiques" qui se dérivent l'un de l'autre par une substitution ordinaire ou bifide; "sous-groupe" le système entier des termes homotypiques à un terme donné; "groupe" le système entier des termes syntypiques à un terme donné. Un groupe peut contenir un seul sous-groupe, ou plusieurs sous-groupes; mais il importe de remarquer que la notion du sous-groupe n'a pas de signification géométrique, et ne sert que comme moyen de former les termes du groupe. Cela étant, le théorème géométrique est celui-ci: "les systèmes de bitangentes représentées par des termes syntypiques (ou autrement dit, par des termes qui appartiennent au même groupe) ont les mêmes propriétés géométriques."

Par exemple, en considérant les bitangentes deux à deux, on a deux sous-groupes, l'un composé de termes homotypiques à 12.13; l'autre, de termes homotypiques à 12.34—ou disons, le sous-groupe 12.13 de 168 termes et le sous-groupe 12.34 de 210 termes; mais ces deux sous-groupes ne forment qu'un seul groupe: pour montrer cela il suffit d'opérer sur 12.13, par exemple avec la substitution 1245.3678, ce qui donne 45.13, terme homotypique à 12.34. Cela veut dire qu'il n'y a pas de combinaison de deux bitangentes qui se distingue d'une manière quelconque de toute autre combinaison de deux bitangentes.

Mais en combinant les bitangentes trois à trois, on a les deux sous-groupes 12.34.56 (420 termes) et 12.23.34 (840 termes) qui forment un groupe de 1260 termes; les trois bitangentes représentées par un quelconque des 1260 termes ont leurs six points de contact sur une même conique. Les trois autres sous-groupes 12.23.31 (56 termes), 12.23.45 (1680 termes) et 12.13.14 (280 termes) forment un groupe de 2016 termes, et pour trois bitangentes représentées par un terme quelconque de ce groupe, les six points de contact ne sont pas situés sur une même conique.

Comme un autre exemple j'explique la constitution des 63 "groupes" de Steiner (voir le mémoire de Steiner, "Eigenschaften der Curven vierten Grades rücksichtlich ihrer Doppeltangenten," t. XLIX. de ce journal, 1855) ou (pour éviter l'emploi de ce mot groupe dans une nouvelle signification) disons les 63 termes G de Steiner, chaque terme composé de 6 paires de bitangentes. On a ici un sous-groupe de 35 termes G, de la forme

12.34; 13.42; 14.23; 56.78; 57.86; 58.67

(pour abréger on peut dénoter ce terme par 1234.5678), et un sous-groupe de 28 termes G, de la forme

13.32; 14.42; 15.52; 16.62; 17.72; 18.82

(pour abréger on peut de même dénoter ce terme par 12.345678), les deux sous-groupes forment le groupe des 63 termes G.

Steiner a de plus considéré les "systèmes" ou disons les termes S_1, S_2, composés chacun de trois termes G; savoir 315 termes S_1 et 336 termes S_2. Les 315 termes S_1 sont ici un groupe composé d'un sous-groupe de 105 termes $3G_1$, de la forme

1234.5678; 1256.3478; 1278.3456

et un sous-groupe de 210 termes $2G_1 + G_2$ de la forme

12.345678; 34.125678 et 1234.5678.

Et de même les 336 termes S_2 sont un groupe composé d'un sous-groupe de 280 termes $2G_1 + G_2$ de la forme

1234.5678; 5234.1678 et 15.234678

et un sous-groupe de 56 termes $3G_2$ de la forme

12.34.5678; 13.24.5678; 31.245678.

Il va sans dire que je me suis servi de l'abréviation 1234.5678 pour dénoter le terme 12.34; 13.42; 14.23; 56.78; 57.86; 58.67; et de même pour les autres termes G_1 ou G_2.

M. Aronhold (dans le mémoire "Ueber den gegenseitigen Zusammenhang der 28 Doppeltangenten einer allgemeinen Curve vierten Grades," *Berl. Monatsber.* Juli 1864), partant de 7 bitangentes données, a trouvé une construction pour les autres 21 bitangentes. Les bitangentes données doivent être indépendantes; savoir pour trois quelconques de ces 7 bitangentes, les six points de contact ne sont pas situés sur une même conique. Les bitangentes représentées par les termes $12, 13, 14, 15, 16, 17, 18$ sont un tel système de bitangentes indépendantes; et en dénotant de cette manière les 7 bitangentes données, la bitangente construite par le moyen de la conique qui touche cinq de ces droites, par exemple les droites $38, 48, 58, 68, 78$, (ou conique 34567) peut être dénotée par 12, et de même pour les autres bitangentes cherchées; on a ainsi le système entier des bitangentes dénotées comme auparavant par $12, 13, 14, \ldots 78$.

J'ajoute que le groupe qui contient $18, 28, 38, 48, 58, 68, 78$ est composé d'un sous-groupe $18, 28, 38, 48, 58, 68, 78$ de 8 termes, et d'un sous-groupe $12, 23, 31, 48, 58, 68, 78$ de 280 termes; le groupe contient donc 288 termes; savoir il y a ce nombre 288 de systèmes de sept bitangentes indépendantes qui peuvent chacun servir à trouver par la construction d'Aronhold les autres 21 bitangentes.

P.S. J'ai trouvé à propos de la méthode de M. Aronhold une forme commode pour l'équation de la conique qui touche cinq droites données; en supposant que l'on ait identiquement $x + y + z + w = 0$, et que les droites données soient $x = 0$, $y = 0$, $z = 0$, $w = 0$, et $ax + by + cz + dw = 0$, l'équation de la conique est

$$(a-d)^2(b-c)^2(xw+yz) + (b-d)^2(c-a)^2(yw+zx) + (c-d)^2(a-b)^2(zw+xy) = 0.$$

J'ajoute qu'en écrivant pour abréger

$$\alpha : \beta : \gamma = (a-d)(b-c) : (b-d)(c-a) : (c-d)(a-b)$$

(d'où $\alpha + \beta + \gamma = 0$) les coordonnées (x, y, z, w) des points de contact avec les droites

$$x = 0,\ y = 0,\ z = 0,\ w = 0 \text{ sont } (0, \gamma, \beta, \alpha),\ (\gamma, 0, \alpha, \beta),\ (\beta, \alpha, 0, \gamma),\ (\alpha, \beta, \gamma, 0)$$

respectivement; et que les coordonnées du point de contact avec la droite $ax + by + cz + dw = 0$ sont

$$x : y : z : w = (bcd) : -(cda) : (dab) : -(abc)$$

où, pour abréger, (bcd) dénote $(b-c)(c-d)(d-b)$, et de même pour $(cda), (dab), (abc)$.

Cambridge, le 23 septembre 1867.

[442] Note sur la Surface du Quatrième Ordre douée de Seize Points Singuliers et de Seize Plans Singuliers.

[From the Journal für die reine und angewandte Mathematik (Crelle), tom. LXXIII. (1871), pp. 292–293.]

L'équation de M. Kummer se transforme sans difficulté en celle-ci

$$\sqrt{\alpha x}\left(\gamma'\gamma''y-\beta'\beta''z-\tfrac{w}{\alpha}\right)+\sqrt{\beta y}\left(\alpha'\alpha''z-\gamma'\gamma''x-\tfrac{w}{\beta}\right)+\sqrt{\gamma z}\left(\beta'\beta''x-\alpha'\alpha''y-\tfrac{w}{\gamma}\right)=0,$$

où

$$\alpha+\beta+\gamma=0,\quad \alpha'+\beta'+\gamma'=0,\quad \alpha''+\beta''+\gamma''=0.$$

Or cette équation rendue rationnelle prend, après toutes les réductions nécessaires, la forme suivante:

$$\begin{aligned}&w^2\big(x^2+y^2+z^2-2yz-2zx-2xy\big)\\&\quad+2w\big(\alpha\alpha'\alpha''(y^2z-yz^2)+\beta\beta'\beta''(z^2x-zx^2)+\gamma\gamma'\gamma''(x^2y-xy^2)+\theta xyz\big)\\&\quad+\big(\alpha\alpha'\alpha''yz+\beta\beta'\beta''zx+\gamma\gamma'\gamma''xy\big)^2=0,\end{aligned}$$

où, pour abréger, l'on a écrit

$$\begin{aligned}\theta&=(\beta-\gamma)\alpha'\alpha''+(\gamma-\alpha)\beta'\beta''+(\alpha-\beta)\gamma'\gamma''\\&=(\beta'-\gamma')\alpha''\alpha+(\gamma'-\alpha')\beta''\beta+(\alpha'-\beta')\gamma''\gamma\\&=(\beta''-\gamma'')\alpha\alpha'+(\gamma''-\alpha'')\beta\beta'+(\alpha''-\beta'')\gamma\gamma'\\&=-\tfrac{1}{2}\big\{(\beta-\gamma)(\beta'-\gamma')(\beta''-\gamma'')+(\gamma-\alpha)(\gamma'-\alpha')(\gamma''-\alpha'')+(\alpha-\beta)(\alpha'-\beta')(\alpha''-\beta'')\big\}.\end{aligned}$$

L'identité de ces différentes valeurs de θ étant facile à vérifier.

En représentant par $Aw^2+2Bw+C=0$ la forme rationnelle de l'équation de la surface, on trouve pour le discriminant $AC-B^2$ de cette équation du second degré en w la valeur

$$AC-B^2=4\alpha\alpha''\beta\beta'\beta''\gamma\gamma'\gamma''-2\theta\,xyz\Big(\tfrac{x}{\alpha}+\tfrac{y}{\beta}+\tfrac{z}{\gamma}\Big)\Big(\tfrac{x}{\alpha'}+\tfrac{y}{\beta'}+\tfrac{z}{\gamma'}\Big)\Big(\tfrac{x}{\alpha''}+\tfrac{y}{\beta''}+\tfrac{z}{\gamma''}\Big).$$

L'équation de la surface rendue rationnelle est symétrique par rapport aux trois systèmes de quantités (α,β,γ), (α',β',γ'), $(\alpha'',\beta'',\gamma'')$; la forme irrationnelle de la même équation peut donc être présentée de trois manières différentes, savoir:

$$\sqrt{\alpha x}\Big(\gamma'\gamma''y-\beta'\beta''z-\tfrac{w}{\alpha}\Big)+\sqrt{\beta y}\Big(\alpha'\alpha''z-\gamma'\gamma''x-\tfrac{w}{\beta}\Big)+\sqrt{\gamma z}\Big(\beta'\beta''x-\alpha'\alpha''y-\tfrac{w}{\gamma}\Big)=0,$$

$$\sqrt{\alpha' x}\Big(\gamma''\gamma y-\beta''\beta z-\tfrac{w}{\alpha'}\Big)+\sqrt{\beta' y}\Big(\alpha''\alpha z-\gamma''\gamma x-\tfrac{w}{\beta'}\Big)+\sqrt{\gamma' z}\Big(\beta''\beta x-\alpha''\alpha y-\tfrac{w}{\gamma'}\Big)=0,$$

$$\sqrt{\alpha'' x}\Big(\gamma\gamma' y-\beta\beta' z-\tfrac{w}{\alpha''}\Big)+\sqrt{\beta'' y}\Big(\alpha\alpha' z-\gamma\gamma' x-\tfrac{w}{\beta''}\Big)+\sqrt{\gamma'' z}\Big(\beta\beta' x-\alpha\alpha' y-\tfrac{w}{\gamma''}\Big)=0;$$

et l'on voit de plus que les équations des seize plans singuliers sont

$$x=0,\quad y=0,\quad z=0,\quad w=0,$$

$$\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=0,\quad \frac{x}{\alpha'}+\frac{y}{\beta'}+\frac{z}{\gamma'}=0,\quad \frac{x}{\alpha''}+\frac{y}{\beta''}+\frac{z}{\gamma''}=0,$$

$$\gamma'\gamma''y-\beta'\beta''z-\tfrac{w}{\alpha}=0,\quad \alpha'\alpha''z-\gamma'\gamma''x-\tfrac{w}{\beta}=0,\quad \beta'\beta''x-\alpha'\alpha''y-\tfrac{w}{\gamma}=0,$$

$$\gamma''\gamma y-\beta''\beta z-\tfrac{w}{\alpha'}=0,\quad \alpha''\alpha z-\gamma''\gamma x-\tfrac{w}{\beta'}=0,\quad \beta''\beta x-\alpha''\alpha y-\tfrac{w}{\gamma'}=0,$$

$$\gamma\gamma' y-\beta\beta' z-\tfrac{w}{\alpha''}=0,\quad \alpha\alpha' z-\gamma\gamma' x-\tfrac{w}{\beta''}=0,\quad \beta\beta' x-\alpha\alpha' y-\tfrac{w}{\gamma''}=0,$$

les quantités α,β,γ, etc. étant liées entre elles par les trois équations

$$\alpha+\beta+\gamma=0,\quad \alpha'+\beta'+\gamma'=0,\quad \alpha''+\beta''+\gamma''=0.$$

Voilà ce me semble la forme la plus simple pour l'équation de cette surface.

Cambridge, le 23 février 1871.

[443] NOTE ON THE SOLUTION OF THE QUARTIC EQUATION $\alpha U + 6\beta H = 0$.

[From the Mathematische Annalen, vol. I. (1869), pp. 54, 55.]

If U denote the quartic function $(a, b, c, d, e \mid x, y)^4$, H its Hessian

$$H = (ac - b^2,\ 2(ad - bc),\ ae + 2bd - 3c^2,\ 2(be - cd),\ ce - d^2 \mid x, y)^4,$$

α and β constants, then we may find the linear factors of the function $\alpha U + 6\beta H$ (or what is the same thing solve the equation $\alpha U + 6\beta H = 0$) by a formula almost identical with that given by me (Fifth Memoir on Quantics, *Phil. Trans.* vol. CXLVIII. (1858), see p. 446, [156]) in regard to the original quartic function U.

In fact (reproducing the investigation) if I, J are the two invariants, $M = \dfrac{I^3}{4J^2}$, Φ the cubicovariant

$$\Phi = (-a^2 d + 3abc - 2b^3,\ \ldots \mid x, y)^6,$$

then the identical equation $JU^3 - IU^2H + 4H^3 = -\Phi^2$, may be written $(1,\ 0,\ -M,\ M \mid IH, JU)^3 = -\frac{1}{4} I^3 \Phi^2$, whence if $\omega_1, \omega_2, \omega_3$ are the roots of the equation $(1,\ 0,\ -M,\ M \mid \omega, 1)^3 = 0$, or what is the same thing $\omega^3 - M(6\omega - 1) = 0$, then the functions

$$IH - \omega_1 JU, \quad IH - \omega_2 JU, \quad IH - \omega_3 JU$$

are each of them a square: writing

$$(\omega_2 - \omega_3)(IH - \omega_1 JU) = X^2,$$
$$(\omega_3 - \omega_1)(IH - \omega_2 JU) = Y^2,$$
$$(\omega_1 - \omega_2)(IH - \omega_3 JU) = Z^2,$$

so that identically $X^2 + Y^2 + Z^2 = 0$, the expression $\alpha X + \beta Y + \gamma Z$ will be a square if only $\alpha^2 + \beta^2 + \gamma^2 = 0$. (To see this observe that in virtue of the equation $X^2 + Y^2 + Z^2 = 0$, we have $X + iY$, $X - iY$ each of them a square, and thence

$$\alpha X + \beta Y + \gamma Z = \tfrac{1}{2}(\alpha + i\beta)(X - iY) + \tfrac{1}{2}(\alpha - i\beta)(X + iY) - \tfrac{1}{2} i\gamma \sqrt{X^2 + Y^2},$$

is a square if the condition in question be satisfied.)

Hence in particular writing

$$\sqrt{\omega_2 - \omega_3}\,\sqrt{\alpha I + 6\beta\omega_1 J},\ \ldots\ \sqrt{\omega_1 - \omega_2}\,\sqrt{\alpha I + 6\beta\omega_3 J},$$

for α, β, γ, we have

$$(\omega_2 - \omega_3)\sqrt{\alpha I + 6\beta\omega_1 J}\,\sqrt{IH + \omega_1 JU} + \ldots + (\omega_1 - \omega_2)\sqrt{\alpha I + 6\beta\omega_3 J}\,\sqrt{IH + \omega_3 JU}$$

a perfect square, and since the product of the four different values is a multiple of $(\alpha U + 6\beta H)^2$ (this is most readily seen by observing that for $\alpha U + 6\beta H = 0$, the irrational expression omitting a factor is $(\omega_2 - \omega_3)(\alpha I + 6\beta\omega_1 J) + \ldots + (\omega_1 - \omega_2)(\alpha I + 6\beta\omega_3 J)$, which vanishes identically) it follows that the expression in question is the square of a linear factor of $\alpha U + 6\beta H$.

It thus appears that the radicals (other than those arising from the solution of $U = 0$) contained in the solution of the equation $\alpha U + 6\beta H = 0$ are the three roots

$$\sqrt{\alpha I + 6\beta\omega_1 J}; \quad \sqrt{\alpha I + 6\beta\omega_2 J}; \quad \sqrt{\alpha I + 6\beta\omega_3 J}.$$

Cambridge, September 2, 1868.

[444] ON THE CENTRO-SURFACE OF AN ELLIPSOID.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 16–18.]

The President [Prof. Cayley] gave an account of his investigations on the centro-surface of an ellipsoid (locus of the centres of curvature of the ellipsoid). The surface has been studied by Dr Salmon, and also by Prof. Clebsch, but in particular the theory of the nodal curve on the surface admits of further development. The position of a point on the ellipsoid is determined by means of the parameters, or elliptic coordinates, h, k; viz., if as usual a, b, c are the semi-axes, and if X, Y, Z are the coordinates of the point in question, then

$$\frac{X^2}{a^2+h}+\frac{Y^2}{b^2+h}+\frac{Z^2}{c^2+h}=1,$$

$$\frac{X^2}{a^2+k}+\frac{Y^2}{b^2+k}+\frac{Z^2}{c^2+k}=1;$$

and hence

$$\begin{aligned}-\beta\gamma X^2&=a^2(a^2+h)(a^2+k),\\ -\gamma\alpha Y^2&=b^2(b^2+h)(b^2+k),\\ -\alpha\beta Z^2&=c^2(c^2+h)(c^2+k),\end{aligned}$$

if for shortness

$$\alpha=b^2-c^2,\quad \beta=c^2-a^2,\quad \gamma=a^2-b^2\qquad(\alpha+\beta+\gamma=0).$$

This being so, the coordinates of the point of intersection of the normal at (X, Y, Z) by the normal at the consecutive point of the curve of curvature

$$\frac{X^2}{a^2+k}+\frac{Y^2}{b^2+k}+\frac{Z^2}{c^2+k}=1$$

are given by the formulae

$$\begin{aligned}-\beta\gamma a^2x^2&=(a^2+h)^2(a^2+k),\\ -\gamma\alpha b^2y^2&=(b^2+h)^2(b^2+k),\\ -\alpha\beta c^2z^2&=(c^2+h)^2(c^2+k);\end{aligned}$$

viz., these equations, considering therein (h, k) as arbitrary parameters, determine the coordinates (x, y, z) of a point on the centre-surface. The principal sections (as is known) consist each of them of an ellipse counting three times, and of an evolute of an ellipse; the evolute and ellipse have four contacts (two-fold intersections) and four simple intersections, but the contacts and intersections respectively are in the different sections real and imaginary; and if (as we may without loss of generality assume) $a^2+c^2>2b^2$, then the form of the principal sections is as shown in the figure (which represents only an octant of the surface); viz., there is a real contact at P in the plane of xz, and a real intersection at Q in the plane of xy. The surface has thus an exterior and an interior sheet, but (instead of meeting in a conical point, as in the wave surface) these intersect in a nodal curve QP. The curve has a cusp at Q, and a node at P; viz., the curve extends beyond P, but from that point is acnodal, or without any real sheet of the surface passing through it. For the nodal curve there must be two values (h, k), (h_1, k_1), giving the same values of (x, y, z); viz., there must exist the relations

$$(a^2+h)^2(a^2+k)=(a^2+h_1)^2(a^2+k_1),$$

$$(b^2+h)^2(b^2+k)=(b^2+h_1)^2(b^2+k_1),$$

$$(c^2+h)^2(c^2+k) = (c^2+h_1)^2(c^2+k_1);$$

from which equations eliminating k and k_1 we should have between h, k a relation which, combined with the expressions of x, y, z in terms of (h, k), determines the nodal curve. But the better course is to eliminate k, k_1, thus obtaining a relation between h and h_1, in virtue whereof h_1 may be regarded as a known function of h; k and k_1 can then be readily expressed in terms of h, h_1; that is, we have k as a function of h, h_1, or in effect as a function of h. The relation between h, h_1 (after a reduction of some complexity) assumes ultimately a form which is very simple and remarkable; viz., writing

$$P = a^2+b^2+c^2, \quad Q = b^2c^2+c^2a^2+a^2b^2, \quad R = a^2b^2c^2,$$

the relation is

$$(QR+3Qh^2+Ph^4)+h_1(3Qh+4Ph^2+3h^3)+h_1^2(P+3h^2)=0;$$

this is a $(2, 2)$ correspondence between the two parameters h, h_1; the united values $h_1 = h$, are given by the equation $6(R+Qh+Ph^2+h^3)=0$, that is

$$(a^2+h)(b^2+h)(c^2+h)=0;$$

viz., the two points on the ellipsoid which have their common centre of curvature on the nodal curve are only situate on the same curve of curvature when this curve is a principal section of the ellipsoid.

[Since the date of the foregoing communication, Prof. Cayley has found that the squared coordinates x^2, y^2, z^2 of a point on the nodal curve can be expressed as rational functions of a single variable parameter σ.]

[445] A MEMOIR ON QUARTIC SURFACES.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 19–69. Read February 10, 1870.]

The present Memoir is intended as a commencement of the theory of the quartic surfaces which have nodes (conical points). A quartic surface may be without nodes, or it may have any number of nodes up to 16. I show that this is so, and I consider how many of the nodes may be given points. Although it would at first sight appear that the number is 8, it is in fact 7; viz., we can, with 7 given points as nodes (but not in a proper sense with 8 or more given points), find a quartic surface; such surface contains in its equation 6 constants, which may be such that the surface has an additional node or nodes. Suppose that the surface has an 8th node: there are two distinct cases; viz., (1) the 8 nodes are the points of intersection of 3 quadric surfaces, or say they are an octad, and the surface is said to be octadic; (2) the 8th node is any point whatever on a certain sextic surface determined by means of the 7 given nodes, and called the dianodal surface of these 7 points; the quartic surface is said to be a dianome. The two cases are in general exclusive of each other; viz., the 7 given points being any points whatever, the dianodal surface does not pass through the 8th point of the octad; and thus the quartic surface with the 8 nodes is either octadic or else a dianome. Assuming it to be a dianome, the constants may be further determined so that there shall be a 9th node; it is necessary to examine whether this forms with 7 of the 8 nodes an octad. Supposing that it does not (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 9th node is then any point whatever on a certain curve of the order 18, determined by means of the 8 nodes, and called the dianodal curve of these 8 points. And, finally, the constants may be further determined so that there shall be a

10th node; supposing, as before, that this does not form an octad with any 7 of the 9 nodes (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 10th node is then any one of a system of 22 [should be 13] points determined by means of the 9 nodes, and called the dianodal system of these 9 points. But the quartic surface is now completely determined; viz., starting with any 7 given points as nodes, we have a dianome with 8 nodes, 9 nodes, or 10 nodes, say, an octodianome, enneadianome, or decadianome, but not with any greater number of nodes; these can only present themselves when particular conditions are satisfied in regard to the 7 given nodes, and to the 8th and 9th node; and the consideration of the quartic surfaces with more than 10 nodes would thus form a separate branch of the subject.

The case of the decadianome (or quartic surface with 10 nodes formed as above with 7 given points as nodes) is peculiarly interesting. I identify this with the surface which I call a symmetroid; viz., the surface represented by an equation $\Delta = 0$, where Δ is a symmetrical determinant of the 4th order the several terms whereof are linear functions of the coordinates (x, y, z, w); this surface is related to the Jacobian surface of 4 quadric surfaces (itself a very remarkable surface), and this theory of the symmetroid and the Jacobian, and of questions connected therewith, forms a large portion of the present Memoir.

The theory of the Jacobian is connected also with the researches in regard to nodal quartic surfaces in general; and, for greater clearness, it has seemed to me proper to commence the Memoir with certain definitions, &c., in regard to this theory. It will be seen in what manner I extend the notion of the Jacobian.

I remark that the present researches on Quartic Surfaces were suggested to me by Professor Kummer's most interesting Memoir "Ueber die algebraischen Strahlensysteme u.s.w.," *Berl. Abh.* 1866, in which, without entering upon the general theory, he is led to consider the quartic surfaces, or certain quartic surfaces, with $16, 15, 14, 13, 12$, or 11 nodes; the last of these, or surface with 11 nodes, being in fact a particular case of the symmetroid.

Considerations in regard to the Jacobian of four, or more or less than four, Surfaces.

1. In the case of any four surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, the differential coefficients of P, Q, R, S in regard to the coordinates (x, y, z, w) may be arranged as a square matrix in either of the ways

$$\begin{array}{c|cccc} & P, & Q, & R, & S \\ \hline \partial_x & & & & \\ \partial_y & & & & \\ \partial_z & & & & \\ \partial_w & & & & \end{array} \quad ; \quad \begin{array}{c|cccc} & \partial_x, & \partial_y, & \partial_z, & \partial_w \\ \hline P & & & & \\ Q & & & & \\ R & & & & \\ S & & & & \end{array}$$

and with either arrangement we may form one and the same determinant, the Jacobian determinant $J(P, Q, R, S)$, or, equating it to zero, the Jacobian surface $J(P, Q, R, S) = 0$ of the four surfaces.

2. In the case of more than four surfaces, adopting the arrangement

$$\begin{array}{c|cccccc} & P, & Q, & R, & S, & T, & \ldots \\ \hline \partial_x & & & & & & \\ \partial_y & & & & & & \\ \partial_z & & & & & & \\ \partial_w & & & & & & \end{array}$$

and considering the several determinants which can be formed with any four columns of the matrix, these equated to zero establish a more than one-fold relation between the coordinates;

viz., in the case of five surfaces, we have $J(P,Q,R,S,T) = 0$ a twofold relation representing a curve; and in the case of six surfaces, $J(P,Q,R,S,T,U) = 0$ a threefold relation representing a point-system; and (since with four coordinates a relation is at most threefold) these are the only cases to be considered.

3. In the case of fewer than four surfaces, adopting the arrangement

$$\begin{array}{c|cccc} & \partial_x, & \partial_y, & \partial_z, & \partial_w \\ \hline P & & & & \\ Q & & & & \end{array}$$

and considering the several determinants which can be formed with any 3 or 2 columns of the matrix and equating these to zero, we have in like manner a more than onefold relation between the coordinates; viz., in the case of three surfaces, we have $J(P,Q,R) = 0$ a twofold relation representing a curve; and in the case of two surfaces $J(P,Q) = 0$ a threefold relation representing a point-system; and (since with four coordinates a relation is at most threefold) these are the only cases to be considered (viz., this denotes the points common to the surfaces $\partial_x P = 0$, $\partial_y P = 0$, $\partial_z P = 0$, $\partial_w P = 0$, that is, the points which are at once a node of the surface $P = 0$ and a node of the surface $Q = 0$).

If the notation were used, $J^4(P) = 0$ would denote the system of four equations $\partial_x P = 0$, $\partial_y P = 0$, $\partial_z P = 0$, $\partial_w P = 0$, which are satisfied simultaneously by the coordinates (x, y, z, w) of any node of the surface $P = 0$. Although in what precedes I have used the sign $=$, it should be understood that, while $J(P,Q,R,S) = 0$ denotes a single equation, $J(P,Q,R,S,T) = 0$ or $J(P,Q) = 0$ are each of them a twofold relation, and $J(P,Q,R,S,T,U) = 0$ or $J(P,Q,R) = 0$ each of them a threefold relation. We should have a fourfold relation $J^4(P) = 0$.

4. It is not asserted that $\ldots J(P,Q,R) = 0$, $J(P,Q,R,S) = 0$, $J(P,Q,R,S,T) = 0$ form a continuous series of analogous relations; and there might even be a hiatus in using, in regard to four or more surfaces, J, and in regard to four or fewer surfaces an inverted J (viz., in regard to four surfaces, either symbol used indifferently); but there is no ambiguity in, and I have preferred to adopt, the use of the single symbol J.

5. Suppose that the orders of the surfaces $P = 0$, $Q = 0$, $\ldots$ are $a+1$, $b+1$, $\ldots$ (so that the orders of the differential coefficients of P, Q, $\ldots$ are a, b, $\ldots$), then we have for the orders of the several loci,

$$\begin{array}{rll} J(P,Q) = 0, & \text{point-system, order} & a^3 + a^2b + ab^2 + b^3; \\ J(P,Q,R) = 0, & \text{curve,} & a^2 + b^2 + c^2 + bc + ca + ab; \\ J(P,Q,R,S) = 0, & \text{surface,} & a + b + c + d; \\ J(P,Q,R,S,T) = 0, & \text{curve,} & ab + ac + \ldots + de; \\ J(P,Q,R,S,T,U) = 0, & \text{point-system,} & abc + abd + \ldots + def; \end{array}$$

see, as to this, Salmon's *Solid Geometry*, Ed. 2, (1865), Appendix IV., "On the Order of Systems of Equations" [not reproduced in the later editions]. In particular, if $a = b = c \ldots = 1$, then the orders are $4, 6, 4, 10, 20$.

As to the Surface obtained by equating to zero a Symmetrical Determinant.

6. It is also shown (Salmon, Ed. 2, p. 495) that the surface obtained by equating to zero any symmetrical determinant has a determinate number of nodes; viz., if the orders of the terms in the diagonal be a, b, c, &c., then the number of nodes is $= \frac{1}{6}(\Sigma a \cdot \Sigma ab - 2\Sigma abc)$, or, as this may also be written, $\frac{1}{2}(\Sigma a^2 b + 2\Sigma abc)$. In particular, the formula applies to the case of the surface

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0,$$

(a, b, c, d) being here the orders of A, B, C, D respectively, and the orders of F, G, &c., being $\frac{1}{2}(b + c)$, $\frac{1}{2}(a + c)$, &c. If the terms are all of them linear functions of the coordinates, or $a = b = c = d = 1$, then the number of nodes is $= 10$.

7. That the surface has nodes is, in fact, clear from the consideration that any point for which the minors of the determinant all vanish will be a node; and that (for the symmetrical determinant), by making the minors all of them vanish, we establish only a threefold relation between the coordinates. The expression for the number of the nodes is, I think, obtained most readily as follows.

The nodes will be points of intersection of the curve and surface

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0, \qquad \begin{vmatrix} B, & F, & M \\ F, & C, & N \\ M, & N, & D \end{vmatrix} = 0,$$

the points

$$\begin{vmatrix} H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0;$$

and not only so, but they touch at the points in question; so that, multiplying together the orders of the curve and surface, and subtracting twice the order of the point-system, we obtain the expression for the number of nodes. In the particular case where the functions are all linear, we have a sextic curve and cubic surface intersecting in 18 points; but the curve and surface touch in 4 points, and the number of nodes is $(18 - 2 \cdot 4) = 10$. And in the same way the formula may be established for the general case.

8. The subsidiary theorem of the contact of the curve and surface requires, however, to be proved. Seeking for the equation of the tangent plane of the surface at any one of the points in question, we have first

$$\begin{vmatrix} \delta B, & \delta F, & \delta M \\ F, & C, & N \\ M, & N, & D \end{vmatrix} + \begin{vmatrix} B, & F, & M \\ \delta F, & \delta C, & \delta N \\ M, & N, & D \end{vmatrix} + \begin{vmatrix} B, & F, & M \\ F, & C, & N \\ \delta M, & \delta N, & \delta D \end{vmatrix} = 0,$$

where, in virtue of the equations

$$\begin{vmatrix} H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0,$$

the last term vanishes. Expanding the other two terms, the equation becomes

$$D(C\delta B + B\delta C - 2F\delta F) - (N^2\delta B - 2MN\delta F + M^2\delta C) + \delta M(FN - CM) + \delta N(BN - MF) = 0;$$

in virtue of the same equations, the coefficients of δM and δN each of them vanish, and we have also

$$N^2\delta B + M^2\delta C - 2MN\delta F = \tfrac{M^2}{D}(C\delta B + B\delta C - 2F\delta F),$$

so that the equation becomes finally $C\delta B + B\delta C - 2F\delta F = 0$. Investigating by a like process the equation of the tangent of the curve

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0,$$

we find between the differentials δA, δB, &c., a twofold linear relation, expressible by means of the foregoing equation $C\delta B + B\delta C - 2F\delta F = 0$, and one other equation; that is, at each of the points in question the tangent of the curve lies in the tangent plane of the surface, or, what is the same thing, the curve and surface touch at these points.

Surfaces represented by an equation $F(P, Q) = 0$, &c.

9. In the remarks which follow as to the surfaces $F(P, Q) = 0$, $F(P, Q, R) = 0$, &c., the function F is a rational and integral function of (P, Q), (P, Q, R), &c., not in general homogeneous in regard to P, Q, R, ... but of such degrees in regard to these functions respectively as to be homogeneous in regard to the coordinates (x, y, z, w).

The surface $F(P, Q) = 0$ has in general a nodal curve $\partial_P F = 0$, $\partial_Q F = 0$; if it has besides any nodes, these are points of the point-system $J(P, Q) = 0$.

The surface $F(P, Q, R) = 0$ has in general nodes $\partial_P F = 0$, $\partial_Q F = 0$, $\partial_R F = 0$; if it has besides any nodes, these are points on the curve $J(P, Q, R) = 0$.

The surface $F(P, Q, R, S) = 0$ has not in general, but it may have, nodes $\partial_P F = 0$, $\partial_Q F = 0$, $\partial_R F = 0$, $\partial_S F = 0$; if it has any other nodes, these are points on the surface $J(P, Q, R, S) = 0$.

Nodes of a Quartic Surface; Circumscribed Cone having its vertex at a Node.

10. A quartic surface may be without nodes; or it may have any number of nodes up to 16. Consider a quartic surface having a node; and take the single node, or (if more nodes than one) any one of the nodes, as the vertex of a circumscribed cone; then, considering any plane through the vertex, the section will be a quartic curve having a node at the vertex, and the generating lines in the plane will be the tangents from the node to the quartic curve; the number of them is therefore 6, and the order of the circumscribed cone is thus $= 6$. Each tangent intersects the quartic curve in the node counting as two intersections, and in the point of contact counting as two intersections; there are consequently no singular tangents; and therefore in the circumscribed cone no singular lines arising from a singular tangency of the generating line. Hence, in the case of a single node on the surface, the circumscribed cone is a cone of the order 6 without nodal or stationary lines; and the class is $= 30$; but in the case of more than one node, say k nodes, the circumscribed cone passes through the remaining $k - 1$ nodes, and the generating line through each of these nodes is a nodal line of the cone; that is, the cone has $k - 1$ nodal lines, and its class is $= 30 - 2k + 2$. The cone is not of necessity a proper cone; the maximum number of nodal lines is when it breaks up into 6 planes, and we have then $k - 1 = 15$; that is, the number of nodes of the surface is at most $= 16$.

11. It is easy to form a table of the different *primâ facie* possible forms of the sextic according to the number of nodes of the surface; viz., writing 6 for a proper sextic cone without nodal lines, $6_1, 6_2, \ldots 6_{10}$ for the proper sextic cone with $1, 2, \ldots$ or 10 nodal lines; and so $5, 5_1, \ldots 5_6$ for the proper quintic cones, $4, 4_1, 4_2, 4_3, 3, 3_1, 2$ for the quartic, cubic, and quadric cones, and 1 for the plane, the table is

Circumscribed Sextic Cone.

Nodes of Surface.	
1	6
2	6_1
3	6_2
4	6_3
5	6_4
6	$6_5;\ 5, 1$
7	$6_6;\ 5_1, 1$
8	$6_7;\ 5_2, 1$
9	$6_8;\ 5_3, 1;\ 4, 2$
10	$6_9;\ 5_4, 1;\ 4_1, 2;\ 4, 1, 1;\ 3, 3$
11	$6_{10};\ 5_5, 1;\ 4_2, 2;\ 4_1, 1, 1;\ 3_1, 3$
12	$5_6, 1;\ 4_3, 2;\ 4_2, 1, 1;\ 3_2, 3;\ 3, 2, 1$
13	$\ldots;\ \ldots;\ 4_3, 1, 1;\ \ldots;\ 3, 2, 1;\ 3, 1, 1, 1;\ 2, 2, 2$
14	$\ldots;\ \ldots;\ \ldots;\ \ldots;\ 3_1, 1, 1, 1;\ 2, 2, 1, 1$
15	$\ldots;\ \ldots;\ \ldots;\ \ldots;\ \ldots;\ 2_1, 1, 1, 1, 1$
16	$\ldots;\ \ldots;\ \ldots;\ \ldots;\ \ldots;\ 1, 1, 1, 1, 1, 1$

and moreover, in the cases where there are two or more forms of the sextic cone, then the k sextic cones may be of the different forms in various combinations. The total number of cases *primâ facie* possible is thus very great; but only a comparatively small number of them actually exist.

12. In the case where there is a plane 1, the sextic cone breaks up into this plane, and into a (proper or improper) quintic cone intersecting the plane in 5 lines; that is, there will be in the plane 6 nodes; the plane is, in fact, a singular tangent plane meeting the surface in a conic twice repeated; and the 6 nodes lie on this conic. Taking any one of these nodes as vertex, the corresponding sextic cone breaks up into the plane, and into a (proper or improper) quintic cone.

13. In the cases $k = 1, 2, 3, 4, 5$, and $k = 15, 16$, there is only one form of sextic cone; so that each node (at least so far as appears) stands in the same relation to the surface. Considering the last mentioned two cases; $k = 16$, each of the 16 nodes gives 6 singular tangent planes, but each of these passes through 6 nodes; therefore the number of planes is $= 16$: similarly, $k = 15$, the number of singular tangent planes is $15 \times 4 \div 6, = 10$.

For $k = 14$, the cones are $3_1, 1, 1, 1$, or $2, 2, 1, 1$: it is easy to see that we have only the three cases

Cones $3_1, 1, 1, 1\ :\ 2, 2, 1, 1$		Singular tangent planes
may be 14, 0	gives	$(14 \cdot 3 + 0 \cdot 6) \div 6, = 7$
8, 6	,	$(8 \cdot 3 + 6 \cdot 2) \div 6, = 6$
2, 12	,	$(2 \cdot 3 + 12 \cdot 2) \div 6, = 5$

and we may in the like manner limit the number of possible cases, for other values of k. But I do not at present further pursue the inquiry.

As to the Number of Constants contained in a Surface.

14. We say that a surface $P = 0$ contains or depends upon a certain number of constants; viz., this is the number of constants contained in the equation $P = 0$ of the surface, taking the coefficient of any one term to be equal to unity; thus the general quadric surface contains 9 constants; the surface can in fact be determined so as to satisfy 9 conditions; or, as we might express it, the Postulation of the surface is $= 9$. [I have elsewhere said *Postulandum* and *Capacity*: I prefer this last expression.] And if, in the general equation so containing 9 constants,

k of these are given, or, what is the same thing, if the quadric surface be made to satisfy any k conditions, then the number of constants, or postulation of the surface, is $= 9 - k$.

15. But a different form of expression is sometimes convenient; the conditions to be satisfied are frequently such that, being satisfied by the surfaces $P = 0$, $Q = 0$, ... they will be satisfied by the surface $\alpha P + \beta Q + \ldots = 0$, where $\alpha, \beta, \ldots$ are any constant multipliers whatever. When this is so, there will be a certain number of solutions $P = 0$, $Q = 0$, ... not connected by any such relation, or say of asyzygetic solutions, such that the general surface satisfying the conditions in question is $\alpha P + \beta Q + \ldots = 0$; and hence, taking one of these coefficients as unity, the number of constants, or postulation of the surface, is equal to the number of the remaining coefficients, or, what is the same thing, it is less by unity than the number of the asyzygetic solutions $P = 0$, $Q = 0, \ldots$. Instead of considering the number of constants, or postulation, we may consider the number of solutions (that is, asyzygetic solutions) or surfaces $P = 0$, $Q = 0$, ... which satisfy the conditions in question.

16. Thus, for the quadric not subjected to any conditions, there are 10 surfaces (for example, these may be taken to be the surfaces $x^2 = 0$, $y^2 = 0$, $z^2 = 0$, $w^2 = 0$, $yz = 0$, $zx = 0$, $xy = 0$, $xw = 0$, $yw = 0$, $zw = 0$); and the general quadric surface is by means of these expressed linearly in the form $(a, \ldots \mid x, y, z, w)^2 = 0$. So for the quadric surfaces through k given points, the number of these is $= 10 - k$; thus for the surfaces through 4 given points, say the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, the 6 given surfaces may be taken to be $yz = 0$, $zx = 0$, $xy = 0$, $xw = 0$, $yw = 0$, $zw = 0$, and every other quadric surface through the 4 points is by means of these expressed linearly in the form $(a, \ldots \mid yz, zx, xy, xw, yw, zw) = 0$; for the quadric surfaces through 8 points there are two surfaces $P = 0$, $Q = 0$; and every quadric surface through the 8 points is by means of these expressed linearly in the form $\alpha P + \beta Q = 0$; and (as the extreme case) if the quadric surface passes through 9 given points, then there is the single quadric surface $P = 0$.

17. In the questions in which such quadric surfaces present themselves, it is in general quite immaterial what particular surfaces are selected as the surfaces $P = 0$, $Q = 0, \ldots$; the selection may be made at pleasure; and, being so made, the surfaces are to be regarded as completely determinate; viz., there would be no gain of generality if these were replaced by any other surfaces $\alpha P + \beta Q + \ldots = 0$. For instance, in the theory of the quartic surfaces with 6 given points as nodes, we have through the 6 given points the 4 quartic surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$, and we consider the quartic functions $(a, \ldots \mid P, Q, R, S)^2$ and $J(P, Q, R, S)$: each of these is unaltered as to its form when P, Q, R, S are replaced each of them by any linear function of these quantities; viz., $(a, \ldots \mid P, Q, R, S)^2$ is changed into a new quadric function $(a', \ldots \mid P, Q, R, S)^2$, and $J(P, Q, R, S)$ into a mere constant multiple of its original value. We have herein a justification of the expressions in question, through 6 given points there are 4 quadric surfaces, &c.

General theory of the Quartic Surface with a given Node or Nodes.

18. A quartic surface contains 34 constants; and the number of conditions to be satisfied in order that a given point may be a node is $= 4$. Hence, if the surface has k given points as nodes, the number of constants is $= 34 - 4k$; and it would at first sight appear that k might be $= 8$, and that with the 8 given points as nodes we should have a quartic surface containing 2 constants. But this is not so in a proper sense; through the 8 given points we have 2 quadric surfaces $P = 0$, $Q = 0$; and we can by means of these form a quartic surface $(a, b, c \mid P, Q)^2 = 0$, containing 2 constants, and having in a sense the 8 points as nodes. This, however, is no proper quartic surface, but is a system of 2 quadric surfaces, each of them passing through the 8 points, and the two quadric surfaces therefore intersecting in a quadri-quadric curve through the 8 points; which curve is therefore a nodal curve on the compound surface; and it is only as points on this nodal curve, and not in a proper sense, that the 8 given points are nodes of the quartic surface. The greatest value of k is thus $k = 7$.

19. Of course, if $k = 0$, we have the general quartic surface $U = 0$, containing 34 constants. The cases $k = 1$, $k = 2$, $k = 3$ (viz., a single given node, 2 given nodes, 3 given nodes), may be at once disposed of; taking for instance the 1st node to be the point $(1, 0, 0, 0)$, the 2nd node the point $(0, 1, 0, 0)$, the 3rd node the point $(0, 0, 1, 0)$, we find at once an equation $U = 0$, with $30, 26$, or 22 constants, having the given node or nodes.

Four given Nodes.

20. The case of 4 given nodes is just as easy; but in reference to what follows, it is proper to consider it more in detail. The equation should contain 18 constants; we have through the 4 given points 6 quadric surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, $T = 0$, $U = 0$, and we can by means of them form a quartic equation $(a, \ldots \mid P, Q, R, S, T, U)^2 = 0$, having the 4 given points as nodes; this contains, however, $(21 - 1 =)20$ constants; the reduction to the right number 18 occurs by reason that the functions (P, Q, R, S, T, U), although linearly independent, are connected by two quadric equations

$$(* \mid P, Q, R, S, T, U)^2 = 0, \qquad (*' \mid P, Q, R, S, T, U)^2 = 0;$$

hence writing the equation of the quartic surface in the form

$$(a, \ldots \mid P_*)^2 - \lambda\, (* \mid P_*)^2 - \mu\, (*' \mid P_*)^2 = 0,$$

the coefficients λ, μ may be so determined as to reduce to zero the coefficients of any two terms of the equation, and the number of constants really is $20 - 2 = 18$, as it should be.

21. In proof, observe that, taking the 4 given nodes to be the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, the quadric surfaces may be taken to be $yz = 0$, $zx = 0$, $xy = 0$, $xw = 0$, $yw = 0$, $zw = 0$; the equation of the quartic surface will thus be

$$(a, \ldots \mid yz, zx, xy, xw, yw, zw)^2 = 0;$$

but we have between the functions xy, &c., the two identical relations

$$xy \cdot zw - xz \cdot yw = 0, \qquad xy \cdot zw - xw \cdot yz = 0;$$

and the number of constants is thus $= 18$.

Five given Nodes.

22. In the case of 5 given nodes, the number of constants should be $= 14$. We have through the 5 given points, 5 quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$, $T = 0$, and we form herewith the quartic equation $(a, \ldots \mid P, Q, R, S, T)^2 = 0$, containing the right number 14 of arbitrary constants. The functions P, Q, &c. are in this case not connected by any quadric relation, and the equation just written down is in fact the general equation of the quartic surface with the 5 given nodes.

23. In verification, take the first 4 nodes to be as above, and the 5th node to be the point $(1, 1, 1, 1)$; we may write

$$(P, Q, R, S, T) = \{x(y - z),\ x(y - w),\ y(x - z),\ y(x - w),\ xy - zw\};$$

and if from the 5 equations $P = x(y - z)$, &c., we eliminate (x, y, z, w), we obtain one, and only one, relation between the functions P, Q, R, S, T; this is found to be

$$PS(Q + R - T) - QR(P + S - T) = 0,$$

or, what is the same thing,

$$R(P - Q)(S - T) - P(R - S)(Q - T) = 0;$$

viz., it is a cubic relation, and there is consequently no quadric relation between the 5 functions.

Six given Nodes.

24. In the case of 6 given nodes, the quartic surface should contain 10 constants. We have through the 6 given points 4 quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$; but if we form herewith the quartic surface $(a, \ldots \mid P, Q, R, S)^2 = 0$, this contains only 9 constants. It is to be shown that the Jacobian surface $J(P, Q, R, S) = 0$ of the 4 quadric surfaces (or say of the 6 points) is a quartic surface having the 6 given points as nodes, and not included in the foregoing form $(a, \ldots \mid P, Q, R, S)^2 = 0$; this being so, we have the quartic surface

$$(a, \ldots \mid P, Q, R, S)^2 + \theta\, J(P, Q, R, S) = 0,$$

having the 6 given points as nodes, and containing the complete number of constants, viz., 10.

25. The 6 given nodes being any points whatever, their coordinates may be taken to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, $(1, 1, 1, 1)$, and $(\alpha, \beta, \gamma, \delta)$. I proceed to find the Jacobian of these 6 points. For this purpose, let (a, b, c, f, g, h) be the 6 coordinates of the line through the points $(1, 1, 1, 1)$ and $(\alpha, \beta, \gamma, \delta)$, viz.,

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, \\ b &= \gamma - \alpha, & g &= \beta - \delta, \\ c &= \alpha - \beta, & h &= \gamma - \delta, \end{aligned}$$

whence $af + bg + ch = 0$, and also

$$\begin{aligned} 0\cdot\ \ -h\cdot\ \ -g\cdot\ \ +a &= 0, \\ -h\cdot\ \ +0\cdot\ \ +f\cdot\ \ +b &= 0, \\ g\cdot\ \ -f\cdot\ \ +0\cdot\ \ +c &= 0, \\ -a\cdot\ \ -b\cdot\ \ -c\cdot\ \ +0 &= 0; \end{aligned}$$

we have through the 6 points the plane pairs

$$\begin{aligned} x\,(\quad \ldots - hy - gz + aw) &= 0, \\ y\,(-hx \quad + fz + bw) &= 0, \\ z\,(\ gx - fy \quad + cw) &= 0, \\ w\,(-ax - by - cz \quad) &= 0, \end{aligned}$$

where, adding the four equations, we have identically $= 0$. For this reason, we cannot take these to be the equations of the 4 quadric surfaces, but we may take the first 3 of them for the surfaces

$P=0$, $Q=0$, $R=0$; and for the 4th surface $S=0$, I take the quadric cone having its vertex at the point $(0,0,0,1)$; viz., the equation is

$$a\alpha\, yz + b\beta\, zx + c\gamma\, xy = 0;$$

that is, I write

$$(P,Q,R,S) = \{x(hy-gz+aw),\ y(-hx+fz+bw),\ z(gx-fy+cw),\ (a\alpha yz+b\beta zx+c\gamma xy)\}.$$

26. The Jacobian is then easily found to be

$$\begin{aligned}&(b\beta zx+c\gamma xy)(-agh, bhf, cfg, abc, -af^2, -gB, hC, aA, bfg, -c^2h \mid x,y,z,w)^2\\ &\quad+(c\gamma xy+a\alpha yz)(agh, -bhf, cfg, abc, fA, -bg^2, -hC, -a^2f, bB, c^2h \mid x,y,z,w)^2\\ &\quad+(a\alpha yz+b\beta zx)(agh, bhf, -cfg, abc, -fA, gB, -ch^2, a^2f, -b^2g, cC \mid x,y,z,w)^2=0;\end{aligned}$$

where for the moment A, B, C denote $bg-ch$, $ch-af$, $af-bg$ respectively. Collecting and reducing, the whole divides by $2abc$; and if finally we replace a,b,c,f,g,h by their values, the result is

$$J = \left\{\begin{array}{l}(\beta-\gamma)yz\,(\alpha w^2-\delta x^2)+(\alpha-\delta)xw\,(\beta z^2-\gamma y^2)\\ \quad+(\gamma-\alpha)zx\,(\beta w^2-\delta y^2)+(\beta-\delta)yw\,(\gamma x^2-\alpha z^2)\\ \quad+(\alpha-\beta)xy\,(\gamma w^2-\delta z^2)+(\gamma-\delta)zw\,(\alpha y^2-\beta x^2)\end{array}\right\}=0.$$

27. It may be shown *à posteriori* that J is not a quadric function of P, Q, R, S. For, attempting to express it in this form, J does not contain the terms x^2w^2, y^2w^2, z^2w^2, and it thence at once appears that the coefficients of P^2, Q^2, R^2 each of them vanish. Hence, introducing for convenience the factor 2, I assume $(0,0,0,D,F,G,H,L,M,N \mid P,Q,R,S)^2 = 2J$. Comparing the terms in $w^2(yz, zx, xy)$, we obtain

$$bcF = a\alpha,\quad caG = b\beta,\quad abH = c\gamma;$$

and comparing the coefficients of $w(y^2z, z^2x, x^2y, yz^2, zx^2, xy^2)$, we obtain

$$\begin{aligned}-Ff + a\alpha M &= \tfrac{a\alpha}{c}\beta, & Ff + a\alpha N &= -\tfrac{a\alpha}{b}\gamma,\\ -Gg + b\beta N &= \tfrac{b\beta}{a}\gamma, & Gg + b\beta L &= -\tfrac{b\beta}{c}\alpha,\\ -Hh + c\gamma L &= -\tfrac{c\gamma}{b}\alpha, & Hh + c\gamma M &= -\tfrac{c\gamma}{a}\beta;\end{aligned}$$

substituting for F, G, H their values, we obtain from the first 3 equations $L, M, N = \frac{f}{bc}, \frac{-g}{ca}, \frac{-h}{ab}$, and from the second 3 equations, $L, M, N = \frac{f}{bc}, \frac{g}{ca}, \frac{h}{ab}$; that is, the equations are inconsistent, and the function J is not expressible in the form in question.

Jacobian Surface of Six given Points.

28. The equation $J=0$ is the locus of the vertices of the quadric cones which pass through the given 6 points; calling these $1,2,3,4,5,6$, we see at once that the surface passes through the 15 lines $12, 13, \ldots 56$, and also through the ten lines 123.456 (viz., line of intersection of the planes through $1,2,3$, and through $4,5,6$), &c. In fact, taking the vertex at any point 0 in the line $1,2$, the lines drawn to the six points are $01=02, 03, 04, 05, 06$; viz., there are only five lines, so that these lie in a quadric cone. And taking the vertex at any point in the line 123.456, the lines to the 6 points lie in these planes 123 and 456 respectively, and the quadric cone is in fact this plane-pair. Moreover, the surface containing the lines $12, 13, 14, 15, 16$, must have the point 1 for a node; and similarly, the points $2,3,4,5,6$ are each of them a node on the surface. It is to be added that the surface contains the skew cubic through the 6 points, or say the skew cubic 123456. See, as to this, *post* No. 108.

29. The surface in question (the Jacobian of the 6 points) is a particular case of the Jacobian of any 4 quadric surfaces. This more general surface will be considered in the sequel; I only remark here that it contains 10 lines, corresponding to the 10 lines 123.456, &c., but it has not any other lines, or any nodes.

Jacobian Curve of Seven given Points, or of an Octad of Points.

30. In connexion with what precedes, we may here consider a curve which presents itself in the sequel; viz., the curve which is the locus of the vertices of the quadric cones which pass through seven given points. The general case is when no one of the points is the vertex of a quadric cone through the other 6 points. We have through the 7 points the three quadric surfaces $P = 0$, $Q = 0$, $R = 0$; hence, forming the equation $\alpha P + \beta Q + \gamma R = 0$ of the general quadric surface through the 7 points, and making this a cone, we find as the locus of the vertex $J(P, Q, R) = 0$; the analytical form shows that this is a sextic curve. It appears, moreover, that the curve is symmetrically related to all the 8 points $P = 0$, $Q = 0$, $R = 0$; and instead of calling it the Jacobian of the 7 points, we may call it the Jacobian of the octad. But in further explanation, take the points to be $1, 2, 3, 4, 5, 6, 7$; the vertex will lie on each of the Jacobian surfaces 123456 and 123457; and it is at present assumed that 7 is not a point on the first surface, nor 6 a point on the second surface. The two surfaces have in common the lines $12, 13, \ldots 45$, and they consequently besides intersect in a curve of the 6th order, or sextic curve, which is the locus in question. At the point 1 there is on the first surface a tangent cone through the lines $12, 13, 14, 15, 16$, and on the second surface a tangent cone through the lines $12, 13, 14, 15, 17$; these two cones have for their complete intersection the lines $12, 13, 14, 15$, which lines belong to the complete intersection of the two surfaces, but not to the sextic curve. It thus appears, *à posteriori*, that the sextic curve does not pass through the point 1; and similarly, that it does not pass through any of the points $2, 3, 4$, or 5. As to the points 6 and 7, each of these is on only one of the quartic surfaces, and therefore the curve of intersection does not pass through either of these points.

31. Suppose, however, that one of the seven points is the vertex of a cone through the other six; it is of course the same thing whether we take this to be one of the points $1, 2, 3, 4, 5$, or one of the points 6 and 7, but the result comes out more easily in the latter case; viz., in the former case, taking 1 to be the point in question, the two tangent cones at 1 are one and the same cone, and all that appears is that there is nothing to hinder a branch or branches of the sextic curve from passing through the point 1. But in the latter case, taking 7 for the point in question, then 7 lies on the surface 123456, being a simple point on this surface, but a node on the surface 123457; and it thus appears that there are through 7 two branches of the sextic curve; so that any one of the seven points, being the vertex of a cone through the other six, is an actual double point on the sextic curve.

32. In the case where two of the points are each of them the vertex of a cone through the other six points, then the seven points lie on a skew cubic; and the sextic curve of the general case becomes this skew cubic twice repeated.

Seven given Nodes.

33. In the case of 7 given nodes, the number of constants should be = 6; the 7 given points determine 3 quadric surfaces $P = 0$, $Q = 0$, $R = 0$: and we have hence the quartic surface $(a, \ldots \mid P, Q, R)^2 = 0$, containing 5 constants only. That this is not the general quartic surface with the 7 given nodes, is also clear from the consideration that the surface in question has 8 nodes; viz., the 8 points of intersection of the three quadric surfaces. Suppose that a particular quartic surface, having the 7 given nodes, but not of the last mentioned form, is $\Delta = 0$; then a quartic surface having the 7 given nodes is

$$(a, \dots \mid P, Q, R)^2 + \theta\,\Delta = 0;$$

and this, as containing 6 constants, will be the general quartic surface with the 7 given nodes.

34. It follows that, if $\Delta' = 0$ be another quartic surface having the 7 given nodes, we must have identically $\Delta' - \rho\Delta = (* \mid P, Q, R)^2$, where ρ is a determinate constant and $(* \mid P, Q, R)^2$ a determinate quadric function of (P, Q, R). The formula extends to the case where $\Delta' = 0$ has the 8 nodes ($P = 0$, $Q = 0$, $R = 0$), but we have then $\rho = 0$, and the meaning is simply that the general quartic surface having the 8 nodes is $(* \mid P, Q, R)^2 = 0$.

35. A particular quartic surface having (in an improper sense) the 7 given nodes, but not having the 8th node, is $M\Omega = 0$, where $M = 0$ in the plane through any 3 of the 7 points and $\Omega = 0$ is the cubic surface through these same 3 points, and having the remaining 4 points as nodes. The equation of the cubic surface, if the 4 points are taken to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, is obviously of the form

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} + \frac{d}{w} = 0, \quad (\text{that is, } ayzw + bzxw + cxyw + dxyz = 0),$$

and by making the surface pass through the 3 points we determine linearly the coefficients (a, b, c, d), that is, their ratios. The equation of the quartic surface thus is

$$(a, \dots \mid P, Q, R)^2 + \theta\, M\Omega = 0,$$

the 7 given points being here proper nodes; and the formula being precisely equivalent to the preceding one containing Δ.

36. We can with the 7 given points form 35 such combinations $M\Omega = 0$ of a plane and a cubic surface, and so present the equation of the quartic surface under 35 different forms; these are of course equivalent in virtue of the before mentioned formula for $\Delta' - \rho\Delta$; viz., we must have identically $M\Omega - \rho' M'\Omega' = (* \mid P, Q, R)^2$: a theorem of some interest, which it might be difficult to verify *à posteriori.*

Investigation of the cases of 8 Nodes.

37. It has already been shown that a quartic surface cannot in a proper sense have 8 given nodes. In regard to the quartic surfaces with 8 nodes, we start from the surface with 7 given nodes; viz.,

$$(a, \dots \mid P, Q, R)^2 + \theta\,\nabla = 0,$$

or, what is the same thing,

$$(a, \dots \mid P, Q, R)^2 + \theta\, M\Omega = 0;$$

and we inquire in what cases this surface has an 8th node. Obviously if $\theta = 0$, that is, if the surface is $(a, \dots \mid P, Q, R)^2 = 0$, the surface will have an 8th node, the remaining intersection of the quadric surfaces $P = 0$, $Q = 0$, $R = 0$ (observe that this is a point in no wise depending on the particular quadric surfaces, but uniquely determined by means of the 7 given points); and we have thus one kind, say the "octadic" surface, of the quartic surfaces with 8 nodes; viz., the nodes are the 8 points of intersection of any 3 quadric surfaces, or they are an octad of points. By what precedes, 7 of the nodes may be given points, and the remaining node is then a uniquely determinate point, the 8th point of the octad.

38. But if θ be not $= 0$, there may still be an 8th node; viz., this must then be a point on the Jacobian surface $J(P, Q, R, \nabla) = 0$, which is of the order 6. It is clear *à priori* that this must be a surface depending only on the 7 points, but independent of the particular surfaces $P = 0$, $Q = 0$, $R = 0$, $\nabla = 0$; to verify this, observe that, substituting for ∇ the function $\nabla_1 = \nabla + (* \mid P, Q, R)^2$, we in fact leave the Jacobian unaltered; I call it the dianodal surface of the 7 points.

39. I say that the 8th node may be any point whatever on the dianodal surface; in fact, regarding for a moment the coordinates of the node as given, and expressing that the point is a node on the quartic surface, we have 4 equations containing

$$aP_0 + hQ_0 + gR_0, \quad hP_0 + bQ_0 + fR_0, \quad gP_0 + fQ_0 + cR_0,$$

(P_0, Q_0, R_0 the values of P, Q, R at the node,) but which, if only the point be on the dianodal surface, reduce themselves to three equations; viz., we have between the coefficients (a, b, c, f, g, h) three equations which being satisfied, the point in question will be a node. And it thus appears that, taking the 8th node to be a given point on the dianodal surface, the equation $(a, \dots \mid P, Q, R)^2 + \theta\,\nabla = 0$ of the quartic surface will contain 3 constants. Observe that we may through the 8 nodes draw 2 quadric surfaces $P = 0$, $Q = 0$; and this being so if $\Delta = 0$ be a particular quartic surface with the 8 nodes, then the general quartic surface will be

$$(a, b, c \mid P, Q)^2 + \theta\,\Delta = 0,$$

containing the right number 3 of constants. But there is not here any simple form of the surface $\Delta = 0$, such as the form $M\Omega = 0$ for the surface through 7 given points.

40. It is clear *à priori* that the relation between the 8 nodes is a symmetrical one; so that the 8th point being situate anywhere on the dianodal surface of the remaining 7 points, each of the 7 points will be situate on the dianodal surface of the remaining 7 points. This is a remarkable property of the dianodal surface, which will have to be again considered.

41. In what precedes, we have the second kind of quartic surfaces with 8 nodes, say the "dianome"; viz., each node is a point on the dianodal surface of the remaining nodes; any 7 of the nodes may be taken to be given points, and the remaining node to be any point whatever on the dianodal surface of the 7 points.

The Dianodal Surface.

42. Consider the seven points $1, 2, 3, 4, 5, 6, 7$. As already mentioned, through three of these say $1, 2, 3$, we may draw a plane $M = 0$; and through the same three points and the remaining points $4, 5, 6, 7$ as nodes ($3 + 4 \cdot 4 = 19$ conditions), a cubic surface $\Omega = 0$; this surface passes through the six lines $45, 46, \dots 67$. Hence we have $\Delta = M\Omega = 0$, a quartic surface with the seven points as nodes. And using this form of Δ it may be shown that the dianodal $J(P, Q, R, \Delta) = 0$ passes through the 21 lines $12, 13, \dots 67$, and through 35 plane cubics such as $M = 0$, $\Omega = 0$; viz., this is a cubic in the plane 123 passing through the points $1, 2, 3$, and through the intersection of the plane with each of the six lines $45, 46, \dots 67$ (nine points determining the cubic); the complete intersection by the plane 123 being therefore composed of this cubic and of the three lines $12, 13, 23$. For the passage through the cubic, we have only to observe that

$$J(P, Q, R, M\Omega) = J(P, Q, R, \Omega)\,M + J(P, Q, R, M)\,\Omega = 0$$

is satisfied by $M = 0$, $\Omega = 0$; and for the passage through the lines, taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the planes $567, 674, 745$, and 456 respectively, each of the functions

P, Q, R is of the form $ayz + bzx + cxy + fxw + gyw + hzw$, and the function Ω is of the form $Ayzw + Bzwx + Cwxy + Dxyz$. Hence, writing in the derived functions $z = 0$, $w = 0$, the first and second lines of the determinant $J(P, Q, R, \Omega)$ will be of the form

$$\begin{pmatrix} cy, & c'y, & c''y, & * \\ cx, & c'x, & c''x, & * \end{pmatrix},$$

or the determinant vanishes for $z = 0$, $w = 0$; that is for any point of the line 45 we have $\Omega = 0$ and also $J(P, Q, R, \Omega) = 0$; consequently $J(P, Q, R, M\Omega) = 0$, and the like for the other lines. The theorem is thus proved.

43. I say that the dianodal surface passes through each of the 7 skew cubics, such as 123456. To prove this, it is only necessary to show that the skew cubic 123456 lies on the dianodal surface. For this purpose it will be enough to show that the skew cubic meets the plane 712 in a point of the surface; for then it will, in like manner, meet each of the 15 planes $712, 713, \ldots 756$ in a point of the surface; that is, we shall have 15 intersections of the curve and surface, and there are, besides, the intersections $1, 2, 3, 4, 5, 6$, in all 21 intersections; that is, the skew cubic must lie on the surface.

44. The plane 712 meets the surface in three lines and in a plane cubic determined by the points $7, 1, 2$ and the six intersections of the plane with the lines $34, 35, \ldots 56$. We have therefore to show that this plane cubic meets the skew cubic 123456. Consider for a moment the points $1, 2, 3, 4, 5, 6$ and another point $7'$. As seen above, we have in general, through the points $1, 2, 7'$ and with the points $3, 4, 5, 6$ as nodes, a determinate cubic surface, which surface passes through the lines $34, 35, \ldots 56$. But the cubic surface becomes indeterminate if the points $1, 2, 7', 3, 4, 5, 6$ are on the same skew cubic; that is, if $7'$ is any point whatever on the skew cubic 123456 (the proof presently). Taking, then, $7'$ as the intersection of the skew cubic by the plane 712, we have in this plane the points $7', 1, 2$, and the intersections of the plane by the lines $34, 35, \ldots 56$, nine points through which there pass an infinity of plane cubics; that is, the plane cubic determined by the points $7, 1, 2$ and the six intersections will pass through the point $7'$; viz., it meets the skew cubic 123456.

45. For the subsidiary theorem, taking X, Y, Z, W as current coordinates, viz., $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ as the equations of the planes $456, 563, 634, 345$ respectively, (x_1, y_1, z_1, w_1) and (x_2, y_2, z_2, w_2) as the coordinates of the points 1 and 2 respectively, and (x, y, z, w) for those of $7'$; the equation of the cubic surface passing through $7', 1, 2$, and having the nodes $3, 4, 5, 6$, is

$$\begin{vmatrix} \frac{1}{X}, & \frac{1}{Y}, & \frac{1}{Z}, & \frac{1}{W} \\ \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{1}{x_1}, & \frac{1}{y_1}, & \frac{1}{z_1}, & \frac{1}{w_1} \\ \frac{1}{x_2}, & \frac{1}{y_2}, & \frac{1}{z_2}, & \frac{1}{w_2} \end{vmatrix} = 0;$$

and this ceases to be a determinate function if only

$$\begin{vmatrix} \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{1}{x_1}, & \frac{1}{y_1}, & \frac{1}{z_1}, & \frac{1}{w_1} \\ \frac{1}{x_2}, & \frac{1}{y_2}, & \frac{1}{z_2}, & \frac{1}{w_2} \end{vmatrix} = 0;$$

viz., considering (x_1, y_1, z_1, w_1), (x_2, y_2, z_2, w_2) as given, this is a twofold relation between the coordinates (x, y, z, w) of the point $7'$. The relation may be represented by the four equations $(yzw) = 0$, $(zwx) = 0$, $(wxy) = 0$, $(xyz) = 0$, if for shortness

$$(yzw) = \begin{vmatrix} yz, & zw, & wy \\ y_1z_1, & z_1w_1, & w_1y_1 \\ y_2z_2, & z_2w_2, & w_2y_2 \end{vmatrix}$$

and the like as to the other symbols. The four equations represent quadric surfaces, each two intersecting in a line [e.g., $(yzw) = 0$, $(zwx) = 0$ in the line $z = 0$, $w = 0$], and the four surfaces besides intersecting in a skew cubic, which is the required locus of the point $7'$, and which, as is seen at once, passes through the points $1, 2, 3, 4, 5, 6$.

46. By what precedes, we have on the dianodal surface through the point 1 the lines $12, 13, 14, 15, 16, 17$, and the skew cubics 123456, &c. The six lines are not on the same quadric cone, and it thus appears that the point 1 must be a cubic-node (point where, instead of the tangent plane, we have a cubic cone) on the surface. It is to be remarked that the lines $12, 13, 14, 15, 16$, and the tangent at 1 to the skew cubic 123456, lie in a quadric cone; viz., this tangent is given as the sixth intersection of the cubic cone with the quadric cone through the lines $12, 13, 14, 15, 16$.

47. I revert to the equation of the dianodal surface as given in the form $J = J(P, Q, R, M\Omega) = 0$, where $M = 0$ is the plane through the points $1, 2, 3$, and $\Omega = 0$ the cubic surface through these points, and having the points $4, 5, 6, 7$, as nodes. We can find the orders of the several functions P, Q, R, M, Ω in the coordinates (x_1, y_1, z_1, w_1), &c., of the several points; viz., writing for shortness x_1^2 to denote the order 2 in regard to (x_1, y_1, z_1, w_1), and so in other cases, we have

$$P = Q = R = x^2(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^2,$$

$$M = x(x_1, x_2, x_3),$$

$$\Omega = x^3(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^3,$$

(where, of course, the x's show in like manner the orders in regard to the current coordinates (x, y, z, w); the proof in regard to Ω is easily supplied.) The order of J is equal that of $PQRM\Omega$, less 4 as regards the current coordinates, by reason of the differentiations; that is, we have $J = x^6(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$; and we thus see that the equation of the dianodal surface as above obtained is encumbered with a constant factor of the form $(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$. In fact, the relation between the 7 points and the current point (x, y, z, w), or say the point 8, as expressing that the 8 points are the nodes of a dianome, should be a symmetrical one in regard to the coordinates of the several points; and being of the order 6 in regard to the coordinates (x, y, z, w), it should be of the same order in regard to the other coordinates; that is, the true form would be $J = (xx_1x_2x_3x_4x_5x_6x_7)^6 = 0$.

48. It is possible that taking the 4 points, say $1, 2, 3, 4$, to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, and the 3 points, say $5, 6, 7$, to be $(1, 1, 1, 1)$, $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, the extraneous factor might exhibit itself, and that the [paper 445 continues on book p. 151].

[443] NOTE ON THE SOLUTION OF THE QUARTIC EQUATION $aU + 6\beta H = 0$ (CONCLUDED).

so that identically $X^2 + Y^2 + Z^2 = 0$, the expression $\alpha X + \beta Y + \gamma Z$ will be a square if only $\alpha^2 + \beta^2 + \gamma^2 = 0$. (To see this observe that in virtue of the equation $X^2 + Y^2 + Z^2 = 0$, we have $X + iY$, $X - iY$ each of them a square, and thence

$$\alpha X + \beta Y + \gamma Z = \tfrac{1}{2}(\alpha + i\beta)(X - iY) + \tfrac{1}{2}(\alpha - i\beta)(X + iY) - \gamma i\sqrt{X^2 + Y^2}$$

is a square if the condition in question be satisfied.)

Hence in particular writing

$$\sqrt{\omega_2 - \omega_3}\,\sqrt{aJ + 6\beta\omega_1 J},\ \ldots,\ \sqrt{\omega_1 - \omega_2}\,\sqrt{aJ + 6\beta\omega_3 J},$$

for α, β, γ, we have

$$(\omega_2 - \omega_3)\sqrt{aJ + 6\beta\omega_1 J}\,\sqrt{H} + (\omega_3 - \omega_1)\sqrt{aJ + 6\beta\omega_2 J}\,\sqrt{H} + (\omega_1 - \omega_2)\sqrt{aJ + 6\beta\omega_3 J}\,\sqrt{H}$$

a perfect square, and since the product of the four different values is a multiple of $(aU + 6\beta H)^2$ (this is most readily seen by observing that for $aU + 6\beta H = 0$, the irrational expression omitting a factor is

$$(\omega_2 - \omega_3)(aJ + 6\beta\omega_1 J) + (\omega_3 - \omega_1)(aJ + 6\beta\omega_2 J) + (\omega_1 - \omega_2)(aJ + 6\beta\omega_3 J),$$

which vanishes identically) it follows that the expression in question is the square of a linear factor of $aU + 6\beta H$.

It thus appears that the radicals (other than those arising from the solution of $U = 0$) contained in the solution of the equation $aU + 6\beta H = 0$ are the three roots

$$\sqrt{aJ + 6\beta\omega_1 J},\quad \sqrt{aJ + 6\beta\omega_2 J},\quad \sqrt{aJ + 6\beta\omega_3 J}.$$

Cambridge, September 2, 1868.

[444] ON THE CENTRO-SURFACE OF AN ELLIPSOID.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 16–18.]

The President [Prof. Cayley] gave an account of his investigations on the centro-surface of an ellipsoid (locus of the centres of curvature of the ellipsoid). The surface has been studied by Dr Salmon, and also by Prof. Clebsch, but in particular the theory of the nodal curve on the surface admits of further development. The position of a point on the ellipsoid is determined by means of the parameters, or elliptic coordinates, h, k; viz., if as usual a, b, c are the semi-axes, and if X, Y, Z are the coordinates of the point in question, then

$$\frac{X^2}{a^2 + h} + \frac{Y^2}{b^2 + h} + \frac{Z^2}{c^2 + h} = 1,$$

$$\frac{X^2}{a^2 + k} + \frac{Y^2}{b^2 + k} + \frac{Z^2}{c^2 + k} = 1;$$

and hence

$$-\beta\gamma X^2 = a^2(a^2 + h)(a^2 + k),$$

$$-\gamma\alpha Y^2 = b^2(b^2+h)(b^2+k),$$
$$-\alpha\beta Z^2 = c^2(c^2+h)(c^2+k),$$

if for shortness

$$\alpha = b^2-c^2, \quad \beta = c^2-a^2, \quad \gamma = a^2-b^2 \qquad (\alpha+\beta+\gamma=0).$$

This being so, the coordinates of the point of intersection of the normal at (X, Y, Z) by the normal at the consecutive point of the curve of curvature

$$\frac{X^2}{a^2+k}+\frac{Y^2}{b^2+k}+\frac{Z^2}{c^2+k}=1$$

are given by the formulae

$$-\beta\gamma a^2x^2 = (a^2+h)^2(a^2+k),$$
$$-\gamma\alpha b^2y^2 = (b^2+h)^2(b^2+k),$$
$$-\alpha\beta c^2z^2 = (c^2+h)^2(c^2+k);$$

viz., these equations, considering therein (h, k) as arbitrary parameters, determine the coordinates (x, y, z) of a point on the centre-surface. The principal sections (as is known) consist each of them of an ellipse counting three times, and of an evolute of an ellipse; the evolute and ellipse have four contacts (two-fold intersections) and four simple intersections, but the contacts and intersections respectively are in the different sections real and imaginary; and if (as we may without loss of generality assume) $a^2+c^2>2b^2$, then the form of the principal sections is as shown in the figure (which represents only an octant of the surface); viz., there is a real contact at P in the plane of xz, and a real intersection at Q in the plane of xy. The surface has thus an exterior and an interior sheet, but (instead of meeting in a conical point, as in the wave surface) these intersect in a nodal curve QP. The curve has a cusp at Q, and a node at P; viz., the curve extends beyond P, but from that point is acnodal, or without any real sheet of the surface passing through it. For the nodal curve there must be two values (h, k), (h_1, k_1), giving the same values of (x, y, z); viz., there must exist the relations

$$(a^2+h)^2(a^2+k) = (a^2+h_1)^2(a^2+k_1),$$
$$(b^2+h)^2(b^2+k) = (b^2+h_1)^2(b^2+k_1),$$
$$(c^2+h)^2(c^2+k) = (c^2+h_1)^2(c^2+k_1);$$

from which equations eliminating k and k_1 we should have between h, k a relation which, combined with the expressions of x, y, z in terms of (h, k), determines the nodal curve. But the better course is to eliminate k, k_1, thus obtaining a relation between h and h_1, in virtue whereof h_1 may be regarded as a known function of h; k and k_1 can then be readily expressed in terms of h, h_1; that is, we have k as a function of h, h_1, or in effect as a function of h. The relation between h, h_1 (after a reduction of some complexity) assumes ultimately a form which is very simple and remarkable; viz., writing

$$P = a^2+b^2+c^2, \quad Q = b^2c^2+c^2a^2+a^2b^2, \quad R = a^2b^2c^2,$$

the relation is

$$\begin{aligned}&(QR+3Qh^2+Ph^4)\\ &+h_1(3Qh+4Ph^2+3h^3)\\ &+h_1^2(P+3h^2)=0;\end{aligned}$$

this is a $(2,2)$ correspondence between the two parameters h, h_1; the united values $h_1 = h$, are given by the equation $6(R + Qh + Ph^2 + h^3) = 0$, that is

$$(a^2 + h)(b^2 + h)(c^2 + h) = 0;$$

viz., the two points on the ellipsoid which have their common centre of curvature on the nodal curve are only situate on the same curve of curvature when this curve is a principal section of the ellipsoid.

[Since the date of the foregoing communication, Prof. Cayley has found that the squared coordinates x^2, y^2, z^2 of a point on the nodal curve can be expressed as rational functions of a single variable parameter σ.]

[445] A MEMOIR ON QUARTIC SURFACES.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 19–69. Read February 10, 1870.]

The present Memoir is intended as a commencement of the theory of the quartic surfaces which have nodes (conical points). A quartic surface may be without nodes, or it may have any number of nodes up to 16. I show that this is so, and I consider how many of the nodes may be given points. Although it would at first sight appear that the number is 8, it is in fact 7; viz., we can, with 7 given points as nodes (but not in a proper sense with 8 or more given points), find a quartic surface; such surface contains in its equation 6 constants, which may be such that the surface has an additional node or nodes. Suppose that the surface has an 8th node: there are two distinct cases; viz., (1) the 8 nodes are the points of intersection of 3 quadric surfaces, or say they are an octad, and the surface is said to be octadic; (2) the 8th node is any point whatever on a certain sextic surface determined by means of the 7 given nodes, and called the dianodal surface of these 7 points; the quartic surface is said to be a dianome. The two cases are in general exclusive of each other; viz., the 7 given points being any points whatever, the dianodal surface does not pass through the 8th point of the octad; and thus the quartic surface with the 8 nodes is either octadic or else a dianome. Assuming it to be a dianome, the constants may be further determined so that there shall be a 9th node; it is necessary to examine whether this forms with 7 of the 8 nodes an octad. Supposing that it does not (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 9th node is then any point whatever on a certain curve of the order 18, determined by means of the 8 nodes, and called the dianodal curve of these 8 points. And, finally, the constants may be further determined so that there shall be a 10th node; supposing, as before, that this does not form an octad with any 7 of the 9 nodes (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 10th node is then any one of a system of 22 [should be 13] points determined by means of the 9 nodes, and called the dianodal system of these 9 points. But the quartic surface is now completely determined; viz., starting with any 7 given points as nodes, we have a dianome with 8 nodes, 9 nodes, or 10 nodes, say, an octodianome, enneadianome, or decadianome, but not with any greater number of nodes; these can only present themselves when particular conditions are satisfied in regard to the 7 given nodes, and to the 8th and 9th node; and the consideration of the quartic surfaces with more than 10 nodes would thus form a separate branch of the subject.

The case of the decadianome (or quartic surface with 10 nodes formed as above with 7 given points as nodes) is peculiarly interesting. I identify this with the surface which I call a symmetroid; viz., the surface represented by an equation $\Delta = 0$, where Δ is a symmetrical determinant of the 4th order the several terms whereof are linear functions of the coordinates (x, y, z, w); this surface is related to the Jacobian surface of 4 quadric surfaces (itself a very remarkable surface), and this theory of the symmetroid and the Jacobian, and of questions connected therewith, forms a large portion of the present Memoir.

The theory of the Jacobian is connected also with the researches in regard to nodal quartic surfaces in general; and, for greater clearness, it has seemed to me proper to commence the Memoir with certain definitions, &c., in regard to this theory. It will be seen in what manner I extend the notion of the Jacobian.

I remark that the present researches on Quartic Surfaces were suggested to me by Professor Kummer's most interesting Memoir "Ueber die algebraischen Strahlensysteme u.s.w.," *Berl. Abh.* 1866, in which, without entering upon the general theory, he is led to consider the quartic surfaces, or certain quartic surfaces, with $16, 15, 14, 13, 12$, or 11 nodes; the last of these, or surface with 11 nodes, being in fact a particular case of the symmetroid.

Considerations in regard to the Jacobian of four, or more or less than four, Surfaces.

1. In the case of any four surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, the differential coefficients of P, Q, R, S in regard to the coordinates (x, y, z, w) may be arranged as a square matrix in either of the ways

$$\begin{array}{c|cccc} & P, & Q, & R, & S \\ \hline \partial_x & & & & \\ \partial_y & & & & \\ \partial_z & & & & \\ \partial_w & & & & \end{array} \qquad \text{or} \qquad \begin{array}{c|cccc} & \partial_x, & \partial_y, & \partial_z, & \partial_w \\ \hline P & & & & \\ Q & & & & \\ R & & & & \\ S & & & & \end{array}$$

according as we make P, Q, R, S the columns, or as we make them the lines, of the matrix; and we then have one and the same determinant, the Jacobian $J(P, Q, R, S) = 0$, regarding as zero the term in question.

2. In the case of more than four surfaces, adopting the arrangement

$$\begin{array}{c|cccccc} & P, & Q, & R, & S, & T, & \ldots \\ \hline \partial_x & & & & & & \\ \partial_y & & & & & & \\ \partial_z & & & & & & \\ \partial_w & & & & & & \end{array}$$

and considering the several determinants which can be formed with any four columns of the matrix, these equated to zero establish a more than one-fold relation between the coordinates, viz., in the case of five surfaces, we have a twofold relation, the intersection of the 5 surfaces in lines, instead of in points; and so in other cases. Or, what is the same thing, the Jacobians $J(P, Q, R, S)$, etc., equated to zero, are not independent equations.

3. In the case of fewer than four surfaces, adopting the arrangement

$$\begin{array}{c|ccc} & P, & Q, & R \\ \hline \partial_x & & & \\ \partial_y & & & \\ \partial_z & & & \\ \partial_w & & & \end{array}$$

and considering the several determinants which can be formed with any 3 of 4 columns of the matrix, we establish a more than ordinary relation, the simultaneous vanishing of J_1, J_2, J_3, J_4 formed by omitting in succession each of the rows; this gives a curve of intersection of higher order than the proper intersection. So in other cases. It will be seen in what manner I extend the notion of the Jacobian.

4. We say that a surface $P = 0$ contains an arbitrary point, or that the equation $P = 0$ is satisfied by the coordinates of an arbitrary point, when there is one or more relations between the coefficients of P, viz., P is regarded as a function of (x, y, z, w) and of a series of coefficients which enter linearly therein. To say that the surface passes through 1 given point is to assert one such relation; to say that it passes through 2 given points is to assert two such relations; and so on.

5. Suppose that the orders of the surfaces $P = 0$, $Q = 0$, $\ldots$ are $a + 1$, $b + 1$, $\ldots$ (so that the orders of the differential coefficients are a, b, $\ldots$); then the Jacobian $J(P, Q, R, S)$ is of the order $a + b + c + d$, and the corresponding surface has $(a + b + c + d)$-tic contact with the surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$ at the points of common intersection of the four surfaces.

6. It is also shown (Salmon, Ed. 2, p. 495) that the surface obtained by equating the discriminant of a quadric surface to zero is the Jacobian of the four polar quadric surfaces with respect to four arbitrary points, viz., the Jacobian determinant $J(P, Q, R, S)$, regarded as a function of (x, y, z, w), expresses the condition that the four planes of polarity should pass through one and the same point. So in other analogous statements about polars and Jacobians.

7. The subsidiary theorem of the contact of the curve and surface above mentioned is, that if for shortness the differential coefficients of P, Q, R, S in regard to (x, y, z, w) be denoted by P_x, $\ldots$, Q_x, $\ldots$, etc., the Jacobian $J(P, Q, R, S) = 0$ may be written, for x, y, z, w replaced by the coordinates of a point on the curve of intersection,

$$\begin{vmatrix} P_x, & Q_x, & R_x, & S_x \\ P_y, & Q_y, & R_y, & S_y \\ P_z, & Q_z, & R_z, & S_z \\ P_w, & Q_w, & R_w, & S_w \end{vmatrix} = 0,$$

and this determinant vanishes identically by virtue of the relation between the partial derivatives at the points common to the surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$. The contact is therefore of the order $a + b + c + d$ at the points of common intersection.

8. The subsidiary theorem of the contact of the curve and surface just mentioned gives rise to the Jacobian as the locus of points the polar planes of which, with respect to P, Q, R, S, pass through one and the same point. We shall in the sequel make considerable use of this theorem in various forms.

9. The Jacobian is then easily reduced to its proper form by a process of elimination, viz., introducing as many auxiliary equations as may be required to render it determinate.

10. We have, in particular, the theorem that the Jacobian of four quadric surfaces is a quartic surface, since each of the differential coefficients is of order 1 and there are four of them.

11. The Jacobian of four quadric surfaces is in fact a quartic surface, and the corresponding surface has 4-tic contact with the four quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$ at the points of common intersection.

12. In the case where there is a plane 1, the sextic cone breaks up into this plane and into a quintic cone, etc.

13. In the cases $k = 1, 2, 3, 4, 5$, and $k = 15, 16$, there is only one figure, and the number of arrangements may, on referring to the table, be at once seen.

Circumscribed Sextic Cone.

Order of Surface	
2	6
3	6, 1
4	6, 2, 1
5	6, 3, 1, 1
6	6, 4, 2, 1, 1
7	6, 5, 2, 1, 1, 1
8	6, 5, 3, 2, 1, 1, 1
9	6, 5, 4, 2, 1, 1, 1, 1
10	6, 5, 4, 3, 2, 1, 1, 1, 1
11	6, 5, 4, 3, 2, 1, 1, 1, 1, 1
12	... 6, 4, 2, 1, 1, 1, 1, 1, 1, 1, 1
13	... 5, 3, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
14	... 3, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
15	... 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
16	... 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

and moreover, in the cases where there are two or more lines of the sextic cone, this the k sextic cones may be of the different forms in various combinations. The total number of cases, proper *facies* (*i.e.* this very gens), but not only a comparatively small number of them actually exist.

14. In the case where there is a plane 1, the sextic cone breaks up into this plane and into a (quintic or improper quintic cone intersecting this plane in 5 lines, etc.). There is, in fact, in the plane 1 a cubic, the plane is, in fact, a singular tangent plane meeting the surface in a conic twice repeated; and the 5 residual lines on the quintic cone touching the surface in a single point of the same kind (the line through the plane and the surface meeting this plane in three points each), and the like for the conic.

15. But a different form of expression is sometimes convenient: the conditions to be satisfied are frequently such that, being satisfied by the surfaces $P = 0$, $Q = 0$, they will be satisfied by the surface $\lambda P + \mu Q = 0$ where, for a given value of $\lambda : \mu$, there is a 1-tuple infinity of surfaces; or, in the same way, satisfied by the surfaces $P = 0$, $Q = 0$, $R = 0$ they will be satisfied by the surface $\lambda P + \mu Q + \nu R = 0$, where there is a 2-tuple infinity of surfaces; and so on.

16. Thus, for the quadric not subjected to any conditions, there are 9 constants (this is, 1 to fix the surface itself, and 8 in the equation thereof; the surface contains 9 constants); and we have therefore one and the same thing in the various forms in which the quartic surface in 9 nodes presents itself.

17. In the questions in which such quadric surfaces present themselves, we have to fix our ideas in regard to the number of constants. If, in such a question, there occur quadric surfaces $P = 0$, $Q = 0$, etc., we are to consider that we may through this or that function of P, Q, etc., have, for any given relations between the surfaces, so many constants; and the calculations are to be conducted accordingly.

18. A quartic surface contains 34 constants. We have, in fact, in the equation of a quartic surface

$$(a, b, c, d, e, f, g, h, i, j, k, l, m, n, p, q, r, s, t, u, v, w, x, y, z, \ldots)$$

35 coefficients, and so on. In a way which it is needless to recapitulate here, we ultimately arrive at the result that the quartic surface contains 34 constants.

19. In particular if we are concerned with a quartic surface having a node, then we may take the node as the point $(0, 0, 0, 1)$, viz., as the point of intersection of three of the four reference planes. The quartic surface then has the term in w^4 absent, etc.

20. The case of 4 given nodes is just as easy. In particular if we are concerned with a quartic surface having 4 nodes, then we may take these as the four points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, viz., as the four vertices of the tetrahedron of reference.

21. In proof, observe that, taking the 4 given nodes to be the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$ as before, the equation of a quartic surface having these four nodes contains a definite number of constants.

22. In the case of 5 given nodes, the number of constants should be reduced still further. We have through the 5 given points $\binom{5}{2} = 10$ lines, the common nodes; the remaining number of constants is $34 - 5 \cdot \text{constraints} = \text{a definite count}$.

23. In verification, take the first 4 nodes to be as above, and let the 5th node be the point $(\alpha, \beta, \gamma, \delta)$.

24. In the case of 6 given nodes, the quartic surface should contain 10 constants. We have through the 6 given points $\binom{6}{2} = 15$ lines, and the calculation proceeds in the manner just indicated.

25. The 6 given nodes being any points whatever, their coordinates may be taken to be $(\alpha_1, \beta_1, \gamma_1, \delta_1), \ldots, (\alpha_6, \beta_6, \gamma_6, \delta_6)$; we may by a linear transformation take the first four of them to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$; and then the remaining two as $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$. The conditions that the quartic surface should have nodes at these 6 points reduce the number of constants to 10.

As to the Surface obtained by equating to zero a Symmetrical Determinant.

(continued). In the above (Salmon, Ed. 2, p. 493) that the surface obtained by equating to zero any symmetrical determinant has a determinate number of nodes; viz., if the order of the determinant is n, the number is $\binom{n+1}{3}$. In particular, the symmetrical determinant of the 4th order (Jacobian) has $\binom{5}{3} = 10$ nodes, as in the text of this Memoir.

26. The Jacobian is then easily found to be the symmetrical determinant

$$\Big| \partial_i \partial_j K \Big|_{i,j=1,\ldots,4} = 0,$$

that is, the corresponding surface is a quartic surface having a determinate number of nodes, viz., 10.

27. It may be shown *a posteriori* that J is not a quadric function of P, Q, R, S; and consequently the equation $J = 0$ represents a quartic surface, the Jacobian of the 4 quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$.

28. The equation $J = 0$ is the locus of the vertices of the quadric cones which can be drawn through the curve of intersection of the surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$; that is, the locus of the vertices of the quadric cones which can be drawn through the 8 points of intersection; in particular, this Jacobian surface passes through the 6 lines 12, 34, 56 (viz., line of intersection of the planes through 1, 2, 3, and through 4, 5, 6); through 13, 24, 56; etc.

29. The surface in question (the Jacobian of the four quadric surfaces, or of the six points) is then a quartic surface having the six points as nodes; the order of the surface is 4 and the number of nodes is 10; we have, in fact, in the equation of the Jacobian surface $34 - 10 \cdot \text{some count} =$ the right number of constants.

30. In connexion with what precedes, we may here consider a curve which presents itself in connexion with the Jacobian surface above mentioned. We have, in fact, through any 7 given points $\binom{7}{2} = 21$ lines, through any 3 of which $4 \cdot \binom{7}{3}/3$ planes are determined; the Jacobian surface contains these lines as part of itself, together with a curve of certain order containing the points.

31. Suppose, however, that one of the seven points, instead of being a node of the dianodal surface, is itself a point on the dianodal surface; the question then arises whether the dianodal surface still passes through this point and whether the remaining structure is preserved.

32. In the case where two of the points are each of them the vertex of a cone passing through the other 5 points, the dianodal surface degenerates further: one or more of the conic relations comes to a duplication.

33. *Particular cases of the dianodal surface.* There are many particular cases of the dianodal surface, but it is unnecessary to enumerate them; in fact, throughout the general theory, the form of the dianodal surface remains the same.

34. It follows that, if $\Delta = 0$ be another quartic surface having the 7 given nodes, then the equation of the dianodal surface (in regard to the symmetrical determinant) must contain the function of Δ, etc.

35. A particular quartic surface having (in an improper sense) the 7 given nodes, viz., in many ways, but in special cases as follows: take any plane $M = 0$ through three of the nodes, say through 1, 2, 3; and any cubic surface $\Omega = 0$ having the other 4 nodes, viz., 4, 5, 6, 7; then the product $M \cdot \Omega$ vanishes at each of the 7 given points.

36. We can with the 7 given points form 35 such combinations $M \cdot \Omega$; these are of course equivalent in virtue of the dianodal relations among them.

37. It has already been shown that a quartic surface cannot in a proper sense have 8 given nodes. In regard to the quartic surfaces with 8 nodes, we start from the dianodal surface as described, and add an 8th node.

38. But if θ be not $= 0$, there may still be an 8th node; viz., this must then lie on the dianodal surface, etc.

39. I say that the 8th node may be any point whatever on the dianodal surface, that is, the dianodal surface is the locus of the 8th node.

40. [*Article 40 develops further the parametric form of the dianodal surface and its relation to the* 7 *given nodes.*]

41. [*Article 41 contains supplementary remarks bearing on the same construction.*]

42. Consider the seven points $1, 2, 3, 4, 5, 6, 7$. As already mentioned, through any 3 of them, say $1, 2, 3$, we have a plane $M = 0$; and through the same 7 points, with the 4 points $4, 5, 6, 7$ as nodes, a cubic surface $\Omega = 0$. We may take the planes through 123, 145, and 456 respectively each as M, and the cubic surfaces having Ω as nodes the remaining 4 points; and similarly, taking other combinations, we have $\Delta = M\Omega = 0$, a quartic surface with the seven points as nodes. And using this form of Δ it may be shown that the dianodal $J(P, Q, R, \Delta) = 0$ passes through the 21 lines $12, 13, \dots 67$, and through 35 plane cubics such as $M = 0$, $\Omega = 0$; viz., this is a cubic in the plane 123 passing through the points $1, 2, 3$, and through the intersection of the plane with each of the six lines $45, 46, \dots 67$ (nine points determining the cubic); the complete intersection by the plane 123 being therefore composed of this cubic and of the three lines $12, 13, 23$. For the passage through the cubic, we have only to observe that

$$J(P, Q, R, M\Omega) = J(P, Q, R, \Omega)\, M + J(P, Q, R, M)\, \Omega = 0$$

is satisfied by $M = 0$, $\Omega = 0$; and for the passage through the lines, taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the planes $567, 674, 745$, and 456 respectively, each of the functions P, Q, R is of the form $ayz + bzx + cxy + fxw + gyw + hzw$, and the function Ω is of the form $Ayzw + Bzwx + Cwxy + Dxyz$. Hence, writing in the derived functions $z = 0$, $w = 0$, the first and second lines of the determinant $J(P, Q, R, \Omega)$ will be of the form

$$\begin{pmatrix} c\,y, & c'\,y, & c''\,y, & * \\ c\,x, & c'\,x, & c''\,x, & * \end{pmatrix},$$

or the determinant vanishes for $z = 0$, $w = 0$; that is for any point of the line 45 we have $\Omega = 0$ and also $J(P, Q, R, \Omega) = 0$; consequently $J(P, Q, R, M\Omega) = 0$, and the like for the other lines. The theorem is thus proved.

43. I say that the dianodal surface passes through each of the 7 skew cubics, such as 123456. To prove this, it is only necessary to show that the skew cubic 123456 lies on the dianodal surface. For this purpose it will be enough to show that the skew cubic meets the plane 712 in a point of the surface; for then it will, in like manner, meet each of the 15 planes $712, 713, \dots 756$ in a point of the surface; that is, we shall have 15 intersections of the curve and surface, and there are, besides, the intersections $1, 2, 3, 4, 5, 6$, in all 21 intersections; that is, the skew cubic must lie on the surface.

44. The plane 712 meets the surface in three lines and in a plane cubic determined by the points $7, 1, 2$ and the six intersections of the plane with the lines $34, 35, \ldots 56$. We have therefore to show that this plane cubic meets the skew cubic 123456. Consider for a moment the points $1, 2, 3, 4, 5, 6$ and another point $7'$. As seen above, we have in general, through the points $1, 2, 7'$ and with the points $3, 4, 5, 6$ as nodes, a determinate cubic surface, which surface passes through the lines $34, 35, \ldots 56$. But the cubic surface becomes indeterminate if the points $1, 2, 7', 3, 4, 5, 6$ are on the same skew cubic; that is, if $7'$ is any point whatever on the skew cubic 123456 (the proof presently). Taking, then, $7'$ as the intersection of the skew cubic by the plane 712, we have in this plane the points $7', 1, 2$, and the intersections of the plane by the lines $34, 35, \ldots 56$, nine points through which there pass an infinity of plane cubics; that is, the plane cubic determined by the points $7, 1, 2$ and the six intersections will pass through the point $7'$; viz., it meets the skew cubic 123456.

45. For the subsidiary theorem, taking X, Y, Z, W as current coordinates, viz., $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ as the equations of the planes $456, 563, 634, 345$ respectively, (x_1, y_1, z_1, w_1) and (x_2, y_2, z_2, w_2) as the coordinates of the points 1 and 2 respectively, and (x, y, z, w) for those of $7'$; the equation of the cubic surface passing through $7', 1, 2$, and having the nodes $3, 4, 5, 6$, is

$$\begin{vmatrix} \frac{1}{X}, & \frac{1}{Y}, & \frac{1}{Z}, & \frac{1}{W} \\ \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{1}{x_1}, & \frac{1}{y_1}, & \frac{1}{z_1}, & \frac{1}{w_1} \\ \frac{1}{x_2}, & \frac{1}{y_2}, & \frac{1}{z_2}, & \frac{1}{w_2} \end{vmatrix} = 0;$$

and this ceases to be a determinate function if only

$$\begin{vmatrix} \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{1}{x_1}, & \frac{1}{y_1}, & \frac{1}{z_1}, & \frac{1}{w_1} \\ \frac{1}{x_2}, & \frac{1}{y_2}, & \frac{1}{z_2}, & \frac{1}{w_2} \end{vmatrix} = 0;$$

viz., considering (x_1, y_1, z_1, w_1), (x_2, y_2, z_2, w_2) as given, this is a twofold relation between the coordinates (x, y, z, w) of the point $7'$. The relation may be represented by the four equations $(yzw) = 0$, $(zwx) = 0$, $(wxy) = 0$, $(xyz) = 0$, if for shortness

$$(yzw) = \begin{vmatrix} yz, & zw, & wy \\ y_1z_1, & z_1w_1, & w_1y_1 \\ y_2z_2, & z_2w_2, & w_2y_2 \end{vmatrix}$$

and the like as to the other symbols. The four equations represent quadric surfaces, each two intersecting in a line [e.g., $(yzw) = 0$, $(zwx) = 0$ in the line $z = 0$, $w = 0$], and the four surfaces besides intersecting in a skew cubic, which is the required locus of the point $7'$, and which, as is seen at once, passes through the points $1, 2, 3, 4, 5, 6$.

46. By what precedes, we have on the dianodal surface through the point 1 the lines $12, 13, 14, 15, 16, 17$, and the skew cubics 123456, &c. The six lines are not on the same quadric cone, and it thus appears that the point 1 must be a cubic-node (point where, instead of the tangent plane, we have a cubic cone) on the surface. It is to be remarked that the lines $12, 13, 14, 15, 16$, and the tangent at 1 to the skew cubic 123456, lie in a quadric cone; viz., this tangent is given as the sixth intersection of the cubic cone with the quadric cone through the lines $12, 13, 14, 15, 16$.

47. I revert to the equation of the dianodal surface as given in the form $J = J(P, Q, R, M\Omega) = 0$, where $M = 0$ is the plane through the points $1, 2, 3$, and $\Omega = 0$ the cubic surface through these points, and having the points $4, 5, 6, 7$, as nodes. We can find the orders of the several functions P, Q, R, M, Ω in the coordinates (x_1, y_1, z_1, w_1), &c., of the several points; viz., writing for shortness x_1^2 to denote the order 2 in regard to (x_1, y_1, z_1, w_1), and so in other cases, we have

$$P = Q = R = x^2(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^2,$$

$$M = x(x_1, x_2, x_3),$$

$$\Omega = x^3(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^3,$$

(where, of course, the x's show in like manner the orders in regard to the current coordinates (x, y, z, w); the proof in regard to Ω is easily supplied.) The order of J is equal that of $PQRM\Omega$, less 4 as regards the current coordinates, by reason of the differentiations; that is, we have $J = x^6(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$; and we thus see that the equation of the dianodal surface as above obtained is encumbered with a constant factor of the form $(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$. In fact, the relation between the 7 points and the current point (x, y, z, w), or say the point 8, as expressing that the 8 points are the nodes of a dianome, should be a symmetrical one in regard to the coordinates of the several points; and being of the order 6 in regard to the coordinates (x, y, z, w), it should be of the same order in regard to the other coordinates; that is, the true form would be $J = (x\, x_1\, x_2\, x_3\, x_4\, x_5\, x_6\, x_7)^6 = 0$.

48. It is possible that taking the 4 points, say $1, 2, 3, 4$, to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, and the 3 points, say $5, 6, 7$, to be $(1, 1, 1, 1)$, $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, the extraneous factor might exhibit itself, and that the calculation thereby be simplified. But I do not at present further pursue the inquiry.

Dianodal Curve of 8 *given Points, or say a Nodo or Section.*

49. The dianodal surface, qua surface having the last-mentioned 8 points for nodes, has on it various special curves, and the discussion of these is here in some sort necessary; viz., we have on it a curve, the section by an arbitrary plane, of the order 6; the curve of intersection of the dianodal surface with another quartic surface through the 7 given nodes and called the dianodal curve of these 8 points. There exists between $P, Q, R, S = M\Omega$ a syzygetic relation

$$\alpha P + \beta Q + \gamma R + \delta\, M\Omega = 0,$$

where in $\alpha, \beta, \gamma, \delta$ are linear functions of the current coordinates, α for instance being a function which vanishes at the points $1, 2, 3$, and which contains a number of arbitrary coefficients; whence the syzygetic relation is one satisfied at every one of the 7 given nodes.

50. This being so, the cubic curve through the last-mentioned six points has its equation of the form $J(P, Q, R, \alpha M\Omega) = 0$; and the dianodal curve of these 8 points is the intersection of the dianodal surface with this cubic curve.

51. Considering the orders in regard to $(a, b, c, d, f, g, h, l, m, n, P, Q, R, S, T, U, V)$, the dianodal curve through the points $1, 2, \ldots, 7$, and an 8th point lies on the dianodal surface, and is of the order 18. It contains in fact one set of intersections (the 21 lines $12, 13, \ldots, 67$ counted as one with multiplicity 2) and the curve proper; this last is the dianodal curve of the 8 points.

52. We have $5 \cdot 6 = 30$ as the order of intersection of the dianodal surface (sextic) and the auxiliary quartic surface; the auxiliary quartic passing through the 7 given points counts as $\binom{7}{2} = 21$, and the dianodal curve is of the order 18. There would be a reduction of 25 if the line through the 6 given points $1, 2, 3, 4, 5, 6$ were in involution with the 6 given quadric surfaces; whence the dianodal curve order would be $30 - 25 = 5$. But this is not the case.

53. In order to find it, taking as above 9 a given point on the Jacobian, we have a tenth point on the surface only if there be a relation between the coordinates of the points 1 through 9. The condition is in fact a determinate condition which fixes the order of the dianodal curve at 18.

54. If there is a 10th node, say 10, this is also a point on the Jacobian curve of the 8 points $1, 2, \ldots, 8$, and similarly a point on the dianodal curves through suitable octads of the 9 remaining nodes.

55. In the case of the surface with 9 nodes, it is clear that this is octadic in one way only: the node 9 cannot form an octad with any 7 of the remaining nodes. But in the case of the surface with 10 nodes, the question arises whether the nodes 9 and 10 may not be such as to form an octad with some six, say with the nodes $1, 2, 3, 4, 5, 6$ of the remaining 8 nodes; that is, whether we can have $1, 2, 3, 4, 5, 6, 7, 8$ forming an octad, and also $1, 2, 3, 4, 5, 6, 9, 10$ forming an octad. I will show that this is impossible if only the points $1, 2, 3, 4, 5, 6$ are given points, that is, points assumed at pleasure and not specially related to each other. For this purpose, assuming that the points form 2 octads as above, take through $1, 2, 3, 4, 5, 6, 7, 9$ the quadric surfaces $P = 0$, $Q = 0$, then each of these passes through 8, 10; take $R = 0$ any other quadric surface through $1, 2, 3, 4, 5, 6, 7, 8$, and $S = 0$ any other quadric surface through $1, 2, 3, 4, 5, 6, 9, 10$. Then $P = 0$, $Q = 0$, $R = 0$ intersect in the 1st octad, [*continues in book p. 154; the memoir then resumes at* § 60 *in the separate retype on book p. 155.*]

(End of chunk: book pages 129–153 of Vol. VII. The memoir on Quartic Surfaces continues with §§ 56–59 on book p. 154, and is then resumed at § 60 in `cayley_vol07_paper_445_RETYPE.tex`.)

[445] A MEMOIR ON QUARTIC SURFACES. *(CONTINUED)*

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 19–69. Read February 10, 1870.*]*

(Continuation from book p. 153*; articles* 56–112.*)*

56. We may through 3 draw the lines LM, QT to meet 14, 26 and 12, 46 respectively; and through 5 the lines RS, NP to meet 14, 26 and 12, 46 respectively. Observe that the points O in the figure are apparent intersections only; viz., NP does not meet QT, nor LM meet RS. In fact, if NP met QT it would be a line in the series of lines meeting 14, QT, 26; or 5 would be situate in a hyperboloid, determined by means of the points 1, 2, 4, 6, 3; viz., 5 would not be an arbitrary point: and so LM does not meet RS. Now the quadrics E, H meet in the lines 14, 26, LM, NP, and the quadrics A, C in the lines 12, 46, QT, RS. Suppose that we had identically $(*\$\, A, C, E, H)^2 = 0$; putting therein $E = 0$, $H = 0$, we should have $(*\$\, A, C)^2 = 0$, viz.,

$$(A + \lambda C)(A + \mu C) = 0;$$

or there would exist quadrics of the forms $A + \lambda C = 0$ containing the lines 14, 26, LM, NP. Now there is no quadric surface $A + \lambda C = 0$ containing the line NP; for $A + \lambda C = 0$ is a quadric containing the sides of the quadrilateral $QRST$; the generating lines of the one kind meet each of the lines RS, QT; those of the other kind neither. Hence NP, which meets RS but not QT, cannot be a generating line of either kind; and we have no identical relation $(A, C, E, H)^2 = 0$.

57. In the octadic surface with 9 nodes: starting with any 7 nodes of the octad, 9 is not the 8th point of the octad, and hence (by the theory of the dianome) it must lie in the dianodal surface of the 7 points; that is, the dianodal surface of the 7 points must pass through 9, viz., through any point whatever of the Jacobian curve of the 7 points, that is, of the octad; or (what is the same thing) the dianodal surface of the 7 points passes through the Jacobian curve of the octad. This is an obvious property of the dianodal surface, the surface $J(P, Q, R, V) = 0$ contains the Jacobian curve $J(P, Q, R) = 0$. But it further appears that, starting with any 6 points of the octad and with the point 9 (that is, any point whatever of the Jacobian curve), the dianodal surface of these 7 points must contain the remaining 2 points of the octad. And in the octadic surface with 10 nodes, starting with any 5 points of the octad and with the points 9 and 10 (that is, any two points on the Jacobian curve) the dianodal surface of these 7 points must contain the remaining three points of the octad. I have not attempted to verify these last properties of the dianodal surface.

Dianomes with 9 or 10 Nodes.

58. I now consider the dianomes with 9 and 10 nodes. Starting from the general form

$$(a, b, c \,\rangle\!\langle\, P, Q)^2 + \theta\Delta = 0,$$

where $\Delta = 0$ is a particular quartic surface having the 8 nodes, it at once appears that if there is a 9th node, say 9, the node must be a point on the Jacobian curve $J(P, Q, \Delta) = 0$; hence, taking it to be one such point, the number of constants in the resulting equation should be 1. Hence if $P = 0$ be the quadric surface through the 9 points, and $\Delta = 0$ a particular quartic surface having the 9 points as nodes, the general equation is $\alpha P^2 + \theta\Delta = 0$.

59. But we may consider the question somewhat differently. Starting with the 7 given points $1, 2, 3, 4, 5, 6, 7$ and with 8 a given point on the dianodal surface of the 7 points; it is clear that 9 must be on the dianodal surface 1234567 and also on the dianodal surface 1234568; the complete intersection is of the order 36, and we have to consider how this breaks up so as to contain a part of itself the dianodal curve of the order 18.

Dianodal Curve of 8 Points.

60. Consider first any 8 points whatever $1, 2, 3, 4, 5, 6, 7, 8$ where 8 is not on the dianodal surface 1234567, nor 7 on the dianodal surface 1234568. The two surfaces have in common the 15 lines $12, 13, \ldots, 56$ and the skew cubic 123456, they therefore besides intersect in a curve of the order 18. At the point 1 the tangent cubic cones of the two surfaces intersect in the lines 12, 13, 14, 15, 16 and the tangent to the skew cubic 123456, 6 lines lying in a quadric cone; they therefore besides intersect in 3 lines lying in a plane; that is, the point 1 is on the curve of the order 18 an actual triple point, the 3 tangents lying *in plano*; and the like of course in regard to each of the points 2, 3, 4, 5, 6. But as 7, 8 lie each of them on only one of the two surfaces, the curve of the order 18 does not pass through 7 or 8.

61. If, however, 8 lies on the dianodal surface 1234567, then each of the 8 points will lie on the dianodal surface of the other 7; and in particular 7 will lie on the dianodal surface 1234568. The surfaces intersect as before in a residual curve of the order 18; the only difference is that 7 and 8 are now points on each surface; viz., each of them is on one of the surfaces an ordinary point, and on the other a cubic node; the points 7 and 8 are thus each of them an actual triple point on the curve; and at each of them the 3 tangents are *in plano*. We thus see that the dianodal curve 12345678 is a curve of the order 18, such that each of the 8 points is a triple point on the curve, the tangents at each of them being *in plano*.

Ten Nodes.

62. Suppose there is a 10th node, say 10; starting from the equation $\alpha P^2 + \theta\Delta = 0$ ($P =$ the quadric surface through the 9 points, $\Delta =$ a particular quartic surface having the 9 points as nodes), it at once appears that the node must be one of the points $J(P, \Delta) = 0$; hence, taking it to be one of these points, we have 4 equations, which, in virtue of the node being one of the points in question, reduce themselves to a single equation determining the ratio $\alpha : \theta$; we have thus a completely determinate surface, say $\square = 0$ having the 10 points as nodes. The number of points $J(P, \Delta)$, writing in the formula No. 5, $a = 1$, $b = 3$, is obtained as $1 + 3 + 9 + 27 = 40$, but it is to be observed that the surface $P = 0$ passes through each of the 9 nodes of the surface $\Delta = 0$; these count twice among the points $J(P, \Delta) = 0$, and the number of residual points (or say the dianodal centres of the 9 points) is $40 - 18 = 22$; viz., this is the number of positions of the node 10. [The nine points count each three times, and the number of residual points, or positions of the node 10, is thus not $40 - 18 = 22$, but $40 - 27, = 13$.]

Dianodal Centres of 9 Points.

63. In further explanation, observe that 9 is any point on the dianodal curve 12345678; the node 10 must lie on this same curve, and also on the dianodal surface 1234569. Take $P = 0$ the quadric through all the 9 points, $Q = 0$ a quadric through all but the point 9, $R = 0$ through all but the point 8, $S = 0$ through all but the point 7. The dianodal curve 12345678 is $J(P, Q, V) = 0$, and the dianodal surface 1234569 is $J(P, R, S, V) = 0$; the total number of intersections is $6 \times 18 = 108$; these include the $4 \times 18 = 72$ points of intersection of the dianodal curve $J(P, Q, \Delta) = 0$ with the Jacobian surface $J(P, Q, R, S) = 0$, except the four points $J(P, Q) = 0$, which are the vertices of the 4 quadric cones through $1, 2, 3, 4, 5, 6, 7, 8$ (which 4 points are not situate on the curve $J(P, R, S) = 0$), and there are besides 40 points $\{108 = (72 - 4) + 40\}$ which are the before-mentioned points $J(P, \Delta) = 0$; viz., these are the 9 points each twice [three times], and the residual 22 [13] points which are the dianodal centres of the 9 points.

General result as to the Dianomes.

64. We have thus established the theory of the dianome quartic surfaces; viz., we have

- The octodianome, 8 nodes, 7 of them arbitrary, and the 8th an arbitrary point on the dianodal surface (order 6) of the 7 points.
- The enneadianome, 9 nodes, the 9th an arbitrary point on the dianodal curve (order 18) of the 8 points.
- The decadianome, 10 nodes, the 10th any one of the 22 [13] dianodal centres of the 9 points.

And as already mentioned, so long as the first 7 nodes are arbitrary, there cannot be more than 10 nodes in all.

The Symmetroid.

The Lineolinear Correspondence of Quartic Surfaces.

65. I consider four equations $S = 0$, $T = 0$, $U = 0$, $V = 0$, lineolinear in regard to the two sets of coordinates (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$; viz., each of these equations is of the form

$$(*\$\, x, y, z, w \,\rangle\!\langle\, \alpha, \beta, \gamma, \delta) = 0.$$

This implies that the point (x, y, z, w) lies on a certain quartic surface $\Theta = 0$, and the point $(\alpha, \beta, \gamma, \delta)$ on a certain quartic surface $\Delta = 0$, and that the two surfaces correspond point to point to each other. In fact, writing the four equations in the form

$$\begin{aligned} L\alpha + M\beta + N\gamma + P\delta &= 0, \\ L'\alpha + M'\beta + N'\gamma + P'\delta &= 0, \\ L''\alpha + M''\beta + N''\gamma + P''\delta &= 0, \\ L'''\alpha + M'''\beta + N'''\gamma + P'''\delta &= 0, \end{aligned}$$

where L, &c., are linear functions of (x, y, z, w), then eliminating $(\alpha, \beta, \gamma, \delta)$, we obtain the equation

$$\Theta = \begin{vmatrix} L & M & N & P \\ L' & M' & N' & P' \\ L'' & M'' & N'' & P'' \\ L''' & M''' & N''' & P''' \end{vmatrix} = 0;$$

and similarly, writing the four equations in the form

$$\begin{aligned} Ax + By + Cz + Dw &= 0, \\ A'x + B'y + C'z + D'w &= 0, \\ A''x + B''y + C''z + D''w &= 0, \\ A'''x + B'''y + C'''z + D'''w &= 0, \end{aligned}$$

where A, &c., are linear functions of $(\alpha, \beta, \gamma, \delta)$, then eliminating (x, y, z, w), we obtain the equation

$$\Delta = \begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix} = 0.$$

Moreover, Θ being $= 0$, the four linear equations in $(\alpha, \beta, \gamma, \delta)$ are equivalent to three equations, and give for instance $(\alpha, \beta, \gamma, \delta)$ proportional to the determinants formed with the matrix

$$\begin{vmatrix} L' & M' & N' & P' \\ L'' & M'' & N'' & P'' \\ L''' & M''' & N''' & P''' \end{vmatrix};$$

and similarly, Δ being $= 0$, the four linear equations in (x, y, z, w) are equivalent to three equations, and give for instance (x, y, z, w) proportional to the determinants formed with the matrix

$$\begin{vmatrix} A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix},$$

which establishes the point-to-point correspondence of the two surfaces.

66. It would at first sight appear that any quartic surface $(*\$\,\alpha, \beta, \gamma, \delta)^4 = 0$ whatever might have its equation expressed in the foregoing determinant form $\Delta = 0$. This equation seems, in fact, to contain homogeneously as many as 64 constants. But if we multiply the determinant line into line by a constant determinant

$$\begin{vmatrix} a & b & c & d \\ a' & b' & c' & d' \\ a'' & b'' & c'' & d'' \\ a''' & b''' & c''' & d''' \end{vmatrix}$$

and then column into column by another constant determinant, the coefficients, all but one of them, of these constant determinants may be used to specialize the form of the resulting equation, [say they are apoclastic constants]; this equation will really contain $64-(2\cdot16-1) = 33$ constants; and in order that the quartic surface $(*\$\,\alpha, \beta, \gamma, \delta)^4 = 0$ may have its equation expressible in the form $\Delta = 0$, a single relation must hold good among the coefficients; but this in passing[1].

67. Returning to the quartic surface

$$\Delta = \begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix} = 0,$$

we may connect this not only with the foregoing surface $\Theta = 0$, but in a similar manner with another quartic surface $\Phi = 0$; viz., taking the current coordinates $(\xi, \eta, \zeta, \omega)$, we may form the

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[1] Applying the same reasoning to a cubic determinant $\Delta = 0$, the number of constants is $36 - (2 \cdot 9 - 1) = 19$; so that a cubic surface is expressible in the form in question. And so for the quadric determinant $\Delta = 0$, the number of constants is $16 - (2 \cdot 4 - 1) = 9$; so that a quadric surface is expressible in the form in question, as is otherwise obvious.

lineolinear equations

$$\begin{aligned}
A\xi + A'\eta + A''\zeta + A'''\omega = 0,\\
B\xi + B'\eta + B''\zeta + B'''\omega = 0,\\
C\xi + C'\eta + C''\zeta + C'''\omega = 0,\\
D\xi + D'\eta + D''\zeta + D'''\omega = 0,
\end{aligned}$$

which, by the elimination of $(\xi, \eta, \zeta, \omega)$, give $\Delta = 0$, and by the elimination of $(\alpha, \beta, \gamma, \delta)$ a determinant quartic equation $\Phi = 0$ between the coordinates $(\xi, \eta, \zeta, \omega)$; and of course the two surfaces $\Delta = 0$, $\Phi = 0$ have a point-to-point correspondence such as exists between the surfaces $\Theta = 0$, $\Delta = 0$. The relation of the point $(\alpha, \beta, \gamma, \delta)$ on the surface $\Delta = 0$ to the point (x, y, z, w) on the surface $\Theta = 0$, and to the point $(\xi, \eta, \zeta, \omega)$ on the surface $\Phi = 0$, may be conveniently indicated by means of the diagram

$$\begin{array}{c|cccc|c}
 & x & y & z & w & \\
\hline
 & A & B & C & D & \xi \\
 & A' & B' & C' & D' & \eta \\
 & A'' & B'' & C'' & D'' & \zeta \\
 & A''' & B''' & C''' & D''' & \omega \\
\hline
\end{array} \qquad \Theta \ / \ \Delta \ / \ \Phi.$$

68. It is to be observed that, writing for A, B, ... their values as linear functions of $(\alpha, \beta, \gamma, \delta)$, we have in all 64 constant coefficients, which we may conceive arranged in the form of a cube, thus

$$\begin{array}{cccc}
a & b & c & d \\
a' & b' & c' & d' \\
\vdots & \vdots & \vdots & \vdots \\
a_1 & b_1 & c_1 & d_1 \\
a'_1 & b'_1 & c'_1 & d'_1 \\
\vdots & \vdots & \vdots & \vdots
\end{array}$$

and taking these in fours height-wise, (a, a_1, a_2, a_3), &c., we compose with them the linear functions $a\alpha + a_1\beta + a_2\gamma + a_3\delta$, &c., which enter into the equation $\Delta = 0$; taking them in fours length-wise, (a, b, c, d), &c., we compose the linear functions $ax + by + cz + dw$, &c., which enter into the equation $\Theta = 0$; and taking them in fours breadth-wise (a, a', a'', a'''), &c., we compose the linear functions $a\xi + a'\eta + a''\zeta + a'''\omega$, &c., which enter into the equation $\Phi = 0$.

69. The process may be indefinitely repeated; we obtain always the same three surfaces over and over again, but on them an indefinite series of corresponding points; viz., we may write

$$\begin{array}{c}
\ldots \Theta,\ \Delta,\ \Phi,\ \Theta,\ \Delta,\ \Phi,\ \Theta,\ \Delta,\ \Phi \ldots \\
\ldots P_1,\ Q_1,\ R_1,\ P,\ Q,\ R,\ P',\ Q',\ R' \ldots
\end{array}$$

viz., a point Q on Δ corresponds to a point P on Θ and to a point R on Φ; R corresponds to Q on Δ and to a new point P' on Θ; P' to R on Φ and to a new point Q' on Δ, and so on. And in the opposite direction P corresponds to Q on Δ and to a new point R_1 on Φ; R_1 to P on Θ and to a new point Q_1 on Δ and so on. And of course the correspondence of any two points of the series, whether belonging to the same surface or to different surfaces, is a one-to-one correspondence.

The Symmetrical Case; Symmetroid and Jacobian.

70. I have established the foregoing general theory; but it is only a particular case of it which connects itself with the theory of nodal quartics; viz., the cube of coefficients is a symmetrically arranged cube

$$\begin{array}{cccc} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{array} \qquad \begin{array}{ccc} a_1 & h_1 & \dots \\ \vdots & & \\ & & \\ & & \end{array}$$

or say its upper face is the symmetrical square matrix

$$\begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{vmatrix}$$

and the other horizontal planes, the like squares with the several terms affected by suffixes. The surface $\nabla = 0$ is here a surface of the form

$$\nabla = \begin{vmatrix} A & H & G & L \\ H & B & F & M \\ G & F & C & N \\ L & M & N & P \end{vmatrix} = 0$$

$[A, B,$ &c. linear functions of $(\alpha, \beta, \gamma, \delta)]$; viz., ∇ is a symmetrical determinant; I call this a *symmetroid*; the surfaces $\nabla = 0$, $\square = 0$ are one and the same surface, the Jacobian of 4 quadric surfaces; moreover the points P and R are one and the same point, and the correspondence R to P' is a reciprocal one; so that, instead of the indefinite series $\dots \Delta, \Theta, \Delta, \Theta, \Delta \dots$, $\dots P', Q, P, Q, P \dots$, we have a surface, the *symmetroid* $\nabla = 0$ which is a surface with 10 nodes, which is clearly not octadic, and which is therefore the decadianome.

71. Consider the quadric surfaces

$$\begin{aligned} S &= (a, b, c, d, f, g, h, l, m, n \between x, y, z, w)^2 = 0, \\ T &= (a_1, \dots) \qquad\qquad\qquad\qquad , \quad y = 0, \\ U &= (a_2, \dots) \qquad\qquad\qquad\qquad , \quad y = 0, \\ V &= (a_3, \dots) \qquad\qquad\qquad\qquad , \quad y = 0, \end{aligned}$$

and a point $(\alpha, \beta, \gamma, \delta)$ in the same or in a different space, such that the surface

$$\alpha S + \beta T + \gamma U + \delta V = 0$$

is a cone, or say for shortness,

$(\alpha, \beta, \gamma, \delta)$ is said to be the determining point, or determinator of the cone.

And if we establish the equations

$$\begin{aligned} \partial_x(\alpha S + \beta T + \gamma U + \delta V) &= 0, \\ \partial_y(\qquad \text{do.} \qquad) &= 0, \\ \partial_z(\qquad \text{do.} \qquad) &= 0, \\ \partial_w(\qquad \text{do.} \qquad) &= 0, \end{aligned}$$

which express that the surface is a cone, then (x, y, z, w) is the vertex of the cone. We have thus 4 equations linearclear in (x, y, z, w) and also in $(\alpha, \beta, \gamma, \delta)$, so that the relation between the Σ points is of the nature of that above considered. The relation between (x, y, z, w) is given by the equation

$$J(S, T, U, V) = 0;$$

viz., the locus is the Jacobian of the 4 quadric surfaces. The relation between $(\alpha, \beta, \gamma, \delta)$ is given by the equation

$$\nabla = \begin{vmatrix} a\alpha + a_1\beta + a_2\gamma + a_3\delta, & h\alpha + \ldots, & g\alpha + \ldots, & l\alpha + \ldots \\ h\alpha + \ldots, & b\alpha + \ldots, & f\alpha + \ldots, & m\alpha + \ldots \\ g\alpha + \ldots, & f\alpha + \ldots, & c\alpha + \ldots, & n\alpha + \ldots \\ l\alpha + \ldots, & m\alpha + \ldots, & n\alpha + \ldots, & d\alpha + \ldots \end{vmatrix} = 0;$$

so that the locus is (by the foregoing definition) the symmetroid. And the determinator and the symmetroid thus corresponds to the cone-vertex on the Jacobian.

72. But the Jacobian may be obtained in a different manner; viz., if we establish the equations

$$\begin{aligned} (\xi\partial_a + \eta\partial_b + \zeta\partial_c + \omega\partial_d)\, S &= 0, \\ (\qquad\qquad \text{do.} \qquad\qquad)\, T &= 0, \\ (\qquad\qquad \text{do.} \qquad\qquad)\, U &= 0, \\ (\qquad\qquad \text{do.} \qquad\qquad)\, V &= 0, \end{aligned}$$

then the elimination of $(\xi, \eta, \zeta, \omega)$ leads to the equation $J(S, T, U, V) = 0$ of the Jacobian. And since each of the equations is symmetrical in regard to (x, y, z, w), C. VII. 21

73. The points (x, y, z, w) of points of $(\alpha, \beta, \gamma, \delta)$, viz., we may either of these latter ones taking the point (x, y, z, w) on the Jacobian surface, the point $(\alpha, \beta, \gamma, \delta)$ on the symmetroid is related to the point (x, y, z, w), and conversely the system of 4 points, Q on the symmetroid, whose centres of these 4 points respectively; or the equation of the symmetroid, when referred to the 4 points, Q on the symmetroid, P and P' on the Jacobian. Q. V. S. I.

74. Consider any 5 pairs of points (x_1, y_1, z_1, w_1), $(\xi_1, \eta_1, \zeta_1, \omega_1)$, &c., related as above; the quartic surfaces $S = 0$, $T = 0$, $U = 0$, $V = 0$ are by the general quadric surface which cuts harmonically the five pairs of points respectively; or say they are quadric surfaces *harmonically related* to the 5 pairs of points respectively. The general quadric surface $\alpha S + \beta T + \gamma U + \delta V = 0$, harmonically related to the 5 pairs, will be a cone if $(\alpha, \beta, \gamma, \delta)$ is on the symmetroid, and the vertex of the cone is then defined by *M.* Chasles (*Comptes Rendus*, tom. LXI., 1865, p. 1147–62), and shown by him to be a variable surface, is thus identified with the Jacobian of any 4 quartic surfaces; and indicated herein we have the particular case, also considered by him, of the locus of the vertices of the quadric cones which pass through 6 given points, or Jacobian of the 6 given points.

75. It is to be shown that there are 10 systems of values $(\alpha, \beta, \gamma, \delta)$, for which the symmetroid $\nabla = 0$ is satisfied; viz., the number of conditions that there should be a node (x, y, z, w) are sufficient: the equation $\alpha S + \beta T + \gamma U + \delta V = 0$ of the symmetroid seems to contain homogeneously 40 constants. But starting with any given symmetrical determinant, the line by a constant determinant, and then column into column by another constant determinant; the coefficients of the constant determinants would in this manner be used, but the resulting product is a symmetrical determinant; and the equation is then with 10 nodes, contains $34 - 10 = 24$ constants.

Symmetroid with given Nodes.

76. A symmetroid can be formed with 7 given points as nodes; but there is no proper symmetroid with 8 given points as nodes. If we endeavour to form such a symmetroid, we obtain a system of 2 quartic cones, each of them passing through the 8 points; the curve of intersection is then a sextic C_2 found in the plane of the conditions of having a given point with the 7 points; the additional condition $(x, y, z, w) = 0$ may be expressed as a linear identification with the conditions of nodality; and also that it is the lineo-linear transformation.

77. Applying the conclusion to the system of quadric surfaces $\alpha S + \beta T + \gamma U + \delta V = 0$, we constants. But starting with this shall be a plane-pair implies a threefold relation between the coefficients $A, B,$ &c., and the required number of nodes is equal to the order of this threefold relation. Establishing the conditions $A, B,$ &c., any 6 linear relations whatever, we should have a ninefold relation between the coefficients of the number of solutions would be equal to the order of the threefold relation; and the number of solutions is clearly $= 10$.

78. The equation $\nabla = 0$ of the symmetroid seems to contain homogeneously 40 constants. But starting with any given symmetrical determinant, we may multiply it line into line by a constant determinant, and then column into column by another constant determinant, in such wise that the resulting product is still a symmetrical determinant; and the coefficients of the constant determinant may then be used to specialize the form of the equation. The equation $\nabla = 0$ of the symmetroid contains, therefore, the constant determinant; the term $(\alpha, \beta, \gamma, \delta)$ $\Sigma\sigma x_i = w^2 - 1$, $\Sigma x x_i = 3a - 3$, but not every rational determinant; the number of constants in $\nabla = 0$ may then be used to specialize the form of the equation. The equation $\nabla = 0$ of the symmetroid contains, therefore, $40 - 16 = 24$ constants; this is as it should be, for the symmetroid, qua quartic surface with 10 nodes, contains $34 - 10 = 24$ constants.

Symmetroid with given Nodes.

79. A symmetroid can be formed with 7 given points as nodes; but there is no *proper* symmetroid with 8 given points as nodes. If we endeavour to form such a symmetroid with 8 given points as nodes, we obtain a system of 2 quartic cones, each of them passing through the 8 points: the curve of intersection is in fact a quartic curve, C_4 in the plane Π, Π $(r = 4)$. It is clear, from this consideration, that any one of the conditions for the nodality of any one of the 8 points is in fact deducible from the others, leaving us with 7 conditions; and a symmetroid with 7 given points as nodes can be formed: but I have not succeeded in effecting this.

80. We have for any node $(\alpha, \beta, \gamma, \delta)$ of the symmetroid,

$$\alpha S + \beta T + \gamma U + \delta V = \text{plane-pair}.$$

If, then, 4 of the given nodes are $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, we must have S, T, U, V each of them a plane-pair. We may without loss of generality assume $S = x^2 + y^2$, $T = x^2 + w^2$; this, however, does not determine the signification of the coordinates (x, y, z, w); viz., we have S to determine the meaning of x, y; and similarly T to determine the meaning of x, w; and similarly T to determine the meaning of x, w; and similarly T remain unaltered if we write

$$x\cos\theta + y\sin\theta\,, \quad x\sin\theta - y\cos\theta \quad \text{for } x,\ y;$$

and similarly T will remain unaltered if we write

$$x\cos\theta_1 + w\sin\theta_1\,,\quad x\sin\theta_1 - w\cos\theta_1\quad \text{for } x,\ w.$$

Hence, if we go on to assume

$$\begin{aligned}U &= k\,(x+my+nz+pw)\,(x+m'z+n'z+p'w),\\ V &= l_1\,(x+m_1y+n_1z+p_1w)\,(x+m_1'z+n_1'z+p_1'w),\end{aligned}$$

formulae which contain the 12 constants

$$(k, m, n, p, m', n', p',\ l_1, m_1, n_1, p_1, m_1', n_1', p_1'),$$

this is right, for the symmetroid containing 24 constants, the symmetroid with 4 given nodes contains $(24 - 4\cdot 3 =)12$ constants. And each additional condition gives a single one of these constants; and so on.

81. But for any 4 new nodes, the equations may be satisfied by writing therein $u = u'$, $p = -p'$, $m = -m_1$, $n_1 = n_1'$; viz., they thus assume the form

$$\begin{aligned}S &= x^2+y^2,\\ T &= x^2+w^2,\\ U &= (\alpha x+cx)^2+(by+dw)^2,\\ V &= (\alpha' x+c'x')^2+(b'y+d'w)^2,\end{aligned}$$

containing 8 constants, which may be determined to make the surfaces have 4 given points. If now with the last mentioned values we form the value of $\alpha S+\beta T+\gamma U+\delta V$, this is a plane-pair if a square if

$$(\alpha+\gamma a^2+\delta a'^2)(\beta+\gamma c^2+\delta c'^2)-(\gamma ac+\delta a'c')^2=0,\quad \text{say this is } \Lambda=0,$$

(continued)

82. (continued)
and the second will be a square if

$$(\alpha+\gamma b^2+\delta b'^2)(\beta+\gamma d^2+\delta d'^2)-(\gamma bd+\delta b'd')^2=0,\quad \text{say this is } \Lambda'=0;$$

so that the condition

$$\alpha S+\beta T+\gamma U+\delta V=\text{cone}$$

will be satisfied if $\Lambda = 0$, or if $\Lambda' = 0$; that is, the equation of the symmetroid will be $\Lambda\Lambda' = 0$, breaks up into the 2 quadric surfaces $\Lambda = 0$, $\Lambda' = 0$, each of which is a cone.

82. It is to be further observed that, considering the first mentioned 4 points $(1, 0, 0, 0)$, &c., and any other 4 given points whatever, the equation of the 4 quadric cones through 4 given points 1, 2, 3, 4 and one of the new points, are obtained by considering the values $(\alpha, \beta, \gamma, \delta)$ which satisfy $\Lambda = 0$, viz., they are equal to be of the form

$$(*\$\,\beta\gamma,\ \gamma\alpha,\ \alpha\beta,\ \beta\delta,\ \gamma\delta)^2=0;$$

this combined with the first form gives, say through $(1, 0, 0, 0)$, &c., and the 5th point of the symmetroid will be satisfied if $\Lambda' = 0$; the equation of the symmetroid will be one of the 4 cones;

viz., we have to express that one of the 4 cones; viz., we have to express the conditions of the constants b, c in the expressions of the foregoing equations; that is the conditions for 4 given points different therefrom by changing a, b, c into $a+\frac{1}{w}(by-gz)$, $b+\frac{1}{w}(-kx+fz)$, $c+\frac{1}{w}(gx-fy)$ respectively; the equation may therefore be written in the symbolic form

$$w^2 \cdot \exp\frac{1}{w}\left[(ky-gz)\,\partial_a + (-kx+fz)\,\partial_b + x^2(gx-fy)\,\partial_c\right] \cdot [(a,b,c)^2] = 0,$$

or, what is the same thing,

$$w^2 \cdot \exp\frac{1}{w}[x(g\partial_b - h\partial_c) + y(h\partial_a - f\partial_c) + z(f\partial_b - g\partial_a)] \cdot [(a,b,c)^2] = 0,$$

where $\exp\theta$ (read exponential) denotes e^θ, and $[(a,b,c)^2]$ represents a determinant as above explained. The equation contains, it is clear, the four terms

$$x^2[(a,-h,g)^2] + y^2[(-h,b,-f)^2] + z^2[(g,-f,c)^2] + w^2[(a,b,c)^2].$$

I am not sure whether this surface of the eighth order has been anywhere considered.

83. In order that the symmetroid may have a 6th given node $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, I observe that the constants may be determined so that $\alpha_1 S + \beta_1 T + \gamma_1 U + \delta_1 V$ shall be equal to the variable factor, writing this as Π^2 and the like for the 5th point; we find $a(\alpha_1 + 5\alpha_1 + 6\alpha_1) + \ldots = 0$, viz., the new node being $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, and putting therein $a_i(1 + i\alpha_i + 6\alpha_1) = 0$, &c., for the various values of values which the new nodes can assume the formulae

$$\alpha_i S + \beta_i T + \gamma_i U + \delta_i V = (a_i, b_i, c_i \rangle\!\langle x, y, z, w)^2,$$

the new variables given through 4 given nodes $(1,0,0,0)$, &c., and the points $1, 2, \ldots, 8$ points; the given plane-pairs at the points $1, 2, \ldots, 7$.

Quartics with 11 or more Nodes.

103. I mention two results which, although they relate to quadric surfaces with more than 10 nodes, present themselves in such immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A & H & G & L \\ H & B & F & M \\ G & F & C & N \\ L & M & N & D \end{vmatrix} = 0$$

of the symmetroid ($A, B, \ldots$ linear functions of the coordinates), we have identically $A = 0$, then the surface has evidently a node in the ordinary manner in addition to the usual 10 nodes; an the surface has in all 11 nodes. And so also if (identically in every case) B is $= 0$, there are 12 nodes; if C is $= 0$, there are 13 nodes. These are, in fact, quartic surfaces with 11, 12, 13, and 14 nodes respectively, mentioned in Kummer's Memoir.

104. We may consider the symmetroid derived from the quadric surfaces which pass through 6 given points: viz., taking as before (see No. 25) the conditions of the 6 points to be $(1,0,0,0)$, $(0,1,0,0)$, $(0,0,1,0)$, $(0,0,0,1)$, $(1,1,1,1)$, (a,b,c,d); the coordinates of the line joining the last-mentioned two points: and, to avoid confusion, taking for the present purpose (X,Y,Z,W) instead of $(\alpha,\beta,\gamma,\delta)$ for the coordinates of a point on the symmetroid, the equation is obtained by arranging in the form of a determinant the coefficients of the quadric form

$$\begin{array}{l}
Xx\,(\qquad\qquad\qquad\qquad\quad)+hy(X-Y)+cyW,\\
h\,(X-Y)+cyW,\\
g\,(Z-X)+by\,W,\quad f(Y-Z)+ax\,W,\\
W\,(ax\,yx+b\beta zx+cyxy\qquad\qquad\qquad);
\end{array}$$

viz., the equation in question is

$$\begin{vmatrix}
\cdot & h(X-Y)+cy\,W & g(Z-X)+by\,W & aX\\
h(X-Y)+cy\,W & \cdot & f(Y-Z)+ax\,W & bY\\
g(Z-X)+by\,W & f(Y-Z)+ax\,W & \cdot & cZ\\
aX & bY & cZ & \cdot
\end{vmatrix}=0,$$

or, as it may be more simply written,

$$\sqrt{aX\,[f(Y-Z)+ax\,W]}+\sqrt{bY\,[g(Z-X)+by\,W]}+\sqrt{cZ\,[h(X-Y)+cy\,W]}=0.$$

This is, in fact, a surface with 16 nodes. It would appear that additional nodes correspond to the six common nodes that the 4 quadric surfaces having in common 1, 2, 3, 4, 5, or 6 points, the corresponding symmetroid would have 11, 12, 13, 14, 15, or 16 nodes. But I reserve this for future consideration.

105. To find the equation to the quadric surface through the three lines $(\alpha_1,\beta_1,\gamma_1,f_1,g_1,h_1)$, $(\alpha_2,\beta_2,\gamma_2,f_2,g_2,h_2)$, $(\alpha_3,\beta_3,\gamma_3,f_3,g_3,h_3)$, $(\alpha_4,\beta_4,\gamma_4,f_4,g_4,h_4)$, $(\alpha,\beta,\gamma,f,g,h)$. Take the equation of a quadric surface through this line will be of the form

$$\begin{vmatrix}
x^2 & y^2 & z^2 & w^2 & yz & zx & xy & yw & zw & xw\\
\alpha^2 & \beta^2 & \gamma^2 & f^2 & \gamma\beta & \gamma\alpha & \alpha\beta & h\gamma & f\beta+g\alpha & f\gamma\\
\alpha_1^2 & \beta_1^2 & \gamma_1^2 & f_1^2 & \gamma_1\beta_1 & \gamma_1\alpha_1 & \alpha_1\beta_1 & h_1\gamma_1 & h_1\alpha_1 & h_1\beta_1\\
2\alpha_1\alpha & 2\beta_1\beta & 2\gamma_1\gamma & 2ff_1 & f\gamma_1+f_1\gamma & f\alpha_1+f_1\alpha & \frac{1}{2}h(g_1+gh_1) & f_1h+fh_1 & f_1g+fg_1 & \frac{1}{2}h(g+\beta h_1)\\
\alpha_2^2 & \beta_2^2 & \gamma_2^2 & f_2^2 & \gamma_2\beta_2 & \gamma_2\alpha_2 & \alpha_2\beta_2 & h_2\gamma_2 & h_2\alpha_2 & h_2\beta_2\\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{vmatrix}=0$$

and we form thus a determinant with three of its lines relating to the line 1, three of them to the line 2, and three to the line 3, and the equation of the quadric surface through the lines 1, 2, 3; and it is clear that the determinant of the order $9\cdot 9$ contains, as a factor, the principal counter-system of the second plane, and the corresponding curves the principal counter-system of the first plane.

106. Taking (ξ,η,ζ,ω) as plane-coordinates, two quadric surfaces

$$(a,b,c,d,f,g,h,l,m,n\,\rangle\!\langle\,\xi,\eta,\zeta,\omega)^2=0,$$
$$(a_1,b_1,c_1,d_1,f_1,g_1,h_1,l_1,m_1,n_1\,\rangle\!\langle\,\xi,\eta,\zeta,\omega)^2=0$$

are said to be intervorts (or interverse) one of the other, when we have between the coefficients the relation

$$(a, b, c, d, f, g, h, l, m, n \rangle\!\!\lozenge\!\!\langle\, a_1, B_1, C_1, F_1, G_1, H_1, L_1, M_1, N_1) = 0,$$

that is,

$$aA + \ldots + 2fF + \ldots = 0.$$

The condition that the two surfaces may be intervorts of each other is linear in regard to the coefficients of each of them; hence, taking a before explained location, intervorts to two given quadrics A quadric, we have 9 quadrics; intervorts to 5 given quadrics is a given quadric surface, we have between the coefficients $10 - k$ quadrics, that is, $5 - k$ moreover, and the question is easily supplied.

If the quadric $(a, \ldots \rangle\!\langle\, \xi, \eta, \zeta, \omega)^2 = 0$ be a plane-pair, the plane-coordinates of the planes are by the equations

$$\begin{array}{rrrr}
 & \multicolumn{3}{r}{(\partial_a + ay + az + pw)\, \partial_\xi + p_1\, \partial_\eta = 0,} \\
hx + \ldots, & bx + \ldots, & fx + \ldots, & mx + \ldots = 0, \\
 & gx + \ldots, & fx + \ldots, & nx + \ldots = 0, \\
lx + \ldots, & mx + \ldots, & nx + \ldots, & dx + \ldots = 0;
\end{array}$$

viz., the locus is (by the foregoing definition) the symmetroid. And the determinator and the symmetroid corresponds to the cone-vertex on the Jacobian.

107. Taking the lines to be those of $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, b_2, c_2, f_2, g_2, h_2)$ $\ldots$ $(a_5, b_5, c_5, f_5, g_5, h_5)$ to be those of three quadric surfaces having no common conditions: But if the conditions $(\xi, \eta, \zeta, \omega)$ belonging to a point, the value is $= 0$.

108. Any 6 points whatever may be regarded as points on a skew cubic; and the coordinates (x, y, z, w) may then be taken to be of the form

$\begin{vmatrix} x, & y, & z \\ y, & z, & w \end{vmatrix} = 0$. This being so, the coordinates of the 6 given points may be taken to be $(1, t_1, t_1^2, t_1^3)$, $(1, t_2, t$

the 6 points; that is, of the surface containing the conditions $(1, t_3, t_3^2, t_3^3)$, $(1, t_4, t_4^2, t_4^3)$; and the equation of the Jacobian surface of

109. Representing as before each line by means of its six coordinates, let $(\alpha, \beta, \gamma, f, g, h)$ be the coordinates of the vertex, the coordinates (X, Y, Z, W) be the coordinates of any one of the lines, the equation (a, b, c, f, g, h) the coordinates of any one of the six lines, the equation of the plane through this line and the vertex for shortness

$$\alpha(zW - wY) + b(yW - wY) + c(xW - wX) = 0;$$

or, what is the same thing, putting for shortness

$$\begin{aligned}
P &= \cdot - hy - gz + aw, \\
Q &= -hx \cdot + fz + bw, \\
R &= gx - fy \cdot + cw, \\
S &= -ax - by - cz \cdot,
\end{aligned}$$

the equation is

$$PX + QY + RZ + SW = 0.$$

The plane in question is a tangent plane to the cone touched by each of the 6 lines. Now when the plane is a tangent plane to the cone touched by each of the 6 lines, the plane in question is a tangent plane to the cone touched by each of the 6 lines. Hence taking the plane $PX + QY + RZ + SW = 0$ on the cone touched by the 6 lines, the equation

$$PX + QY + RZ + SW = 0.$$

To ascertain the form of this, write for a moment $y = 0$, $x = 0$; the equation is

$$[(aw + bz + bw, \quad yz + ew)^2] = 0,$$

or attending only to the highest and lowest powers of w, this is

$$w^2\,[(\alpha, b, c)^2] + \ldots + w^4\,[(a, -h, g)^2]\ldots$$

and it is thence easy to infer that the whole equation divides by w^2, so that, omitting this factor, the form of the equation is

$$[(a, b, c \,\rangle\!\langle\, g, y, z, w)^2 \ldots] = 0;$$

viz., this equation is satisfied if z denote any one of the quantities $(\alpha_1, \beta_1, \gamma_1, f_1, g_1, h_1)$, &c., that is, if any one of the 6 lines pass through the assumed point in the direction of any vertex of the cone, and if so, the equation of the cone, regarded as a function of w, omitting this factor, the form of the equation is

$$((a, b, c, f, g, h \,\rangle\!\langle\, x, y, z, w)^2 = 0,$$

viz., the equation is satisfied if z denote any of the quantities $\alpha_1, b_1, c_1, f_1, g_1, h_1$; and if so, the equation of the cone, regarded as a function of x, y, z, w, is

$$\left|\begin{array}{ccc} x, & y, & z \\ a, & b, & c \end{array}\right| = 0.$$

Locus of the vertex of a Quadric Cone which touches each of Six given Lines.

110. Representing as before each line by means of its six coordinates, let $(\alpha, \beta, \gamma, f, g, h)$ be the coordinates of the vertex, and (X, Y, Z, W) the current coordinates of any one of the lines, then the equation of the plane through this line and the vertex is

$$\alpha\,(zW - wZ) + b\,(yW - wY) + c\,(xW - wX) + f\,(yZ - zY) + g\,(zX - xZ) + h\,(xY - yX) = 0;$$

or, in the same notation as before, for shortness

$$\begin{aligned} P &= \cdot - hy - gz + aw, \\ Q &= -hx \cdot + fz + bw, \\ R &= gx - fy \cdot + cw, \\ S &= -ax - by - cz \cdot, \end{aligned}$$

the equation is

$$PX + QY + RZ + SW = 0.$$

The plane in question is a tangent plane to the cone touched by each of the 6 lines. Hence taking the plane $PX + QY + RZ + SW = 0$ on the cone touched by the 6 lines, attending only to the highest and lowest powers of w, this is

$$w^2[(a, b, c)^2] + \ldots + w^4[(a, -h, g)^2]\ldots,$$

and it is thence easy to infer that the whole equation divides by w^2, so that, omitting this factor, the form of the equation is

$$((a, b, c, f, g, h \,\rangle\!\langle\, x, y, z, w))^2 = 0,$$

viz., the equation in question is of the order 8 in the coordinates (x, y, z, w) and of the degree 2 in the coordinates of each of the lines. It would not be very

111. difficult to actually develope the equation; in fact, starting from the term $w^2[(a, b, c)^2]$ the other terms are obtained therefrom by changing a, b, c into $a+\dfrac{1}{w}(hy-gz)$, $b+\dfrac{1}{w}(-hx+fz)$, $c+\dfrac{1}{w}(gx-fy)$ respectively; the equation may therefore be written in the symbolic form

$$w^2 \cdot \exp\frac{1}{w}\big[(hy-gz)\,\partial_a + (-hx+fz)\,\partial_b + (gx-fy)\,\partial_c\big] \cdot \big[(a, b, c)^2\big] = 0,$$

or, what is the same thing,

$$w^2 \cdot \exp\frac{1}{w}\big[x(g\,\partial_b - h\,\partial_c) + y(h\,\partial_a - f\,\partial_c) + z(f\,\partial_b - g\,\partial_a)\big] \cdot \big[(a, b, c)^2\big] = 0,$$

where $\exp\theta$ (read exponential) denotes e^{θ}, and $[(a, b, c)^2]$ represents a determinant as above explained.

112. I am not sure whether this surface of the eighth order has been anywhere considered.

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[446] On the Mechanical Description of a Nodal Bicircular Quartic.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 100–106.]

The ingenious method, devised by Mr S. Roberts (*Proceedings*, vol. II. p. 133) for the description of a nodal bicircular quartic suggests a further investigation. We have a quadrilateral $OAA'O'$, in which the adjacent sides OA, AA' are equal to each other, and the other two adjacent sides OO', $O'A'$ are also equal to each other; O, O' are fixed points; and we have thus a link AA', the extremities of which are connected by means of the radii OA, $O'A'$ respectively, and consequently describe circles about the centres O, O' respectively, the radius OA of the one circle being equal to the length AA' of the link, and the radius $O'A'$ of the other circle being equal to the distance OO' of the centres. The theorem is, that any point C, rigidly connected with the link AA', describes a nodal bicircular quartic, that is, a quartic curve with three nodes (or unicursal quartic), two of the nodes being the circular points at infinity. Any such curve is the inverse of a conic, and it is also the antipode of a conic; viz., if at such point of the curve we draw a line at right angles to the radius vector from the node, then envelope of the line passing through this point is a circle. If the equalities $OA = AA'$, $O'A' = OO'$ did not subsist, then to a given position of the link AA' corresponds two positions of A, A' respectively, on the circles; the radius OA of the one circle being equal to the length AA' of the link, and the radius $O'A'$ of the other circle being equal to the distance OO'. The theorem is, that any point C, rigidly connected with the link AA', describes a nodal bicircular quartic, that is, a quartic curve with three nodes (or unicursal quartic), two of the nodes being the circular points at infinity.

I have a curve AA': the link; OO' may be called the bar; and the locus of the radius AA' produced; moreover AA' may be called the link; the radius OA of the one circle being equal to the length AA', and the radius $O'A'$ of the other circle being equal to the distance OO' of the centres of the two circles respectively. Suppose the link AA' to slide along the bar OO'; the link AA' remains parallel to itself in this its motion; viz., on this slide-motion the bar OO' moves about its centre, and the link AA' traverses the constant triangle; and, producing OA, $O'A'$ to meet in K, the locus of K is called the constant triangle; and the locus of C will be a higher order than is in the actual problem.

I have called AA' the link; OO' may be called the bar. OA is then the link-radius, $O'A'$ the bar-radius; moreover AA' may be called the constant triangle; and, producing OA, $O'A'$ to meet in K, then AKA' is the constant triangle. Since at any instant the motion of A is normal to KA, and the motion of A' to KA', it is clear that the motion of any point on KA or on KA' is also normal to KA, KA' respectively; and in like manner the motion of A', in its position on the line KA', is in like manner the motion of any point on the line KA'. We have thus an instantaneous rotation about the point K.

Imagine any two positions of the link; say these are A_1A_1', and A_2A_2'. Join A_1A_2 and at its mid-point draw a perpendicular thereto; join in like manner $A_1'A_2'$, and at its mid-point draw a perpendicular thereto; and let these two perpendiculars meet in T; we have the two equal triangles $TA_1 = TA_2$, $TA_1' = TA_2'$, and the angles $A_1TA_1' = A_2TA_2'$, so that there is a rotation about the centre T, which conducts A_1A_1' into A_2A_2'. Now, in the limit, when the two positions A_1A_1', A_2A_2' are made to coincide with the triangles $A_1A_1'T$ and $A_2A_2'T$ respectively; that is, T will be a node in the locus described by this point, but in the present case, this point K is a fixed point. It is indifferent any point will pass with the link AA' in some such manner, and not indifferent any point of the locus C, of which the same considerations may be applied to itself; it is, for any point of the locus, when the link describes the ordinary points of curvature. And in the linkage describing the bicircular quartic, the centres are at the same distance, a from the centre O of the linkage. The quartic, when described by the two points A_1A_2 describes a circle of the same diameter centred at O of the linkage and centres at the same distance from the centre O of the linkage; the quartic curve, when described by the two points A_1A_2 describes a circle of the same diameter a from the centre O of the linkage and centres at the same distance from O.

Passing now to the analytical investigation, I take the origin at O, the axis of x along the direction from O to O', and the axis of y, at right angles thereto, upwards from O. The inclinations of OA, AA', $O'A'$ to the axis Ox are taken to be θ, ϕ, θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and the angles θ, θ' are connected by the relation $\sin\theta'/\sin\theta = a/a'$, and finally $AB = b$, $BC = c$.

Observing that the angle $AA'O'$ is $= \theta$, we have $\theta' = \pi - \theta - \phi$, or $\theta - \phi$ is the deviation from O, and the axis of y, at right angles, upwards from O. The inclinations of OA, AA', $O'A'$ to the axis Ox are taken to be θ, ϕ, θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and finally $AB = b$, $BC = c$.

Hence, if we go on to assume

$$m = 1 : 1 + m : 1 - m = a' : a : a' + a : 2a' : 2a;$$

and finally $AB = b$, $BC = c$.

Observing that the angle $AA'O'$ is $= \theta$, we have $\theta' = \pi - \theta - \phi$, upwards from O. The inclinations of OA, AA', $O'A'$ to the axis Ox are θ, ϕ, θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and $OO' = a$, $OO' = a'$; and finally $AB = b$, $BC = c$.

(Algebraic developments.) For the locus of C we have

$$x = a\cos\theta + b\cos\phi + c\sin\phi,$$

$$y = a\sin\theta + b\sin\phi + c\cos\phi,$$

or, instead of θ, ϕ introducing u, we have

$$x = -a\,\frac{u^2-1}{u^2+1} + b\,\frac{u^2-m^2}{u^2+m^2} - c\,\frac{2mu}{u^2+m^2},$$

$$y = a\,\frac{2u}{u^2+1} + b\,\frac{2mu}{u^2+m^2} + c\,\frac{u^2-m^2}{u^2+m^2}.$$

[Further reductions and the curve obtained as an inverse-of-a-conic and antipode-of-a-conic are given in the remainder of the paper.]

(Closing remarks.) There remains to be observed, further, that the equation of the line in question obtained by writing in the foregoing equations, $u = u'$ and eliminating the remaining quantities δx, δy, &c., and the corresponding expressions δx_i, δy_i, &c.; the result of the elimination will be the same as if, writing $u = u_1$, we have in fact only to consider the equation of the line in the form

$$ax' + by' + c = 0 \quad \text{(in fact, for } A,\ B \text{ or of } (x, y).)$$

The remainder of the paper carries out this elimination and shows that the locus is in fact a circular quartic with the prescribed nodal properties.

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[447] ON THE RATIONAL TRANSFORMATION BETWEEN TWO SPACES.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 127–180. *Account of the Paper given at the Meeting* 11 *March* 1869.*]*

Two figures are rationally transformable each into the other (or, say, there is a rational transformation between the two figures) when to a variable point of each of them there corresponds a single variable point of the other. The figures may be either lost in a space, or loci of any number of dimensions; or they may be lost in plane spaces themselves. Thus the figures may be each a line of one space (or space of one dimension); and there will be a rational transformation between two lines, or they may be each a plane (or space of two dimensions); and there will be a rational transformation between two planes; or, finally, the figures may be each surfaces (or spaces of three dimensions), and there will be a rational transformation between two spaces. The problem of the rational transformation between two planes (or generally between two plane spaces of equal dimensions) is the only one which has been hitherto considered, and it is in fact thereto, that every rational transformation between two planes, and to be more particular still, between two plane spaces, may be reduced. The general problem in three dimensions is here treated of; the most general transformation between two plane spaces of equal dimensions is, in fact, the transformation (which is already mentioned) in the only one peculiar transformation between two plane spaces; and consequently a rational transformation between two planes; viz., in regard hereto I examine the general theory, but attend mainly to what I call the lineo-linear transformation; viz., it is assumed that the coordinates

The General Principle of the Rational Transformation between Two Spaces.

1. In all that follows, the two spaces (lines, planes, or three dimensional spaces, as the case may be), to any rational corresponding loci in the two spaces respectively, are referred to as the first and second figures respectively; viz., between two lines, between two planes, or between two three-dimensional spaces, as the case may be. The notation of the first and second figures and the corresponding loci, when considered, not as superimposed or situate in a common space, but as existing, each independently of the other, but as together mentioned in the only one peculiar way, will be referred to as the first and the second figures and the corresponding loci. Writing as before

$$x : y : z = X : Y : Z,$$

the unicursal correspondence (which is, in fact, the only one between two lines) unicursal correspondence; viz., if as before, we refer to the lines, planes, or three-dimensional spaces, the correspondence between them is essentially unsymmetrical.

2. If, to fix the ideas, we attend to the case of two planes, or take the sets

$$x : y : z = X : Y : Z, \qquad x : y : z = X' : Y' : Z',$$

[The opening articles, §§ 2–35 on PDF pages 212–225, develop the general theory of the rational transformation between two spaces, in particular establishing the basic formulæ of the lineo-linear (or, as Cayley calls it, the homographic) transformation between two planes; the construction of the constants of the transformation; the conditions under which a system of equations of the rational correspondence is satisfied; the count of free constants; the various special and degenerate cases (when the transformation breaks up into a product of a linear and a rational correspondence, or when certain equations of correspondence become identically satisfied); the use of these formulæ for the calculation of the number of intersections of a curve in the first plane with its transform in the second plane; the count of free parameters in the transformation; and, finally, the discussion of the system of *principal curves* and *principal counter-curves* (or *réseaux* of Cremona) which arise in the transformation. The detailed algebraic developments occupy pages 189–203 of the book, and the memoir is continued from book page 204 onward in the next chunk of the typesetting.]

(End of chunk: PDF pages 176–225*, book pages* 154–203*.*
Paper 445 *§§* 56–112 *(tail of the Memoir on Quartic Surfaces),*
paper 446 *in full,*
paper 447 *§§* 1–35 *(opening of the Rational Transformation memoir).)*

[447] On the Rational Transformation between Two Spaces.

(continued from book page 203 / scan page 225.)

[Book pages 204–211 are occupied almost entirely by Cremona's tables of principal systems for the rational transformations of orders $n = 2, 3, 4, 5, 6, 7, 8, 9, 10$ and of the generic types $n = p$, $n = 2p$, $n = 2p+1$, $n = 3p$, $n = 3p+1$, $n = 3p+2$, $n = 4p$, $n = 4p+1$, $n = 4p+2$, $n = 4p+3$. Each table is a rectangular grid: the top row lists the columns of partitions, indexed by an outer label n; the body shows the integers $a_1, a_2, \ldots, a_{n-1}$ that constitute the partition. The tables are sibi-reciprocal, the conjugate pairs being marked a_r and a'_r. In what follows I retype the body text and reproduce the smaller auxiliary tables verbatim; for the long Cremona tables of article 41 on book pp. 204–207 I give a faithful but compact representation, preserving every entry as it stands in the printed page.]

Article 41 (continued): Cremona's Tables of Partitions.

The tables are read as follows: in each block the topmost row gives the column-label n (the order of the transformation), then under it stand successive columns, each headed by the values $a_1, a_2, \ldots, a_{n-1}$ written one below the other. A blank entry denotes 0.

Table $n = 7$, $n = 8$, $n = 9$, $n = 10$. *(Book p. 204.)*

Block I, $n = 7$, 8, 9:

	$n=7$						$n=8$								$n=9$									
	3	4	6	8	10	12	14	3	1	4	6	8	12	12	16	4	2	0	3	7	1	3	0	1
a_1	3	4	6	8	10	12	14	3	1	4	6	8	12	12	16	4	2	0	3	7	1	3	0	1
a_2		1	3	3	5	5		1	3	6	0	2	3	1		2	4	0	3	7	1	3	0	1
a_3			1	1	2	3	2		1	0	5	0	0	0		0	1	0	0	0	0	0	0	0
a_4				1		1	1	1			0	5	1	1				0	0	0	0	0	0	0
a_5					1	1	1						1					0	0	0	0	0	0	0
a_6							1											0	0	0	0	0	0	0

Block II, $n = 10$. Last column marked by Cremona as "omitted by Cremona."

	18	5	1	0	0	3	6	2	4	1	2	1	3	3	3	0	1
a_1	18	5	1	0	0	3	6	2	4	1	2	1	3	3	3	0	1
a_2		4	2	0	3	0	3	1	3	3	0	6	1	0	0	0	0
a_3			2	3	0	1	0	2	0	5	0	1	0	5	0	0	0
a_4				2	3	0	1	0	2	1	3	0	0	0	5	0	1
a_5						1	0	2	1	0	2	0	0	0	0	5	0
a_6							1	1	0	0	1	0	0	0	0	0	0
a_7								1	0	0	0	0	0	0	0	0	0
a_8									0	0	0	0	0	0	0	0	0
a_9		1								0	0	0	0	0	0	0	0

[Footnote on p. 204: "Omitted by Cremona."]

Tables for the generic forms $n = p$, $n = 2p$, $n = 2p+1$, $n = 3p$. *(Book p. 205.)*

Form $n = p$.

	a_1	a_{p-1}
$a_1 = 2p-2$	$2p-2$	1
$a_{p-1} = 1$	1	1

Form $n = 2p$.

	a_1	a_2	a_{p-1}	a_p
$a_1 = 3$	3			$2p-2$
$a_2 = 2p-2$		$2p-2$	1	
$a_{p-1} = 1$		1		
$a_p = 4$	$2p-2$			1

Form $n = 2p + 1$.

	a_1	a_p	a_{p+1}	a_{2p-1}
$a_1 = 3$	3			$2p-1$
$a_2 = 2p-1$				
$a_p = 4$	$2p-1$	1	1	
$a_{p+1} = 1$	4	1		
$a_{2p-1} = 1$		$2p-3$	1	1

Form $n = 3p$.

	a_1	a_2	a_p	a_{p+1}	a_{2p-1}
$a_1 = 4$	4				$2p-2$
$a_2 = 2p-2$		$2p-2$			
$a_p = 3$					
$a_{p+1} = 4$	$2p-2$				4
$a_{2p-1} = 1$					1
$a_{3p-2} = 1$	1	$2p-2$	1	4	1

Tables for the generic forms $n = 3p + 1$**,** $n = 3p + 2$**,** $n = 4p$**,** $n = 4p + 1$**,** $n = 4p + 2$**,** $n = 4p + 3$**.** *(Book p.* 206.*)*

Form $n = 3p + 1$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2$	2	$2p-2$		
$a_2 = 7$	7			$2p-1$
$a_p = 1$		1		
$a_{2p-1} = 3$	$2p-1$		1	3

Form $n = 3p + 2$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2p-4$	$2p-4$	1		$2p-1$
$a_2 = 7$		3	4	1
$a_p = 1$			1	3
$a_{p+1} = 3$	$2p-2$		1	4
$a_{2p} = 1$				1

Form $n = 4p$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 9$	9	6	2	$2p-1$
$a_2 = 3$	3		3	5
$a_3 = 3$		$2p-3$	1	2
$a_4 = 2p-2$	1		$2p-1$	1
$a_p =$				
$a_{p+1} =$				
$a_{2p+1} = 1$				1
$a_{4p-1} = 1$	4	1	1	

Form $n = 4p + 1$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2$	2	$2p-2$	3	$2p-1$
$a_2 = 3$	3		4	7
$a_3 = 2p-3$		1		4
$a_4 = 2p-3$				1
$a_p =$	1	3		
$a_{p+1} =$				
$a_{4p-3} = 1$			1	

Form $n = 4p + 2$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2p-4$	1	$2p-2$	1	$2p-1$
$a_2 = 7$	3		3	4
$a_3 =$	2		4	3
$a_4 = 2p-4$		$2p-2$		4
$a_p =$			3	1
$a_{p+1} =$				
$a_{4p-3} = 1$			1	1

Form $n = 4p + 3$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2p-2$	9	$2p-3$	2	$2p-1$
$a_2 = 3$	6		3	5
$a_3 = 3$		$2p-3$	1	2
$a_4 = 2p-2$	1		$2p-1$	5
$a_p =$				1
$a_{p+1} = 3$		$2p+1$	1	1
$a_{2p+1} =$	1		$2p+3$	1
$a_{4p-1} = 1$		$2p+1$	1	1

42. The system $(a_1, a_2, \ldots, a_{n-1})$ geometrically determines completely the system $(a'_1, a'_2, \ldots, a'_{n-1})$; it ought therefore to determine it arithmetically; that is, given the one series of numbers, we ought to be able to determine, or at least to select from the table, the other series of numbers. Cremona has shown that the two series consist of the same numbers in the same or a different order. By examination of the tables, it appears that there are certain columns which are single (that is, no other column contains in a different order the same numbers), others that occur in pairs, the two columns of a pair containing the same numbers in a different order. Where the column is single, it is clear that this must give as well the values of $(a'_1, a'_2, \ldots, a'_{n-1})$ as of $(a_1, a_2, \ldots, a_{n-1})$. Where there is a pair of columns, as far as Cremona has examined, if the one column is taken to be $(a_1, a_2, \ldots, a_{n-1})$ the other column is $(a'_1, a'_2, \ldots, a'_{n-1})$; it appears, however, not to be shown that this is universally the case; viz., it is not shown but that the two columns, instead of being reckoned as a pair, might be reckoned as two separate columns, each by itself representing the values as well of $(a_1, a_2, \ldots, a_{n-1})$ as of $(a'_1, a'_2, \ldots, a'_{n-1})$; neither is it shown that there are not, in any case, more than two columns having the same numbers in different orders. It seems, however, natural to suppose that the law, as exhibited in the tables, holds good generally; viz., that the tables contain only single columns, each giving the values as well of $(a_1, a_2, \ldots, a_{n-1})$ as of $(a'_1, a'_2, \ldots, a'_{n-1})$; or else pairs of columns, one giving the values of $(a_1, a_2, \ldots, a_{n-1})$, and the other those of $(a'_1, a'_2, \ldots, a'_{n-1})$; or, say, that the partitions are either *sibi-reciprocal*, or else *conjugate*.

43. Assuming that the two systems $(a_1, a_2, \ldots, a_{n-1})$ and $(a'_1, a'_2, \ldots, a'_{n-1})$ are each known, there is still a question of grouping to be settled; viz., the Jacobian of the first plane consists of a'_1 lines, a'_2 conics, $\ldots$ a'_{n-1} unicursal $(n-1)$-thics; each line, each conic, &c., passes a certain number of times through certain of the points $a_1, a_2, \ldots, a_{n-1}$: but through which of them? For instance, each of the a'_1 lines will pass through two of the points $a_1, a_2, \ldots, a_{n-1}$: will these be points a_1, or points a_2, &c., or a point a_1 and a point a_2, &c.? The mere symmetry of the different groups of points determines certain conditions of the solution[1]; for instance, if any particular one of the a'_1 lines passes through two points a_1, then each of the a'_1 lines must pass through two points a_1; and since the points a_1 are symmetrical, we must in this way use all the pairs of points a_1; that is, if $a'_1 = \frac{1}{2}a_1(a_1 - 1)$, but not otherwise, it may be that each of the a'_1 lines passes through two of the points a_1. In the case of an equality $a_1 = a_2$ we could not hereby decide whether the line passed through two points a_1 or through two points a_2. So, again, if any one of the a'_1 lines pass through a point a_1 and a point a_2, then each of the a'_1 lines must do so likewise, and we must hereby exhaust the combinations of a point a_1 with a point a_2; viz., the assumed relation can only hold

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[1] It is by such considerations of symmetry that Cremona has demonstrated the before-mentioned theorem of the identity of the numbers $(a_1, a_2, \ldots, a_{n-1})$ and $(a'_1, a'_2, \ldots, a'_{n-1})$.

good if $a_1' = a_1 a_2$. Similarly, each of the a_2' conics will pass through five of the points $a_1, a_2, \dots, a_{n-1}$; each of the a_3' nodal cubics will pass twice through one (have a double point there) and through six others of the points $a_1, a_2, \dots, a_{n-1}$;—which are the points so passed through? I do not know how a general solution is to be obtained, but most of the cases within the limits of the foregoing table have been investigated by Cremona.

The results may conveniently be stated in a tabular form; the tables exhibit in the outside upper line the values of $a_1, a_2, \dots, a_{n-1}$, and in the outside left-hand line the values of $a_1', a_2', \dots, a_{n-1}'$, and they are to be read as

$$\left.\begin{array}{c} a_1' \text{ lines} \\ a_2' \text{ conics} \\ \text{\&c.} \end{array}\right\} \text{passes ( ) times through ( ) of the points } a_1, a_2, \dots, a_{n-1}$$

respectively; the numbers in the table being those of the points passed through, and the indices in the table (index = 1 when no index is expressed) showing the number of times of passage, that is, showing whether the point is a simple, double, triple, &c., point on the curve referred to.

44. Thus (in the tables which follow) the last of the tables $n = 6$ gives the constitution of the Jacobian of the first plane, where the principal system is $(3, 4, 0, 1, 0)$; and it is to be read:

> Each of the 4 lines passes through 1 of the points a_1 and through the point a_4;
> The 1 conic „ „ 4 of the points a_1 and through the point a_4;
> Each of the 3 cubics „ „ 2 of the points a_1, 4 of the points a_2, and twice through the point a_4 (that is, a_4 is a double point on each cubic).

It is hardly necessary to remark that the tables are sibi-reciprocal, or else conjugate, as appears by the outer lines of each table.

Table $n = 2$.

	a_1
	$\frac{1}{3}$
$a_1' = 3$	2 [was originally printed, 3.]

Table $n = 3$.

	a_1	a_2
	$\frac{4}{4}$	$\frac{1}{1}$
$a_1' = 4$	1	1
$a_2' = 1$	4	1

Tables $n = 4$.

	a_1	a_2
	6	1
$a_1' = 6$	1	1

	a_1	a_2	a_3
	3	3	1
$a_1' = 3$	2	1	
$a_2' = 3$	2	3	
$a_3' = 1$	1	1	1

Tables $n = 5$.

	a_1	a_2	a_3	a_4
	8	0	0	1
$a_1' = 8$	1			1
$a_4' = 1$	8			1^6

	a_1	a_2	a_3	a_4
	2	3	1	0
$a_1' = 2$	1	1		
$a_2' = 3$	1	3	1	
$a_3' = 1$	1	3	3	1^*

	a_1	a_2	a_3	a_4
	0	6	0	0
$a_2' = 6$		5		

Tables $n = 6$.

	a_1	a_2	a_3	a_4	a_5
	10	0	0	0	1
$a_1' = 10$	1				1
$a_5' = 1$	10				1^8

	a_1	a_2	a_3	a_4	a_5
	0	5	0	0	0
$a_2' = 5$		2			
$a_3' = 2$		4	1		

	a_1	a_2	a_3	a_4	a_5
	4	1	3	0	0
$a'_1 = 4$	1		1		
$a'_2 = 1$	4	1	3		

	a_1	a_2	a_3	a_4	a_5
	3	4	0	1	0
$a'_1 = 3$	2	4		1	
$a'_2 = 4$	1	4		1	
$a'_4 = 1$	2	4		1^2	

Footnote: "Read, 'Each of the two cubics passes through the point a_4, the four points a_1, and, $(1^2, 1)$, twice through one and once through the other of the points a_2.'"

45. It is to be remarked upon the tables—first, as regards the lines; if we add the numbers in each line, reckoning m^p as mp (that is, each multiple point, according to the number of branches through it), the sums for the successive lines are $2, 5, 8, 11, 14$, &c.; that is, each line passes through 2 points, each conic through 5 points, each cubic through 8 points, each quartic through 11 points, &c. But if we add the numbers reckoning m^p as $\frac{1}{2}mp(p+1)$ (that is, each multiple point according to its effect in the determination of the curve), then the sums are $2, 5, 9, 14, 20$, &c., that is, all the curves are completely determined, viz., the line by 2 conditions, the conic by 5 conditions, the cubic by 9 conditions, &c. Secondly, as regards the columns, if for any column, reckoning m^p as mp, we multiply each number by the corresponding outside left-hand number, add, and divide the sum by the outside number at the head of the column, the successive results are $2, 5, 8, 11, 14$, &c.; this merely expresses the known circumstance that the Jacobian passes $3r - 1$ times through each point a_r.

46. The analogous tables showing the passage of the Jacobian through the principal system, in the solutions belonging to certain special forms of n, are

Table $n = p$.

	a_1	a_{p-1}
	$2p-2$	1
$a'_1 = 2p-2$	1	1
$a'_{p-1} = 1$	$2p-2$	1^{p-2}

Tables $n = 2p$.

	a_1	a_2	a_{p-1}	a_p	a_{2p-2}
	3	$2p-2$	0	0	1
$a'_1 = 3$	1				1
$a'_{2p-2} = 1$	2	$2p-2$			1^{p-2}

	a_1	a_2	a_{p-1}	a_p	a_{2p-2}
	$2p-2$	1	1	3	0
$a'_2 = 3$		2			
$a'_{p-1} =$	$2p-2$	1	1	3	
$a'_p =$					
$a'_{2p-1} =$	1	$2p-2$	$p-3$	3^{p-1}	

Tables $n = 2p+1$.

	a_1	a_p	a_{p+1}	a_{2p-1}
	3	$2p-1$	0	1
$a'_1 = 3$	1			1
$a'_p =$				
$a'_{p+1} =$				
$a'_{2p-1} =$				

	a_1	a_p	a_{p+1}	a_{2p-1}
	$2p-1$	0	3	0
$a'_p = 3$	1			1
$a'_p = 2p-1$	1	$2p-1$	1	1
$a'_{p+1} =$				
$a'_{2p-1} = 1$	$2p-1$	3^{p-1}	1^p	

Tables $n = 3p$.

	a_1	a_2	a_p	a_{2p-1}	a_{3p-2}	a_{3p-1}
	1	4	0	$2p-3$	0	1
$a'_1 = 2p-3$			1			1
$a'_2 = 0$						
$a'_p = 0$						
$a'_p = 4$		3	$2p-3$			1^{p-1}
$a'_{p+1} = 1$	1	4	$2p-3$			1^p
$a'_{2p-1} = 1$	1	4	$(2p-3)^2$			1^{2p-3}
$a'_{2p} =$						

Further tables (book p. 212).

	a_1	a_2	a_p	a_{2p-1}	a_{3p-1}	a_{3p-1}
	$2p-3$	0	0	4	1	0
$a'_1=1$				1	1	
$a'_2=4$			3	1	1	
$a'_p=2p-3$	1			4	1	1^2
$a'_p=0$						
$a'_{p+1}=0$						
$a'_{2p-1}=0$						
$a'_{3p-2}=1$	$2p-3$			4^{p-1}	1^p	1^{p-2}

	a_1	a_2	a_p	a_{p+1}	a_{2p}	a_{3p-1}
	4	1	$2p-2$	0	0	1
$a'_1=2p-2$			1			1
$a'_2=0$						
$a'_p=0$						
$a'_{p-1}=1$			$2p-2$			1^{p-2}
$a'_p=4$	1	1	$2p-2$			1^{p-1}
$a'_{2p}=1$	4	1	$(2p-2)^2$			1^{2p-2}
$a'_{3p-1}=0$						

Final Jacobian table (book p. 213).

	a_1	a_2	a_3	a_{p-1}	a_{2p}	a_{3p-1}
	$2p-2$	0	0	1	4	1
$a'_2=4$				1	1	1
$a'_3=1$					4	1
$a'_{p-1}=2p-2$	1			1	4	1^p
$a'_{2p}=$						
$a'_{2p-1}=$						
$a'_{3p}=$		$p-2$			4^{p-1}	1^{2p-2}
$a'_{3p-2}=1$	$2p-2$					

47. The before-mentioned theorem, that $(a_1, a_2, \ldots, a_{n-1})$ and $(a'_1, a'_2, \ldots, a'_{n-1})$ are the same series of numbers, of course implies $\Sigma a_r = \Sigma a'_r$; this relation Cremona demonstrates independently, by consideration of the pencil of curves

$$(aX + bY + cZ) + \theta(a_1X + b_1Y + c_1Z) = 0,$$

(θ a variable parameter), which corresponds in the first plane to the pencil of lines

$$(ax' + by' + cz') + \theta(a_1x' + b_1y' + c_1z') = 0,$$

which pass through a fixed point ($ax' + by' + cz' = 0$, $a_1x' + b_1y' + c_1z' = 0$) in the second figure. In general, in the pencil $U + \theta V = 0$ (U, V given functions of the order n) there are $3(n-1)^2$ values of θ, each giving a nodal curve. But in the present case each of the curves $U = 0$, $V = 0$ has multiple points at the principal points a_r of the first plane: the question is to obtain the number of values which give a curve having one new double point; and this is found to be

$$= 3(n-1)^2 - \Sigma(r-1)(3r+1)a_r.$$

We have $\Sigma r^2 a_r = n^2 - 1$, $\Sigma r a_r = 3n - 3$; or, substituting, the value of θ is $= \Sigma a_r$. But the curves which have an additional double point are those which correspond to the lines which in the second figure pass through one of the principal points a'_r; viz., these are the lines drawn from the point ($ax' + by' + cz' = 0$, $a_1x' + b_1y' + c_1z' = 0$) to the several principal points a'_r; and the number of them is $= \Sigma a'_r$. We have thus the required relation $\Sigma a_r = \Sigma a'_r$.

The Quadric Transformation between Two Planes.

48. This is of course given by what precedes. The principal system in each plane is a set of three points; and the Jacobian of the same plane is the set of three lines joining each pair of points; that is, the three lines of either plane are the principal counter-system of the other plane. But to give the analytical investigation directly: taking the coordinates (x, y, z) to refer to the principal system of the first plane (viz., taking the three points to be the vertices of the triangle formed by the lines $x = 0$, $y = 0$, $z = 0$), then $X = 0$, $Y = 0$, $Z = 0$ being conics through the three points, the functions X, Y, Z will be each of them of the form $fyz + gzx + hxy$; x', y', z' being proportional to three such functions, there will be linear functions of x', y', z' proportional to yz, zx, xy; or taking these linear functions of the original (x', y', z') for the coordinates (x', y', z') of a point in the second plane, the formulæ of transformation will be

$$x' : y' : z' = yz : zx : xy,$$

and we have then conversely

$$x : y : z = y'z' : z'x' : x'y'.$$

That is, the formulæ for the transformation in question are

$$x' : y' : z' = yz : zx : xy, \qquad x : y : z = y'z' : z'x' : x'y'.$$

We at once verify *à posteriori* that the Jacobian in the first plane is $xyz = 0$, and that in the second plane is $x'y'z' = 0$.

The equations may be written

$$x' : y' : z' = \frac{1}{x} : \frac{1}{y} : \frac{1}{z}, \qquad x : y : z = \frac{1}{x'} : \frac{1}{y'} : \frac{1}{z'},$$

or, if we please, $xx' = yy' = zz'$; the transformation is thus given as an inverse transformation.

49. With respect to the metrical interpretation and actual construction of the transformation, it is to be observed that if x, y, z be taken to be proportional (not to given multiples of the perpendicular distances, but) to the perpendicular distances of P from the sides of the triangle in the first plane, and similarly x', y', z' to be proportional to the perpendicular distances of P' from the sides of the triangle in the second plane, then in general the equations of transformation must be written, not as above, but in the form

$$\frac{x'}{f} = \frac{y'}{g} = \frac{z'}{h} = \frac{1}{xyz} \text{ a scalar multiple,}$$

involving arbitrary multipliers $f : g : h$. We may imagine in the second plane a point P'' determined by coordinates (x'', y'', z''), the same coordinates as (x, y, z), that is, proportional to the perpendicular distances of P'' from the sides of the triangle in the second plane, which point P'' corresponds homographically to P in such wise that

$$\frac{x}{f} : \frac{y}{g} : \frac{z}{h} = x'' : y'' : z''.$$

We have then, in the second plane, the two points P', P'' corresponding to each other in such wise that

$$x'x'' = y'y'' = z'z'';$$

and either of these points being given, the other can at once be constructed; viz., it is obvious that, joining P', P'' with any vertex, say A', of the triangle $A'B'C'$, the lines $A'P'$, $A'P''$ are equally inclined to the bisectors of the angle A'; and consequently, P being given, we have the three lines $A'P'$, $B'P''$, $C'P''$ intersecting in a common point P'', which is therefore determined by means of any two of these lines. We have thus a geometrical construction of the transformation between P and P''.

50. The analysis assumes that the principal points A, B, C of the first figure are three distinct points; but they may two of them, or all three, coincide. In the first case, say if B, C coincide, the line BC is still to be regarded as having a definite direction; and taking $x = 0$ for this line, $y = 0$ for the line joining A with (BC), and $z = 0$ an arbitrary line through A, the functions X, Y, Z will be each of them of the form $\frac{1}{2}fy^2 + \frac{1}{2}gzx + \frac{1}{2}hxy$; and replacing, as before, the original x', y', z' by linear functions of these quantities, these linear functions being taken for the coordinates (x', y', z'), we may write

$$x : y : z = y^2 : xy : xz.$$

Forming the converse system, the equations for the transformation are

$$x' : y' : z' = y^2 : xy : xz, \qquad x : y : z = y'^2 : x'y' : x'z';$$

so that the points A', B', C' in the second plane are related as the points in the first plane; viz., B', C' coincide, the line $B'C'$ being definite.

It is easy to verify that the Jacobian in the first plane is $xy^2 = 0$, and the Jacobian in the second plane is $x'y'^2 = 0$.

51. Secondly, if A, B, C all coincide, these being however consecutive points on a curve of finite curvature, or say on a conic; then, taking $x = 0$ for the tangent at (ABC), $z = 0$ for any other tangent, and $y = 0$ for the chord of contact, the functions X, Y, Z will be of the form $ax^2 + b(y^2 - zx) + 2hxy$; whence we may write

$$x' : y' : z' = x^2 : xy : y^2 - xz.$$

Forming the converse equations, the equations of transformation are

$$x' : y' : z' = x^2 : xy : y^2 - xz, \qquad x : y : z = y'^2 - x'y' : x'y' : x'^2.$$

So that the points A', B', C' in the second plane are related as those of the first plane; viz., they are the consecutive points of a curve of continuous curvature.

We may verify that the Jacobian of the first plane is $x^3 = 0$, and the Jacobian of the second plane $x'^3 = 0$.

The Lineo-linear Transformation between Two Planes.

52. We have two equations of the form

$$(a, \ldots \between x, y, z \between x', y', z') = 0,$$
$$(a_1, \ldots \between x, y, z \between x', y', z') = 0;$$

writing these in the form

$$Px' + Qy' + Rz' = 0, \qquad P_1x' + Q_1y' + R_1z' = 0,$$

where (P, Q, R, P_1, Q_1, R_1) are linear functions of (x, y, z), we have

$$x' : y' : z' \text{ proportional to } \begin{vmatrix} P, & Q, & R \\ P_1, & Q_1, & R_1 \end{vmatrix},$$

that is to $X : Y : Z$, where $X = 0$, $Y = 0$, $Z = 0$ are conics each passing through the same three points in the first plane.

And conversely, writing the equations in the form

$$P'x + Q'y + R'z = 0, \qquad P'_1x + Q'_1y + R'_1z = 0,$$

where $(P', Q', R', P'_1, Q'_1, R'_1)$ are linear functions of (x', y', z'), we have

$$x : y : z \text{ proportional to } \begin{vmatrix} P', & Q', & R' \\ P'_1, & Q'_1, & R'_1 \end{vmatrix},$$

that is to $X' : Y' : Z'$, where $X' = 0$, $Y' = 0$, $Z' = 0$ are conics each passing through the same three points in the second plane.

53. The lineo-linear transformation is thus the same thing as the quadric transformation. It is, moreover, clear that the equations must, by linear transformations on the two sets of variables respectively, and by linear combination of the two equations, be reducible into forms giving the before-mentioned values of $x : y : z$ and $x' : y' : z'$ respectively. Thus, in the general case, where in each plane the three points are distinct points, the lineo-linear equations will be reducible to

$$xx' - yy' = 0, \qquad xx' - zz' = 0;$$

in the case where B, C in the first plane, and B', C' in the second plane respectively coincide, the forms will be

$$xx' - yy' = 0, \qquad yz' - y'z = 0;$$

and in the case where A, B, C in the first plane, and A', B', C' in the second plane respectively coincide, the forms will be

$$xy' - yx' = 0, \qquad xz' - yy' + zx' = 0.$$

The determination of the actual formulæ for these reductions would, it is probable, give rise to investigations of considerable interest.

The General Rational Transformation between Two Planes (resumed).

54. Consider, as above, the first plane or figure with a principal system $(a_1, a_2, \ldots, a_{n-1})$, and the second plane or figure with a principal system $(a'_1, a'_2, \ldots, a'_{n-1})$. To a line in the second plane there corresponds in the first plane a curve of the order n passing 1 time through each of the points a_1, 2 times through each of the points a_2, 3 times through each of the points a_3, &c.; or, as we may write this

First figure.	Second figure.
Points $a_1\ a_2\ a_3\ \ldots\ a_{n-1}$	Points $a'_1\ a'_2\ a'_3\ \ldots\ a'_{n-1}$
$1\ 2\ 3\ \ldots\ n-1$ curve order n	1 curve order 1

viz., the 1's denote the number of times which the curve of the order n passes through the several points a_1 respectively; the 2's the number of times which the curve passes through the several points a_2 respectively; and so on.

55. We may, in the second figure, in the place of a line consider a curve of the order k'. If the equation hereof is $(*\$x', y', z')^{k'} = 0$, then the corresponding curve in the first figure is $(*\$X, Y, Z)^{k'} = 0$; viz., this is a curve of the order $k = nk'$. If, however, the curve in the second figure passes once or more times through all or any of the points $a'_1, a'_2, \ldots, a'_{n-1}$, then there will be a depression in the order of the corresponding curve in the first figure; and, moreover, this curve will pass a certain number of times through all or some of the points $a_1, a_2, \ldots, a_{n-1}$. The diagram of the correspondence will be

First figure.	Second figure.
$a_1\ a_2\ a_3\ \ldots\ a_{n-1}$	$a'_1\ a'_2\ a'_3\ \ldots\ a'_{n-1}$
$a_1\ a_2\ a_3\ \ldots\ a_{n-1}$	$a'_1\ a'_2\ a'_3\ \ldots\ a'_{n-1}$
$b_1\ b_2\ b_3\ \ldots\ b_{n-1}$ curve order k	$b'_1\ b'_2\ \ldots\ b'_{n-1}$ curve order k'
$c_1\ c_2\ \ldots$	$c'_1\ c'_2\ \ldots$
$\vdots\ \vdots\ \ldots$	$\vdots\ \vdots\ \ldots$

where a_1, b_1, $c_1 \ldots$ denote the number of times that the curve of the order k passes through the several points a_1 respectively, (viz., the number of the letters a_1, b_1, $c_1 \ldots$ is $= a_1$, any or all of them being zeros,) a_2, b_2, $c_2 \ldots$ the number of times that the curve passes through the several points a_2 respectively, (viz., the number of the letters a_2, b_2, $c_2 \ldots$ is $= a_2$, any or all of them being zeros,) and so on; and the like for the curve in the second figure.

56. By what precedes, it is easy to see that, if the curve k' passes through a point a'_1, then the curve k throws off a line, and the depression of order is $= 1$; so, if the curve passes 2 times, 3 times, … or a'_1 times through the point in question, then the curve throws off the line repeated 2 times, 3 times, … a'_1 times, or the depression of order is $= 2, 3, \ldots$ or a'_1; and the like for each of the points a'_1; so that, writing for shortness $a'_1 + b'_1 + c'_1 + \ldots = \Sigma a'_1$, the depression of order on account of the passages through the several points a'_1 is $= \Sigma a'_1$. Similarly, for each time of passage through a point a'_2, there is thrown off a conic; or if $a'_2 + b'_2 + \ldots = \Sigma a'_2$, then the depression of order is $2\Sigma a'_2$; and so on; and the like for the figure in the other plane; and we thus arrive at the equations

$$k = k'n - \Sigma(a'_1 + 2a'_2 + 3a'_3 \ldots + (n-1)a'_{n-1}),$$

$$k' = kn - \Sigma(a_1 + 2a_2 + 3a_3 \ldots + (n-1)a_{n-1}).$$

57. The simplest case is when the curve k' does not pass through any of the points $a'_1, a'_2, \ldots, a'_{n-1}$. We have then

$$a'_1 = b'_1 = c'_1 = \ldots = 0, \quad a'_2 = b'_2 = \ldots = 0, \quad \ldots \quad a'_{n-1} = b'_{n-1} = \ldots = 0;$$

consequently $k = k'n$. And, moreover, it is easy to see that

$$a_1 = b_1 = \ldots = k', \quad a_2 = b_2 = \ldots = 2k', \quad \ldots \quad a_{n-1} = b_{n-1} = \ldots = (n-1)k';$$

so that the correspondence is:

First Figure.	Second Figure.
$k'\ 2k'\ 3k'\ \ldots\ (n-1)k'$	0
$k'\ 2k'\ 3k'\ \ldots\ (n-1)k'$ curve order $k = nk'$	k'
⋮	curve order k'

We have

$$\Sigma a_1 = k'a_1, \quad \Sigma a_2 = 2k'a_2, \quad \tfrac{1}{2}a_1(a_1 - 1) = k'(k'-1)a_1, \text{ \&c.},$$

and the formulæ for k, k' become

$$k = k'n, \quad k' = kn - k'(a_1 + 4a_2 + \ldots + (n-1)^2 a_{n-1});$$

viz., the second equation is here $k' = kn - k'(n^2 - 1)$; that is, $k'n^2 = kn$, agreeing, as it should do, with the first equation.

58. Moreover, the deficiency-relation is

$$\tfrac{1}{2}(k-1)(k-2) - \Sigma\tfrac{1}{2}\left[k'(k'-1) + 2k'(2k'-1) \ldots + (n-1)k'((n-1)k'-1)\right] = \tfrac{1}{2}(k'-1)(k'-2);$$

or, what is the same thing, this is

$$(nk'-1)(nk'-2) - (k'-1)(k'-2) = k'^2\{a_1 + 4a_2 + \ldots + (n-1)^2 a_{n-1}\} - k'\{a_1 + 2a_2 + \ldots + (n-1)a_{n-1}\}.$$

The right-hand side is

$$k'^2(n^2-1) - k'(3n-3) = (n-1)k'\{(n+1)k' - 3\},$$

and we have thus the identical equation

$$(nk'-1)(nk'-2) - (k'-1)(k'-2) = (n-1)k'\{(n+1)k' - 3\}.$$

59. It should be possible, when the nature of the correspondence between the two planes is completely given, to express each of the numbers $a_1, b_1, c_1, \ldots, a_{n-1}, b_{n-1}, \ldots$ in terms of k', $a'_1, b'_1, c'_1, \ldots, a'_{n-1}, b'_{n-1}, \ldots$; and reciprocally each of the numbers $a'_1, b'_1, c'_1, \ldots, a'_{n-1}, b'_{n-1}, \ldots$ in terms of k, $a_1, b_1, c_1, \ldots, a_{n-1}, b_{n-1}, \ldots$, thus completing a system of relations between the two sets

$$(k,\, a_1, b_1, c_1, \ldots, a_{n-1}, b_{n-1}, \ldots), \quad (k',\, a'_1, b'_1, c'_1, \ldots, a'_{n-1}, b'_{n-1}, \ldots);$$

but even if the theory was known, there would be considerable difficulty in forming a proper algorithm for the expression of these relations.

60. The two curves must have each of them the same deficiency. It is to be noticed, that if the curve in the first plane passes any number of times through a point P, which is not one of the points $a_1, a_2, \ldots, a_{n-1}$, then the corresponding curve in the second plane will pass the same number of times through the corresponding point P', which point will not be one of the points $a'_1, a'_2, \ldots, a'_{n-1}$. The points P, P' will therefore contribute equal values to the deficiencies of the two curves respectively; so that, in equating the two deficiencies, we may disregard P, P', and attend only to the points $a_1, a_2, \ldots, a_{n-1}$ of the first plane, and $a'_1, a'_2, \ldots, a'_{n-1}$ of the second plane. The required relation thus is

$$\begin{aligned} &\tfrac{1}{2}(k-1)(k-2) - \tfrac{1}{2}\{a_1(a_1-1) + a_2(a_2-1) + \ldots + a_{n-1}(a_{n-1}-1)\} \\ = &\tfrac{1}{2}(k'-1)(k'-2) - \tfrac{1}{2}\{a'_1(a'_1-1) + a'_2(a'_2-1) + \ldots + a'_{n-1}(a'_{n-1}-1)\}. \end{aligned}$$

61. In the case of the quadric transformation $n = 2$, we have in the first plane the three points a_1, say these are A, B, C; and in the second plane the three points a'_1, say these are A', B', C'. And if in the first plane the curve of the order k passes a, b, c times through the three points respectively, and in the second plane the corresponding curve of the order k' passes a', b', c' times through the three points respectively, then it is easy to obtain

$$\begin{array}{ll} k' = 2k - a - b - c, & k = 2k' - a' - b' - c', \\ a' = k - b - c, & a = k' - b' - c', \\ b' = k - c - a, & b = k' - c' - a', \\ c' = k - a - b, & c = k' - a' - b'. \end{array}$$

The Quadric Transformation any number of times repeated.

62. We may successively repeat the quadric transformation according to the type

First Fig.	Second Fig.	Third Fig.	Fourth Fig.
$A,\ B,\ C$	$A',\ B',\ C'$	$A'',\ B'',\ C''$	$A''',\ B''',\ C'''$
	$D',\ E',\ F'$	$D'',\ E'',\ F''$	$D''',\ E''',\ F'''$
		$G'',\ H'',\ I''$	$G''',\ H''',\ I'''$

viz., in the transformation between the first and second figures, the principal systems are ABC and $A'B'C'$ respectively; in that between the second and third figures, they are $D'E'F'$ and $D''E''F''$ respectively; in that between the third and fourth figures, they are $G''H''I''$ and $G'''H'''I'''$; and so on. And it is then easy to see that between the first and any subsequent figure we have a rational transformation of the order 2 for the second figure, 4 for the third figure, 8 for the fourth figure, and so on.

63. But to further explain the relation, we may complete the diagram, by taking, in the transformation between the second and third figures, A'', B'', C'' to correspond to A', B', C'; similarly, in that between the third and fourth, A''', B''', C''' to correspond to A'', B'', C''; and D''', E''', F''' to correspond to D'', E'', F''. And so in the transformation between the second and third figure, we may make G', H', I' to correspond to G'', H'', I'', and between the first and second figures make D, E, F to correspond to D', E', F', and G, H, I to G', H', I', the diagram being thus:

First Fig.	Second Fig.	Third Fig.	Fourth Fig.
A, B, C	A', B', C'	A'', B'', C''	A''', B''', C'''
D, E, F	D', E', F'	D'', E'', F''	D''', E''', F'''
G, H, I	G', H', I'	G'', H'', I''	G''', H''', I'''

Observe that in the principal systems (for instance, A, B, C and A', B', C') the points A, B, C correspond, not to the points A', B', C', but to the lines $B'C'$, $C'A'$, $A'B'$ respectively; and so in the other case.

64. Consider now a line in the first figure: there corresponds hereto in the second figure a conic through the points A', B', C'; and to this conic there corresponds in the third figure a quartic curve passing through each of the points A'', B'', C'' once, and through each of the points D'', E'', F'' twice. And conversely, to a line in the third figure corresponds in the second figure a conic through the points D', E', F'; and hereto in the first figure a quartic through the points D, E, F, once and through the points A, B, C twice; that is, we have between the first and third figures a quartic transformation wherein $a_1 = a_2 = 3$ and $a'_1 = a'_2 = 3$, or say a quartic transformation $3_1 3_2$ and $3_1 3_2$. In like manner, passing to the fourth figure, to a line in the first figure corresponds in the fourth figure an octic curve passing through A''', B''', C''' once, through D''', E''', F''' twice, and through G''', H''', I''' four times; and conversely, to a line in the fourth figure there corresponds in the first figure an octic curve passing through the points G, H, I once, the points D, E, F twice, and the points A, B, C four times; that is, between the first and fourth figures we have an octic transformation, wherein $a_1 = a_2 = a_4 = 3$, $a'_1 = a'_2 = a'_4 = 3$, or say a transformation, order 8, of the form $3_1 3_2 3_4$ and $3_1 3_2 3_4$. And so between the first and fifth figures there is a transformation, order 16, of the form $3_1 3_2 3_4 3_8$ and $3_1 3_2 3_4 3_8$.

65. It is, moreover, easy to find the Jacobians or counter-systems in the several transformations respectively. Thus, in the transformation between the first and second figures, in the second figure the Jacobian consists of 3 lines such as $B'C'$ (viz., these are, of course, the lines $B'C'$, $C'A'$, $A'B'$). Hence, in the transformation between the first and third figures, the Jacobian in the third figure consists of

3 conics $B''C''\,(D''E''F'')$,
3 lines $D''E''$;

viz., one of the conics is that through the five points B'', C'', D'', E'', F'', one of the lines that through the two points D'', E''. And so in the fourth figure, the Jacobian consists of

3 quartics $B'''C'''\,(D'''E'''F''')\,(G'''H'''I''')$,
3 conics $D'''E'''\,(G'''H'''I''')$,
3 lines $G'''H'''$;

viz., one of the quartics passes through B''', C'''; through D''', E''', F''' each once; and through G''', H''', I''' each twice. And so in the fifth figure the Jacobian consists of

3 octics $B''''C''''\,(D''''E''''F'''')\,(G''''H''''I'''')\,(J''''K''''L'''')$,
3 quartics $D''''E''''\,(G''''H''''I'''')\,(J''''K''''L'''')$,
3 conics $G''''H''''\,(J''''K''''L'''')$,
3 lines $J''''K''''$,

and so on.

66. The conditions are in each case sufficient for the determination of the curve. This depends on the numerical relation

$$4 + 3\{1 \cdot 2 + 2 \cdot 3 + 4 \cdot 5 + 8 \cdot 9 + \ldots + 2^q(2^q - 1)\} = \tfrac{1}{2} \cdot 2^{q+1}(2^{q+1} + 3).$$

The term in { } is

$$1 + 4 + 16 + \ldots + 2^{2q} + 1 + 2 + 4 + \ldots + 2^q,$$

that is

$$\frac{2^{2q+2}-1}{2^2-1}+\frac{2^{q+1}-1}{2-1},$$

which is

$$=\tfrac{1}{3}[2^{2q+2}+3\cdot 2^{q+1}-4];$$

and the relation is thus identically true.

67. Conversely, in the transformation between the first figure and the several other figures respectively, the Jacobian of the first figure is

3 lines AB and so on; for order 2, between first and second figures;
3 conics $DE\,(ABC)$,
3 lines AB for order 4, between first and third figures;
3 quartics $GH\,(DEF)\,(ABC)$,
3 conics $DE\,(ABC)$,
3 lines AB for order 8, between first and fourth figures;
3 octics $JK\,(GHI)\,(DEF)\,(ABC)$,
3 quartics $GH\,(DEF)\,(ABC)$,
3 conics $DE\,(ABC)$,
3 lines AB for order 16, between first and fifth figures;

and so on.

Special Cases—Reduction of the General Rational Transformation to a Series of Quadric Transformations.

68. It was remarked by Mr Clifford that any Cremona-transformation whatever may be obtained by this method of repeated quadric transformations, if only the principal systems, instead of being completely arbitrary, are properly related to each other. To take the simplest instance; suppose that we have

First figure.	Second figure.	Third figure.
$A,\ B,\ C$	$A',\ B',\ C'$	$B'',\ C''$
		$D'',\ E'',\ F''$
$E,\ F$	$D'=A',\ E',\ F'$	

viz., in the transformation between the first and second figures, we have the principal systems ABC and $A'B'C'$ (arbitrary as before); but in the transformation between the second and third figures, the principal systems are $D'E'F'$ and $D''E''F''$, where D', instead of being arbitrary, coincides with A'. And we then have B'', C'' in the third figure corresponding to B', C' in the second figure, and E, F in the first figure corresponding to E', F' in the second figure. This being so, to a line in the first figure corresponds in the second figure a conic through A', B', C'. But $A'=D'$; viz., this conic passes through a point D' of the principal system of the second figure, in regard to the transformation between the second and third figures. That is, (k, a, b, c referring to the second figure, and k', a', b', c' to the third figure, $k=2$, $a=1$, $b=0$, $c=0$, and therefore $k'=3$, $a'=2$, $b'=1$, $c'=1$,) corresponding to the conic we have in the third figure a curve, order 3 (cubic curve), passing twice through D'', but once through E'' and F'' respectively; this cubic curve passes also through the points B'', C'' which correspond to B', C' respectively; that is,

cubic passes through E'', F'', B'', C'' each 1 time
" " D'' 2 times

or, corresponding to a line in the first figure, we have in the third figure a curve, order 3, passing through four fixed points each 1 time, and through one fixed point 2 times. That is, we have $n=3$, $a'_1=4$, $a'_2=1$. And in the same manner, to a line in the third figure there corresponds in the first figure a cubic through four fixed points (viz., B, C, E, F) each 1 time, and through one fixed point, A, 2 times; so that also $a_1=4$, $a_2=1$. The transformation is thus of the order 3, and the form $4_1\,1_2$ and $4_1\,1_2$ (this is in fact the only cubic transformation; see the Tables, *ante*, No. 41).

69. Mr Clifford has also devised a very convenient algorithm for this decomposition of a transformation of any order into quadric transformations. The quadric transformation is denoted by [3], the cubic transformation by [41], the quartic transformations by [601], [330], the quintic ones by [8001], [3310], [0600], and so on; see the Tables just referred to. (This is substantially the same as a notation employed above, the zeros enabling the omission of the suffixes; viz., $[8001] = 8_1 1_4$; and so in other cases.)

70. The foregoing result is represented thus $[4, 1] = [3\$0, 0, 1)$, which I proceed to explain. Consider in the first figure a line; the symbol [3] denotes that in the second figure we have a conic with three points (a_1'). We are about to apply to this a quadric transformation $(0, 0, 0)$; would denote that the three points of the principal system in the second figure were all of them arbitrary; $(0, 0, 1)$ that one of these points was a point a_1'; $(0, 1, 1)$ that two of them were points a_1'; $(1, 1, 1)$ that all three of them were points a_1'; $(0, 0, 2)$ would denote that one of the points was a point a_2'; only in the present case we can have no such symbol, by reason that there are no points a_2'. Hence [3$001) denotes that the conic has applied to it a quadric transformation such that, in the transformation thereof, one point of the principal system coincides with one of the points (a_1') on the conic. To [3], qua quadric transformation, belongs the number 2; and from 2, (001) we derive 3, (112), (in general k, (a, b, c) gives k', (a', b', c'), where $k' = 2k - a - b - c$, $a' = k - b - c$, $b' = k - c - a$, $c' = k - a - b$). $k = 2$ corresponds to a symbol [3] of one number, $k' = 3$ to a symbol of two numbers; viz., we change [3] into [30]; we then, in the symbols (112) and (001), consider the frequencies of the several numbers $1, 2, \ldots$ taking those in the first symbol as positive, and those in the second symbol as negative; or, what is the same thing, representing the frequency as an index, we have $1^2\, 2^1\, 1^{-1}$; or, combining, $1^{2-1}\, 2^1$; these indices are then added on to the numbers of [30]; viz., the index of 1 to the first number, the index of 2 to the second number (and, in the case of more numbers, so on); [30] is thus converted into [41], and we have the required equation

$$[41] = [3\$001),$$

where the rationale of this algorithmic process appears by the explanation, *ante*, No. 68.

71. As another example take [8001] = [601$003). To [601], *quâ* quartic transformation, belongs the number 4; and from 4, (003) we form 5, (114); where the 5 indicates that [601] is to be changed into [6010]; then (114), (003), writing them in the form $1^2\, 2^0\, 3^{-1}\, 4^1$, show that to the numbers of [6010] we are to add 2, 0, -1, 1; thus changing the symbol into [8001], so that we have the required relation.

72. Mr Clifford calculated in this way the following table, showing how any transformation of an order not exceeding 8 can be expressed by means of a series of quadric transformations; the symbols Cr. 3, Cr. 4.1, 4.2, &c., refer to the order and number of Cremona's tables, *ante*, No. 41.

(Cremona / Clifford reduction table, books pp. 223–224.*)*

$$\begin{aligned}
\text{Cr. 3.}\quad [41] &= [3\$001),\\
\text{Cr. } 4.1 = [601] &= [41\$002) = [3\$001\$002),\\
4.2 = [330] &= [3\$000) = [41\$011) = [3\$001\$011),\\
\text{Cr. } 5.1 = [8001] &= [601\$003) = [3\$001\$002\$003),\\
5.2 = [3310]&\\
5.3 = [0600] &= [41\$001) = [3\$001\$001),\\
&= [330\$111)\\
\text{Cr. } 6.1 = [10,0001] &= [3303\$111) = [3\$001\$011\$111),\\
6.2 = [14200] &= [8001\$004)\\
6.3 = [41300] &= [330\$011) = [3\$001\$002\$003)[004),\\
&= [601\$011)\\
6.4 = [34010] &= [3310\$022) = [3\$001\$011) = [3\$001\$011\$011),\\
&= [3303\$002)\\
&= [3310\$013) = [3\$001\$002\$011)\\
&= [3\$001\$001\$022)\\
&= [3\$000\$002)\\
&= [3\$001\$001\$013)\\
\text{Cr. } 7.1 = [12,00001] &= [10,0001\$005) = [3\$001\$002\$003\$004\$005),\\
7.2 = [330\$001) = [3\$000\$001) = [232100],&\\
7.3 = [034000] &= [3310\$111) = [3\$001\$001\$111),\\
7.4 = [503100] &= [601\$001) = [3\$001\$002\$001),\\
7.5 = [350010] &= [3310\$003) = [3\$001\$001\$003),\\
\text{Cr. } 8.1 = [14,000001] &= [3\$001\$002\$003\$004\$005\$006),\\
8.2 = [3230100] &= [3310\$002) = [3\$001\$001\$002),\\
8.3 = [1322000] &= [3310\$011) = [3\$001\$001\$011),\\
8.4 = [0070000] &= [0340\$222) = [3\$001\$001\$111\$222),\\
8.5 = [3600010] &= [34010\$004) = [3303\$002\$004) = [3\$000\$002\$004),\\
8.6 = [6013000]&\\
&= [601\$000) = [3\$001\$002\$000),\\
8.7 = [0520100] &= [0600\$002) = [3\$000\$111\$222),\\
8.8 = [2051000] &= [4130\$112) = [3\$001\$000\$112),\\
8.9 = [3303000] &= [3\$000\$000\$000).
\end{aligned}$$

73. The reduction as above of a transformation to a series of quadric transformations, enables the determination of the reciprocal transformation; or, what is the same thing, the determination of the Jacobian of the first figure; see the example, *ante*, No. 67, where it appears that the reciprocal transformation of [41] is [41]. But I do not see any easy algorithmic process for the determination of the reciprocal transformation, or still less any general form in which the result can be expressed; and I do not at present pursue the inquiry.

The Rational Transformation between Two Spaces.

74. The general principles have been already explained: the two systems

$$x : y : z : w = X : Y : Z : W \quad\text{and}\quad x : y : z : w = X' : Y' : Z' : W'$$

must be derivable the one from the other; and starting with the first system, this will be the case if only the surfaces $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ have a common intersection equivalent to $n^3 - 1$ points of intersection, but not equivalent to a complete common intersection of n^3 points. The last-mentioned circumstance would arise, if the condition of the common intersection should impose upon the surface more than $\frac{1}{6}(n + 1)(n + 2)(n + 3) - 4$ conditions; viz., the surfaces would then be connected by an identical equation or syzygy $\alpha X + \beta Y + \gamma Z + \delta W = 0$. The common intersection is a figure composed of points and curves: say it is the principal system in the first space; the problem is, to determine a principal system equivalent to $n^3 - 1$ points of intersection but such that the number of conditions to be satisfied by a surface passing through it is not more than

$$\tfrac{1}{6}(n + 1)(n + 2)(n + 3) - 4.$$

75. The following locutions are convenient. We may say that the number of conditions imposed upon a surface of the order n which passes through the common intersection is the *Postulation* of this intersection; and that the number of points represented by the common intersection (in regard to the points of intersection of any three surfaces each of the order n which pass through it) is the *Equivalence* of this intersection. The conditions above referred to are thus

$$\text{Equivalence} = n^3 - 1,$$

$$\text{Postulation} \not> \tfrac{1}{6}(n+1)(n+2)(n+3) - 4.$$

76. It would appear by the analogy of the rational transformation between two planes, that the only cases to be considered are those for which

$$\text{Postulation} = \tfrac{1}{6}(n+1)(n+2)(n+3) - 4;$$

but I cannot say whether this is so.

77. In the transformation between two planes, the two conditions lead, as was seen, to the result that the curve $aX + bY + cZ = 0$ is unicursal. I do not see that in the present case of two spaces, the two conditions lead to the corresponding result that the surface $aX + bY + cZ + dW = 0$ is unicursal; that this is so, appears, however, at once from the general notion of the rational transformation. In fact, the equation in question $aX + bY + cZ + dW = 0$ is satisfied by $x : y : z : w = X' : Y' : Z' : W'$ and $ax' + by' + cz' + dw' = 0$; the last equation determines the ratios $x' : y' : z' : w'$ in terms of two arbitrary parameters (say these are $x' : y'$ and $x' : z'$), and we have then $x : y : z : w$ proportional to rational functions of these two parameters; that is, the surface $aX + bY + cZ + dW = 0$ is unicursal. And similarly the surface $aX' + bY' + cZ' + dW' = 0$ is unicursal.

78. In the most general point of view, the principal system will contain a given number of points which are simple points, a given number which are quadriconical points, a given number which are cubiconical points, &c. &c., on the surfaces; and similarly a given number of curves which are simple curves, a given number which are double curves, &c. &c., on the surfaces. But, to simplify, I will consider that it includes only points which are simple points, and a curve which is a simple curve on the surfaces: this curve may, however, break up into separate curves, and we thus, in fact, admit the case where there are any number of separate curves—each of them a simple curve on the surfaces. It is right to remark that we cannot assert *à priori*—and it is not in fact the case—that the principal system in the second space will be subject to the like restrictions: starting with such a principal system in the first space, we may be led in the second space to a principal system including a curve which is a double curve on the surfaces: an instance of this will in fact occur.

79. It is shown (Salmon's *Solid Geometry*, 2nd ed., p. 283; [Ed. 4, p. 321]), that in the intersection of three surfaces of the orders μ, ν, ρ respectively, a curve of intersection of the order m and class r counts as $m(\mu + \nu + \rho - 2) - r$ points of intersection. For a curve without actual double points or stationary points, we have $r = m(m-1) - 2h$, where h is the number of apparent double points; or, substituting, we have the curve counting for $m(\mu + \nu + \rho - 2) - m(m-1) + 2h$ points of intersection; this is in fact a more general form of the formula, inasmuch as it extends to the case of a curve with actual double points and stationary points. Or, what is the same thing, the three surfaces intersecting in the curve of the order m with h apparent double points, will besides intersect in $\mu\nu\rho - m(\mu + \nu + \rho - 2) + m(m-1) - 2h$ points; viz., the curve may, besides the apparent double points, have actual double points and stationary points; but these do not affect the formula.

80. Some caution is necessary in the application of the theorem. For instance, to consider cases that will present themselves in the sequel: let the surfaces be cubics ($\mu = \nu = \rho = 3$); the number of remaining intersections is given as $27 - 7m + m(m-1) - 2h$. Suppose that the curve consists of four non-intersecting lines, $m = 4$, $h = 6$, the number is given as -1. But observe in this case there are two lines each meeting the four given lines; that is, any cubic surface passing through the four given lines meets these two lines each of them in four points, that is, the cubic passes also through each of the two lines; the complete *curve*-intersection of the surfaces is made up of the six lines $m = 6$, $h = 7$ (since each of the two lines, as intersecting the four lines, gives actual double points, but the two lines together give one apparent double point), and the expression for the number of the remaining points of intersection becomes $27 - 42 + 30 - 14 = 1$, which is correct.

81. Similarly, if the given curve of intersection be a conic and two non-intersecting lines, there is here in the plane of the conic a line meeting each of the two given lines, and therefore meeting the cubic surface, in four points, that is, lying wholly in the cubic surface: the complete *curve*-intersection consists of the conic, the two given lines, and the last-mentioned line, $m = 5$, $h = 5$, and the number of points of intersection $= 27-35+20-10 = 2$, which is correct. Again, if the given curve of intersection be two conics, here the line of intersection of the planes of the conics lies in the cubic surface; or, for the complete *curve*-intersection we have $m = 5$, $h = 4$; and the number of points is $27 - 35 + 20 - 8 = 4$. If in this last case each or either of the conics become a pair of intersecting lines, or if in the preceding case the conic becomes a pair of intersecting lines, the results remain unaltered.

82. If a surface of the order μ pass through a curve of the order m and class r without stationary points or actual double points, this imposes on the surface a number of conditions $= m(\mu + 1) - \frac{1}{2}r$. In the case in question, the value of r is $= m(m-1) - 2h$; or, substituting, the number of conditions is $= (\mu+1)m - \frac{1}{2}m(m-1) + h$; and the formula in this form holds good even in the case where the curve has stationary points and actual double points. Thus $\mu = 3$, the number of conditions is $4m - \frac{1}{2}m(m-1) + h$. If the curve be a line, $m = 1$, $h = 0$, number of conditions is $= 4$; if the curve be a pair of non-intersecting lines, $m = 2$, $h = 1$, number of conditions is $= 8$. And so in general, if the curve consist of k non-intersecting lines ($k = 4$ at most), then $m = k$, $h = \frac{1}{2}k(k-1)$, and the number of conditions is $= 4k$. If the curve be a conic, or a pair of intersecting lines, $m = 2$, $h = 1$, and the number of conditions is $= 7$. If the curve consist of k lines, such that there are θ pairs of intersecting lines, then $m = k$, $h = \frac{1}{2}k(k-1) - \theta$, and the number of conditions is $= 4k - \theta$. It is obvious that, the number of conditions for a line being $= 4$, that for the k lines with θ intersecting pairs must have the foregoing value $-\theta$. In fact, when the lines do not intersect, we take on each line 4 points, and the cubic surface passing through any such 4 points will contain the line; but for two lines which intersect, taking this point, and on each of the intersecting lines 3 other points, the cubic surface through the 7 points will pass through the two lines; and so in other cases.

83. The formula must, in some instances, be applied with caution. Thus, given five non-intersecting lines $k = 5$, $\theta = 0$, and the number of conditions is $= 20$; and a cubic surface cannot be, in general, made to pass through the lines. But if the five lines are met by any other line, then a cubic surface, if it pass through the five lines, will pass through this sixth line; for the six lines $k = 6$, $\theta = 5$, and the number of conditions is $24 - 5 = 19$; so that there is a determinate cubic surface through the six lines, and therefore through the five lines related in the manner just referred to.

84. Recurring to the problem of transformation, it appears by what precedes, that if the principal system in the first plane consists of a_1 points, and of a curve of the order m_1 with h_1 apparent double points (the a_1 points being simple points, and the curve a simple curve on the surfaces), then the conditions for a transformation are

$$(3n-2)m_1 - m_1(m_1-1) + 2h_1 + a_1 = n^3 - 1,$$

$$(n+1)m_1 - \tfrac{1}{2}m_1(m_1-1) + h_1 + a_1 = \tfrac{1}{6}(n+1)(n+2)(n+3) - 4,$$

where, in the second line, instead of $\not>$ I have written $=$. I remark, in passing, that I have ascertained that an actual triple point counts as an apparent double point; or, what is the same thing, that if the curve has

t_1 actual triple points, then we may, instead of h_1, write $h_1 + t_1$. The equations give

$$m_1(4n - 5 - m_1) = \tfrac{1}{3}(n-1)(5n^2 - n - 12) - 2h_1,$$

$$(n-4)m_1 - a_1 = \tfrac{1}{3}(n-1)(2n^2 - 4n - 15),$$

to which may be joined

$$(3n+8)m_1 - 2m_1(m_1 - 1) + 4h_1 + 5a_1 = (n-1)(6n+17).$$

The first two equations for the successive values of n give

$$\begin{array}{lll}
n = 2, & m_1(3 - m_1) = 2 - 2h_1, & 2m_1 + a_1 = 5 \\
n = 3, & m_1(7 - m_1) = 20 - 2h_1, & m_1 + a_1 = 6 \\
n = 4, & m_1(11 - m_1) = 64 - 2h_1, & a_1 = -1 \\
n = 5, & m_1(15 - m_1) = 144 - 2h_1, & -m_1 + a_1 = -20 \\
n = 6, & m_1(19 - m_1) = 270 - 2h_1, & -2m_1 + a_1 = -55 \\
\&\text{c.} & \&\text{c.} & \&\text{c.}
\end{array}$$

85. It is remarkable that for $n = 4$ there is no solution of the geometrical problem; in fact, $a_1 = -1$, a negative value of a_1, shows that this is so. For the higher values of n, there seem to be solutions with large values of m_1, h_1, a_1; for example, $n = 5$, we have $m_1 = 20 + a_1$, is $= 20$ at least. Writing $m_1 = 20$, we have $-100 = 144 - 2h_1$, or $2h_1 = 244$. The highest value of $2h_1$ is $= (m_1 - 1)(m_1 - 2)$, which for $m_1 = 20$ is $= 342$; or the foregoing value $2h_1 = 244$ is admissible. Thus $m_1 = 20$, $h_1 = 122$, a_1 gives a solution; and, moreover, any larger value of m_1, say $m_1 = 20 + \alpha$, gives an admissible solution, $m_1 = 20 + \alpha$, $h_1 = 122 + \frac{1}{2}\alpha(\alpha + 25)$, $a_1 = \alpha$. And so for $n = 6$, &c.; but I have not further examined any of these cases, and do not understand them.

There remain the cases $n = 2$, $n = 3$. For $n = 2$, since $2m_1 + a_1 = 5$, we have $m_1 = 0$, 1, or 2; $m_1 = 0$ gives $h_1 = 0$, which is not admissible. The remaining solutions are $m_1 = 1$, $h_1 = 0$, $a_1 = 3$; and $m_1 = 2$, $h_1 = 0$, $a_1 = 1$.

For $n = 3$, since $m_1 + a_1 = 6$, we have $m_1 = 0, 1, 2, 3, 4, 5$, or 6. $m_1 = 0$ gives $h_1 = 10$; $m_1 = 1$ gives $h_1 = 7$; $m_1 = 2$ gives $h_1 = 5$; $m_1 = 3$ gives $h_1 = 4$: these values are not geometrically admissible. The remaining cases are $m_1 = 4$, $h_1 = 4$, $a_1 = 2$; $m_1 = 5$, $h_1 = 5$, $a_1 = 1$; $m_1 = 6$, $h_1 = 7$, $a_1 = 0$.

86. The reciprocal transformation is in every case of the order $n' = n^2 - m_1$. Hence considering the quadric transformations:

First, the case $n = 2$, $m_1 = 1$, $h_1 = 0$, $a_1 = 3$; the reciprocal transformation is of the order $n' = 3$. Suppose for a moment that the principal system in the second space is of the same nature as that above considered in the first space, consisting of a'_1 points, and a curve of the order m'_1 with h'_1 apparent double points (the a'_1 points each a simple point, and the curve a simple curve on the surfaces $X' = 0$, &c.). Passing back to the original transformation, we should have $2 = \frac{9 - m'_1}{}$, that is, $m'_1 = 7$. But it has just been seen that, for $n = 3$, the only values of m_1 are 4, 5, 6; hence for $n' = 3$ we cannot have $m'_1 = 7$. The explanation is, that the principal system in the second space is not of the form in question; it, in fact, consists (as will appear) of three lines each a simple line, and of another line which is a double line on the surfaces $X' = 0$, &c. In the intersection of any two of these surfaces, the three lines count each once, the double line four times, and the order of the curve of intersection is thus $3 + 4 = 7$, as it should be. The principal system may be characterized $a'_1 = 0$, $m'_1 = 3$, $\nu = 3$, $h'_1 = 0$, where here ν is the number of simple lines and a separate count is made of the double line.

(End of chunk: PDF pages 226–250, book pages 204–228. Paper 447, Cremona-tables of article 41 through the opening of Article 86. The text resumes in the next chunk at the detailed enumeration of the principal systems in three-space.)

Cayley — Collected Mathematical Papers, Vol. VII

Modern reset of pages 251–275 (paper 447 continued, papers 448, 449, 450 begin)

Next, the case $n = 2$, $m_1 = 2$, $h_1 = 0$, $a_1 = 1$: the reciprocal transformation is of the order $n' = 2$; it is evidently not of the form above considered (for this would make the original transformation to be of the order 3). Hence, assuming (as it seems allowable to do) that the principal system does not contain any multiple point or curve, the reciprocal transformation will be of the same form as the original one, viz., we shall have $n' = 2$, $m'_1 = 2$, $h'_1 = 0$, $a'_1 = 1$.

87. Considering next the cubic transformations, or those belonging to $n = 3$; in the case $m_1 = 4$, $h_1 = 4$, $a_1 = 2$, the reciprocal transformation is of the order $9 - 4, = 5$; and in the case $m_1 = 5$, $h_1 = 5$, $a_1 = 1$, the reciprocal transformation is of the order $9 - 5, = 4$: I do not consider these cases. But $m_1 = 6$, $h_1 = 7$, $a_1 = 0$, the reciprocal transformation is of the order $-9+6, = 3$; and assuming (as seems allowable) that the principal system does not contain any multiple point or curve, it must be of the same form as the original transformation, that is, we must have $n' = 3$, $m'_1 = 6$, $h'_1 = 7$, $a'_1 = 0$.

88. The transformations to be studied are thus, 1° The quadri-quadric transformation $n = 2$, $m_1 = 2$, $h_1 = 0$, $a_1 = 1$, and $n' = 2$, $m'_1 = 2$, $h'_1 = 0$, $a'_1 = 1$; the principal system in each space consists of a point and of a conic (which may be a pair of intersecting lines); and the surfaces are quadrics. 2° The quadri-cubic transformation $n = 2$, $m_1 = 1$, $h_1 = 0$, $a_1 = 3$, and $n' = 3$, $a'_1 = 0$, $h'_2 = 3$, $m'_2 = 1$, $h'_2 = 0$: in the first space the principal system consists of three points and a line, and the surfaces are quadrics: in the second figure, the principal system consists of three simple lines and a double line; and the surfaces are cubic surfaces passing through this principal system, that is, they are cubic scrolls. 3° The cubo-cubic transformation $n = 3$, $a_1 = 0$, $m_1 = 6$, $h_1 = 7$, and $n' = 3$, $a'_1 = 0$, $m'_1 = 6$, $h'_1 = 7$; in each space the principal system is a sextic curve with seven apparent double points (but there are different cases to be considered according as the sextic curve does or does not break up into inferior curves), and the surfaces are cubic surfaces through the sextic curve.

The Quadri-quadric Transformation between Two Spaces.

89. Starting from the equations $x' : y' : z' : w' :: X : Y : Z : W$, we have here $X = 0$, &c., quadric surfaces passing through a given point and a given conic (which may be a pair of intersecting lines). Take $x = 0$, $y = 0$, $z = 0$ for the coordinates of the given point; $w = 0$ for the equation of the plane of the conic; the conic is then given as the intersection of this plane by a cone having the given point for its vertex; or say the equations of the conic are $w = 0$, $(a, \ldots \mid x, y, z)^2 = 0$; the general equation of a quadric through the point and conic is $w(\alpha x + \beta y + \gamma z) + h(a, \ldots \mid x, y, z)^2 = 0$; and it hence appears that the equations of the transformation may be taken to be

$$x' : y' : z' : w' \;=\; xw : yw : zw : (a, \ldots \mid x, y, z)^2;$$

these give at once a reciprocal system of the same form; viz., the two sets are

$$x' : y' : z' : w' \;=\; xw : yw : zw : (a, \ldots \mid x, y, z)^2,$$

and

$$x : y : z : w \;=\; x'w' : y'w' : z'w' : (a, \ldots \mid x', y', z')^2.$$

90. The Jacobian of the first space is at once found to be

$$w^2(a, \ldots \mid x, y, z)^2 = 0;$$

that of the second space is of course

$$w'^2(a, \ldots \mid x', y', z')^2 = 0.$$

The two spaces are similar to each other; we may say that there is in each of them a principal point and a principal conic; that the plane of the conic is the principal plane, and the cone having its vertex at the point and passing through the conic is the principal cone. To the principal point of either space corresponds any point whatever in the principal plane of the other space; and conversely. More definitely, the points of the one principal plane and the infinitesimal elements of direction through the principal point of the other space correspond according to the equations $x : y : z = x' : y' : z'$. To any point on the principal conic of either space corresponds in the other space, not a mere element of direction through the principal point of the other space, but a line of the principal cone; that is, to the points of the principal conic of the one space correspond the lines of the principal cone of the other space. The Jacobian of either space, consisting of the principal plane twice, and of the principal cone, is thus the principal counter-system of the other space.

91. {Writing $(a, \ldots \mid x, y, z)^2 = x^2 + y^2 + z^2$, $w = w' = 1$, the equations of transformation become

$$x' : y' : z' : 1 \ = \ x : y : z : x^2 + y^2 + z^2,$$

and

$$x : y : z : 1 \ = \ x' : y' : z' : x'^2 + y'^2 + z'^2,$$

or, what is the same thing, if for shortness

$$x^2 + y^2 + z^2 = r^2, \quad x'^2 + y'^2 + z'^2 = r'^2,$$

the equations are

$$x' = \frac{x}{r^2}, \quad y' = \frac{y}{r^2}, \quad z' = \frac{z}{r^2}; \quad \text{and} \quad x = \frac{x'}{r'^2}, \quad y = \frac{y'}{r'^2}, \quad z = \frac{z'}{r'^2},$$

whence also $rr' = 1$; this is the well known transformation by reciprocal radius vectors.}

92. The principal conic may be a pair of intersecting lines; taking its equations to be $w = 0$, $xy = 0$, the equations of transformation here become

$$x' : y' : z' : w' \ = \ yw : xw : xy : zw,$$

and

$$x : y : z : w \ = \ y'w' : x'w' : x'y' : z'w'.$$

There is no difficulty in the further development of the theory.

The Quadri-cubic Transformation between Two Spaces.

93. It will be convenient to have the unaccented letters (x, y, z, w) referring to the cubic surfaces. I will therefore take the quadric surfaces in the second figure; viz., I will start from the equations $x : y : z : w = X' : Y' : Z' : W'$, where $X' = 0$, $Y' = 0$, $Z' = 0$, $W' = 0$ are quadric surfaces passing through three fixed points (say the principal points) and through a fixed line (say the principal line) in the second figure. Taking $x' = 0$, $y' = 0$ for the planes passing through the principal line and through two of the principal points respectively; $z' = 0$ for the plane passing through the three principal points, $w' = 0$ for an arbitrary plane passing through the first mentioned two principal points, the implicit factors of x', y', w' may be so determined that for the third principal point $x' = y' = -w'$. That is, we shall have

for principal line $x' = 0, \ y' = 0,$
for principal points $(x' = 0, \ z' = 0, \ w' = 0),$
$(y' = 0, \ z' = 0, \ w' = 0),$
$(x' = y' = -w', \ z' = 0),$

and this being so, the equation of a quadric surface through the principal points and line will be

$$(\alpha x' + \beta y')z' + \gamma x'(y' + w') + \delta y'(x' + w'),$$

and the equations of transformation may be taken to be

$$x : y : z : w \ = \ x'z' : y'z' : x'(y'+w') : y'(x'+w').$$

94. Writing these in the extended form

$$x : y : z : w : z-w : x-y \ = \ x'z' : y'z' : x'(y'+w') : y'(x'+w') : z'(x'-y') : w'(x-y)$$

and forming also the equation

$$xy : (xw-yz) \ = \ z' : x'-y',$$

we at once derive the reciprocal system of equations

$$x' : y' : z' : w' \ = \ x(xw-yz) : y(xw-yz) : (x-y)xy : (z-w)xy,$$

so that this is a cubic transformation. And the cubic surface in the first space (corresponding to an arbitrary plane $ax'+by'+cz'+dw'=0$ of the second space) is $(ax+by)(xw-yz)+c(x-y)xy+d(z-w)xy=0$; viz., this is a cubic surface having the fixed double line $(x=0,\ y=0)$, the fixed simple lines $(x=0,\ z=0)$, $(y=0,\ w=0)$, and $(x-y=0,\ z-w=0)$; it has also the variable simple line $(dz+cx=0,\ dw+cy=0)$. The principal figure of the first space thus consists of the three simple lines $(x=0,\ z=0)$, $(y=0,\ w=0)$, $(x-y=0,\ z-w=0)$, and of the line $(x=0,\ y=0)$, a double line counting four times in the intersection of two of the cubic surfaces.

95. The cubic surface as having the double line $(x=0,\ y=0)$ is a cubic scroll, and this line is the nodal directrix thereof; the line $(dz+cx=0,\ dw+cy=0)$ is the simple directrix; the lines $(x=0,\ z=0)$, $(y=0,\ w=0)$, $(x-y=0,\ z-w=0)$ are at once seen to be lines meeting each of these directrix lines; and they are generating lines of the scroll. To explain the generation of the scroll, observe that the section by any plane is a cubic curve having a given double point (viz., the intersection of the plane with the nodal directrix); and three other given points (viz., the intersections of the plane with the three generating lines respectively); this cubic also passes through the intersection of the plane with the simple directrix. Conversely, if the plane be assumed at pleasure, and if, taking for the simple directrix any line which meets the given generating lines, we draw a cubic as above, then the scroll is the scroll generated by a line which meets each of the directrix lines, and also the cubic.

If the plane be taken to pass through any generating line, then the cubic section breaks up into this line, and a conic; the conic does not meet the simple directrix, but it meets the nodal directrix; and any such conic will serve as a directrix; viz., the scroll is generated by the lines which meet the two directrix lines and the conic.

96. Any two scrolls as above meet in the three fixed generating lines, and in the nodal directrix counting four times; they consequently meet besides in a curve of the second order, which is a conic (one of the conics just referred to). In order to further explain the theory, suppose for a moment that the two scrolls had only a common nodal directrix; they would besides meet in a quintic curve; this curve would meet the nodal directrix in four points, viz., the points at which the two scrolls have a common tangent plane. Now if at any point of the nodal directrix the two scrolls have a common generating line, then the plane through this line and the nodal line is one of the two tangent planes of each scroll; that is, the scrolls have this plane for a common tangent plane. Hence, in the case of the common three generating lines, the points where these meet the nodal line are three of the four points just referred to; there remains therefore one point, which is the point where the conic meets the nodal line; through this point there are for each of the scrolls two generating lines; one of these for the first scroll, and one for the second scroll, lie in a plane with the nodal line; the other two determine the plane of the conic; and the tangent to the conic at its intersection with the nodal line is the intersection of the plane of the conic with the plane of the first-mentioned two generating lines.

97. Analytically we have the two equations

$$c(x-y)xy+(ax+by)(xw-yz)+d(z-w)xy=0,$$
$$c'(x-y)xy+(a'x+b'y)(xw-yz)+d'(z-w)xy=0;$$

or, combining these equations so as to eliminate successively the terms in $x(xw-yz)$ and $y(xw-yz)$, and for this purpose writing

$$(bc'-b'c,\ ca'-c'a,\ ab'-a'b,\ ad'-a'd,\ bd'-b'd,\ cd'-c'd) = (a, b, c, f, g, h),$$

and therefore

$$af + bg + ch = 0,$$

we have

$$\begin{aligned} h(x-y)x - c(xw-yz) - f(z-w)x = 0, \\ -h(x-y)y + c(xw-yz) - g(z-w)y = 0, \end{aligned}$$

and multiplying the first of these by $c+g$ and the second by $c-f$, and adding, the whole divides by $x-y$, and the final result is

$$(c+g)(hx-fy) - (c-f)(ay+gx) = 0;$$

viz., this is the equation of the plane of the conic.

98. Any two scrolls as above meeting in a conic, a third scroll will meet the conic in six points; but these include the point on the nodal directrix twice, and the points on the three fixed generating lines each once; there is left a single point of intersection, viz., this is the one variable point of intersection of the three scrolls which is in accordance with the theory.

99. For the Jacobian of the second space, we have

$$\left|\begin{array}{ccccc} z' & . & x', & . & \\ . & z' & y' & . & \\ y'+w' & x'+w' & . & x' & \\ . & . & x'+w' & 0 & y' \end{array}\right| = 0;$$

that is, $x'y'z'(x'-y') = 0$; viz., $z' = 0$ is the plane containing the three principal points; and $x' = 0$, $y' = 0$, $x'-y' = 0$ are the planes which pass through the principal line and the three principal points respectively.

100. For the Jacobian of the first space, we have

$$\left|\begin{array}{cccc} 2xw-yz & -xz & -xy & x^2 \\ yw & xw-2yz & -y^2 & xy \\ y(z-w) & x(z-w) & xy & -xy \\ (z-w)y & (z-w)x & xy & -xy \end{array}\right| = 0;$$

that is, $3x^2y(x-y)^2(xw-yz) = 0$; viz., $x = 0$, $y = 0$, $x-y = 0$ are the planes through the nodal directrix and the three fixed generators respectively (each plane therefore occurring twice); and $xw-yz = 0$ is the quadric scroll generated by the lines which meet each of the three generators ($x = 0,\ z = 0$), ($y = 0,\ w = 0$), ($x-y = 0,\ z-w = 0$); this scroll passing also through the nodal directrix $x = 0$, $y = 0$.

The Cubo-cubic Transformation between Two Spaces.

101. The principal system in the first space is a sextic curve with 7 apparent double points; but this curve may be either a single curve, or it may break up into inferior curves. I have not examined all the cases which may arise; but the two extreme cases are (A) The sextic curve breaks up into six lines, viz., two non-intersecting lines, and four other lines each meeting each of the two lines (this implies that no two of the four lines meet each other): here the two lines give 1 apparent double point, and the four lines give 6 apparent double points; total number is = 7, as it should be. (B) The curve is a proper sextic curve, with 7 apparent double points: this gives, as will be shown, the general lineo-linear transformation. The two cases are each of them symmetrical.

(A) The Principal System consists of Six Lines.

102. Taking in the first space, for the equations of the two lines, $(x=0,\ y=0)$ and $(z=0,\ w=0)$, and for the equations of the four lines, $(x=0,\ z=0)$, $(y=0,\ w=0)$, $(x-y=0,\ z-w=0)$, $(x-py=0,\ z-qw=0)$, then, if the equations of transformation are taken to be

$$\begin{aligned} x'-py' : x'-y' : z'-qw' : z'-w' = {} & (x-py)(xw-yz) \\ & :\ (x-y)(qxw-pyz) \\ & :\ (z-qw)(xw-yz) \\ & :\ (z-w)(qxw-pyz); \end{aligned}$$

these lead conversely (see post, No. 104) to a like system,

$$\begin{aligned} x-py : x-y : z-qw : z-w = {} & (x'-py')M' \\ & :\ (x'-y')N' \\ & :\ M'(z'-qw') \\ & :\ (z'-w')N', \end{aligned}$$

where for shortness

$$\begin{aligned} M' &= p(q-1)^2x'w' - q(p-1)^2y'z' + (pq-1)(p-q)\,y'w', \\ N' &= (q-1)^2x'w' - (p-1)^2y'z' + (pq-1)(p-q)\,y'w'; \end{aligned}$$

or, as these are better written,

$$\begin{aligned} M' &= -q(p-1)y'\{(p-1)z'-(pq-1)w'\} + p(q-1)w'\{(q-1)x'-(pq-1)y'\}, \\ N' &= -(p-1)y'\{(p-1)z'-(pq-1)w'\} + (q-1)w'\{(q-1)x'-(pq-1)y'\}. \end{aligned}$$

Hence the principal system in the second plane is composed of the two non-intersecting lines $(x'=0,\ y'=0)$, $(z'=0,\ w'=0)$ and the four lines $\{(p-1)z'-(pq-1)w'=0,\ (q-1)x'-(pq-1)y'=0\}$, $(y'=0,\ w'=0)$, $(x'-y'=0,\ z'-w'=0)$, $(x'-py'=0,\ z'-qw'=0)$, each meeting each of the two lines.

103. The Jacobian of the first space is

$$\begin{vmatrix} 2xw-yz-pyw, & -xz-pxw+2pyz, & -y(x-py), & x(x-py) \\ 2qxw-pyz-qyw, & -pxz-qxw+2pyz, & -py(x-y), & -x(x-y) \\ w(z-qw), & -z(z-qw), & xw-2yz+qyw, & xz-2qxw+qyz \\ qw(z-w), & -pz(z-w), & qxw-2pyz+pyw, & qxz-2qxw+pyz \end{vmatrix} = 0,$$

viz., this is $xyzw(x-y)(x-py)(z-w)(z-qw)=0$, the equation of the planes each passing through one of the four lines and one of the two lines.

Similarly, the Jacobian of the second space is

$$\{(q-1)x'-(pq-1)y'\}\{(p-1)z'-(pq-1)w'\}(x'-y')(z'-w')(x'-py')(z'-qw')\,y'w'=0;$$

viz., this is the equation of the eight planes each passing through one of the four lines and one of the two lines.

104. To effect the foregoing transformation, writing

$$\begin{aligned} x' : y' : z' : w' = {} & (x-py)(xw-yz) \\ & :\ (x-y)(qxw-pyz) \\ & :\ (z-qw)(xw-yz) \\ & :\ (z-w)(qxw-pyz); \end{aligned}$$

or what will ultimately be the same thing, but it is more convenient for working with,

$$\begin{aligned} x' &= (x-py)(xw-yz), \\ y' &= (x-y)(qxw-pyz), \\ z' &= (z-qw)(xw-yz), \\ w' &= (z-w)(qxw-pyz), \end{aligned}$$

these give

$$\begin{aligned} x-py &= M'x', \\ x-y &= N'y', \\ z-qw &= M'z', \\ z-w &= N'w', \end{aligned}$$

where M', N' are quantities which have to be determined; and thence

$$\begin{aligned} (1-p)x &= M'x'-pN'y', \\ (1-p)y &= M'x'-N'y', \\ (1-q)z &= M'z'-qN'w', \\ (1-q)w &= M'z'-N'w'; \end{aligned}$$

whence also

$$\begin{aligned} (1-p)(1-q)(xw-yz) &= N'\big[\{(q-1)x'w'-(p-1)y'z'\}M'+(p-q)\,y'w'N'\big], \\ (1-p)(1-q)(qxw-pyz) &= M'\big[-(p-q)x'z'M'+\{(pq-q)x'w'-(pq-p)y'z'\}N'\big]; \end{aligned}$$

but we have

$$\frac{xw-yz}{qxw-pyz}=\frac{x'}{y'}\cdot\frac{x-py}{x-y}=\frac{x'}{y'}\cdot\frac{M'x'}{N'y'}=\frac{N'}{M'},$$

or, substituting,

$$M'\{(q-1)x'w'-(p-1)y'z'\}+N'(p-q)y'w'=M'\{-(p-q)x'z'\}+N'\{(pq-q)x'w'-(pq-p)y'z'\};$$

$$M'\{(q-1)x'w'-(p-1)y'z'+(p-q)x'z'\}=N'\{(pq-q)x'w'-(pq-p)y'z'-(p-q)y'w'\};$$

or, what is the same thing,

$$\begin{aligned} M' &= (pq-q)x'w'-(pq-p)y'z'-(p-q)y'w', \\ N' &= (q-1)x'w'-(p-1)y'z'+(p-q)x'z'; \end{aligned}$$

viz., M', N' having these values, the original equations

$$\begin{aligned} x':y':z':w' = {} & (x-py)(xw-yz) \\ & :\ (x-y)(pxw-qyz) \\ & :\ (z-qw)(xw-yz) \\ & :\ (z-w)(pxw-qyz), \end{aligned}$$

give

$$x-py:x-y:z-qw:z-w \;=\; M'x':N'y':M'z':N'w'.$$

If, in these equations, in place of (x',y',z',w') we write $(x'-py',\ x'-y',\ z'-qw',\ z'-w')$, the new values of M', N' are found to be

$$\begin{aligned} M' &= p(q-1)^2x'w'-q(p-1)^2y'z'+(pq-1)(p-q)\,y'w', \\ N' &= (q-1)^2x'w'-(p-1)^2y'z'+(pq-1)(p-q)\,y'w', \end{aligned}$$

and we have the formulae of No. 102.

(B) The Principal System of a Proper Sextic Curve; the Lineo-linear Transformation between Two Spaces.

105. I start with the lineo-linear transformation, and show that this is in fact a transformation such that the principal system in either space is a sextic curve with seven apparent double points. I do not attempt any formal proof, but assume that the lineo-linear transformation is the most general one which gives rise to such a principal system.

We have between (x, y, z, w), (x', y', z', w') three lineo-linear equations; writing these first under the form

$$\begin{aligned}(P_1, Q_1, R_1, S_1 \mid x', y', z', w') &= 0,\\ (P_2, Q_2, R_2, S_2 \mid x', y', z', w') &= 0,\\ (P_3, Q_3, R_3, S_3 \mid x', y', z', w') &= 0,\end{aligned}$$

we have $x' : y' : z' : w' = X : Y : Z : W$, where X, Y, Z, W are the determinants (each with its proper sign) formed out of the matrix

$$\begin{pmatrix} P_1 & Q_1 & R_1 & S_1 \\ P_2 & Q_2 & R_2 & S_2 \\ P_3 & Q_3 & R_3 & S_3 \end{pmatrix}.$$

106. Each of the surfaces $X = 0$, $Y = 0$, $Z = 0$, $W = 0$, or generally any surface $aX+bY+cZ+dW = 0$, is thus a cubic surface passing through the curve

$$\begin{vmatrix} P_1 & Q_1 & R_1 & S_1 \\ P_2 & Q_2 & R_2 & S_2 \\ P_3 & Q_3 & R_3 & S_3 \end{vmatrix} = 0,$$

which is at once seen to be of the order 6. In fact any two of these surfaces, for instance

$$\begin{vmatrix} P_1 & Q_1 & R_1 \\ P_2 & Q_2 & R_2 \\ P_3 & Q_3 & R_3 \end{vmatrix} = 0 \quad \text{and} \quad \begin{vmatrix} P_1 & Q_1 & S_1 \\ P_2 & Q_2 & S_2 \\ P_3 & Q_3 & S_3 \end{vmatrix} = 0$$

have in common a curve

$$\begin{vmatrix} P_1 & Q_1 \\ P_2 & Q_2 \\ P_3 & Q_3 \end{vmatrix} = 0,$$

which is of the order 3; they consequently besides intersect in a curve of the order 6, which is the before mentioned curve of intersection of all the surfaces. And it further appears that the number of the apparent double points is $= 7$; in fact the formula in the case of two surfaces of the orders μ, ν, the complete intersection of which consists of a curve of the order m with h apparent double points, and of a curve of the order m' with h' apparent double points, the numbers m, m', h, h' are connected by the equation $m'-m-2(h-h') = (m-\mu)(\nu-1)$. (Salmon's *Solid Geometry*, 2nd Ed., p. 273 [Ed. 4, p. 311]). Hence, in the case of the two cubic surfaces intersecting as above (since for the cubic curve we have $m' = 3$, $h' = 1$, and for the sextic $m = 6$), the formula becomes $2(h-1) = 12$, that is $h = 1+6 = 7$; or the number of apparent double points is $= 7$.

107. It thus appears that the principal system in the first plane is a curve of the order 6, with seven apparent double points: it is to be added that there are not in general any actual double points or stationary points, so that the class of the curve is $6\cdot5-2\cdot7, = 16$, and its deficiency is $\frac{1}{2}\cdot5\cdot4-7, = 3$. For convenience I will refer to this as the curve Σ.

The transformation is obviously a symmetrical one; hence the principal system in the second space is in like manner a curve of the order 6, with seven apparent double points; say it is the curve Σ'.

108. Consider in the first space any point P on the curve Σ; for this point the three equations

$$\begin{aligned}(P_1, Q_1, R_1, S_1 \mid x', y', z', w') &= 0,\\ (P_2, Q_2, R_2, S_2 \mid \;\;,\;\; \;\;) &= 0,\\ (P_3, Q_3, R_3, S_3 \mid \;\;,\;\; \;\;) &= 0,\end{aligned}$$

are not independent, but are equivalent to two linear equations in (x', y', z', w'); that is, to the point P on the curve Σ there corresponds in the second space, not a determinate point P', but any point whatever on a certain line L'; or say to the point P on Σ there corresponds a line L'; and as P describes the curve Σ, L' describes a scroll Π'; that is, to the curve Σ there corresponds a scroll Π', the principal counter-system in the second space. Similarly to the curve Σ' there corresponds a scroll Π, the principal counter-system of the first space.

109. The scroll Π is the Jacobian of the first space; and as such it is of the order 8, having the curve Σ for a triple line; and it thus appears that the Jacobian of the first space is a scroll (a theorem the analytical verification of which seems by no means easy). But without assuming the identity of the scroll Π with this Jacobian, or taking the order of the scroll to be known, I proceed to show that the scroll Π is the scroll generated by the lines each of which meets the curve Σ three times; it will thereby appear that the order is $= 8$, and that the curve is a triple line on the scroll.

Consider a point P' on Σ', and the corresponding line L of the first space; take a plane Θ' in the second space; corresponding to it the cubic surface Θ in the first space. By imposing a single relation on the coefficients (a, b, c, d) in the equation $ax' + by' + cz' + dw' = 0$ of the plane Θ', we make it pass through the point P'; therefore by imposing this same single relation on the coefficients (a, b, c, d) of the cubic surface Θ, we make it pass through the line L. Θ is a cubic surface through Σ; and it is easy to see that the effect will be as above only if the line L cuts the curve Σ three times; this being so, the general cubic surface Θ meets L in three points (viz., the three intersections of L with Σ), and if Θ be made to pass through a fourth point on the line L, it will pass through the line L; it thus appears that the line L meets Σ three times, and consequently that the scroll Π is generated by the lines which meet Σ three times.

110. The theory of a scroll so generated is considered in my "Memoir on Skew Surfaces, otherwise Scrolls"(1). Writing $m = 6$, $h = 7$ and therefore $M\ [= \tfrac{1}{2}m(m-1) - h],\ = 8$, the order of the scroll is $(\tfrac{1}{6}[m]^3 + (m-2)M = 40 - 32)\ = 8$; but calculating the values of

$$NG(m^2) = \tfrac{1}{8}[m]^4 + 6m + M(3[m]^2 - 12m + 33) + M^2 \cdot 3,$$
$$NR(m^2) = \tfrac{1}{30}[m]^5 + \tfrac{1}{4}[m]^4 - \tfrac{2}{3}[m]^3 - 3m + M(\tfrac{1}{2}[m]^3 - 4m^2 + 8m - 20) + M^2(\tfrac{1}{2}[m]^2 - 27m)$$

these are found to be respectively $= 0$; viz., there are no nodal generators, and no nodal residue; the sextic curve Σ is a triple curve on the surface, and there is not any other multiple line.

111. It may be remarked that any plane Θ' meets the sextic curve Σ' in six points; hence the corresponding cubic surface Θ contains six lines, generatrices of Π, and, therefore, each meeting the curve Σ three times; say six lines L. Through one of these lines L, draw to the cubic surface Θ a triple tangent plane meeting it in the line L and in two other lines, say M, N; this plane must meet Σ in three new points which must lie on the lines M, N; viz., one of these lines must pass through two of the points, and the other line through the third point.

Addition — September, 1870.

[Some corrections have been made in accordance with the concluding paragraph of a paper "Note on the Rational Transformation and on Special Systems of Points," 450.]

The formulae of No. 84 are included in the following more general formulae; viz., if the principal system consist of α_1 points, each a simple point, α_2 points each a quadri-conical point, α_3 points each a cubi-conical point, &c., and of a simple curve order m_1 with h_1 apparent double points, a double curve order m_2 with h_2 apparent double points, and so on; and if moreover, the curves m_1, m_2 intersect in $k_{1,2}$ points, the curves m_1, m_3 in $k_{1,3}$ points, &c.; then writing in

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[1] *Phil. Trans.* vol. CLIII. 1863, pp. 453–483, [339]. See the Table $S(m^2)$ &c., p. 457; in the value of $NR(m^2)$ instead of term $+3m$ read $-3m$. [This correction should have been made in the present Reprint.]

general $p = \frac{1}{2}m(m-1) - h$; that is, $p_1 = \frac{1}{2}m_1(m_1-1) - h_1$, $p_2 = \frac{1}{2}m_2(m_2-1) - h_2$, &c., I find that the general condition of equivalence is

$$\left.\begin{aligned}
&\alpha_1 + (3n-2)\alpha_2\\
&\quad + 8\alpha_3 + (12n-16)\alpha_3\\
&\quad + r^2\alpha_r + (3r^2n - 2r^2)\alpha_r - 2r^2p_r\\
&\quad - 5k_{1,2} - 8k_{1,3} \ldots - (3r-1)k_{1,r}\\
&\qquad - 28k_{2,3} - \ldots\\
&\qquad\qquad - r^2s^2(r,s)\,k_{r,s}\ (s<r)
\end{aligned}\right\} = n^2 - 1;$$

and that the general condition of postulation is

$$\left.\begin{aligned}
&\alpha_1 + (n+1)\alpha_1 + p_1\\
&\quad + \tfrac{1}{6}r(r+1)(r+2)\alpha_r\\
&\quad + [\tfrac{1}{2}r(r+1)n - \tfrac{1}{6}r(r+1)(2r-5)]m_r\\
&\quad - \tfrac{1}{24}\big[(r-1)(r-2)(r-3)(r-4)\\
&\qquad + 4r(r+1)(2r+1)\big]p_r\\
&\quad - \tfrac{5}{2}k_{1,2} - 5k_{1,3} \ldots - rk_{1,r}\\
&\qquad - 8k_{2,3}\\
&\qquad - \tfrac{1}{6}r(r+1)(r+1-\tfrac{1}{2}(r+2))\,k_{r,s}\ (s<r)
\end{aligned}\right\} = \tfrac{1}{24}(n+1)(n+2)(n+3) - 4;$$

in which formulae it is however assumed that the curves have not any actual multiple points. This implies that if any one of the curves, say m_r, break up into two or more curves, the component curves do not intersect each other; for, of course, any such point of intersection would be an actual double point on the curve m_r. I believe, however, that the formulae will extend to this case by admitting for s the value $s = r$; viz., if we suppose the curve m_r to be the aggregate of the two curves m'_r, m''_r intersecting in κ_r points, then that the corresponding terms in the equivalence-equation are

$$\begin{aligned}
&r^2(n^2 - (3rn - 2r^2))\\
&\quad + r^2(m'_r + m''_r) - 2r^2(p'_r + p''_r) - r^2\kappa_r,
\end{aligned}$$

and that those in the postulation-equation are

$$\begin{aligned}
&-[\tfrac{1}{2}r(r+1)n - \tfrac{1}{6}r(r+1)(2r-5)](m'_r + m''_r)\\
&\quad + \tfrac{1}{24}[(r-1)(r-2)(r-3)(r-4) - 4r(r+1)(2r+1)](p'_r + p''_r)\\
&\qquad - \tfrac{1}{2}r(r+1)(2r+1)\kappa_r.
\end{aligned}$$

Let the r-tuple curve consist of three right lines meeting in a point: this is an actual triple point, and the formulae do not apply. But calculating the postulation-terms by the formula, we have $m_r = 3$, $p_r = \frac{1}{2}\cdot 3\cdot 2 - 0, = 3$; and the terms are

$$-[\tfrac{1}{2}r(r+1)n - \tfrac{1}{6}r(r+1)(2r-5)]\cdot 3 + \tfrac{1}{24}[(r-1)(r-2)(r-3)(r-4) + 4r(r+1)(2r+1)],$$

which are $= \frac{1}{2}r(r+1)(3n-4r+4) - \frac{1}{24}(r-1)(r-2)(r-3)(r-4)$, or say $= \frac{1}{2}r(r+1)(3n-4r+4) + \frac{1}{24}(-r^4 + 10r^3 - 35r^2 + 50r - 24)$.

I have found by an independent investigation that this value requires the correction

$$+\tfrac{1}{24}[r^4 - 8r^3 + 30r^2 - 56r + 24 + \tfrac{1}{2}\{1 - (-)^r\cdot 1\}],$$

and that the true value of the postulation is

$$H = \tfrac{1}{2}r(r+1)(3n-4r+6) + \tfrac{1}{12}[2r^3 - 5r^2 - 6r + \tfrac{1}{2}\{1 - (-)^r\cdot 1\}],$$

viz., that this is the number of the conditions to be satisfied that a surface of the order n may have for an r-tuple curve three given right lines meeting in a point.

448.

NOTE ON THE CARTESIAN WITH TWO IMAGINARY AXIAL FOCI.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 181, 182. Read June 9, 1870.]

Let A, A', B, B' be a pair of points and antipoints; viz.,

$$\begin{aligned}&(A, A') \text{ the two imaginary points, coordinates } (\pm\beta i, 0),\\ &(B, B') \text{ the two real points, coordinates } (0, \pm\beta);\end{aligned}$$

and write ρ, ρ', σ, σ' for the distances of a point (x, y) from the four points respectively; viz.,

$$\begin{aligned}\rho &= \sqrt{(x+\beta i)^2+y^2}, &\sigma &= \sqrt{x^2+(y+\beta)^2},\\ \rho' &= \sqrt{(x-\beta i)^2+y^2}, &\sigma' &= \sqrt{x^2+(y-\beta)^2}.\end{aligned}$$

We have

$$\rho^2+\rho'^2 = 2x^2+2y^2-2\beta^2 = 2y^2-\sigma^2-\sigma'^2+4\beta^2,$$
$$\rho\rho' = \sqrt{(x+\beta i+yi)(x+\beta i-yi)(x-\beta i+yi)(x-\beta i-yi)} = \sigma\sigma';$$

and thence

$$\begin{aligned}(\rho+\rho')^2 &= (\sigma+\sigma')^2-4\beta^2,\\ (\rho-\rho')^2 &= (\sigma-\sigma')^2-4\beta^2,\end{aligned}$$

or say

$$\begin{aligned}\rho+\rho' &= \sqrt{(\sigma+\sigma')^2-4\beta^2},\\ i(\rho-\rho') &= \sqrt{4\beta^2-(\sigma-\sigma')^2}.\end{aligned}$$

The equation of a Cartesian having the two imaginary axial foci A, A' is

$$(p+qi)\rho+(p-qi)\rho'+2k^2 = 0;$$

viz., this is

$$p(\rho+\rho')+qi(\rho-\rho')+2k^2 = 0;$$

or, what is the same thing, it is

$$p\sqrt{(\sigma+\sigma')^2-4\beta^2}+q\sqrt{4\beta^2-(\sigma-\sigma')^2}+2k^2 = 0,$$

which is the equation expressed in terms of the distances σ, σ' from the non-axial real foci B, B'. Of course, the radicals are to be taken with the signs ±. This equation gives, however, the Cartesian in combination with an equal curve situate symmetrically therewith in regard to the axis of y.

The distances σ, σ' may conveniently be expressed in terms of a single variable parameter θ; in fact, we may write

$$\begin{aligned}\pm p\sqrt{(\sigma+\sigma')^2-4\beta^2} &= -k^2-k\theta,\\ \pm q\sqrt{4\beta^2-(\sigma-\sigma')^2} &= -k^2+k\theta;\end{aligned}$$

that is

$$\begin{aligned}(\sigma+\sigma')^2-4\beta^2 &= \frac{k^2}{p^2}(k+\theta)^2,\\ 4\beta^2-(\sigma-\sigma')^2 &= \frac{k^2}{q^2}(k-\theta)^2;\end{aligned}$$

and therefore

$$\sigma + \sigma' = \pm\sqrt{4\beta^2 + \frac{k^2}{p^2}(k+\theta)^2},$$

$$\sigma - \sigma' = \pm\sqrt{4\beta^2 - \frac{k^2}{q^2}(k-\theta)^2};$$

so that, assigning to θ any given value, we have σ, σ', and thence the position of the point on the curve. We may draw the hyperbola $y^2 = 4\beta^2 + \frac{k^2}{p^2}x^2$, and the ellipse $y^2 = 4\beta^2 - \frac{k^2}{q^2}x^2$; and then measuring off in these two curves respectively the ordinates which belong to the abscissae $k+\theta$ for the hyperbola, $k-\theta$ for the ellipse, we have the values $\sigma+\sigma'$ and $\sigma-\sigma'$, which determine the point on the curve. Considering k, p, q, β as disposable quantities, the conics may be any ellipse and hyperbola whatever, having a pair of vertices in common; and the complete construction is,—From the

fixed point K in the axis of x, measure off in opposite directions the equal distances KM, KN, and take

$\frac{1}{2}(\sigma+\sigma')$ the ordinate at M in the hyperbola,
$\frac{1}{2}(\sigma-\sigma')$ N in the ellipse,

where σ, σ' denote the distances of the required point from the fixed points B and B' respectively, the distance of each of these from the origin being $= \frac{1}{2}$ the common semi-axis. We may imagine N travelling from one extremity of the x-axis of the ellipse to the other, the value of $\sigma+\sigma'$ will be real and greater than BB', that of $\sigma-\sigma'$ real and less than BB', and the point (σ, σ') will be real. The construction gives, it will be observed, the two symmetrically situated curves.

The x-semi-axis of the ellipse is $\frac{q}{k}\cdot 2\beta$, and the form of the curve depends chiefly on the value of the ratio $k : \frac{q}{k}\cdot 2\beta$; or, what is the same thing, $k^2 : 2\beta q$. We see, for instance, that, in order that the curve may meet the axis of y in two real points between the foci, the value $\sigma - \sigma' = \mathrm{a} - \mathrm{a}'$; viz., that we must give a real value of must have $\frac{k^4}{4\beta^2} > 4k^2$; that is, $\beta^2 q^2 > k^2$, or $k^2 < 2\beta q$. If k has this value, viz., $k = \frac{q}{k}\cdot 2\beta = \frac{1}{2}$ semi-axis, the curve touches the axis of y at the origin; if $k < \frac{1}{2}$ semi-axis, the curve cuts the axis of y in two real points between the foci; if $k > \frac{1}{2}$ semi-axis, the curve does not cut the axis of y between the foci.

449.

SKETCH OF RECENT RESEARCHES UPON QUARTIC AND QUINTIC SURFACES.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 186–195. Read Nov. 10, 1870.]

The classification of quartic surfaces is even as to its highest divisions incomplete; and it is by no means easy to make it at once exhaustive and precise; an enumeration of all the prima facie possible cases would include forms which do not really exist. Thus the singular curve (if any) is of the order 1, 2, or 3; but in the case where the order is = 3, the curve, as is at once evident, cannot be a plane cubic, nor (among other excluded forms) a system of three non-intersecting lines. And certain forms of the singular curve, e.g. all but one of the admissible forms of a curve of the order 3, make the surface to be a scroll, so that, if (as is convenient) we wish to separate the scrolls, certain forms otherwise admissible must be excluded. The expression "singular" means double or cuspidal, or refers to a higher singularity, but the cases of higher singularity are very special. I will, at the cost of some inaccuracy, use the expression "nodal" as meaning, in general, double, but as including the signification "cuspidal"; and, if there are any cases of higher singularity, as extending to cases of higher singularity: and I provisionally arrange the non-scrolar quartic surfaces as follows:

1. Without a nodal curve.
2. With a nodal line.
3. With a nodal conic, or line-pair (pair of intersecting lines).
4. With three nodal lines (not in the same plane) meeting in a point.

(Observe that the omitted cases are cases which, as I believe, ought to be omitted; thus the case of a nodal skew cubic is omitted, because the surface is then of necessity a scroll.) And to these I join

5. The quartic scrolls;

omitting altogether the torse and cones.

The references, by the name of the author, and number (if any) of his paper, are to the subjoined list of Memoirs.

As to the scrolls, we have Cayley (3) and (4), and Cremona; the division into 12 species is, I believe, complete: see post, the remarks upon Schwarz's paper on quintic scrolls.

As regards the non-scrolar surfaces:

1. Without a singular curve. The surface may be without a cnicnode (conical point), or it may have any number of cnicnodes up to 16, Cayley (7): the cases of singularity higher than a cnicnode are probably very numerous, but they have been scarcely at all examined. The memoir just referred to relates chiefly to the several cases of not more than 10 nodes; the cases of 11, 12, 13, 14, 15, 16 nodes are considered incidentally, Kummer (2), but it was not the object of his paper to make an enumeration, and there may be cases which are not considered; the discussion of the cases considered is very full and interesting. The case of 16 nodes is also considered, Kummer (1). As to the surface with 16 nodes, it is to be remarked that the wave-surface, or generally the surface obtained by the homographic deformation of the wave-surface called, Cayley (1), the "tetrahedroid"—is a special form of surface with 16 nodes: its relation to the general surface is explained, Cayley (2).

2. Quartic surface with nodal line: considered incidentally, Clebsch (2) and (3). There are through the nodal line 8 planes, each meeting the surface in a line-pair; considering any 7 of these, and taking out of each of them a line, the 7 lines are met by a conic which also meets a determinate line out of the remaining line-pair; there are thus on the surface 2^7, $= 128$, conics; viz., these form 64 pairs, each pair lying in a plane, and being the complete intersection of the surface by such plane; the number of these planes is of course $= 64$.

Although not properly included in the present case, I mention the quartic surface which is the reciprocal of the cubic surface $XIX = 12 - B_3 - C_2$, Cayley (5): the nodal curve is here an oscnodal line counting as three nodal lines.

3. Quartic surfaces with nodal conic. Such a surface may be without cnicnodes, or it may have 1, 2, 3, or 4 cnicnodes; the cases, other than that of 3 cnicnodes, are mentioned, Kummer (3); but the question is examined, and the remaining case of 3 cnicnodes established, Cayley (6).

The general case of the nodal conic without cnicnodes is elaborately considered, Clebsch (1): it is shown that there are on the surface 16 lines, each meeting the conic, and which in their arrangement are strikingly analogous to the 27 lines on a cubic surface; viz., if on a cubic surface we select at pleasure any one of the 27 lines, and through this line draw a plane which besides meets the cubic surface in a conic; then, disregarding the line in question and the 10 lines which meet it, the remaining 16 lines each meet the conic, and are related to it and to each other in the same manner that the 16 lines of the quartic surface are related to the nodal conic and to each other. And the ground hereof appears, Geiser; viz., it is shown that the quartic surface with the nodal conic, is rationally transformable into a cubic surface, the 16 lines and the nodal conic corresponding respectively to the 16 lines and the conic of the cubic surface.

The several cases of 1, 2, 3, and 4 cnicnodes are considered, Korndorfer.

In the case where the nodal conic is the circle at infinity, the surfaces have been termed "anallagmatic" (perhaps "bicircular" would be a more convenient name), and a great deal has been written upon these surfaces by Moutard, Clifford, and others. Such a surface may of course have 1, 2, 3, or 4 cnicnodes; these surfaces, viz. the cnicnodal anallagmatics, in fact arise from the inversion of a quadric surface by the method of reciprocal radius vectors,—that is, by the change of x, y, z into $\left(\frac{x}{r^2}, \frac{y}{r^2}, \frac{z}{r^2}\right)$; the centre of inversion is a node on the quartic surface. If the quadric surface is a cone, there is another node, the inverse point of the vertex; if the quadric surface is one of revolution, there are two other nodes; and if it is a cone of revolution, there are three other nodes—viz., in all, four nodes. The last-mentioned surface is, or includes, the Cyclide; viz., this is a quartic surface having the circle at infinity for a nodal curve, and having besides four nodes, which are a system of skew antipoints. The surface was first considered by Dupin (*Applications de Géométrie* &c., 1822) as the envelope of a sphere touching three given spheres—its lines of curvature are thus circles; and the surface has been very frequently considered in reference to this property and otherwise: see Maxwell, where a classification (not quite complete) is made of the different forms of the surface, and also stereographic drawings given. It is to be observed, that one interesting form, the parabolic cyclide, is not a quartic but a cubic surface.

In the class of surfaces which have been under consideration, the cnicnodes have been points not on the nodal conic—in fact, a point on a nodal curve cannot be, properly speaking, a cnicnode, though it may be a point of higher singularity in the nature of a cnicnode; viz., there may be on the nodal curve points which, in the classification of the surfaces, must be counted as cnicnodes. Such a case presents itself in the "Conic Torus," or surface generated by the rotation of a conic about a line whether not in or in the plane of the conic. The surface has been considered, De la Gournerie (1), although more in reference to the constructions of descriptive geometry than as a theory of pure geometry, and Cayley (6). The surface has a nodal circle, and upon it two singular points, the circular points at infinity; so that it belongs to the case of a nodal conic with two cnicnodes. In the particular case where the axis of rotation is in the plane of the conic, then there are on the axis two cnicnodes; so that the case is that of a nodal conic and four cnicnodes; and when the generating conic is a circle, viz., when the surface is the ordinary torus, or anchor ring, generated by the rotation of a circle about a line in its own plane, then the nodal conic is the circle at infinity having upon it two cnicnodal points (its intersections by the planes at right angles to the axis) and the surface has also two cnicnodes on the axis: the surface, although presenting considerable peculiarity, may be

regarded as a particular case of the cyclide. In reference to the plane sections of the conic torus and its various particular cases, see De la Gournerie (2); the ordinary torus has been the subject of numerous papers by Darboux and others, and possesses very interesting properties.

In connexion with the foregoing, I speak of the surfaces having a cuspidal conic: the general case is briefly referred to, Cayley (6); viz., this is the surface (AA) the equation of which is $V^2 - x^2y = 0$, and which it is shown has a reciprocal of the order 6. A special case is the surface (AB) having a reciprocal of the order 3; viz., the quartic surface is here the reciprocal of the cubic surface $XX = 12 - U_8$, Cayley (5). And it appears from the memoir last referred to, that there is another cubic surface, $XVII = 12 - 2B_3 - C_2$, the reciprocal of which is a quartic surface having a cuspidal conic. But the theory of the quartic surfaces with a cuspidal conic has been hardly at all considered.

I do not know that anything has been done in regard to the quartic surfaces where the nodal conic becomes a line-pair; that is, where we have two intersecting nodal lines. Although not properly belonging to the case in question, I mention here the quartic surface which is the reciprocal of the cubic surface $XVIII = 12 - B_4 - 2C_2$, Cayley (5); the nodal curve consists of two intersecting lines, but one of them is tacnodal, counting as two nodal lines.

4. Quartic surface with three nodal lines (not in the same plane) meeting in a point. This is, in fact, Steiner's quartic surface; and it has been the subject of numerous investigations.

The equation of the surface may be taken to be $\sqrt{x} + \sqrt{y} + \sqrt{z} + \sqrt{w} = 0$; and the surface thus presents itself as the reciprocal of the cubic surface $\frac{1}{w} + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$, (XVI $= 12 - 4C_2$) with four cnicnodes(1).

It was convenient to make the foregoing enumeration before speaking of Kummer's paper (3), and of the several memoirs which relate to the Abbildung of certain quartic and quintic surfaces.

As regards Kummer's paper, the object appears by the title, viz., he considers in what cases a quartic surface has upon it a system of conics; or, what is the same thing, in what cases there is a system of planes each intersecting the surface in two conics. It is, in the first place, remarked that there is no proper quartic surface cut by every plane in a pair of conics, or even a proper quartic surface cut in a pair of conics by every plane through a fixed point. The cases considered are I., where the planes are non-tangent planes; II., where they are single tangent planes; and III., where they are double tangent planes. The case I. is—(1) when there is a nodal conic and two cnicnodes; viz., any plane through the 2 cnicnodes gives a section with 4 nodes, therefore a pair of conics, (and the special case of 4 cnicnodes is noticed

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[1] The Author exhibited, and pointed out some of the properties of, a model of Steiner's surface.

incidentally); (2) when there is a nodal line; any plane through the nodal line besides meets the surface in a conic; (3) when the surface has two "Selbstberührungspuncte"—viz., either of these is a point where the tangent plane is replaced by two coincident planes, and which, when a plane passes through it, gives in the section a tacnode, = 2 nodes; the section by a plane through two such points, consists of two conics touching each other at the point in question: the equation of such a surface is $\phi^2 = (* \mid p, q)^2$, where ϕ is a quadric function, and p, q linear functions of the coordinates. In all the cases the planes pass through a fixed line, and the surface may be considered as the locus of a variable conic, the plane of which always passes through such line. II. is (1) Steiner's surface, where every tangent plane meets the surface in a pair of conics; and (2) surface with a nodal conic and one cnicnode, where every tangent plane through the cnicnode meets the surface in a pair of conics. And III. is (1) the surface with a nodal conic, where every double tangent plane meets the surface in a pair of conics: it is shown that there are 5 quadric cones, such that a tangent plane of any one of these cones is always a double tangent plane of the surface. Or the surface is (2) a quartic scroll; any plane through two intersecting lines of the surface besides meets the surface in a conic.

It is in the paper, Cayley (6), remarked, that the quartic surface $(* \mid U, V, W)^2 = 0$ can also be expressed in the form $UW - V^2 = 0$; under which form the surface is seen to be the envelope of the series of quadric surfaces $(U, V, W \mid \theta, 1)^2 = 0$; and by reason of this property it is very easy to find the equations of the reciprocal surfaces, or plane-equations of the quartic surfaces in question. And, in the same paper, it is noticed that the surfaces of the form in question include the reciprocals of several interesting surfaces of the orders 6, 8, 9, 10, and 12; viz., order 6, parabolic ring: order 8, elliptic ring: order 9, centro-surface of paraboloid: order 10, parallel surface of paraboloid; envelope of planes through the points of an ellipsoid at right angles to the radius vectors from the centre: order 12, centro-surface of ellipsoid; parallel surface of ellipsoid.

It will be noticed that several of the papers by Clebsch and others refer in their titles to the "Abbildung" of a surface; viz., they show that a (1, 1) correspondence exists between the points of the surface and the points of a plane. The most simple instance is the quadric surface; here, taking any fixed point O on the surface, the line OP drawn to any point P on the surface meets a plane in a point P', and the points P, P' have, it is clear, a (1, 1) correspondence. And, of course, to any curve on the quadric surface there corresponds a curve on the plane, and the discussion of the nature of the plane curves which correspond to the different curves on the quadric surface would constitute a theory of the Abbildung of the quadric surface.

Similarly, as remarked, Clebsch (2), for a cubic surface, taking upon it any two lines which do not meet, if from a point P on the surface we draw, meeting each of the two lines, a line to meet the plane in P', then the point P on the cubic surface and the point P' on the plane will have a (1, 1) correspondence; and we have thus a like theory for the cubic surface. The Abbildung of a cubic surface had been, however, previously effected by Clebsch {in the paper "Die Geometrie auf den Flächen

dritter Ordnung," *Crelle*, t. LXV. (1866), pp. 359–380}, and by Cremona, in a different and really the most simple manner (1), but having a less obvious geometrical signification.

For surfaces of the higher orders, it is only certain surfaces which admit of an Abbildung, or (1, 1) correspondence of the points thereof with the points of a plane; viz. (in the same way as a plane curve, in order to its being unicursal, must have a sufficient number of nodes or cusps) a surface, in order that it may thus correspond with the plane (or say, in order that it may be unicursal), must have a sufficient singularity in the way of a nodal or cuspidal curve. The quartic and quintic surfaces considered in the enumerated memoirs are there considered for the sake of the Abbildung theory which they give rise to; whereas, in the present sketch, the Abbildung theory is considered only for the sake of the quartic and quintic surfaces to which the theory has been applied. But the methods of the theory furnish results in relation to these surfaces; and it is proper to give some account of them.

Clebsch's memoirs (2) and (3) relate to the same subject, which is elaborately treated in the latter of them: the former of them contains, however, some valuable remarks which are not reproduced in the other. In these memoirs (2) and (3), after explaining the above method of the transformation of a cubic surface by means of two of the lines thereof, the author goes on to notice that the like method is applicable to certain quartic and quintic surfaces; viz., (1) quartic surface with a nodal conic: there are here, as already mentioned, 16 lines, each meeting the conic; if, selecting any one of these, from a point P on the surface we draw, meeting the line and the conic, a line to cut the plane in P', then the points P on the surface and P' on the plane have a (1, 1) correspondence. (2) Quartic surface with a nodal line: as already mentioned, there are on the surface 128 conics, each meeting the nodal line; selecting any one of these, if from a point P of the surface we draw, meeting the nodal line and the conic, a line cutting the plane in P', then the point P on the surface, and the point P' on the plane, have a (1, 1) correspondence.

Similarly, (3), for a quintic surface having a nodal skew cubic; then if from a point P on the surface we draw, meeting the skew cubic twice, a line to cut the plane in P', the point P on the surface and the point P' on the plane have a (1, 1) correspondence. The nodal skew cubic may break up into a conic and line which meets it, or into three lines, two of them not meeting each other, but each met by the third line; and the like theory applies to these quintic surfaces.

It is to be noticed that (as for the cubic surface) the above methods of Abbildung, although they have the most obvious geometrical significations, are (as explained in the foregoing foot-note) not the most simple ones; but for each of the foregoing cases (1), (2), (3), the most simple transformation is established in the memoirs now under consideration. The memoir [Memoirs (1) and (2)] of Korndorfer, as indicated by its title, relates to the Abbildung of a quartic surface having a nodal conic and 1, 2, 3, or 4 cnicnodes.

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[1] Any method of transformation leads to an expression of the coordinates of a point on the surface as proportional to rational and integral functions of a given degree ν of the coordinates (x, y, z) of a point on the plane, and that transformation is the more simple for which ν has the smaller value: for the method of the text, the value is $\nu = 3$, but for the methods previously given by Clebsch and Cremona, it is $\nu = 2$.

Clebsch's paper (4) relates to the Abbildung of a quartic scroll.

As regards quintic surfaces (not being scrolls), we have, so far as I am aware, only the before-mentioned paper, Clebsch (3), relating to quintic surfaces with a nodal skew cubic; and the paper, Clebsch (5), which relates to the Abbildung of a quintic surface having a nodal quadriquadric. The method employed is that of a preliminary Abbildung upon a twofold plane (2-blättrige Ebene); that is, it consists in establishing, in the first instance, a (1, 2) correspondence between the surface and the plane; and by means hereof it is shown that there exist on the surface the conics K and G presently referred to, and which give, ultimately, an ordinary Abbildung or (1, 1) correspondence of the points of the surface with those of the plane; viz., this final result is as follows:

There is on the surface a system of conics K, such that their planes pass through a point and envelope a quadricone; and also 64 conics G each meeting each of the conics K in a single point: we select one of these and call it the conic G.

Take now the plane B_1 of a conic K_1 of the series of conics K, which plane B_1 passes, of course, through the vertex V of the cone enveloped by the planes of the conics K; viz., these planes intersect the plane B_1 in a series of lines passing through the point V.

Take a point P on the surface; this lies on a conic K meeting the conic G in a point ξ; and if we draw the line ξP to meet the plane B_1 in P', then P on the quintic surface, and P' on the plane B_1, will have a (1, 1) correspondence; in fact, it appears that, given P, there exists a single position of P'; and conversely, given P', this lies on a line VP' through which there passes the plane B_1 and one other tangent plane of the cone: this tangent plane contains a conic K meeting the conic G in a point ξ; and joining $\xi P'$, this meets the conic K in one other point; viz., given P', there is a single position of P; and there is thus a (1, 1) correspondence.

There are, as originally shown by Schläfli, and as further appears by my memoir on cubic surfaces, Cayley (5), 3 kinds of cubic surfaces of the class 5; viz., these are the surfaces $XIII = 12 - B_3 - 2C_2$, $XIV = 12 - B_4 - C_2$, and $XV = 12 - U_7$; for each of these the reciprocal surface is a quintic surface of the class 3, having a nodal line and a cuspidal quartic curve. For the reciprocal of $XIII$, the cuspidal curve is a quadriquadric; for that of XIV, the cuspidal curve breaks up into the nodal line (viz., this is a cuspnodal line) and into a skew cubic; for that of XV, the cuspidal curve is a cuspidal quadriquadric, or curve of intersection of two quadric surfaces with singular contact.

It only remains to speak of Schwarz's memoir on quintic scrolls: it is to be remarked that the theory of scrolls is allied more closely with that of plane curves than with that of surfaces; viz., considering any plane section of the scroll, the lines of the scroll have, in general, a (1, 1) correspondence with the points of the plane section, and the scrolls of any given order are properly arranged according to the deficiency of the plane section. This is what is done by Cremona in the memoir on quartic scrolls above referred to; viz., for a quartic scroll the deficiency is either 0 or 1; and of the 12 species, there are 10 for which the deficiency is = 0 (or which are unicursal), and 2 for which the deficiency is = 1. And this is the principle of classification in Schwarz's memoir; viz., for a quintic scroll the deficiency is = 0, 1, or 2; the number of species established being 10, 4, and 1 for these deficiencies respectively.

List of Memoirs.

Cayley. 1. Sur la surface des ondes. *Liouv.* t. XI. (1846), pp. 291–296, [47].

2. Sur un cas particulier de la surface du quatrième ordre avec seize points singuliers. *Crelle*, t. LXV. (1866), pp. 284–290, [356].

3. Second Memoir on Skew Surfaces, otherwise Scrolls. *Phil. Trans.*, vol. CLIV. (1864), pp. 559–577, [340].

4. Third Memoir on Skew Surfaces, otherwise Scrolls. *Phil. Trans.*, vol. CLIX. (1869), pp. 111–126, [410].

5. Memoir on Cubic Surfaces. *Phil. Trans.*, vol. CLIX. (1869), pp. 231–326, [412].

6. On the Quartic Surfaces $(* \mid U, V, W)^2 = 0$. *Quart. Math. Journ.* vol. X. (1868), pp. 24–34; and vol. XI. (1870), pp. 15–25, and pp. 111–113.

7. Memoir on Quartic Surfaces. *Proc. Lond. Math. Soc.*, vol. III. (1870), pp. 19–69, [445].

Clebsch. 1. Ueber die Flächen vierter Ordnung welche eine Doppelcurve zweiten Grades besitzen. *Crelle*, t. LXIX. (1868), pp. 355–358.

2. Intorno alla rappresentazione di superficie algebriche sopra un piano. *Atti del R. Ist. Lomb.* (12 Nov. 1868), 13 p.

3. Ueber die Abbildung algebraischer Flächen, insbesondere der vierten und fünften Ordnung. *Math. Ann.*, t. I. (1869), pp. 253–316.

4. Ueber die ebene Abbildung der geradlinigen Flächen vierter Ordnung, welche eine Doppelcurve dritten Grades besitzen. *Math. Ann.*, t. II. (1870), pp. 445–466.

5. Ueber die Abbildung einer Classe von Flächen fünfter Ordnung. *Gött. Abh.*, t. XV. 64 p.

Cremona. Sulle superficie gobbe di quarto grado. *Mem. di Bologna*, t. VIII. (30 April, 1868), 15 p.

De la Gournerie. 1. Mémoire sur la surface engendrée par la révolution d'une conique autour d'une droite située d'une manière quelconque dans l'espace. *Jour. de l'Éc. Polyt.*, t. XXIII. (1863), pp. 1–74.

2. Mémoire sur les lignes spiriques. *Liouv.*, t. XIV. (1869), 92 p.

Geiser. Ueber die Flächen vierten Grades welche eine Doppelcurve zweiten Grades haben. *Crelle*, t. LXX. (1869), pp. 249–257.

Korndorfer. [There was only a general reference; the several memoirs are:

1. Die Abbildung einer Fläche vierter Ordnung mit einer Doppelcurve zweiten Grades und einem oder mehreren Knotenpunkten. *Math. Ann.*, t. I. (1869), pp. 592–626.
2. Fortsetzung dieses Aufsatzes. t. II. (1870), pp. 41–64.
3. Ueber diejenigen Raumcurven deren Coordinaten sich als rationale Functionen eines Parameters darstellen. t. III. (1871), pp. 415–423.
4. Die Abbildung einer Fläche vierter Ordnung mit zwei sich schneidenden Doppelgeraden. t. III. (1871), pp. 496–522.
5. Die Abbildung einer Fläche vierter Ordnung mit einer Doppelcurve zweiten Grades welche aus zwei sich schneidenden unendlich nahen Geraden besteht. t. IV. (1871), pp. 117–134.]

Kummer. 1. {Surfaces of the fourth order with sixteen conical points.} *Berl. Monatsb.* (1864), pp. 246–260, and 495–499.

2. Ueber die algebraischen Strahlensysteme, insbesondere über die der ersten und zweiten Ordnung. *Berl. Abh.* (1866), pp. 1–120.

3. Ueber die Flächen vierten Grades auf welchen Schaaren von Kegelschnitten liegen. *Berl. Monatsb.* (July, 1868). *Crelle*, t. LXIV. (1864), pp. 66–76.

Maxwell. On the Cyclide. *Quart. Math. Journ.*, t. IX. (1867), pp. 111–126.

Schwarz. Ueber die geradlinigen Flächen fünften Grades. *Crelle*, t. LXVII. (1867), pp. 23–57.

450.

NOTE ON THE THEORY OF THE RATIONAL TRANSFORMATION BETWEEN TWO PLANES, AND ON SPECIAL SYSTEMS OF POINTS.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 196–198. Read December 8, 1870.]

In Prof. Cremona's theory of the transformation of plane curves, the fundamental equations are taken to be

$$\alpha_1 + 4\alpha_2 + 9\alpha_3 + \ldots = n^2 - 1, \tag{1}$$

$$\alpha_1 + 3\alpha_2 + 6\alpha_3 + \ldots = \tfrac{1}{2}(n^2 + 3n) - 2; \tag{2}$$

and from these we have as a consequence

$$\alpha_1 + 3\alpha_2 + \ldots = \tfrac{1}{2}(n-1)(n-1); \tag{3}$$

viz., the first equation expresses that any two curves of the system intersect in a single variable point; the second, that the curves form a réseau, or system containing two arbitrary parameters; and the third, that the curves are unicursal.

In the equivalent theory of the rational transformation between two planes, as given in my "Memoir on the Rational Transformation between Two Spaces," [447], we have the equation (1); but instead of the equation (2), it would *primâ facie* appear to be sufficient if we had the inequation

$$\alpha_1 + 3\alpha_2 + 6\alpha_3 + \ldots \leqq \tfrac{1}{2}(n^2 + 3n) - 2$$

but on the ground there explained, the case

$$\alpha_1 + 3\alpha_2 + 6\alpha_3 + \ldots < \tfrac{1}{2}(n^2 + 3n) - 2$$

is excluded, and we thus have the equation (2), giving with (1) the equation (3).

[450] NOTE ON THE THEORY OF THE RATIONAL TRANSFORMATION BETWEEN TWO PLANES, AND ON SPECIAL SYSTEMS OF POINTS. *(CONCLUDED)*

(Continuation from the previous chunk; closing pages of paper 450.*)*

I believe the better course is to assume (1) and (3) as the fundamental equations, from them deducing (2); and we thus also get over a difficulty presently referred to, but which did not occur to me when the memoir was written.

In fact, starting with the equations

$$x' : y' : z' = \frac{X}{x} : \frac{Y}{y} : \frac{Z}{z}$$

(which are to give $x : y : z = X' : Y' : Z'$), we have in the first instance the equation (1). Moreover, establishing for x', y', z' a linear equation $ax' + by' + cz' = 0$, we have corresponding hereto a curve $aX + bY + cZ = 0$, and the coordinates x, y, z of a point on this curve are proportional to $X' : Y' : Z'$; that is, substituting for z' the value $-\frac{1}{c}(ax' + by')$, they are proportional to rational and integral (homogeneous) functions of (x', y'), that is, to rational and integral functions of the single parameter $x' : y'$; wherefore the curve $aX + bY + cZ = 0$ is unicursal; whence the equation (3). The like change may be made in the theory of the rational transformation between two spaces; and it is in this case a more important one.

The difficulty is as follows: It is not self-evident that we are at liberty to assume

$$a_1 + 3a_2 + 8a_3 + \cdots \leq \tfrac{1}{2}(n^2 + 3n) - 2;$$

for imagine that we had a system of $(a_1, a_2, a_3, \ldots)$ points, such that $a_1 + 4a_2 + \cdots$ being $= n^2 - 1$, and $a_1 + 3a_2 + \cdots$ being $> \frac{1}{2}(n^2 + 3n) - 2$, the points were such that the conditions in question (viz., the condition that the curve passes once through each of the points a_1, twice through each of the points a_2, ...) should be less than $a_1 + 3a_2 + \cdots$, and in fact $=$ or $< \frac{1}{2}(n^2 + 3n) - 2$; then the functions X, Y, Z would not of necessity be connected by a linear relation $\lambda X + \mu Y + \nu Z = 0$, and the ground for the assumption in question, $a_1 + 3a_2 + \cdots < \frac{1}{2}(n^2 + 3n) - 2$, would no longer exist. And except by the process now adopted of deriving the equation (2) from the equations (1) and (3), I do not know how the impossibility of such a system is to be established; viz., I do not know how we are to prove the following theorem: *There is not any system of* $(a_1, a_2, a_3, \ldots)$ *points*, where

$$a_1 + 4a_2 + 9a_3 + \cdots = n^2 - 1,$$
$$a_1 + 3a_2 + 6a_3 + \cdots > \tfrac{1}{2}(n^2 + 3n) - 2,$$

such that (for a curve of the order n passing once through each point a_1*, twice through each point* a_2*, ...) the number of conditions actually imposed on the curve is = or* $< \frac{1}{2}(n^2 + 3n) - 2$.

A system of $(a_1, a_2, \ldots)$ points such that the number of actually imposed conditions is less than $a_1 + 3a_2 + \cdots$, may be termed a *special* system; we have, of course, the well-known case $(a_1 = n^2)$ of a system of n^2 points, such that any curve of the order n passing through $\frac{1}{2}(n^2 + 3n) - 1$ of these passes through all the remaining points {or what is the same thing, where the number of conditions actually imposed is $= \frac{1}{2}(n^2 + 3n) - 1$}; and we have the following special system, which presented itself to Dr Clebsch, in his researches on the *Abbildung* of a quintic surface with two non-intersecting nodal lines; viz., "we may have $a_1 = 12$, $a_2 = 2$. We may have 12 points and 2 points such that, for a quintic curve passing once through each of the 12 points and twice through each of the 2 points, the number of conditions actually imposed (instead of being $12 + 3 \cdot 2, = 18$) is $= 17$." The construction is as follows: viz., starting with the 2 points and any 10 points, we may draw a quartic passing twice through the first of the 2 points, once through the second of them, and through the 10 points; and another quartic passing twice through the second of the 2 points, once through the first of them, and through the 10 points: the two quartics intersect in the 2 points each twice, in the 10 points, and in 2 new points, forming, with the 10 points, a system of 12 points; and the first-mentioned 2 points and the 12 points form the system in question.

A more complicated case, $a_1 = 10$, $a_2 = 6$, $a_3 = 1$, occurs in Dr Nöther's paper, "Ueber Flächen, welche Schaaren rationaler Curven besitzen," [*Math. Ann.*, t. iii. (1871), pp. 161–227]. Except these two, I do not know any other case of a special system for which $a_2, a_3, \ldots$ are not all $= 0$; the investigation of such systems would, I think, be very interesting.

[A concluding paragraph of seven lines gave some corrections to the "Memoir on the Rational Transformation between Two Spaces," 447, which corrections are made in the present reprint of that paper.]

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[451] A Second Memoir on Quartic Surfaces.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 198–202. *Read December* 8, 1870.*]*

In my Memoir on Quartic Surfaces, *ante* pp. 19–69, [445], although remarking (see No. 79) that the identification was not completely made out, I tacitly assumed that the symmetroid and the decadianome (each of them a quartic surface with 10 nodes) were in fact identical. There is yet a good deal which I cannot completely explain; but the truth appears to be, that the decadianome includes two cases of coordinate generality, say the *sextic decadianome*, and the *bicubic decadianome = symmetroid*: viz., in the first of these the circumscribed cone, having for vertex any one of the 10 nodes, is a proper sextic cone with 9 double lines; in the second it is a system of two cubic cones, intersecting, of course, in 9 lines, which are double lines of the aggregate sextic cone: or, in the notation of the Table No. 11, in the case of the sextic decadianome, the circumscribed cones are each of them 6_9; in that of the bicubic decadianome = symmetroid, they are each of them $(3, 3)$. We thus arrive at a very remarkable system of 10 points in space, viz., giving the name "ennead" to any 9 points *in plano*, which are the intersections of two cubic curves, or to any 9 lines through a point which are the intersections of two cubic cones; the 10 points in space are such that, taking any one whatever of them as vertex, and joining it with the remaining points, the 9 lines form an ennead. I purpose in the present short Memoir to consider the theories in question; the paragraphs are numbered consecutively with those of the Memoir on Quartic Surfaces.

Plane Sextic Curve with 9 Nodes.

110. A sextic curve contains 27 constants; and the number of conditions to be satisfied in order that a given point may be a node is = 3. Hence it would at first sight appear that the curve could be found so as to have 9 given nodes; this would be $9 \times 3, = 27$ conditions, or the curve would be completely determinate. But observe that through the 9 given points we have a determinate cubic curve $U = 0$; we have therefore U^2 = a sextic curve, and the only sextic curve with the 9 given nodes; that is, there is not in a proper sense any sextic curve with the 9 given nodes. The number of given nodes is thus = 8 at most.

111. The sextic curve with 8 given nodes should contain $27 - 3 \cdot 8 = 3$ constants. We may through the 8 given points draw the two cubics $P = 0$, $Q = 0$; and we have then $(a, b, c \mid P, Q)^2 = 0$, a bicubic, or improper sextic curve having the 8 nodes, and also a ninth node, viz., the remaining point of intersection of the two cubic curves, or say the remaining point of the ennead. Hence if $V = 0$ be any particular sextic curve having the 8 given nodes, we have

$$(a, b, c \mid P, Q)^2 + \theta V = 0,$$

a proper sextic curve having the 8 given nodes; and this, as containing the right number (= 3) of constants, will be the general sextic curve having the 8 given nodes.

112. There will be a ninth node if $\theta = 0$; viz., the curve is then $(a, b, c \mid P, Q)^2 = 0$, a bicubic, or improper sextic curve, having for nodes the 9 points of the ennead. Observe that the ninth node is here a point completely and uniquely determined by means of the given 8 nodes. Moreover the number of constants is = 2, so that we have here a general (improper) solution of the question of finding a sextic curve with 9 nodes, 8 of them given.

113. But if θ is not = 0, then the ninth node must be a point on the curve $J(P, Q, V) = 0$; viz., this is a curve of the order 9, determined by means of the given 8 points; say it is the "dianodal curve" of these 8 points, and, as is easy to see, it has each of these 8 points for a node. The ninth node of the sextic may be any point whatever on the dianodal curve; and regarding it as a given point, the sextic will still contain 1 constant; that is, we have the general solution of the problem of finding a sextic curve with 9 nodes, 8 of them given, and the 9th a given point on the dianodal curve.

114. So long as the 8 points are arbitrary, the dianodal curve does not pass through the 9th point of the ennead, and the two cases above considered are mutually exclusive. It will be noticed how closely analogous this theory of the plane sextic with 9 nodes, is to that of the quartic surface with 8 nodes.

115. Of course, instead of the plane sextic curve, we may have a sextic cone; such a cone has at most 8 given double lines; and if there be a 9th double line, then there are the two cases of coordinate generality; viz., (1), the new double line is the ninth line of the ennead, the cone being in this case not a proper sextic cone, but a bicubic cone; (2), the new double line may be any line whatever on the dianodal cone, (cone of the order 9 determined by the 8 given lines, and having each of these for a double line,) and regarding it as a given line on the dianodal cone, the sextic cone contains 1 constant.

Each circumscribed cone of the Symmetroid is $(3, 3)$.

116. Using (x, y, z, w) as current coordinates of a point of the symmetroid, I take S, T, U, V to be quadric functions of the coordinates $(\alpha, \beta, \gamma, \delta)$; the equation of the symmetroid is therefore given by

$$xS + yT + zU + wV = \text{cone},$$

and the nodes thereof are determined by

$$xS + yT + zU + wV = \text{plane-pair}.$$

Suppose that a node is $(x = 0, y = 0, z = 0)$; the condition for this is $V =$ plane-pair, and we may without loss of generality write $V = \gamma^2 + \delta^2$. Hence, putting for shortness $\Omega = xS + yT + zU$, that is a quadric function

$$(a, b, c, d, f, g, h, l, m, n \,|\, \alpha, \beta, \gamma, \delta)^2,$$

wherein the coefficients $a, b, \ldots$ are arbitrary linear functions of (x, y, z), but not containing w, the equation of the symmetroid is given by

$$\Omega + w(\gamma^2 + \delta^2) = \text{cone}.$$

117. It follows that the equation is

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c+w, & n \\ l, & m, & n, & d+w \end{vmatrix} = 0;$$

viz., this is

$$V + w(\partial_c + \partial_d)V + w^2\partial_c\partial_d V = 0,$$

where V denotes the foregoing determinant, writing therein $w = 0$. Or, observing that V contains c and d each only linearly, the equation may be written

$$V + w(\partial_c V + \partial_d V) + w^2\partial_c\partial_d V = 0,$$

which is a quartic surface having, as it should have, the point $(0, 0, 0)$ for one of its ten nodes.

118. The equation of the circumscribed cone is

$$(\partial_c V - \partial_d V)^2 + 4(\partial_c V \cdot \partial_d V - V \cdot \partial_c\partial_d V) = 0,$$

or, what is the same thing, it is

$$(\partial_c V - \partial_d V)^2 + 4(\partial_c V \cdot \partial_d V - V \cdot \partial_c\partial_d V) = 0.$$

But we have identically

$$\partial_c V \cdot \partial_d V - (\tfrac{1}{2}\partial_{cd} V)^2 = V \cdot \partial_c\partial_d V;$$

so that the equation is

$$(\partial_c V - \partial_d V)^2 + (\partial_{cd} V)^2 = 0;$$

a sextic cone breaking up into the two cubic cones

$$\partial_c V - \partial_d V \pm i\,\partial_{cd} V = 0,$$

so that the cone is $(3, 3)$. And since clearly the point $(0, 0, 0)$ may be regarded as representing any one whatever of the 10 nodes, it follows that for any node whatever of the symmetroid, the circumscribed cone is $(3, 3)$, so that, as stated above, bicubic decadianome = symmetroid.

Deductions from the foregoing theory.

119. Referring to No. 85 of the original memoir, it appears that, with 6 given points as nodes, we can actually find for the symmetroid an equation containing 6 constants. I cannot discover any ground for doubting that 3 of these may be determined so as to give to the symmetroid a seventh given node; and I therefore assume that with 7 given points as nodes, an equation can be found with 3 constants. The symmetroid is certainly not octadic, hence the eighth node must lie on the dianodal surface of the 7 given points. I can discover no ground for doubting but that two of the constants may be determined so that the eighth node shall be any given point whatever on the dianodal surface of the 7 points; and (this being so) that further the remaining constant may be determined so that the ninth node shall be any given point whatever on the dianodal curve of the 8 points. But if all this be so, the consequence is very remarkable; the tenth node is not any one whatever of the 22 dianodal centres of the 9 points, but it is a uniquely determinate "enneadic centre," viz., we must have the following theorem:

120. "Take any 7 points; an eighth point at pleasure on the dianodal surface of the 7 points; a ninth point at pleasure on the dianodal curve of the 8 points. In the system of 9 points so determined, take any one as vertex, and joining it with the remaining 8, construct the ninth line of the ennead. Performing this construction with each of the 9 points successively as vertex, we obtain 9 lines passing through the 9 points respectively. These 9 lines meet in a point which is the 'enneadic centre' of the 9 points: and further, the 10 points form a completely symmetrical system, so that each one of them is the enneadic centre of the remaining 9."

121. Assuming that the 9 lines do intersect so as to give rise to an enneadic centre, there is no difficulty in conceiving that the loci, which by their intersection determine the dianodal centres, do each of them pass through the enneadic centre; so that this enneadic centre counts once or more among the dianodal centres, and the number of proper dianodal centres, instead of being $= 22$, will be suppose $= 22 - \omega$; and if, further, the 9 points, together with the enneadic centre, are the nodes of a symmetroid, but the 9 points together with any one of the $22 - \omega$ dianodal centres are the nodes of a sextic decadianome, then we must also have as follows:

122. "Considering any 9 points as above; taking any one as vertex, and joining it with the remaining 8, these 8 lines determine a dianodal ninthic cone. We have thus 9 dianodal cones, which cones pass all of them through the same $22 - \omega$ points."

123. I am not able to verify these theorems *à posteriori*. It appears to me that the theorem in regard to the enneadic centre subsists for a system of 9 points such as referred to in the statement; but that if by possibility the statement be too general, the theorem must, at all events, subsist for a more special system of 9 points; and that there certainly exist systems of 10 points, such that each 9 of the points have as an enneadic centre the tenth point. (*I have since ascertained that if a quartic surface with* 10 *nodes has a single node* $(3, 3)$, *the surface is a symmetroid; whence, by what precedes, the remaining nine nodes are each of them* $(3, 3)$. *Added* 25 *March,* 1871.)

124. I notice, as a subject of investigation, the following system of correspondence; viz., given any 8 points in space: then to every point in space corresponds a line through this point, viz., the ninth line of the ennead obtained by joining the point with the 8 given points respectively; and to each line in space a point or points on the line, viz., the point or points for each of which the line is the ninth line of the ennead obtained by joining the point with the eight given points respectively.

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[452] On an Analytical Theorem from a New Point of View.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 220, 221. *Read February* 9, 1871.*]*

The theorem is a well-known one, derived from the equation

$$(az^2 + 2bz + c)w^2 + 2(a'z^2 + 2b'z + c')w + a''z^2 + 2b''z + c'' = 0;$$

viz., considering this equation as establishing a relation between the variables z and w, and writing it in the forms

$$0 = u = Aw^2 + 2Bw + C = A'z^2 + 2B'z + C' = 0,$$

(where, of course, A, B, C are quadric functions of z, and A', B', C' quadric functions of w,) we have

$$0 = \frac{du}{dw}\,dw + \frac{du}{dz}\,dz = (Aw + B)\,dw + (A'z + B')\,dz;$$

but in virtue of the equation $u = 0$, we have $Aw+B = \sqrt{B^2 - AC}$, and $A'z+B' = \sqrt{B'^2 - A'C'}$, and the differential equation thus becomes

$$\frac{dw}{\sqrt{B'^2 - A'C'}} + \frac{dz}{\sqrt{B^2 - AC}} = 0,$$

where $B'^2 - A'C'$ and $B^2 - AC$ are quartic functions of w and z respectively. This is, of course, integrable (viz., the integral is the original equation $u = 0$); and it follows, from the theory of elliptic functions, that the two quartic functions must be linearly transformable into each other; viz., they must have the same absolute invariant $I^3 \div J^2$. It is, in fact, easy to verify, not only that this is so, but that the two functions have the same quadrinvariant I, and the same cubinvariant J.

The new point of view is, that we take the coefficients a, b, &c., to be homogeneous functions of (x, y), their degrees being such that the equation $u = 0$ is a quartic equation $(*\,|\,x, y, z, w)^4 = 0$: viz., this equation now represents a quartic surface having a node (conical point) at the point $(x = 0, y = 0, z = 0)$, and also a node at the point $(x = 0, y = 0, w = 0)$, say, these points are O, O' respectively. The equation $B^2 - AC = 0$ gives the circumscribed sextic cone having O for its vertex, and the equation $B'^2 - A'C' = 0$ the circumscribed sextic cone having O' for its vertex; each of these cones has the line OO' $(x = 0, y = 0)$ for a nodal line, as appears geometrically, and also by the equations containing z, w respectively in the degree 4. Considering $B^2 - AC$ as a quartic function of z, its quadrinvariant is a function $(x, y)^4$, and its cubinvariant a function $(x, y)^6$; and similarly, considering $B'^2 - A'C'$ as a quartic function of w, its invariants are functions $(x, y)^4$ and $(x, y)^6$. We have thus, between the two cones, a geometrical relation answering to the analytical one of the identity of the invariants; but the nature of this geometrical relation is not obvious; and it presents itself as an interesting subject of investigation.

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[453] ON A PROBLEM IN THE CALCULUS OF VARIATIONS.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 221, 222. *Read February* 9, 1871.*]*

The problem is, $z = \frac{1}{2}(3x - y^2)\,y$, to find y a function of x such that $\int z\,dx =$ max. or min., subject to a given condition $\int y\,dx = c$ (the limits of each integral being x_0, x_1, where these quantities are each positive, and $x_1 > x_0$). The ordinary method of solution gives $y^2 = x + \lambda$, where $(x_1 + \lambda)^{3/2} - (x_0 + \lambda)^{3/2} = \frac{3}{2}c$; so long as c is not less than $(x_1 - x_0)^{3/2}$, there is a real value of λ, but for a smaller value of c there is no real value. The difficulty arising in this last case is somewhat illustrated by replacing the original problem by a like problem of ordinary maxima and minima; viz., $x_1, x_2, \ldots x_n$ being given positive values of x, in the order of increasing magnitude; and if, in general, $z_i = \frac{1}{2}(3x_i - y_i^2)\,y_i$, then the problem is to find y_i a function of x_i, such that $\Sigma z_i =$ max. or min., subject to the condition $\Sigma y_i = c$. We have here $y_i^2 = x_i + \lambda$ where λ is to be determined by the condition $\Sigma y_i = c$; the remainder of the investigation turns on the question of the sign $y_i = +\sqrt{x_i + \lambda}$ or $y_i = -\sqrt{x_i + \lambda}$, to be taken for the several values of i respectively.

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[454] A THIRD MEMOIR ON QUARTIC SURFACES.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 234–266. *Read April* 13, 1871.*]*

The present Memoir is a continuation of my former researches on Nodal Quartic Surfaces, [445, 451]. The leading idea is, that for a quartic surface with k-nodes, given the nature of the circumscribed, $(k-1)$ nodal, sextic cone belonging to any one node of the surface (for instance, $k = 10$, that it is a cone $(3, 3)$ composed of two cubic cones), we thereby determine the equation of the quartic surface, and consequently the nature of the remaining $(k-1)$ nodes thereof. By means of this general theory I complete, in an essential point, the theory of the Symmetroid; viz., I show that a 10-nodal quartic surface having a single node $(3, 3)$ is a Symmetroid; whence, as appears by my second Memoir, [451], each of the remaining nine nodes is also a node $(3, 3)$; and we have the theory of the remarkable system of ten points in space such that, joining any one of them with the remaining nine, the nine lines thus obtained are the intersections of two cubic cones. A large part of the Memoir is devoted to the consideration of the surfaces with 16, 15, 14, and 13 nodes: this is substantially a reproduction of the results obtained by Kummer in the Memoir "Ueber die algebraischen Strahlensysteme, &c.," already referred to; but the results in question are brought into connexion with the theory of the present Memoir, and they are, by a change of the constants, exhibited in a form of much greater symmetry and elegance. I attach importance also to the square diagrams by means of which I have exhibited, in a compendious form, the relation between the several nodes and circumscribed sextic cones.

The paragraphs are numbered consecutively with those of the first and second Memoirs.

Preliminary Considerations and Classification.

125. I call to mind that if a quartic surface has a node (conical point), then there is for this node a tangent quadricone and a circumscribed sextic cone; viz., if the surface has $(k-1)$ other nodes, or in all k nodes, then the sextic cone has $(k-1)$ nodal lines (passing through the other nodes respectively), and we have thus for the different forms of the sextic the table No. 11; viz., this is

CIRCUMSCRIBED SEXTIC CONE.

Nodes of Surface	Nodal Lines of Cone	Forms
1	0	6
2	1	6_1
3	2	6_2
4	3	6_3
5	4	6_4
6	5	$6_5,\ 5$
7	6	$6_6,\ 5_1$
8	7	$6_7,\ 5_2,\ 4,\ 2$
9	8	$6_8,\ 5_3,\ 4_1, 2,\ 4, 1, 1$
10	9	$6_9,\ 5_4,\ 4_2, 2,\ 4_1, 1, 1,\ 3, 3$
11	10	$6_{10},\ 5_5,\ \ldots,\ 4_3, 1, 1,\ 3_1, 3$
12	11	$5_6,\ \ldots,\ 4_3, 1, 1,\ 3, 2, 1$
13	12	$\ldots,\ 3, 2, 1,\ 3, 1, 1, 1,\ 2, 2, 2$
14	13	$\ldots,\ 3_1, 1, 1, 1,\ 2, 2, 1, 1$
15	14	$\ldots,\ 2, 1, 1, 1, 1$
16	15	$1, 1, 1, 1, 1, 1$

viz., 6 denotes a proper sextic cone without nodal lines; 6_1 a proper sextic cone with one nodal line; $5, 1$ a proper quintic cone and a plane, &c.

We may distinguish the nodes according to the sextic cones; thus, a node 6 means a node for which the circumscribed cone is a proper sextic cone, $(1, 1, 1, 1, 1, 1)$ a node where the circumscribed cone breaks up into six planes, &c.

126. A 16-nodal surface has 16 nodes $(1, 1, 1, 1, 1, 1)$, and a 15-nodal surface has 15 nodes $(2, 1, 1, 1, 1)$; but, for a 14-nodal surface, the question arises how many nodes are $(3_1, 1, 1, 1)$, and how many $(2, 2, 1, 1)$. It was remarked, No. 13, that the only possible cases were $14, 0$; $8, 6$; or $2, 12$; and that we might, in like manner, limit the number of possible cases for other values of k; but that the inquiry was not then further pursued. I resume this inquiry, but without obtaining as yet a complete answer.

127. It is to be observed that a line joining any two nodes is not, in general, a line on the surface, but that it may be so; the surfaces for which this is so (viz., any surface which contains upon it a line through two nodes) form, however, a class by themselves, which at present I altogether exclude from consideration. This being so, it will appear in the sequel that there is but one kind of surface having a node $(2, 2, 1, 1)$, and but one kind of surface having a node $(3_1, 1, 1, 1)$. Now there is a surface, Kummer's 14-nodal, the nodes of which are $8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$; wherefore the two kinds are identical, and are each of them Kummer's 14-nodal surface. Similarly, for the 13-nodal surfaces, there is but one kind having a node $(4_1, 1, 1)$, but one kind having a node $(3, 1, 1, 1)$, and moreover but one kind having a node $(3_1, 2, 1)$; and we have Kummer's 13-nodal surface with the nodes $3(4_1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$; hence the three kinds are identical with each other and with Kummer's. Moreover, there is but one kind having a node $(2, 2, 2)$; hence all the other nodes must be $(2, 2, 2)$, and we have a surface $13(2, 2, 2)$ not given by Kummer. And in like manner for the 12-nodal surfaces, we have the two kinds given by Kummer, and a third kind $12(4_1, 1, 1)$ not given by him; the arrangement thus far being

No. of Nodes.	**Character of Surface.**
16	$16(1, 1, 1, 1, 1, 1)$,
15	$15(2, 1, 1, 1, 1)$,
14	$8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$,
13 (α)	$3(4_1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$,
(β)	$13(2, 2, 2)$.
12 (α)	$12(4_1, 2)$,
(β)	$2(5_1, 1) + 6(3_1, 3_1) + 4(3, 2, 1)$,
(γ)	$12(4_1, 1, 1)$.

128. But in the next following case we have Kummer's surface, viz.

$$11\,(\alpha)\quad 1(6_{10}) + 10(3_1, 3),$$

and I do not know whether one, two, or three kinds of surface having nodes $(4_1, 1, 1)$, $(4_2, 2)$, and $(5_5, 1)$. And in the next case we have (as will appear) the Symmetroid, viz.,

$$10\,(\alpha)\quad 10(3, 3),$$

and I do not know how many kinds of surfaces having a node or nodes 6_9, $(5_4, 1)$, $(4_2, 2)$, $(4, 1, 1)$.

It will be observed that the present division has nothing to do with the octadic and disinodal division in the former Memoir.

129. I consider a conic $A = 0$, and any six tangents thereof, $t_1 = 0$, $t_2 = 0$, $t_3 = 0$, $t_4 = 0$, $t_5 = 0$, $t_6 = 0$; we have an identical equation which might be written $AC - B^2 = t_1t_2t_3t_4t_5t_6$, but it will be convenient, introducing a constant factor K, to write it

$$AC - B^2 = K\, t_1t_2t_3t_4t_5t_6,$$

B being a cubic function and C a quartic function of the coordinates.

Consider now the series of factors, such as

$$\begin{aligned}
&lA + mt_1t_2,\\
&sA + mt_1t_2t_3,\\
&UA + mt_1t_2t_3t_4,\\
&VA + mt_1t_2t_3t_4t_5,\\
&WA + mt_1t_2t_3t_4t_5t_6,
\end{aligned}$$

where m is a constant, l a constant, s a linear function, U, V, W functions of the degrees 2, 3, 4 respectively; and compose with one or more such factors an expression involving the term $Kt_1t_2t_3t_4t_5t_6$; for instance, such an expression is

$$\frac{mm'}{K}\,(sA + mt_1t_2t_3)(l'A + m't_4t_5t_6);$$

this is, of the form $A\Omega + Kt_1t_2t_3t_4t_5t_6$, viz., $A\Omega + (AC - B^2)$, or $A(\Omega + C) - B^2$, say $A\Gamma - B^2$; or what is the same thing, introducing a new coordinate w, we have a quadric function

$$Aw^2 + 2Bw + \Gamma,$$

the discriminant of which, $A\Gamma - B^2$, is equal to the expression in question.

130. In the sequel (x, y, z, w) are considered as the coordinates of a point in space; $A = 0$ is thus a quadric cone, $t_1 = 0$, $t_2 = 0$, $\ldots$, $t_6 = 0$ any six tangent planes thereof; and hence $Aw^2 + 2Bw + \Gamma = 0$ is a quartic surface, having the point $(x = 0, y = 0, z = 0)$ for a node, whereof the circumscribed cone $A\Gamma - B^2 = 0$ breaks up in the assumed manner.

Thus, in order that the circumscribed cone may be as above

$$(sA + mt_1t_2t_3)(l'A + m't_4t_5t_6),$$

we have only to assume

$$\Gamma = C + \frac{mm'}{K}(sl'A + sm't_4t_5t_6 + l'mt_1t_2t_3),$$

and so in other cases. Observe that $sA + mt_1t_2t_3 = 0$ is a cubic cone, which, so long as m, s are arbitrary, has no nodal line; but establishing a single relation (say s remains arbitrary, but a proper value is assigned to m) it will be a cubic cone having a nodal line. And so $UA + mt_1t_2t_3t_4 = 0$ is a quartic cone without any nodal line, but by particularising the constants it may be made to have one, two, or three nodal lines. Such nodal determinations are obviously required in order that the formula may extend to all the before-mentioned forms of the circumscribed cone. The foregoing analysis is the foundation of the whole theory: I have given it, as above, apart from the theory, in order that the nature of it may be the better perceived; but I have now to bring it into connexion with the theory.

On the Sextic Curves, $AC - B^2 = 0$.

131. I revert to the consideration of plane curves. The equation of a sextic curve $(*\,|\,x, y, z)^6 = 0$ cannot be in general expressed in the form $AC - B^2 = 0$, where the degrees of A, B, C are 2, 3, 4 respectively; in fact, the existence of such a form implies that there is a conic $A = 0$ touching the sextic 6 times; and since a conic can only be made to satisfy 5 conditions, there is not in general any such conic.

132. Such conic, when it exists, is said to be *inscribed in* the sextic, and the sextic to be *circumscribed about* the conic, or to be an "amphigram;" and then, $A = 0$ being the equation of the conic, that of the sextic is expressible in the form in question $AC - B^2 = 0$. It is clear that $B = 0$ is a cubic curve passing through the 6 points of contact of the conic with the sextic, and that any such curve may be taken for the curve B; in fact if a particular cubic through the 6 points is $B = 0$, and the equation of the sextic is $AC - B^2 = 0$, then taking p an arbitrary linear function of the coordinates, the equation of the general cubic is $B' = B + pA = 0$; and then writing

$$\begin{aligned} A &= A, \\ B' &= B + pA, \\ C' &= C + 2Bp + Ap^2, \end{aligned}$$

we have $AC' - B'^2 = AC - B^2$; so that the original form $AC - B^2 = 0$ becomes $AC' - B'^2 = 0$. But the cubic $B = 0$ being assumed at pleasure, the quartic $C = 0$ is a determinate curve.

133. It is to be observed that a sextic curve may be an amphigram in more than one way: certainly in two, three, or four, and possibly in a greater number of ways. For the equation of the curve contains 27 constants, and hence determining the sextic so as to touch 4 given conics each of them 6 times, there are still 3 constants; and the curve will be an amphigram in regard to each of the 4 conics; say it is a quadruple amphigram. But in the sequel we are only concerned with a sextic curve considered as an amphigram in regard to a given conic $A = 0$ (no attention being paid to the other inscribed conics, if any); and then, by what precedes, taking $B = 0$ any cubic whatever through the 6 points of contact, we have a determinate quartic curve $C = 0$, and the equation of the sextic curve assumes the form $AC - B^2 = 0$.

134. The curves $A = 0$, $B = 0$, $C = 0$ contain respectively 5, 9, 14 constants; whence considering the function B as containing an arbitrary constant factor, for the curve $AC-B^2 = 0$, the number of constants is *primâ facie* $5+9+14+1 = 29$; but on account of the arbitrary linear function p, the real number is $29 - 3 = 26$: this is right, for a sextic curve contains 27 constants; and the curve being an amphigram, there is one relation between the constants, $27 - 1 = 26$.

135. Suppose now that the sextic curve $AC - B^2 = 0$ breaks up into two or more separate curves, say into the two curves $P = 0$, $Q = 0$ of the orders f, g respectively $(f + g = 6)$. We have

$$AC - B^2 = PQ = 0;$$

and the conic $A = 0$ touching the sextic six times, must, it is clear, touch the curves $P = 0$, $Q = 0$, f and g times respectively. And so when the sextic breaks up into any number of curves, each component curve $P = 0$ of the order f must touch the sextic f times.

136. It follows that if the sextic break up into six lines, say $AC - B^2 = t_1t_2t_3t_4t_5t_6 = 0$, then that each of the lines $t_1 = 0$, $t_2 = 0$, …, $t_6 = 0$ is a tangent to the conic. And conversely, starting with the conic $A = 0$ and any six tangents thereof $t_1 = 0$, $t_2 = 0$, …, $t_6 = 0$, we have an identity of the form in question. In fact, taking any two of the tangents, say $t_1 = 0$ and $t_2 = 0$, then, if $p = 0$ be the equation of the line joining their points of intersection, the equation of the conic will be of the form $t_1t_2 + p^2 = 0$, that is, we may write $A = t_1t_2 + p^2$, or what is the same thing, $t_1t_2 = A-p^2$. (Considering A as a given quadric function of the coordinates, this of course implies that the implicit constant factors of t_1, t_2, p are properly determined.) Similarly, $q = 0$ being the line through the points of contact of t_3, t_4, and $r = 0$ that through the points of contact of t_5, t_6, we have $t_3t_4 = A - q^2$ and $t_5t_6 = A - r^2$; whence, to satisfy the equation

$$AC - B^2 = t_1t_2t_3t_4t_5t_6,$$

we have only to assume $B = lA + pqr$, l an arbitrary linear function of the coordinates, and the equation then gives

$$C = l^2A - A(p^2 + q^2 + r^2) + (q^2r^2 + r^2p^2 + p^2q^2) + l^2A + 2l\,pqr.$$

137. It will be observed that the grouping of the six tangents into pairs is arbitrary. By altering this grouping, we merely alter the linear function l, but do not obtain any new solution. Thus, say that the new form is $B' = l'A + p'q'r'$, then, by properly determining the linear function l, we can reduce this to the original form $B = lA+pqr$; viz., we can satisfy identically the equation $(l - l')A + pqr - p'q'r' = 0$; or what is the same thing, $\lambda A + pqr - p'q'r' = 0$, where λ is a linear function of the coordinates. We have, in fact, the conic $A = 0$ and the cubic $pqr = 0$ intersecting in the six points of contact; any other cubic through these six points; and consequently the cubic $p'q'r'$ must be expressible in the form $\lambda A + pqr = 0$, and we have thus the identity in question.

138. We have just seen that the value of B is necessarily of the form $B = lA + pqr$, but we are not concerned with its expression in this particular form. What we require in the sequel is a value of B, and thence one of C, satisfying the identity in question, $AC - B^2 = t_1t_2t_3t_4t_5t_6$; or what is the same thing, introducing for convenience a constant factor K, the identity

$$AC - B^2 = Kt_1t_2t_3t_4t_5t_6.$$

139. Instead of C, I write Γ, and consider the sextic amphigram $A\Gamma - B^2 = 0$ touched by the conic $A = 0$ in the points of contact of the conic with the six tangents $t_1 = 0$, $t_2 = 0$, $\dots$, $t_6 = 0$. Suppose the sextic curve breaks up into factors; if one of these factors is a line, it is one of the six tangents, say the tangent $t_6 = 0$. If there is a conic factor, this is a conic touching the conic $A = 0$ at its points of contact with two of the tangents, say the equation is $lA+mt_1t_2 = 0$. Similarly, if there is a cubic, quartic, or quintic factor, then the equation hereof is $sA+mt_1t_2t_3 = 0$, $UA+mt_1t_2t_3t_4 = 0$, or $VA+mt_1t_2t_3t_4t_5 = 0$. Or going on to the next case of a sextic factor (being of course the whole curve), we may say that this is $WA+mt_1t_2t_3t_4t_5t_6 = 0$. (Observe that since $AC - B^2 = Kt_1t_2t_3t_4t_5t_6$, this means only that the equation of the sextic amphigram is of the assumed form $A\Gamma - B^2 = 0$.)

140. By what precedes we can, for a sextic amphigram which breaks up in any assigned manner, determine the value of Γ. For instance, let the amphigram break up into two cubic curves; say these are $sA + mt_1t_2t_3 = 0$, $s'A + m't_4t_5t_6 = 0$. Assume

$$A\Gamma - B^2 = \frac{mm'}{K}(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6),$$

then this equation is

$$A\Gamma - B^2 = \frac{mm'}{K}\Big[ss'A^2 + (sm't_4t_5t_6 + s'mt_1t_2t_3)A\Big] + AC - B^2;$$

that is, we have

$$\Gamma = C + \frac{mm'}{K}(ss'A + sm't_4t_5t_6 + s'mt_1t_2t_3),$$

and so in any other case.

I have already adverted to the question of the "nodal determination" of the formulae, and it might be properly here considered; viz., the question is as to the determination of the constants in such manner that, for instance, $sA + mt_1t_2t_3 = 0$ may be a nodal cubic, $UA + mt_1t_2t_3t_4 = 0$ a nodal, binodal, or trinodal quartic, &c.; but I defer it for the moment in order first to apply the theory to the quartic surfaces.

Application to Quartic Surfaces.

141. If a quartic surface has a node or nodes, we may take for a node the point $x = 0$, $y = 0$, $z = 0$; the equation of the quartic surface is then of the form

$$Aw^2 + 2Bw + \Gamma = 0,$$

where A, B, Γ are functions of x, y, z of the degrees 2, 3, 4 respectively. $A = 0$ is the tangent quadricone at the node in question; and the circumscribed cone is $A\Gamma - B^2 = 0$. By what precedes, this is an amphigram touching the quadricone along six generating lines thereof; say the tangent planes of the cone $A = 0$ along these six lines respectively are $t_1 = 0$, $t_2 = 0$, …, $t_6 = 0$. We have then an identical equation

$$AC - B^2 = Kt_1t_2t_3t_4t_5t_6,$$

viz., regarding for a moment this equation as an equation for the determination of B, and B' as any particular solution thereof, then its general solution is $B = B' + tA$, where t is an arbitrary linear function of (x, y, z), and the B in the equation of the surface is properly $= B' + tA$. But by the substituting $w - t$ in place of w, the B of the equation of the surface would then be made $= B'$; and it thus appears that we may, without loss of generality, take the B of this equation to be any particular value B satisfying the identity in question; and then, B having such particular value, C is a quartic function of (x, y, z) completely determined by the same identity. And we then, by what precedes, at once determine Γ so that the circumscribed cone $A\Gamma - B^2 = 0$ may be a cone breaking up in any assigned manner; for instance, if it be a cone $(3, 3)$, then, as just mentioned, the two cubic cones are $sA + mt_1t_2t_3 = 0$, $s'A + m't_4t_5t_6 = 0$; and Γ has the value

$$\Gamma = C + \frac{mm'}{K}(ss'A + l's't_4t_5t_6 + ls't_4t_5t_6),$$

above obtained.

On the Nodal Determination.

142. I am not able to discuss with much completeness the question of nodal determination. We have to consider a cubic curve $sA + mt_1t_2t_3 = 0$, a quartic curve $UA + mt_1t_2t_3t_4 = 0$, &c., as the case may be, and to determine the constants so that this shall have a node or nodes. Consider for a moment the form $PA + t_1\Omega = 0$, where Ω denotes the product $mt_2t_3\cdots$ of all or any of the tangents $t_2, \ldots, t_n$; the orders of PA, $t_1\Omega$ are of course equal, that is, the order of P is less by unity than that of Ω. I say that, by establishing a single relation between the constants, this may be made to have a node at the point of contact $A = 0$, $t_1 = 0$. In fact, writing $\Delta = \lambda\partial_x + \mu\partial_y + \nu\partial_z$, where λ, μ, ν are arbitrary, there will be a node at any point if for that point $\Delta(PA + t_1\Omega) = 0$. But for the point $A = 0$, $t_1 = 0$ this becomes $P\Delta A + \Omega\Delta t_1 = 0$; moreover, if t be any other tangent of the conic $A = 0$, and if $p = 0$ be the line joining the points of contact of the tangents t, t_1, then we may write $A = tt_1 - p^2$, and thence (since at the point in question, $A = 0$, $t_1 = 0$, we have also $p = 0$) we find $\Delta A = t\Delta t_1$, and the foregoing equation thus becomes $(tP + \Omega)\Delta t_1 = 0$; viz., this equation is satisfied irrespectively of the values of λ, μ, ν, if only at the point in question (that is, for the values of the coordinates which belong to the point $t_1 = 0$, $A = 0$) we have $tP + \Omega = 0$, which is a single relation between the constants.

143. In particular the cubic curve $sA + mt_1t_2t_3 = 0$ may be made to have a node at the point of contact of any one of the three tangents; the quartic curve $UA + mt_1t_2t_3t_4 = 0$, a node, or two or three nodes, at the point or points of contact of any one, two, or three of the four tangents; and so in other cases. These are not the only solutions, and they are in fact solutions which (as afterwards explained) I propose to reject, attending in each case only to the remaining or proper solutions of the problem.

144. To obtain in a different manner the foregoing result, consider again the cubic curve $sA + mt_1t_2t_3 = 0$; regarding this as a given curve, the conic $A = 0$ is a conic determined (not of course completely) as a conic having therewith 3 points of 2-pointic intersection; viz., if the cubic has a node, then the cone $A = 0$ is either a conic passing through the node and besides touching the curve twice, or else it is a conic touching the curve 3 times; the former is of course the above mentioned case where there is a node at one of the points of contact on the conic $A = 0$; the latter is regarded as the proper solution. So in the case of a quartic curve $UA + mt_1t_2t_3t_4 = 0$, regarding this as a given curve, the conic $A = 0$ is a conic having therewith four points of 2-pointic intersection; viz., if the quartic curve has one, two, or three nodes, then the conic is either a conic passing through one, two, or three nodes, and besides touching the quartic thrice, twice, or once; or else it is a conic touching the quartic four times. The former is the above mentioned case where there is a node or nodes at a point or points of contact with the conic $A = 0$; the latter is regarded as the proper solution.

145. To fix the ideas, and at the same time obtain a result which will be afterwards useful, I work out the formulae for the cubic curve $sA + mt_1t_2t_3 = 0$, taking this equation under the form

$$s\left(\tfrac{x}{\lambda} + \tfrac{y}{\mu} + \tfrac{z}{\nu}\right) + \tfrac{m}{\lambda\mu\nu}\left(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy\right) + \tfrac{m}{\lambda\mu\nu}\, xyz = 0.$$

This may have a node in two different ways; viz.,

1°. At the point of contact of one of the tangents $x = 0$, $y = 0$, $z = 0$ with the conic $A = 0$; say at the point of contact of $x = 0$, that is, the point $x = 0$, $y = z = 0$. The value of m is $\dfrac{4}{\mu} + \dfrac{4}{\nu}$; hence $\dfrac{m}{s} = \dfrac{4}{\mu} + \dfrac{4}{\nu}$; and, substituting, the equation of the curve becomes

$$\tfrac{s\,x}{\lambda}\left[\left(\tfrac{x}{\lambda}\right) + \tfrac{1}{\mu\nu}\left\{x^2 - 2x(y + z) + (y - z)^2\right\}\right] + \tfrac{1}{\lambda\mu\nu}\, m\,(x + y - z)^2 = 0,$$

which has obviously a node at the point in question.

2°. The node may be at a point not on the conic $A = 0$, viz. the value of $\dfrac{m}{s}$ is $-\dfrac{4\lambda}{\mu\nu}$; the equation is

$$s\left(\tfrac{x}{\lambda} + \tfrac{y}{\mu} + \tfrac{z}{\nu}\right) - \tfrac{m}{\lambda\mu\nu}\left(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy\right) + \tfrac{m}{\lambda\mu\nu}\, xyz = 0.$$

In fact, writing for shortness

$$\begin{aligned} -\lambda + \mu + \nu &= L, \\ \lambda - \mu + \nu &= M, \\ \lambda + \mu - \nu &= N, \\ \lambda + \mu + \nu &= P, \end{aligned}$$

the node is at the point $x : y : z = L\lambda : M\mu : N\nu$; which is at once verified, if we remark that, writing for convenience $x, y, z = L\lambda, M\mu, N\nu$, then we have

$$\begin{aligned} -x + y + z &= MN, \\ x - y + z &= NL, \\ x + y - z &= LM, \\ \tfrac{x}{\lambda} + \tfrac{y}{\mu} + \tfrac{z}{\nu} &= P, \quad x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = -LMNP \ (= A). \end{aligned}$$

For, of the three equations for the coordinates of a node, the first is

$$-\frac{s}{\lambda}LMNP + P(-2MN) + \frac{m}{\lambda\mu\nu}\,MN\mu\nu = 0,$$

that is, for the values in question,

$$-LMNP - 2\lambda MNP + P^2MN = 0,$$

or, finally, $-P - 2\lambda + P = 0$, which is satisfied; and similarly the other two equations are satisfied.

Quartic Surfaces resumed.

146. Passing now from curves to cones, and to the theory of the quartic surface, suppose that there is a component cone having a nodal line, say the cubic cone $sA + mt_1t_2t_3 = 0$: if the remaining factor is $t_4t_5t_6$, then we have

$$\Gamma = C + \frac{m}{K}\, s\, t_4t_5t_6.$$

Suppose the nodal line is a line of contact with the cone $A = 0$, say its equations are $t_1 = 0$, $p = 0$ (p a linear function), then $sA + mt_1t_2t_3$ is a quadric function $(*|\,t_1, p)^2$ (of course with variable coefficients); hence $A\Gamma - B^2$ is a quadric function; and B being a linear function $(*\,|\,t_1, p)$, it follows that Γ is a linear function, and thence that Γ is also a linear function; that is, A, B, Γ are each of them a linear function $(*\,|\,t_1, p)$, or the line in question (viz. the line of contact $A = 0$, $t_1 = 0$) is a line on the quartic surface $Aw^2 + 2Bw + \Gamma = 0$. As already mentioned, *I exclude from consideration the surfaces which have upon them a line through two nodes*; that is, I exclude from consideration the case in question where any component cone, or say where the sextic cone, has a nodal line which is a line on the tangent cone $A = 0$.

147. Now, excluding the case just referred to, *I assume as a postulate* that there is but one way in which the cubic cone $sA+mt_1t_2t_3 = 0$ can be made to have a nodal line, or the quartic cone $UA+mt_1t_2t_3t_4 = 0$ one, two, or three nodal lines &c., as the case may be. It is to be understood that this does not mean that the constants are in any of these cases completely determined, but that there is between them a relation or relations constituting a general solution which includes in itself every particular solution whatever. I have no doubt that as regards the cubic cone at least the assumption is correct. This being so, the character of a single node determines the nature of the surface; for instance, if there is a node $(3_1, 3)$, then taking this as the point $(x = 0, y = 0, z = 0)$ the equation of the surface is $Aw^2 + 2Bw + \Gamma = 0$, where

$$\Gamma = C + \frac{mm'}{K}(ss'A + l's\, t_4t_5t_6 + ls't_1t_2t_3),$$

a surface of a determinate nature; so that the character of all the remaining nodes is completely determined.

148. The point to be attended to is, that if for instance there were two essentially distinct ways of giving the cubic cone $sA + mt_1t_2t_3 = 0$ a nodal line (such as there would be if the excluded case were considered admissible), then the foregoing equation of the surface would or might include two distinct forms of equation applying to different kinds of surface. The conclusion is that there is but one kind of quartic surface having a node $(3_1, 3)$. Admitting this, and similarly that there is but one kind of quartic surface having a node 6_{10}, it follows that if (as the fact is) there is a surface having the nodes $1(6_{10}) + 10(3_1, 3)$ (Kummer's 11-nodal surface), then that the two first-mentioned kinds are in fact each of them this last-mentioned kind of surface; and it was in this manner that I arrived at the enumeration given near the beginning of the present Memoir.

149. The reasoning is, of course, in place of a direct demonstration which would consist in showing that a surface having a node $(3_1, 3)$ has 9 other like nodes, and also a node 6_{10}; and that a surface having a node 6_{10} has 10 other nodes $(3_1, 3)$; and that, starting from either form of equation, we could, by passing to a node of the other kind, obtain the other form of equation.

Enumeration of the Cases.

150. I collect the results as follows: I call to mind that we have always the identical equation $AC - B^2 = Kt_1t_2t_3t_4t_5t_6$, that the equation of the surface is $Aw^2 + 2Bw + \Gamma = 0$, and that the circumscribed cone is $A\Gamma - B^2 = 0$. The equation of a surface having different kinds of nodes will assume different forms according as the origin (or point $x = 0$, $y = 0$, $z = 0$) is taken to be at a node of one or other of these kinds; these forms of the equations are distinguished as "node-forms," viz., we speak of the node-form $(3_1, 3)$ when the origin is a node $(3_1, 3)$, and so in other cases.

The 16-nodal surface $16(1,1,1,1,1,1)$.
node-form $(1,1,1,1,1,1)$,
cone is $t_1t_2t_3t_4t_5t_6 = 0$,
and $\Gamma = C$;
viz., equation is $Aw^2 + 2Bw + C = 0$.

The 15-nodal surface $15(2,1,1,1,1)$.
node-form $(2,1,1,1,1)$,
cone is $(lA + mt_1t_2)\,t_3t_4t_5t_6 = 0$,
and $\Gamma = C + \dfrac{Kl}{m}\,t_3t_4t_5t_6$.

The 14-nodal surface $8(3_1,1,1,1) + 6(2,2,1,1)$.
node-form $(3_1,1,1,1)$,
cone is $(sA + mt_1t_2t_3)\,t_4t_5t_6 = 0$,
where $sA + mt_1t_2t_3 = 0$ is a nodal cubic 3_1,
and $\Gamma = C + \dfrac{Ks}{m}\,t_4t_5t_6$.

node-form $(2,2,1,1)$,
cone is $(lA + mt_1t_2)(l'A + m't_3t_4)\,t_5t_6 = 0$,
and $\Gamma = C + \dfrac{mm'}{K}(ll'A + lm't_3t_4 + l'mt_1t_2)\,t_5t_6$.

The 13(α)-nodal surface $3(4_1,1,1) + 1(3,1,1,1) + 9(3_1,2,1)$.
node-form $(4_1,1,1)$,
cone is $(UA + mt_1t_2t_3t_4)\,t_5t_6 = 0$,
where $UA + mt_1t_2t_3t_4 = 0$ is a trinodal quartic 4_1,
and $\Gamma = C + \dfrac{KU}{m}\,t_5t_6$.

node-form $(3,1,1,1)$,
cone is $(sA + mt_1t_2t_3)\,t_4t_5t_6 = 0$,
and $\Gamma = C + \dfrac{Ks}{m}\,t_4t_5t_6$.

node-form $(3_1,2,1)$,
cone is $(sA + mt_1t_2t_3)(l'A + m't_4t_5)\,t_6 = 0$,
where $sA + mt_1t_2t_3 = 0$ is a nodal cubic 3_1,
and $\Gamma = C + \dfrac{mm'}{K}(sl'A + m's\,t_4t_5 + ml't_1t_2t_3)\,t_6$.

The 13(β)-nodal surface $13(2,2,2)$.
node-form $(2,2,2)$,
cone is $(lA + mt_1t_2)(l'A + m't_3t_4)(l''A + m''t_5t_6) = 0$,
and

$$\Gamma = C + \frac{mm'm''}{K^2}\big[ll'l''A^2 + (l'l''mt_1t_2 + l''lm't_3t_4 + ll'm''t_5t_6)A \\ + lm'm''t_3t_4t_5t_6 + l'm''m\,t_5t_6t_1t_2 + l''mm'\,t_1t_2t_3t_4\big].$$

The 12(α)-nodal surface $12(4_1,2)$.
node-form $(4_2,2)$,
cone is $(UA + mt_1t_2t_3t_4)(l'A + m't_5t_6) = 0$,

where $UA + mt_1t_2t_3t_4 = 0$ is a trinodal quartic 4_3,
and $\Gamma = C + \dfrac{mm'}{K}(l'UA + m'Ut_5t_6 + l'm\,t_1t_2t_3t_4)$.

The $12(\beta)$-nodal surface $2(5_1, 1) + 6(3_1, 3_1) + 4(3, 2, 1)$.
node-form $(5_1, 1)$,
cone is $(VA + mt_1t_2t_3t_4t_5)\,t_6 = 0$,
where $VA + mt_1t_2t_3t_4t_5 = 0$ is a 6-nodal quintic 5_1,
and $\Gamma = C + \dfrac{KV}{m}\,t_6$.

node-form $(3_1, 3_1)$,
cone is $(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6) = 0$,
where $sA + mt_1t_2t_3 = 0$ and $s'A + m't_4t_5t_6 = 0$ are each of them a nodal cubic 3_1,
and $\Gamma = C + \dfrac{mm'}{K}(ss'A + m's\,t_4t_5t_6 + ms't_1t_2t_3)$.

node-form $(3, 2, 1)$,
cone is $(sA + mt_1t_2t_3)(l'A + m't_4t_5)\,t_6 = 0$,
and $\Gamma = C + \dfrac{mm'}{K}(sl'A + m's\,t_4t_5 + l'm\,t_1t_2t_3)\,t_6$.

The $12(\gamma)$-nodal surface, $12(4_1, 1, 1)$.
node-form $(4_1, 1, 1)$,
cone is $(UA + mt_1t_2t_3t_4)\,t_5t_6 = 0$,
where $UA + mt_1t_2t_3t_4 = 0$ is a binodal quartic 4_3,
and $\Gamma = C + \dfrac{KU}{m}\,t_5t_6$.

The $11(\alpha)$-nodal surface, $1(6_{10}) + 10(3_1, 3)$.
node-form (6_{10}),
cone is $WA + mt_1t_2t_3t_4t_5t_6 = 0$,
where this is a 10-nodal sextic 6_{10},
and $\Gamma = C + \dfrac{KW}{m}$.

node-form $(3_1, 3)$,
cone is $(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6) = 0$,
where $sA + mt_1t_2t_3 = 0$ is a nodal cubic 3_1,
and $\Gamma = C + \dfrac{mm'}{K}(ss'A + m's\,t_4t_5t_6 + ms't_1t_2t_3)$.

Other 11-nodal surfaces.
node-form $(5_1, 1)$,
cone is $(VA + mt_1t_2t_3t_4t_5)\,t_6 = 0$,
where $VA + mt_1t_2t_3t_4t_5 = 0$ is a 5-nodal quintic 5_1,
and $\Gamma = C + \dfrac{KV}{m}\,t_6 = 0$;

node-form $(4_2, 2)$,
cone is $(UA + mt_1t_2t_3t_4)(l'A + m't_5t_6) = 0$,
where $UA + mt_1t_2t_3t_4 = 0$ is a binodal quartic 4_2,
and $\Gamma = C + \dfrac{mm'}{K}(l'UA + m'Ut_5t_6 + l'm\,t_1t_2t_3t_4)$;

node-form $(4, 1, 1)$,
cone is $(UA + mt_1t_2t_3t_4)\,t_5t_6 = 0$,
where $UA + mt_1t_2t_3t_4$ is a nodal quartic 4_1,
and $\Gamma = C + \dfrac{KU}{m}\,t_5t_6$,
but whether these node-forms belong to the same or to different surfaces is not ascertained.

The enumeration is not extended to the 10-nodal surfaces, but I consider one case of these surfaces.

The $10(\alpha)$-nodal surface $10(3,3)$.

151. I assume only that there is a single node $(3,3)$: taking the cone to be

$$(sA+mt_1t_2t_3)(s'A+m't_4t_5t_6)=0;$$

then for the equation of the surface, in the node-form $(3,3)$ in question, we have

$$\Gamma=C+\frac{mm'}{K}(ss'A+sm't_4t_5t_6+s'm\,t_1t_2t_3).$$

But I present this result under a different form, as follows: I write

$$A=p^2+ft_1t_2=q^2+gt_3t_4=r^2+ht_5t_6;$$

where f, g, h are constants, and, as before, $p=0$, $q=0$, $r=0$ are the lines joining the points of contact of t_1, t_2; t_3, t_4; and t_5, t_6 respectively: we have

$$sA+mt_1t_2t_3=sA+mt_3\Big(\frac{A-q^2}{g}\Big),\quad\text{ and }\quad s'A+m't_4t_5t_6=s'A\Big(\cdots\Big),$$

or in place of s, s' introducing new linear functions σ, σ', the cubic curves may be taken to be $\sigma A-\dfrac{m}{g}\,q^2t_3$, $\sigma'A-\dfrac{m'}{h}\,r^2t_5$, so that we have

$$\frac{mm'}{K}\Big(\sigma A-\tfrac{m}{g}\,q^2t_3\Big)\Big(\sigma'A-\tfrac{m'}{h}\,r^2t_5\Big)+\cdots=A\Big(\cdots\Big)-\frac{mm'}{fgh}\Big(\cdots\Big);$$

whence $B=\dfrac{mm'}{fgh}\,(pqr+tA)$, where t is a linear function of the coordinates; and we then have

$$\Gamma=C+\frac{mm'}{K}[\sigma\sigma'A+\cdots],$$

where A may be considered as standing for $q^2+gt_3t_4$. The equation $Aw^2+2Bw+\Gamma=0$ of the surface, substituting throughout for A its value, is therefore

$$\Big(\tfrac{1}{g}\Big)(q^2+gt_3t_4)\,w^2+2\big\{pqr+t(q^2+gt_3t_4)\big\}\,w+(\cdots)=0,$$

where the cone is $\cdots=0$.

152. Writing in the equation of the surface $w\sqrt{f}$ instead of w, it becomes

$$(q^2+gt_3t_4)\,w^2+2\big[pqr+t(q^2+gt_3t_4)\big]\,w+\tfrac{fgh}{mm'}\Big[\tfrac{m'}{f}(q^2+gt_3t_4)+2tpqr\Big]=0,$$

and then writing $\dfrac{m}{f}\sigma$ and $\dfrac{m'}{h}\sigma'$ for σ and σ' respectively, this is

$$(q^2+gt_3t_4)\,w^2+2pqrw+2tw(q^2+gt_3t_4)+g\big[\cdots\big]+t^2(q^2+gt_3t_4)+2tpqr=0.$$

We may consider t_3, t_4 as denoting not the functions originally so represented, but these functions each multiplied by a suitable constant, and thereupon write $g=-1$; viz., $t_3=0$, $t_4=0$, will now denote any two tangents to the conic $A = 0$, the implicit factors being so determined that $A = q^2 - t_3t_4$. The equation of the surface is

$$(q^2 - t_3t_4)\,w^2 + 2pqrw + 2tw(q^2 - t_3t_4)$$

$$-\sigma\sigma'(q^2 - t_3t_4) + \sigma r^2t_5 + \sigma' p^2t_3 + p^2r^2 + t^2(q^2 - t_3t_4) + 2tpqr = 0;$$

viz., this is

$$-(q^2 - t_3t_4)\big[(w+t)^2 - \sigma\sigma'\big] + 2pqr(w+t) + \sigma r^2t_5 + \sigma' p^2t_3 + p^2r^2 = 0,$$

the sextic cone being

$$-\big[\sigma(q^2 - t_3t_4) - p^2t_3\big]\cdot\big[\sigma'(q^2 - t_3t_4) - r^2t_5\big] = 0.$$

153. But the foregoing equation of the surface is

$$\begin{vmatrix} -\sigma', & w+t, & \cdot, & r \\ w+t, & -\sigma, & p, & \cdot \\ \cdot, & p, & -q, & t_3 \\ r, & \cdot, & t_3, & -q \end{vmatrix} = 0,$$

as is at once seen by developing the determinant; the functions $w+t$, σ, σ', p, q, r, t_3, t_5 are all of them linear; and the determinant is thus a symmetrical quartic determinant the terms whereof are linear functions of the coordinates; viz. the surface is a symmetroid. That is, a surface having a single node $(3,3)$ is a symmetroid; but I have shown (Second Memoir, No. 116) that a symmetroid has each of its ten nodes $(3,3)$; wherefore the surface having a single node $(3,3)$ is the $10(\alpha)$-nodal surface, nodes $10(3,3)$.

154. Start from two cubic cones $U = 0$, $V = 0$, having each the same vertex $(x = 0, y = 0, z = 0)$; we may in a variety of ways determine the two cones $\alpha U + \beta V = 0$, $\gamma U + \delta V = 0$, having a common inscribed quadric cone $A = 0$ (viz., $\alpha : \beta$ being assumed at pleasure, then $\gamma : \delta$ will be determined; not, I believe, uniquely, but I do not know what the multiplicity is). This being so, the quadric cone $A = 0$ is uniquely determined; and then, assuming at pleasure the plane $w = 0$, the $10(\alpha)$-nodal surface $Aw^2 + 2Bw + \Gamma = 0$ is uniquely determined: consequently the remaining nine nodes are determinate points on the nine lines $U = 0$, $V = 0$ respectively. And we have thus a system of ten points in space such that, joining any one of them with the remaining nine, the nine lines so obtained are the intersections of two cubic cones, or say that they are an ennead of lines.

Notation for the Cases afterwards considered.

155. I proceed to further develop the theory of some of the different surfaces. The same node-form of equation will, of course, assume different shapes according to the actual expressions in terms of the coordinates (x, y, z) of the several functions A, B, &c., which enter into it. I have found it convenient to attribute to A and B certain specific values which are not in every case those of the coefficients of w^2, w in the equation of the surface: this means that we must, in the equation of the surface, substitute new symbols for these coefficients, and write the equation say in the form $A'w^2 + 2B'w + \Gamma' = 0$; the change of notation, when it occurs, will be duly explained.

156. It is in general (but not always) convenient to take the equation of the tangent cone to be $x^2+y^2+z^2-2yz-2zx-2xy=0$; for then any plane $\dfrac{x}{\alpha}+\dfrac{y}{\beta}+\dfrac{z}{\gamma}=0$, where $\alpha+\beta+\gamma=0$, will be a tangent plane; so that six tangent planes may be represented by $x=0$, $y=0$, $z=0$, and by three equations of the form just referred to. And in reference to this assumed form of the equation of the tangent cone, and to what follows, I write

$$\begin{aligned}
&\alpha+\beta+\gamma=0,\\
&\alpha'+\beta'+\gamma'=0,\\
&\alpha''+\beta''+\gamma''=0,\\
&P\;=\tfrac{x}{\alpha}+\tfrac{y}{\beta}+\tfrac{z}{\gamma},\\
&P'=\tfrac{x}{\alpha'}+\tfrac{y}{\beta'}+\tfrac{z}{\gamma'},\\
&P''=\tfrac{x}{\alpha''}+\tfrac{y}{\beta''}+\tfrac{z}{\gamma''},
\end{aligned}$$

$$\begin{aligned}
X&=\alpha\,(\gamma'\gamma''y-\beta'\beta''z),\\
Y&=\beta\,(\alpha'\alpha''z-\gamma'\gamma''x),\\
Z&=\gamma\,(\beta'\beta''x-\alpha'\alpha''y),\\
X'&=\alpha'\,(\gamma''\gamma y-\beta''\beta z),\\
Y'&=\beta'\,(\alpha''\alpha z-\gamma''\gamma x),\\
Z'&=\gamma'\,(\beta''\beta x-\alpha''\alpha y),\\
X''&=\alpha''\,(\gamma\gamma'y-\beta\beta'z),\\
Y''&=\beta''\,(\alpha\alpha'z-\gamma\gamma'x),\\
Z''&=\gamma''\,(\beta\beta'x-\alpha\alpha'y),
\end{aligned}$$

$$\begin{gathered}
A=x^2+y^2+z^2-2yz-2zx-2xy,\\
B=\alpha\alpha'\alpha''\,(y^2z-yz^2)+\beta\beta'\beta''\,(z^2x-zx^2)+\gamma\gamma'\gamma''\,(x^2y-xy^2)+M\,xyz,\\
C=(\alpha\alpha'\alpha''yz+\beta\beta'\beta''zx+\gamma\gamma'\gamma''xy)^2-\tfrac{1}{4}\cdots
\end{gathered}$$

where

$$\begin{aligned}
M&=(\beta-\gamma)\alpha'\alpha''+(\gamma-\alpha)\beta'\beta''+(\alpha-\beta)\gamma'\gamma'',\\
&=(\beta'-\gamma')\alpha''\alpha+(\gamma'-\alpha')\beta''\beta+(\alpha'-\beta')\gamma''\gamma,\\
&=(\beta''-\gamma'')\alpha\alpha'+(\gamma''-\alpha'')\beta\beta'+(\alpha''-\beta'')\gamma\gamma',
\end{aligned}$$

$$\tfrac{1}{2}\cdots=\big[(\beta-\gamma)(\beta'-\gamma')(\beta''-\gamma'')+(\gamma-\alpha)(\gamma'-\alpha')(\gamma''-\alpha'')+(\alpha-\beta)(\alpha'-\beta')(\alpha''-\beta'')\big];$$

also

$$A^2=4\alpha\alpha'\alpha''\beta\beta'\beta''\gamma\gamma'\gamma'';$$

and we have identically [an identity follows from this notation].

The 16-nodal Surface 16(1, 1, 1, 1, 1, 1).

157. Kummer starts from an irrational equation, which is readily converted into the following

$$\sqrt{x(X-w)}+\sqrt{y(Y-w)}+\sqrt{z(Z-w)}=0,$$

and then, rationalizing, we have

$$Aw^2+2Bw+C=0,$$

where as above

$$AC-B^2=K\,xyz\,PP'P''.$$

This agrees with the foregoing theory; viz., the point $(x=0, y=0, z=0)$ being a node, the rationalized equation must, of course, be in the node-form $(1,1,1,1,1,1)$, (being the only node-form); and the symmetry of the formulae enables us at once to write down the equations of the 16 singular planes, and thence to deduce the coordinates of the 16 nodes; viz.,

the singular planes are	**and the nodes are**
(1) $x=0$,	(1) $(0,\ -\beta,\ \gamma,\ \alpha'\alpha''\beta\gamma)$,
(2) $y=0$,	(2) $(\alpha,\ \cdot,\ -\gamma,\ \beta'\beta''\gamma\alpha)$,
(3) $z=0$,	(3) $(-\alpha,\ \beta,\ \cdot,\ \gamma'\gamma''\alpha\beta)$,
(4) $w=0$,	(4) $(\alpha'\alpha'',\ \beta'\beta'',\ \gamma'\gamma'',\ \cdot)$,
(5) $X-w=0$,	(5) $(1,\ \cdot,\ \cdot,\ \cdot)$,
(6) $Y-w=0$,	(6) $(\cdot,\ 1,\ \cdot,\ \cdot)$,
(7) $Z-w=0$,	(7) $(\cdot,\ \cdot,\ 1,\ \cdot)$,
(8) $P=0$,	(8) $(\cdot,\ \cdot,\ \cdot,\ 1)$,
(9) $X'-w=0$,	(9) $(\cdot,\ -\beta',\ \gamma',\ \alpha''\alpha\beta'\gamma')$,
(10) $Y'-w=0$,	(10) $(\alpha',\ \cdot,\ -\gamma',\ \beta''\beta\gamma'\alpha')$,
(11) $Z'-w=0$,	(11) $(-\alpha',\ \beta',\ \cdot,\ \gamma''\gamma\alpha'\beta')$,
(12) $P'=0$,	(12) $(\alpha''\alpha,\ \beta''\beta,\ \gamma''\gamma,\ \cdot)$,
(13) $X''-w=0$,	(13) $(\cdot,\ -\beta'',\ \gamma'',\ \alpha\alpha'\beta''\gamma'')$,
(14) $Y''-w=0$,	(14) $(\alpha'',\ \cdot,\ -\gamma'',\ \beta\beta'\gamma''\alpha'')$,
(15) $Z''-w=0$,	(15) $(-\alpha'',\ \beta'',\ \cdot,\ \gamma\gamma'\alpha''\beta'')$,
(16) $P''=0$,	(16) $(\alpha\alpha',\ \beta\beta',\ \gamma\gamma',\ \cdot)$,

where the nodes and planes are numbered as by Kummer; and by means of his (differently arranged) diagram of the relation between the several nodes and planes, I was enabled to form the following square diagram, which exhibits this relation in, I think, the most convenient form. To explain this, observe that in the upper and left-hand margins, the numbers refer to the nodes; in the body of the table, and in the right-hand margin to the planes, the table shows that for the node 1, the circumscribed cone is made up of the planes $1,6,7,8,9,13$; and that the remaining 15 nodes are situate on the nodal lines of this cone, the node 2 on the intersection of the planes $7,8$; the node 3 on the intersection of the planes $6,8$, and so on; and the like as regards the other lines of the table.

[The original page 283 here gives a 16×16 square diagram (Table No. 12) exhibiting the incidence of the 16 nodes with the 16 singular planes of the 16-nodal surface. Each row i shows the planes through node i (the six planes whose intersection is the node), and the body of the table records, in row i column j, the pair of planes whose intersection at node j lies on the sextic cone of node i. The right-hand margin lists the planes meeting at the node of that row: $1\ldots\ 1,6,7,8,9,13$; $2\ldots\ 2,6,8,10,14,\ldots$; etc. (See PDF page 283 of the volume for the full table.)]

158. The before mentioned irrational equation may be written

$$\sqrt{1\cdot 5}+\sqrt{2\cdot 6}+\sqrt{3\cdot 7}=0,$$

and by symmetry we see that also

$$\sqrt{1\cdot 9}+\sqrt{2\cdot 10}+\sqrt{3\cdot 11}=0,$$

$$\sqrt{1\cdot 13}+\sqrt{2\cdot 14}+\sqrt{3\cdot 15}=0;$$

viz., these are three equations each containing the planes 1, 2, 3, which are three of the planes belonging to the node 1; the other three planes in any such equation (for instance, the planes 5, 6, 7, in the first equation) being three planes belonging to another node. Instead of the planes 1, 2, 3, we may have any other three planes belonging to the node 1; and instead of the node 1, any other node; but each equation belongs to two nodes: the number of equations is thus

$$\tfrac{6\cdot 5\cdot 4}{1\cdot 2\cdot 3} \times 16 \times 3 \div 2, \; = 480.$$

159. To obtain the planes belonging to any such equation, combine any two of the outside right-hand lines of the diagram, these contain in common two numbers the places of which are interchanged; striking these out, we have four columns, and taking out of these any three columns, we have the corresponding sets of planes. For instance, lines 1 and 2 contain 7 8 and 8 7 respectively; striking these out, the lines are

$$1, \quad 9, 13, \quad 6;$$

$$2, \quad 10, 14, \quad 5;$$

whence we have the sets $(1, 9, 13)$ and $(2, 10, 14)$; viz., there is an irrational equation of the form

$$\sqrt{1\cdot 2} + \sqrt{9\cdot 10} + \sqrt{13\cdot 14} = 0,$$

but it is probably necessary to introduce constant factors along with the products $1\cdot 2$, $9\cdot 10$, and $13\cdot 14$ respectively. There are $\frac{1}{2}16\cdot 15$, $= 120$ pairs of lines, and each line gives 4 equations; in all 120×4, $= 480$ equations, as above.

160. I stop to remark that Kummer gives for his 13-nodal surface an equation containing three arbitrary constants, say λ, μ, ν, such that, putting one of these $= 0$, we have the 14-nodal surface; putting two of them each $= 0$, the 15-nodal surface; putting all three of them $= 0$, the 16-nodal surface. The equations for the 16-nodal surface and that for the 14-nodal surface, made use of by Kummer, are, in fact, those deduced as above from the equation of the $13(\alpha)$-nodal surface; and the like form might have been used for the 15-nodal surface. But the form actually used by Kummer, as presently appearing, is an equivalent form not thus deducible from the equation of the $13(\alpha)$-nodal surface.

The 15-nodal Surface $15(2, 1, 1, 1, 1)$.

161. Kummer's equation is readily converted into the following:

$$Aw^2 + 2Bw + C + \tfrac{Kl}{m}\, xyzP = 0,$$

the circumscribed cone being thus

$$(lA + mP'P'')\, xyzP = 0,$$

and the equation being in the node-form $(2, 1, 1, 1, 1)$.

The formulae for the 15 nodes and the 10 singular planes depend upon a quadric equation, for the symmetrical expression of which I write

$$\omega = \alpha'\alpha'' - \beta'\gamma'' = \beta'\beta'' - \gamma''\alpha' = \gamma'\gamma'' - \alpha''\beta',$$

$$\varpi = \alpha'\alpha'' - \beta'\gamma'' = \beta'\beta'' - \gamma'\alpha'' = \gamma'\gamma'' - \alpha'\beta'';$$

so that

$$\omega - \varpi = \frac{\beta'\gamma'' - \beta''\gamma'}{\alpha} = \frac{\gamma'\alpha'' - \gamma''\alpha'}{\beta} = \frac{\alpha'\beta'' - \alpha''\beta'}{\gamma},$$

$$\omega + \varpi = \alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'' + \cdots ;$$

the equation in question then is

$$(\rho - \omega)(\rho - \varpi) - \frac{Kl}{4\alpha\beta\gamma m} = 0;$$

so that, calling the roots of it ρ_1, ρ_2, we have

$$\rho_1 + \rho_2 = \omega + \varpi, \quad \rho_1\rho_2 = \omega\varpi + \frac{Kl}{4\alpha\beta\gamma m};$$

or we may write

$$\rho_1 = \tfrac{1}{2}(\omega + \varpi + \sqrt{\Omega}), \quad \rho_2 = \tfrac{1}{2}(\omega + \varpi - \sqrt{\Omega}),$$

if for shortness

$$\Omega = (\omega - \varpi)^2 - \frac{Kl}{\alpha\beta\gamma m}.$$

162. I write also for shortness

$$\begin{aligned}
\mathfrak{a} &= (\beta - \gamma)\alpha'\alpha'' + \alpha(\beta'\beta'' - \gamma'\gamma''),\\
\mathfrak{b} &= (\gamma - \alpha)\beta'\beta'' + \beta(\gamma'\gamma'' - \alpha'\alpha''),\\
\mathfrak{c} &= (\alpha - \beta)\gamma'\gamma'' + \gamma(\alpha'\alpha'' - \beta'\beta'');
\end{aligned}$$

and I say that the singular planes are

$$\begin{aligned}
&(1)\ x = 0, \quad (2)\ y = 0, \quad (3)\ z = 0,\\
&(9)\ \ -\tfrac{\gamma'\beta'}{\alpha}\, p_1(Z' - w) + (\rho_1 - \omega)\left(\tfrac{x}{\beta} + \tfrac{y}{\gamma}\right) = 0,\\
&(10)\ \ -\tfrac{\beta'\alpha'}{\gamma}\, p_1(Y' - w) + (\rho_1 - \omega)\left(\tfrac{z}{\alpha} + \tfrac{x}{\beta}\right) = 0,\\
&\quad \cdots\\
&P = 0;
\end{aligned}$$

[The full lists of 15 *nodes and* 10 *singular planes, with their pairings to Kummer's numbering, occupy book pages* 285–286*. The formulae are routine consequences of the symmetric functions* $\rho_1, \rho_2, \mathfrak{a}, \mathfrak{b}, \mathfrak{c}$ *above. See the PDF pages* 307–308 *of the scan for the literal expressions.]*

163. The small reference numbers are those used by Kummer. It is, I think, better to retain the reference numbers belonging to the case of the 16-nodal surface; viz., there are here given, large, $1, 2, 3, 8, 9, 10, 11, 13, 14, 15$ for the planes, and $1, 2, 3, 5, 6, \ldots$ to 16 for the nodes. Belonging to each node (that is, with the node as vertex) there is a quadric cone passing through 8 other nodes; and each node lies (exclusively of the cone whose vertex it is) in 8 such cones. We have thus the following square diagram:

[The original page 287 *here gives a* 15×15 *square diagram exhibiting, for the* 15*-nodal surface, the incidence of nodes with planes and quadric cones* C*, where the cone* C *replaces a pair of the planes of the* 16*-nodal table. See PDF page* 287 *for the table.]*

164. The arrangement is the same as in the 16-nodal square diagram; only, in the right-hand margin, a bracket $(6, 7)$ denotes that instead of the planes $(6, 7)$ we have a quadric cone; which cones are, in the body of the table, denoted by C. Thus for the node 1 the sextic cone is made up of the planes 1, 8, 9, 13 and of a quadric cone $(6, 7)$, $= C$: the remaining 14 nodes lie on the nodal lines of the sextic cone, viz., the node 2 on an intersection of the cone C with the plane 8, the node 3 on an intersection of the same cone and plane, the node 5 on the intersection of the planes 9, 13, and so on.

The Equation of the 15-nodal Surface, as deduced from that of the 13(α)-nodal.

165. If, in the equation hereafter given for the 13-nodal surface, we write $\nu = 0$, $\mu = 0$, or (what is the same thing) in that of the 14-nodal surface we write $\mu = 0$, the form is

$$\big[A + 4\lambda yz\big]\, w^2 + 2w\big[B - 2\lambda yzX\big] + C = 0.$$

The circumscribed sextic cone is thus $(2, 1, 1, 1, 1)$,

$$(\alpha\alpha''\beta\beta''yz + \beta\gamma\alpha''zx + \gamma\alpha\beta''xy + \alpha\beta xz)\, yz\, (P' - p_1P'')(P' - p_2P'') = 0,$$

where p_1, p_2 now denote the roots of the equation

$$PP'\,(p\alpha - \alpha''\gamma' - \alpha'\gamma'')\,(\cdots) + (\cdots)\, p\alpha'\alpha'' = 0.$$

The singular planes are

$$\begin{aligned}
&w = 0,\\
&P' - p_1P'' = 0,\\
&P' - p_2P'' = 0,\\
&z = 0,\\
&y = 0,\\
&\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_2\,(Z' - w) - (Z'' - w) = 0,\\
&\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_1\,(Z' - w) - (Z'' - w) = 0,\\
&\tfrac{\beta'\gamma''}{\alpha'\alpha''}\, p_2\,(Y' - w) - (Y'' - w) = 0,\\
&\tfrac{\beta'\gamma''}{\alpha'\alpha''}\, p_1\,(Y' - w) - (Y'' - w) = 0,\\
&Z - w = 0,
\end{aligned}$$

and we have then the same square table as before: the coordinates of the 15 nodes may be obtained without difficulty.

166. The form is really equivalent to that first considered in regard to the 15-nodal surface. To show that this is so, we have only to arrange according to powers of x; viz., the equation thus becomes

$$\begin{aligned}
&x^2\big[(\gamma'\gamma'')^2y^2 + (\beta'\beta'')^2z^2 + w^2 - 2w\beta'\beta''z + 2w\gamma'\gamma''y + 2\beta'\beta''\gamma'\gamma''yz\big]\\
&\quad + 2x\big[\cdots\big] + \big[aa'\alpha''yz + wy - wz\big]^2 - 4\lambda yzw(X - w) = 0,
\end{aligned}$$

where, if for a moment A denotes the coefficient of x^2, we have $y = 0$, $z = 0$, $w = 0$, $X - w = 0$, four tangent planes of the quadric cone $A = 0$.

14-nodal Surface $8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$, Node-form $(3_1, 1, 1, 1)$.

167. In the equation hereafter given for the 13-nodal surface, writing $\nu = 0$, the sextic cone becomes

$$4z(\lambda\alpha y^2 + \mu\beta zx + \beta'\gamma''xy + \gamma\alpha xy^2 + \alpha\beta xyz)\,(P' - p_1P'')(P' - p_2P'') = 0;$$

viz., this is of the form in question $(3_1, 1, 1, 1)$; and the equation of the surface is

$$w^2\big\{A + 4(\lambda y + \mu x)z - 4\lambda\mu z^2\big\} + 2w\big\{B - 2(\lambda yzX + \mu zxY)\big\} + C = 0.$$

The singular planes are

$$\begin{aligned}
&w = 0, && (4)\\
&P' - p_1 P'' = 0, && (12)\\
&P' - p_2 P'' = 0, && (16)\\
&z = 0, && (3)\\
&\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_2\,(Z' - w) - (Z'' - w) = 0, && (15)\\
&\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_1\,(Z' - w) - (Z'' - w) = 0, && (11)
\end{aligned}$$

and the nodes are

$$(0, 0, 0, 1), \quad (1, 0, 0, 0), \quad (0, 1, 0, 0), \quad (0, 0, 1, 0),$$

together with the 11 other nodes corresponding (with Kummer's small numbers in parentheses) to the table on page 289 of the scan.

168. In these formulae, p_1, p_2 denote the roots of the equation

$$\begin{aligned}
&\bigl[\lambda(\beta'\gamma'')^2 + \mu(\alpha\beta'')^2\bigr](p\alpha\beta\gamma)^2\\
&\quad + \bigl[\lambda(\beta'\gamma')^2 + \mu(\gamma'\alpha')^2\bigr](\alpha''\beta'\gamma'')^2\\
&\quad - 2\bigl[\lambda\beta'\beta'' + \mu\alpha'\alpha'' + 2\gamma'\gamma''(\omega - \varpi)^2\bigr]\, p\alpha\beta\gamma\, \alpha''\beta'\gamma'' = 0.
\end{aligned}$$

169. The relation of the nodes and sextic cones is given by means of the square diagram on the opposite page:

[The original page 291 here gives a 14×14 square diagram of the 14-nodal surface, with notations C for the nodal cubic cone (a cubic replacing planes $5, 6, 8$), D, D' for two quadric cones (replacing the planes $2, 5$ and $9, 13$ respectively), and $\times$ denoting the nodal line of the cubic. Vacant squares are understood as (D, D'). See PDF page 291 for the full table.]

170. The equation of the surface may be presented in the irrational form

$$\begin{aligned}
&\sqrt{(P' - p_1 P'')}\bigl\{\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_2\,(Z' - w) - \gamma'\alpha''\beta''(Z'' - w)\bigr\}\\
&\quad + \sqrt{(P' - p_2 P'')}\bigl\{\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_1\,(Z' - w) - \gamma'\alpha''\beta''(Z'' - w)\bigr\}\\
&\quad + (p_2 - p_1)\,\mu\alpha'\alpha''\beta'\sigma\sqrt{\cdots}\,(z_0 + \cdots)\, zw = 0.
\end{aligned}$$

In fact the norm of the left-hand side is

$$\begin{aligned}
&(p_1 - p_2)^2(\omega - \varpi)^2\bigl\{w^2\bigl[A + 4z(\lambda y + \mu x) - 4\lambda\mu z^2\bigr]\\
&\quad + 2w\bigl[B - 2z(\lambda y X + \mu x Y)\bigr]\\
&\quad + C\bigr\}.
\end{aligned}$$

To partially verify this, observe that, writing the equation under the form $w\sqrt{R} + \sqrt{S} + \sqrt{T} = 0$, on writing therein $w = 0$, we ought to have

$$(R - S)^2 = (p_1 - p_2)^2(\omega - \varpi)^2(\alpha\alpha'\alpha'' yz + \beta\beta'\beta'' zx + \gamma\gamma'\gamma'' xy)^2.$$

But writing $w=0$, we have

$$\begin{aligned} R-S &= (P'-p_2P'')\,(\gamma'\alpha'\beta''p_1Z'-\gamma'\alpha''\beta''Z'') \\ &\quad -(P'-p_1P'')\,(\gamma'\alpha'\beta''p_2Z'-\gamma'\alpha''\beta''Z'') \\ &= (p_2-p_1)\,(\gamma'\alpha''\beta''P''Z'-\gamma'\alpha''\beta''P'Z''), \\ &= (p_2-p_1)\big[\gamma''(\beta'\gamma'x+\gamma'\alpha''y+\alpha'\beta'z)\,(\beta\gamma\beta x-\alpha\alpha''y) \\ &\quad -\gamma'(\beta''\gamma''x+\gamma''\alpha''y+\alpha''\beta''z)\,(\beta\beta'x-\alpha\alpha'y)\big], \end{aligned}$$

which is easily found to be

$$=-(p_2-p_1)(\omega-\varpi)(\alpha\alpha'\alpha''yz+\beta\beta'\beta''zx+\gamma\gamma'\gamma''xy);$$

and thus $(R-S)^2$ has the value in question.

171. In further verification, observe that, writing $x,y,z=\alpha'\alpha'',\beta'\beta'',\gamma'\gamma''$, and therefore $P'=0$, $P''=0$, we ought to have

$$\tfrac{1}{4}(p_1-p_2)^2(\alpha'\beta'\gamma')^2\Big(\cdots\Big)z^2w^2=(p_1-p_2)^2(\omega-\varpi)^2w^2\big[A+4z(\lambda y+\mu x)-4\lambda\mu z^2\big],$$

observing that, for the values in question, B, X, Y, C all vanish.

This is

$$(p_1-p_2)^2(\alpha'\beta'\gamma')^2\Big(\tfrac{\lambda}{\alpha}+\tfrac{\mu}{\beta}\Big)(\gamma'\gamma'')^2z^2w^2=(\omega-\varpi)^2\big[(\omega-\varpi)^2+4\gamma'\gamma''(\lambda\beta'\beta''+\mu\alpha'\alpha''-\lambda\mu\gamma'\gamma'')\big],$$

which is in fact the value of $(p_1-p_2)^2$ obtained from the equation in p.

The $13(\alpha)$-nodal Surface $3(4_1,1,1)+1(3,1,1,1)+9(3_1,2,1)$.

172. The equation, node-form $(4_3,1,1)$, is

$$\begin{gathered} w^2\big\{A+4(\lambda yz+\mu zx+\nu xy)-4(\mu\nu x^2+\nu\lambda y^2+\lambda\mu z^2)\big\} \\ +2w\big\{B-2(\lambda yzX+\mu zxY+\nu xyZ)\big\}+C=0; \end{gathered}$$

for the circumscribed cone we have

$$\begin{aligned} &\big\{A+4(\lambda yz+\mu zx+\nu xy)-4(\mu\nu x^2+\nu\lambda y^2+\lambda\mu z^2)\big\}\,C \\ &\quad -\big\{B-2(\lambda yzX+\mu zxY+\nu xyZ)\big\}^2 \\ &=4\big\{\lambda\alpha yz^2+\mu\beta zx^2+\nu\gamma xy^2+\beta\gamma\alpha xy^2+\gamma\alpha\beta yz^2+\alpha\beta\gamma z^2xy\big\} \\ &\quad\cdot\Big\{-\lambda\frac{X^2}{\alpha^2}-\mu\frac{Y^2}{\beta^2}-\nu\frac{Z^2}{\gamma^2}+\frac{1}{\alpha'\alpha''\beta'\beta''\gamma'\gamma''}\,P'P''\Big\}, \end{aligned}$$

where, on the right-hand side, the first factor, equated to zero, represents a trinodal quartic cone, the nodal lines whereof are $(y=0,z=0)$, $(z=0,x=0)$, $(x=0,y=0)$.

173. As regards the second factor, it is to be observed that, writing as above

$$\omega-\varpi=\frac{\beta'\gamma''-\beta''\gamma'}{\alpha}=\frac{\gamma'\alpha''-\gamma''\alpha'}{\beta}=\frac{\alpha'\beta''-\alpha''\beta'}{\gamma},$$

we have identically

$$\alpha'P'-\alpha''P''=\frac{\cdots}{\beta'\beta''}\,x-\cdots\,,$$

$$\beta'P'-\beta''P''=\cdots\,,\quad \gamma'P'-\gamma''P''=\cdots\,,$$

so that the second factor is

$$\tfrac{1}{(\omega-\varpi)^2}\Big[-\lambda\big(\beta'\beta''\gamma'\gamma''\big)^2(\alpha'P'-\alpha''P'')^2-\mu\big(\gamma'\gamma''\alpha'\alpha''\big)^2(\beta'P'-\beta''P'')^2$$
$$-\nu\big(\alpha'\alpha''\beta'\beta''\big)^2(\gamma'P'-\gamma''P'')^2+(\omega-\varpi)^2\,\alpha'\alpha''\beta'\beta''\gamma'\gamma''P'P''\Big];$$

or, what is the same thing,

$$\tfrac{1}{(\omega-\varpi)^2}\,\alpha'\alpha''\beta'\beta''\gamma'\gamma''\Big\{\cdots(\alpha'\beta'\gamma'P')^2+\cdots(\alpha''\beta''\gamma''P'')^2+2[\cdots]\,\alpha'\beta'\gamma'P'\,\alpha''\beta''\gamma''P''\Big\},$$

so that, equating to zero, we have a pair of planes, each passing through the line $P'=0$, $P''=0$, and which it is clear must be the tangent planes from this line to the quadric cone $A'=0$. I will presently return to this, but I consider first the foregoing identical equation in regard to the circumscribed cone.

174. In verification hereof, observe first that, if $x=0$, the equation becomes

$$\begin{aligned}&\big\{(y-z)^2+4\lambda yz-4\lambda(\beta\gamma y^2+\mu z^2)\big\}\,(\alpha\alpha'\alpha'')^2y^2z^2\\&\quad-\big[(\alpha\alpha'\alpha'')(y^2z-yz^2)-2\lambda yz\alpha(\beta'\beta''y-\gamma'\gamma''z)\big]^2\\&=4\lambda y^2z^2\big[-\lambda\alpha^2(\beta'\beta''y-\gamma'\gamma''z)^2-(\alpha\alpha'\alpha'')^2(\nu y^2+\mu z^2)-\alpha^2\alpha'\alpha''(\gamma'y+\beta'z)(\gamma''y+\beta''z)\big],\end{aligned}$$

viz., omitting terms which destroy each other, this is

$$\big[(y-z)^2+4\lambda yz\big](\alpha\alpha'\alpha'')^2y^2z^2-\big[\alpha\alpha'\alpha''(y^2z-yz^2)-2\lambda yz\,\alpha(\beta'\beta''y-\gamma'\gamma''z)\big]^2$$
$$=4\lambda y^2z^2\big[-\lambda\alpha^2(\beta'\beta''y-\gamma'\gamma''z)^2-\alpha^2\alpha'\alpha''(\gamma'y+\beta'z)(\gamma''y+\beta''z)\big].$$

Or again, this is

$$4\lambda yz\,(\alpha\alpha'\alpha'')^2y^2z^2+4\lambda yz\,\alpha^2\alpha'\alpha''(y^2z-yz^2)(\beta'\beta''y-\gamma'\gamma''z)$$
$$-4\lambda^2y^2z^2\alpha^2(\beta'\beta''y-\gamma'\gamma''z)^2=-4\lambda yz\,\alpha^2\alpha'\alpha''(\gamma'y+\beta'z)(\gamma''y+\beta''z);$$

viz., this is

$$\alpha'\alpha''yz+(y-z)(\beta'\beta''y-\gamma'\gamma''z)=(\gamma'y+\beta'z)(\gamma''y+\beta''z),$$

which is at once seen to be true.

175. Again, compare the terms which contain x^2yz. On the right-hand side, we have

$$\big(-K\nu-\mu-\nu-\tfrac14\cdots\big)4\beta\gamma yz\ \cdots\ \text{(a term in } x^2yz \text{ of } \alpha'\alpha''\beta\beta'\gamma\gamma''P'P'')^2,$$

viz., the coefficient is

$$4\beta\gamma\big\{-\mu(\gamma''\gamma)^2-\nu(\beta'\beta'')^2+\beta'\beta''\gamma'\gamma''\big\}.$$

On the left-hand side, that is in $A'C-B'^2$, the only terms which give rise to the terms in question are

$$\text{in } A',\ (1-4\mu\nu)x^2;\ \text{in } C,\ 2\beta\beta'\beta''\gamma\gamma'\gamma''\,x^2yz;$$

and in B^2

$$x^2z\big(-\gamma\gamma'\gamma''-2\mu\gamma'\beta'\beta''\big)-x^2y\big(\beta\beta'\beta''-2\nu\beta\gamma'\gamma''\big),$$

whence the coefficient is

$$-2\beta\beta'\beta''\gamma\gamma'\gamma''(1-4\mu\nu)+2\big(\gamma\gamma'\gamma''+2\mu\beta'\beta''\big)\big(\beta\beta'\beta''+2\nu\beta\gamma'\gamma''\big),$$

which is in fact

$$=4\beta\gamma\big\{-\mu(\gamma'\gamma'')^2-\nu(\beta'\beta'')^2+\beta'\beta''\gamma'\gamma''\big\},$$

which is right; and the verification may be completed without difficulty.

176. The singular planes are

$$\begin{array}{ll} w=0, & (4) \\ P'-p_1P''=0, & (12) \\ P'-p_2P''=0, & (16) \\ z=0, & (3) \end{array}$$

and the nodes are

$$\begin{array}{lr} (0,0,0,1), & (8) \\ (1,0,0,0), & (5) \\ (0,1,0,0), & (6) \\ (0,0,1,0), & (7) \\ \left(\frac{\alpha\alpha'\alpha''}{\alpha'-\alpha p_1},\ \frac{\beta\beta'\beta''}{\beta'-\beta p_1},\ \frac{\gamma\gamma'\gamma''}{\gamma''-\gamma p_1},\ 0\right), & (16) \\ \left(\frac{\alpha\alpha'\alpha''}{\alpha'-\alpha p_2},\ \frac{\beta\beta'\beta''}{\beta'-\beta p_2},\ \frac{\gamma\gamma'\gamma''}{\gamma''-\gamma p_2},\ 0\right), & (12) \\ (\alpha'\alpha'',\ \beta'\beta'',\ \gamma'\gamma'',\ 0), & (4) \\ \text{three nodes on line } P'-p_1P''=0,\ z=0 & (13,14,15) \\ \text{three nodes on line } P'-p_2P''=0,\ z=0 & (9,10,11) \end{array}$$

where the large numbers are those referring to the 16-nodal surface, and are here adopted. In the foregoing formulae p_1, p_2 are the roots of

177.

$$\begin{aligned} &\left[\lambda(\beta'\gamma'')^2+\mu(\gamma'\alpha')^2+\nu(\alpha'\beta')^2\right](p\alpha'\beta'\gamma')^2 \\ &\quad+\left[\lambda(\beta''\gamma'')^2+\mu(\gamma''\alpha'')^2+\nu(\alpha''\beta'')^2\right](\alpha''\beta''\gamma'')^2 \\ &\quad-2\left[\lambda\beta'\beta''\gamma'\gamma''+\mu\gamma'\alpha'\gamma''\alpha''+\nu\alpha'\beta'\alpha''\beta''+\tfrac{1}{4}(\omega-\varpi)^2\right]p\alpha'\beta'\gamma'\alpha''\beta''\gamma''=0. \end{aligned}$$

We have the square diagram in the following page:

[The original page 296 here gives a 13×13 square diagram for the $13(\alpha)$-nodal surface, with notations including the cubic cone C (or G) replacing the planes $5,6,7$ at node 4, the trinodal quartic cone $(8;\,1,2,3)$ at node 8, the nodal cubic $(2,3;\,5)$ at node 5, &c. The vacant places are understood to be CC. See PDF page 296 for the full table.]

where for greater clearness I have omitted the symbol CC, which is to be understood as occupying each of the vacant squares.

The arrangement is the same as before; the right-hand margin shows the sextic cone; viz., for the node 4 this is made up of the singular planes 4, 12, 16 and of a cubic cone represented by $(5,6,7)$ (as replacing the planes 5, 6, 7 in the case of the 16-nodal surface). Similarly for the node 5, the sextic cone is made up of the singular plane 4, the nodal cubic cone $(2,3;\,5)$, and the quadric cone $(9,13)$ (the numbers in these last symbols indicating the planes in the case of the 16-nodal surface, which are here replaced by cones). So for the node 8, the sextic cone is made up of the singular planes 12, 16 and of the trinodal quartic cone $(8;\,1,2,3)$. As regards a nodal cubic cone, for example $(2,3;\,5)$, the semicolon is used to indicate that the nodal line replaces the intersection of the planes 2, 3; the other intersections 2, 5 and 3, 5 having disappeared. And so for a trinodal quartic cone $(8;\,1,2,3)$, the semicolon is used to indicate that the nodal lines replace the intersection $(1,2)$, the intersection $(1,3)$, and the intersection $(2,3)$ respectively; the other intersections 1, 8; 2, 8; and 3, 8 having disappeared. Finally, in the body of the table, C

is used to denote the cubic or the quartic cone (as the case may be); × to denote a nodal line of either of these cones; and C' the quadric cone; as already mentioned, the vacant places are considered to be CC.

The reading of the table is then as follows; viz., for the node 4, the remaining twelve nodes lie on the nodal lines of the sextic cone 4, 12, 16, $(5, 6, 7)$, as shown; viz., 5, 6, 7 are each of them on the intersection of the cubic cone with the plane 4; 8 is on the intersection of the planes 12 and 16; and so on.

I reserve for another Memoir the discussion of the $13(\beta)$-nodal surface, and the surfaces with less than 13 nodes.

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[455] On Plücker's Models of Certain Quartic Surfaces.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 281–285. *Read June* 8, 1871.*]*

The Society possesses a series of 14 wooden models of surfaces, constructed under the direction of the late Prof. Plücker, in illustration of the theory developed in his posthumous work, "Neue Geometrie des Raumes gegründet auf die Betrachtung der geraden Linie als Raumelemente," Leipzig, 1869. These all of them represent, I believe, Equatorial Surfaces; viz., models 1 to 8 represent cases of the 78 forms of equatorial surfaces "deren Breiten-Curven eine feste Axenrichtung besitzen," vol. ii. pp. 352–363, the remaining models, Nos. 9–14, I have not completely identified. I propose to go into the theory only so far as is required for the explanation of the models.

In a "Complex," or triply infinite system of lines, there is in any plane whatever a singly infinite system of lines enveloping a curve; and if we attend only to the curves the planes of which pass through a given fixed line, the locus of these curves is a "complex surface." Similarly, there is through any point whatever a single infinite series of lines generating a cone; and if we attend only to the cones which have their vertices in the given fixed line, then the envelope of these cones is the same complex surface. In the case considered of a complex of the second degree, the curves and cones are each of them of the second order; the fixed line is a double line on the surface, so that (attending to the first mode of generation) the complete section by any plane through the fixed line is made up of this line twice, and of a conic; the surface is thus of the order 4: it is also of the class 4; the surface has, in fact, the nodal line, and also 8 nodes (conical points), and we have thus a reduction = 32 in the class of the surface.

In the particular case where the nodal line is at infinity, the complex surface becomes an equatorial surface; viz., (attending to the first mode of generation) we have here a series of parallel planes each containing a conic, and the locus of these conics is the equatorial surface.

It is convenient to remark that, taking a, b, h to be homogeneous functions of (x, w) of the order 2; f, g of the order 1; and c of the order 0 (a constant); then the equation of a complex surface (see vol. i. p. 162) is

$$\begin{vmatrix} \cdot, & y, & z, & 1 \\ y, & a, & h, & g \\ z, & h, & b, & f \\ 1, & g, & f, & c \end{vmatrix} = 0;$$

and that, writing $w = 1$, or considering a, h, b; f, g; c as functions of x of the orders 2, 1, 0 respectively, we have an equatorial surface.

A particular form of equatorial surface is thus $bcy^2 + caz^2 + ab = 0$, or taking $c = 1$, this is $by^2 + az^2 + ab = 0$, where a, b are quadric functions of x.

The surface is still, in general, of the fourth order: it may however degenerate into a cubic surface, or even into a quadric surface; the last case is however excluded from the enumeration. The section by any plane parallel to that of yz is a conic; the section by the plane $y = 0$ is made up of the pair of lines $a = 0$, and of the conic $z^2 + b = 0$; that by the plane $z = 0$ is made up of the pair of lines $b = 0$, and of the conic $y^2 + a = 0$; the last-mentioned planes may be called the principal planes, and the conics contained in them principal conics. The surface is thus the locus of a variable conic, the plane of which is parallel to that of yz, and which has for its vertices the intersections of its plane with the two principal conics respectively. And we have thus the particular equatorial surfaces considered by Plücker, vol. ii. pp. 346–363 (as already mentioned), under the form

$$\frac{y^2}{Ex^2 + 2Ux + G} + \frac{z^2}{Fx^2 + 2Rx + B} + 1 = 0,$$

and of which he enumerates 78 kinds; viz., these are

1 to 17	Principal conics each proper.
18 „ 29	One of them a line-pair.
30 „ 32	Each a line-pair.
33 „ 39	Principal conics, each proper, but having a common point.
40 „ 43	One of them a line-pair, its centre on the other principal conic.
44 „ 61	One principal conic a parabola.
62 „ 73	One principal conic a pair of parallel lines.
74 „ 76	Principal conics each a parabola.
77 and 78	Principal conics, one of them a parabola, the other a pair of parallel lines.

The models 1 to 8 correspond hereto, as follows:

Mod. 1 is 13,		i.e. 2 is Mod. 9,
„ 2	9,	„ 3 40,
„ 3	40,	„ 4 34,
„ 4	34,	„ 5 2,
„ 5	2,	„ 6 3,
„ 6	3,	„ 7 4,
„ 7	4,	„ 8 32,
„ 8	32,	„ 13 1.

Mod. 5 is 2: the form of the equation is here

$$\frac{y^2}{l^2\left[(x-\alpha)^2 + \beta^2\right]} + \frac{z^2}{l'^2\left[(x-\alpha')^2 + \beta'^2\right]} = 1;$$

viz., the principal conics are one of them a hyperbola, the other imaginary; hence the generating conic has always two, and only two, real vertices, viz., it is always a hyperbola; there are no real lines.

Mod. 6 is 3: the form of the equation is

$$\frac{y^2}{l^2\left[(x-\alpha)^2 + \beta^2\right]} + \frac{z^2}{l'^2\left[(x-\alpha')^2 + \beta'^2\right]} = 1;$$

viz., the principal conics are each of them a hyperbola; the generating conic has four real vertices, viz., it is always an ellipse: there are no real lines.

Mod. 7 is 4: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2[(x-\alpha')^2+\beta'^2]} + 1 = 0.$$

The principal conics are one of them an ellipse, the other imaginary; for values of x between γ and δ, the variable conic has two real vertices or it is a hyperbola; for any other values it is imaginary, so that the surface lies wholly between the planes $x=\gamma$, $x=\delta$: the surface contains the real lines $y=0$, $x=\gamma$ and $y=0$, $x=\delta$.

Mod. 2 is 9: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta')} + 1 = 0,$$

where, say the values γ, δ lie between the values γ', δ': the principal conics are each of them an ellipse, the vertices (on the axis or line $y=0$, $z=0$) of the one ellipse lying between those of the other ellipse. The variable conic for values of x between γ, δ has four real vertices, or it is an ellipse; for values beyond these limits, but within the limits γ', δ', say from γ to γ', and from δ to δ', there are two real vertices, or the conic is a hyperbola; and for values beyond the limits γ', δ', the variable conic is imaginary.

There are four real lines $(y=0, x=\gamma)$, $(y=0, x=\delta)$, $(z=0, x=\gamma')$, $(z=0, x=\delta')$. The surface consists of a central pillow-like portion, joined on by two conical points to an upper portion, and by two conical points to an under portion, the whole being included between the planes $x=\gamma'$, $x=\delta'$.

Mod. 1 is 13: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta')} + 1 = 0;$$

the values γ', δ' lying between γ, δ; the principal conics are one of them a hyperbola, the other an ellipse, the vertices (on the axis or line $y=0$, $z=0$) of the hyperbola lying between those of the ellipse.

The variable conic, for values of x between γ', δ', has two real vertices, or it is a hyperbola; for the values, say, from γ' to γ, and δ' to δ, there are four real vertices, or the conic is an ellipse; for values beyond the limits γ, δ, there are two real vertices, and the conic is a hyperbola. There are the four real lines $(y=0, x=\gamma)$, $(y=0, x=\delta)$, and $(z=0, x=\gamma')$, $(z=0, x=\delta')$. The surface consists of 8 portions joined to each other by 8 conical points, but the form can scarcely be explained by a description.

Mod. 8 is 32: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)^2} + \frac{z^2}{l'^2(x-\gamma')^2} = 1;$$

viz., the principal conics are each of them a line-pair, the variable conic is always an ellipse.

There are the two real nodal lines $(y=0, x=\gamma)$ and $(z=0, x=\gamma')$, each of these being in the neighbourhood of the axis crunodal, and beyond certain limits acnodal; the surface is a scroll, being, in fact, the well-known surface which is the boundary of a small circular pencil of rays obliquely reflected, and consequently passing through two focal lines.

Mod. 4 is 34: the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta)} + 1 = 0,$$

where $x = \delta$ is not intermediate between the values $x = \gamma$ and $x = \gamma'$; say the order is δ, γ, γ'. The surface is thus a cubic surface; the principal conics are ellipses having on the axis a common vertex at the point $x = \delta$, and the remaining two vertices on the same side of the last-mentioned one. The variable conic for values between δ and γ has four real vertices, or it is an ellipse; for values between γ and γ' two real vertices, or it is a hyperbola; and for values beyond the limits δ, γ' it is imaginary. There are on the surface the two real lines $(y = 0, x = \gamma)$ and $(z = 0, x = \gamma')$. The surface consists of a finite portion joined on by two conical points to the remaining portion.

Mod. 3 is 40: the form of equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\delta)^2} + 1 = 0.$$

The surface is thus a cubic surface; the principal conics are, one of them an ellipse, the other a pair of imaginary lines intersecting on the ellipse; for values of x between γ and δ, the variable conic has thus two real vertices, and it is a hyperbola; for values beyond these limits it is imaginary, and the whole surface is thus included between the planes $x = \gamma$ and $x = \delta$. There are the two real lines $(y = 0, x = \gamma)$ and $(z = 0, x = \delta)$.

Taking $l^2 = l'^2 = 1$, the surface is

$$\frac{y^2}{(x-\gamma)(x-\delta)^2(x-\delta)^2} + \frac{z^2}{\cdots} + 1 = 0,$$

which is a particular case of the parabolic cyclide.

As already mentioned, I have not completely identified the remaining models 9 to 14, but I will say a few words about them.

The equatorial surfaces, not included in the preceding 78 cases, Plücker distinguishes (vol. ii. p. 363) as "gedrehte" or "tordirte," say as twisted equatorial surfaces; the equation of such a surface is

$$by^2 + 2hyz + az^2 + ab - h^2 = 0,$$

where

$$\begin{aligned} b &= Fx^2 - 2Rx + B, \\ a &= Ex^2 + 2Ux + G, \\ h &= Kx^2 - Ox - G \text{ (or in particular} = -Ox - G). \end{aligned}$$

Mod. 13 is such a surface, being a twisted form of model 2.

Mod. 9 and Mod. 14 belong, I think, to the case $a = 0$; viz., the form of the equation is $by^2 + 2hyz - h^2 = 0$. The variable conic is a hyperbola, the direction of one of the asymptotes being constant (vol. ii. p. 368).

There are moreover (p. 372) equatorial surfaces in which the variable conic is always a parabola, and where there are on the surface four real or imaginary lines.

Mods. 10, 11, and 12 seem to represent such surfaces.

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[456] NOTE ON THE DISCRIMINANT OF A BINARY QUANTIC.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), p. 23.]

It is well known that the discriminant of a binary quantic $(a, b, c, d, \ldots | t, 1)^n$ is of the form

$$Ma + Nb^2,$$

but it is further to be remarked that if $b = 0$, then the form is

$$a(Ma + Nc^3),$$

if $b = 0$, $c = 0$, the form is

$$a^2(Ma + Nd^4),$$

and so on, until only the lowest two coefficients are not put $= 0$. Or, what is the same thing, if in the discriminant of the original function we put $a = 0$, then the discriminant divides by b^2; if $b = 0$, the discriminant divides by a, and, omitting this factor, if we then write $a = 0$, it divides by c^3; if $b = 0$, $c = 0$, the discriminant divides by a^2, and omitting this factor, if we then write $a = 0$, it divides by d^4; and so on, until as before.

Thus if $b = 0$, the discriminant of $(a, 0, c, d, e \,|\, t, 1)^4$, divides by a, and omitting this factor it is

$$\begin{aligned} & a^2e^2 \\ & -\ 18\,ace^2 \\ & +\ 54\,acd^2e \\ & -\ 27\,ad^4 \\ & +\ 81\,c^4e \\ & -\ 54\,c^3d^2, \end{aligned}$$

which for $a = 0$ has the factor c^3; if $b = 0$, $c = 0$, the discriminant of $(a, 0, 0, d, e \,|\, t, 1)^4$ has the factor a^2, and omitting this factor it is

$$ae^4 - 27\,d^4,$$

which for $a = 0$ has the factor d^4; the series of theorems here terminates, since the lowest two coefficients d, e are not to be put $= 0$.

(End of chunk: PDF pages 276–325*, book pages* 254–303*.*
Paper 450 *(tail, two pages),*
paper 451 *in full,*
paper 452 *in full,*
paper 453 *in full,*
paper 454 *in full,*
paper 455 *in full,*
paper 456 *opening (one page).)*

[457] ON THE QUARTIC SURFACES $(*\,|U, V, W)^2 = 0$.

*[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (*1870*), pp.* 24–34.*]*

I propose to myself for investigation the quartic surfaces represented by an equation

$$(*\,|U,\,V,\,W)^2 = 0,$$

where U, V, W are quadric functions of the coordinates.

Such a surface has 8 nodes (conical points), viz., these are the points of intersection of the quadric surfaces $U = 0$, $V = 0$, $W = 0$. It is to be observed, that not every quartic surface with 8 nodes is included under the above form; in fact the equation of a quartic surface contains (homogeneously) 35 coefficients, or say 34 arbitrary parameters; in order that a given point may be a node, 4 conditions must be satisfied, and it is consequently possible to find a quartic surface having 8 given points as nodes (and having in its equation $(34 - 8\cdot 4 =)$ 2 arbitrary parameters): but 8 given points are not in general the intersections of three quadric surfaces, and such a quartic surface is therefore not in general included under the above form. I think, however, that it may be assumed that the above form includes all the quartic surfaces having 8 nodes, points of intersection of three quadric surfaces. It will presently appear that, included in the form, we have surfaces where (instead of the 8 nodes) there is a nodal or cuspidal conic; and that these are the most general forms of such quartic surfaces.

A quartic surface has at most 16 nodes, and the general form with 8 nodes must admit of being particularised so that the surface shall acquire any number not exceeding 8 of additional nodes. This does not show, but it is probable, that the above special form with 8 nodes can be particularised so that the surface shall in like manner acquire any number not exceeding 8 of additional nodes. Similarly, a quartic surface with a nodal conic may have besides 1, 2, 3, or 4 nodes; and it will be shown in the sequel how the form, particularised so as to give a nodal conic, may be further particularised so as to give the 1, 2, 3, or 4 nodes. So a quartic surface with a cuspidal conic may besides have 1 node, and it will be shown how the form, particularised so as to give a cuspidal conic, may be further particularised so as to give 1 node.

Starting from the equation $(*\,|\,U, V, W)^2 = 0$, we may, by substituting for U, V, W linear functions of these expressions, transform the equation precisely in the manner of a conic, and therefore into any of the forms under which the equation of a conic can be exhibited; for instance, in the forms

$$aU^2 + bV^2 + cW^2 = 0, \qquad fVW + gWU + hUV = 0, \qquad UW - V^2 = 0, \quad \text{\&c.}$$

I attend at present only to the last-mentioned form $UW - V^2 = 0$, which, it thus appears, is equally general with the original form $(*\,|\,U, V, W)^2 = 0$.

The quartic surface

$$UW - V^2 = 0,$$

where U, V, W are any quadric functions of the coordinates, may be considered as the envelope of the quadric surface

$$(U, V, W\,|\,\theta, 1)^2 = 0,$$

where θ is an arbitrary parameter. And it thus appears that it is very easy to reciprocate (in regard to any given quadric surface) the quartic surface. For the reciprocal of the quartic surface is clearly the envelope of the reciprocal of the variable quadric surface; this reciprocal is itself a quadric surface, and the reciprocal of the quartic surface is thus given in the same form as the original surface, viz., as the envelope of a quadric surface the equation whereof contains rationally the variable parameter θ; the equation of the reciprocal surface is consequently obtained by equating to zero the discriminant in regard to θ, of the equation of the reciprocal quadric surface.

It is to be observed that, inasmuch as the equation of the reciprocal quadric surface is of the third degree in the coefficients of the original quadric, it is in general of the degree 6 in the parameter; we have thus a sextic function of θ, the coefficients whereof are quadric functions of the coordinates; and the discriminant is a function of the order 10 in these coefficients, that is, of the order 20 in the coordinates. The reciprocal of the quartic surface is thus a surface of the order 20; this is right, for in a general quartic surface the order of the reciprocal surface is 36, and the 8 nodes reduce the order by 16; $36 - 16 = 20$.

In the equation $UW - V^2 = 0$, if U reduce itself to the square of a linear function, $= P^2$, the equation becomes $V^2 - P^2W = 0$, which is the general form of the quartic surface having the nodal conic $V = 0$, $P = 0$. And if, moreover, W be the product of this same linear function P by another linear function Q, $W = PQ$, then the form is $V^2 - P^3Q = 0$, which is the general form of the quartic surface having the cuspidal conic $V = 0$, $P = 0$.

Writing for greater convenience x, y in the place of P, Q respectively, we have the quartic

$$V^2 - x^3y = 0, \tag{AA}$$

having the cuspidal conic $V = 0$, $x = 0$; and which has besides the conic of plane contact $V = 0$, $y = 0$. In virtue of the cuspidal conic the reciprocal surface should be of the order 6; and by the foregoing method of obtaining the equation of the reciprocal surface, I will verify that this is so. To effect this as simply as possible, I fix the remaining coordinates z, w as follows. The line $x = 0$, $y = 0$ is not in general a tangent to the surface $V = 0$; it therefore meets this surface in two points, and we may take $z = 0$, $w = 0$ to be the equations of the tangent planes at these two points respectively; we have thus $V = ax^2 + 2hxy + by^2 + 2nzw$. Introducing for convenience the numerical factor 2, and taking the equation of the surface to be

$$(ax^2 + 2hxy + by^2 + 2nzw)^2 - 2x^3y = 0,$$

this is the envelope of the quadric surface

$$\theta^2 \cdot x^2 + 2\theta(ax^2 + 2hxy + by^2 + 2nzw) + 2xy = 0,$$

which is a surface $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n} \,|\, x, y, z, w)^2 = 0$, where $\mathfrak{a} = \theta^2 + 2a\theta$, $\mathfrak{b} = 2b\theta$, $\mathfrak{h} = 2\theta h + 1$, $\mathfrak{n} = 2\theta n$, and where all the other coefficients vanish. Assuming, as usual, that the reciprocation is effected in regard to the surface $x^2 + y^2 + z^2 + w^2 = 0$, the general equation is

$$\begin{aligned}
&x^2 \cdot \mathfrak{d}(\mathfrak{b}\mathfrak{c} - \mathfrak{f}^2) - \mathfrak{c}\mathfrak{m}^2 - \mathfrak{b}\mathfrak{n}^2 + 2\mathfrak{f}\mathfrak{m}\mathfrak{n} \\
&+ y^2 \cdot \mathfrak{d}(\mathfrak{c}\mathfrak{a} - \mathfrak{g}^2) - \mathfrak{a}\mathfrak{n}^2 - \mathfrak{c}\mathfrak{l}^2 + 2\mathfrak{g}\mathfrak{n}\mathfrak{l} \\
&+ z^2 \cdot \mathfrak{d}(\mathfrak{a}\mathfrak{b} - \mathfrak{h}^2) - \mathfrak{b}\mathfrak{l}^2 - \mathfrak{a}\mathfrak{m}^2 + 2\mathfrak{h}\mathfrak{l}\mathfrak{m} \\
&+ w^2 \cdot \mathfrak{a}\mathfrak{b}\mathfrak{c} - \mathfrak{a}\mathfrak{f}^2 - \mathfrak{b}\mathfrak{g}^2 - \mathfrak{c}\mathfrak{h}^2 + 2\mathfrak{f}\mathfrak{g}\mathfrak{h} \\
&+ 2yz \cdot \mathfrak{d}(\mathfrak{g}\mathfrak{h} - \mathfrak{a}\mathfrak{f}) + \mathfrak{l}^2\mathfrak{f} + \mathfrak{a}\mathfrak{m}\mathfrak{n} - \mathfrak{h}\mathfrak{n}\mathfrak{l} - \mathfrak{g}\mathfrak{l}\mathfrak{m} \\
&+ 2zx \cdot \mathfrak{d}(\mathfrak{h}\mathfrak{f} - \mathfrak{b}\mathfrak{g}) + \mathfrak{m}^2\mathfrak{g} + \mathfrak{b}\mathfrak{n}\mathfrak{l} - \mathfrak{f}\mathfrak{h}\mathfrak{n} - \mathfrak{h}\mathfrak{m}\mathfrak{n} \\
&+ 2xy \cdot \mathfrak{d}(\mathfrak{f}\mathfrak{g} - \mathfrak{c}\mathfrak{h}) + \mathfrak{n}^2\mathfrak{h} + \mathfrak{c}\mathfrak{l}\mathfrak{m} - \mathfrak{g}\mathfrak{m}\mathfrak{n} - \mathfrak{f}\mathfrak{n}\mathfrak{l} \\
&+ 2xw \cdot -\mathfrak{l}(\mathfrak{b}\mathfrak{c} - \mathfrak{f}^2) - \mathfrak{m}(\mathfrak{h}\mathfrak{f} - \mathfrak{b}\mathfrak{g}) - \mathfrak{n}(\mathfrak{f}\mathfrak{g} - \mathfrak{c}\mathfrak{h}) \\
&+ 2yw \cdot -\mathfrak{m}(\mathfrak{c}\mathfrak{a} - \mathfrak{g}^2) - \mathfrak{l}(\mathfrak{f}\mathfrak{g} - \mathfrak{c}\mathfrak{h}) - \mathfrak{n}(\mathfrak{g}\mathfrak{h} - \mathfrak{a}\mathfrak{f}) \\
&+ 2zw \cdot -\mathfrak{n}(\mathfrak{a}\mathfrak{b} - \mathfrak{h}^2) - \mathfrak{m}(\mathfrak{g}\mathfrak{h} - \mathfrak{a}\mathfrak{f}) - \mathfrak{l}(\mathfrak{h}\mathfrak{f} - \mathfrak{b}\mathfrak{g}),
\end{aligned}$$

(I write down this general result as it will be useful for reference in other cases); in the present case this becomes simply

$$x^2 \cdot b\mathfrak{n}^2 + y^2 \cdot a\mathfrak{n}^2 + 2xy \cdot \mathfrak{n}\mathfrak{h}^2 + 2zw(-\mathfrak{n}ab + \mathfrak{n}\mathfrak{h}^2) = 0,$$

where a, b, $\mathfrak{n}$, $\mathfrak{h}$ have the foregoing values; the equation is thus only of the order 4 in regard to θ; but it in fact divides by $\mathfrak{n}\,(=2\theta n)$ and thus reduces itself to the third order, viz. it becomes

$$\mathfrak{n}(bx^2+ay^2)-2\mathfrak{h}^2xy+2(ab-\mathfrak{h}^2)zw=0,$$

or, substituting for $\mathfrak{a}$, $\mathfrak{h}$, $\mathfrak{n}$, $\mathfrak{b}$ their values, this is

$$x^2\cdot 4bn\theta^3+2y^2(2n\theta^3+4an\theta^2)+2zw(2b\theta^2+4ab\theta)-2(xy+zw)(2\theta h+1)^2=0,$$

say this is

$$(\mathfrak{A},\mathfrak{B},\mathfrak{C},\mathfrak{D}\,|\,\theta,1)^3=0,$$

where $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are each of them a quadric function of the coordinates; it being observed that $\mathfrak{C}$ and $\mathfrak{D}$ are respectively numerical multiples of the same function $(xy+zw)$. Hence equating the discriminant to zero, we have

$$\mathfrak{A}^2\mathfrak{D}^2+4\mathfrak{A}\mathfrak{C}^3+4\mathfrak{B}^3\mathfrak{D}-3\mathfrak{B}^2\mathfrak{C}^2-6\mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D}=0,$$

which equation, inasmuch as every term contains either $\mathfrak{C}$ or $\mathfrak{D}$ as a factor, divides by $xy+zw$, and thus becomes an equation of the order 6 in the coordinates: that is, the order of the reciprocal surface is 6. Multiplying by $\frac{1}{8}$ to avoid fractions, the actual values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are

$$\begin{aligned}\mathfrak{A}&=b\,n^2+3(ny^2+2hzw),\\ \mathfrak{B}&=2\{hnx^2+any^2+2ahzw-2h^2(xy+zw)\},\\ \mathfrak{C}&=-4fh\,(xy+zw),\\ \mathfrak{D}&=-3\,(xy+zw),\end{aligned}$$

or say $\mathfrak{A}=3\alpha$, $\mathfrak{B}=2\beta$, $\mathfrak{C}=-4h\gamma$, $\mathfrak{D}=-3\gamma$; where $\gamma=xy+zw$; substituting these values and omitting the factor 8γ, the equation is

$$27\alpha^3\gamma-256h^4\alpha\gamma^3-32\beta^3-64h^3\beta^2\gamma^2-144h\alpha\beta^2\gamma=0,$$

which is an equation of the form $(*\,|\,\alpha,\beta,\gamma)^3=0$. The sextic surface has thus singular points $\alpha=0$, $\beta=0$, $\gamma=0$, viz. these are the two points $(x=0,y=0,z=0)$, $(x=0,y=0,w=0)$ each four times. The further discussion of the sextic surface is reserved for another occasion.

I do not at present attempt to enumerate the particular cases of the surface $V^2-x^3y=0$, but content myself with the discussion of a particular case in which the order of the reciprocal surface is $=3$. Suppose that $V=0$ is a cone, $y=0$ a tangent plane to the cone (so that the conic $y=0$, $V=0$ breaks up into a line twice repeated), $x=0$ an arbitrary plane (so that we have still the proper cuspidal conic $x=0$, $V=0$). Any other tangent plane of the cone may be taken for the plane $z=0$; the plane containing the lines of contact of the two tangent planes for the plane $w=0$; the equation of the conic then is $V=dw^2+2fyz=0$; and the equation of the surface is

$$(dw^2+2fyz)^2-x^3y=0. \tag{AB}$$

For convenience of comparison, I change x, y, w, z into y, w, z, x, and assign numerical values to the coefficients, writing the equation under the form

$$27(4aw+z^2)^2-64y^3w=0. \tag{AB}$$

The quartic is here the envelope of the quadric surface

$$\theta^2\cdot 4yw+\theta\cdot 9(4aw+z^2)+12yz=0,$$

viz., comparing with the general form

$$0 = (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n} \,|\, x, y, z, w)^2,$$

we have $\mathfrak{b} = 12$, $\mathfrak{c} = 9\theta$, $\mathfrak{l} = 18\theta$, $\mathfrak{m} = 2\theta^2$, all the other coefficients vanishing. The reciprocal equation is

$$\mathfrak{d}\mathfrak{m}^2 + x^2(-\mathfrak{c}\mathfrak{m}^2) + y^2(-\mathfrak{c}\mathfrak{l}^2) + z^2(-\mathfrak{b}\mathfrak{l}^2) + 2xw \cdot \mathfrak{m}^2\mathfrak{g} + 2xw(-\mathfrak{l}\mathfrak{b}\mathfrak{c}) = 0,$$

or substituting for $\mathfrak{b}$, $\mathfrak{c}$, $\mathfrak{l}$, $\mathfrak{m}$ their values, this is found to be

$$\theta(\theta x - 9y)^2 + 108(z^2 + xw) = 0.$$

Representing this by $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \,|\, \theta, 1)^3 = 0$, the discriminant in regard to θ would, in virtue of the values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, contain $\mathfrak{D}$ as a factor; the reason of this appears from the original form; in fact, forming the derived equation in regard to θ, this is found to be $(\theta x - 9y)(\theta x - 3y) = 0$; the value $\theta x - 9y = 0$ gives as a factor of the discriminant $z^2 + xw$; the value $\theta x - 3y = 0$ gives $3y(-6y)^2 + 108x(z^2 + xw)$, that is the factor $y^3 + x(z^2 + xw)$; the complete value of the discriminant as obtained by substitution of the values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ being $x^2(z^2 + xw)\{y^3 + x(z^2 + xw)\}$; the equation of the reciprocal surface is

$$y^3 + x(z^2 + xw) = 0,$$

viz. this is a cubic surface. Prof. Schläfli's Case XX., having a uniplanar point $x = 0$, $y = 0$, $z = 0$ reducing the class by 8, and so giving a reciprocal surface of the order $(12 - 8 =)\, 4$, viz. the surface $27(4xw + z^2)^2 - 64y^3w = 0$. See the Memoir, Schläfli, "On the distribution of surfaces of the third order into species in reference to the absence or presence of singular points and the reality of their lines," *Phil. Trans.*, vol. CLIII. (1853), pp. 193–241.

I pass to the case of a surface

$$V^2 - P^2U = 0$$

having a nodal conic $V = 0$, $P = 0$, but not having in general any nodes. And I propose to show how the constants may be determined so that the surface shall have 1, 2, 3, or 4 nodes. It is to be remarked that in the above equation the plane $P = 0$ is a determinate plane, but the quadric surface $U = 0$ is not a determinate quadric, we may in fact substitute for it the quadric $U + \lambda P^2 = 0$, writing the equation under the form

$$V^2 + 2\lambda VP^2 + \lambda^2P^4 = P^2(U + \lambda^2P^2),$$

so that we may without loss of generality, by means of the disposable constant λ, subject the surface $V = 0$ to any single condition; for instance, take it to be a cone, or to pass through a given point, &c.

Taking the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$ to be arbitrary planes, the implicit constant factors in these equations may be determined in such wise that the equation of the given plane $P = 0$ shall be $x + y + z + w = 0$. The equation of the surface will then be

$$\begin{aligned} &\{(a, b, c, d, f, g, h, l, m, n \,|\, x, y, z, w)^2\}^2 \\ &\qquad\qquad = (x + y + z + w)^2 \cdot (a', b', c', d', f', g', h', l', m', n' \,|\, x, y, z, w)^2, \end{aligned}$$

and we may assume that the node or nodes (if any) lie at a vertex or vertices of the tetrahedron $x = 0$, $y = 0$, $z = 0$, $w = 0$, say at the points A, B, C, D. The conditions for a node at each of these points are at once found to be

Node at A,	Node at B,	Node at C,	Node at D,
$2a^2 = a'$,	$2bh = h' + b'$,	$2cg = g' + c'$,	$2dl = l' + d'$,
$2ah = h' + a'$,	$2b^2 = b'$,	$2cf = f' + c'$,	$2dm = m' + d'$,
$2ag = g' + a'$,	$2hf = f' + h'$,	$2c^2 = c'$,	$2dn = n' + d'$,
$2al = l' + a'$,	$2bm = m' + b'$,	$2cn = n' + c'$,	$2d^2 = d'$.

The first set of equations gives

$$a' = a^2, \quad h' = 2ah - a^2, \quad g' = 2ag - g^2, \quad l' = 2al - a^2.$$

If the first and second sets are satisfied simultaneously, we have $2(a-b)h = a^2 - b^2$, that is $a = b$, or else $h = \frac{1}{2}(a+b)$; that is, the two sets may be satisfied in two different ways according as a and b are equal or unequal. Similarly the first, second, and third sets may be satisfied in three different ways and the four sets in five different ways according as there are or are not any equalities between a, b, c, and between a, b, c and d respectively. The several solutions are shown in the annexed table, viz., in the line I no set is satisfied; in the line II only the first set; in the lines III and IV the first and second sets; in the lines V to VII the first, second, and third sets; and in the lines VIII to XII the four sets.

[See next page for this Table, which should come in here.]

I is the general case, $V^2 = P^2U$, of a quartic and a nodal conic but without nodes. (AC).

II is the case of a single node; writing, as without loss of generality we may do, $a = 0$, the equation is

$$\{(0, b, c, d, f, g, h, l, m, n \,|\, x, y, z, w)^2\}^2 = (x + y + z + w)^2 \cdot (b, c, d, n, m, f \,|\, y, z, w)^2,$$

viz., the quadric $U = 0$ is here a cone having its vertex on the quadric $V = 0$. (AD).

III and IV are two cases each of them with two nodes, viz. III, the equation is

$$\begin{aligned}\{(ax + by)(x + y) + cz^2 + 2nzw + dw^2 + 2x(gz + lw) + 2y(fz + mw)\}^2 \\ = (x + y + z + w)^2\,[(ax + by)^2 + c'z^2 + 2n'zw + d'w^2 \\ + 2x\{(2ag - a^2)z + (2al - a^2)w\} + 2y\{(2bf - b^2)z + (2bm - b^2)w\}], \quad (AE)\end{aligned}$$

where it is to be observed that the line $z = 0$, $w = 0$ joining the two nodes $(y = 0, z = 0, w = 0)$ and $(w = 0, z = 0, x = 0)$ is a line on the surface. Writing, as we may do, $a = 0$, the equation assumes the more simple form

$$\begin{aligned}\{by(x + y) + cz^2 + 2nzw + dw^2 + 2x(gz + lw) + 2y(fz + mw)\}^2 \\ = (x + y + z + w)^2\,[b^2y^2 + c'z^2 + 2n'zw + d'w^2 + 2y\{(2bf - b^2)z + (2bm - b^2)w\}]. \quad (AE)\end{aligned}$$

[Page 310 of the original carries here a full-page sideways table classifying the twelve cases I–XII according to the equalities between a, b, c, d and the consequent values of a', b', c', d', h', g', l', f', m', n'. The table is reproduced in schematic form below.]

Line	Conditions on a, b, c, d	Number of nodes
I	all distinct, no set satisfied	0
II	1st set only; $a = 0$	1
III	1st and 2nd sets; a, b distinct	2
IV	1st and 2nd sets; $a = b$	2
V	1st, 2nd, 3rd sets; a, b, c all distinct	3
VI	1st, 2nd, 3rd sets; one equality	3
VII	1st, 2nd, 3rd sets; $a = b = c$	3
VIII	all four sets; a, b, c, d all distinct	4
IX	all four sets; one pair equal	4
X	all four sets; two pairs equal	4
XI	all four sets; three equal	4
XII	all four sets; $a = b = c = d$	4

In IV the equation is

$$\begin{aligned}&\{a(x^2+y^2)+2hxy+cz^2+dw^2+2nzw+2x(gz+lw)+2y(fz+mw)\}^2\\&\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2)+2(2ah-a^2)xy+c'z^2+2n'zw+d'w^2\\&\qquad\qquad+2x\{(2ag-a^2)z+(2al-a^2)w\}+2y\{(2af-a^2)z+(2am-a^2)w\}],\quad(AF)\end{aligned}$$

where it will be observed that the line $z=0$, $w=0$ joining the two nodes is not a line on the surface.

Writing, as we may do, $a=0$, the equation becomes

$$\begin{aligned}&\{2hxy+cz^2+2nzw+dw^2+2x(gz+lw)+2y(fz+mw)\}^2\\&\qquad=(x+y+z+w)^2(c'z^2+2n'zw+d'w^2),\quad(AF)\end{aligned}$$

viz. the form is $V^2=P^2QR$, the quadric surface $U=0$ breaking up into the two planes $Q=0$, $R=0$; and the nodes being situate at the intersections of the line $Q=0$, $R=0$ with the surface $V=0$.

V, VI, VII are apparently cases with three nodes, but in fact VI is the only case of a proper quartic surface with three nodes. For in V the equation is

$$\begin{aligned}&\{(ax+by+cz)(x+y+z)+dw^2+2w(lx+my+nz)\}^2\\&\qquad=(x+y+z+w)^2\,[(ax+by+cz)^2+d'w^2\\&\qquad\qquad+2w\{(2al-a^2)x+(2bm-b^2)y+(2cn-c^2)z\}],\end{aligned}$$

which is satisfied by $w=0$, and the surface thus breaks up into the plane $w=0$ and a cubic surface.

And in VII the equation is

$$\begin{aligned}&\{a(x^2+y^2+z^2)+dw^2+2fyz+2gzx+2hxy+2lxw+2myw+2nzw\}^2\\&\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2+z^2)+d^2w^2\\&\qquad+2\{(2af-a^2)yz+(2ag-a^2)zx+(2ah-a^2)xy\\&\qquad\qquad+(2al-a^2)xw+(2am-a^2)yw+(2an-a^2)zw\}],\end{aligned}$$

which putting $a=0$ is

$$(dw^2+2fyz+2gzx+2hxy+2lxw+2myw+2nzw)^2=(x+y+z+w)^2\,d'^2w^2,$$

viz., this is a pair of quadric surfaces.

In the remaining case VI the equation is

$$\begin{aligned}&\{a(x^2+y^2)+2hxy+cz^2+dw^2+(a+c)(yz+zx)+2w(lx+my+nz)\}^2\\&\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2)+c'^2z^2+d'w^2+2acz(x+y)+2(2ah-a^2)xy\\&\qquad\qquad+2w\{(2al-a^2)x+(2am-a^2)y+(2cn-c^2)z\}],\quad(AG)\end{aligned}$$

which putting therein $a=0$ is

$$\begin{aligned}&\{2hxy+dw^2+cz(x+y+z)+2w(lx+my+nz)\}^2\\&\qquad=(x+y+z+w)^2\,\{c'^2z^2+d'w^2+2(2cn-c^2)zw\},\quad(AG)\end{aligned}$$

which is a surface having the nodes A, B, C; and it is to be observed that the lines CA, CB, but not the line AB, are lines on the surface.

IX, X, XI, XII are apparently cases with four nodes, but it is only XI which is a proper quartic with four nodes. In fact IX is

$$\{(ax+ay+cz+dw)(x+y+z+w)+2(h-a)xy\}^2 = (x+y+z+w)^2\{(ax+ay+cz+dw)^2+4a(h-a)xy\},$$

which is satisfied if $x=0$ or if $y=0$; that is, the surface breaks up into the two planes $x=0$, $y=0$, and a quadric surface.

X is

$$\begin{aligned}\{a(x^2+y^2+z^2)+dw^2+2fyz+2gzx+2hxy+(a+d)(xw+yw+zw)\}^2&\\ = (x+y+z+w)^2\,\{a^2(x^2+y^2+z^2)+d^2w^2&\\ +2(2af-a^2)yz+2(2ag-a^2)zx+2(2ah-a^2)xy+2ad(xw+yw+zw)\}&,\end{aligned}$$

which putting therein $a=0$ is

$$\{dw(x+y+z+w)+2fyz+2gzx+2hxy\}^2 = (x+y+z+w)^2\,d^2w^2,$$

and thus breaks up into two quadrics.

And XII is

$$\begin{aligned}\{a^2(x^2+y^2+z^2+w^2)+2fyz+2gzx+2hxy+2lxw+2myw+2nzw\}^2&\\ = (x+y+z+w)^2\,[a^2(x^2+y^2+z^2+w^2)&\\ +2(2af-a^2)yz+2(2ag-a^2)zx+2(2ah-a^2)xy&\\ +2(2al-a^2)xw+2(2am-a^2)yw+2(2an-a^2)zw\}&,\end{aligned}$$

which putting $a=0$ is

$$(2fyz+2gzx+2hxy+2lxw+2myw+2nzw)^2=0,$$

and is thus a quadric surface twice repeated.

There remains XI, and here the equation is

$$\begin{aligned}\{(ax+ay+cz+cw)(x+y+z+w)+2(h-a)xy+2(n-c)zw\}^2&\\ = (x+y+z+w)^2\,\{(ax+ay+cz+cw)^2+4a(h-a)xy+4c(n-c)zw\}&,\end{aligned}$$

or writing $h+a$, $n+c$ in place of a, c respectively, this is

$$\begin{aligned}\{(ax+ay+cz+cw)(x+y+z+w)+2hxy+2nzw\}^2&\\ = (x+y+z+w)^2\,\{(ax+ay+cz+cw)^2+4ahxy+4cnzw\}&, \qquad (AH)\end{aligned}$$

or putting herein $a=0$ it is

$$\{c(z+w)(x+y+z+w)+2hxy+2nzw\}^2 = c(x+y+z+w)^2\{c(z+w)^2+4nzw\}, \qquad (AH)$$

which may also be written

$$c(x+y+z+w)\{hxy(z+w)-nzw(x+y)\}+(hxy+nzw)^2=0, \qquad (AH)$$

the equation of a quartic surface with the four nodes A, B, C, D; it is to be observed that the lines AC, AD, BC, BD are, the lines AB and CD are not, lines on the surface.

A more simple form may be given to the equation as follows; using the second of the above forms, multiplying the equation by 4, and writing therein

$$\begin{aligned} p &= \sqrt{c}\,(x+y+z+w), \\ qr &= c(z+w)^2 + 4nzw, \\ st &= c(x+y)^2 - 4hxy, \end{aligned}$$

q, r, and s, t being the linear factors of the two quadric functions respectively, we have

$$qr - st = c(x+y+z+w)(-x-y+z+w) + 4hxy + 4nzw,$$

and thence

$$p^2 + qr - st = 2c(z+w)(x+y+z+w) + 4hxy + 4nzw,$$

wherefore the equation is

$$\tfrac{1}{4}(p^2+qr-st)^2 = st \cdot \tfrac{1}{4}\,p^2 - qr, \qquad (AH)$$

or, what is the same thing,

$$p + \sqrt{qr} + \sqrt{st} = 0, \qquad (AH)$$

where p, q, r, s, t are any linear functions of the coordinates; this is the equation of a quartic surface having the nodal conic $p=0$, $qr-st=0$; and the four nodes $(q=0, r=0, p-\sqrt{st}=0)$ and $(s=0, t=0, p-\sqrt{qr}=0)$. It includes the Cyclide, the equation of which may be written

$$\theta = \sqrt{(ax-ek)^2 + b^2z^2} + \sqrt{(ex-ak)^2 - b^2z^2}.$$

I remark that Prof. Kummer in his most valuable Memoir, "Ueber die Flächen vierten Grades auf welchen Schaaren von Kegelschnitten liegen," *Crelle*, t. LXVI. (1864), pp. 66–76, has considered several of the cases of a quartic surface with a nodal conic, viz. no node, (AC); a single node, (AD); two nodes (the case AF); and four nodes, (AH); but he has not considered two nodes, the case (AE); nor three nodes, (AG).

In reference to the general case of a quartic surface with a nodal conic, some most interesting properties have recently been obtained by Prof. Clebsch, see *Berl. Monatsh.*, April 30, 1868, where it is shown that there are on the surface 16 right lines forming 20 systems of double-fours, analogous in some respect to the 27 lines and 36 systems of double-sixes of a cubic surface.

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[458] On the Anharmonic-Ratio Sextic.

*[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (*1870*), pp.* 56, 57.*]*

Mr Walker's equation is

$$4A\left(\lambda^2+\lambda+1\right)^3 - 27I^2\left(\lambda^2-\lambda\right)^2 = 0;$$

changing the sign of λ, and also the numerical multipliers of I, A (so as to convert the discriminant equation into its standard form $\Delta = I^3 - 27J^2$), the equation is

$$4\Delta\left(\lambda^2+\lambda+1\right)^3 - 27J^2\left(\lambda^2+\lambda\right)^2 = 0.$$

I remark that this is most readily obtained as follows; writing

$$\begin{aligned} A &= (a-d)(b-c), \\ B &= (b-d)(c-a), \\ C &= (c-d)(a-b), \end{aligned}$$

then we have $A+B+C=0$,

$$\begin{aligned} I &= \tfrac{1}{6}(A^2+B^2+C^2) = -\tfrac{1}{3}(BC+CA+AB), \\ J &= \tfrac{1}{216}(B-C)(C-A)(A-B), \\ \sqrt{\Delta} &= \tfrac{1}{12}ABC, \end{aligned}$$

see my Fifth Memoir on Quantics, *Phil. Trans.*, vol. CXLVIII. (1858), pp. 429–460, [156].

And observe also, that in virtue of the relation $A+B+C=0$, we have

$$12I = A^2+AB+B^2 = A^2+AC+C^2 = B^2+BC+C^2.$$

Hence writing

$$u = \frac{A}{B} + \frac{1}{4}\frac{B}{A} + \frac{1}{4},$$

when λ has any one of the values $\frac{A}{B}$, $\frac{B}{A}$, $\frac{A}{C}$, $\frac{C}{A}$, $\frac{B}{C}$, $\frac{C}{B}$, we see that u assumes only the values A, B, C, and u is thus determined by the equation

$$u^2 - 12Iu - 16\sqrt{\Delta} = 0.$$

Eliminating u, we obtain

$$16\sqrt{\Delta}\left\{\frac{4\Delta}{27J^2}\left(\lambda+\tfrac{1}{4}\cdot\tfrac{1}{\lambda}+1\right)^2 - \left(\lambda+\tfrac{1}{4}\cdot\tfrac{1}{\lambda}+1\right) - 1\right\} = 0,$$

or, what is the same thing,

$$4\Delta\left(\lambda^2+\lambda+1\right)^3 - 27J^2\lambda^2\left(\lambda+1\right)^2 = 0,$$

that is

$$4\Delta\left(\lambda^2+\lambda+1\right)^3 - 27J^2\left(\lambda^2+\lambda\right)^2 = 0,$$

the required equation.

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[459] On the Double-sixers of a Cubic Surface.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), pp. 58–71.]

The 27 lines on a cubic surface include, and that in 36 different ways, a double-sixer; viz. a system of two sets of six lines $1, 2, 3, 4, 5, 6$; $1', 2', 3', 4', 5', 6'$, such that every line of the one set intersects all the non-corresponding lines of the other set, thus

	1	2	3	4	5	6
$1'$	·	●	●	●	●	●
$2'$	●	·	●	●	●	●
$3'$	●	●	·	●	●	●
$4'$	●	●	●	·	●	●
$5'$	●	●	●	●	·	●
$6'$	●	●	●	●	●	·

there being in all 30 intersections.

Any line say 4, of the one set, intersects five lines $1', 2', 3', 5', 6'$ of the other set; and these six lines being given the double-sixer may be constructed; viz. (besides the line 4) we have a line 1 meeting the lines $2', 3', 5', 6'$; a line 2 meeting the lines $3', 5', 6', 1'$; a line 3 meeting the lines

$5', 6', 1', 2'$; a line 5 meeting the lines $6', 1', 2', 3'$; and a line 6 meeting the lines $1', 2', 3', 5'$; and then the lines $1, 2, 3, 5, 6$ are all of them met by a single line $4'$, which completes the system.

We may, if we please, consider the lines 4, 2 as given, and then $1', 3', 5', 6'$ will be any four lines each of them meeting the two given lines 4, 2; $2'$ will be any line meeting 4; and we have to determine a line $4'$ meeting 2, such that there may exist the lines $1, 3, 5, 6$, completing the system as above. Or what is the same thing, we have a skew quadrilateral $1', 2, 3', 4$; $5'$ and $6'$ meet 2 and 4; $2'$ meets 4, and $4'$ meets 2: 5 and 6 meet $1'$ and $3'$; 1 meets $3'$ and 3 meets $1'$; and the two sets $2', 4', 5', 6'$ and $1, 3, 5, 6$ meet thus

	1	3	5	6
$2'$	·	·	●	●
$4'$	·	·	●	●
$5'$	●	●	·	·
$6'$	●	●	·	·

Hence, starting with the skew quadrilateral $1'23'4$, and taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the four planes $41'$, $1'2$, $23'$, $3'4$ respectively; or what is the same thing $x = 0$, $y = 0$ for the equations of the line $1'$; $y = 0$, $z = 0$ for those of the line 2; $z = 0$, $w = 0$ for those of the line $3'$; and $w = 0$, $x = 0$ for those of the line 4; the several lines may be determined, each of them by means of its six coordinates, as follows:

	a	b	c	f	g	h
$1'$	·	·	·	·	·	1
2	·	·	·	1	·	·
$3'$	·	·	·	·	1	·
4	1	·	·	·	·	·
$2'$	A_1	B_1	C_1	·	G_1	H_1
$4'$	·	B_2	C_2	F_2	G_2	H_2
$5'$	·	B_3	C_3	F_3	G_3	H_3
$6'$	·	B_4	C_4	F_4	G_4	H_4
1	a_1	b_1	c_1	f_1	g_1	·
3	·	·	·	f_2	g_2	·
5	·	·	·	f_3	g_3	·
6	a_4	b_4	·	f_4	g_4	·

where

$$\begin{aligned} B_1G_1 + C_1H_1 &= 0, \\ B_2G_2 + C_2H_2 &= 0, \\ B_3G_3 + C_3H_3 &= 0, \\ B_4G_4 + C_4H_4 &= 0, \end{aligned}$$

the conditions in regard to the lines $2', 4', 5', 6'$ and $1, 3, 5, 6$ are formed by means of the diagram:

	1	3	5	6
$2'$	f_1	·	·	·
$4'$	$b_1g_2 - g_1b_4$	·	·	·
$5'$	·	·	·	·
$6'$	·	·	·	·

viz. we have the equations

first set,

$$\begin{aligned} f_1A_1 + g_1B_1 + h_1C_1 + a_1F_1 + b_1G_1 + c_1H_1 &= 0,\\ g_1B_1 + a_1F_1 + b_1G_1 + c_1H_1 &= 0,\\ g_1B_1 + b_1G_1 + c_1H_1 &= 0,\\ g_1B_1 + b_1G_1 + c_1H_1 &= 0; \end{aligned}$$

second set,

$$\begin{aligned} f_2A_1 + g_2B_1 + h_2C_1 + b_2G_1 + c_2H_1 &= 0,\\ g_2B_1 + c_2H_1 &= 0,\\ g_2B_1 + c_2H_1 &= 0,\\ g_2B_1 + c_2H_1 &= 0; \end{aligned}$$

third set,

$$\begin{aligned} f_3A_1 + g_3B_1 + h_3C_1 + b_3G_1 + c_3H_1 &= 0,\\ g_3B_1 + b_3G_1 + c_3H_1 &= 0,\\ g_3B_1 + b_3G_1 + c_3H_1 &= 0; \end{aligned}$$

fourth set,

$$\begin{aligned} f_4A_1 + g_4B_1 + h_4C_1 + a_4F_1 + b_4G_1 + c_4H_1 &= 0,\\ g_4B_1 + b_4G_1 + c_4H_1 &= 0,\\ g_4B_1 + b_4G_1 + c_4H_1 &= 0; \end{aligned}$$

and it is to be shown, that taking as given the coordinates of $2'$, $5'$, $6'$, that is $(A_1, B_1, C_1, G_1, H_1)$, $(B_3, C_3, F_3, G_3, H_3)$, $(B_4, C_4, F_4, G_4, H_4)$, we can find the coordinates of the remaining lines $4'$, 1, 3, 5, 6.

The first set of equations gives

$$g_1\, b_1 = b_1 \cdot g_1\,; \quad \begin{vmatrix} B_1 & C_1 & \cdot \\ B_3 & C_3 & H_3 \end{vmatrix};$$

viz. g_1, b_1, a_1 are proportional, but so only the ratios are material, they may be taken equal to the determinants $G_1H_3 - C_1H_3$, $B_3H_1 - B_1H_3$, $B_1C_3 - B_3C_1$, &c.; that is to signify these values respectively, the first equation gives f_1A_1, and the second equation gives a_1F_1; multiplying together these values, and writing $a_1f_1 = k_1g_1$, we find

$$(B_1B_3{\cdot}G_1G_3{\cdot}H_1H_3{\cdot}G_1H_3{\cdot}G_3H_1){\cdot}k_1(A_1, B_1, C_1, F_1, G_1, H_1) + B_3(B_1C_3 - B_3C_1)F_2{\cdot}A_1 + B_1(B_1C_3 - B_3C_1)F_2 = 0.$$

Proceeding in a similar manner with the second set of equations, we have first

$$g_2,\ b_2,\ b_2 = \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix} \quad \text{(observe } k_2 = G_1B_3 - G_3B_1,\ = -c_1\text{)},$$

and then

$$(B_3B_1 \cdot G_1C_1 \cdot G_4C_4 \cdot G_4G_1 \cdot C_4C_1 \cdot G_1G_4 \cdot G_4G_1 + A_1F_1 \cdot B_3C_2 + B_1C_2 \cdot G_3 \cdot B_3G_1)k_2\, b_2 = 0.$$

The third set gives in like manner with the second set of equations, this becomes

$$g_3^2B_1B_3 + g_3b_3(B_1G_3 + B_3G_1 + A_1F_2) + b_3^2G_1G_3 = 0,$$

or since $g_3 \cdot b_3 = G_1 \cdot -B_1$, this is

$$G_1^2B_1B_3 - G_1B_1(B_1G_3 + B_3G_1 + A_1F_2) + B_1^2G_1G_3 = 0,$$

and similarly, the fourth set gives

$$g_4^2B_1B_3 + g_4b_4(B_1G_3 + B_3G_1 + A_1F_2) + b_4^2G_1G_3 = 0,$$

or since $g_4 \cdot b_4 = G_1 \cdot -B_1$, this is

$$G_1^2 B_1 B_3 - G_1 B_1 (B_1 G_3 + B_3 G_1 + A_1 F_2) + B_1^2 G_1 G_3 = 0,$$

and these last two results lead to the values of the ratios of B_3B_4, $B_3G_4 + B_4G_3$, G_3G_4 in the values containing the common factor B_1G_1, viz. these are proportional to expressions containing the common factor $B_3G_3 - B_4G_4$, and taking them equal instead of merely proportional to the resulting expressions (which is allowable, since the absolute values are not material), we have

$$B_3B_4,\ B_3G_4 + B_4G_3 + A_1F_2,\ G_3G_4 = B_1B_3,\ B_1G_3 + B_3G_1,\ G_1G_3,$$

$$(B_3B_4,\ G_3G_4,\ G_3B_4 + B_3G_4;\ p_4,\ q_4,\ r_4) = (B_1B_3, G_1G_3, B_1G_3 + G_1B_3; H_3,\ G_1,\ -H_1);$$

but the terms containing g_1, b_1 are $(B_1g_1 + G_1b_1)(B_4g_2 + G_4b_4), = -H_1c_1 \cdot -H_4c_4$, that is $H_1H_4c_1c_4$; the whole equation is thus divisible by c_1, and omitting this factor, it becomes

$$g_1(H_1B_4 + H_4B_1) + h_1(G_1H_4 + G_4H_1) + a_1(H_1F_4 + H_4F_1) = 0.$$

Proceeding in like manner with the result obtained from the second set of equations, this becomes

$$(B_1B_3, C_2C_4, C_1G_1, C_1G_4 + C_4G_1, B_2G_4 + B_4G_2, B_4G_4 + B_2G_4 + B_2C_1(G_1, b_4, c_4)) = 0,$$

where the terms containing g_3, b_3 are $(B_2g_3+C_2b_3)(B_4g_2+C_4b_4)$, viz. the whole equation divides by h_3, and it then becomes

$$g_3(B_3C_4 + B_4C_3) + h_3(C_3G_4 + C_4G_3) + b_3(G_3B_4 + G_4B_3) = 0.$$

Considering B_3, G_3, F_3 as given by the equations

$$B_3B_4 = B_1B_3, \quad G_3G_4 = G_1G_3, \quad B_3G_4 + G_3B_4 = A_1F_3 = B_1G_3 + B_3G_1,$$

But we have

$$\begin{aligned} g_1B_1B_4 - b_1H_4H_1H_4 &= B_4H_1(G_1H_4 - G_4H_1) - B_1H_4(G_1H_4 - G_4H_1), \\ &= B_4H_4(B_3G_1 + G_3B_1) - B_1H_1(B_3G_4 + G_3B_4) + C_1B_3 = 0, \end{aligned}$$

and similarly

$$\begin{aligned} g_1C_1C_4 - b_1G_4G_1G_4 &= 0, \\ g_1C_1B_4 - b_1H_4G_1G_4 &= 0, \\ h_1C_1G_4 - g_1B_4H_1G_4 &= 0. \end{aligned}$$

Moreover, writing $p_4B_4 + q_4G_4 + r_4H_4$, we have

$$\begin{aligned} p_4B_4G_4 &= b_4(G_4^2 - G_1G_4), \\ q_4G_4G_1 &= b_4(G_1G_4 - G_4^2), \\ r_4G_4 &= -b_4(G_4 - G_1), \end{aligned}$$

and the few terms of the equation in question thus separately vanish; and the equation is consequently verified.

We may collect the results as follows:

Data are lines 1′, 2, 3′, 4; 5′, 6′;

and then, for the remaining lines 1, 3, 5, 6 at the common factor omitted in each case, the coordinates are as follows:

For $4'$,

$$B_1B_4 = B_1B_3, \quad G_3G_4 = G_1G_3, \quad B_3G_4 + G_3B_4 = A_1F_3 = B_1G_3 + B_3G_1,$$

$$A_1F_2 = B_1G_1 \cdot B_3G_3 - B_3G_1 \cdot B_1G_3, \quad A_1 = 0,$$

$$\begin{vmatrix} B_1 & C_1 & H_1 \\ B_3 & C_3 & H_3 \end{vmatrix}, \quad \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix}, \quad H_1H_3, \quad B_1B_3, \quad C_1C_3, \quad G_1G_3;$$

$$\begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix} + \begin{vmatrix} C_1 & G_1 & H_1 \\ C_3 & G_3 & H_3 \end{vmatrix} \quad \Big(B_1G_3 + B_3G_1 = -2A_1F_2, \text{ identity}\Big).$$

For 1,

$$(g_1, b_1, a_1) = \begin{vmatrix} B_1 & C_1 & H_1 \\ B_3 & C_3 & H_3 \end{vmatrix}, \quad A_1F_1 = \begin{vmatrix} B_3 & C_3 & H_3 \\ B_1 & C_1 & G_1 \end{vmatrix},$$
$$(a_1f_1 + k_1g_1, \text{ identity}).$$

For 3,

$$(g_2, b_2, b_2) = \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix}, \quad A_1F_2 = \begin{vmatrix} B_3 & C_3 & G_3 \\ B_1 & C_1 & C_1 \end{vmatrix},$$
$$(a_2f_2 + k_2g_2, \text{ identity}).$$

For 5,

$$\begin{aligned} &g_3 = G_3, \quad b_3 = B_3, \quad A_1F_3 = B_1G_3 - B_3G_1, \\ &c_3 = G_3, \quad b_3 = 0, \qquad a_3 = \frac{A_1F_3G_3 + b_3(B_1G_1 + C_1H_1)}{B_3G_1 + b_1H_3}, \quad k_3 = 0. \end{aligned}$$

For 6,

$$\begin{aligned} &g_4 = G_4, \quad b_4 = -B_4, \quad A_1F_4 = B_1G_4 - B_4G_1, \\ &c_4 = 0, \qquad b_4 = 0, \qquad\quad a_4 = 0, \end{aligned}$$

and, for actual calculation, it is convenient to remark that as only the ratios are material, we may suppose without loss of generality that $c_1 = 1$, &c., respectively.

But results may be further reduced. Writing

$$(g_1, b_1, c_1) = \begin{vmatrix} B_3 & C_3 & H_3 \\ B_4 & C_4 & H_4 \end{vmatrix},$$

we have

$$-(g_1B_4 + b_1G_4 + h_1C_4) = C_1\left(\frac{B_3G_3}{B_1G_1}\right) + h_4C_4,$$

and many further simplifications follow, leading to compact final forms for C_1, H_3, etc.

[The remaining algebraic deductions on book pp. 321–325 *develop in the same manner the explicit formulas for the coordinates of* $2'$, $4'$, $5'$, $6'$ *and* 1, 3, 5, 6, *the closing display exhibiting the four quantities*

$$C_1 = \frac{C_1G_4\,p_4B_4 + b_4G_4 \cdot k_4 + B_4G_4}{B_4 \cdot p_4B_4 + h_4b_4 + k_4a_4},$$

together with the parallel expressions for G_1, F_1, H_1, H_3, *and the like for the alternative subscripts. The explicit verification is carried out on book p.* 325.*]*

I have thought it worth while to effect the numerical calculations for enabling the construction of a drawing or model. For this purpose taking X, Y, Z as ordinary rectangular coordinates, I write

$$\begin{aligned} x &= X + Y + Z - 10, \\ y &= Z, \\ z &= -X + Y + Z - 2, \\ w &= Y, \end{aligned}$$

that is, I take $1'$ and 2 to be lines in the plane of XY, defined by the equations $X+Y=10$ and $X-Y=-10$ respectively, and X and $3'$ to be lines in the plane of YZ defined by the equations $Z=10$, $Z+Z-2=12$ respectively. And I take $5'$ to be the line joining the points $(9,0,7)$ and $(-9,2,0)$; $4'$ the line joining the points $(9,0,7)$ and $(-2,8,0)$; $5'$ the line joining the points $(9,0,7)$ and $(-2,8,0)$; $6'$ the line joining the points $(9,0,7)$ and $(-2,8,0)$. We calculate then for each of the lines the upper coordinates of two points, and thence the six coordinates of the line; thus

	x	y	z	w	p	yw	A	B	C	F	G	H
$1'$	$0,0,-1,0$	$-10,0,0,1$	$0,0,0,0$	$0,0,0,0$			0,	10,	50,	0,	0,	-10
2	$0,0,-1,0$	$-10,0,0,1$	$0,0,0,0$				0,	10,	50,	0,	0,	-10
$3'$	$0,0,1,0$	$-10,0,0,1$					0,	10,	50,	0,	0,	-10
4	$0,0,1,0$	$-10,0,0,1$					76,	30,	0,	0,	10,	0

and, effecting the calculations for the remaining lines, we have

	A	B	C	F	G	H
$2'$	$\cdot$	24	$\frac{144}{15}$	$\cdot$	$\cdot$	$\frac{8}{15}$
1	$\frac{380}{15}$	60	35	$\frac{380}{15}$	0	3
3	127500	-720	0	$\frac{21}{15}$	140	-30
5	304	-54	6	0	7	0
6	$\frac{333}{5}$	-10	0	$\frac{12}{5}$	0	0

or reducing to integers, the values are

	A	B	C	F	G	H
$2'$	0	48	36	0	7	8
$4'$	76	26	2	0	0	1
1	1444	13452	6720	16443	-1131	0
3	463384	-2738	0	5	581	-114
5	9772	-552	0	21	132	0
6	5772	-3033	0	84	226	0

The line $(a,b,c;f,g,h)$ is given as the intersection of the two of the four planes

$$\begin{vmatrix} \cdot & h & -g & a \\ -h & \cdot & f & b \\ g & -f & \cdot & c \\ -a & -b & -c & \cdot \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0,$$

or substituting for x, y, w the values $X+Y+Z-10$, Z, $-X+Y+Z-2$, Y, these become

$$\begin{vmatrix} \cdot & h & -g & a \\ -h & \cdot & f & b \\ g & -f & \cdot & c \\ -a & -b & -c & \cdot \end{vmatrix} \begin{Bmatrix} X+Y+Z-10 \\ Z \\ -X+Y+Z-2 \\ Y \end{Bmatrix} = 0,$$

viz., what is the same thing,

$$\begin{vmatrix} \cdot & 2y-a-c & 2y-f-h & \cdot & -2y \\ -2y+a+c & \cdot & a+b+c+f-h & \cdot & -c \\ -2y+f+h & -a-b-c-f+h & \cdot & \cdot & -f-h \\ 2y & c+c & f+h & \cdot & \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0.$$

And substituting, we have the equations of the several lines, viz.

$$\begin{array}{rl}
(1') & X+Y=10, \quad Z=0; \\
(2) & -X+Y=10, \quad Z=0; \\
(3') & -X+Y=2, \quad Z=10; \\
(4) & X+Y=10, \quad Z=0;
\end{array}$$

$$(2') \quad \left|\begin{matrix} \cdot & \cdot & 49 & 5 \\ -49 & \cdot & \cdot & -36 \\ -5 & 36 & \cdot & \cdot \end{matrix}\right| \left\{\begin{matrix} X \\ Y \\ Z \\ 1 \end{matrix}\right\} = 0;$$

$$(4') \quad \left|\begin{matrix} \cdot & -55 & 26 & -7 \\ -55 & \cdot & \cdot & -3 \\ 26 & \cdot & \cdot & -1 \\ 7 & 26 & -3 & \cdot \end{matrix}\right| \left\{\begin{matrix} X \\ Y \\ Z \\ 1 \end{matrix}\right\} = 0;$$

and so on for (5′), (6′), (1), (3), (5), (6) with the corresponding numerical matrices given on book p. 328.

[Page 329 of the original reproduces the numerical coordinates of the various intersection points $11', 12', 13', \ldots$, in fractional form (numerators and denominators displayed on separate lines), and adds:]

viz., the coordinates of 12′ (intersection of lines 1 and 2′) are $\left(\frac{611}{132}, \frac{-554}{132}, \frac{-216}{132}\right)$, and so in other cases; where there is no divisor the coordinates are integers. I find however, on laying down the figure, that the lines 3 and 4, 5′ and 6 come so close together, that the figure cannot be obtained with any accuracy.

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[460] NOTE ON MR FROST'S PAPER ON THE DIRECTION OF LINES OF CURVATURE IN THE NEIGHBOURHOOD OF AN UMBILICUS.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), pp. 111–113.]

I remark as follows:

1. In regard to a quartic surface, it is not, I think, correct to say that the generation through an umbilic (= curve of curvature) notwithstanding that, in fact in the present case there will generate 5 in one plane and consequently intersect. The way in which these generatrices as quasi-curves-of-curvature present themselves is as follows:

The curve of curvature satisfy a certain differential equation, the complete integral of which gives those curves as the intersections of the given quadric surface by the series of confocal quadrics

$$\frac{x^2}{a^2-h} + \frac{y^2}{b^2-h} + \frac{z^2}{c^2-h} = 1,$$

h being the constant of integration. The singular solution of the differential equation, or envelope of the series of curvatures denoted as above, gives the umbilical generatrices.

2. In regard to a surface in general, I think it must be considered, not that there pass through the umbilic three distinct curves, but that there pass through the umbilic a single point, or rather a point at which there are in general three distinct directions of the curve. The umbilicar curve of curvature is in fact present itself as the curve belonging to a certain value of the constant of integration h, in order that the curve of curvature may pass through a given point on the

surface; a most simple example will, in order that there are in general distinct directions of the curve.

But, as an umbilic is a point for which (as in effect shown, p. 81, for the particular case of a quadric surface) the two values of h become equal; that is, through the umbilic only a singular curve of curvature; but $\dfrac{dy}{dx}$ is determined by a cubic equation, and the umbilic is thus (as just mentioned) a point at which there are in general three distinct directions of the curve.

3. Some researches on the subject are contained in my paper "On Differential Equations and Umbilici," *Phil. Mag.*, vol. XXVI. (1863), pp. 373–379 and 441–452, [230]. It is noticeable that in the integral equations which I have there obtained for the differential equation $dy(p^3-1)+(a-c)pq=0$, and the more general form $(\mathrm{d}e+p)(p+q)=0$, and the umbilic itself, the differential equation breaks up into three distinct curves; and the same is the case with the cubic equation, that the umbilic does not occur in the surface $xyz=1$ presently referred to.

4. In the paper "Mémoire sur les surfaces orthogonales," *Liouv.*, t. XII. (1847), pp. 241–254, M. Serret has given two very remarkable cases of three systems of surfaces intersecting each other at right angles, and consequently in the curves of curvature of each system. It was only on referring to this paper, in connexion with that of Mr Frost, that I perceived an obvious enough simplification of M. Serret's formulae, whereby it appears that the curves of curvature on the surface $xyz=1$ are given as the intersection of this surface with the series of surfaces

$$h=(x^2+xy^2+xz^2)^{\frac{1}{3}}+(x^2+xy^2+xz^2)^{\frac{1}{3}},$$

where ω is an imaginary cube root of unity; the rationalised equation is of the twelfth order in (x,y,z), and for the particular value $h=0$ is $(y^3-x)^3+(x^3-y^2)^3+(xy^3-1)^3$. The point $x=y=z=1$ is obviously an umbilic on the surface $xyz=1$, and the corresponding value of h being $h=0$, the equation just obtained determines the umbilicar curves of curvature, viz., combining therewith the equation $xyz=1$, we infer that the surface have here hyperbolic curves

$$(y=z,\ xy^2=1),\quad (z=x,\ yz^2=1),\quad (x=y,\ zx^2=1).$$

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[461] On the Geometrical Interpretation of the Covariants of a Binary Cubic.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), pp. 148–149.]

Consider the binary cubic $U=(a,b,c,d\,|\,x,y)^3$, and the discriminantal (invariant)

$$\nabla=a^2d^2-6abcd+4ac^3+4b^3d-3b^2c^2,$$

the Hessian

$$H=(ac-b^2)x^2+(ad-bc)xy+(bd-c^2)y^2,$$

and the cubicovariant

$$\Phi=\begin{array}{l}(\ \ a^2d-3abc+2b^3)x^3\\ +(3abd+6ac^2-9b^2c)x^2y\\ -(3acd+6b^2d-9bc^2)xy^2\\ -(a^2d-3acd+2c^3)y^3,\end{array}$$

connected by the identical relation

$$\Phi^2=-\nabla U^2-4H^3.$$

Then if we regard (a, b, c, d) as the coordinates of a point in space, but (x, y) as variable parameters, the equation

$$\nabla = 0$$

represents a quartic torse, having for its cuspidal curve the plane cubic $ad - bc = 0$, $bd - c^2 = 0$, the equation

$$U = 0$$

is that of the tangent plane to the torse along the line $ad + 2bc + cy = 0$, $bd^2 + 2acy + dy^2 = 0$; the line is the cuspidal curve in the above considered as line coordinates; viz. $a : b : c : d = ay^3 : -y^2 : y : -1$, the equation

$$H = 0$$

is that of a quadric surface having the first point of the osses point in regard to the torse.

The equation $\Phi^2 = \nabla U^2 - 4H^3$, gives $\Phi = -4H^3$, a result which implies that $\Phi = 0$ is a certain torse required twice, and that $U = 0$, $\Phi = 0$ is the same equator repeated twice. The curve in question is at any rate on the surface $\Phi = 0$; and conversely, the curve is on the surface $\Phi = 0$ regarded as the envelope of the plane $U = 0$, viz. in $(2, 3)$-plane, &c., what is the same thing.

The equation $\Phi^2 + \nabla U^2 + 4H^3 = 0$, gives $\Phi^2 = -4H^3$, a result which implies that $\Phi = 0$ is a certain torse repeated twice, that is, $H = 0$, $\Phi = 0$, the whole equation in this divisible by c, and omitting this factor, it becomes

$$g(H_1B_4 + H_4B_1) + h(G_1H_4 + G_4H_1) + a(H_1F_4 + H_4F_1) = 0.$$

Proceeding in like manner with the result obtained from the second set of equations, this becomes

$$(B_1B_3, C_2C_4, C_1G_1, C_1G_4 + C_4G_1, B_2G_4 + B_4G_2, B_4G_4 + B_2G_4) = 0.$$

[End of paper 461; the manipulations on book p. 333 verify the geometrical interpretation by reducing to the form $MU = (a^2, b, c, d \mid x, y)^3 \cdot M$, $\Phi = U$ and complete the discussion of the cubicovariant.]

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[462] A NINTH MEMOIR ON QUANTICS.

[From the Philosophical Transactions of the Royal Society of London, vol. CLXI. (for the year 1871), pp. 17–50. Received April 7. —Read May 19, 1870.]

It was shown not long ago by Professor Gordan that the number of the irreducible covariants of a binary quantic of any order is finite (see his memoir “Beweis dass jede Covariante und Invariante einer binären Form eine ganze Function mit numerischen Coefficienten einer endlichen Anzahl solcher Formen ist,” *Crelle*, t. LXIX. (1869), Memoir dated 8 June 1868), and in particular that for a binary quintic the number of irreducible covariants (including the quintic and the invariants) is = 23, and that for a binary sextic the number is = 26. From the theory given in my “Second Memoir on Quantics,” *Phil. Trans.*, 1856, [141], I derived the conclusion, which, as it now appears, was erroneous, that for a binary quintic the number of irreducible covariants was infinite. The theory requires, in fact, a modification, by reason that certain linear relations, which I had assumed to be independent, are really not independent, but, on the contrary, linearly connected together; the interconnexion in question does not occur in regard to the quadric, cubic, or quartic; and for these cases respectively the theory is as it stands; for the quintic the interconnexion first presents itself in regard to the degree 8 in the coefficients and order 14 in the variables, viz. the theory gives correctly the number of covariants of any degree not exceeding 7, and also those of the degree 8 and order less than 14; but for the order 14 the theory as it stands gives a non-existent irreducible covariant $(a, \dots \,|\, x, y)^{14}$, viz. of the form in question $10 - 6, = 4$ asyzygetic covariants; but the number of asyzygetic covariants being $= 5$, there is left, according to the theory, 1 irreducible covariant of the form in question. The fact is that in applying the theory, the composite covariants are equivalent to 5 independent syzygies only, the composite covariants are equivalent to $10 - 5, = 5$, the full number of the asyzygetic covariants. And similarly the theory as it stands gives a non-existent of the asyzygetic covariants.

[Article No. 326 *continues on book pp.* 334–335 *with Cayley's revised treatment of the issue; thence Articles* 327–365 *develop, on book pp.* 335–353*, the systematic proof. The articles fall into the following sections.]*

Article Nos. 328 **to** 332**.** ***Reproduction of my original Theory as to the Number of the Irreducible Covariants.***

328. I reproduce to some extent the considerations by which, in my Second Memoir on Quantics, I endeavoured to obtain the number of the irreducible covariants of a given binary quantic $(a, b, \dots | x, y)^n$.

Considering in the first instance the covariants as functions of the coefficients $(a, b, c, \dots)$ without regarding the variables (x, y), and attending only to the following properties — 1°, without regarding the variables[1], a covariant is a rational and integral homogeneous function of the coefficients $(a, b, c, \dots)$ of the properties — 1°, a covariant is a rational and integral homogeneous function of the coefficients $(a, b, c, \dots)$; and 2°, the coefficients of a given degree μ are linearly independent of the coefficients of a less degree; and that a covariant not connected by linear relations as above; and it is assumed that we know the number of the *irreducible* covariants; and it is in the same manner that we know the number of the asyzygetic covariants in regard to a given quantic, and that as the degree μ is increased successively to 1, 2, 3, … there is a certain function P, Q, R, … for each degree, and that a function not so obtained must be an *irreducible* covariant; and it is assumed that we know the number of the asyzygetic covariants of the degree μ as a given function P_μ of the coefficients a_i, ….

[The text continues through Articles 329–346 *on book pp.* 336–345*, treating successively: the binary cubic with its three syzygies; the binary quartic with its three covariants and one syzygy; the quintic with its* 23 *irreducible covariants (Table No.* 87*, identifying the* 23 *in terms of Gordan's notation* A, B, C, …, W*); Table No.* 88*, giving the actual generation, by successive transvectants, of each of the* 23 *forms (degrees* 1 *through* 8*); the related Tables Nos.* 89 *and* 90 *exhibiting the relations and complete linear expressions of the syzygies among the products of the covariants. The reasoning is throughout that of Gordan's Beweis but cast in the notation* $(f, f), (f, (f, f)), \dots$ *of the Second Memoir.]*

Article Nos. 347 **to** 365**.** ***Sketch of Professor Gordan's proof for the finite Number,*** $= 23$***, of the Covariants of a Binary Quintic.***

347. I propose to reproduce the leading points of Professor Gordan's proof that the binary quintic $(a, b, c, d, e, f \,|\, x, y)^5$ has a finite system of 23 covariants, viz. a system such that every other covariant whatever is a rational and integral function of these 23 covariants.

348. Derivation. Consider for a moment any two binary quantics ϕ, ψ of the same or different orders, and which may be either independent or, as will be the case here, identical or connected

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[1] For this case of covariants, α_1 is of course $= 1$; but in the investigation the term covariant stands for any function satisfying the conditions 1° and 2°.

by some relation f. We may form the derivatives

$$\begin{aligned}&(\phi,\psi) = \phi\psi,\\ &(\phi,\psi)' = 1{\cdot}2\ \phi\psi,\ \partial_x\phi{\cdot}\partial_y\psi,\ -\partial_y\phi{\cdot}\partial_x\psi,\\ &(\phi,\psi)'' = 1{\cdot}2\ \phi\psi,\ \partial_x\phi,\ \partial_y\phi{\cdot}\partial_x\psi,\ -2\partial_x\partial_y\phi{\cdot}\partial_x\partial_y\psi + \partial_y^2\phi{\cdot}\partial_x^2\psi,\end{aligned}$$

where, however, there is no occasion to use the notation $(\phi,\psi)'$ (as this is simply the product $\phi\psi$), and the succeeding derivatives may (when there is no risk of ambiguity) be written more shortly $(\phi\psi)$, $(\phi\psi)'$, &c.; in all that follows the term "derivative" (Gordan's *Ueberschiebung*) is to be understood in this special sense.

349. The degree of the derivative $(\phi\psi)^r$ is the sum of the degrees of ϕ, ψ; the order of the derivative is the sum of the orders of the constituents ϕ, ψ; the order of the derivative is less than this by $2r$; it being understood throughout that the next degree refers to the coefficients, and the order to the variables.

350. THEOREM A. The covariants of a quantic f of a given degree m can be all of them obtained by derivation from f and the covariants of the next inferior degree $(m-1)$.

In particular for the degree 1 the only covariant is the quantic f itself; the degree 2 the covariants of the third degree consist of the quantic f itself in succession, the covariants of the third degree are each of them obtained, by derivation from f and the covariants of the second degree.

351. Suppose that the covariants of the second degree $(f,f)^r$, $(f,f,f)^r$, $(f,(f,f))^r$, … are in this order represented by $\beta_1,\beta_2,\beta_3,\ldots$, then the covariants of the third degree written in the order

$$(\beta_1 f)^r,\ (\beta_1 f)^{r'},\ \ldots,\ (\beta_2 f)^r,\ (\beta_2 f)^{r'},\ \ldots,\ (\beta_3 f)^r,\ (\beta_3 f)^{r'}\ldots$$

may be represented by $\delta_1,\delta_2,\ldots$, and the covariants of the fourth degree written in the order

$$(\gamma_1 f)^r,\ (\gamma_1 f)^{r'},\ \ldots,\ (\gamma_2 f)^r,\ (\gamma_2 f)^{r'},\ \ldots,\ (\gamma_3 f)^r,\ (\gamma_3 f)^{r'},\ldots$$

may be represented by $\delta_1,\delta_2,\ldots$, and so on.

Observe that each term μ_r is a derivative $(\lambda f)^r$, the derivatives of λ with f being assumed to be an earlier term; the derivative with a less index of derivation is earlier than that with a greater index of derivation, or, what is the same thing, those, those are earlier which have a less index r, and which have a greater index of the higher order.

352. The series $\mu_1,\mu_2,\mu_3,\ldots$ is not asyzygetic; we make this so by considering in succession whether the several terms $\mu_1,\mu_2,\ldots$ respectively are expressible as linear functions of the other terms, or omitting every term which is so expressible. The reduced series obtained is a series of terms which is in fact a complete asyzygetic series of covariants of the degrees in question.

353. The quadricovariants $(a,b,c\,|\,x,y)^2$ are proportional, but as only the ratios are material, they may be taken equal to the determinants $G_1H_3 - G_3H_1$, $B_3G_1 - B_1G_3$, $B_1C_3 - B_3C_1$, &c. They form a system T of T-series, and the second equation gives $\alpha_1 f_1$; multiplying together these values, and writing $a_1 f_1 = k_1 g_1$, we find

$$(B_1B_3\cdot G_1G_3\cdot H_1H_3\cdot G_1H_3\cdot G_3H_1)\cdot k_1(A_1,B_1,C_1,F_1,G_1,H_1) = 0.$$

354. For the cubic $(a,b,c,d\,|\,x,y)^3$ the number of asyzygetic covariants $a^\mu x^\nu$ is

$$\text{coeff. } a^\mu x^\nu \text{ in } \frac{1-\frac{1}{x^2}}{(1-a)(1-ax)\left(1-\frac{a}{x}\right)},$$

or transforming as before, this is

$$\text{coeff. } a^\mu x^\nu \text{ in } \frac{1-x^2}{(1-ax)(1-ax^3)(1-a^3)}.$$

The development is

$$\begin{array}{l|l}
1 & \\
+\,ax^3 & -\,\tfrac{1}{x^2} \\
+\,a^2(x^2+1) & +\,a\cdot\big(\tfrac{1}{x}\big) \\
+\,a^3(x^3+x+1) & +\,a^2\big(\tfrac{1}{x}+1\big) \\
+\,a^4(x^2+x^4+1) & +\,a^3\big(\tfrac{1}{x}+\tfrac{1}{x^3}\big) \\
\vdots & +\,a^4\big(\tfrac{1}{x}+\tfrac{1}{x^3}+1\big)
\end{array}$$

which is

$$= A(x) - \tfrac{1}{x^2}\,A\big(\tfrac{1}{x}\big)\,,$$

where

$$A(x) = \frac{1}{(1-ax^3)(1-ax)},$$

and, since $\tfrac{1}{x^2}\,A\big(\tfrac{1}{x}\big)$ contains only negative powers, the required number is

$$= \text{coeff. } a^\mu x^\nu \text{ in } \frac{1}{(1-ax^3)(1-ax)},$$

indicating that the covariants are powers and products of (ax^3 and ax), the quadric itself, and the discriminant. Compare Second Memoir, No. 40, according to which, writing therein a for a, the number of asyzygetic covariants is

$$= \text{coeff. } a^\mu x^\nu \text{ in } \frac{1}{(1-ax)(1-x)}.$$

[Article numbers 355 *onwards (New formula for the number of Asyzygetic Covariants) continue on book pp.* 338–345 *in the same vein, expressing the number of asyzygetic covariants of the quadric, cubic, quartic and quintic as the coefficient of* $a^\mu x^\nu$ *in a generating fraction with a quadric or quartic Hessian-like denominator; these articles culminate in the explicit verification (book p.* 344*) that for the quintic the number is finite and* = 23*, and in the introduction (book p.* 345*) of* ***Article No.*** **337*****. Identification of the*** **23** ***Fundamental Covariants****, with Table No.* 87 *relating each of the* 23 *irreducible covariants to its Gordan symbol A, B, C, . . . , W.]*

[Tables Nos. 87*,* 88 *and the supplementary Table No.* 89 *(giving the syzygies and interconnexions of the syzygies among the products of the* 23 *covariants) occupy book pp.* 341–346*. They are intricate purely typographical exhibits whose intelligent reproduction is deferred to a later chunk; the present chunk closes with the opening of Articles* 347–365*, "Sketch of Professor Gordan's proof for the finite Number,* = 23*, of the Covariants of a Binary Quintic," as begun above.]*

ARTICLE NO. 363. For U equal to any one of the last eleven values, the form is $(Q, S)R$ which is $= R(QS) + S(QR)$, and is thus a function of covariants of a lower degree; there remain the derivatives formed with two of the functions f, ϕ, i, j, p, τ, or of one of these with α or γ. But these are all U other than the derivatives

$$(fj),\ (\phi j),\ (\phi p),\ (\phi\tau),\ (p\tau);\ (f\alpha),\ (\phi\alpha),\ (j\alpha),\ (p\alpha);\ (f\gamma),\ (\phi\gamma),\ (j\gamma),\ (p\gamma),\ (\tau\gamma),$$

and since $\gamma = (\tau\alpha)$, the derivatives containing γ will depend upon covariants of a lower degree; there remain therefore only (fj), (ϕj), (ϕp), $(\phi\tau)$, $(p\tau)$; $(f\alpha)$, $(\phi\alpha)$, $(j\alpha)$, $(p\alpha)$: each of these can be actually calculated in the form $F(U)$.

Hence finally, assuming that every covariant of a degree inferior to m is $F(U)$, it follows that every covariant of the degree m is $F(U)$; whence every covariant whatever is $F(U)$, viz. it is a rational and integral function of the 23 covariants U.

364. It will be observed that, writing A, B, C for P, Q, i, the proof depends on the theorems

$$\begin{array}{rl} ((AB),\ C), & \text{a linear function of } A(BC)',\ B(CA)',\ C(AB)', \\ (AB,\ C)' & \text{do.}\quad \text{do.}\quad \text{do.} \\ ((AB),\ C)' & \text{do.}\quad \text{do.}\quad ((AC)',B),\ ((BC)',A),\ B(AC)',\ C(AB)', \end{array}$$

which are theorems relating to any three functions A, B, C whatever.

365. I remark upon the proof that the really fundamental theorem seems to be that which I have called theorem A. As to the forms W it is difficult to see *à priori* why such forms are to be considered, or what the essential property involved in their definition is; and in fact in a more recent paper, “Die simultanen Systeme binären Formen” (*Math. Annalen*, t. II. (1869), see p. 256), Professor Gordan has modified the definition of the forms W by omitting the condition that the order of the function shall exceed n; if it were possible further to omit the condition of at least one index being $=$ or $< \frac{1}{2}n$, and so only retain the conditions $-n-i$, $n-j$, &c., each of them > 0, then the essential property of the forms W would be that any such form was a rational and integral function of the special covariants formed, as above, by means of the quantic of the next inferior order. And moreover, as regards the theorem B, there seems something indirect and artificial in the employment of such a property; one sees no reason why, when a system of irreducible covariants is once written down, it should not be possible to show that the derivatives of $F(U)$ with the original quantic are each of them $F(U)$, instead of having to show this in regard to the derivatives of $F(U)$ with the several covariants χ: as regards the quintic, where there is a single covariant χ, the quadric function i, there is obviously a great abbreviation in this employment of i in place of f; but for the higher orders, assuming that the proof could be conducted by means of the quantic itself, it does not appear that there would be even an abbreviation in the employment in its stead of the several covariants χ. The like remarks apply to the proof in the last-mentioned paper. I cannot but hope that a more simple proof of Professor Gordan's theorem will be obtained — a theorem the importance of which, in reference to the whole theory of forms, it is impossible to estimate too highly.

(End of chunk: PDF pages 326–375*, book pages* 304–353*.*
Paper 457 *in full,*
paper 458 *in full,*
paper 459 *in full,*
paper 460 *in full,*
paper 461 *in full,*
paper 462 *opening (Articles* 326–365*, book pp.* 334–353*).)*

463. NOTE ON A DIFFERENTIAL EQUATION.

[From the Memoirs of the Literary and Philosophical Society of Manchester, vol. II. *(*1865*), pp.* 111–114. *Read February* 18, 1862.*]*

The following investigation was suggested to me by Mr Harley's "Remarks on the Theory of the Transcendental Solution of Algebraic Equations," communicated to the Society at the Meeting of the 4th of February.

Mr Harley's equation

$$y^n - ny + (n-1)\,x = 0$$

may be written

$$y = \frac{n-1}{n}\,x + \frac{1}{n}\,y^n;$$

or putting

$$\frac{n-1}{n}\,x = u, \qquad \frac{1}{n} = a,$$

it becomes

$$y = u + ay^n,$$

which equation may be considered instead of the original equation; and it is to be shown that y, regarded as a function of u, satisfies a certain linear differential equation of the order $n-1$. In fact, expanding y by Lagrange's theorem, we have

$$\begin{aligned} y &= u + au^n + \frac{a^2}{1\cdot 2}\,(u^{2n})' + \frac{a^3}{1\cdot 2\cdot 3}\,(u^{3n})'' + \&\text{c.} \\ &= u + au^n + \frac{a^2}{1\cdot 2}\,2n\,u^{2n-1} + \frac{a^3}{1\cdot 2\cdot 3}\,3n\,(3n-1)\,u^{3n-2} + \&\text{c.}, \end{aligned}$$

the law whereof is obvious, and using the ordinary notation of factorials, viz.

$$[u]^r = u\,(u-1)\,(u-2)\,\ldots\,(u-r+1),$$

we may write

$$y = S_\theta\,\frac{[n\theta]^{\theta-1}}{[\theta]^\theta}\,a^\theta\,u^{n\theta-\theta+1},$$

where θ extends from 0 to ∞.

It is now very easy to show that y satisfies the differential equation

$$\left[u\,\frac{d}{du}\right]^{n-1} y \;=\; na\,\frac{d}{du}\cdot\left[\frac{n}{n-1}\,u\,\frac{d}{du} - \frac{2n-1}{n-1}\right]^{n-1} u^{n-1}\,y.$$

In fact, using on the left-hand side the foregoing value of y, and on the right-hand side the following value of $u^{n-1}\,y$, obtained from that of y by writing $\theta-1$ in the place of θ, viz.

$$u^{n-1}\,y \;=\; S_\theta\,\frac{[n\theta-n]^{\theta-2}}{[\theta-1]^{\theta-1}}\,a^{\theta-1}\,u^{n\theta-\theta+n},$$

and observing that in general the symbol $u\,\dfrac{d}{du}$, as regards u^m, is $= m$, the equation in question will be satisfied, if only

$$\frac{[n\theta]^{\theta-1}}{[\theta]^\theta}\,\{(n-1)\theta+1\}^{n-1} \;=\; \frac{n\,[n\theta-n]^{\theta-2}}{[\theta-1]^{\theta-1}}\left[\frac{n}{n-1}\,\{(n-1)\theta+1\} - \frac{2n-1}{n-1}\right]^{n-1},$$

where the right-hand side is

$$= \frac{n\,[n\theta-n]^{\theta-2}}{[\theta-1]^{\theta-1}}\,\{n\theta-1\}^{n-1};$$

and the equation may be written

$$\frac{n\theta\,[n\theta-1]^{\theta-1}}{\theta\,[\theta-1]^{\theta-1}}\,\{(n-1)\theta+1\}^{n-1} \;=\; \frac{n\,[n\theta-n]^{\theta-2}}{[\theta-1]^{\theta-1}}\,\{n\theta-1\}^{n-1};$$

that is,

$$\{n\theta-1\}^{\theta-1}\,\{(n-1)\theta+1\}^{n-1} \;=\; [n\theta-1]^{n-1}\,\{n\theta-n\}^{\theta-2},$$

which, since each side of the equation is $[n\theta-1]^{\theta+n-2}$, is obviously true.

The foregoing differential equation is developable in the form

$$\left[a_0+a_1\,u\,\frac{d}{du}+a_2\,u^2\left(\frac{d}{du}\right)^2+\ldots+a_{n-1}\,u^{n-1}\left(\frac{d}{du}\right)^{n-1}\right]y \;=\; \frac{1}{na}\left(\frac{d}{du}\right)^{n-1}y;$$

but to find the coefficients $a_0, a_1, \ldots, a_{n-1}$, I start from this form, and proceed to substitute in the equation the value of y, which on the left-hand side I use in the original form, and on the right-hand side in the form obtained by writing $\theta+1$ in the place of θ, viz.

$$y \;=\; S_\theta\,\frac{[n(\theta+1)]^\theta}{[\theta+1]^{\theta+1}}\,a^{\theta+1}\,u^{n\theta+n-\theta}.$$

The equation to be satisfied is

$$\frac{[n\theta]^{\theta-1}}{[\theta]^\theta}\Big(a_0+a_1[(n-1)\theta+1]^1+a_2[(n-1)\theta+1]^2+\ldots+a_{n-1}[(n-1)\theta+1]^{n-1}\Big) \;=\; \frac{1}{n}\,\frac{[n(\theta+1)]^\theta}{[\theta+1]^{\theta+1}}\,\{(n-1)\theta+n\}^{n-1},$$

or, what is the same thing,

$$\frac{1}{[\theta]^\theta}\Big(a_0[n\theta]^{\theta-1}+a_1[n\theta]^\theta+a_2[n\theta]^{\theta+1}+\ldots+a_{n-1}[n\theta]^{\theta+n-2}\Big) \;=\; \frac{1}{n}\,\frac{[n(\theta+1)]^{\theta+n-1}}{[\theta+1]^{\theta+1}}.$$

Observing that the right-hand side may be written

$$\frac{1}{n}\,\frac{n(\theta+1)\,[n\theta+n-1]^{\theta+n-2}}{(\theta+1)\,[\theta]^\theta},$$

the equation becomes

$$a_0[n\theta]^{\theta-1}+a_1[n\theta]^\theta+a_2[n\theta]^{\theta+1}+\ldots+a_{n-1}[n\theta]^{\theta+n-2} \;=\; [n\theta+n-1]^{\theta+n-2},$$

or, what is the same thing,

$$a_0+a_1[(n-1)\theta+1]^1+a_2[(n-1)\theta+1]^2+\ldots+a_{n-1}[(n-1)\theta+1]^{n-1} \;=\; [n\theta+n-1]^{n-1};$$

so that $a_0, a_1, \ldots, a_{n-1}$ are the coefficients of the expansion of $[n\theta+n-1]^{n-1}$ (which is a rational and integral function of θ, of the degree $n-1$) in a factorial series, as shown by the left-hand side of the equation.

To determine the actual values, write

$$(n-1)\theta+1=\phi;$$

this gives

$$n\theta+n-1 \;=\; \frac{n\phi+n^2-3n+1}{n-1},$$

and we have therefore

$$\left[\frac{n\phi+n^2-3n+1}{n-1}\right]^{n-1} \;=\; a_0+a_1[\phi]^1+a_2[\phi]^2+\ldots+a_{n-1}[\phi]^{n-1};$$

and thus the general expression is

$$a_r \;=\; \frac{1}{[r]^r}\,\Delta^r\left(\frac{n\phi+n^2-3n+1}{n-1}\right)^{n-1},$$

where Δ denotes the difference in regard to ϕ ($\Delta U_\phi = U_{\phi+1}-U_\phi$), and, after the operation Δ^r is performed, ϕ is to be put equal to zero.

464. NOTE ON PLANA'S LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXIII. *(*1862:1863*), pp.* 211–215.*]*

I have been much surprised to find that there is an error of the order $m^2\gamma^2$, arising from the omission of a factor $(1+\gamma^2)^{-1}$, in the expression for $\dfrac{d^2\delta u}{dv^2}+\delta u$, as given by the equation (II.)' (*Théorie de la Lune*, t. I., p. 267), being the equation made use of in the theory for the determination of δu, the perturbation of the reciprocal of the radius vector. This error may probably be the cause of some of the discrepancies in the terms of the fourth and higher orders, between Plana's results and those of Pontécoulant and Delaunay.

Plana's equation (6), t. I., p. 260, is

$$\begin{aligned}\frac{d^2\delta u}{dv^2}+\delta u = aR''&\\ &+f(e,\gamma)\,Q'e\cos\left(cv-\int\varpi\,dv\right)\\ &-\left\{f(e,\gamma)(1+\gamma^2)(1+s_1^2)^{-\frac{3}{2}}-\frac{a(1+s^2)^{-\frac{3}{2}}}{a_1\,\psi(e,\gamma)}\right\}\\ &+f(e,\gamma)\,P\gamma^2\,(1+s_1^2)^{-\frac{3}{2}}\,\Theta,\end{aligned}$$

if for shortness

$$\Theta \;=\; \tfrac{1}{2}\gamma^2-\left(1+\tfrac{1}{2}\gamma^2\right)\cos\left(2gv-2\int\theta\,dv\right)+\tfrac{1}{2}\gamma^2\cos\left(4gv-4\int\theta\,dv\right).$$

R'' (p. 256) should be

$$R'' \;=\; \frac{1}{aa_1\,\psi(e,\gamma)}\cdot\frac{\Omega_1\,\dfrac{du}{dv}\,\dfrac{d\Omega}{u\,dv}-\pi\,(1+s^2)^{-\frac{3}{2}}\,2\textstyle\int U\,dv}{1+2\textstyle\int U\,dv},$$

but, by an error which is implicitly corrected, the σ which multiplies $(1+s^2)^{-\frac{3}{2}}\,2\textstyle\int U\,dv$ is omitted. Hence the equation (6) becomes

$$\begin{aligned}\left(1+2\int U\,dv\right)\left(\frac{d^2\delta u}{dv^2}+\delta u\right) = \frac{a}{aa_1\,\psi(e,\gamma)}&\left\{\Omega_1-\frac{du}{dv}\,\frac{d\Omega}{u\,dv}-\sigma\,(1+s^2)^{-\frac{3}{2}}\,2\int U\,dv\right\}\\ &+\left(1+2\int U\,dv\right)f(e,\gamma)\,Q'e\cos\left(cv-\int\varpi\,dv\right)\\ &-\left(1+2\int U\,dv\right)\left\{f(e,\gamma)(1+\gamma^2)(1+s_1^2)^{-\frac{3}{2}}-\frac{a(1+s^2)^{-\frac{3}{2}}}{a_1\,\psi(e,\gamma)}\right\}\\ &+\left(1+2\int U\,dv\right)f(e,\gamma)\,P\gamma^2(1+s_1^2)^{-\frac{3}{2}}\,\Theta,\end{aligned}$$

in which equation

$$\Omega_1=\frac{d\Omega}{du}+\frac{s}{u}\,\frac{d\Omega}{dv},\quad \text{pp. 26, 245,}\qquad \frac{d\Omega}{u^t\,dv}=\frac{\sigma\sigma_t}{\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}}\,U,\quad \text{p. 265,}$$

$$\psi(e,\gamma)=\lambda^{-\frac{1}{2}}(1+\gamma^2)^{-\frac{1}{2}},\qquad f(e,\gamma)=\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}},\quad \text{p. 261.}$$

But retaining for greater convenience the function $f(e,\gamma)$ in two of the terms, we have

$$\begin{aligned}\left(1+2\int U\,dv\right)\left(\frac{d^2\delta u}{dv^2}+\delta u\right) = \frac{a}{aa_1\,\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}}\Big[&\frac{d\Omega}{du}+\frac{s}{u}\,\frac{d\Omega}{dv}\\ &-\frac{\sigma\sigma_t}{\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}}\,U\,\frac{du}{dv}-\pi(1+s^2)^{-\frac{3}{2}}\,2\int U\,dv\Big]\end{aligned}$$

$$
\begin{aligned}
&+\left(1+2\int U\,dv\right) f(e,\gamma)\,Q'e\cos\Big(cv-\int\varpi\,dv\Big)\\
&-\left(1+2\int U\,dv\right)\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{3}{2}}\left\{(1+s_1^2)^{-\frac{3}{2}}-\frac{a}{a_1}\,(1+s^2)^{-\frac{3}{2}}\right\}\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\,P\gamma^2(1+s_1^2)^{-\frac{3}{2}}\,\Theta,
\end{aligned}
$$

which is

$$
\begin{aligned}
&=\frac{a}{aa_1}\,\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}\left(\frac{d\Omega}{du}+\frac{s}{u}\,\frac{d\Omega}{dv}\right)\\
&-a\,U\,\frac{du}{dv}\\
&-\frac{a}{a_1}\,\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}\,(1+s^2)^{-\frac{3}{2}}\,2\int U\,dv \quad \text{(destroyed by term } \textit{infra}\text{)}\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\,Q'e\cos\Big(cv-\int\varpi\,dv\Big)\\
&-\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{3}{2}}\left\{(1+s_1^2)^{-\frac{3}{2}}-\frac{a}{a_1}(1+s^2)^{-\frac{3}{2}}\right\}\\
&-\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{3}{2}}\,(1+s_1^2)^{-\frac{3}{2}}\,2\int U\,dv\\
&+\frac{a}{a_1}\,\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}\,(1+s^2)^{-\frac{3}{2}}\,2\int U\,dv \quad \text{(destroyed by term } \textit{supra}\text{)}\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\,P\gamma^2(1+s_1^2)^{-\frac{3}{2}}\,\Theta;
\end{aligned}
$$

or, putting $u=\dfrac{1}{a}\,(\nu+\delta u)$, this becomes

$$
\begin{aligned}
\left(1+2\int U\,dv\right)\left(\frac{d^2\delta u}{dv^2}+\delta u\right)=&\ \frac{a}{aa_1}\,\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}\left(\frac{d\Omega}{du}+\frac{s}{u}\,\frac{d\Omega}{dv}\right)\\
&-U\left(\frac{du}{dv}+\frac{d\delta u}{dv}\right)\\
&-\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{3}{2}}\,(1+s_1^2)^{-\frac{3}{2}}\,2\int U\,dv\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\,Q'e\cos\Big(cv-\int\varpi\,dv\Big)\\
&-\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{3}{2}}\left\{(1+s_1^2)^{-\frac{3}{2}}-\frac{a}{a_1}(1+s^2)^{-\frac{3}{2}}\right\}\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\,P\gamma^2(1+s_1^2)^{-\frac{3}{2}}\,\Theta,
\end{aligned}
$$

agreeing with the Formula II. p. 265, except that in Plana's last term, instead of the factor $f(e,\gamma)$ $(=\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}})$, we have the factor $\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{3}{2}}$. That is, the last term, as given by Plana, should be divided by $1+\gamma^2$. And this error is introduced from the formula (II.) into the formula (II.)', p. 267, viz. the incorrect factor $\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}$ is there replaced by its value q; whereas, the true value being $\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{3}{2}}$, the factor in (II.)' should be $\dfrac{q}{1+\gamma^2}$.

The corrected formula (II.)' is

$$
\begin{aligned}
-\frac{d^2\delta u}{dv^2}-\delta u=&-Q'\,\frac{qe}{1+\gamma^2}\,\cos\Big(cv-\int\varpi\,dv\Big)\\
&+q\left\{(1+s_1^2)^{-\frac{3}{2}}-\frac{a}{a_1}(1+s^2)^{-\frac{3}{2}}\right\}\\
&+\mu^2(R_2+R_4)-\mu^2\left(\frac{du}{dv}+\frac{d\delta u}{dv}\right)R_1
\end{aligned}
$$

$$\begin{aligned}
&- 2\mu^2\left[\frac{d^2\delta u}{dv^2} + \delta u + q\,(1+s_1^2)^{-\frac{3}{2}} - \frac{Q'qe}{1+\gamma^2}\cos\left(cv - \int \varpi\, dv\right)\right]\int R_1\, dv \\
&- Pq\gamma^2\,(1+\gamma^2)^{-1}\,(1+s_1^2)^{-\frac{3}{2}}\,(1 - 2\mu^2\int R_1\, dv)\times \\
&\quad\left\{\tfrac{1}{2}\gamma^2 - (1+\tfrac{1}{2}\gamma^2)\cos(2gv - 2\int\theta dv) + \tfrac{1}{2}\gamma^2\cos(4gv - 4\int\theta dv)\right\}.
\end{aligned}$$

Observing that P is of the order m^2, and that q is approximately equal to unity, the error in $\dfrac{d^2\delta u}{dv^2} + \delta u$ is of the order $m^2\gamma^2$, as noticed above. It may be right to mention that I obtained the correction in the first instance by starting from the fundamental equations, and not as here from the intermediate equation (6), so that there is not in that equation any error afterwards implicitly corrected in the transformation to (II.)'.

465. NOTE ON THE LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXV. *(*1864–1865*), pp.* 182–189.*]*

I attend, in the expressions for the lunar coordinates, only to the coefficients independent of m. Plana's values, taken to the fourth order only, are as follows; for greater simplicity I write $a = 1$; and, instead of $nt +$ constant, $cnt +$ constant, $gnt +$ constant, I write g, l, c, respectively; viz., l is the mean longitude, c the mean anomaly, g the mean distance from node: this being so, then r, v, y, denoting the radius vector, longitude and latitude respectively, we have

$$\tfrac{1}{r}\,(\text{Plana}) = \begin{array}{rcl}
 & & \cos\ c \\
+e - \tfrac{1}{8}e^3 - \tfrac{1}{2}\gamma^2 e & ,, & c \\
+e^2 - \tfrac{1}{3}e^4 - \tfrac{1}{2}\gamma^2 e^2 & ,, & 2c \\
+\tfrac{3}{8}e^3 & ,, & 3c \\
+\tfrac{1}{3}e^4 & ,, & 4c \\
-\tfrac{1}{2}\gamma^2 & ,, & 2g \\
-\tfrac{1}{8}\gamma^2 e & ,, & c-2g
\end{array}$$

(but I omit Plana's term $+\frac{3}{8}\gamma^2 e\, \cos(2c+2g)$ which should be $= 0$).

$$v\,(\text{Plana}) = l + \begin{array}{rcl}
 & & \sin\ c \\
+2e - \tfrac{1}{4}e^3 - \tfrac{1}{2}\gamma^2 e & ,, & c \\
+\tfrac{5}{4}e^2 - \tfrac{11}{24}e^4 - \tfrac{3}{4}\gamma^2 e^2 & ,, & 2c \\
+\tfrac{13}{12}e^3 & ,, & 3c \\
+\tfrac{103}{96}e^4 & ,, & 4c \\
-\tfrac{1}{4}\gamma^2 - \tfrac{1}{16}\gamma^2 e^2 + \tfrac{1}{4}\gamma^4 & ,, & 2g \\
+\tfrac{1}{4}\gamma^2 e & ,, & c-2g \\
-\tfrac{1}{8}\gamma^2 e & ,, & c+2g \\
-\tfrac{1}{16}\gamma^2 e^2 & ,, & 2c-2g \\
+\tfrac{11}{48}\gamma^2 e^2 & ,, & 2c+2g \\
+\tfrac{1}{32}\gamma^4 & ,, & 4g
\end{array}$$

$$y\,(\text{Plana}) = \begin{array}{rcl}
 & & \sin\ g \\
\gamma - \gamma e^2 - \tfrac{3}{8}\gamma^4 & ,, & g \\
+\gamma e - \tfrac{1}{8}\gamma e^3 & ,, & c-g \\
+\gamma e - \tfrac{7}{8}\gamma e^3 - \tfrac{1}{2}\gamma^3 e & ,, & c+g \\
+\tfrac{17}{8}\gamma e^2 & ,, & 2c-g \\
+\tfrac{5}{8}\gamma e^2 & ,, & 2c+g \\
+\tfrac{17}{12}\gamma e^3 & ,, & 3c-g \\
+\tfrac{5}{12}\gamma e^3 & ,, & 3c+g \\
-\tfrac{1}{8}\gamma^3 & ,, & 3g \\
+\tfrac{1}{16}\gamma^3 e & ,, & c-3g \\
-\tfrac{1}{16}\gamma^3 e & ,, & c+3g.
\end{array}$$

To compare these with the elliptic values, it is necessary to write $e\,(1+\frac{1}{2}\gamma^2)$ in place of e. Making this change, or say reducing Plana's (e,γ) to the elliptic (e,γ), I write down in a first column the transformed coefficients, and in a second column the elliptic coefficients, as follows:

Plana, with Elliptic e,γ	Elliptic	
$\frac{1}{r}=$	$\frac{1}{r}=$	
1	1	cos c
$+e-\frac{1}{8}e^3$	$+e-\frac{3}{8}e^3$	„ c
$+e^2-\frac{1}{3}e^4$	$+e^2-\frac{1}{3}e^4$	„ $2c$
$+\frac{3}{8}e^3$	$+\frac{3}{8}e^3$	„ $3c$
$+\frac{1}{3}e^4$	$+\frac{1}{3}e^4$	„ $4c$
$-\frac{1}{2}\gamma^2$	0	„ $2g$
$-\frac{1}{8}\gamma^2 e$	0	„ $c-2g$.

Plana, with Elliptic e,γ	Elliptic	
$v=$	$v=$	
l	l	sin c
$+2e-\frac{1}{4}e^3$	$+2e-\frac{1}{4}e^3$	„ c
$+\frac{5}{4}e^2-\frac{11}{24}e^4-\frac{1}{16}\gamma^2e^2$	$+\frac{5}{4}e^2-\frac{11}{24}e^4$	„ $2c$
$+\frac{13}{12}e^3$	$+\frac{13}{12}e^3$	„ $3c$
$+\frac{103}{96}e^4$	$+\frac{103}{96}e^4$	„ $4c$
$-\frac{1}{4}\gamma^2-\frac{9}{16}\gamma^2e^2+\frac{1}{8}\gamma^4$	$-\frac{1}{4}\gamma^2+\gamma^2e^2+\frac{1}{8}\gamma^4$	„ $2g$
$+\frac{3}{4}\gamma^2e$	$-\frac{1}{2}\gamma^2e$	„ $c-2g$
$-\frac{1}{2}\gamma^2e$	$-\frac{1}{2}\gamma^2e$	„ $c+2g$
$-\frac{1}{8}\gamma^2e^2$	$+\frac{3}{16}\gamma^2e^2$	„ $2c-2g$
$-\frac{13}{16}\gamma^2e^2$	$-\frac{13}{16}\gamma^2e^2$	„ $2c+2g$
$+\frac{1}{32}\gamma^4$	$+\frac{1}{32}\gamma^4$	„ $4g$.

Plana, with Elliptic e,γ	Elliptic	
$y=$	$y=$	
$\gamma-\gamma e^2-\frac{3}{8}\gamma^3$	$+\gamma e-\gamma e^2-\frac{1}{2}\gamma^3$	sin g
$+\gamma e-\frac{3}{8}\gamma e^3-\frac{3}{8}\gamma^3e$	$+\gamma e-\frac{17}{8}\gamma e^3-\frac{1}{2}\gamma^3e$	„ $c+g$
$+\gamma e-\frac{7}{8}\gamma e^3+\frac{1}{2}\gamma^3e$	$+\gamma e-\frac{17}{8}\gamma^3e$	„ $c-g$
$+\frac{2}{3}\gamma e^2$		
$+\frac{1}{8}\gamma e^2$	$+\frac{1}{8}\gamma e^2$	„ $2c-g$
$+\frac{1}{16}\gamma e^2$		
$+\frac{17}{12}\gamma e^3$	$+\frac{17}{12}\gamma e^3$	„ $3c-g$
$+\frac{5}{12}\gamma e^3$	$+\frac{5}{12}\gamma e^3$	„ $3c+g$
$-\frac{1}{8}\gamma^3$	$-\frac{1}{8}\gamma^3$	„ $3g$
$+\frac{1}{16}\gamma^3e$	$-\frac{1}{16}\gamma^3e$	„ $c-3g$
$-\frac{1}{16}\gamma^3e$	$+\frac{1}{16}\gamma^3e$	„ $c+3g$,

where, for greater clearness, I remark that the values called "elliptic" of e, γ, c, g, refer to an ellipse, such that the longitude of the node, and the longitude (in orbit) of the pericentre, vary uniformly with the time, viz., we have mean distance $=1$, excentricity $=e$, tangent of inclination $=\gamma$, mean longitude $=l$, mean anomaly $=c$, distance from node $=g$.

We have therefore

$$
\delta\frac{1}{r} = \begin{array}{rcl}
-\tfrac{1}{2}\gamma^2e^2 & \cos & 2g \\
-\tfrac{1}{8}\gamma^2e & ,, & c-2g, \\
\delta v = -\tfrac{1}{16}\gamma^2e^2 & \sin & 2c \\
-\tfrac{11}{48}\gamma^2e^2 & ,, & 2g \\
+\tfrac{1}{4}\gamma^2e & ,, & c-2g \\
-\tfrac{1}{16}\gamma^2e^2 & ,, & 2c-2g, \\
\delta y = -\tfrac{3}{8}\gamma^3 + \tfrac{1}{8}\gamma e^2 & \sin & c-g \\
+\tfrac{2}{3}\gamma e^2 & ,, & 2c-g \\
+\tfrac{2}{3}\gamma e^3 & ,, & 3c-g \\
+\tfrac{1}{8}\gamma^3e & ,, & c-3g,
\end{array}
$$

viz., these are the increments to be added to the elliptic values of $\frac{1}{r}$, v, y, respectively, in order to obtain the disturbed values of $\frac{1}{r}$, v, y, attending only to the coefficients independent of m; *they represent, in fact, the Lunar inequalities which rise two orders by integration.*

The elliptic values of $\frac{1}{r}$ and y are functions, and that of v, is equal to $l+$ a function, of e, γ, c, g, and the foregoing disturbed values may be obtained by affecting each of the quantities e, γ, c, g, and l, with an inequality depending on the argument $2c-2g$, viz., these inequalities are

$$
\begin{aligned}
\delta e &= -\tfrac{3}{8}\gamma^2 e\cos 2c-2g \\
\delta c &= \tfrac{1}{8}\gamma^2\sin 2c-2g \\
\delta\gamma &= -\tfrac{3}{8}\gamma e^2\cos 2c-2g \\
\delta g &= \tfrac{3}{8}e^2\sin 2c-2g \\
\delta l &= -\tfrac{1}{16}\gamma^2e^2\sin 2c-2g.
\end{aligned}
$$

The verification may be effected without difficulty; thus, for instance, starting from the elliptic value of $\frac{1}{r}$, we have to the fourth order

$$
\begin{aligned}
\delta\frac{1}{r} &= \delta\begin{pmatrix} e\cos c \\ +e^2\cos 2c\end{pmatrix}\begin{pmatrix} -e\sin c \\ -2e^2\sin 2c\end{pmatrix}\delta c' + \begin{pmatrix}\cos c \\ +2e\cos 2c\end{pmatrix}\delta e \\
&= -\tfrac{1}{8}\gamma e\,(-\sin c\sin 2c-2g\cos c\cos 2c-2g) \\
&\quad + \tfrac{3}{8}\gamma^2\,(-\sin 2c\sin 2c-2g - \cos 2c\cos 2c-2g) \\
&= -\tfrac{1}{2}\gamma^2e\cos c-g \\
&\quad - \tfrac{3}{8}\gamma^2\cos 2g,
\end{aligned}
$$

which is right; and the verification of the values of δv, δy, may be effected in a similar manner.

I have, in order to fix the ideas, preferred to give in the first instance the foregoing *à posteriori* proof; but I now inquire generally as to the form of the values of $\frac{1}{r}$, v, y, or say of r, v, y, taking account only of coefficients independent of m; and I proceed to show that these may be obtained from the elliptic values expressed as above in terms of l, e, γ, c, g, by affecting l, e, γ, c, g, each with an inequality depending on the multiple sines or cosines of $c-g$.

Writing for greater simplicity $n=1$, we have $l=t+L$, $c=ct+C$, $g=gt+G$, where $c=1-\frac{3}{4}m^2+\&c.$, $g=1+\frac{3}{4}m^2+\&c.$; viz., c, g, are constants which differ from unity by terms involving m^2.

The required values of r, v, y, satisfy the undisturbed equations of motion, if after the differentiations we write in the coefficients (which coefficients are functions of m through c, g) $m=0$; that is, if we write in the coefficients $c=1$, $g=1$. In fact, the required values of r, v, y, are what the complete values become, upon writing in the coefficients of the complete values $m=0$; that is, the required values of r, v, y, differ from the complete values by terms the coefficients whereof contain m as a factor; and the disturbed equations differ from the undisturbed equations in that they contain the differential coefficients of the disturbing function; that is, terms the coefficients whereof have the factor m^2. Imagine the complete values of r, v, y, substituted in the disturbed equations of motion; the resulting equations are satisfied identically; and, therefore, whatever be the value of m; that is, they are satisfied if in these equations respectively we write $m = 0$: it requires a little consideration to see that this is so, if in *the coefficients only* we write $m = 0$; but recollecting that c, g, stand for functions $ct + C$, $gt + G$, so that, for example, $c - g, = (c - g)t + C - G$, upon writing therein $m = 0$, becomes equal, not to zero, but to the constant value $C - G$, the identity subsists in regard to the coefficient of the sine or cosine of each separate argument $\alpha c \pm \beta g$, and, consequently, it subsists notwithstanding that in the arguments c and g, instead of being each put $= 1$, are left indeterminate. And granting this (viz. that the equations are satisfied if in the coefficients only we write $m = 0$), then it is clear that, as above stated, the required values of r, v, y, satisfy the undisturbed equations of motion, if after the differentiations we write in the coefficients $c = 1$, $g = 1$.

The required values of r, v, y, are of the form $r = \phi(c, g),\ y = \psi(c, g),\ v = l + \chi(c, g)$, but writing $w = v + c - l, = c + \chi(c, g)$, the last mentioned property will equally subsist in regard to the functions r, w, y: in fact, v enters into the differential equations only through its differential coefficient $\frac{dv}{dt}$, and the differential coefficients of v and w, that is, of $l + \chi(c, g)$ and $c + \chi(c, g)$, differ only by the quantity $c - 1$, which becomes $= 0$, in virtue of the assumed relations $c = 1$, $g = 1$.

Hence the undisturbed equations are satisfied by the values $r = \phi(c, g),\ y = \psi(c, g),\ w = c + \chi(c, g)$, when after the differentiations we write in the coefficients $c = 1$, $g = 1$; the foregoing values contain t through the quantities c, g, only; and we have, therefore,

$$\frac{d}{dt} = \frac{d}{dc} + \frac{d}{dg}.$$

Hence, writing in the coefficients $c = 1$, $g = 1$, we have $\frac{d}{dt} = \frac{d}{dc} + \frac{d}{dg}$; that is, the values $r = \phi(c, g),\ y = \psi(c, g),\ w = \chi(c, g)$, regarding r, v, y, as functions of c, g, satisfy the partial differential equations obtained from the undisturbed equations of motion by writing therein $\frac{d}{dc} + \frac{d}{dg}$ in place of $\frac{d}{dt}$. Hence also, considering r, w, y, as functions of c and $c - g$, then observing that $\left(\frac{d}{dc} + \frac{d}{dg}\right)(c - g) = 0$, the values of r, v, y, satisfy the partial differential equations obtained by writing $\frac{d}{dc}$ in place of $\frac{d}{dt}$; and inasmuch as these partial differential equations do not contain $\frac{d}{dg}$, they are to be integrated as ordinary differential equations in regard to c as the independent variable, the constants of integration being replaced by arbitrary functions of $c - g$.

Consider the pure elliptic values of r, v, y, in an elliptic orbit with the following elements: A, the mean distance; N, the mean motion ($N^2A^3 = 1$ and therefore $A = N^{-\frac{2}{3}}$); E, the excentricity; $Nt + D$, the mean anomaly; $Nt + H$, the mean distance from node; $Nt + K$, the mean longitude; then writing c in place of t, we have

$$\begin{aligned} r &= N^{-\frac{2}{3}} \operatorname{elqr}(E,\ Nc + D), \\ v\,(= l - c + w) &= l - c + Nc + K + P(E,\ \Gamma,\ Nc + D,\ Nc + H), \\ y &= Q(E,\ \Gamma,\ Nc + D,\ Nc + H), \end{aligned}$$

where N, E, Γ, D, H, K, are arbitrary functions of $c - g$: P and Q denote given functional expressions. But, in order that r, v, y, considered as functions of c and g, may be of the proper form, it is necessary as regards N to write simply $N = 1$; we have then

$$\begin{aligned} r &= \operatorname{elqr}(E,\ c + D), \\ v &= l + K + P(E,\ \Gamma,\ c + D,\ c + H), \\ y &= Q(E,\ \Gamma,\ c + D,\ c + H), \end{aligned}$$

where E, Γ, D, H, K, are arbitrary functions of $c - g$; or, what is the same thing, writing for these quantities respectively $e + \delta e,\ \gamma + \delta\gamma,\ c + \delta c,\ g + \delta g,\ l + \delta l$, where δe, $\delta\gamma$, δc, δg, δl are arbitrary functions of $c - g$, we have

$$\begin{aligned} r &= \operatorname{elqr}(e + \delta e,\ c + \delta c), \\ v &= l + \delta l + P(e + \delta e,\ \gamma + \delta\gamma,\ c + \delta c,\ g + \delta g), \end{aligned}$$

$$y = Q(e + \delta e,\ \gamma + \delta\gamma,\ c + \delta c,\ g + \delta g),$$

that is, the values of r, v, y, are obtained from the elliptic values

$$\begin{aligned} r &= \text{elqr}\,(e,\ c), \\ v &= l + P(e,\ \gamma,\ c,\ g), \\ y &= Q(e,\ \gamma,\ c,\ g), \end{aligned}$$

by affecting each of the quantities $e,\ \gamma,\ c,\ g,\ l$, with an inequality which is a function of $c - g$.

466. SECOND NOTE ON THE LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXV. (1864–1865), pp. 203–207.]

The elliptic values of

r, the radius vector, v, the longitude, y, the latitude,

are functions of

a, the mean distance,
e, the excentricity,
γ, the tangent of the inclination,
l, the mean longitude,
c, the mean anomaly,
g, the mean distance from node;

see my Note in the last Monthly Notice, p. 182, [465], where, for the present purpose, $\frac{a}{r}$ should be written instead of $\frac{1}{r}$; and it is there shown that the disturbed values, attending only to the coefficients independent of m, are obtained by affecting $a,\ e,\ \gamma,\ c,\ g,\ l$, with the inequalities

$$\begin{aligned} \delta a &= 0 \\ \delta e &= -\tfrac{3}{8}\gamma^2 e \cos\, 2c - 2g \\ \delta\gamma &= +\tfrac{3}{8}\gamma e^2 \quad ,, \quad 2c - 2g \\ \delta c &= +\tfrac{1}{8}\gamma^2 \sin\, 2c - 2g \\ \delta g &= +\tfrac{3}{8}e^2 \quad ,, \quad 2c - 2g \\ \delta l &= -\tfrac{1}{16}\gamma^2 e^2 \,,, \quad 2c - 2g, \end{aligned}$$

or, what is the same thing, adding to the elliptic values the inequalities

$$\begin{aligned} \delta\frac{a}{r} &= -\tfrac{1}{2}\gamma^2 \cos\, 2g \\ &\quad - \tfrac{1}{8}\gamma^2 e \quad ,, \quad c - 2g, \\ \delta v &= -\tfrac{1}{16}\gamma^2 e^2 \sin\, 2c \\ &\quad - \tfrac{11}{48}\gamma^2 e^2 \quad ,, \quad 2g \\ &\quad + \tfrac{1}{4}\gamma^2 e \quad ,, \quad c - 2g \\ &\quad - \tfrac{1}{16}\gamma^2 e^2 \quad ,, \quad 2c - 2g, \\ \delta y &= -\tfrac{3}{8}\gamma^3 + \tfrac{1}{8}\gamma e^2 \sin\, c - g \\ &\quad + \tfrac{2}{3}\gamma e^2 \quad ,, \quad 2c - g \\ &\quad + \tfrac{2}{3}\gamma e^3 \quad ,, \quad 3c - g \\ &\quad + \tfrac{1}{8}\gamma^3 e \quad ,, \quad c - 3g. \end{aligned}$$

I propose to show how these results may be obtained by the method of the variation of the elements. For this purpose, treating a, e, γ, c, g, l, as elements, the proper formulae are obtained very readily from those given in my "Memoir on the Problem of Disturbed Elliptic Motion," Mem. R. Ast. Soc., vol. XXVII. (1859), pp. 1–29, [212]; viz., writing c in place of g, the formulae, p. 25, give the variations of a, e, c, η, θ, ϕ; we have then

$$g = c + \mathfrak{C}, \qquad l = c + \mathfrak{C} + \theta,$$

and therefore

$$\gamma = \tan\phi,$$

$$\begin{aligned} dg &= dc + d\mathfrak{C}, \\ dl &= dc + d\mathfrak{C} + d\theta, \\ d\gamma &= (1+\gamma^2)\, d\phi, \end{aligned}$$

which give for the transformation of the differential coefficients of Ω,

$$\begin{aligned} \frac{d\Omega}{dc} &= \frac{d\Omega}{dc} + \frac{d\Omega}{dg} + \frac{d\Omega}{dl}, \\ \frac{d\Omega}{d\mathfrak{C}} &= \frac{d\Omega}{dg} + \frac{d\Omega}{dl}, \\ \frac{d\Omega}{d\theta} &= \frac{d\Omega}{dl}, \\ \frac{d\Omega}{d\phi} &= (1+\gamma^2)\,\frac{d\Omega}{d\gamma}, \end{aligned}$$

and the formulae finally become

$$\begin{aligned} \frac{da}{dt} &= \frac{2}{na}\,\frac{d\Omega}{dc} + \frac{2}{na}\,\frac{d\Omega}{dg} + \frac{2}{na}\,\frac{d\Omega}{dl}, \\ \frac{de}{dt} &= \frac{1-e^2}{na^2e}\,\frac{d\Omega}{dc} + \frac{1-e^2-\sqrt{1-e^2}}{na^2e}\,\frac{d\Omega}{dg} + \frac{1-e^2-\sqrt{1-e^2}}{na^2e}\,\frac{d\Omega}{dl}, \\ \frac{d\gamma}{dt} &= \frac{1+\gamma^2}{na^2\sqrt{1-e^2}\,\gamma}\,\frac{d\Omega}{dg} + \frac{(1+\gamma^2)(1-\sqrt{1+\gamma^2})}{na^2\sqrt{1-e^2}\,\gamma}\,\frac{d\Omega}{dl}, \\ \frac{dc}{dt} &= -\frac{2}{na}\,\frac{d\Omega}{da} - \frac{1-e^2}{na^2e}\,\frac{d\Omega}{de}, \\ \frac{dg}{dt} &= -\frac{2}{na}\,\frac{d\Omega}{da} - \frac{1-e^2-\sqrt{1-e^2}}{na^2e}\,\frac{d\Omega}{de} - \frac{1+\gamma^2}{na^2\sqrt{1-e^2}\,\gamma}\,\frac{d\Omega}{d\gamma}, \\ \frac{dl}{dt} &= -\frac{2}{na}\,\frac{d\Omega}{da} - \frac{1-e^2-\sqrt{1+e^2}}{na^2e}\,\frac{d\Omega}{de} - \frac{(1+\gamma^2)(1-\sqrt{1+\gamma^2})}{na^2\sqrt{1-e^2}\,\gamma}\,\frac{d\Omega}{d\gamma}. \end{aligned}$$

The disturbing function contains the term

$$m^2a^2a''\left(+\tfrac{3}{8}e^2\gamma^2\right)\cos 2c-2g.$$

If after the differentiations we write for greater simplicity $a=1$, $n=1$, we have

$$\begin{aligned} \frac{d\Omega}{da} &= +\tfrac{15}{4}\,m^2e^2\gamma^2\cos 2c-2g, \\ \frac{d\Omega}{de} &= +\tfrac{3}{4}\,m^2e\gamma^2 \;\;,\,,\;\; 2c-2g, \\ \frac{d\Omega}{d\gamma} &= +\tfrac{3}{4}\,m^2e^2\gamma \;,\,,\;\; 2c-2g, \\ \frac{d\Omega}{dc} &= -\tfrac{3}{4}\,m^2e^2\gamma^2\sin 2c-2g, \\ \frac{d\Omega}{dg} &= -\tfrac{3}{4}\,m^2e^2\gamma^2 \;\;,\,,\;\; 2c-2g, \end{aligned}$$

$$\frac{d\Omega}{dl} = 0,$$

and the formulae for the variations give

$$\begin{aligned}
\frac{da}{dt} &= \frac{2}{na}\left(\frac{d\Omega}{dc} + \frac{d\Omega}{dg}\right) = 0,\\
\frac{de}{dt} &= -\frac{1}{e}\,\frac{d\Omega}{dc} = -\tfrac{3}{8}\,m^2e\gamma^2 \sin 2c - 2g,\\
\frac{d\gamma}{dt} &= -\frac{1}{\gamma}\,\frac{d\Omega}{dg} = -\tfrac{3}{8}\,m^2\gamma e^2 \;\;,,\;\; 2c - 2g,\\
\frac{dc}{dt} &= -\frac{1}{e}\,\frac{d\Omega}{de} = -\tfrac{3}{8}\,m^2\gamma^2 \cos 2c - 2g,\\
\frac{dg}{dt} &= -\frac{1}{\gamma}\,\frac{d\Omega}{d\gamma} = -\tfrac{3}{8}\,m^2e^2 \;\;,,\;\; 2c - 2g,\\
\frac{dl}{dt} &= -2\,\frac{d\Omega}{da} + \tfrac{1}{2}\,e\,\frac{d\Omega}{de} + \tfrac{1}{2}\,\gamma\,\frac{d\Omega}{d\gamma} = \left(-\tfrac{15}{2} + \tfrac{3}{8} + \tfrac{3}{8}\right) m^2e^2\gamma^2 \;\;,,\;\; 2c - 2g,
\end{aligned}$$

but this value of $\frac{dl}{dt}$ is, as will presently be seen, incomplete.

Writing $a + \delta a$, $e + \delta e$, &c., in place of a, e, &c., and observing that the divisor for the integration of the term in $2c - 2g$ is $2(c - g)$, $= -3m^2$, the first five equations give respectively

$$\begin{aligned}
\delta a &= 0,\\
\delta e &= -\tfrac{3}{8}\,\gamma^2 e \cos 2c - 2g,\\
\delta\gamma &= +\tfrac{3}{8}\,\gamma e^2 \;\;,,\;\; 2c - 2g,\\
\delta c &= +\tfrac{1}{8}\,\gamma^2 \sin 2c - 2g,\\
\delta g &= +\tfrac{3}{8}\,e^2 \;\;,,\;\; 2c - 2g.
\end{aligned}$$

The constant term in Ω is

$$= \; m^2a^2a''\left(\tfrac{1}{2} + \tfrac{3}{4}e^2 - \tfrac{3}{4}\gamma^2\right),$$

and this gives in

$$\frac{dl}{dt} \;=\; -2\,\frac{d\Omega}{da} + \tfrac{1}{2}e\,\frac{d\Omega}{de} + \tfrac{1}{2}\gamma\,\frac{d\Omega}{d\gamma},$$

a term

$$m^2\left(-1 - \tfrac{9}{4}e^2 + \tfrac{9}{4}\gamma^2\right) + \tfrac{3}{4}e^2 - \tfrac{3}{4}\gamma^2,$$

which is

$$= \; m^2\left(-1 - \tfrac{3}{2}e^2 + \tfrac{3}{2}\gamma^2\right).$$

Substituting for e, γ, their correct values $e + \delta e$, $\gamma + \delta\gamma$, it appears that $\frac{dl}{dt}$ contains the term

$$m^2\left(-\tfrac{3}{2}e\,\delta e + \tfrac{3}{2}\gamma\,\delta\gamma\right),$$

which is

$$= m^2\left(\tfrac{9}{16} + \tfrac{9}{16}\right) e^2\gamma^2 \cos 2c - 2g = \tfrac{9}{8}\,m^2e^2\gamma^2 \;\;,,\;\; 2c - 2g,$$

and joining to this the before-mentioned term

$$= \; -\tfrac{27}{8}\,m^2e^2\gamma^2 \;\;,,\;\; 2c - 2g,$$

we find

$$\frac{dl}{dt} \;=\; \left(\tfrac{9}{8} - \tfrac{27}{8}\right) m^2e^2\gamma^2 \;\;,,\;\; 2c - 2g,$$

whence, writing as above $l + \delta l$ for l, and integrating, we have

$$\delta l \;=\; -\tfrac{1}{16}\,e^2\gamma^2 \sin 2c - 2g,$$

and it thus appears that the values of δa, δe, $\delta\gamma$, δc, δg, δl, agree with those obtained in my former Note.

467. EXPRESSIONS FOR PLANA'S e, γ IN TERMS OF THE ELLIPTIC e, γ.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXV. *(*1864–1865*), pp.* 265–271.*]*

The coefficient of $\sin cnt$ in Plana's expression for the true longitude v (see Plana, t. I. p. 574), putting therein $E' = e' = e'$, that is, neglecting the terms which depend on the variation of the solar excentricity, is

$$\begin{aligned}
&= e\left(2 + \tfrac{3}{4}m^2 - \tfrac{11}{48}m^4 - \tfrac{2363}{1152}m^6 - \tfrac{19259}{4608}m^6 - \tfrac{511469}{36864}m^8\right)\\
&\quad + e^3\left(-\tfrac{1}{4} - 17m^2 - \tfrac{415}{32}m^4 - \tfrac{84539}{1152}m^6\right)\\
&\quad + e^5\left(\tfrac{1}{96} + \tfrac{1717}{96}m^2\right)\\
&\quad + e^7\left(\tfrac{6727}{1152}\right)\\
&\quad + e\gamma^2\left(-\tfrac{1}{4} - \tfrac{11}{16}m^2 + \tfrac{1487}{96}m^4 + \tfrac{7212}{144}m^6\right)\\
&\quad + e\gamma^4\left(\tfrac{11}{32} - \tfrac{8231}{384}m + \tfrac{4923}{72}m^2\right)\\
&\quad + e^3\gamma^2\left(-\tfrac{183}{16}\right)\\
&\quad + e\gamma^6\left(-\tfrac{1331}{144}\right)\\
&\quad + e\gamma^4\left(-\tfrac{3}{8} + \tfrac{135}{32}m + \tfrac{37}{12}m^2\right)\\
&\quad + e^3\gamma^4\left(\tfrac{4233}{144}\right)\\
&\quad + e\gamma^6\left(-\tfrac{157}{72}\right)\\
&\quad + ee'^2\left[\left(-\tfrac{3}{8} + \tfrac{3}{2} = \tfrac{9}{8}\right) - 9m^2 + \left(-\tfrac{1855}{128} + \tfrac{339}{32}\right) = \tfrac{3}{8}\,\tfrac{1}{32}m^4 + \left(-\tfrac{3045153}{1024} - \tfrac{267}{32}\right) = \tfrac{1}{32}\,\tfrac{3}{1024}m^6\right]\\
&\quad + e^3e'^2\left[\left(-\tfrac{4923}{96} - \tfrac{3}{8}\right) = -\tfrac{4131}{96}m^2\right]\\
&\quad + ee'^2\gamma^2\left[\left(\tfrac{8551}{192} - \tfrac{3309}{16}\right) = -\tfrac{15825}{96}m^2\right]\\
&\quad + ee'^4\left[\left(\tfrac{15}{8} - \tfrac{2925}{32}\right) = -\tfrac{1791}{32}m^4\right]\\
&\quad + ee^2\left(-\tfrac{4593}{1024}m^2\right)\\
&\quad + ee'^6\left(-\tfrac{25}{32}\right).
\end{aligned}$$

Taking this to the fifth order only, and comparing it with the coefficient in the elliptic theory, we have

Plana.	Elliptic.
$e\ \left(2 - \tfrac{7}{12}m^2 - \tfrac{1027}{1152}m^4 - \tfrac{8235}{4608}m^6\right)$	$= e\ (2)$
$+e^3\ \left(-\tfrac{1}{4} - 17m^2\right)$	$+e^3\left(-\tfrac{1}{4}\right)$
$+e^5\ \left(\tfrac{1}{96}\right)$	$+e^5\left(\tfrac{1}{96}\right)$
$+e\gamma^2\ \left(-\tfrac{1}{4} - \tfrac{23}{12}m^2\right)$	
$+e\gamma^4\ \left(-\tfrac{3}{8}\right)$	
$+e^3\gamma^2\ \left(\tfrac{11}{16}\right)$	
$+ee'^2\ \left(-9m^2\right).$	

The coefficient of $\sin gnt$ in Plana's expression for the latitude (see t. I. p. 704) is

$$\begin{aligned}
&= \gamma\left(1 + \tfrac{15}{16}m^2 + \tfrac{121}{48}m^4 - \tfrac{1023}{1152}m^6\right)\\
&\quad + \gamma e^2\left(-\tfrac{1}{2} - \tfrac{15}{32}m^2 - \tfrac{257}{96}m^4\right)\\
&\quad + \gamma e^4\left(\tfrac{5}{16} + \tfrac{339}{96}m^2\right)\\
&\quad + \gamma e^6\left(\tfrac{3}{16}\right)\\
&\quad + \gamma^3\left(-\tfrac{1}{8} - \tfrac{31}{96}m^2 + \tfrac{1119}{96}m^4\right)\\
&\quad + \gamma^3e^2\left(\tfrac{4}{16} - \tfrac{4133}{96}m^2\right)\\
&\quad + \gamma^5\left(\tfrac{3}{32}\right)\\
&\quad + \gamma e'^2\left(\tfrac{27}{32}m^2 - m^4\right).
\end{aligned}$$

But according to the calculation of Prof. Adams (quoted by M. Delaunay, *Comptes Rendus*, t. LIV. (1862), this should be

$$\begin{aligned}
&= \gamma\left(1 + \tfrac{13}{16}m^2 - \tfrac{1}{12}m^4 - \tfrac{6135}{1152}m^6 - \tfrac{4351477}{2304}m^6\right) \\
&\quad + \gamma e^2\left(-\tfrac{1}{2} - \tfrac{15}{32}m^2 - \tfrac{249}{96}m^4\right) \\
&\quad + \gamma e^4\left(\tfrac{5}{16} + \tfrac{15}{96}m^2\right) \\
&\quad + \gamma e^6\left(\tfrac{3}{16}\right) \\
&\quad + \gamma^3\left(-\tfrac{1}{8} - \tfrac{2}{3} + \tfrac{1119}{96}m^4\right) \\
&\quad + \gamma^3 e^2\left(\tfrac{4}{16} + \tfrac{4133}{96}m^2\right) \\
&\quad + \gamma^5\left(\tfrac{3}{32}\right) \\
&\quad + \gamma e'^2\left(\tfrac{27}{32}m^2 - \tfrac{1}{16}m^4\right).
\end{aligned}$$

Adopting this as the true expression according to Plana's theory, taking it to the fifth order only, and comparing with the elliptic value of the same coefficient, we have

Plana.	Elliptic.
$\gamma\ \left(1 + \frac{13}{16}m^2 - \frac{1}{12}m^4\right)$	$= \gamma\ (1)$
$+\gamma e^2\ \left(-\frac{1}{2} - \frac{15}{32}m^2\right)$	$+\gamma e^2\ \left(-\frac{1}{2}\right)$
$+\gamma e^4\ \left(\frac{5}{16}\right)$	$+\gamma e^4\ \left(\frac{5}{16}\right)$
$+\gamma^3\ \left(-\frac{1}{8} - \frac{2}{3}\right)$	$+\gamma^3\ \left(-\frac{1}{8}\right)$
$+\gamma^3 e^2\ \left(\frac{4}{16}\right)$	$+\gamma^3 e^2\ \left(\frac{4}{16}\right)$
$+\gamma^5\ \left(\frac{3}{32}\right)$	$+\gamma^5\ \left(\frac{3}{32}\right)$
$+\gamma e'^2\ \left(\frac{27}{32}m^2\right)$	.

We have thus two equations for the determination of Plana's e, γ in terms of the elliptic e, γ. And the solution of these equations give

$$\begin{aligned}
&\qquad\qquad \text{Elliptic.} \\
e\,(\text{Plana}) = e\quad &\left(1 - \tfrac{3}{8}\,m^2 + \tfrac{27}{32}m^4 + \tfrac{521}{1152}m^6\right) \\
&+e^3\ \left(\tfrac{17}{16}m^2\right) \\
&+\gamma^2 e\ \left(\tfrac{1}{4} + \tfrac{23}{12}m^2\right) \\
&+\gamma^2 e^3\ \left(-\tfrac{1}{2}\right) \\
&+\gamma e\ \left(\tfrac{1}{4}\right) \\
&+e e'^2\ \left(\tfrac{9}{2}m^2\right), \\
\gamma\,(\text{Plana}) = \gamma\quad &\left(1 - \tfrac{3}{32}m^2 + \tfrac{2}{3}m^4\right) \\
&+\gamma e^2\ \left(-\tfrac{3}{16}\right) \\
&+\gamma e^4\ \left(-\tfrac{1}{16}m^2\right) \\
&+\gamma e^6\ \left(\tfrac{11}{16}\right) \\
&+\gamma^3\ \left(\tfrac{2}{3}\right) \\
&+\gamma^5 e^2\ \left(-\tfrac{1}{2}m^2\right).
\end{aligned}$$

I annex the verification of these expressions; we have

Plana.	Elliptic.
$e\left(2+\frac{3}{4}m^2-\frac{11}{48}m^4-\frac{2363}{1152}m^6\right)$	$=e\big(2-\frac{3}{8}m^2+\frac{27}{32}m^4+\frac{521}{1152}m^6$ $+\frac{4}{3}m^4-\frac{8}{3}m^6$ $-\frac{19}{12}m^6-\frac{6135}{1152}m^6\big)$
$+e^3\left(\frac{17}{16}m^2\right)$	$=e^3\left(\frac{17}{16}m^2\right)$
$+e\gamma^2\left(\frac{1}{4}+\frac{23}{12}m^2\right)$	$=e\gamma^2\left(\frac{1}{4}+\frac{23}{12}m^2\right)$ $+\frac{4}{3}m^4$
$+e\gamma^4\left(-\frac{3}{8}\right)$	$=e\gamma^4\left(-\frac{3}{8}\right)$
$+e\gamma^6\left(\frac{11}{16}\right)$	$=e\gamma^6\left(\frac{11}{16}\right)$
$+ee'^2\left(-9m^2\right)$	$=ee'^2\left(-9m^2\right),$
$e^3\left(-\frac{1}{4}-17m^2\right)$	$=e^3\left(-\frac{1}{4}+\frac{17}{16}m^2\right)$ $-17m^2$
$e^5\left(\frac{1}{96}\right)$	$=e^5\left(\frac{1}{96}\right)$
$e^3\gamma^2\left(-\frac{1}{8}-\frac{23}{16}m^2\right)$	$=e^3\gamma^2\left(-\frac{1}{8}-\frac{23}{16}m^2\right)$ $+\frac{4}{3}m^4$
$e^3\gamma^4\left(\frac{11}{16}\right)$	$=e^3\gamma^4\left(\frac{11}{16}\right)$
$e^5\gamma^2\left(-\frac{1}{8}\right)$	$=e^5\gamma^2\left(-\frac{1}{8}\right)$
$ee'^2\left(-9m^2\right)$	$=ee'^2\left(-9m^2\right),$

whence, adding, we have the first equation.

And, moreover,

Plana.	Elliptic.
$\gamma\left(1+\frac{13}{16}m^2-\frac{1}{12}m^4\right)$	$=\gamma\big(1-\frac{3}{32}m^2+\frac{2}{3}m^4$ $+\frac{4}{3}m^4-\frac{2}{3}m^4$ $+\gamma e^2\left(-\frac{23}{12}m^2\right)$ $+\gamma e^2\left(\frac{2}{3}\right)$ $+\gamma e^4\left(-\frac{1}{2}m^2\right)$ $+\gamma^3\left(-\frac{2}{3}\right),$
$\gamma e^2\left(\frac{3}{32}\right)$	$=\gamma e^2\left(\frac{3}{32}\right)$
$\gamma\left(-\frac{1}{2}+\frac{15}{32}m^2\right)$	$=\gamma\left(-\frac{1}{2}+\frac{15}{32}m^2\right)$
$\gamma^3e^2\left(\frac{4}{16}\right)$	$=\gamma^3e^2\left(\frac{4}{16}\right)$
$\gamma^5\left(\frac{3}{32}\right)$	$=\gamma^5\left(\frac{3}{32}\right)$
$\gamma e^2\left(\frac{27}{32}m^2\right)$	$=\gamma e^2\left(\frac{27}{32}m^2\right),$

whence, adding, we have the second equation.

It may be noticed that, taking the foregoing expressions only as far as the third order, we have

Plana.		Elliptic.
e	$=$	$e\,(1+\frac{1}{4}\gamma^2-\frac{3}{8}m^2),$
γ	$=$	$\gamma.$

And moreover that, attending only to the terms which are independent of m, we have

$$e=e\,(1+\tfrac{1}{4}\gamma^2-\tfrac{3}{8}e^2\gamma^2+\tfrac{1}{4}\gamma^4),$$
$$\gamma=\gamma\,(1-\tfrac{3}{16}e^2+\tfrac{4}{16}e^2\gamma^2-\tfrac{1}{2}\gamma^4).$$

which are formulae that may be found useful.

468. ADDITION TO SECOND NOTE ON THE LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXVII. *(*1866–1867*), pp.* 267–269.*]*

Writing as in my Second Note, *Monthly Notices*, Vol. XXV., pp. 203–207 (May 1865), [466], for the Moon,

a, the mean distance,
e, the excentricity,
γ, the tangent of the inclination,
l, the mean longitude,
c, the mean anomaly,
g, the mean distance from node,

I obtained by the ordinary method of the variation of the elements, from the constant term of R and the term involving $\cos(2c - 2g)$, the following expressions of the variations,

$$\begin{aligned}
\delta a &= 0,\\
\delta e &= -\tfrac{3}{8}\gamma^2 e \cos 2c - 2g,\\
\delta\gamma &= +\tfrac{3}{8}\gamma e^2 \;\;,,\;\; 2c - 2g,\\
\delta c &= +\tfrac{1}{8}\gamma^2 \sin 2c - 2g,\\
\delta g &= +\tfrac{3}{8}e^2 \;\;,,\;\; 2c - 2g,\\
\delta l &= +\tfrac{1}{16}\gamma^2 e^2 \;\;,,\;\; 2c - 2g,
\end{aligned}$$

viz. if in the elliptic expressions of the radius vector, longitude, and latitude, we apply to a, e, γ, c, g, l, the foregoing increments, we obtain to the fourth order in (e, γ) the portions independent of m in the expressions of the radius vector, latitude, and longitude. I wish to notice that the results, to the very limited extent to which they go, agree with those obtained by M. Delaunay in his "Théorie du Mouvement de la Lune," from his 49th operation, the object of which is to take away the term (63) of R, that is the term involving $\cos(2c - 2g)$. The formulae (see vol. I. p. 788), taken only to the necessary degree of approximation are

$$\begin{array}{rcl}
a & \text{replaced by} & a,\\
e^2 & ,, & e^2 - \tfrac{3}{4}\gamma e^2 \cos 2g,\\
\gamma^2 & ,, & \gamma^2 + \tfrac{3}{4}\gamma^2 \;\;,,\;\; 2g,\\
l & ,, & l - \tfrac{1}{8}\gamma^2 \sin 2g,\\
h + g + l & ,, & h + g + l + \tfrac{3}{8}\gamma^2 e^2 \;\;,,\;\; 2g,\\
h & ,, & h + \tfrac{3}{8}e^2 \;\;,,\;\; 2g,
\end{array}$$

which, observing that

$$\begin{aligned}
\gamma\,(\text{Del.}) &= \tfrac{1}{2}\gamma \text{ (for present purpose)},\\
l &= c,\\
g + h &= g,\\
h + g + l &= l,
\end{aligned}$$

and therefore

$$g = -(c - g),$$

become

$$\begin{array}{rcl}
a & \text{replaced by} & a,\\
e^2 & ,, & e^2 - \tfrac{3}{8}\gamma^2 e^2 \cos 2c - 2g,
\end{array}$$

$$\begin{array}{rcl} \gamma^2 & ,, & \gamma^2+\tfrac{3}{4}\gamma^2e^2\ ,,\ 2c-2g, \\ c & ,, & c+\tfrac{1}{8}\gamma^2\ \sin\,2c-2g, \\ l & ,, & l-\tfrac{1}{16}\gamma^2e^2\ ,,\ 2c-2g, \\ l-g & ,, & l-g-\tfrac{3}{8}e^2\ ,,\ 2c-2g, \end{array}$$

the last of which may be changed into

$$g\quad ,,\quad g+\tfrac{3}{8}e^2\ ,,\ 2c-2g,$$

or if the new values of a, e, γ, c, g, l, are called $a+\delta a,\ e+\delta e,\ \gamma+\delta\gamma,\ c+\delta c,\ g+\delta g,\ l+\delta l$, then the increments δa, δe, $\delta\gamma$, δc, δg, δl, have the values given above. The process of my Second Note, taken as a first transformation, has in fact the object of removing the term $\cos(2c-2g)$, and to the degree of approximation regarded, the result is not affected by the previous transformations, or by the substitution, t. II. p. 800, introducing for a, e, γ, their standard elliptic values.

469. ON AN EXPRESSION FOR THE ANGULAR DISTANCE OF TWO PLANETS.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXVII. *(*1866–1867*), pp.* 312–315.*]*

If for the planet m, referred to any fixed plane and origin of longitudes, we have

v, the longitude in orbit,
θ, the longitude of node,
ϕ, the inclination,

and similarly for the planet m' referred to the same fixed plane and origin of longitudes, if the corresponding quantities are v', θ', ϕ'; then the angular distance of the two planets will of course be expressible in terms of v, θ, ϕ, v', θ', ϕ', but I am not aware that the actual expression has been given. To obtain it in the most simple manner, I write further for the planet m:

$\theta+x$, the reduced longitude,
y, the latitude,
z, the distance from node,

so that z $(=v-\theta)$, x, y, are the hypothenuse, base, and perpendicular of a right-angled spherical triangle, the base angle of which is $=\phi$. And similarly $\theta'+x'$, y', z', have the like significations for the planet m'. I write also r, r', for the distances of the two planets respectively.

This being so, the rectangular coordinates of the planet m are

$$\begin{aligned} & r\cos y\,\cos(\theta+x), \\ & r\cos y\,\sin(\theta+x), \\ & r\sin y. \end{aligned}$$

But observing that from the right-angled triangle we have

$$\begin{aligned} \cos z &= \cos x\,\cos y, \\ \cos\phi &= \tan x\,\cot z, \\ \sin x &= \cot\phi\,\tan y, \\ \sin y &= \sin\phi\,\sin z, \end{aligned}$$

and therefore also

$$\sin x\cos y\ =\ \cot\phi\,\sin y,\ =\ \cos\phi\,\sin z,$$

the expressions for the coordinates become

$$\begin{aligned}
&r\,(\cos z\,\cos\theta - \sin\phi\,\sin\theta\,\cos\phi),\\
&r\,(\cos z\,\sin\theta + \sin z\,\cos\theta\,\cos\phi),\\
&r\,(\sin z\,\sin\phi).
\end{aligned}$$

Forming the analogous expressions for the coordinates of m', then if H be the angular distance of the two planets, we deduce at once the expression for $\cos H$, viz. this is

$$\begin{aligned}
rr'\cos H = {}&(\cos z\,\cos\theta - \sin z\,\sin\theta\,\cos\phi)\,(\cos z'\,\cos\theta' - \sin z'\,\sin\theta'\,\cos\phi')\\
&+ (\cos z\,\sin\theta + \sin z\,\cos\theta\,\cos\phi)\,(\cos z'\,\sin\theta' + \sin z'\,\cos\theta'\,\cos\phi')\\
&+ (\sin z\,\sin\phi)\,(\sin z'\,\sin\phi'),
\end{aligned}$$

or, multiplying out, this is

$$\begin{aligned}
rr'\cos H = {}&\cos z\,\cos z'\,\cos(\theta-\theta')\\
&+ \cos z\,\sin z'\,\sin(\theta-\theta')\,\cos\phi'\\
&- \sin z\,\cos z'\,\sin(\theta-\theta')\,\cos\phi\\
&+ \sin z\,\sin z'\,(\cos(\theta-\theta')\,\cos\phi\,\cos\phi' + \sin\phi\,\sin\phi'),
\end{aligned}$$

say this is

$$rr'\cos H \;=\; A\,\cos z\,\cos z' + B\,\cos z\,\sin z' + C\,\sin z\,\cos z' + D\,\sin z\,\sin z',$$

viz. it is

$$\begin{aligned}
&\cos(z-z')\cdot\tfrac{1}{2}A + \tfrac{1}{2}D\\
&\quad + \sin(z-z')\cdot\tfrac{1}{2}B - \tfrac{1}{2}C\\
&\quad + \cos(z+z')\cdot\tfrac{1}{2}A - \tfrac{1}{2}D \qquad z+z' = v+v'-\theta-\theta',\\
&\quad + \sin(z+z')\cdot\tfrac{1}{2}B + \tfrac{1}{2}C.
\end{aligned}$$

But we have

$$z-z' \;=\; v-v'-\theta+\theta',$$

[Paper 469 continues on subsequent pages, beyond the present chunk.]

[Paper 476, continued: *On the Determination of the Orbit of a Planet from Three Observations*, Articles 64–97.]

[p. 429]

we obtain the six coordinates of the three rays respectively

$$\begin{aligned}(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1) &= (\,0,\ \sqrt{3},\ -1,\ \ \ 0,\ 1,\ \sqrt{3}),\\ (a_2,\, b_2,\, c_2,\, f_2,\, g_2,\, h_2) &= (\,3,\ \sqrt{3},\ \ 2,\ -\sqrt{3},\ 1,\ -2\sqrt{3}),\\ (a_3,\, b_3,\, c_3,\, f_3,\, g_3,\, h_3) &= (-3,\ \sqrt{3},\ \ 2,\ -\sqrt{3},\ 1,\ -2\sqrt{3}),\end{aligned}$$

whence the intersections with the orbit-plane are given by

$$\begin{aligned}x_1' \,:\, y_1' \,:\, 1 &= \beta'\sqrt{3} - \gamma' \ :\ -\beta'\sqrt{3} + \gamma \ :\ \beta'' + \gamma''\sqrt{3},\\ x_2' \,:\, y_2' \,:\, 1 &= 3\alpha' + \beta'\sqrt{3} + 2\gamma' \ :\ -3\alpha - \beta\sqrt{3} - 2\gamma \ :\ \alpha''\sqrt{3} + \beta'' - 2\sqrt{3}\gamma'',\\ x_3' \,:\, y_3' \,:\, 1 &= -3\alpha' + \beta'\sqrt{3} + 2\gamma' \ :\ 3\alpha - \beta\sqrt{3} - 2\gamma \ :\ -\alpha''\sqrt{3} + \beta'' - 2\sqrt{3}\gamma'',\end{aligned}$$

where if (as before) the position of the orbit-plane be determined by means of the longitude b and colatitude c of the orbit-pole, we have

$$\begin{aligned}\alpha\ ,\ \beta\ ,\ \gamma &= \sin b\ ,\ -\cos b\ ,\ 0\ ,\\ \alpha',\ \beta',\ \gamma' &= \cos b\cos c,\ \sin b\cos c,\ -\sin c,\\ \alpha'',\ \beta'',\ \gamma'' &= \cos b\sin c,\ \sin b\sin c,\ \cos c,\end{aligned}$$

and the passage from the coordinates x', y', to x, y, is given by

$$\begin{aligned}x' &= x\sin b - y\cos b,\\ y' &= +x\cos b + y\sin b,\end{aligned}$$

or conversely

$$\begin{aligned}x &= x'\sin b + y'\cos b,\\ y &= -x'\cos b + y'\sin b.\end{aligned}$$

65. To develope the results, I consider the orbit-pole as passing through certain series of positions. The locus may be a meridian circle: by reason of the symmetry of the system, the results are not altered by a change of 120° in the longitude of the meridian; so that, by considering the two meridians 0°–180° and 90°–270°, we, in fact, consider twelve half meridians at the intervals of 30°. An illustration is afforded by Plate I.; the orbit-pole describes successively the meridians 0°, 30°, 60°, 90°, and the line 1, by its intersection with the orbit-plane, traces out on this plane a series of hyperbolas shown in the figure; the hyperbola for the meridian 90° is a right line, but (except for the position where the orbit-plane passes through the line 1) the locus is a determinate point on this line. Pianogram No. 1 (Plate II.) refers to the meridian 90°–270°, and Pianogram No. 2 (Plate III.) to the meridian 0°–180°. Next, if the orbit-pole be at one of the points A, that is, if the orbit-plane pass through a ray though the position of the orbit-pole be here determinate, yet as there is a series of orbits, this also will give rise to a pianogram: I call it Pianogram No. 3. The orbit-pole may pass along a separator circle (viz. the orbit-plane be parallel to a ray), this is Pianogram No. 4. And, lastly, the orbit-pole may pass along the ecliptic (or the orbit-plane may pass through the axis SZ), I call this Pianogram No. 5. But the last three planograms are not considered in the like detail as the first two, and I have not, in regard to them, tabulated the results, nor given any Plates.

[p. 430]

Article Nos. 66 *to* 82. *Planogram No.* 1, *the Meridian* 90°–270° *(see Plate II.).*

66. Supposing that the orbit-plane rotates about the axis $S1$ (fig. 6, see No. 64) in the plane of the ecliptic, the orbit-pole will describe the meridian 90°–270°, the position of the orbit-pole being $b = 90°$, $c = 0°$ to 90°, or else $b = 270°$, $c = 0°$ to 90°. But the same analytical formula extends to the two half meridians, viz., we may take $b = 90°$, and extend c over 180°, in the final results making c an arc between 0° and 90°, and $b = 90°$, or $= 270°$, as the case requires.

67. Assuming then $b = 90^\circ$, we have

$$\begin{array}{lll} \alpha,\ \beta,\ \gamma = 1, & 0, & 0, \\ \alpha',\ \beta',\ \gamma' = 0, & -\cos c, & \sin c, \\ \alpha'',\ \beta'',\ \gamma'' = 0, & \sin c, & \cos c, \end{array}$$

and, moreover, $x' = x$, $y' = y$: so that instead of $(x'_1,\ y'_1)$, &c., we may write at once $(x_1,\ y_1)$, &c. The formulae become

$$\begin{aligned} x_1 : y_1 : 1 &= \sqrt{3}\cos c + \sin c \ :\ 0 \ :\ \sin c + \sqrt{3}\cos c, \\ x_2 : y_2 : 1 &= -\sqrt{3}\cos c - 2\sin c \ :\ -3 \ :\ \sin c - 2\sqrt{3}\cos c, \\ x_3 : y_3 : 1 &= \ \ \sqrt{3}\cos c - 2\sin c \ :\ -3 \ :\ \sin c - 2\sqrt{3}\cos c, \end{aligned}$$

that is

$$x_1 = 1, \qquad y_1 = 0.$$

(viz. the orbit-plane, as is evident, meets the ray 1 in a fixed point, its intersection with the plane of xy)

$$x_2 = \frac{-\sqrt{3}\cos c - 2\sin c}{\sin c - 2\sqrt{3}\cos c}, \qquad y_2 = \frac{-3}{\sin c - 2\sqrt{3}\cos c},$$

$$x_3 = \frac{\sqrt{3}\cos c - 2\sin c}{\sin c - 2\sqrt{3}\cos c} = -x_2, \qquad y_3 = \frac{-3}{\sin c - 2\sqrt{3}\cos c} = -y_2.$$

[Note: in the printed text the formula for x_3 is given with an obvious typographical sign error; the corrected expression makes $x_3 = -x_2$ and $y_3 = y_2$ symmetric, but the printed page gives $y_3 = -y_2$ which we have preserved verbatim.]

and writing

$$\frac{2\sqrt{3}}{\sqrt{13}}\cos\omega = \frac{1}{\sqrt{13}}, \qquad \frac{1}{\sqrt{13}}\sin\omega = \frac{1}{2\sqrt{3}}\tan\omega,$$

(whence $\omega = 16^\circ\,6'$) we find

$$\begin{aligned} x_2 &= -\tfrac{1}{2} + \tfrac{3\sqrt{3}}{\sqrt{13}}\tan(c+\omega), \\ y_2 &= +\tfrac{3}{\sqrt{13}}\sec(c+\omega), \end{aligned}$$

and we thence have for the hyperbola, the locus of $(x_2,\ y_2)$ and $(x_3,\ y_3)$

$$\left(x + \tfrac{1}{2}\right)^2 = \tfrac{3}{12}\left(y^2 - \tfrac{3}{4}\right),$$

[p. 431]

viz. the points $(x_2,\ y_2)$ and $(x_3,\ y_3)$ are situate on the hyperbola, symmetrically on opposite sides of the axis of x. For $c = 0$, we have $x_2 = -\frac{1}{2}$, $y_2 = \frac{1}{2}\sqrt{3}$, $(x_2^2 + y_2^2 = 1)$, and the hyperbola at this point touches the circle $x^2 + y^2 = 1$; and similarly for x_3, y_3. The inclination of the asymptotes to the axis of y is given by $\tan\eta = \sqrt{\frac{4}{9}}$, $\eta = 22^\circ\,56'$.

68. The orbits are conics, focus S and vertex 1. It will be convenient to consider c as passing from 0° to $90^\circ - \omega$, and from 0° to $-(90^\circ + \omega)$; that is, from 0° to $73^\circ\,54' - \epsilon$, and from 0° to $-106^\circ\,6' + \epsilon$, if ϵ be indefinitely small: the point 2 will thus traverse the upper branch (alone shown in the Plate) of the guide-hyperbola, viz., for $c = 0^\circ$ it will be at the point of contact with the circle; for $c = 73^\circ\,54' - \epsilon$ it will be at ∞, and for $c = -106^\circ\,6' + \epsilon$ at ∞'. For $c = 0^\circ$ the orbit is the circle; as c increases positively, it becomes an ellipse of increasing eccentricity and major axis, until for a certain value ($c = 46^\circ\,48'$ as will appear) it becomes a parabola; it then becomes a hyperbola (concave branch); for $c = 52^\circ\,45'$ it becomes the hyperbola Σ' subsequently referred to; and for $c = 60^\circ$ (the point 2 being then on the line shown in the figure) the orbit becomes this right line. As c continues to increase, the orbit becomes a hyperbola (convex branch); and ultimately for $c = 73^\circ\,54' - \epsilon$, the point 2 goes to ∞, and the orbit becomes a hyperbola (convex) Σ, having an asymptote parallel to that of the guide-hyperbola: the inclination to the axis of x being thus $90^\circ - 22^\circ\,56' = 67^\circ\,4'$.

69. Next as c increases negatively, the point 2 moves from the point of contact in the other direction to ∞': for $c = 0^\circ$ the orbit is of course the circle, and as c increases negatively the orbits are at first the very same series of orbits as those belonging to the positive values[1], viz., they are first ellipses, of increasing eccentricity and major axis; then for $c = -92^\circ\, 54'$ the orbit is the parabola; the orbits are then hyperbolas (concave), and finally for $c = -106^\circ\, 6' + \epsilon$, when 2 is at ∞', the orbit is a hyperbola Σ', the asymptote of which is parallel to that of the guide-hyperbola, viz., the inclination to the axis of x is $67^\circ\, 4'$.

70. It will be observed that the orbits from the circle to the hyperbola Σ' each intersect the guide-hyperbola (that is, the branch shown in the figure) in two points, the one corresponding to a positive, the other to a negative value of c; in the positive series, the remaining orbits from the hyperbola Σ', through the right line to the convex hyperbola Σ, each intersect the guide-hyperbola (same branch) in a single point only, for which c is positive.

71. There is, in the passage of the orbit-pole from $c = -106^\circ\, 6' + \epsilon$ to $c = 73^\circ\, 54' - \epsilon$, say at $c = 73^\circ\, 54'$, a discontinuity of orbit, viz., an abrupt change from the concave hyperbola Σ' to the concave hyperbola Σ; observe that the direction of the asymptotes being the same in each, the eccentricity e has the same value.

[p. 432]

The point in question ($b = 90^\circ$, $c = 73^\circ\, 54'$) is one of the points B of the spherogram, and the hyperbolas Σ, Σ' are two of the four orbits belonging to this point. And, by what precedes, it appears that as the orbit-pole passes through this point along a meridian downwards to the ecliptic the change is from a concave to a convex orbit.

72. On account of the symmetry in regard to the axis of x, the equation of the orbit will be of the form $r = \frac{Ax + B}{1}$; viz., the equation is at once found to be

$$r = \frac{x_2 - 1}{x_2 - 1}\,(x - 1).$$

73. The eccentricity is the coefficient A taken positively ($e = \pm A$): it is in the present case proper to attend to the value of the coefficient itself,

$$A = \frac{r_2 - 1}{x_2 - 1},$$

the sign of A will then indicate the position of the centre of the orbit, viz., according as A is positive or negative the centre will be on the negative or the positive side of the focus S. To investigate the variation of A, we may express it as a function of $\tan c$, $= \lambda$ suppose. We have

$$x_2 = \frac{\sqrt{3} - 2\lambda}{\lambda - 2\sqrt{3}}, \qquad y_2 = \frac{-3}{\lambda - 2\sqrt{3}},$$

and thence

$$r_2 = \frac{1}{\lambda - 2\sqrt{3}}\,R_2, \qquad R_2 = \pm\sqrt{12 - 4\sqrt{3}\lambda + 13\lambda^2},$$

viz., r_2 must be positive, that is, R_2 is positive or negative according to the sign of $\lambda - 2\sqrt{3}$; negative if $\lambda < 2\sqrt{3}$ or $c < 73^\circ\, 54'$, positive if $\lambda > 2\sqrt{3}$ or $c > 73^\circ\, 54'$. And we have then

$$A = \frac{\lambda - 2\sqrt{3} - R_2}{3(\lambda - \sqrt{3})}.$$

But a more convenient formula is obtained by writing

$$\theta = -\cot c + \frac{1}{2\sqrt{3}}, \qquad a = \frac{1}{2\sqrt{3}} - 1,$$

---

[1] Of course, as corresponding to different values of c, they are not the same orbits in space, but they are only the same curves in the planogram.

we then have

$$\sqrt{1+\theta^2} = \theta r_2,$$

which determines the sign of the radical, viz., this must have the same sign as θ; and then for the coefficient

$$A = \frac{2}{3(\theta+a)}\big(-\sqrt{1+\theta^2}+\theta\big).$$

[p. 433]

74. For c a small arc $= \epsilon$, θ is large and negative, and $\sqrt{1+\theta^2}$, having the same sign as θ, is $= \theta$ nearly; we have therefore

$$A = \frac{2}{3\theta}\cdot\frac{-1}{2\theta} = \frac{-1}{3\theta^2} \quad \text{approximately.}$$

For c nearly $= 60°$, say $c = 60° \pm \epsilon$,

$$\cot c = \cot 60° \pm \epsilon \csc^2 60° = \tfrac{1}{\sqrt{3}} \pm \frac{4\epsilon}{3},$$

$$\theta = -\frac{1}{2\sqrt{3}} \pm \frac{4\epsilon}{3}, \qquad \theta + a = \pm\frac{4\epsilon}{3}, \qquad \sqrt{1+\theta^2} = -\frac{\sqrt{13}}{2\sqrt{3}},$$

and thence

$$A = \frac{-1+\sqrt{13}}{2\sqrt{3}} \pm \frac{4\epsilon}{3} = \pm\frac{\sqrt{3}}{8}\cdot\frac{-1+\sqrt{13}}{\epsilon},$$

viz., this is $-\infty$ for $c = 60° - \epsilon$, and $+\infty$ for $c = 60° + \epsilon$.

For c nearly $= 90° - \omega$, say first $c = 73°\,54' - \epsilon$, we have

$$\cot c = \frac{1}{2\sqrt{3}} + \tfrac{13}{12}\epsilon, \qquad \theta = -\tfrac{13}{12}\epsilon, \qquad \theta + a = -\frac{1}{2\sqrt{3}}, \qquad \sqrt{1+\theta^2} = -1,$$

whence

$$A = \frac{4}{\sqrt{3}} = 2{\cdot}30940;$$

but if $c = 73°\,54' + \epsilon$, then $\theta = \tfrac{13}{12}\epsilon$, $\theta + a = -\dfrac{1}{2\sqrt{3}}$, $\sqrt{1+\theta^2} = +1$, and

$$A = -\frac{4}{\sqrt{3}} = -2{\cdot}30940,$$

viz., there is an abrupt change from $A = +\dfrac{4}{\sqrt{3}}$ to $A = -\dfrac{4}{\sqrt{3}}$; corresponding to the discontinuity of orbit already referred to. We may diminish c by 180°, and consider the last-mentioned value, $A = -\dfrac{4}{\sqrt{3}}$, as belonging to $c = 90° - \omega + \epsilon = -(106°\,6' - \epsilon)$.

75. Consider next that c passes from 0° to $-(106°\,6' - \epsilon)$. First if c is a small negative quantity $c = -\epsilon$, θ is large and positive, and $\sqrt{1+\theta^2}$ having the same sign as θ (positive) is $= \theta + \dfrac{1}{2\theta}$ nearly, we have therefore $A = \dfrac{2}{3\theta}\cdot\dfrac{-1}{2\theta} = \dfrac{-1}{3\theta^2}$ (same as for $c = +\epsilon$). And it is easy to see that as c increases negatively, A is always increasing negatively, its value for $c = -90°$ being $A = \dfrac{1-\sqrt{13}}{3} = -{\cdot}8685$, and for $c = -106°\,6' + \epsilon$ A being $= -2{\cdot}30940$ as above. We have a diagram of A (see next page).

[p. 434]

76. It thus appears that from $A=-\dfrac{4}{\sqrt{3}}$ to $A=-\dfrac{4}{\sqrt{3}}$, there are always to any given value of A two values of c, or positions of the orbit-pole. In particular if A be $=-1$, the curve will be a parabola; the values of c lying between 0°, 60° and between 73° 54′, 90° respectively.

[Figure 7: schematic graph of A as a function of c, showing the asymptotes at $c=60^\circ$ and at $c=-106^\circ\,6'$, the level $A=-1$ marked along the horizontal axis, with the abscissa scale running from -100° on the left through 0° at centre to 90° on the right.]

To find them, writing $A=-1$, we have

$$-3\theta-3a=2\theta-2\sqrt{1+\theta^2},\quad \text{that is,}\quad 5\theta+3a=2\sqrt{1+\theta^2},$$

or

$$21\theta^2+30a\theta+9a^2-4=0,$$

that is, substituting for a its value $=\dfrac{1}{2\sqrt{3}}$,

$$21\theta^2+5\sqrt{3}\theta-\tfrac{13}{4}=0,\qquad (14\theta\sqrt{3}+5)^2=116,$$

or

$$\theta=\frac{-5\pm\sqrt{116}}{14\sqrt{3}},$$

giving

$$\theta=-{\cdot}23797,\quad\text{or}\quad \theta=-{\cdot}65034,$$

$$\cot c=+{\cdot}93902,\quad\text{or}\quad \cot c=+{\cdot}05071,$$

$$c=46^\circ\,48',\quad\text{or}\quad c=87^\circ\,6'.$$

77. It has been seen that $c=73^\circ\,54'+\epsilon$ gives $A=-\dfrac{4}{\sqrt{3}}=-2{\cdot}30940$; there will be between 0° and 60° another value of c, giving for A this same value; to find this value write $A=-\dfrac{4}{\sqrt{3}}$, then we have

$$2\theta\Big(1+\frac{2}{\sqrt{3}}\theta\Big)-4=\frac{4}{\sqrt{3}}\Big(\sqrt{1+\theta^2}-\theta\Big)\cdot\sqrt{3},$$

[p. 435]

that is

$$(1+2\sqrt{3})\theta+1=\sqrt{1+\theta^2},$$

or

$$\theta^2(12+4\sqrt{3})+(2+4\sqrt{3})\theta=0,$$

satisfied as it should be by $\theta=0$, and also by

$$\theta=-\frac{1}{2(3+\sqrt{3})},$$

giving

$$\cot c={\cdot}76038\quad\text{or}\quad c=52^\circ\,45'.$$

78. Representing the equation of the orbit by

$$r=Ax\pm a(1-A^2),$$

we have for the point 1,

$$1=A\pm a(1-A^2),$$

that is

$$a=\frac{\pm1}{1+A},$$

where the sign is to be taken so that a shall be positive.

79. With a view to the calculation of the times of passage, I calculate a series of values of x_2, y_2, r_2, A, a, for values of c at the intervals of 5° and for a few intermediate values; we have x_3, y_3, $r_3 = x_2$, y_2, r_2, so that these are known; so long as the orbit is an ellipse, the time of passage between the points 2 and 3, say T_{23}, may be calculated by Lambert's equation, the length of the chord $= y_2 - y_3$, $= 2y_2$ being known without any fresh calculation. And then the times T_{12} and T_{31} being equal, and the sum $T_{12} + T_{23} + T_{31}$ being equal to the whole periodic time (reckoned as $= 3a^{\frac{3}{2}}$), the times T_{12} and T_{31} are also known. But when the orbit is a concave hyperbola there is no time T_{23}, and the other two times T_{12}, $= T_{31}$ must be calculated. For the reason referred to (ante, No. 39) I did not use Lambert's equation, and it was less necessary to do so, by reason that, the transverse axis coinciding with the axis of x, the other formula could be employed without difficulty.

80. The formulae for x_2, y_2 adapted to logarithmic calculation are

$$\log(x_2 + \cdot 615.39) = 11\cdot 60174 + \log\tan(c + 16^\circ\, 6'),$$
$$\log y_2 = n\cdot 92015 + \log\sec(c + 16^\circ\, 6'),$$

where y_2 is always positive, but the sign of x_2 must be attended to. The values of r_2 and its inclination ϕ_2 to the axis of x are then to be calculated from

$$\frac{y_2}{x_2} = \tan\phi_2; \quad r_2 = x_2 \sec\phi_2 \text{ or } = y_2 \csc\phi_2,$$

(viz. for r_2 it is proper to use the first or the second value, according as x_2 is greater or less than y_2).

[p. 436]

We have then $e\,(= \pm A)$ and a from the foregoing formulae

$$A = \frac{r_2 - 1}{x_2 - 1}, \qquad a = \frac{\pm 1}{1 + A},$$

where a, e are each of them positive.

And then for the Times

$$T_{31} = T_{12} = \tfrac{3}{2\pi}\, a^{\frac{3}{2}} \left(\chi' - \chi - \sin\chi' + \sin\chi\right); \qquad (\log \tfrac{3}{2\pi} = \bar{1}\cdot 67894)$$

where

$$a\cos\chi = a - r_2 - y_2,$$
$$a\cos\chi' = a - r_2 + y_2,$$

and attention is necessary in order to the selection of the proper values of the angles χ, χ'.

And finally

$$T_{23} = T_{12} - \tfrac{3}{2\pi}\,(3a^{\frac{3}{2}} - T_{12}).$$

81. I subjoin a specimen; the characteristics of the logarithms are (as in the actual calculations) omitted.

[Specimen calculation, $b = 90°$, $c = 20°$. The page presents a column-arithmetic worked example. The numerical content from the scan is reproduced below in alignment.]

$b = 90^\circ \quad c = 20^\circ$

$c + 16^\circ\, 6' = 36^\circ\, 6'$

$\log\sec$ ·09259	$\log\tan$ ·86285
·92015	·60174
·01274	·46459
$y_2 = 1{\cdot}0297$	·61539
	·01274
	·29147
	·51044
	$\therefore x_2 = -{\cdot}32{\cdot}392$
	$\log =$ ·51044
	·02046
	·01274
·50230	·03320
$\phi_2 = 72^\circ\, 33'$	$r_2 = 1{\cdot}0794$
·0794 $\log = \bar{1}{\cdot}89982$	·94003 $\log = \bar{1}{\cdot}97314$
1·3239 $\log =$ ·12185	comp = ·02686
	·77797

[p. 437]

$A = -{\cdot}059975$	$a = 1{\cdot}0638$
	1·0638
	1·0794

$A - r_2 = -{\cdot}0156$

$y_2 = +1{\cdot}0297$

$1{\cdot}0141 = a\cos\chi'$

$-1{\cdot}0453 = a\cos\chi$

·00608	·01924	
·02686	·02686	
·97922	·99238	
$\chi' = 17^\circ\, 35'$	χ (= Supp. $10^\circ\, 42'$) $= 169^\circ\, 18'$	$\chi - \chi' = 151^\circ\, 43'$

151°	2·63544	·02686
43′	·01250	·01343
$-\sin\chi$	$-$ ·18566	·47712
$\sin\chi'$	·30209	·51741
	2·95003	$3a^{\frac{3}{2}} = 3{\cdot}2916$
	·18566	1·4482
	2·76437 $\log =$ ·44160	1·8434
	·02686	$T_{12} = T_{31} =$ ·9217
	·01343	
	·67894	
$T_{23} = 1{\cdot}4482$	·16083	

82. For the Time in a hyperbola, we have

$$T_{31} = T_{12} = \frac{3}{2\pi}\, a^{\frac{3}{2}} \big\{ e\tan u - h{\cdot}l.\tan(45^\circ + \tfrac{1}{2}u) \big\},$$

where

$$\tan u = \frac{y_2}{a\sqrt{e^2-1}}.$$

[p. 438]

Taking as a specimen the case $c = 75^\circ$, we have here

$$\begin{array}{lll} a = \cdot 9004 & e = 2\cdot 1106 & y_2 = 43\cdot 341 \\ \log = \bar{1}\cdot 95444 & \log = \cdot 32441 & \log = 1\cdot 63690 \end{array}$$

$$\begin{aligned} a(e^2-1) &= 3\cdot 1106 \\ \log &= \cdot 49284 \end{aligned}$$

and then the calculation is

$$\begin{array}{ll}
\log a = \bar{1}\cdot 95444 & \\
\log a(e^2-1) = \cdot 49284 & \\
\overline{}\ \cdot 44728 & \\
\log a\sqrt{e^2-1} = \cdot 22364 & \\
\log y_2 = 1\cdot 63690 & \\
\log \tan u = 1\cdot 41326 & u = 87^\circ\, 47' \\
 & h\cdot l \tan(45^\circ + \tfrac{1}{2}u) = 3\cdot 95140 \\
\cdot 32441 & e \tan u = 54\cdot 660 \\
\overline{}\ \cdot 73767 & \quad 3\cdot 951 \\
 & \quad 50\cdot 709 \\
 & \log = \cdot 70508 \\
 & \quad \cdot 95444 \\
 & \quad \cdot 47722 \\
 & \quad \cdot 67894 \\
 & \\
 & \quad 3\cdot 1568 \\
 & T_{12} = T_{31} = 20\cdot 686
\end{array}$$

83. In the case of the parabola $p = 1$, and the expression for the Times is

$$T_{12} = T_{31} = \frac{1}{6\sqrt{\pi}}\left\{(\rho + \rho' + \gamma)^{\frac{3}{2}} - (\rho + \gamma)^{\frac{3}{2}}\right\},$$

where for

$$\begin{aligned} c &= 46^\circ\, 48' \\ c &= 87^\circ\, 6' \end{aligned}$$

we have

$$\begin{aligned} T_{12} = T_{31} &= \cdot 787, \\ T_{12} = T_{31} &= 2\cdot 588. \end{aligned}$$

[p. 439]

Planogram No. 1, first part, $b = 90^\circ$.

	c	x_2	y_2	ϕ_2	r_2	A	a	T_{12}	T_{23}	T_{31}
Circle	0°	−·500	+·866	60°	1·000	0	1·000	1·000	1·000	1·000
	5	·461	·892	62° 39′	1·004	−·003	1·003			
	10	·420	·927	65 38	1·017	−·012	1·012	·960	1·135	·960
	15	·374	·972	68 56	1·041	−·030	1·031			
Ellipses	20	·324	1·030	72 33	1·079	−·060	1·064	·922	1·448	·922
	25	·267	1·104	76 26	1·136	−·107	1·120	·887	2·275	·887
	30	·200	1·200	80 32	1·216	−·180	1·220			
	35	·120	1·325	84 49	1·330	−·295	1·418			
	40	−·021	1·492	90 48	1·492	−·482	1·931	·838	6·371	·838
	40° 54′	·000	1·515	90	1·515	−·515	2·061			
	45	+·109	1·722	93 37	1·725	−·814	5·362			
Parab.	46° 48′	·166	1·826	95 11	1·834	−1·000	∞	·787	∞	·787
	50	·287	2·054	97 57	2·074	−1·505	1·981	·750	∼	·750
Hyperbs. {	52° 45′	·418	2·306	100 16	2·344	−2·309	·764	·628	∼	·628
	55	·552	2·569	102 8	2·627	−3·632	·380			
Line	60	1·000	3·464	106 6	3·605	∓∞	·000	·000	∼	·000
	65	1·937	5·378	109 48	5·716	+5·032	·166		Convex orbit.	
Convex	70	5·248	12·233	113 13	13·311	+2·898	·257			
	73° 54′ − ε	+∞	∞	115 39	∞	+2·309	·302			
	73° 54′ + ε	−∞	∞	64 21	∞	−2·309	·764	∞	∼	∞
	76	−21·432	43·341	63 42	48·346	−2·111	·900	20·68	∼	20·68
Hyperbs.	80	−4·356	7·830	60 55	8·960	−1·486	2·056	3·856	∼	3·856
	85	−2·653	4·322	58 27	5·072	−1·115	8·718	2·912	∼	2·912
Parab.	87° 6′	−2·320	3·644	57 31	4·320	−1·000	∞	2·588	∞	2·588
Ellipse	90	−2·000	3·000	56 18	3·606	−·869	7·622	2·255	58·62	2·255

The mark ∼ in the T_{23} column shows that there is no Time T_{23}.

[p. 440]

Planogram No. 1, *second part,* $b = 270°$.

	c	x_2	y_2	ϕ_2	r_2	A	a	T_{12}	T_{23}	T_{31}
Circ.	0°	−·500	+·866	60°	1·000	0	1·000	1·000	1·000	1·000
	5	·537	·848	57 39	1·003	−·002	1·002			
	10	·573	·837	55 37	1·014	−·009	1·009	1·044	·951	1·044
	15	·608	·832	53 52	1·030	−·019	1·019			
	20	·643	·834	52 23	1·053	−·032	1·033	1·091	·969	1·091
	25	·678	·842	51 10	1·081	−·048	1·051			
	30	·714	·857	50 11	1·116	−·068	1·073	1·145	1·043	1·145
All Ellipses	35	·752	·879	49 27	1·157	−·090	1·098			
	40	·793	·910	48 51	1·207	−·115	1·130	1·207	1·192	1·207
	45	·836	·950	48 40	1·266	−·145	1·169			
	50	·884	1·002	48 36	1·336	−·179	1·217	1·283	1·464	1·283
	55	·938	1·069	48 45	1·442	−·218	1·278			
	60	1·000	1·154	49 7	1·527	−·264	1·358	1·377	1·983	1·377
	65	1·074	1·266	49 42	1·660	−·318	1·466			
	70	1·164	1·412	50 31	1·830	−·383	1·622	1·506	3·036	1·506
	75	1·280	1·611	51 35	2·056	−·464	1·864			
	80	1·431	1·891	52 53	2·372	−·564	2·295	1·771	6·888	1·771
	85	1·651	2·311	54 28	2·840	−·694	3·269			
	90	−2·000	+3·000	56 18	3·606	−·869	7·622	2·255	58·62	2·255

[p. 441]

Article Nos. 84 *to* 94. *Planogram No.* 2, *the Meridian* 0°–180° *(see Plate III.).*

84. The orbit-plane here rotates about an axis in the plane of the ecliptic at right angles to $S1$ (Fig. 6). The entire meridian is given by $b = 0°$, $c = 0°$ to $90°$, and $b = 180°$, $c = 0°$ to $90°$, but it is sufficient to consider one of these half meridians, say the latter of them, as the series of values is the same for each of them, with only an interchange of the points 2, 3. I write therefore, $b = 180°$, so that we have

$$\begin{gathered}\alpha,\ \beta,\ \gamma = 0,\ 1,\ 0,\\ \alpha',\ \beta',\ \gamma' = -\cos c,\ 0,\ -\sin c,\\ \alpha'',\ \beta'',\ \gamma'' = -\sin c,\ 0,\ \cos c,\end{gathered}$$

consequently

$$\begin{aligned}x_1' &: y_1' : 1 = \sin c \ :\ -\sqrt{3}\ :\ \sqrt{3}\cos c,\\ x_2' &: y_2' : 1 = -1 - 3\cos c - 2\sin c\ :\ -\sqrt{3}\ :\ -\sqrt{3}\sin c - 2\sqrt{3}\cos c,\\ x_3' &: y_3' : 1 = -1 - 3\cos c - 2\sin c\ :\ -\sqrt{3}\ :\ \sqrt{3}\sin c - 2\sqrt{3}\cos c,\end{aligned}$$

and moreover $x = y'$, $y = -x'$; so that, introducing into the formulae (x_1, y_1), &c., in place of the (x_1', y_1'), &c., we have

$$\begin{aligned}x_1 &= \sec c, \qquad y_1 = -\tfrac{1}{\sqrt{3}}\tan c,\\ x_2 &= \frac{-1}{\sin c - 2\cos c}(\sin c + 2\cos c) + \tfrac{1}{\sqrt{3}}\cdot\frac{1 + 2\sin c - 3\cos c}{\sqrt{3}\sin c - 2\cos c},\\ x_3 &= \frac{-1}{\sin c - 2\cos c}(\sin c + 2\cos c) - \tfrac{1}{\sqrt{3}}\cdot\frac{1 - 2\sin c - 3\cos c}{\sqrt{3}\sin c - 2\cos c},\end{aligned}$$

which, putting

$$\cos\delta = \tfrac{2}{\sqrt{5}},\quad \sin\delta = \tfrac{1}{\sqrt{5}},\quad \tan\delta = \tfrac{1}{2},\quad \delta = 26°\,34',$$

become

$$\begin{aligned}x_1 &= \sec c, & y_1 &= -\tfrac{1}{\sqrt{3}}\tan c,\\ x_2 &= -\tfrac{1}{\sqrt{5}}\sec(c-\delta), & y_2 &= \tfrac{1}{\sqrt{3}}\big\{\tfrac{2}{5} + \tfrac{1}{2}\tan(c-\delta)\big\},\\ x_3 &= -\tfrac{1}{\sqrt{5}}\sec(c+\delta), & y_3 &= \tfrac{1}{\sqrt{3}}\big\{-\tfrac{2}{5} + \tfrac{1}{2}\tan(c+\delta)\big\}\end{aligned}$$

so that the guide-hyperbolas are (angle of asymptotes = 30°):

$$\begin{aligned}&x_1^2 - 3y_1^2 = 1,\\ &\quad x_2^2 = 15y_2^2 - 16\sqrt{3}\,y_2 + 13, \qquad \tan^{-1}\tfrac{1}{\sqrt{15}} = 14°\,28'\\ &\quad x_3^2 = 15y_3^2 + 16\sqrt{3}\,y_3 + 13,\end{aligned}$$

[p. 442]

It is easy to verify that
Hyperbola 2 passes through $x_2 = -\frac{1}{2}$, $y_2 = +\frac{1}{2}\sqrt{3}$, and touches there circle $x^2 + y^2 = 1$,
3 $x_3 = -\frac{1}{2}$, $y_3 = -\frac{1}{2}\sqrt{3}$ „
and we thus have the figure in the Plate.

85. The figure shows the motion of the points 1, 2, 3, along their respective hyperbolas, viz. $c = 0°$ to $90°$, the point 1 moves from contact with the circle, along a half branch to infinity: 2 moves from contact along a small portion of the half branch; 3 moves from contact, along the half branch to infinity for $c = \tan^{-1} 2 = 63°\,26'$, and then reappearing at the opposite infinity, as c increases to 90° describes a portion of the opposite half branch.

86. For $c = 0$, the orbit is the circle; as c increases the orbit becomes elliptic, then parabolic, $c = 51°$, and afterwards hyperbolic (concave); until for $c = 60°$, the three points are on the horizontal line of the figure, and the orbit is this right line; it is to be noticed that the arrangement of the points on these orbits is 1, 2, 3; so that for the parabola, T_{23} is $= \infty$, and for the hyperbolas and right line T_{23} does not exist.

87. For $c < 60^\circ$ until $c = 63^\circ\,26'$ the orbit is a convex hyperbola, the arrangement of the points being still 1, 2, 3: say for $c = 63^\circ\,26' - \epsilon$, the orbit is the convex hyperbola Ω. At $c = 63^\circ\,26'$ there is an abrupt change of orbit; say for $c = 63^\circ\,26' + \epsilon$ the orbit is a concave hyperbola Ω_1; and for $c = 65^\circ\,52'$ the orbit is a parabola; the arrangement of the points on these orbits is 2, 1, 3; so that for the hyperbolas T_{23} does not exist, and T_{21} is $= \infty$ for the parabola. Observe also that for the hyperbola Ω_1, the point 3 is at infinity, or we have $T_{31} = \infty$. As c continues to increase, the orbit becomes an ellipse, the eccentricity having a minimum value = ·628 (about), for $c = 69^\circ$ (about). For $c = 89^\circ\,20'$ the orbit is again a parabola, and then until $c = 90^\circ$ it is a hyperbola; the order of the points on the last-mentioned parabola and hyperbolas being 1, 3, 2; so that for the parabola T_{32} is $= \infty$, and for the hyperbolas T_{32} does not exist. In the hyperbola for $c = 90^\circ$, say the hyperbola Ω', the point 1 is at infinity, or we have $T_{12} = \infty$. The foregoing results, obtained (except as to the numerical values) by consideration of the figure, will be confirmed by means of the calculated values of e.

88. The equation of the orbit may be written

$$\left|\begin{array}{cccc}
\dfrac{r}{\sqrt{3}} & \dfrac{x}{\sqrt{3}} & y & \dfrac{1}{\sqrt{3}} \\
r_1\cos c & ,\quad 1, & \sin c & \cos c \\
+r_2(\sin c + 2\cos c), & -1, & +2\sin c - 3\cos c, & \sin c + 2\cos c \\
 & & & \sin c - 2\cos c \\
-r_3(\sin c + 2\cos c), & 1, & +2\sin c - 3\cos c, &
\end{array}\right| = 0,$$

[p. 443]

or developing, this is

$$\begin{aligned}
&\tfrac{r}{\sqrt{3}}\, 6(\sin^2 c - 3\cos^2 c)\\
&- \tfrac{x}{\sqrt{3}}\Big\{ 4r_1(\sin^2 c - 3\cos^2 c)\cos c\\
&\qquad\qquad - r_2(\sin c + 2\cos c)(\sin^2 c - 3\cos^2 c)\\
&\qquad\qquad + r_3(\sin c - 2\cos c)(\sin^2 c - 3\cos^2 c)\Big\}\\
&+ y\,\Big\{ -2r_1 \sin c\cos c\\
&\qquad\qquad + r_2(-\sin^2 c + \sin c\cos c + 6\cos^2 c)\\
&\qquad\qquad + r_3(\ \sin^2 c + \sin c\cos c - 6\cos^2 c)\Big\}\\
&- \tfrac{1}{\sqrt{3}}\Big\{ r_1 \cdot 6\cos^2 c\\
&\qquad\qquad + 3r_2(\sin^2 c + \sin c\cos c + 2\cos^2 c)\\
&\qquad\qquad + 3r_3(\sin^2 c - \sin c\cos c - 2\cos^2 c)\Big\} = 0\,;
\end{aligned}$$

(observe that the orbit will be a right line if $\sin^2 c - 3\cos^2 c = 0$, that is if $c = 60°$, which is right, since $60°$ is the angular radius of the regulator circle).

89. Putting in the equation $\tan c = \lambda$, and therefore $\cos c = \dfrac{1}{\sqrt{1+\lambda^2}}$, the equation becomes

$$r = \tfrac{1}{6\sqrt{1+\lambda^2}}\big(4r_1 - (\lambda+2)r_2 + (\lambda-2)r_3\big)x + \tfrac{1}{2\sqrt{3}(\lambda^2-3)}\big(2\lambda r_1 + (\lambda-3)(\lambda-2)r_2 - (\lambda+3)(\lambda+2)r_3\big)y + \tfrac{1}{2(\lambda^2-3)}\big(-2r_1 + (\lambda-1)(\lambda+2)r_2 + (\lambda+1)(\lambda-2)r_3\big).$$

We have

$$\begin{gathered}
x_1 = \sqrt{1+\lambda^2},\quad x_2 = \frac{-\sqrt{1+\lambda^2}}{\lambda+2},\quad x_3 = \frac{-\sqrt{1+\lambda^2}}{\lambda-2},\\
y_1 = \tfrac{1}{\sqrt{3}}\,\lambda,\quad y_2 = \tfrac{1}{\sqrt{3}}\cdot\frac{2\lambda+3}{\lambda+2},\quad y_3 = \tfrac{1}{\sqrt{3}}\cdot\frac{2\lambda-3}{\lambda-2},
\end{gathered}$$

and thence, writing for shortness

$$\begin{aligned}
R_1 &= \sqrt{1+\tfrac{4}{3}\lambda^2},\\
R_2 &= \tfrac{1}{\sqrt{3}}\sqrt{7\lambda^2+12\lambda+12},\\
R_3 &= \tfrac{1}{\sqrt{3}}\sqrt{7\lambda^2-12\lambda+12},
\end{aligned}$$

[p. 444]

we have

$$r_1 = R_1, \quad r_2(\lambda+2) = R_2, \quad r_3(\lambda-2) = R_3,$$

where r_1, r_2, r_3 are positive, and the signs of R_1, R_2, R_3 must be determined accordingly; viz., R_1 is always positive, and ($c = 0°$ to $c = 90°$, as here supposed) R_2 is also positive but R_3 has the same sign as $\lambda - 2$; viz., $c = 0°$ to $c = 63° \, 26'$, R_3 is negative; and $c = 63° \, 26'$ to $c = 90°$, R_3 is positive. It is to be observed that this position, $c = \tan^{-1} 2 = 63° \, 26'$, of the pole is the intersection of the meridian $b = 180°$ by a separator circle, and corresponds to an intersection at infinity on the ray 3.

90. Substituting the foregoing values of r_1, r_2, r_3, the equation of the orbit becomes

$$r = \frac{1}{6\sqrt{1+\lambda^2}}(4R_1 - R_2 + R_3)x + \frac{1}{2\sqrt{3}(\lambda^2-3)}\{\lambda(2R_1 + R_2 - R_3) - 3(R_2+R_3)\}\, y + \frac{1}{2(\lambda^2-3)}[\lambda(R_2+R_3) - 2R_1 - R_2 + R_3],$$

where $\lambda = \tan c$; and the equation of the orbit may thence be calculated for any given value of c.

91. The analytical expression for the eccentricity is

$$e = \sqrt{A^2 + B^2},$$

[Figure 8: graph of the eccentricity e versus c, showing the values $e = 0$ at $c = 0°$, the parabolic levels $e = 1$ marked, the vertical asymptote near $c = 63° \, 26'$ between the convex and concave hyperbola branches, the minimum near $c = 69°$ at $e \approx {\cdot}628$, and the second parabolic level near $c = 89° \, 20'$. Abscissa runs $0°$ to $90°$ with 1-unit ordinate marks.]

where, as above,

$$A = \frac{1}{6\sqrt{1+\lambda^2}}\big(4R_1 - R_2 + R_3\big),$$
$$B = \frac{1}{2\sqrt{3}(\lambda^2-3)}\big\{\lambda(2R_1 + R_2 - R_3) - 3(R_2+R_3)\big\};$$

[p. 445]

but this expression is too complicated to allow of an analytical discussion of the series of values of e (such as was given for A, ϵ, in planogram No. 1). The numerical calculation gives the results mentioned ante No. 87, viz., $c = 0$, $e = 0$; $c = 51°$, $e = 1$; $c = 60°$, $e = \infty$; $c = 63° \, 26' - \epsilon$, $e = 4{\cdot}912$; $c = 63° \, 26' + \epsilon$, $e = 1{\cdot}853$; $c = 69°$, $e = {\cdot}628$ (min.); $c = 89° \, 20'$ (viz. $\lambda = 86{\cdot}176$), $e = 1$; $c = 90°$, $e = 1{\cdot}018$; values which are exhibited in the diagram in the preceding page.

92. It may be further remarked, in reference to the formula $r = Ax + By + C$, that for $c = 60°$, that is $\lambda = \sqrt{3}$, we have A and C each infinite, but B finite, and of opposite signs; viz., the equation becomes $r = -{\cdot}2242\, x + \infty(y-1)$, that is $y = 1$, orbit a right line as above.

The abrupt change at $c = 63° \, 26'$, $\lambda = 2$, arises from the change of sign of R_3; viz., $c = 63° \, 26' - \epsilon$, $R_3 = -2{\cdot}309$, but $c = 63° \, 26' + \epsilon$, $R_3 = +2{\cdot}309$; the two orbits are

$$c = 63° \, 26' - \epsilon, \quad r = +{\cdot}234\, x + 4{\cdot}906\, y - 3{\cdot}671, \qquad e = 4{\cdot}912, \quad a = {\cdot}159,$$
$$c = 63° \, 26' + \epsilon, \quad r = {\cdot}578\, x - 1{\cdot}761\, y + 3{\cdot}257, \qquad e = 1{\cdot}853, \quad a = 1{\cdot}338.$$

For $c = 90°$ the equation is

$$r = {\cdot}770\, x + {\cdot}666\, y + 1{\cdot}527$$

and therefore $e = \sqrt{} = 1{\cdot}018$ as above; $a = \frac{9}{2}\sqrt{21} = 41{\cdot}243$.

It is to be added that for c nearly $= 90°$, or λ very large, we have

$$R_1 = \tfrac{2}{\sqrt{3}}\lambda, \quad R_2 = \sqrt{\tfrac{7}{3}}\lambda + 2\sqrt{3}, \quad R_3 = \sqrt{\tfrac{7}{3}}\lambda - 2\sqrt{3},$$

and thence

$$A = \tfrac{4}{3\sqrt{3}} - \tfrac{2}{\sqrt{21}} \cdot \frac{1}{\lambda} = {\cdot}770 - {\cdot}430\frac{1}{\lambda},$$
$$B = \tfrac{2}{\sqrt{7}}\frac{1}{\lambda} - \tfrac{6}{\sqrt{3}}\frac{1}{\lambda^2} = {\cdot}666 - 1{\cdot}890\frac{1}{\lambda^2},$$
$$C = \sqrt{\tfrac{7}{3}} - \tfrac{4}{\sqrt{3}}\frac{1}{\lambda} = 1{\cdot}527 - 1{\cdot}555\frac{1}{\lambda}.$$

It was, in fact, by means of these expressions that the value $\lambda = 86{\cdot}176$ $(c = 89^\circ\,20')$ corresponding to the last-mentioned parabolic orbit was obtained.

[p. 446]

93. For the calculation of the table we have

$$\begin{aligned} \log x_1 &= \overline{10} + \log\sec c, \\ \log y_1 &= \overline{10}{\cdot}76144 + \log\tan c, \\ \log x_2 &= \overline{10}{\cdot}65052 + \log\sec(c - 26^\circ\,34'), \\ \log(y_2 - {\cdot}92376) &= \overline{10}{\cdot}06247 + \log\tan(c - 26^\circ\,34'), \\ \log x_3 &= \overline{10}{\cdot}65052 + \log\sec(c + 26^\circ\,34'), \\ \log(y_3 + {\cdot}92376) &= \overline{10}{\cdot}06247 + \log\sec(c + 26^\circ\,34'), \end{aligned}$$

the values of r_1, r_2, r_3, are then calculated from

$$x_1 = r_1\cos\phi_1, \quad y_1 = r_1\sin\phi_1,$$

or say

$$\frac{y_1}{x_1} = \tan\phi_1, \quad r_1 = x_1\sec\phi_1, \ \&c.$$

and those of the chords γ_{12}, γ_{23}, γ_{31}, from

$$x_1 - x_2 = \gamma_{12}\cos\theta_{12}, \quad y_1 - y_2 = \gamma_{12}\sin\theta_{12},$$

or say

$$\frac{y_1 - y_2}{x_1 - x_2} = \tan\theta_{12}, \quad \gamma_{12} = (x_1 - x_2)\sec\theta_{12}.$$

We have then to find the equation of the orbit $r = Ax + By + C$; this might be done by substituting in the determinant expression the numerical values of x_1, y_1, r_1, x_2, y_2, r_2, and x_3, y_3, r_3, so calculating the result, but I have preferred to employ the formula of No. 90, using only the calculated values of r_1, r_2, r_3; viz. we have

$$r_1 = R_1, \quad r_2(\lambda + 2) = R_2, \quad r_3(\lambda - 2) = R_3.$$

which gives the values of R_1, R_2, R_3. And then we have e, ϖ, a, from the equations

$$A = e\cos\varpi, \quad B = e\sin\varpi, \quad a = \frac{\pm C}{1 - e^2},$$

e and a being each regarded as positive. The times in the elliptic, and parabolic orbits are then calculated from Lambert's equation, as explained in regard to Planogram No. 1, but for the hyperbolic orbits, the other formulae were made use of.

[p. 447]

94. I annex a specimen; the characteristics of the logarithms are omitted.

[$c = 20^\circ$. The columnar specimen calculation gives, in summary, the following intermediate and final results.]

$x_1 = 1{\cdot}06418$	$x_2 = -{\cdot}45017$	$x_3 = -{\cdot}65049$
$y_1 = {\cdot}21014$	$y_2 = +{\cdot}91038$	$y_3 = {\cdot}80171$
$\phi_1 = 11^\circ\,10'$	$\phi_2\,(= 63^\circ\,41') = 116^\circ\,19'$	$\phi_3\,(= 50^\circ\,57') = 230^\circ\,57'$
$r_1 = 1{\cdot}0847$	$r_2 = 1{\cdot}0157$	$r_3 = 1{\cdot}0323.$

The calculation of the equation of the orbit is then as follows:

$$\lambda = \cdot 36397,$$

$$\lambda^2 = \cdot 13248,$$

$$\lambda^2 - 3 = -2\cdot 86752,$$

$$\sqrt{1+\lambda^2} = 1\cdot 06477,$$

$$6\sqrt{1+\lambda^2} = 6\cdot 38862,$$

$$2\sqrt{3}(\lambda^2 - 3) = -9\cdot 93330,$$

$$2(\lambda^2 - 3) = -5\cdot 73504,$$

$$R_1 = 1\cdot 0847, \quad R_2 = +2\cdot 4010, \quad R_3 = -1\cdot 6889,$$

$$4R_1 = 4\cdot 3388, \quad 2R_1 = 2\cdot 1694,$$

$$A = \cdot 038982, \quad B = -\cdot 014265, \quad C = +\cdot 10464,$$

$$\varpi\,(= 20^\circ\, 6') = 160^\circ\, 52', \quad e = \cdot 04151, \quad 1 - e^2 = \cdot 998277, \quad a = 1\cdot 0481.$$

[p. 448]

[The page is occupied entirely by columnar arithmetic continuing the specimen calculation begun on p. 447. The principal intermediate values are reproduced above; the column totals yield the same $A = \cdot 038982$, $B = -\cdot 014265$, $C = \cdot 10464$, *and lead through* $e^2 = \cdot 001723$, $1 - e^2 = \cdot 998277$, *to* $a = 1\cdot 0481$.*]*

[p. 449]

[The greater part of this page is blank in the original.]

The calculation of the Times is similar to that for the first planogram, and requires no further illustration.

The Table for Planogram No. 2 is as follows:

[pp. 450–451]

Planogram No. 2, $b = 180^\circ$, $c = 0^\circ$ *to* 90°.

[The original lays the table across two facing pages: a left half (p. 450) carrying columns c, x_1, y_1, x_2, y_2, x_3, y_3, r_1, ϕ_1, r_2, ϕ_2, r_3, ϕ_3; *and a right half (p. 451) carrying the Equation of Orbit* ($r = Ax + By + C$), *the eccentricity e, the angle* ϖ, *a, the chords* γ_{12}, γ_{23}, γ_{31}, *and the times* T_{12}, T_{23}, T_{31}, *repeating c at the right margin. Heading note: "The mark* $\sim$ *in any of the T columns shows that the Time does not exist." The numerical entries are reproduced below as a single combined table; columns marked all + or all − retain the printed signs.]*

	c	x_1	y_1	x_2	y_2	x_3	y_3	r_1	ϕ_1	r_2	ϕ_2	r_3	ϕ_3	Eq. of Orbit	e	ϖ	a	γ_{12}	γ_{23}	γ_{31}	T_{12}	T_{23}	T_{31}
Circle	0°	1·000	·000	−·500	+·866	−·500	−·866	1·000	0° 0′	1·000	120° 0′	1·000	240° 0′	·000 ·000 + 1·000	·000	ind.°	1·000	1·732	1·732	1·732	1·000	1·000	1·000
	5	1·004	·051	·481	·878	−·525	·853	1·005	2 52	1·001	118 42	1·001	238 40	+·0101 − ·0015 1·0104	·010	171 19	1·010	1·729	1·832	1·678	·987	1·106	·953
	10	1·015	·102	·467	·889	·557	·838	1·020	5 44	1·004	117 41	1·006	236 40	·039 ·014 1·046	·041	160 52	1·048	1·724	1·991	1·668	·956	1·316	·946
	15	1·035	·156	·456	·900	·598	·821	1·047	8 30	1·009	116 54	1·016	233 30										
	20	1·064	·210	·450	·910	·650	·802	1·085	11 10	1·016	116 19	1·032	230 30	·083 ·061 1·126	·103	143 48	1·138	1·718	2·244	1·710	·924	1·777	·962
	25	1·103	·269	·447	·921	·719	·778	1·136	13 43	1·024	115 55	1·060	227 30										
Ellipses	30	1·155	·333	·448	·931	·812	·749	1·202	16 6	1·033	115 42	1·104	222 30	·135 ·209 1·317	·248	122 54	1·404	1·740	2·684	1·826	·878	3·238	·966
	35	1·221	·404	·452	·941	·939	·710	1·286	18 19	1·044	115 40	1·178	217 30										
	40	1·305	·484	·460	·951	1·125	·657	1·392	20 21	1·056	116 48	1·302	210 30	·161 ·395 1·527	·426	112 12	1·867	1·805	3·055	1·925			
	45	1·414	·577	·471	·962	1·414	·577	1·527	22 12	1·071	116 6	1·528	202 30	·186 ·815 1·972	·836	102 50	6·554	2·016	3·659	2·063	·878	48·60	·849
Parab.	50	1·556	·688	·487	·974	1·925	·440	1·701	23 51	1·088	116 35	1·975	192 30	·191 ·982 2·140	1·000	101 14	∞	2·100	3·831	2·097	·879	∞	·820
	51° 0′	1·589	·713	·491	·976	2·077	·400	1·741	24 9	1·093	116 43	2·115	190 30	·196 1·150 2·319	1·166	99 39	6434						
	52	1·624	·739	·495	·978	2·256	·353	1·787	24 28	1·097	116 51	2·283	188 30										
	54	1·701	·795	·504	·984	2·729	·229	1·878	25 2	1·105	117 7	2·738	184 30	·203 1·719 2·898	1·720	96 44	1·481						
Hyperbs.	55	1·743	·824	·509	·986	3·049	·145	1·928	25 18	1·109	117 16	3·053	182 30	·207 2·187 3·366	2·192	95 25	·855	2·781	4·890	2·258	·895	$\sim$	·665
	56° 18′	1·802	·866	·515	·990	3·601	·000	1·999	25 39	1·116	117 30	3·601	180 30	·212 3·074 4·227	3·081	93 56	·498						
	59	1·942	·961	·530	·997	5·786	+·566	2·166	26 20	1·129	118 59	5·813	174 30	·221 − 1·415 + 15·42	1·415	90 30	·077						
Line	60	2·000	1·000	·536	1·000	7·468	1·000	2·236	26 34	1·134	118 11	7·534	172 30	·224 ∞ $\infty(y-1)$	∞	90	·000	6·932	9·468	2·536	·000	$\sim$	·000
	61	2·063	1·042	·542	1·003	−10·53	+1·793	2·311	26 48	1·140	118 24	10·68	170 30					Convex Orbits.					
Convex {	$63^\circ\, 26' - \epsilon$	2·236	1·155	·559	1·010	∞	∞	2·517	27 19	1·155	118 57	∞	165 30	·234 + 4·906 − 3·671	4·912	87 17	·159						
	$63^\circ\, 26' + \epsilon$					$-\infty$	∞					$-\infty$	134 30	·578 − 1·761 + 3·257	1·853	108 11	1·338	∞	∞	2·799	$\sim$	∞	·909
	64	2·281	1·184	·563	1·012	+45·22	−12·60	2·570	27 26	1·157	119 6	46·94	344 30										
Hyperbs.{	65	2·366	1·238	·571	1·015	16·36	5·146	2·670	27 37	1·165	119 21	17·15	342 30	·587 ·979 2·134	2·494	120 21	8·666	18·014	15·380	2·945			
Parab.	65° 52′	2·446	1·289	·578	1·019	10·12	3·552	2·765	27 47	1·171	119 35	10·80	340 30	·591 ·805 2·257	1·000	126 19	∞	11·633	9·072	3·036	∞	7·746	1·386
	66	2·459	1·297	·579	1·019	9·987	3·500	2·779	27 48	1·172	119 37	10·59	340 30	·593 ·779 2·221	·979	127 15	5·383						
	68	2·669	1·429	·596	1·026	5·617	2·369	3·028	28 10	1·186	120 11	6·090	337 30	·606 ·338 1·894	·693	150 53	3·645						
	70	2·924	1·586	·616	1·033	3·912	1·927	3·326	28 29	1·202	120 48	4·360	335 30	·619 − ·120 1·708	·630	169	2·834	5·409	3·649	3·584	5·735	6·343	2·685
Ellipses	72	3·237	1·777	·638	1·041	3·008	1·694	3·693	28 47	1·221	121 29	3·455	330 30	·635 + ·027 1·599	·636	182 25	2·674						
	75	3·864	2·155	·674	1·054	2·230	1·488	4·424	29 9	1·251	122 36	2·681	325 30	·654 ·185 1·497	·680	195 47	2·783	3·859	3·981	4·644	2·62	6·68	4·64
	80	5·759	3·274	·751	1·079	1·568	1·312	6·624	29 37	1·315	124 49	2·045	320 30	·692 ·366 1·439	·783	207 52	3·716	3·327	6·212	6·870	1·97	10·21	9·343
	85	11·47	6·599	·854	1·112	1·217	1·216	13·25	29 54	1·402	127 32	1·720	315 30	·740 ·514 1·455	·892	214 47	7·721	3·115	12·895	13·480			
Parab.	89° 20′	86·41	49·79	·979	1·148	1·024	1·161	99·5	29 56	1·508	130 24	1·548	311 30	·764 ·645 1·505	1·000	219 51	∞	3·055	9·943	102·7	1·200	225·4	∞
Hyperbs.	$90 - \epsilon$	∞	∞	1·000	1·155	+1·000	−1·155	∞	30	1·527	130 54	1·527	310 30	·770 + ·666 + 1·527	1·018	220 6	41·000	3·055	∞	∞	1·148	∞	$\sim$

[p. 452]

Article Nos. 95 *to* 98. *Planogram No.* 3, *the Orbit-pole at one of the points A.*

95. When the orbit-pole is at one of the points A, the orbit-plane passes through one of the rays, and as there is no longer on this ray any determinate point of intersection, the orbit (as was seen) becomes indeterminate. Thus consider the point A for which $b = 270°$, $c = 60°$: we have

$$\begin{gathered}\alpha,\ \beta,\ \gamma = -1,\ 0,\ 0,\\ \alpha',\ \beta',\ \gamma' = 0,\ -\tfrac{1}{2},\ -\tfrac{1}{2}\sqrt{3},\\ \alpha'',\ \beta'',\ \gamma'' = 0,\ -\tfrac{1}{2}\sqrt{3},\ \tfrac{1}{2},\end{gathered}$$

[Figure 9: a circle in the (x, y)-plane of unit radius, with the four cardinal positions of points 1, 2, 3, *B marked. Point* 1 *at the top,* 2 *at lower left,* 3 *at lower right, B at the right.]*

and consequently the formula gives

$$\begin{aligned} x_1' : y_1' : 1 &= 0 : 0 : 0,\\ x_2' : y_2' : 1 &= -\tfrac{1}{2}\sqrt{3} - \sqrt{3} : 3 : -\tfrac{1}{2}\sqrt{3} - \sqrt{3},\\ x_3' : y_3' : 1 &= -\tfrac{1}{2}\sqrt{3} - \sqrt{3} : -3 : -\tfrac{1}{2}\sqrt{3} - \sqrt{3},\end{aligned}$$

and, moreover, $x = -x'$, $y = -y'$. From the formula the value of x_1' or x_1 is given as $\frac{0}{0}$, but the true value is obviously $x_1 = 1$; the value of y_1 is actually indeterminate. The formulae give the values of $(x_2,\ y_2)$, $(x_3,\ y_3)$, viz. the system is

$$\begin{aligned} &x_1 = 1, && y_1 = \text{ind.}\\ &x_2 = -1, && y_2 = \tfrac{2}{\sqrt{3}}, \qquad \text{whence } r_2 = r_3 = \sqrt{\tfrac{7}{3}},\\ &x_3 = -1, && y_3 = -\tfrac{2}{\sqrt{3}},\end{aligned}$$

[p. 453]

so that the orbits in the planogram are the whole series of conics having a given focus, S, and passing through two fixed points, 2, 3, having the common abscissa $x = -1$, and at equal distances $\frac{2}{\sqrt{3}}$ (= 1·15470) on opposite sides of the axis. The axis of x is obviously the common transverse axis for all the orbits; that is, the equation of the orbit will be of the form $r = Ax + B$; and writing $x = -1$, we have $\sqrt{\frac{7}{3}} = -A + B$, viz. the equation is $r - \sqrt{\frac{7}{3}} = A(x + 1)$; the value of A will be determined if we assume for the point 1 a determinate position on the line $x = 1$, say its ordinate is $= y_1$; for then if $r_1 = \sqrt{1 + y_1^2}$ we have $r_1 - \sqrt{\frac{7}{3}} = 2A$, and the equation is $r - \sqrt{\frac{7}{3}} = \frac{1}{2}(r_1 - \sqrt{\frac{7}{3}})(x + 1)$. In particular if $y_1 = 0$, we have $r_1 = 1$, and the equation of the orbit is $r - \sqrt{\frac{7}{3}} = \frac{1}{2}(1 - \sqrt{\frac{7}{3}})(x + 1)$: this is the orbit, eccentricity $= \frac{1}{2}(\sqrt{\frac{7}{3}} - 1) = {\cdot}264$, belonging to the point A as a point in planogram No. 1: for the value of y_1 being in that planogram originally assumed = 0, is of course = 0 when the orbit-pole comes to be the point A.

96. We may conversely take the equation of the orbit, or say the value of $A\,(= \pm e)$ in the equation $r - \sqrt{\frac{7}{3}} = A(x + 1)$, to be given; and then writing $x = x_1 = 1$, we have

$$r_1 = \sqrt{\tfrac{7}{3}} + 2A, \quad \text{that is} \quad y_1^2 = (\sqrt{\tfrac{7}{3}} + 2A)^2 - 1;$$

for

$$r_1 = 1 \quad \text{or} \quad y_1 = 0, \quad A = \tfrac{1}{2}(1 - \sqrt{\tfrac{7}{3}}) = -{\cdot}264,$$

and as r_1 increases to $r_1 = \sqrt{\frac{7}{3}}$, or y_1 increases to $+\frac{2}{\sqrt{3}}$, A diminishes from $-{\cdot}264$ to 0; viz., for $r_1 = \sqrt{\frac{7}{3}}$ or $y_1 = \frac{2}{\sqrt{3}}$, the orbit is a circle; as r_1 increases from $\sqrt{\frac{7}{3}}$, or y_1 from $+\frac{2}{\sqrt{3}}$, A increases from 0 positively;

for $r_1 = \sqrt{\frac{7}{3}} + 2 = 3{\cdot}527$, or $y_1 = +\frac{1}{\sqrt{3}}\surd = 2{\cdot}896$, A becomes $= 1$; that is, the orbit is a parabola; and for larger positive values of r_1, or positive or negative values of y_1, the orbit is a hyperbola (concave); and ultimately for $r_1 = \infty$ or $y_1 = \pm\infty$, the orbit is the right line $x + 1 = 0$. Thus A extends from $-{\cdot}264$ to 0, and thence from 0 positively to ∞.

97. In further illustration, suppose that the orbit-pole, instead of being at A, is a point in the immediate neighbourhood of A, say that the rectangular spherical coordinates, measured from A in the direction of the meridian and perpendicular thereto, are ξ and η, the colatitude and longitude of the orbit-pole being thus $c = 60° + \xi$, and $b = 270° + \dfrac{2}{\sqrt{3}}\eta$; we have then, ξ, η being indefinitely small,

$$\begin{aligned}
\alpha,\ \beta,\ \gamma &= -1,\ -\tfrac{2}{\sqrt{3}}\eta,\ 0, \\
\alpha',\ \beta',\ \gamma' &= \tfrac{2}{\sqrt{3}}\eta\,,\ -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\xi\,,\ -\tfrac{\sqrt{3}}{2} + \tfrac{1}{2}\xi\,, \\
\alpha'',\ \beta'',\ \gamma'' &= \eta\,,\ -\tfrac{\sqrt{3}}{2} + \tfrac{1}{2}\xi\,,\ \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\xi\,.
\end{aligned}$$

[Scan pages 501–509: Plates I–V (frontispiece plates referenced from later papers — Guide Hyperbolas $h = 0°$ to $90°$; Planogram No. 1, $b = 90°$; Planogram No. 2, $b = 180°$; x-Spherogram; xy-Spherogram). Reproductions are not embedded in the typeset edition; see the IA scan for the original lithographic plates.]

477.

ON THE GRAPHICAL CONSTRUCTION OF A SOLAR ECLIPSE.

[From the Memoirs of the Royal Astronomical Society, vol. XXXIX. (1870–1871), pp. 1–17. Read January 13, 1871.]

[p. 479]

The present Memoir contains the explanation of a Graphical Construction of a Solar Eclipse, which (it appears to me) is at once easy, and susceptible of considerable accuracy: I think that if made on the suggested scale (radius = 12 inches) we might by means of it construct a diagram such as the eclipse-diagrams of the *Nautical Almanac*, with at least as much accuracy as could be exhibited in a diagram on that scale.

Article Nos. 1 to 9. General Explanation of the Construction.

1. We may imagine the celestial sphere as seen from the centre of the Earth stereographically projected at each instant during the eclipse—the radius of the bounding circle of the projected hemisphere being a given length, say twelve inches, which is taken as unity—in such wise that the centre of the Moon is always at the centre of the projection, say M, and the pole (suppose the north pole, say N) of the Earth on a given radius: its position on this radius will in strictness be variable, viz. distance from centre = projection of Moon's N.P.D. $= \tan\frac{1}{2}\Delta$. Suppose, for a moment, that the position at each instant of the Sun's centre were also laid down on the projection, so as to obtain the projection of the Sun's relative orbit; this will be a terminated short line $A'B'$ (fig. 1), nearly straight, and lying near the centre of the projection (this relative orbit is not to be actually laid down, but it is replaced, as will presently be explained, by a relative orbit on a very enlarged scale); if at any instant the position of the Sun on the relative orbit be denoted by S', then the straight line MS' is the projection of the arc of great circle through the centres of the Moon and Sun, so that E being the angular distance of the centres, the length of the line MS' is $= \tan\frac{1}{2}E$, or (E being small) it is $= \frac{1}{2}E$.

[p. 480]

2. Produce $S'M$ through the centre M to a point Z, and consider Z as representing a point on the Earth's surface: to determine the geographical position of Z, we must consider the projected meridian NZ which passes through Z: the arc NZ, regarded as a projection, represents the N.P.D. or colatitude of Z, and the actual angle at N which the tangent of NZ makes with the line NM is equal to the celestial angle ZNM which is = Moon's hour-angle from Z, or what is the same thing = difference of Moon's hour-angle from Greenwich and of the longitude of Z (as the figure is drawn, $\angle ZNM$ = Moon's hour-angle E. of Greenwich, less E. longitude of Z).

[Figure 1: stereographic projection in a circle centred at M; N marked above M on the vertical radius; S' below M on the same line, Q' slightly off S'; Z on the produced line beyond M, on the bounding circle; arc NZ shown.]

3. Now, considering the Moon and Sun as seen from Z, we may disregard the parallactic depression of the Sun, and attribute to the Moon a displacement equal to the difference of the parallactic displacements of the Moon and Sun; that is, regarding the zenith distances ZM, ZS' as equal, we may consider the Moon's centre as depressed by parallax in the direction of the arc MS' through an arc $MQ' = \sin^{-1} P' \sin ZM$, where $P' = \cdot 99837\,(\varpi - \varpi')$ is the quantity thus designated in the Appendix to the *Nautical Almanac* for 1836, viz. it is $= \varpi - \varpi'$, the difference of the equatoreal horizontal parallaxes at the time of the eclipse, multiplied by a factor ·99837, which answers to a distance of Z from the Earth's centre = Earth's radius for latitude 45°. And if we take Q' such that its angular distance from S' = sum of angular semidiameters of the Sun and Moon, the locus of Q' is very nearly a circle about the centre S', and the corresponding positions of Z give the positions on the Earth where the limbs are in exterior contact, or, what is the same thing, give the penumbral curve on the Earth's surface for the position S' of the Sun.

4. Instead of

$$\text{Arc}\, MQ' = \sin^{-1} P' \sin ZM,$$

we may write

$$\text{Arc}\, MQ' = P' \sin ZM,$$

[p. 481]

or, using ρ to denote the linear distance ZM in the projection, we have $\rho = \tan \frac{1}{2} ZM$, and therefore $\sin ZM = \dfrac{2\rho}{\rho^2+1}$; hence

$$\text{Arc}\, MQ' = P' \, \frac{2\rho}{\rho^2+1},$$

and the linear distance MQ' in the projection is $\tan \frac{1}{2}$ arc MQ', say this is $= \frac{1}{2}$ arc MQ', or calling this linear distance r' we have

$$r' = P' \, \frac{\rho}{\rho^2+1}.$$

5. Hence, if instead of the original representation of the Sun's relative orbit we consider an enlarged representation thereof and of the depressed positions Q' of the Moon, obtained by increasing the several distances from the centre of the projection in the ratio $\frac{1}{P'}$ to 1, and if instead of A', B', S', Q', we use A, B, S, Q, as referring to this enlarged representation, then representing by r the linear distance MQ, we have $r = \dfrac{1}{P'} r'$, and consequently

$$r = \frac{2\rho}{\rho^2+1}.$$

We have here r representing the parallactic depression corresponding to the zenith distance ZM, where $\rho = \tan \frac{1}{2} ZM$; that is, $ZM = 90°$, $\rho = 1$, and therefore $r = 1$; but for $ZM = 90°$ the parallactic depression is $= P'$; that is, the scale of the enlarged representation of the Sun's relative orbit, or say simply the scale of the relative orbit (for on the original scale it was never actually constructed at all) is such that we have P' (= about 60′) represented by the radius of the bounding circle of the projected hemisphere, = 12 inches.

6. The process is, construct the relative orbit on the scale P' = radius of bounding circle: take S for the position at any given instant of the Sun in the relative orbit, and with centre S and radius $= s + \sigma$ (sum of the angular semidiameters, of course on the same scale) describe a circle. The positions A and B of the Sun at the beginning and end of the eclipse respectively are such that this circle just touches the bounding circle externally, viz. the distances of A and B from the centre of the projection are each = radius of bounding circle $-(s+\sigma)$. At any intermediate instant the circle, radius $s + \sigma$, lies wholly or partly within the bounding circle; in the latter case we attend only to the arc thereof which lies within the bounding circle. Taking then Q any point whatever on the circle or arc in question, we join Q with the centre M of the projection, and produce this line through M to a point Z, such that the distances MQ, MZ, being r, ρ respectively, we have as above

$$r = \frac{2\rho}{\rho^2+1},$$

or, what is the same thing, writing θ in place of z, and regarding this angle θ as a variable parameter, the relation between r, ρ, may be expressed by means of the two equations, $\rho = \tan \frac{1}{2}\theta$, $r = \sin\theta$.

C. VII. 61

[p. 482]

7. Practically the construction may be performed very easily by means of a straight edge twenty-four inches long, graduated from the centre, one half of it for the values of r, and the other half for the corresponding values of ρ (that is, the first half is graduated for $\sin\theta$, and the second half for $\tan\frac{1}{2}\theta$): we have thus, corresponding to the circle or arc of circle which is the locus of Q, a closed curve, or arc thereof terminated each way at the bounding circle, for the locus of Z: which curve or arc of a curve is the penumbral curve on the Earth's surface for the position S of the Sun in the relative orbit.

8. The north pole of the Earth occupies in the projection a given position, viz. it is situate on a given radius at a distance $= \tan\frac{1}{2}$ Moon's N.P.D.; which N.P.D. may be considered as being throughout the eclipse constant, and equal to its value at the middle of the eclipse. But in order to arrive at the geographical signification of the figure it is necessary to lay down on the projection the position of the meridian of Greenwich; which position, it will be remembered, varies according to the position of S. Supposing this done, we could of course (at least theoretically) draw the whole series of meridians and parallels, and thereby determine the latitudes and longitudes of the several points of the penumbral curve, or (if need is) transfer it to a different projection of the Earth's surface. The actual description of the meridians and parallels would, however, be very laborious, and fortunately it can be avoided by means of a single blank projection and a slight modification of the foregoing process, as will be explained.

9. But before considering how this is, it is proper to remark that constructing as above a figure of the penumbral curves corresponding to the several positions of the Sun: by what precedes these different curves may indeed be considered as belonging to the same position of the north pole in the projection, but they belong to different positions of the meridian of Greenwich; and thus they do not constitute a representation of the penumbral curves each in its proper terrestrial position, but only a representation in which the penumbral curves are affected each of them by a different displacement in longitude.

Article Nos. 10 to 13. Modification in order to the Applicability of a Single Blank Projection.

10. Imagine a stereographic projection of the meridians and parallels on the plane of a meridian, radius of this meridian, that is of the bounding circle of the projected hemisphere, being = 12 inches as before; and the poles N, S being of course opposite points on the circumference of the bounding circle—the meridians and parallels are, however, to be produced outside the bounding circle; say this is the "blank projection," and let its centre be denoted by M_1. Then, if at any point M on the radius M_1N, we draw the chord CD at right angles to M_1N, and on CD as diameter describe a circle, this will cut out from the blank projection a new projection having the last-mentioned circle for its bounding circle, and in which N is the north pole; viz. the meridians of the blank projection will be meridians, and the parallels of the blank projection will be parallels, in this new projection. And, moreover, if the longitudes are reckoned from the meridian NMM_1, then the meridian of a given longitude in the blank projection will in the new projection be the meridian of the same

[p. 483]

longitude—but the parallel of a given colatitude c' in the blank projection will, in the new projection, be the parallel of a different colatitude c,—the relation of c, c' being, however, a very simple one, as presently explained.

[Figure 2: a circle with centre M_1 and a chord CD perpendicular to the radius M_1N; the chord defines a smaller circle of diameter CD (the new bounding circle) inscribed within the blank projection. Points N, C, M, D, S, Σ, R, Q marked as described in the text.]

11. The blank projection thus at once gives a projection in which the north pole N has any assumed position whatever; and it is easy to see that in order that its distance MN from the centre of the projection may represent a given angle Δ, we have only to take $M_1M = \cos\Delta$ (that is = 12 inches $\times\cos\Delta$), the corresponding value of MC being $MC = \sin\Delta$ (that is = 12 inches $\times\sin\Delta$). Hence Δ denoting the Moon's N.P.D. at the middle of the eclipse, we can by means of the blank projection construct a projection such as that above referred to, only the radius of its bounding circle, instead of being unity (12 inches), is in the reduced ratio of $1 : \sin\Delta$.

12. The figure of the penumbral curves as originally constructed requires, therefore, to be reduced in the ratio $1 : \sin\Delta$, viz. each of the distances from the centre M should be reduced in this ratio; this could of course be done easily enough with a pair of proportional compasses; but by means of a different graduation of the straight edge we may, in the first instance, construct the penumbral curves on the proper reduced scale; viz. assuming that we have on the proper scale a proportional-scale figure such as is here shown, the line Mr ($= 12$ inches) being graduated for $\sin\theta$, and the line MA (also $= 12$ inches) for $\tan\frac{1}{2}\theta$, and a set of parallel lines being drawn through the last-mentioned graduations—then taking the distance $M\rho = \sin\Delta$, that is $= 12$ inches $\times \sin\Delta$, and drawing the line $M\rho$, this line will, it is clear, be graduated for $\sin\Delta\tan\frac{1}{2}\theta$: so that we may from the figure graduate the straight edge, the one half of it by means of the line Mr, and the other half of it by means of the line $M\rho$; and with the straight edge thus graduated, at once lay down the penumbral curve on the scale now in question. And we thus obtain a figure containing as well

61–2

[p. 484]

the penumbral curves, as the meridians and parallels which serve to fix their terrestrial position.

[Figure 3: triangular proportional-scale figure with apex M at the bottom; horizontal top edge labelled with points r', ρ, A; parallel lines through the graduations of Mr and MA used to construct the corresponding values on the reduced scale.]

13. It remains in the new projection to find the colatitude belonging to any given parallel. Supposing that the colatitude in the blank projection is $= c'$, then it may be shown that the colatitude c of the same parallel in the reduced projection is given by means of the equation

$$\tan\tfrac{1}{2}c = \cot\tfrac{1}{2}\Delta\,\tan\tfrac{1}{2}c',$$

from which c might be calculated numerically: but the required value may also be obtained graphically. In fact, considering the parallel which cuts $N\Sigma$ (see fig. 2) in a point R, then, if by lines drawn from C as a centre we project N, R, Σ, on the circumference of the bounding circle of the new projection—say the projections of these points are n, r, s, respectively, the arc ns is a semicircle, and the arcs nr, sr, are respectively the N.P.D. and the S.P.D. of the parallel in question. It may be added that in the new projection the equator is represented by the parallel through the points C, D; so that if this cuts $N\Sigma$ in Q, and the point Q be in like manner projected on the bounding circle—say its projection is q, then the arcs nq, sq, will be each of them a quadrant, and the arc qr will be the latitude of the parallel in question.

Article Nos. 14 to 18. As to the Construction of the Relative Orbits.

14. It is convenient to notice that if e, e', be the values of the equation of time at the preceding and following Greenwich Mean Noons (viz. e or e' = G.M.T. of apparent Noon) then that the Sun's hour-angle E. of Greenwich at the Greenwich mean time t is

$$h' = e + t\left(1 + \frac{e' - e}{24^{\mathrm{h}}}\right),$$

[p. 485]

and that if a, a', are the R.A.'s of the Moon and Sun respectively, then $h' - h = a' - a$, which is also of the form $A + Bt$. In the reduced projection, the Moon is always at the centre M; by means of the values of $h' - h$ we lay down at any instant the Sun's position in R.A. and then by means of the values of h', the position of the meridian of Greenwich; and we thus at any instant read off the terrestrial longitude of any point of the reduced projection, or say, of a point on the penumbral curve.

15. With regard to the construction of the relative orbit, it is to be observed that if at any instant the hour-angle and N.P.D. of the Moon are h, Δ, and those of the Sun, h', Δ', then taking M as origin, and the axes Mx, My, in the direction of NM

[Figure 4: small rectangular axes diagram with M at the origin, N above, S' below and slightly to one side; horizontal axis y to the right of M, vertical axis x downward in the direction of NM produced.]

produced, and perpendicular thereto to the right (or eastwards), then the rectangular coordinates of S' are approximately $x = \frac{1}{2}(\Delta' - \Delta)$, $y = \frac{1}{2}(h' - h)\sin\Delta$, where $h' - h$ is equal to the difference of R.A. of the Sun and Moon. Hence, in the adopted relative orbit, the coordinates of S would be

$$x = \frac{\Delta' - \Delta}{P'}\ 12\,\text{in.}, \qquad y = \frac{h' - h}{P'}\sin\Delta\ \ 12\,\text{in.},$$

where, P' being reckoned in minutes, $\Delta' - \Delta$ and $h' - h$ are also reckoned in minutes.

16. Moreover, Δ may be considered as constant during the eclipse: and the relative orbit, assumed to be a straight line, will be determined by means of two points thereof; viz. knowing the values of $\Delta' - \Delta$, and $h' - h$ at about the time of the beginning and at about the time of the end of the eclipse, we construct by these formulæ two points of the orbit, and joining them by a straight line, we have the orbit. Also the position at any instant of the Sun in this relative orbit will be obtained by considering its motion therein as being uniform. I think there is no advantage in the adoption of a more accurate construction: for although we may for any given instant use the accurate values of h, Δ, h', Δ', and so construct the position in the relative orbit, and the corresponding penumbral curve, yet if in the deter-

[p. 486]

mination of the geographical significance thereof, we were to use for each curve a different value of Δ, the simplicity of the construction would disappear; and it is, moreover, doubtful whether the trifling corrections would not be within the limits of the necessary errors of the drawing.

17. But if $MS' = E$, and $\angle xMS' = \theta$, the accurate values for the coordinates of S' are $x = \tan\frac{1}{2}E\cos\theta$, $y = \tan\frac{1}{2}E\sin\theta$, and the values for the coordinates of S are

$$x = \frac{2}{P'\cdot\text{arc}\,1'}\ \tan\tfrac{1}{2}E\cos\theta\cdot 12\,\text{in.}, \qquad y = \frac{2}{P'\cdot\text{arc}\,1'}\ \tan\tfrac{1}{2}E\sin\theta\cdot 12\,\text{in.},$$

where P' is still reckoned in minutes, and of course $\text{arc}\,1' = \dfrac{\pi}{10800}$. As the scale is considerable, it is worth while to inquire whether the employment of the accurate formulæ would produce an appreciable difference in the position of S.

We have $\sin\theta = \sin\Delta'\sin(h' - h) \div \sin E$, that is, $\sin E\sin\theta = \sin\Delta'\sin(h' - h)$, and $\cos E = \cos\Delta\cos\Delta' + \sin\Delta\sin\Delta'\cos(h' - h)$: or putting for shortness $\Delta' - \Delta = \alpha$, $h' - h = \beta$, we have $\sin E\sin\theta = \sin\Delta'\sin\beta$, and $\cos E = \cos\alpha - \sin\Delta\sin\Delta'\cdot 2\sin^2\frac{1}{2}\beta$. Hence, attending to the equations $\cos^2\frac{1}{2}E = \frac{1}{2}(\cos^2\frac{1}{2}\alpha - \sin\Delta\sin\Delta'\sin^2\frac{1}{2}\beta)$ and $\sin^2\frac{1}{2}E = \frac{1}{2}(\sin^2\frac{1}{2}\alpha + \sin\Delta\sin\Delta'\sin^2\frac{1}{2}\beta)$, we find

$$\tan\tfrac{1}{2}E\sin\theta = \frac{\sin\frac{1}{2}\beta\cos\frac{1}{2}\beta\sin\Delta'}{\cos^2\frac{1}{2}\alpha - \sin\Delta\sin\Delta'\sin^2\frac{1}{2}\beta},$$

and

$$\tan\tfrac{1}{2}E\cos\theta = \sqrt{\frac{\sin^2\frac{1}{2}\alpha + \sin\Delta\sin\Delta'\sin^2\frac{1}{2}\beta}{\cos^2\frac{1}{2}\alpha - \sin\Delta\sin\Delta'\sin^2\frac{1}{2}\beta} - \frac{\sin^2\frac{1}{2}\beta\cos^2\frac{1}{2}\beta\sin^2\Delta'}{(\cos^2\frac{1}{2}\alpha - \sin\Delta\sin\Delta'\sin^2\frac{1}{2}\beta)^2}},$$

whence, considering α, β as small quantities of the same order, and neglecting terms of the third order, we have $\tan\frac{1}{2}E\sin\theta = \sin\frac{1}{2}\beta\cos\frac{1}{2}\beta\sin\Delta'$, or what is the same thing, $\sin\frac{1}{2}\beta\sin\Delta$, or finally, $\frac{1}{2}\beta\sin\Delta$, that is $\frac{1}{2}(h' - h)\sin\Delta$, which is the foregoing approximate value, and thus in the adopted orbit $y = \dfrac{h' - h}{P'}\sin\Delta\cdot 12\,\text{in.}$ = approx. value. As regards the expression for $\tan\frac{1}{2}E\cos\theta$, writing for a moment $\Omega = \sec^2\frac{1}{2}\alpha\sin\Delta\sin\Delta'\sin^2\frac{1}{2}\beta$, the quantity under the radical sign is

$$\frac{\tan^2\frac{1}{2}\alpha + \Omega}{1 - \Omega} - \frac{\sin^2\frac{1}{2}\beta\cos^2\frac{1}{2}\beta\sin^2\Delta'}{\cos^4\frac{1}{2}\alpha\,(1 - \Omega)^2},$$

and, taking this to the third order, it is

$$= \tan^2\tfrac{1}{2}\alpha + \Omega(1 + \tan^2\tfrac{1}{2}\alpha) - \frac{\sin^2\frac{1}{2}\beta\cos^2\frac{1}{2}\beta\sin^2\Delta'}{\cos^4\frac{1}{2}\alpha},$$

which, substituting for Ω its value, is

$$= \tan^2 \tfrac{1}{2}\alpha + \frac{\sin^2 \frac{1}{2}\beta \sin\Delta'}{\cos^4 \frac{1}{2}\alpha}\bigl(\sin\Lambda - \sin\Delta' \cos^2 \tfrac{1}{2}\beta\bigr),$$

where $\sin\Lambda = \sin\Delta' \cos^2 \frac{1}{2}\beta + \sin\Delta = \sin(\Delta+\beta)\cos^2 \frac{1}{2}\beta$;

[p. 487]

or neglecting herein terms of the second order, this is

$$= (\sin\Delta + \sin\alpha\cos\Delta - \sin\Delta)\cos^2 \tfrac{1}{2}\beta,$$

$$= \sin\alpha\cos\Delta \;=\; 2\tan\tfrac{1}{2}\alpha\cos^2\tfrac{1}{2}\alpha\cos\Delta,$$

so that to the third order the quantity under the radical sign is

$$\tan^2\tfrac{1}{2}\alpha + \frac{2\tan\frac{1}{2}\alpha\sin^2\frac{1}{2}\beta\sin\Delta'\cos\Delta}{\cos^2\frac{1}{2}\alpha},$$

and to the second order, that is finally neglecting terms of the third order,

$$\tan\tfrac{1}{2}E\cos\theta = \tan\tfrac{1}{2}\alpha + \frac{\sin^2\frac{1}{2}\beta\sin\Delta'\cos\Delta}{\tan\frac{1}{2}\alpha},$$

or, what is the same thing,

$$= \tfrac{1}{2}\alpha + \sin\Delta\cos\Delta\cdot\tfrac{1}{2}\beta^2.$$

18. Hence, writing $\alpha = (\Delta'-\Delta)\,\mathrm{arc}\,1'$, $\beta = (h'-h)\,\mathrm{arc}\,1'$, and passing to the adopted orbit, we have

$$x = \frac{\Delta'-\Delta}{P'}\,12\,\text{in.} + \tfrac{1}{2}\sin\Delta\cos\Delta\cdot\frac{h'-h}{P'}\,12\,\text{in.}\times(h'-h)\,\mathrm{arc}\,1',$$

viz. putting

$$y = \frac{h'-h}{P'}\sin\Delta\cdot 12\,\text{in.},$$

we have

$$x = \frac{\Delta'-\Delta}{P'}\,12\,\text{in.} - y\cdot\tfrac{1}{2}\cos\Delta\cdot(h'-h)\,\mathrm{arc}\,1',$$

or say

$$x = \frac{\Delta'-\Delta}{P'}\,12\,\text{in.} - y\cdot\tfrac{1}{2}\cos\Delta\cdot(h'-h)\,\frac{\pi}{10800}.$$

The value of the second term may amount to about $\frac{1}{12}$ of an inch, and thus be sensible, but there is no difficulty in taking account of it.

Article No. 19. As to the Equation $r = \dfrac{2\rho}{\rho^2+1}$.

19. It may be remarked that the equation $r = \dfrac{2\rho}{\rho^2+1}$, which served for the graduation of the straight edge, was in effect obtained from the equations

$$r = P'\tan\tfrac{1}{2}E, \qquad P'\sin z = \sin E, \qquad \rho = \tan\tfrac{1}{2}z$$

by assuming therein $\tan\frac{1}{2}E = \frac{1}{2}E$ and $\sin E = E$ respectively. But the elimination of E and z can be effected without this assumption, viz. we have $\sin E = \dfrac{2\tan\frac{1}{2}E}{1+\tan^2\frac{1}{2}E} = \dfrac{P'r}{1+\frac{1}{4}P'^2r^2}$,

[p. 488]

and then as before, $\sin z = \dfrac{2\tan\frac{1}{2}z}{1+\tan^2\frac{1}{2}z} = \dfrac{2\rho}{1+\rho^2}$, whence the relation between r and ρ is found to be

$$\frac{r}{1+\frac{1}{4}P'^2r^2} = \frac{2\rho}{1+\rho^2},$$

which however assumes that P' is reckoned in parts of the radius; reckoning it as before in minutes, we must, instead of P', write $P' \operatorname{arc} 1' = \dfrac{P'\pi}{10800}$, viz. the numerical value is about $\frac{1}{120}$, and taking it to be this number, the formula is

$$\frac{r}{1+\dfrac{1}{14400}r^2} = \frac{2\rho}{1+\rho^2},$$

where r, ρ are reckoned in parts of the radius ($=$ 12 inches). Supposing that r_1 is calculated from the formula $r_1 = \dfrac{2\rho}{1+\rho^2}$, then we have very nearly $\dfrac{r}{r_1} = \left(1+\dfrac{r_1^2}{14400}\right)$, and r_1 being $=1$ at most, the correction is inappreciable: if however this were not the case, the more accurate formula might have been used; the only difference being that the making of the graduation would have been more laborious.

Article No. 20. Remark as to the Geometrical Theory of the Projection of the Penumbral Curve.

20. The stereographic projection of the penumbral curve on the Earth's surface (assumed to be spherical) is, as I have elsewhere shown, a bicircular quartic. It may be shown that the stereographic projection, as given by the foregoing approximate method, is a bicircular quartic: we have, in fact a circle, the equation of which in the polar coordinates r, θ is

$$(r\cos\theta-\alpha)^2 + r^2\sin^2\theta = \beta^2,$$

and where (θ being unaltered) r is changed into ρ, where $r = \dfrac{2\rho}{\rho^2+1}$, that is $\dfrac{1}{r} = \frac{1}{2}\left(\rho+\dfrac{1}{\rho}\right)$. The equation of the circle is

$$r^2 - 2\alpha r\cos\theta + \alpha^2 - \beta^2 = 0,$$

or say

$$1 - 2\alpha\cos\theta\cdot\frac{1}{r} + (\alpha^2-\beta^2)\cdot\frac{1}{r^2} = 0,$$

and the transformed equation is therefore

$$1 - \alpha\cos\theta\left(\rho+\frac{1}{\rho}\right) + \tfrac{1}{4}(\alpha^2-\beta^2)\left(\rho+\frac{1}{\rho}\right)^2 = 0,$$

that is

$$(\alpha^2-\beta^2)(\rho^2+1)^2 - 4\alpha\cos\theta\,\rho\,(\rho^2+1) + 4\rho^2 = 0,$$

or in rectangular coordinates

[p. 489]

that is

$$\rho^4 - \frac{4\alpha}{\alpha^2-\beta^2}\,x\rho^2 + \left(2+\frac{4}{\alpha^2-\beta^2}\right)\rho^2 - \frac{4\alpha}{\alpha^2-\beta^2}\,x + 1 = 0,$$

where $\rho^2 = x^2+y^2$; the form of the equation shows that the curve is a bicircular quartic. Writing for shortness $\dfrac{2}{\alpha^2-\beta^2} = m$, the equation is

$$\rho^4 - 2max\rho^2 + (2+2m)\rho^2 + 1 - 2max = 0,$$

that is

$$[\rho^2 - m(ax-1)+1]^2 - m^2(ax-1)^2 - 2m\alpha = 0,$$

or, what is the same thing,

$$\{(x-\tfrac{1}{2}ma)^2+y^2-\tfrac{1}{4}m^2a^2+m+1\}^2-m^2(ax-1)^2-2m\alpha=0,$$

which putting $x+\frac{1}{2}ma$ for x is

$$\{x^2+y^2-\tfrac{1}{4}m^2a^2+m+1\}^2-m^2(\alpha x+\tfrac{1}{2}ma^2-1)^2-2m\alpha=0,$$

viz. the terms of the fourth order being $(x^2+y^2)^2$, and there being no terms of the third order, the curve represented by this equation is a bicircular quartic.

Article Nos. 21 to 30. Practical Details and Application to Eclipse of December 21–22, 1870.

21. There are some practical details which it is proper to explain, using to fix the ideas the eclipse of December 21–22, 1870: the constant value of Δ (see *infrà*) is taken to be $+90° + 22°35'(\cdot)$.

I have a blank projection (radius 12 in. as above) with the meridians and parallels each at intervals of 5°. And also another blank form which has on it merely a circle, radius 12 in., graduated as to one quadrant thereof with lines about $1\frac{1}{2}$ in. long, inwards towards the centre. It contains also in a corner the foregoing proportional-scale figure.

22. On the blank projection I measure off, downwards from the centre, a distance $M_1M = 12\sin 22°35'$ (= 4·61), distances all in inches; and then with the centre M and radius $MC = 12\cos 22°35'$ (= 11·08), describe a circle which is the bounding circle of the reduced projection. With this same radius I describe on the second form, concentric with the 12-inch circle and above the horizontal diameter thereof, a semicircle: and, cutting out the included area, replace it with tracing cloth. The form thus prepared is placed over the blank projection, so that the semicircle shall coincide with the corresponding semicircle on the blank projection, and the two sheets are fixed together by their lower edges, and by folding down the remaining sides. We have thus the upper half of the reduced projection, represented by the semicircle, with the

Footnote: See Plate, which exhibits in dotted lines the blank projection under the other blank form; the part within the red semicircle, as seen through the tracing cloth, the rest really hidden.

[p. 490]

meridians and parallels marked out thereon by lines seen through the tracing cloth. See the Plate; the dotted line shows a paper scale afterwards affixed to the second form or upper sheet. Observe that so far the only eclipse-datum made use of is the value $\Delta = 90° + 22°35'$.

23. We have for the eclipse in question, taking t for the G.M.T. in hours, positive or negative according as the time is after or before G.M. Noon, Dec. 22, and h' also in hours,

$$h' = 0^{\text{h}}{\cdot}02 + t\cdot{\cdot}9996,$$

and then taking the values of a, a', Δ, Δ' from the N.A. we have as follows:

G.M.T. 1870, Dec.		$h'+2^{\text{h}}=$	$a=$	$a'=$	$\Delta=90°+$	$\Delta'=90°+$	$\Delta'-\Delta$ in Minutes of Arc.	$h'-h$ in ditto.
d	h	h m s	h m s	h m s	° ′ ″	° ′ ″		
21	22	0 1 13·42	17 56 1·84	18 1 48·72	22 27 8·5	23 27 17·1	7 0·30 60·143	−86·620 +100·632
22	3	5 1 7·37	18 9 26·74	18 2 44·27	22 43 12·5	23 27 13·9	4 54 24·90	44·023

and moreover

$$\begin{aligned}\text{Moon's Parallax}\quad & \varpi = 60'\,38''{\cdot}6\\ \text{Sun's ditto}\quad & \varpi' = \quad 9''{\cdot}1\\ & \varpi - \varpi' = 60'\,29''{\cdot}5 = 60'{\cdot}49\\ & P' = 60'{\cdot}39\\ \text{Moon's Semidiam.}\quad & s = 16'\,33''{\cdot}2\\ \text{Sun's ditto}\quad & s' = 16\ 17{\cdot}9\\ & s + s' = 32'\,51''{\cdot}2 = 32'{\cdot}85\end{aligned}$$

whence

$$\frac{s+s'}{P'}\,12\,\text{in.} = 6{\cdot}53$$

viz. this is the radius of the circles used in the construction of the penumbral curves.

24. We have for x, y the formulæ

$$x = \frac{\Delta' - \Delta}{P'}\,12\,\text{in.} + y \cdot (h' - h) \cdot 0{\cdot}00006,$$

$$y = \frac{h' - h}{P'}\,12\,\text{in.} \times \sin\Delta,$$

viz. I find Dec. 21, 22^{h},

$$\begin{aligned} x &= 11{\cdot}95 - 0{\cdot}06 = 11{\cdot}89,\\ y &= -15{\cdot}80,\end{aligned}$$

[p. 491]

and Dec. 22–3^{h},

$$\begin{aligned} x &= 8{\cdot}75 + 0{\cdot}11 = 8{\cdot}85,\\ y &= +18{\cdot}45,\end{aligned}$$

where I have taken account of the small corrections to the approximate values of x: it may be added that, using the conjunction-value $52'\,9''{\cdot}4$ of $\Delta' - \Delta$, we have at conjunction,

$$x = 10{\cdot}36, \quad y = 0.$$

25. We thus lay down on the relative orbit the two points 22^{h} and 3^{h}, and the point of conjunction or intersection with the axis of x; the three points are found to be sensibly in a straight line: the distance between the extreme points is about 34 inches, representing 5 hours, so that the scale is nearly 7 inches to an hour: the line is then graduated to quarters of an hour. We then, by means of the distance $12+6{\cdot}53 = 18{\cdot}53$, mark off on the relative orbit, the points B, E, which correspond to the beginning and end of the eclipse respectively: the times as read off from my figure, and compared with the true times given in the N.A. are

	from figure	N.A.
Beginning	$22^{\text{h}}\,12^{\text{m}}{\cdot}5$	22 13·6
End	2 40·5	2 41·1

26. With centre B describing a circle radius 6·53 this will of course just touch the 12-inch circle, and the penumbral curve will be a mere point, viz., this is the point B' on the bounding circle, opposite to the point of contact. And so with centre E describing a circle of the same radius 6·53, that will just touch the 12-inch circle, and the penumbral curve will be a mere point, viz., the point E' on the bounding circle, opposite the point of contact.

27. I draw the circles corresponding to the times $22^{\text{h}}\,30^{\text{m}}$, 45^{m}, $23^{\text{h}}\,0^{\text{m}}$, viz., so much of each as lies within the 12-inch circle. Each of these is then transformed into a penumbral curve, drawn in the upper semicircle on the tracing cloth. For this purpose we construct a straight edge of paper, the one half graduated for $12\sin\theta$, the other half for $11{\cdot}08\tan\frac{1}{2}\theta$, by means of the proportional-scale figure, as already explained: $\theta = 0°$ to $90°$ at intervals of $5°$, is quite sufficient; the points on any particular penumbral curve are laid down in pairs with the utmost facility, and the curve is traced by hand from 4 or 5 pairs of points.

28. We then graduate for latitude; viz., we see through the tracing cloth, the equator cutting the vertical radius in Q, and a parallel cutting the same radius, say in R; drawing lines from C, we refer these to the points q, r on the bounding circle, viz., on the quadrant thereof which is graduated by means of the graduation-lines of the 12-inch circle; and we thus read off the latitude of the parallel in question; this latitude is then marked for each parallel on the vertical radius from Q up to the bounding circle, viz., not on the tracing cloth, but on the paper affix; and we then on this same radius (on the paper affix) interpolate the positions where this would be intersected by the parallels for the colatitudes, 5°, 10°, 15°, &c. Or

[p. 492]

(what is perhaps better) we may without marking the latitudes of the parallels of the blank form, construct directly the last-mentioned graduations; viz., marking off on the bounding circle from the point q, equal intervals of 5°, and from any such mark drawing to C, a line to meet the vertical radius, the point of intersection is the point belonging to the parallel, latitude equal to the corresponding multiple of 5°.

29. Finally, we must (not on the tracing cloth but on the paper affix) graduate an arc of the equator for the position of the meridian of Greenwich, that is for h. We have

$$\begin{aligned} \text{At } 22^{\text{h}} \quad h &= -2^{\text{h}} + 7\ 0{\cdot}30 = -1^{\text{h}}\,52^{\text{m}}\,52^{\text{s}}{\cdot}30 = -28°\,13'{\cdot}08, \\ \text{At } 3^{\text{h}} \quad h &= -2^{\text{h}} + 4\ 54\ 24{\cdot}90 = +2^{\text{h}}\,54^{\text{m}}\,24^{\text{s}}{\cdot}90 = +43°\,36'{\cdot}20. \end{aligned}$$

The equator is already graduated in longitude by means of the meridians of the blank projection: hence we lay down the marks for 22^{h} and 3^{h} in the positions belonging to $-28°\,13'$, and $+43°\,36'$ respectively. And then since the interval of 5 hours answers to $71°\,49'$, that of 1 hour will answer to $14°\,22'$, so that, measuring off these intervals of longitude, we have the marks for the intermediate times 23^{h}, 0^{h}, 1^{h}, 2^{h}; or it might be proper to find in this way the marks corresponding to each interval of 20^{m} of time, answering to about 5° of longitude; the further subdivisions would be proportional to the intervals of time.

30. I have in this way read off the positions of the points B' and E' belonging to the beginning and end of the eclipse; the values, as compared with the true ones, are

		From Figure	N.A.
B'	latitude N.	34°	35° 37′
	longitude W.	47°	45° 44′
E'	latitude N.	26°	26° 5′
	longitude W.	$38\frac{1}{2}°$	37° 16′

I remark that my figure, although drawn carefully, is not drawn with anything like the degree of accuracy which would be easily attainable; and I think that far better results might be obtained. I merely from a scale lay down tenths and estimate hundredths of an inch, but certainly fiftieths might be laid down from a scale.

[Scan page 525: Plate VI — graduated straight edge "Scale for graduation of straight edge," with the 12-inch circle and the proportional-scale figure used in the construction. Reproduction not embedded in the typeset edition; see the IA scan for the original.]

[Scan page 526 is blank.]

478.

ON THE GEODESIC LINES ON AN ELLIPSOID.

[From the Memoirs of the Royal Astronomical Society, vol. XXXIX. *(1872), pp. 31–53. Read January 13, 1871.]*

[p. 493]

The fundamental equations, in regard to the geodesic lines on an ellipsoid, were established by Jacobi, viz., representing by a, b, c, the squares of the semiaxes, that is, taking the ellipsoid to be

$$\frac{x^2}{a}+\frac{y^2}{b}+\frac{z^2}{c}=1$$

(where $a>b>c$), if we introduce the elliptic coordinates h, k, and write

$$\frac{x^2}{a+h}+\frac{y^2}{b+h}+\frac{z^2}{c+h}=1,$$

$$\frac{x^2}{a+k}+\frac{y^2}{b+k}+\frac{z^2}{c+k}=1,$$

or, what is the same thing,

$$x^2=\frac{a\,(a+h)(a+k)}{(a-b)(a-c)},\qquad y^2=\frac{b\,(b+h)(b+k)}{(b-c)(b-a)},\qquad z^2=\frac{c\,(c+h)(c+k)}{(c-a)(c-b)};$$

then, if β be an arbitrary constant, the differential equation of a geodesic line is

$$(1)\qquad \text{const.}=\int dh\sqrt{\frac{h}{(a+h)(b+h)(c+h)(\beta+h)}}+\int dk\sqrt{\frac{k}{(a+k)(b+k)(c+k)(\beta+k)}};$$

and the expression for the length of any arc of the curve is given by

$$(2)\qquad s=\int dh\sqrt{\frac{h\,(\beta+h)}{(a+h)(b+h)(c+h)}}+\int dk\sqrt{\frac{k\,(\beta+k)}{(a+k)(b+k)(c+k)}}.$$

I propose in the present Memoir to develope the theory to the extent of showing how we can, by means of the first of these equations, explain the course of the geodesic lines; and for given numerical values of a, b, c, calculate, construct, and exhibit in a drawing the course of these lines: I attend more particularly to the series of geodesic lines through an umbilicus (which lines pass also through the opposite umbilicus), and to the case where the semiaxes are connected by the equation $ac-b^2=0$, a relation which simplifies the formulae.

General Considerations as to the Course of the Lines.

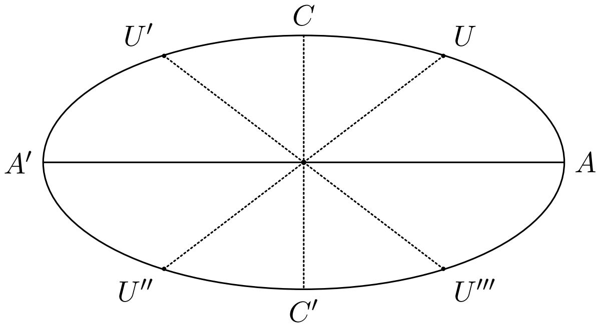
1. It will be observed that h and k enter into the formulae symmetrically: it will be convenient to distinguish between these coordinates by considering h as extending between the values $-a$, $-b$; and k as extending between the values $-b$, $-c$. Thus:

$h=$ const. denotes a curve of curvature of the one kind, viz.

$h=-a$, the principal section ABA' (or major-mean section), $h=-b$, the curves UU' and $U''U'''$ (or portions of the umbilicar section $ACA'C'$); similarly,

$k=$ const. denotes a curve of curvature of the other kind, viz.

$k=-c$, the principal section CBC' (or minor-mean section), $k=-b$, the curves UU''' and $U'U''$ (remaining portions of the umbilicar section $ACA'C'$).



2. To any given (admissible) values of h, k there correspond eight points, situate in the eight octants of the surface respectively; but, unless the contrary is expressed, it is assumed that the coordinates x, y, z, are positive, and that the point is situate in the octant ABC.

3. The constant β may have any value from $+a$ to $+c$; viz., if it has a value between a and b, or say, if $-\beta$ has an h-value, then the geodesic lines wholly

[p. 494]

between the two ovals of the curve of curvature $h=-\beta$ (being in general an indefinite undulating curve touching each oval an indefinite number of times). Similarly, if β has any value between b and c, or say, if $-\beta$ has a k-value, then the geodesic line lies wholly between the two ovals of the curve of curvature $k=-\beta$ (being in general an indefinite undulating curve touching each oval an indefinite number of times). The intermediate case is when $\beta=b$, or say when $-\beta$ has the umbilicar value: here the geodesic line is in general an indefinite undulating curve passing an infinite number of times through the opposite umbilici U, U'', or U', U'''; to fix the ideas, say through U, U''.

Lines through an Umbilicus.

4. I attend in particular to the last-mentioned case, and thus write $\beta=b$. We may in the formula (1) fix at pleasure a limit of each integral; and writing for convenience

$$\Pi(h)=\int_{-a}^{h}\frac{-dh}{b+h}\sqrt{\frac{h}{(a+h)(c+h)}},\qquad \Psi(k)=\int_{b}^{k}\frac{-dk}{b+k}\sqrt{\frac{k}{(a+k)(c+k)}},$$

the equation (1) becomes

$$\text{Const.}=\Pi(h)+\Psi(k).$$

5. It is to be observed, in regard to these integrals, that writing $h=-a+u$, we have

$$\Pi(h)=\int_{0}^{u}\frac{du}{a-b-u}\sqrt{\frac{a-u}{u\,(a-c-u)}},$$

which, for u small, is

$$=\frac{1}{a-b}\sqrt{\frac{a}{a-c}}\int_{0}^{u}\frac{du}{\sqrt{u}}=\frac{2\sqrt{u}}{a-b}\sqrt{\frac{a}{a-c}}.$$

By the assistance of this formula the value of the integral may be calculated by quadratures; viz., the formula gives the integral for any small value of u, and we can then proceed by the method of quadratures. The integral becomes infinite for $h=-b$: suppose that we have by quadratures calculated it up to $h=-b-m$ (m small), then to calculate it up to any value $-b-m+u$ nearer to $-b$, we have

$$\begin{aligned}\Pi(h)&=\Pi(-b-m)+\int_{m}^{m-u}\frac{du}{m-u}\sqrt{\frac{b+m-u}{(a-b-m+u)(b-c+m-u)}}\\&=\Pi(-b-m)+\sqrt{\frac{b}{(a-b)(b-c)}}\left[\log\frac{m}{m-u}\right]\\&=\Pi(-b-m)-\sqrt{\frac{b}{(a-b)(b-c)}}\log\left(1-\frac{u}{m}\right),(1)\end{aligned}$$

where the second term is positive, and the value thus increases slowly with u, becoming as it should do $=\infty$ for $u=m$ or $h=-b$.

[1] Except when the contrary is stated, the symbol "log" denotes throughout the hyperbolic logarithm.

[p. 495]

6. Similarly in the second integral writing $k=-c-v$, we have

$$\Psi(k)=\int_0^v \frac{dv}{b-c-v}\sqrt{\frac{c+v}{(a-c-v)\,v}},$$

which, for v small, is

$$=\frac{1}{b-c}\sqrt{\frac{c}{a-c}}\int_0^v \frac{dv}{\sqrt{v}}=\frac{2\sqrt{v}}{b-c}\sqrt{\frac{c}{a-c}},$$

which is of the like assistance in regard to the calculation by quadratures. And if we have by quadratures calculated the integral up to $k=-b+n$ (n small), then, to calculate it up to any value $-b+n-v$ nearer to $-b$, we have

$$\begin{aligned}\Psi(k)&=\Psi(-b+n)+\int_n^{n-v}\frac{dv}{n-v}\sqrt{\frac{b-n+v}{(a-b+n-v)(b-c-n+v)}}\\&=\Psi(-b+n)+\sqrt{\frac{b}{(a-b)(b-c)}}\left[\log\frac{n}{n-v}\right]\\&=\Psi(-b+n)-\sqrt{\frac{b}{(a-b)(b-c)}}\log\Big(1-\frac{v}{n}\Big),\end{aligned}$$

where the second term is positive, and the value thus increases slowly with v, becoming as it should do $=\infty$ for $v=n$, or $k=-b$.

7. It may be remarked that in $\Pi(h)$ and $\Psi(k)$ respectively the coefficient of the logarithmic term has in each case the same value $=\sqrt{\dfrac{b}{(a-b)(b-c)}}$. As regards the initial terms $\sqrt{u}$ and $\sqrt{v}$, the coefficients are $\dfrac{1}{a-b}\sqrt{\dfrac{a}{a-c}}$ and $\dfrac{1}{b-c}\sqrt{\dfrac{c}{a-c}}$ respectively, which are equal if $\dfrac{a}{(a-b)^2}=\dfrac{c}{(b-c)^2}$, or $ac-b^2=0$.

8. We may consider the two geodesic lines $\Pi(h)\pm\Psi(k)=\text{const.}$; suppose that these each of them pass through the point P, coordinates (h_0,k_0) in the ABC octant of the ellipsoid; then for one of them we have $\Pi(h)-\Psi(k)=\Pi(h_0)-\Psi(k_0)$, and for the other of them we have $\Pi(h)+\Psi(k)=\Pi(h_0)+\Psi(k_0)$: I attend first to the former: of these, say $\Pi(h)-\Psi(k)=C$ (where C is $=\Pi(h_0)-\Psi(k_0)$); and I say that this denotes the curve UPU''. In fact, by reason of the equation $\Pi(h)$ and $\Psi(k)$ must both increase or both diminish; they both increase as h passes from h_0 to $-b$, and as k passes from k_0 to $-b$: we may have $h=-b+u$, $k=-b+v$ where u and v are both indefinitely small, the functions Π and Ψ being then indefinitely large, but $\Pi-\Psi=C$; and we have thus a series of points nearer and nearer to the umbilicus U; that is, we have the portion PU of the curve. Tracing the curve in the opposite direction, or considering h as passing from h_0 to $-a$, and k as passing from k_0 to $-c$, then if C be positive, k will attain the value $-c$, before h attains the value $-a$, say that we have simultaneously $h=h_1$, $k=-c$; the equation is $\Pi(h_1)-\Psi(-c)=C$, that is, $\Pi(h_1)=C$; and the geodesic line then arrives at a point P_1 on the arc CB of the minor-mean

[p. 496]

principal section. The function Ψ then changes its sign, viz., considering it as always positive, the equation is now $\Pi(h)+\Psi(k)=C$, k passing from the value $-c$ towards $-b$, that is, $\Psi(k)$ increasing, and therefore $\Pi(h)$ diminishing, or h passing from h_1 towards the value $-a$; until at last, say for $k=k_2$, we have $h=-a$, that is, $C=\Pi(-a)+\Psi(k_2)$, or $C=\Psi(k_2)$; the geodesic line here arrives at a point P_2 on the arc BA' of the major mean principal section. The function Π then changes its sign, viz., Π denoting a positive function as before, the equation is $-\Pi(h)+\Psi(k)=C$; h passes from $-a$ towards $-b$, that is $\Pi(h)$ increases, and therefore $\Psi(k)$ must also increase, or k pass from k_2 towards $-b$: we have at length $h=-b-u$, $k=-b+v$, u and v being each indefinitely small; and therefore Π and Ψ each indefinitely large (but $-\Pi+\Psi=C$): that is, we arrive at the umbilicus U'', completing the geodesic line UPU''.

9. If instead of $C = +$ we have $C = -$, everything is similar, but the geodesic line proceeding from U in the direction UP will first cut the arc BA of the major mean section at a point P_1; then the arc BC of the minor mean section at a point P_2; and, finally, arrive as before at the umbilicus U''.

10. The intermediate case is when $C = 0$, viz., we have here $\Pi(h) - \Psi(k) = 0$; the geodesic line here passes from U in the direction UP to B (extremity of the mean axis, $h = -a$, $k = -c$); Π and Ψ then each change their sign, so that, considering them as positive, the equation still is $\Pi(h) - \Psi(k) = 0$, and the geodesic line at last arrives at the umbilicus U''. It will be easily understood how in the like manner $\Pi(h) + \Psi(k) = 0$ refers to the line $U'PU'''$.

11. Reverting to the equation $\Pi(h) - \Psi(k) = C$, or as I will now write it

$$\Pi(h) - \Psi(k) = \Pi(h_0) - \Psi(k_0),$$

which belongs to the portion UP of the geodesic line UPU'', we require when h is $= -b-u$, and $k = -b+v$ (u and v indefinitely small) to know the ratio of the increments u, v; this in fact serves to determine the direction at U of the geodesic line through the given point (h_0, k_0).

12. For this purpose writing $h = -b - u$, we find

$$\Pi(h) = \int_u^{b-c} \frac{du}{u} \sqrt{\frac{b+u}{(a-b-u)(b-c+u)}},$$

which is

$$= \int_u^{b-c} \frac{du}{u} \left\{ \sqrt{\frac{b+u}{(a-b-u)(b-c+u)}} - \sqrt{\frac{b}{(a-b)(b-c)}} \right\} + \sqrt{\frac{b}{(a-b)(b-c)}} \log \frac{b-c}{u},$$

and, when u is indefinitely small, this is

$$= \int_0^{b-c} \frac{du}{u} \left\{ \sqrt{\frac{b+u}{(a-b-u)(b-c+u)}} - \sqrt{\frac{b}{(a-b)(b-c)}} \right\} + \sqrt{\frac{b}{(a-b)(b-c)}} \log \frac{b-c}{u}.$$

[p. 498]

Similarly, when $k = -b + v$, where v is indefinitely small

$$\Psi(k) = \int_0^{b-c} \frac{dv}{v} \left\{ \sqrt{\frac{b-v}{(a-b+v)(b-c-v)}} - \sqrt{\frac{b}{(a-b)(b-c)}} \right\} + \sqrt{\frac{b}{(a-b)(b-c)}} \log \frac{b-c}{v}.$$

13. Each of the integrals is of the dimension $-\frac{1}{2}$ in a, b, c, and the difference of the integrals may be represented by

$$M\sqrt{\frac{b}{(a-b)(b-c)}};$$

we have therefore

$$\Pi(h) - \Psi(k) = \sqrt{\frac{b}{(a-b)(b-c)}} \left\{ M + \log \frac{u}{v} \right\},$$

where

$$\sqrt{\frac{b}{(a-b)(b-c)}}\, M = \int_0^{b-c} \frac{du}{u} \left\{ \sqrt{\frac{b+u}{(a-b-u)(b-c+u)}} - \sqrt{\frac{b}{(a-b)(b-c)}} \right\}$$
$$- \int_0^{b-c} \frac{dv}{v} \left\{ \sqrt{\frac{b-v}{(a-b+v)(b-c-v)}} - \sqrt{\frac{b}{(a-b)(b-c)}} \right\}.$$

14. Suppose the inferior limits replaced by the indefinitely small positive quantities ϵ, ϵ' respectively; and for the variable in the second integral write $-u$; then

$$M = \int_{-(b-c)}^{-\epsilon'} \frac{du}{u} + \int_{\epsilon}^{b-c} \frac{du}{u} \left\{ \sqrt{\frac{b+u}{(a-b-u)(b-c+u)}} - \sqrt{\frac{b}{(a-b)(b-c)}} \right\};$$

it being understood that the values $u = -\epsilon'$ to $u = +\epsilon$ are omitted from the integration: this is

$$= \int_{-(b-c)}^{b-c} \frac{du}{u} \left\{ \sqrt{\frac{b+u}{(a-b-u)(b-c+u)}} - \sqrt{\frac{b}{(a-b)(b-c)}} \right\} + \sqrt{\frac{b}{(a-b)(b-c)}} \log \frac{\epsilon}{\epsilon'} \cdot \frac{a-b}{b-c}.$$

with the same convention as to the integral; or if $\epsilon' = \epsilon$, then

$$M = M' - \sqrt{\frac{b}{(a-b)(b-c)}} \log \frac{a-b}{b-c},$$

where

$$\sqrt{\frac{b}{(a-b)(b-c)}}\, M' = \int_{-(b-c)}^{b-c} \frac{du}{u} \sqrt{\frac{b+u}{(a-b-u)(b-c+u)}} = \int_{-a}^{-c} \frac{dh}{b+h} \sqrt{\frac{h}{(a+h)(c+h)}},$$

the omitted elements being from $u = -\epsilon$ to $u = +\epsilon$; that is (in the language of Cauchy) we take for the integral its principal value. And hence

$$\Pi(h) - \Psi(k) = \sqrt{\frac{b}{(a-b)(b-c)}} \left\{ M' + \log \frac{u}{v} \right\}.$$

15. By what precedes this is $= \Pi(h_0) - \Psi(k_0)$; or if we write simply (h, k) instead of (h_0, k_0), that is, consider the geodesic line UP, which is drawn from the

[p. 499]

point P, coordinates (h, k), to the umbilicus U, the coordinates of a point consecutive to the umbilicus are $-b-u$, $-b+v$, where u, v are connected by the last-mentioned equation, in which M' is a transcendental function depending on (a, b, c) but independent of the particular geodesic line.

16. If for the geodesic line through the point B, or say for the B-geodesic

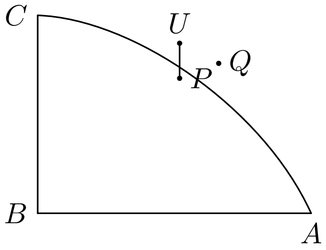
$$\frac{u}{u_0} = \frac{v}{v_0}, \text{ then } M' = -\log\frac{u_0}{v_0},$$

and we have in general

$$\Pi(h) - \Psi(k) = \sqrt{\frac{b}{(a-b)(b-c)}}\log\frac{v\,u_0}{u\,v_0},$$

a result which I proceed to further transform as follows:

If x_0, y_0, z_0 refer to the umbilicus U, then considering first the consecutive point P on the geodesic line (coordinates $-b-u$, $-b+v$) and next the consecutive point Q on the umbilicar section, we have for these two points respectively,



$$dx_0 = \frac{\frac{1}{2}\sqrt{a}\,(v-u)}{\sqrt{(a-b)(a-c)}},$$
$$dy_0 = \frac{\sqrt{b}\sqrt{uv}}{\sqrt{(a-b)(b-c)}},$$
$$dz_0 = \frac{\frac{1}{2}\sqrt{c}\,(v-u)}{\sqrt{(b-c)(a-c)}};$$
$$\delta x_0 = \frac{\frac{1}{2}\sqrt{a}}{\sqrt{(a-b)(a-c)}},$$
$$\delta y_0 = 0,$$
$$\delta z_0 = \frac{-\frac{1}{2}\sqrt{c}}{\sqrt{(b-c)(a-c)}};$$

[p. 500]

say these are α, β, γ, and α', β', γ'; and then

$$\begin{aligned}\alpha\alpha'+\beta\beta'+\gamma\gamma' &= \tfrac{1}{4}\left\{\frac{a}{(a-b)(a-c)}+\frac{c}{(b-c)(a-c)}\right\}(v-u)\\ &= \tfrac{1}{4}\cdot\frac{(v-u)}{a-c}\left\{\frac{a}{a-b}+\frac{c}{b-c}\right\}=\tfrac{1}{4}\cdot\frac{b\,(v-u)}{(a-b)(b-c)},\\ \alpha^2+\beta^2+\gamma^2 &= \tfrac{1}{4}\left\{\frac{a}{(a-b)(a-c)}+\frac{c}{(b-c)(a-c)}\right\}(v-u)^2+\frac{b\,uv}{(a-b)(b-c)}\\ &= \tfrac{1}{4}\cdot\frac{b\,(v-u)^2}{(a-b)(b-c)}+\frac{b\,uv}{(a-b)(b-c)}\\ &= \frac{b}{(a-b)(b-c)}\cdot\tfrac{1}{4}(u+v)^2,\\ \alpha'^2+\beta'^2+\gamma'^2 &= \frac{b}{(a-b)(b-c)}\cdot\tfrac{1}{4},\end{aligned}$$

whence

$$\sqrt{\alpha^2+\beta^2+\gamma^2}\sqrt{\alpha'^2+\beta'^2+\gamma'^2}=\frac{\tfrac{1}{4}(u+v)\,b}{(a-b)(b-c)},$$

and hence

$$\cos\phi=\frac{\alpha\alpha'+\beta\beta'+\gamma\gamma'}{\sqrt{\alpha^2+\beta^2+\gamma^2}\sqrt{\alpha'^2+\beta'^2+\gamma'^2}}=\frac{v-u}{u+v},$$

that is,

$$\cos(180^\circ-\phi)=\frac{u-v}{u+v},\quad\text{or}\quad\tan^2\tfrac{1}{2}\phi=\frac{v}{u},$$

where it U is the umbilicus, P the consecutive point $-b-u$, $-b+v$, and UQ the element of the umbilicar principal section, $\phi=\angle PUQ$, $180^\circ-\phi=\angle PUQ'$. For the B-geodesic we have

$$2\log\tan\tfrac{1}{2}\phi_0=\log\frac{v_0}{u_0}=M'.$$

17. The foregoing equation for $\Pi(h)-\Psi(k)$ now becomes

$$\Pi(h)-\Psi(k)=\sqrt{\frac{b}{(a-b)(b-c)}}\log\frac{\tan^2\tfrac{1}{2}\phi_0}{\tan^2\tfrac{1}{2}\phi},$$

viz., ϕ_0 is the south azimuth of the B-geodesic at the umbilicus, a mere function of (a,b,c) and ϕ is the south azimuth at the umbilicus, of the geodesic line under consideration, so that we may consider the geodesic line to be determined by the south azimuth ϕ as its parameter.

[p. 501]

Formulæ for the case $ac - b^2 = 0$.

18. I annex the following investigation in regard to the case $ac - b^2 = 0$.

We have in general

$$\frac{1}{\sqrt{b\,(a-b)(b-c)}}\,\frac{d}{dx}\log\frac{\sqrt{-x\,(a-b)(b-c)}+\sqrt{-b\,(a+x)(c+x)}}{\sqrt{-x\,(a-b)(b-c)}-\sqrt{-b\,(a+x)(c+x)}} = -\frac{1}{b}\cdot\frac{1}{\sqrt{x\,(a+x)(c+x)}} + \frac{1}{b}\cdot\frac{1}{b+x}\sqrt{\frac{x}{(a+x)(c+x)}} - \frac{1}{bx+ac}\sqrt{\frac{x}{(a+x)(c+x)}}.$$

In fact, denoting the logarithm by $\log\dfrac{P+Q}{P-Q}$, we have

$$\frac{d}{dx}\log\frac{P+Q}{P-Q} = \frac{2(PQ'-P'Q)}{P^2-Q^2},$$

where

$$\begin{aligned}2(PQ'-P'Q) = 2PQ\left(\frac{Q'}{Q}-\frac{P'}{P}\right) &= \sqrt{x\,(a+x)(c+x)}\,b\,(a-b)(b-c)\left(\frac{Q'}{Q}-\frac{P'}{P}\right)\\ &= \frac{\sqrt{b\,(a-b)(b-c)}}{\sqrt{x\,(a+x)(c+x)}}(x^2-ac),\end{aligned}$$

and

$$P^2 - Q^2 = -x\,(a-b)(b-c) + b\,(a+x)(c+x) = (bx+ac)(b+x);$$

that is

$$\begin{aligned}\frac{2(PQ'-P'Q)}{P^2-Q^2} &= \frac{\sqrt{b\,(a-b)(b-c)}}{\sqrt{x\,(a+x)(c+x)}}\cdot\frac{x^2-ac}{(bx+ac)(b+x)}\\ &= \frac{\sqrt{b\,(a-b)(b-c)}}{\sqrt{x\,(a+x)(c+x)}}\left\{-\frac{1}{b}+\frac{x}{b(b+x)}+\frac{x}{bx+ac}\right\},\end{aligned}$$

which proves the theorem.

19. Hence in the particular case $ac = b^2$ we have

$$\frac{1}{\sqrt{b\,(a-b)(b-c)}}\log\frac{\sqrt{-h\,(a-b)(b-c)}+\sqrt{-b\,(a+h)(c+h)}}{\sqrt{-h\,(a-b)(b-c)}-\sqrt{-b\,(a+h)(c+h)}} = -\frac{1}{b}\int_{-a}^{h}\frac{dh}{\sqrt{h\,(a+h)(c+h)}} + \frac{2}{b}\int_{-a}^{h}\frac{dh}{b+h}\sqrt{\frac{h}{(a+h)(c+h)}},$$

[p. 502]

that is

$$\Pi(h) = \tfrac{1}{2}\int_{-a}^{h}\frac{dh}{\sqrt{h\,(a+h)(c+h)}} + \tfrac{1}{2}\sqrt{\frac{b}{(a-b)(b-c)}}\log\frac{\sqrt{-h\,(a-b)(b-c)}+\sqrt{-b\,(a+h)(c+h)}}{\sqrt{-h\,(a-b)(b-c)}-\sqrt{-b\,(a+h)(c+h)}}$$

or say

$$\Pi(h) = \tfrac{1}{2}\int_{-a}^{h}\frac{dh}{\sqrt{h\,(a+h)(c+h)}} + \tfrac{1}{2}\sqrt{\frac{b}{(a-b)(b-c)}}\log\frac{1+H}{1-H},$$

where

$$H^2 = \frac{b}{(a-b)(b-c)} \cdot \frac{(a+h)(c+h)}{h},$$

viz. we see that $\Pi(h)$ depends on the more simple integral

$$\int \frac{dh}{\sqrt{h\,(a+h)(c+h)}}.$$

20. Similarly

$$\frac{1}{\sqrt{b\,(a-b)(b-c)}} \log \frac{\sqrt{-k\,(a-b)(b-c)} + \sqrt{-b\,(a+k)(c+k)}}{\sqrt{-k\,(a-b)(b-c)} - \sqrt{-b\,(a+k)(c+k)}} = -\frac{1}{b}\int_b^{-c} \frac{dk}{\sqrt{k\,(a+k)(c+k)}} + \frac{2}{b}\int_b^{-c} \frac{dk}{b+k}\sqrt{\frac{k}{(a+k)(c+k)}},$$

that is

$$\Psi(k) = \tfrac{1}{2}\int_b^{-c} \frac{dk}{\sqrt{k\,(a+k)(c+k)}} + \tfrac{1}{2}\sqrt{\frac{b}{(a-b)(b-c)}} \log \frac{\sqrt{-k\,(a-b)(b-c)} + \sqrt{-b\,(a+k)(c+k)}}{\sqrt{-k\,(a-b)(b-c)} - \sqrt{-b\,(a+k)(c+k)}},$$

or say

$$\Psi(k) = \tfrac{1}{2}\int_b^{-c} \frac{dk}{\sqrt{k\,(a+k)(c+k)}} + \tfrac{1}{2}\sqrt{\frac{b}{(a-b)(b-c)}} \log \frac{1+K}{1-K},$$

where

$$K^2 = \frac{b}{(a-b)(b-c)} \cdot \frac{(a+k)(c+k)}{k},$$

that is, $\Psi(k)$ depends on the more simple integral,

$$\int \frac{dk}{\sqrt{k\,(a+k)(c+k)}}.$$

Write $h = -b - u$, $k = -b + v$, where u and v are indefinitely small, then

$$\Pi(h) - \Psi(k) = \tfrac{1}{2}\int_{-a}^{-c} \frac{dh}{\sqrt{h\,(a+h)(c+h)}} + \tfrac{1}{2}\sqrt{\frac{b}{(a-b)(b-c)}} \log \frac{1+U}{1-U} \cdot \frac{1-V}{1+V},$$

[p. 503]

where

$$U^2 = \frac{\left(1+\dfrac{u}{a-b}\right)\left(1+\dfrac{u}{b-c}\right)}{1+\dfrac{u}{b}} = 1 - \frac{bu^2}{(a-b)(b-c)(b+u)} \quad (\text{attending to } ac = b^2),$$

and

$$V^2 = \frac{\left(1+\dfrac{v}{a-b}\right)\left(1-\dfrac{v}{b-c}\right)}{1-\dfrac{v}{b}} = 1 - \frac{bv^2}{(a-b)(b-c)(b-v)}.$$

21. Comparing with the result obtained for the general case the two agree, if only

$$\int_{-a}^{-c} \frac{dh + b\sqrt{1+\dfrac{(a+h)(c+h)}{h}}}{b+h} = \tfrac{1}{2}\int_{-a}^{-c} \frac{dh}{\sqrt{h\,(a+h)(c+h)}},$$

where on the left-hand side the integral has its principal value: a result which must therefore hold good when $ac = b^2$.

Calculation of the Umbilicar Geodesies for Ellipsoid $a:b:c=4:2:1$.

22. As a specimen of the way in which we may, on a given ellipsoid, calculate the course of a geodesic line, I take the semiaxes to be as $2:\sqrt{2}:1$, or, for convenience, $a=1000$, $b=500$, $c=250$; and, considering the geodesic lines through the umbilicus, I calculate by quadratures the functions

$$\Pi(h)=100{,}000\int_{-1000}^{h}\frac{-dh}{500+h}\sqrt{\frac{h}{(1000+h)(250+h)}},$$

$$\Psi(k)=100{,}000\int_{b}^{k}\frac{-dk}{500+k}\sqrt{\frac{k}{(1000+k)(250+k)}}.$$

The results do not pretend to minute accuracy: I have not attempted to estimate or correct for any error occasioned by the intervals (10 units) being too large; and there may possibly be accidental errors.

[p. 504]

TABLE I.

$-h$	Π'	$\Pi(h)$	$-h$	Π'	$\Pi(h)$	$-h$	Π'	$\Pi(h)$
1000	∞	0	840	27.6	6746	630	51.5	13972
999	231.4	462	830	27.8	7023	620	54.1	14499
998	164	659	820	27.9	7301	610	59.2	15066
997	134.2	809	810	28.1	7582	600	65.5	15689
996	116.5	934	800	28.4	7865	590	72.1	16377
995	104.4	1044	790	28.7	8151	580	80.9	17142
990	74.6	1492	780	29.2	8440	570	93.0	18011
980	54.0	2135	770	29.7	8735	560	106.8	19010
970	45.1	2630	760	30.3	9035	550	127.6	20183
960	40.1	3056	750	31.0	9341	540	159.1	21616
950	36.6	3439	740	31.8	9655	530	211.5	23469
940	34.2	3794	730	32.6	9977	520	316.7	26111
930	32.5	4127	720	33.6	10308	510	632.7	30858
920	31.2	4446	710	34.7	10650	505	1265	35602
910	30.2	4753	700	36.0	11004	504	1581.2	37014
900	29.4	5051	690	37.4	11371	503	2107.7	38834
890	28.8	5342	680	39.0	11754	502	3162.3	41398
880	28.4	5628	670	40.8	12153	501	6324.1	45792
870	28.1	5911	660	42.0	12567	500	∞	∞
860	27.9	6190	650	45.4	13005			
850	27.8	6469	640	48.2	13473			

[p. 505]

TABLE II.

$-k$	Ψ'	$\Psi(k)$	$-k$	Ψ'	$\Psi(k)$	$-k$	Ψ'	$\Psi(k)$
250	∞	0	320	45.5	4655	440	107.1	12207
251	232.2	462	330	46.1	5114	450	127.9	13383
252	165.5	661	340	47.3	5581	460	159.2	14818
253	136.0	811	350	48.9	6062	470	211.6	16673
254	118.6	939	360	51.1	6562	480	316.7	19314
255	106.8	1051	370	53.8	7086	490	632.7	24062
260	78.1	1514	380	57.2	7641	495	1265.0	28806
270	60.5	2207	390	61.4	8235	496	1581.2	30218
280	51.7	2768	400	66.7	8875	497	2108.1	32037
290	48.2	3268	410	73.2	9575	498	3162.3	34602
300	46.3	3741	420	81.6	10349	499	6234.1	38995
310	45.5	4200	430	91.4	11214	500	∞	∞

23. But it is obviously convenient to revert these Tables so as to have for the common arguments a series of uniformly increasing values of Π or Ψ, viz., we obtain by interpolation the values of h and k belonging to the given values of Π or Ψ, and thus obtain the following Table. Here, in any line of the Table the values of h, k are such that $\Pi(h) - \Psi(k) = 0$, viz., the values in question belong to successive points of the B-geodesic. And to obtain the values for any other geodesic line $\Pi(h) - \Psi(k) = \pm 500m$, we have only to take each value of k from the line m lines above or below the line from which h is taken; and similarly the table gives at once the values belonging to a geodesic line $\Pi(h) + \Psi(k) = 500m$.

[p. 506]

TABLE III.

$\Pi = \Psi =$	h	D.	k	D.	$\Pi = \Psi =$	h	D.	k	D.
0	1000		250		13000	650.1		446.7	
		1.2		1.4			20.6		7.8
500	998.8		251.4		14000	629.5		454.5	
		3.4		3.1			18.3		6.5
1000	995.4		254.5		15000	611.2		461.0	
		5.5		5.2			15.7		5.3
1500	989.9		259.7		16000	595.5		466.3	
		7.8		7.3			13.6		4.9
2000	982.1		267.0		17000	581.9		471.2	
		9.5		8.2			11.8		3.8
2500	972.6		275.2		18000	570.1		475.0	
		11.5		9.4			10.0		3.8
3000	961.1		284.6		19000	560.1		478.8	
		12.8		10.3			8.5		2.6
3500	948.3		294.9		20000	551.6		481.4	
		14.5		10.7			7.3		2.2
4000	933.8		305.6		21000	544.3		483.6	
		15.6		11.0			6.2		2.0
4500	918.2		316.6		22000	538.1		485.6	
		16.5		10.9			4.9		2.2
5000	901.7		327.5		23000	533.2		487.8	
		17.2		10.8			4.6		2.1
5500	884.5		338.3		24000	528.6		489.9	
		17.7		10.4			8.1		2.1
6000	866.8		348.7		26000	520.5		492.0	
		17.9		10.1			4.5		2.1
6500	848.9		358.8		28000	516.0		494.1	
		18.1		9.5			4.2		1.7
7000	830.8		368.3		30000	511.8		495.8	
		17.9		9.2			3.0		1.2
7500	812.9		377.5		32000	508.8		497.0	
		17.6		8.6			2.1		0.8
8000	795.3		386.1		34000	506.7		497.8	
		17.3		8.0			2.0		0.5
8500	778.0		394.1		36000	504.7		498.3	
		16.8		7.7			1.2		0.5
9000	761.2		401.8		38000	503.5		498.8	
		16.3		7.1			1.0		
9500	744.9		408.9		39000			499.0	
		15.6		6.6					
10000	729.3		415.5		40000	502.5			
		14.9		6.2			0.6		
10500	714.4		421.7		42000	501.9			
		14.3		5.9			0.6		
11000	700.1		427.6		44000	501.4			
		13.5		5.3			0.4		
11500	686.6		432.9		45800	501.0			
		12.8		5.0					
12000	673.8		437.9		∞	500		500	
		23.7		8.8					

[p. 507]

Graphical Construction: Projection on the Umbilicar Plane.

24. The most convenient mode of delineation of the geodesic lines is obtained by projecting them orthogonally on the umbilicar plane: the contour of the figure is here the umbilicar section, or ellipse

$$\frac{x^2}{a}+\frac{z^2}{c}=1\,;$$

and the curves of curvature of each series are projected into elliptic arcs lying within the ellipse in question, the one set cutting at right angles the axes AA', the other cutting at right angles the axes CC'; the equations of the complete ellipses being

$$x^2\,\frac{a-b}{a\,(a+h)}+z^2\,\frac{c-b}{c\,(c+h)}-1=0$$

and

$$x^2\,\frac{a-b}{a\,(a+k)}+z^2\,\frac{c-b}{c\,(c+k)}-1=0.$$

25. I constructed, by means of the table, a drawing of this kind for the ellipsoid a, b, $c=1000$, 500, 250, the lengths $\sqrt{a}$ and $\sqrt{c}$ being taken to be 12 inches and 6 inches respectively: the process consists in taking from the table for a series of values $\Pi=\Psi$ (say $\Pi=\Psi=1000$, $=2000$ &c.), the values of h and k, laying down for such values the elliptic arcs which represent the two curves of curvature respectively, thus dividing the bounding ellipse into a series of curvilinear rectangles, and then obtaining the geodesic lines by drawing the diagonals of these rectangles, and of course rounding off the corners so as to form continuous curves. The Plate shows on a reduced scale so much of the drawing as is comprised within a quadrant of the bounding ellipse (viz. it is a representation of an octant of the ellipsoid).

Elliptic-Function Formulæ.

26. I have in all that precedes abstained from the use of elliptic functions, since obviously the form $\sqrt{1-k^2\sin^2}$ of the radical of an elliptic function is in nowise specially appropriate to the present question. But (more particularly in the above-mentioned case $ac-b^2=0$, where the radical is $\sqrt{h\,(a+h)(c+h)}$ without any exterior factor $b+h$ in the denominator) the formulae are expressible easily and elegantly by elliptic functions, and it is desirable to make the transformation. Reverting to the formulae which, in the case in question (viz. when $ac-b^2=0$), give the values of $\Pi(h)$ and $\Psi(k)$; and writing therein $h=-a+(a-c)\sin^2\phi$, $k=-a+(a-c)\sin^2\psi$, also

$$\kappa=\sqrt{1-\frac{c}{a}},\quad\text{or }\frac{c}{a}=1-\kappa^2=\kappa'^2,$$

we have

$$\int_{-a}^{h}\frac{dh}{\sqrt{h\,(a+h)(c+h)}}=2\int_{0}^{\phi}\frac{d\phi}{\sqrt{a-(a-c)\sin^2\phi}}=\frac{2}{\sqrt{a}}\,F(\kappa,\phi),$$

$$\int_{b}^{k}\frac{dk}{\sqrt{k\,(a+k)(c+k)}}=2\int_{\frac{1}{2}\pi}^{\psi}\frac{d\psi}{\sqrt{a-(a-c)\sin^2\psi}}=\frac{2}{\sqrt{a}}\{F_1(\kappa)-F(\kappa,\psi)\}.$$

[p. 508]

27. Hence

$$\Pi(h)=\frac{1}{\sqrt{a}}\,F(\kappa,\phi)+\frac{1}{2\sqrt{a\,(1-\kappa')}}\log\frac{1+H}{1-H},$$

where

$$H=\frac{(1-\kappa')\sin\phi\cos\phi}{\sqrt{1-\kappa^2\sin^2\phi}};$$

(observe, as h passes from $-a$ to $-b$, ϕ passes from $\phi=0$ to $\sin^2\phi=\dfrac{1}{1+\kappa'}$, and H from $H=0$ to $H=1$).

Similarly

$$\Psi(k)=\frac{1}{\sqrt{a}}\{F_1(\kappa)-F(\kappa,\psi)\}+\frac{1}{2\sqrt{a\,(1-\kappa')}}\log\frac{1+K}{1-K},$$

where

$$K=\frac{(1+\kappa')\sin\psi\cos\psi}{\sqrt{1-\kappa^2\sin^2\psi}};$$

and as k passes from $-c$ to $-b$, ψ passes from $\frac{1}{2}\pi$ to $\sin^2\psi=\dfrac{1}{1+\kappa'}$, and K from 0 to 1.

28. The before-mentioned identical equation

$$\int_{-a}^{-c}\frac{dh}{b+h}\sqrt{\frac{h}{(a+h)(c+h)}}=\tfrac{1}{2}\int_{-a}^{-c}\frac{dh}{\sqrt{h\,(a+h)(c+h)}}$$

is by the same transformation converted into

$$\int_0^{\frac{1}{2}\pi}\frac{1-(1-\kappa')\sin^2\phi}{1-(1+\kappa')\sin^2\phi}\cdot\frac{d\phi}{\sqrt{1-\kappa^2\sin^2\phi}}=0.$$

To prove this, I remark that the equation is

$$0=\int_0^{\frac{1}{2}\pi}d\phi\,\frac{\dfrac{1-\kappa'}{1+\kappa'}+\dfrac{2\kappa'}{1+\kappa'}\cdot\dfrac{1}{1-(1+\kappa')\sin^2\phi}}{\Delta\phi},$$

viz. this is

$$0=\frac{1-\kappa'}{1+\kappa'}\,F_1+\frac{2\kappa'}{1+\kappa'}\,\Pi_1(-1-\kappa'),$$

or, what is the same thing,

$$\Pi_1(-1-\kappa')=-\frac{1-\kappa'}{2\kappa'}\,F_1,$$

where $\Pi_1(-1-\kappa')$ denotes the principal value of the integral

$$\int_0^{\frac{1}{2}\pi}d\phi\,\frac{1}{1-(1+\kappa')\sin^2\phi}\cdot\frac{1}{\Delta\phi}.$$

[p. 509]

Now (Leg. *Fonct. Ellip.*, t. i., p. 71), we have

$$\Pi_1(-\kappa^2\sin^2\theta)+\Pi_1\left(-\frac{1}{\sin^2\theta}\right)=F_1,$$

where, upon examination, it will appear that $\Pi_1\left(-\dfrac{1}{\sin^2\theta}\right)$ in fact represents the principal value of the integral.

Writing herein $\sin^2\theta=\dfrac{1}{1+\kappa'}$, and therefore $\cos^2\theta=\dfrac{\kappa'}{1+\kappa'}$, or $\tan^2\theta=\kappa'$, this is

$$\Pi_1(-1+\kappa')+\Pi_1(-1-\kappa')=F_1,$$

and the formula (p'), p. 141, attributing therein to θ the foregoing value, becomes

$$\Pi_1(-1-\kappa') = \theta_1 + \frac{1}{\kappa'}\{F_1 E(\theta) - E_1 F(\theta)\}.$$

But θ is the value for the bisection of the function F_1, viz., we have

$$\begin{aligned} 2F(\theta) &= F_1, \\ 2E(\theta) &= E_1 + 1 - \kappa', \end{aligned}$$

whence

$$F_1 E(\theta) - E_1 F(\theta) = \tfrac{1}{2}(1-\kappa')F_1,$$

or the formula in question gives

$$\Pi_1(-1+\kappa') = \frac{1+\kappa'}{2\kappa'}\,F_1,$$

whence

$$\Pi_1(-1-\kappa') = -\frac{1-\kappa'}{2\kappa'}\,F_1,$$

the result which was to be proved.

29. The value of M' (observing that $\sqrt{(a-b)(b-c)} = \dfrac{b}{\sqrt{a-c}} = \dfrac{b}{\sqrt{a}\,\kappa}$) is

$$\frac{1}{\sqrt{a\,(1-\kappa')}}\,M' = \tfrac{1}{2}\int_{-a}^{-c} \frac{dh}{\sqrt{h\,(a+h)(c+h)}},$$

which is

$$= \tfrac{1}{2}\cdot\frac{2}{\sqrt{a}}\,F_1(\kappa),$$

that is we have

$$M' = (1-\kappa')F_1(\kappa),$$

or, what is the same thing,

$$\log\tan\tfrac{1}{2}\phi_0 = \frac{1-\kappa'}{2}\,F_1(\kappa),$$

that is

$$\tan\tfrac{1}{2}\phi_0 = e^{\frac{1-\kappa'}{2}\,F_1(\kappa)}$$

(ϕ_0 the South azimuth of the B-geodesic at the umbilicus).

30. I purposely calculated the Table by quadratures as being a method available where the equation $ac - b^2 = 0$ is not satisfied; but in the present case, where this

[p. 510]

equation is satisfied, the table might have been calculated from Legendre's Tables of Elliptic Integrals. Observe that $a = 1000$, $b = 500$, $c = 250$, gives $\kappa = \frac{1}{2}\sqrt{3}$ or angle of modulus $= 60°$. As an instance of the comparison (^{1}), suppose $h = -800$, then $\sin^2\phi = \dfrac{200}{750}$, $\log\sin\phi = 9{\cdot}71298$, $\phi = 31°\,5'$.

$$\sqrt{\frac{b}{(a-b)(b-c)}} = \sqrt{\frac{500}{500\cdot 250}} = \frac{\sqrt{10}}{50} = {\cdot}06325,$$

$$H^2 = \frac{500\cdot 200\cdot 550}{800\cdot 500\cdot 250} = \frac{110}{200}, \quad \log = \bar{1}{\cdot}87018,$$

$$H = {\cdot}7416\;\frac{1+H}{1-H} = \frac{1{\cdot}7416}{{\cdot}2584} = 6{\cdot}7582,$$

$$\Pi(h) = {\cdot}03163\,F(31°\,5') + {\cdot}03163\,\mathrm{h.\,l.}\,6{\cdot}7582,$$

$$\begin{array}{rl}
F\,31^\circ \;\; = & \cdot 56166 \\
 & \;\;\;163 \\
F\,31^\circ\,5' = & \cdot 56329 \\
\text{h. l. } 6{\cdot}7582 = & 1{\cdot}91075 \\
 & 2{\cdot}47404 \\
\times \text{ by} & \cdot\,03163 \\
 & \cdot\,0782043
\end{array}$$

or multiplying by 100,000 (factor introduced into my Table) this is $= 7820{\cdot}43$. The value $\Pi(-800) = 7864$ given by my Table agrees sufficiently well with this, the correct value.

31. I calculate also the angle ϕ_0, viz. we have

$$\text{h. l. } \tan\tfrac{1}{2}\phi_0 = \frac{1-\kappa'}{2}\,F_1\kappa = \tfrac{1}{4}F_1(60^\circ). \quad \text{Leg. vol. iii. Table viii.}$$

$$= \tfrac{1}{4}\cdot 2{\cdot}15651 = {\cdot}53913,$$

whence by Leg. Table iv.

$$\begin{aligned}
\tfrac{1}{2}\phi_0 &= 45^\circ + \tfrac{1}{2}\cdot 29^\circ 29'{\cdot}64 \\
&= 59^\circ\, 44'{\cdot}82
\end{aligned}$$

or

$$\phi_0 = 119^\circ 29'{\cdot}64.$$

This exceeds 90°, and since at the umbilicus the tangent plane is at right angles to the plane of projection, the B-geodesic should in the drawing proceed (as it in fact does) from U in the sense UC, touching the bounding ellipse at the point U.

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[1] In the present calculation, log denotes an ordinary logarithm, the hyperbolic logarithm being distinguished as **h. l.**

[Plate VII: graphical projection of geodesic lines on an ellipsoid $a:b:c = 4:2:1$ *within a quadrant of the umbilicar ellipse, showing the network of curves of curvature and their diagonal geodesics; numerical labels along the bounding ellipse mark the values* $\Pi = \Psi = 1000$, 2000, …. *Cayley's Papers VII.]*

[p. 511]

479.

THE SECOND PART OF A MEMOIR ON THE DEVELOPMENT OF THE DISTURBING FUNCTION IN THE LUNAR AND PLANETARY THEORIES.

[From the Memoirs of the Royal Astronomical Society, vol. XXXIX. *(1872), pp. 55–74. Read January 12, 1872.]*

The present communication is a sequel to my paper, "The First Part of a Memoir on the Development of the Disturbing Function in the Lunar and Planetary Theories," *Memoirs R.A.S.*, vol. XXVIII. (1859), pp. 187–215, [214], and I have therefore entitled it as above, but it, in fact, relates only to the Planetary Theory. In the First Part, I gave in effect, but not explicitly, an expression for the general coefficient $D(j,j')$ in terms of the coefficients of the multiple cosines of θ in the expansions of the several powers $(r^2+r'^2-2rr'\cos\theta)^{-s-\frac{1}{2}}$, or say $(a^2+a'^2-2aa'\cos\theta)^{-s-\frac{1}{2}}$; viz., at the foot of page 208 I speak of the term involving $\cos(jU+j'U')$ as having a certain given value; the term in question is $D(j,j')\cos(jU+j'U')$; and consequently the expression for $D(j,j')$ is

$$D(j,j')=\Sigma\,\frac{\Pi_{\frac{1}{2}}(x-\frac{1}{2})}{\Pi x}\,\eta^{2x}\,\Sigma M_x^{\xi}\,R_x^{\xi};$$

the omission was, however, a material one, inasmuch as this expression for the general coefficient serves to connect my formulae with Leverrier's development, *Annales de l'Observ. de Paris*, t. i. (1855), pp. 275–330 and 358–383, and I resume the question for the purpose of applying it.

Formula for the general Coefficient $D(j,j')$.

In the First Part, the reciprocal of the distance of the two planets, or function

$$\{r^2+r'^2-2rr'(\cos U\cos U'+\sin U\sin U'\cos\Phi)\}^{-\frac{1}{2}}$$

[p. 512]

is taken to be developed in multiple cosines of U, U', the general term being

$$D(j,j')\cos(jU+j'U'),$$

where j, j' have each of them any integer value from $-\infty$ to $+\infty$ (zero not excluded), but so that j, j' are simultaneously even or simultaneously odd. We have $D(-j,-j')=D(j,j')$ and $D(j',j)=D(j,j')$; and it hence appears that the really distinct values of the coefficient may be taken to be those for which j is not negative, and as regards absolute magnitude is not less than j'; and for such values of j, j' we have the above-mentioned expression

$$D(j,j')=\Sigma\,\frac{\Pi_{\frac{1}{2}}(x-\frac{1}{2})}{\Pi x}\,\eta^{2x}\,\Sigma M_x^{\xi}\,R_x^{\xi},$$

which I proceed to explain and develope.

$\Pi_{\frac{1}{2}}(x-\frac{1}{2})$ and Πx (x being a positive integer) denote respectively $\frac{1}{2}\cdot\frac{3}{2}\ldots(x-\frac{1}{2})$, and $1\cdot 2\cdot 3\ldots x$; in particular for $x=0$, the value of each factorial is $=1$.

η denotes $\sin\frac{1}{2}\Phi$.

The coefficients R_x^{ξ} are those of the multiple cosines in certain developments, viz. we have

$$r^s r'^s(r^2+r'^2-2rr'\cos(U-U'))^{-s-\frac{1}{2}}=\Sigma R_x^{\xi}\cos i(U-U'),$$

where, as usual, i extends from $-\infty$ to ∞ and $R_x^{-i} = R_x^i$. Writing with Leverrier

$$\begin{aligned}(a^2+a'^2-2aa'\cos H)^{-\frac{1}{2}} &= \tfrac{1}{2}\Sigma A^i\cos iH,\\ aa'(a^2+a'^2-2aa'\cos H)^{-\frac{3}{2}} &= \tfrac{1}{2}\Sigma B^i\cos iH,\\ a^2a'^2(a^2+a'^2-2aa'\cos H)^{-\frac{5}{2}} &= \tfrac{1}{2}\Sigma C^i\cos iH,\\ a^3a'^3(a^2+a'^2-2aa'\cos H)^{-\frac{7}{2}} &= \tfrac{1}{2}\Sigma D^i\cos iH,\end{aligned}$$

then $2R_0^i$, $2R_1^i$, $2R_2^i$, $2R_3^i$ are the same functions of r, r' that A^i, B^i, C^i, D^i respectively are of a, a'.

The expression of M_x^ξ is

$$M_x^\xi = (-)^{\frac{1}{2}(j+j'-\xi)}\frac{\Pi(x)}{\Pi\frac{1}{2}(x-j-\xi)\,\Pi\frac{1}{2}(x+j'+\xi)}\cdot\frac{\Pi(x)}{\Pi\frac{1}{2}(x-j+\xi)\,\Pi\frac{1}{2}(x+j'-\xi)},$$

and, finally, in the expression for $D(j,j')$, x has every integer value from 0 to ∞ and, for any given value of x, ξ extends by steps of two units from the inferior value $-(x-j')$ to the superior value $x-j$.

It is convenient to write $x = \frac{1}{2}(j+j')+s$; we have then ξ extending from $-\frac{1}{2}(j-j')-s$ to $-\frac{1}{2}(j-j')+s$, or writing $\xi = -\frac{1}{2}(j-j')+\vartheta$, ϑ has the $s+1$ values s, $s-2$, $s-4$, $\ldots -s$, viz. for $s=2p+1$ the values are ±1, ±3, $\ldots\pm(2p+1)$, and for $s=2p$ they are 0, ±2, ±4, $\ldots\pm 2p$.

[p. 513]

Making these changes we have

$$D(j,j') = \Sigma\frac{\Pi_{\frac{1}{2}}\{\frac{1}{2}(j+j')+s-\frac{1}{2}\}}{\Pi\{\frac{1}{2}(j+j')+s\}}\,\eta^{j+j'+2s}\,\Sigma M_{\frac{1}{2}(j+j')+s}^{-\frac{1}{2}(j-j')+\vartheta}\,R_{\frac{1}{2}(j+j')+s}^{-\frac{1}{2}(j-j')+\vartheta},$$

where

$$M_{\frac{1}{2}(j+j')+s}^{-\frac{1}{2}(j-j')+\vartheta} = (-)^{\vartheta}\frac{\Pi\{\frac{1}{2}(j+j')+s\}}{\Pi\frac{1}{2}(s-\vartheta)\,\Pi\frac{1}{2}(j+j'+s+\vartheta)}\cdot\frac{\Pi\{\frac{1}{2}(j+j')+s\}}{\Pi\frac{1}{2}(s+\vartheta)\,\Pi\frac{1}{2}(j+j'+s-\vartheta)},$$

viz. this is $(-)^\vartheta$ into the product of two binomial coefficients, each belonging to the exponent $\frac{1}{2}(j+j')+s$.

Particular Cases, $j+j' = 0$, 2, 4, 6, *being those required in the Planetary Theory.*

Considering successively the cases $j+j'=0$, 2, 4, 6, we have, first,

$$D(j,-j) = \Sigma\frac{\Pi_{\frac{1}{2}}(s-\frac{1}{2})}{\Pi s}\,\eta^{2s}\,\Sigma(-)^{\vartheta}\left\{\frac{\Pi s}{\Pi\frac{1}{2}(s-\vartheta)\,\Pi\frac{1}{2}(s+\vartheta)}\right\}^2 R_s^{-j+\vartheta},$$

which, developed as far as η^6, is

$$\begin{aligned}(*)\quad D(j,-j) &= \tfrac{1}{2}A^{-j}\\ &\quad-\tfrac{1}{2}\eta^2(B^{-j-1}+B^{-j+1})\\ &\quad+\frac{1\cdot3}{2\cdot4\cdot6}\,\eta^4(\ldots)\\ &\quad+\ldots,\end{aligned}$$

where, and in what immediately follows, A, B, C, D are used to denote functions (not of (a,a'), but) of r, r'.

Secondly,

$$D(j,-j+2) = \Sigma\frac{\Pi_{\frac{1}{2}}(s-\frac{1}{2})}{\Pi(s+1)}\,\eta^{2s+2}\,\Sigma(-)^{\vartheta}\frac{\Pi(s+1)}{\Pi\frac{1}{2}(s-\vartheta)\,\Pi\frac{1}{2}(s+\vartheta)+2}\cdot\frac{\Pi(s+1)}{\Pi\frac{1}{2}(s+\vartheta)\,\Pi\frac{1}{2}(s-\vartheta)+2}\,R_{s+1}^{-j+1+\vartheta},$$

which, developed to η^6, is

$$\begin{aligned}(*)\quad D(j,-j+2) = \eta^2\big\{&\tfrac{1}{2}B^{-j+1}\\ &-\frac{1\cdot3}{2\cdot4\cdot6}\,\eta^4\,(\ldots)\big\}.\end{aligned}$$

[p. 514]

Thirdly,

$$D(j,-j+4) = \Sigma \frac{\Pi_{\frac{1}{2}}(s+\frac{3}{2})}{\Pi(s+2)}\,\eta^{2s+4}\,\Sigma(-)^{\vartheta}\left\{ \frac{\Pi(s+2)}{\Pi\frac{1}{2}(s-\vartheta)\,\Pi\frac{1}{2}(s+\vartheta)+2} \cdot \frac{\Pi(s+2)}{\Pi\frac{1}{2}(s+\vartheta)\,\Pi\frac{1}{2}(s-\vartheta)+2} \right\} R_{s+2}^{-j+2+\vartheta},$$

which, developed to η^6, is

$$(*)\quad D(j,-j+4) = \eta^4\left\{ \frac{1\cdot 3}{2\cdot 4}\,C^{-j+2} - \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\,\eta^2\,\tfrac{1}{4}(3D^{-j+1}+3D^{-j+3}) \right\};$$

and, fourthly,

$$D(j,-j+6) = \Sigma \frac{\Pi_{\frac{1}{2}}(s+\frac{5}{2})}{\Pi(s+3)}\,\eta^{2s+6}\,\Sigma(-)^{\vartheta}\left\{ \frac{\Pi(s+3)}{\Pi\frac{1}{2}(s-\vartheta)\,\Pi\frac{1}{2}(s+\vartheta)+3} \cdot \frac{\Pi(s+3)}{\Pi\frac{1}{2}(s+\vartheta)\,\Pi\frac{1}{2}(s-\vartheta)+3} \right\} R_{s+3}^{-j+3+\vartheta},$$

which, developed to η^6, is simply

$$(*)\quad D(j,-j+6) = \eta^6\,\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\,D^{-j+3}.$$

The foregoing formulae, although obtained on the supposition $j=0$, or positive, apply without alteration to the case $j=$ negative, and the entire series of terms of an order not exceeding 6 as regards η may be written,

$$\begin{aligned} &D(j,-j)\cos(jU-jU') \\ &+2D(j,-j+2)\cos(jU+(-j+2)U') \\ &+2D(j,-j+4)\cos(jU+(-j+4)U') \\ &+2D(j,-j+6)\cos(jU+(-j+6)U'), \end{aligned}$$

where j has every integer value from $-\infty$ to $+\infty$.

Comparison with Leverrier.

This is in fact what Leverrier's expression becomes on putting therein $e=e'=0$. To verify this, observe that Leverrier having defined his A^i, B^i, C^i, D^i, as above, writes further

$$\begin{aligned} E^i &= \tfrac{1}{2}(B^{i-1}+B^{i+1}), \\ \mathfrak{E}^i &= \tfrac{1}{4}(C^{i-2}+4C^i+C^{i+2}), \\ \mathfrak{F}^i &= \tfrac{1}{8}(D^{i-3}+9D^{i-1}+9D^{i+1}+D^{i+3}), \\ F^i &= \tfrac{1}{2}(C^{i-1}+C^{i+1}), \\ \mathfrak{G}^i &= \tfrac{1}{8}(D^{i-2}+2D^i+D^{i+2}). \end{aligned}$$

[p. 515]

(consequently $E^{-i}=E^i$, $G^{-i}=G^i$, $H^{-i}=H^i$, $L^{-i+k}=L^i$, $S^{-i+k}=S^i$, $T^{-i+k}=T^i$), and that the terms in question, putting in the coefficients $=\theta\theta'=0$, are with him

$$\begin{aligned}&\{(1)^i+(11)^i\eta^2+(17)^i\eta^4+(20)^i\eta^6\}\cos(W-i\lambda),\\ &+\{(212)^i\eta^2+(218)^i\eta^4+(221)^i\eta^6\}\cos[W-(i-2)\lambda-2\varpi'],\\ &+\{(372)^i\eta^4+(375)^i\eta^6\}\cos[W-(i-4)\lambda-4\varpi'],\\ &+\{(449)^i\eta^6\}\cos[W-(i-6)\lambda-6\varpi'],\end{aligned}$$

where, substituting for $(1)^i$, $(11)^i$, &c., their values, the coefficients are

$$\tfrac{1}{4}A^i=\eta^i\,\tfrac{1}{4}E^i+\eta^4\cdot\tfrac{1}{4}G^i-\eta^6\,\tfrac{1}{4}H^i;$$

$$\begin{aligned}&-\tfrac{1}{4}A^i=\eta^2\cdot\tfrac{1}{4}(B^{i-1}+B^{i+1})+\eta^4\cdot\tfrac{1}{16}(C^{i-2}+4C^i+C^{i+2})-\eta^6\cdot\tfrac{1}{64}(D^{i-3}+9D^{i-1}+9D^{i+1}+D^{i+3});\\ &\eta^2\cdot\tfrac{1}{4}B^{i-1}-\eta^4\cdot L^i+\eta^6S^i,\ =\eta^4\cdot\tfrac{1}{8}L^{i-1}-\eta^6\,\tfrac{1}{4}(D^{i-2}+C^i)+\eta^6\cdot\tfrac{1}{16}(D^{i-3}+3D^{i-1}+D^{i+1});\\ &\eta^4\cdot\tfrac{1}{8}C^{i-2}-\eta^6T^i,\ =\eta^6\cdot\tfrac{1}{8}C^{i-1}-\tfrac{1}{16}(D^{i-3}+D^{i-1});\end{aligned}$$

and

$$\eta^6\cdot\tfrac{1}{64}D^{i-3}.$$

Writing herein j in place of i, and for A^j, B^{j-1}, &c., the equal values A^{-j}, B^{-j+1}, &c., we have precisely the foregoing coefficients $D(j,-j),\ldots D(j,-j+6)$.

The Development in Powers of e, e'.

The complete expression of the reciprocal of the distance is obtained from

$$\begin{aligned}&D(j,-j)\cos(jU-jU')\\ &\ +2D(j,-j+2)\cos\{jU+(-j+2)U'\}\\ &\ +2D(j,-j+4)\cos\{jU+(-j+4)U'\}\\ &\ +2D(j,-j+6)\cos\{jU+(-j+6)U'\},\end{aligned}$$

by writing therein for r, r', U, U', instead of the circular, the elliptic values, that is the values

$$\begin{aligned}r&=a\cdot\mathrm{elqr}(e,L-\Pi),\ =a(1+x),\\ r'&=a'\cdot\mathrm{elqr}(e',L'-\Pi'),\ =a'(1+x'),\\ U&=\Pi-\Theta+\mathrm{elta}(e,L-\Pi),\ =U-\Theta+f,\\ U'&=\Pi'-\Theta'+\mathrm{elta}(e',L'-\Pi'),\ =U'-\Theta'+f';\end{aligned}$$

L, Π, Θ = the mean longitude in orbit, longitude of perihelion in orbit, and longitude of node; and the like for L', Π', Θ'; "elqr"=elliptic quotient radius, "elta"=elliptic true anomaly; or, what is the same thing, if we write $\mathrm{elta}(e,L-\Pi)=L-\Pi+\mathrm{eltt}(e,L-\Pi)$, and the like for $\mathrm{elta}(e',L'-\Pi')$, then

$$\begin{aligned}U&=L-\Theta+\mathrm{eltt}(e,L-\Pi),\ =L-\Theta+y,\\ U'&=L'-\Theta'+\mathrm{eltt}(e',L'-\Pi'),\ =L'-\Theta'+y'.\end{aligned}$$

[p. 516]

The process for doing this is explained, First Part, pp. 205–207, [214], viz., writing $r=a(1+x)$, $r'=a'(1+x')$, and restoring j' (instead of its value $-j,\ldots-j+6$, as the case may be), we have a general term

$$D(j,j')\,x^\alpha x'^{\alpha'}\cos[jU+j'U'],$$

where $D(j, j')$ now denotes the value obtained by writing a, a' in place of r, r' and f, f' are the true anomalies elta$(e, L-\Pi)$ and elta$(e', L'-\Pi')$. And the second factor, $x^\alpha x'^{\alpha'}$ into the cosine, is given as a series

$$\Sigma\Sigma\,([\cos]^\alpha + [\sin]^\alpha)\,([\cos]^{\alpha'} + [\sin]^{\alpha'})\cos[i(L-\Pi) + i'(L'-\Pi') + j(\Theta-\Theta') - j'(\Theta'-\Theta')],$$

where $[\cos]^\alpha$, $[\sin]^\alpha$ are functions of e, $[\cos]^{\alpha'}$, $[\sin]^{\alpha'}$ functions of e'. Or, what is better, the term $x^\alpha x'^{\alpha'}$ into the cosine may be written $x^\alpha x'^{\alpha'}\cos[j(L-\Theta+y) + j'(L'-\Theta'+y')]$, and the expansion then is

$$\Sigma\Sigma\,([\cos]^\alpha + [\sin]^\alpha)\,([\cos]^{\alpha'} + [\sin]^{\alpha'})\cos[i(L-\Pi) + i'(L'-\Pi') + j(L-\Theta) + j'(L'-\Theta')],$$

where as before $[\cos]^\alpha$, $[\sin]^\alpha$ are functions of e, $[\cos]^{\alpha'}$, $[\sin]^{\alpha'}$ are the same functions of e', viz. the e-functions are those given in the two "datum-tables" $(\alpha^0 \ldots id')\cos jy$ and $(\alpha^0 \ldots \alpha s')\sin jy$, taken from Leverrier, which I have given in my "Tables of the Developments of Functions in the Theory of Elliptic Motion," *Memoirs R.A.S.* vol. xxix. (1861), pp. 191–306, [216]. In order to better show which are the symbols referred to, we may, instead of $[\cos]^\alpha$, &c., write $[x^\alpha \cos jy]^\alpha$, &c., the formula will then be

$$\begin{aligned} &x^\alpha x'^{\alpha'}\cos[j(L-\Theta+y) + j'(L'-\Theta'+y')] \\ &\quad = \Sigma\Sigma\,\big([x^\alpha\cos jy]^\alpha + [x^\alpha \sin jy]^\alpha\big)\big([x'^{\alpha'}\cos j'y']^{\alpha'} + [x'^{\alpha'}\sin j'y']^{\alpha'}\big) \\ &\qquad \times \cos[i(L-\Pi) + i'(L'-\Pi') + j(L-\Theta) + j'(L'-\Theta')], \end{aligned}$$

and if we attribute to i, i' any given values, that is, attend to any particular multiple cosine,

$$\cos[i(L-\Pi) + i'(L'-\Pi') + j(L-\Theta) + j'(L'-\Theta')],$$

the coefficient hereof will be

$$\Sigma\Sigma\,\tfrac{1}{1\cdot2\cdot\ldots\alpha}\,\alpha^\alpha\left(\frac{d}{d\alpha}\right)^\alpha \frac{1}{1\cdot2\cdot\ldots\alpha'}\,\alpha'^{\alpha'}\left(\frac{d}{d\alpha'}\right)^{\alpha'} D(j, j')\,\{[x^\alpha\cos jy]^\alpha + [x^\alpha \sin jy]^\alpha\}\,\{[x'^{\alpha'}\cos j'y']^{\alpha'} + [x'^{\alpha'}\sin j'y']^{\alpha'}\},$$

where α, α' each extend from zero to infinity, but to obtain the expression up to a given order p in e, e', we take only the values up to $\alpha + \alpha' = p$.

Particular Case.

Thus, for instance, in $\cos[j(L-\Theta) - j'(L'-\Theta')]$ the terms independent of e' are

$$\begin{aligned} &D(j, -j)\,\{[x^0\cos jy]^0 + [x^0\sin jy]^0\} \\ &\quad + \tfrac{1}{1}\,\alpha\frac{d}{d\alpha}D(j, -j)\,\{[x\cos jy]^1 + [x\sin jy]^1\} \\ &\quad + \tfrac{1}{1\cdot2}\,\alpha^2\left(\frac{d}{d\alpha}\right)^2 D(j, -j)\,\{[x^2\cos jy]^2 + [x^2\sin jy]^2\} \\ &\quad + \&\text{c.} \end{aligned}$$

[p. 517]

which, observing that in the present case the sine terms vanish, is

1	$\frac{x^2}{8}$	$\frac{x^4}{384}$	$\frac{x^6}{46080}$	
1	$-8j^2$ $-54j^4$ $-3440j^6$	$+96j^2$ $+3920j^4$	$-1280j^2$	
	$+4$	$-48j^2-360j^4$	$-700j^2$	$\frac{1}{1}\cdot\alpha\frac{d}{d\alpha}$
	$+4$	$-96j^2+1920j^4$ $-1320j^6$		$\frac{1}{1\cdot 2}\alpha^2\left(\frac{d}{d\alpha}\right)^2$
		$+144-2880j^2$		$\frac{1}{1\cdot 2\cdot 3}\alpha^3\left(\frac{d}{d\alpha}\right)^3$
		$+144-5760j^2$		$\frac{1}{1\cdot 2\cdot 3\cdot 4}\alpha^4\left(\frac{d}{d\alpha}\right)^4$
			$+14400$	$\frac{1}{1\cdot 2\cdot\ldots 5}\alpha^5\left(\frac{d}{d\alpha}\right)^5$
			$+14400$	$\frac{1}{1\cdot 2\cdot\ldots 6}\alpha^6\left(\frac{d}{d\alpha}\right)^6$
				$\frac{1}{1\cdot 2\cdot\ldots 7}\alpha^7\left(\frac{d}{d\alpha}\right)^7$

viz. the term in e^0 is

$$e^0\left\{-j^2+\tfrac{1}{2}\alpha\frac{d}{d\alpha}+\tfrac{1}{8}\alpha^2\left(\frac{d}{d\alpha}\right)^2\right\}D(j,-j),$$

viz. writing $\eta=0$, and therefore $D(j,-j)=\frac{1}{2}A^{-j}$, the term in e^2 is

$$e^2\left\{-j^2+\tfrac{1}{2}\alpha\frac{d}{d\alpha}+\tfrac{1}{8}\alpha^2\left(\frac{d}{d\alpha}\right)^2\right\}\tfrac{1}{2}A^{-j},$$

which, conformably with Leverrier's subscript notation

$$A_1^j=\tfrac{1}{1}\alpha\frac{d}{d\alpha}A^j,\quad A_2^j=\tfrac{1}{1\cdot 2}\alpha^2\left(\frac{d}{d\alpha}\right)^2A^j,\ \&\text{c.},$$

[p. 518]

I write

$$e^2\left[-j^2+\tfrac{1}{2}(1+2)A_1^{-j}+\tfrac{1}{8}\cdot 2\cdot 2\,A_2^{-j}\right]\tfrac{1}{2}A^{-j}+\tfrac{1}{2}A_1^{-j}.$$

The term in question is given by Leverrier as $(\frac{1}{2}e)^2(2)^i$, $=e^2\cdot\frac{1}{4}(2)$, $h=i$ and $K^i=A^i$,

$$=e^2\cdot\tfrac{1}{4}\{-2i^2A^i+A_1^i+A_2^i\},$$

which agrees.

Similarly the term in e^4 is

$$\tfrac{e^4}{384}\left[96j^2-54j^4-48j^2(\ )_1-96j^4(\ )_1+144(\ )_2+144(\ )_3\right]\tfrac{1}{2}A^{-j},$$

$$=\tfrac{e^4}{768}\{(96j^4-54j^4)A^{-j}-48j^2A_1^{-j}-96j^4A_1^{-j}+144A_2^{-j}+144A_3^{-j}\},$$

and the term in question is given by Leverrier as $(\frac{1}{2}e)^4(4)^i$, $=e^4\cdot\frac{1}{16}(4)$, $h=i$ and $K^i=A^i$, which agrees. I have not made the comparison of any more terms.

Leverrier's Results expressed in terms of the Arguments, $L' - \Theta'$, $L' - \Pi'$, $L - \Theta$, $L - \Pi$.

The angles which Leverrier uses in his arguments are l', λ, ϖ, ϖ', and τ', viz. we have,

$$\begin{aligned} l' &= \Theta' + (L' - \Theta'), \\ \lambda &= \Theta' + (L - \Theta), \\ \varpi &= \varpi' + (L' - \Theta') - (L' - \Pi'), \\ \varpi &= \Theta' + (L - \Theta) - (L - \Pi), \\ \tau' &= \Theta', \end{aligned}$$

where L, Π, Θ are the mean longitude of the planet m, its perihelion and the mutual node, all in the orbit of m; and similarly L', Π', Θ' are the mean longitude of the planet m', of its perihelion and of the mutual node, all in the orbit of m'. On substituting the foregoing values of l', λ, &c., Θ', as it should do, disappears, and the arguments are all of them linear functions of $L' - \Theta'$, $L' - \Pi'$, $L - \Theta$, $L - \Pi$; or, if we please, of $L' - \Theta'$, $L - L' + \Pi'$, $L - \Theta$, $L - \Pi$, that is of the distances of each planet from its own perihelion and from the mutual node. It is, I think, convenient to use these last angular distances, and accordingly in Leverrier's arguments, I write,

$$\begin{aligned} l' &= \Theta' + (L' - \Theta'), \\ \lambda &= \Theta' \qquad\qquad\qquad\qquad\qquad\qquad + (L - \Theta), \\ \varpi &= \Theta' + (L - \Theta) - (L - \Pi), \\ \varpi' &= \Theta' + (L' - \Theta') - (L' - \Pi'), \\ \tau' &= \Theta', \end{aligned}$$

[p. 519]

and for the purpose of reference form as it were an Index to his result as follows:

Reciprocal of Distance = as follows

Terms of order zero: terms of orders 2, 4, 6, *having the same arguments.*

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(1)^i$	(1 … 20) cos	i	0	$-i$	0
$(21)^i(\frac{1}{2}e)\,(\frac{1}{2}e')$	(21 … 30) ,,	i	+1	$-i$	−1
$(31)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2$	(31 … 34) ,,	i	+2	$-i$	−2
$(35)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^3$	(35 … 35) ,,	i	+3	$-i$	−3
$(36)^i(\frac{1}{2}e)^2\eta^2$	(36 … 39) ,,	i	0	$-i+2$	−2
$(40)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^2$	(40 … 43) ,,	i	−1	$-i+2$	−1
$(44)^i(\frac{1}{2}e')^2\eta^2$	(44 … 47) ,,	i	−2	$-i+2$	0
			+1	$-i+2$	−3
$(48)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	(48 … 48) ,,	i			
$(49)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(49 … 49) ,,	i	−3	$-i+2$	+1

Terms of the first order: terms of orders 3, 5, 7, *having the same arguments.*

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(50)^i\frac{1}{2}e$	$50\,(\ldots\ 69)$ cos	i	0	$-i$	$+1$
$(70)^i\frac{1}{2}e'$	$70\,(\ldots\ 89)$, ,	i	$+1$	$-i$	0
$(90)^i(\frac{1}{2}e)^2(\frac{1}{2}e')$	$90\,(\ldots\ 99)$, ,	i	$+1$	$-i$	-2
$(100)^i(\frac{1}{2}e)(\frac{1}{2}e')^2$	$(100\ \ldots\ 109)$, ,	i	$+2$	$-i$	-1
$(110)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^2$	$(110\ \ldots\ 113)$, ,	i	$+2$	$-i$	-3
$(114)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^3$	$(114\ \ldots\ 117)$, ,	i	$+3$	$-i$	-2
$(118)^i(\frac{1}{2}e)^4(\frac{1}{2}e')^3$	$(118\ \ldots\ 118)$, ,	i	$+3$	$-i$	-4
$(119)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^4$	$(119\ \ldots\ 119)$, ,	i	$+4$	$-i$	-3
$(120)^i(\frac{1}{2}e)\eta^2$	$(120\ \ldots\ 129)$, ,	i	0	$-i+2$	-1
$(130)^i(\frac{1}{2}e')\eta^2$	$(130\ \ldots\ 139)$, ,	i	-1	$-i+2$	0
$(140)^i(\frac{1}{2}e')^2\eta^2$	$(140\ \ldots\ 143)$, ,	i	0	$-i+2$	-3
$(144)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^2$	$(144\ \ldots\ 147)$, ,	i	$+1$	$-i+2$	-2
$(148)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^2$	$(148\ \ldots\ 151)$, ,	i	-1	$-i+2$	-2
$(152)^i(\frac{1}{2}e)^2\eta^2$	$(152\ \ldots\ 155)$, ,	i	-2	$-i+2$	-1
$(156)^i\frac{1}{2}e\,(\frac{1}{2}e')^2\eta^2$	$(156\ \ldots\ 159)$, ,	i	-2	$-i+2$	$+1$
$(160)^i(\frac{1}{2}e')^3\eta^2$	$(160\ \ldots\ 163)$, ,	i	-3	$-i+2$	0
$(164)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	$(164\ \ldots\ 164)$, ,	i	$+1$	$-i+2$	-4
$(165)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	$(165\ \ldots\ 165)$, ,	i	$+2$	$-i+2$	-3
$(166)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	$(166\ \ldots\ 166)$, ,	i	-3	$-i+2$	$+2$
$(167)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	$(167\ \ldots\ 167)$, ,	i	-4	$-i+2$	$+1$
$(168)^i(\frac{1}{2}e)\eta^4$	$(168\ \ldots\ 168)$, ,	i	0	$-i+2$	-3
$(169)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^4$	$(169\ \ldots\ 169)$, ,	i	-1	$-i+4$	-2
$(170)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^4$	$(170\ \ldots\ 170)$, ,	i	-2	$-i+4$	-1
$(171)^i(\frac{1}{2}e')^3\eta^4$	$(171\ \ldots\ 171)$, ,	i	-3	$-i+4$	0

[p. 520]

Terms of second order: terms of orders 4, 6, *having the same arguments.*

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(172)^i(\frac{1}{2}e)^2$	(172 … 181) cos	i	0	$-i$	+2
$(182)^i(\frac{1}{2}e)(\frac{1}{2}e')$	(182 … 191) , ,	i	+1	$-i$	+1
$(192)^i(\frac{1}{2}e')^2$	(192 … 201) , ,	i	+2	$-i$	0
$(202)^i(\frac{1}{2}e)^3(\frac{1}{2}e')$	(202 … 205) , ,	i	+1	$-i$	−3
$(206)^i(\frac{1}{2}e)(\frac{1}{2}e')^3$	(206 … 209) , ,	i	+3	$-i$	−1
$(210)^i(\frac{1}{2}e)^4(\frac{1}{2}e')^2$	(210 … 210) , ,	i	+2	$-i$	−4
$(211)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^4$	(211 … 211) , ,	i	+4	$-i$	−2
$(212)^i\eta^2$	(212 … 221) , ,	i	0	$-i+2$	0
$(222)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^2$	(222 … 225) , ,	i	+1	$-i+2$	−1
$(226)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^2$	(226 … 229) , ,	i	−1	$-i+2$	+1
$(230)^i(\frac{1}{2}e)^2\eta^2$	(230 … 230) , ,	i	0	$-i+2$	−4
$(231)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^2$	(231 … 231) , ,	i	−1	$-i+2$	−3
$(232)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(232 … 232) , ,	i	−2	$-i+2$	−2
$(233)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(233 … 233) , ,	i	−3	$-i+2$	−1
$(234)^i(\frac{1}{2}e')^4\eta^2$	(234 … 234) , ,	i	−4	$-i+2$	0
$(235)^i(\frac{1}{2}e')^3(\frac{1}{2}e)\eta^2$	(235 … 235) , ,	i	+2	$-i+2$	−2
$(236)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(236 … 236) , ,	i	−2	$-i+2$	+2
$(237)^i(\frac{1}{2}e)^2\eta^4$	(237 … 237) , ,	i	0	$-i+4$	−2
$(238)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^4$	(238 … 238) , ,	i	−1	$-i+4$	−1
$(239)^i(\frac{1}{2}e')^2\eta^4$	(239 … 239) , ,	i	−2	$-i+4$	0

Terms of third order: terms of orders 5, 7, *having the same arguments.*

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(240)^i(\frac{1}{2}e)^3$	(240 … 249) cos	i	0	$-i$	+3
$(250)^i(\frac{1}{2}e)^2(\frac{1}{2}e')$	(250 … 259) , ,	i	+1	$-i$	+2
$(260)^i(\frac{1}{2}e)(\frac{1}{2}e')^2$	(260 … 269) , ,	i	+2	$-i$	+1
$(270)^i(\frac{1}{2}e')^3$	(270 … 279) , ,	i	+3	$-i$	0

[p. 521]

Terms of third order (concluded):

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(280)^i(\frac{1}{2}e)^4(\frac{1}{2}e')$	(280 … 283) cos	i	+1	$-i$	−4
$(284)^i(\frac{1}{2}e)(\frac{1}{2}e')^4$	(284 … 287) ,,	i	+4	$-i$	−1
$(288)^i(\frac{1}{2}e)^5(\frac{1}{2}e')^2$	(288 … 289) ,,	i	+2	$-i$	−5
$(290)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	(290 … 299) ,,	i	0	$-i+2$	+1
$(300)^i(\frac{1}{2}e')\eta^2$	(300 … 309) ,,	i	+1	$-i+2$	0
$(310)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^2$	(310 … 313) ,,	i	−1	$-i+2$	+2
$(314)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^2$	(314 … 317) ,,	i	+2	$-i+2$	−1
$(318)^i(\frac{1}{2}e)^2\eta^2$	(318 … 318) ,,	i	0	$-i+2$	−5
$(319)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^2$	(319 … 319) ,,	i	−1	$-i+2$	−4
$(320)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(320 … 320) ,,	i	−2	$-i+2$	−3
$(321)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(321 … 321) ,,	i	−3	$-i+2$	−2
$(322)^i(\frac{1}{2}e)(\frac{1}{2}e')^4\eta^2$	(322 … 322) ,,	i	−4	$-i+2$	−1
$(323)^i(\frac{1}{2}e')^5\eta^2$	(323 … 323) ,,	i	−5	$-i+2$	0
$(324)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^3\eta^2$	(324 … 324) ,,	i	−2	$-i+2$	+3
$(325)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^2\eta^2$	(325 … 325) ,,	i	+3	$-i+2$	−2
$(326)^i(\frac{1}{2}e)^3\eta^4$	(326 … 329) ,,	i	0	$-i+4$	−1
$(330)^i(\frac{1}{2}e')^3\eta^4$	(330 … 333) ,,	i	−1	$-i+4$	0
$(334)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^4$	(334 … 334) ,,	i	+1	$-i+4$	−2
$(335)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^4$	(335 … 335) ,,	i	−2	$-i+4$	+1

Terms of fourth order: terms of order 6*, and of same argument.*

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(336)^i(\frac{1}{2}e)^4$	(336 … 339) cos	i	0	$-i$	+4
$(340)^i(\frac{1}{2}e)^3(\frac{1}{2}e')$	(340 … 343) ,,	i	+1	$-i$	+3
$(344)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2$	(344 … 347) ,,	i	+2	$-i$	+2
$(348)^i(\frac{1}{2}e)(\frac{1}{2}e')^3$	(348 … 351) ,,	i	+3	$-i$	+1
$(352)^i(\frac{1}{2}e')^4$	(352 … 355) ,,	i	+4	$-i$	0
$(356)^i(\frac{1}{2}e)^5(\frac{1}{2}e')$	(356 … 356) ,,	i	+1	$-i$	−5
$(357)^i(\frac{1}{2}e)(\frac{1}{2}e')^5$	(357 … 357) ,,	i	+5	$-i$	−1

[p. 522]

Terms of fourth order (concluded):

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(358)^i(\frac{1}{2}e)^2\eta^2$	(358 … 358) cos	i	0	$-i+2$	+2
$(362)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^2$	(362 … 364) , ,	i	+1	$-i+2$	+1
$(366)^i(\frac{1}{2}e')^2\eta^2$	(366 … 369) , ,	i	+2	$-i+2$	0
$(370)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	(370 … 370) , ,	i	−1	$-i+2$	+3
$(371)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(371 … 371) , ,	i	+3	$-i+2$	−1
$(372)^i\eta^4$	(372 … 375) , ,	i	0	$-i+4$	0
$(376)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^4$	(376 … 376) , ,	i	+1	$-i+4$	−1
$(377)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^4$	(377 … 377) , ,	i	−1	$-i+4$	+1

Terms of fifth order: terms of order 7 *having the same arguments.*

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(378)^i(\frac{1}{2}e)^5$	(378 … 381) cos	i	0	$-i$	+5
$(382)^i(\frac{1}{2}e)^4(\frac{1}{2}e')$	(382 … 385) , ,	i	+1	$-i$	+4
$(386)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^2$	(386 … 389) , ,	i	+2	$-i$	+3
$(390)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^3$	(390 … 393) , ,	i	+3	$-i$	+2
$(394)^i(\frac{1}{2}e)(\frac{1}{2}e')^4$	(394 … 397) , ,	i	+4	$-i$	+1
$(398)^i(\frac{1}{2}e')^5$	(398 … 401) , ,	i	+5	$-i$	0
$(402)^i(\frac{1}{2}e')^6(\frac{1}{2}e)$	(402 … 402) , ,	i	+1	$-i$	−6
$(403)^i(\frac{1}{2}e)(\frac{1}{2}e')^6$	(403 … 403) , ,	i	+6	$-i$	−1
$(404)^i(\frac{1}{2}e)^3\eta^2$	(404 … 407) , ,	i	0	$-i+2$	+3
$(408)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^2$	(408 … 411) , ,	i	+1	$-i+2$	+2
$(412)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^2$	(412 … 415) , ,	i	+2	$-i+2$	+1
$(416)^i(\frac{1}{2}e')^3\eta^2$	(416 … 419) , ,	i	+3	$-i+2$	0
$(420)^i(\frac{1}{2}e)^4(\frac{1}{2}e')\eta^2$	(420 … 420) , ,	i	−1	$-i+2$	+4
$(421)^i(\frac{1}{2}e)(\frac{1}{2}e')^4\eta^2$	(421 … 421) , ,	i	+4	$-i+2$	−1
$(422)^i(\frac{1}{2}e)\eta^4$	(422 … 425) , ,	i	0	$-i+4$	+1
$(426)^i(\frac{1}{2}e')\eta^4$	(426 … 429) , ,	i	+1	$-i+4$	0
$(430)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^4$	(430 … 430) , ,	i	−1	$-i+4$	+2
$(431)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^4$	(431 … 431) , ,	i	+2	$-i+4$	−1
$(432)^i(\frac{1}{2}e)\eta^6$	(432 … 432) , ,	i	0	$-i+6$	−1
$(433)^i(\frac{1}{2}e')\eta^6$	(433 … 433) , ,	i	−1	$-i+6$	0

[p. 523]

Terms of sixth order.

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(434)^i(\frac{1}{2}e)^6$	(434 ... 434) cos	i	0	$-i$	+6
$(435)^i(\frac{1}{2}e)^5(\frac{1}{2}e')$	(435 ... 435) ,,	i	+1	$-i$	+5
$(436)^i(\frac{1}{2}e)^4(\frac{1}{2}e')^2$	(436 ... 436) ,,	i	+2	$-i$	+4
$(437)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^3$	(437 ... 437) ,,	i	+3	$-i$	+3
$(438)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^4$	(438 ... 438) ,,	i	+4	$-i$	+2
$(439)^i(\frac{1}{2}e)(\frac{1}{2}e')^5$	(439 ... 439) ,,	i	+5	$-i$	+1
$(440)^i(\frac{1}{2}e')^6$	(440 ... 440) ,,	i	+6	$-i$	0
$(441)^i(\frac{1}{2}e)^2\eta^2$	(441 ... 441) ,,	i	0	$-i+2$	+4
$(442)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	(442 ... 442) ,,	i	+1	$-i+2$	+3
$(443)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(443 ... 443) ,,	i	+2	$-i+2$	+2
$(444)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(444 ... 444) ,,	i	+3	$-i+2$	+1
$(445)^i(\frac{1}{2}e')^4\eta^2$	(445 ... 445) ,,	i	+4	$-i+2$	0
$(446)^i(\frac{1}{2}e)^2\eta^4$	(446 ... 446) ,,	i	0	$-i+4$	+2
$(447)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^4$	(447 ... 447) ,,	i	+1	$-i+4$	+1
$(448)^i(\frac{1}{2}e')^2\eta^4$	(448 ... 448) ,,	i	+2	$-i+4$	0
$(449)^i\eta^6$	(449 ... 449) ,,	i	0	$-i+6$	0

Terms of seventh order.

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(450)^i(\frac{1}{2}e)^7$	(450 ... 450) cos	i	0	$-i$	+7
$(451)^i(\frac{1}{2}e)^6(\frac{1}{2}e')$	(451 ... 451) ,,	i	1	$-i$	+6
$(452)^i(\frac{1}{2}e)^5(\frac{1}{2}e')^2$	(452 ... 452) ,,	i	2	$-i$	+5
$(453)^i(\frac{1}{2}e)^4(\frac{1}{2}e')^3$	(453 ... 453) ,,	i	3	$-i$	+4
$(454)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^4$	(454 ... 454) ,,	i	4	$-i$	+3
$(455)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^5$	(455 ... 455) ,,	i	5	$-i$	+2
$(456)^i(\frac{1}{2}e)(\frac{1}{2}e')^6$	(456 ... 456) ,,	i	6	$-i$	+1
$(457)^i(\frac{1}{2}e')^7$	(457 ... 457) ,,	i	7	$-i$	0
$(458)^i(\frac{1}{2}e)^3\eta^2$	(458 ... 458) ,,	i	0	$-i+2$	+5

[p. 524]

Terms of seventh order (concluded).

		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$(459)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	(459 … 459) cos	i	$+1$	$-i+2$	$+4$
$(460)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(460 … 460) , ,	i	$+2$	$-i+2$	$+3$
$(461)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^3\eta^2$	(461 … 461) , ,	i	$+3$	$-i+2$	$+2$
$(462)^i(\frac{1}{2}e)(\frac{1}{2}e')^4\eta^2$	(462 … 462) , ,	i	$+4$	$-i+2$	$+1$
$(463)^i(\frac{1}{2}e')^5\eta^2$	(463 … 463) , ,	i	$+5$	$-i+2$	0
$(464)^i(\frac{1}{2}e)^3\eta^4$	(464 … 464) , ,	i	0	$-i+4$	$+3$
$(465)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^4$	(465 … 465) , ,	i	$+1$	$-i+4$	$+2$
$(466)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^4$	(466 … 466) , ,	i	$+2$	$-i+4$	$+1$
$(467)^i(\frac{1}{2}e')^3\eta^4$	(467 … 467) , ,	i	$+3$	$-i+4$	0
$(468)^i(\frac{1}{2}e)\eta^6$	(468 … 468) , ,	i	0	$-i+6$	$+1$
$(469)^i(\frac{1}{2}e')\eta^6$	(469 … 469) , ,	i	$+1$	$-i+6$	0

Here the several coefficients are ultimately given in terms of the before-mentioned quantities A^i, B^i, C^i, D^i, E^i, G^i, H^i, L^i, S^i, T^i (functions of a, a'), and their differential coefficients in regard to a,

$$\left(A^i_1 = \tfrac{1}{1}\alpha\frac{d}{d\alpha}A^i,\quad A^i_2 = \tfrac{1}{1\cdot2}\alpha^2\left(\frac{d}{d\alpha}\right)^2 A^i,\ \text{\&c.}\right),$$

as follows:—we have Leverrier, pp. 299–330, a list of functions $(1), (2), \ldots (154)$ of the form

$$(1) = \tfrac{1}{2}K^i,\quad (2) = -2h^2K^i + K^i_1 + K^i_2,\quad (3) = -2v^2K^i + K^i_1 + K^i_2,\ \text{\&c.},$$

involving i, h, and K^i and its derived functions K^i_1, K^i_2, &c. The coefficients of the several cosines are given by means of the functions in question, thus, first coefficient, above denoted as $(1)^i(1\ldots 20)$, is

$$(1)^i + (2)^i(\tfrac{1}{2}e)^2 + (3)^i(\tfrac{1}{2}e')^2 + \cdots + (20)^i\eta^6$$

where $(1)^i = (1)$, $(2)^i = (2) \ldots$ writing in the functions $(1), (2) \ldots (10)$, $h = i$, and $K^i = A^i$;

$$(11)^i = (1),\quad (12)^i = (2),\ \text{\&c., writing } h = i \text{ and } K^i = E^i;$$

$$(20)^i = (1),\ \text{writing } h = i \text{ and } K^i = -\tfrac{1}{4}S^i;$$

and so on for the various component coefficients $(1)^i, (2)^i, \ldots (469)^i$.

But the resulting expressions, for the several integer values $i = -10$ to $+10$, are worked out in the Addition II. (*Numerical Tables for the Calculation of the Coefficients of the Development of the Disturbing Function*), pp. 358–383. And this Addition contains also, indicated by the letters δ and Δ respectively, the expressions of the

[p. 525]

terms which experience an alteration in passing from the development of the reciprocal of the distance to those of the disturbing functions m' upon m, and m upon m' respectively.

We have

Disturbing Function m' upon m

$$= m'\left\{-\frac{r\cos H}{r'^2} + \frac{1}{\rho}\right\}.$$

Disturbing Function m upon m'

$$= m\left\{-\frac{r'\cos H}{r^2} + \frac{1}{\rho}\right\}.$$

The expressions of $-\dfrac{r\cos H}{r'^2}$ and $-\dfrac{r'\cos H}{r^2}$, developed to the third order in the eccentricities and inclination, are given, Leverrier, pp. 272 and 274. Expressed in the terms of the foregoing arguments $L-\Theta'$, &c., and in terms of a, a' in place of a and $\mathbf{a}$, these are as follows:

$\dfrac{-r\cos H}{r'^2} = \dfrac{a}{a'^3}$ into		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$-1+\frac{3}{2}(e^2+e'^2)+\eta^2$	cos	1	0	-1	0
$-e^2$	"	$+1$	$+1$	-1	-1
$+\frac{3}{4}e^2-\frac{3}{4}ee'^2-\frac{3}{4}e\eta^2$	"	$+1$	0	-1	$+1$
$-\frac{1}{4}e+\frac{3}{8}ee^2+\frac{3}{8}e^3+\frac{1}{4}e\eta^2$	"	$+1$	0	-1	-1
$-2e'+e^2e'+\frac{3}{8}e'^3+2e'\eta^2$	"				
$-\frac{3}{4}ee'^2$	"	$+1$	$+1$	-1	-2
$+\frac{1}{4}e'^2$	"	$+1$	$+1$	-1	-2
$-\frac{3}{4}e'^2$	"	-1	$+2$	$+1$	-1
$+\frac{1}{4}e'\eta^2$	"	$+1$	$+2$	-1	-1
	"	$+1$	0	$+1$	-1

[p. 526]

$\frac{-r\cos H}{r'^2} = \frac{a}{a'^3}$ into		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$-\frac{1}{8}e^3$	cos	$+1$	0	-1	$+2$
$-\frac{3}{8}e^3$	„	$+1$	0	-1	-2
$+3\,ee'^2$	„			-1	$+1$
$-3e^2e'$	„		$+1$	$+1$	$+1$
	„		$+2$	-1	
$-\frac{15}{8}e^2e'$	„		-2	$+1$	-1
$-\frac{1}{4}e\eta^2$	„		$+1$	-1	$+3$
$-\frac{17}{8}e\eta^2$	„		$+2$	-1	
$-\frac{3}{4}e\eta^2$	„		$+2$	$+1$	-3
$+\frac{1}{8}e\eta^2$	„			-1	
	„		$+3$	$+1$	$+2$
$-\frac{1}{8}e^2e'$	„			-1	$+1$
	„		$+3$	$+1$	$+1$
$-\frac{1}{8}ee'^2$	„		$+1$		$+1$
$+\frac{3}{8}e^2e'$	„				$+1$
$-\frac{1}{8}e'^3$	„				
$-\frac{1}{8}e'\eta^2$	„	$+1$			
$-\frac{3}{8}e'\eta^2$	„				
$-2\,e'\eta^2$	„			$+$	
	„	$+1$			
	„		$-$		
	„		$+$		
	„		$-$		
	„		$+$		
	„		$+1$		
	„		$+$		

$\frac{r'\cos H}{r^2} = \frac{a'}{a^3}$ into		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$-1+\frac{3}{2}(e^2+e'^2)+\eta^2$	cos	1	0	-1	0
$-e'^2$	„	$+1$	$+1$	-1	-1
$-2e+ee'^2+\frac{3}{8}e^3+2e\eta^2$	„	$+1$	0	-1	-1
$+\frac{3}{4}e'-\frac{3}{4}e^2e'-\frac{3}{4}e\eta^2$					
$-\frac{1}{4}e'+\frac{3}{8}e^2e'+\frac{3}{8}e'^3+\frac{1}{4}e'\eta^2$	„			$+1$	$+1$
$+\frac{1}{4}e\eta^2$	„		$+$	$+1$	-1
$-\frac{3}{4}e^2e'$	„	$+2$	-1	-2	$+2$
$-e^2$	„		$+$	$+1$	-1
	„	$+1$	$+2$	-1	-1
$+3\,e^2e'$	„				
	„	$+1$	-1	$+1$	
	„	$+1$	0	-1	$+2$
	„	$+1$		-1	-2
	„	-1	$+1$	$+1$	$+1$
	„		$+2$	$+1$	

[p. 527]

$\dfrac{r'\cos H}{r^2} = \dfrac{a'}{a^3}$ into		$L'-\Theta'$	$L'-\Pi'$	$L-\Theta$	$L-\Pi$
$-\frac{1}{8}e'^3$	cos	$+1$	0	$+2$	-1
$-\eta^2$	”		$+$		$+1$
$-\frac{1}{8}e^3$	”				
$-\frac{3}{8}e^3$	”			$+$	$-1+3$
	”				$-1-3$
	”		$+$		
$+\frac{15}{8}e^2e'$	”		$-$	$+$	$+1+2$
$-\frac{3}{4}e^2e'$	”	$+$		$+$	$-1+2$
$-\frac{1}{4}ee'^2$	”	-1	$+2$	$+1$	$+1$
$-\frac{1}{8}e'^3$	”		$-$	$+3$	$+1$
$-\frac{1}{8}e^3$	”				
	”	$+1$		$+3$	-1
$-2\,e\eta^2$	”		$+$		$+1+1$
$-\frac{1}{8}e'\eta^2$	”	$+1$		$+1$	$+1$

It is hardly necessary to observe that, to obtain the expressions of the Disturbing Functions, these additional terms are to be combined with the corresponding terms in the expression of the reciprocal of the distance: thus, in the Disturbing Function Ω (m' upon m), the entire term depending on $\cos[(L'-\Theta')-(L-\Theta)]$ is

$$= m'\cdot\{2(1,\ldots 20)_{i=1} + \tfrac{a}{a'^3}\left(-1+\tfrac{3}{2}(e^2+e'^2)+\eta^2\right)\}\cos[(L'-\Theta')-(L-\Theta)],$$

where, however, the supplemental term is taken to the third order only.

[p. 528]

480.

ON THE EXPRESSION OF DELAUNAY'S l, g, h, IN TERMS OF HIS FINALLY ADOPTED CONSTANTS.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXXII. (1871–72), pp. 8–16.]

We have in Delaunay's lunar theory,

l, the mean anomaly of the Moon,
g, the mean distance of perigee from ascending node,
h, the mean longitude of ascending node,

quantities which vary directly as the time, the coefficients of t, or values of $\frac{dl}{dt}$, $\frac{dg}{dt}$, $\frac{dh}{dt}$ being given in his *Théorie du Mouvement de la Lune*, vol. II. pp. 237, 238. But these values are not expressed in terms of his constants a (or n), e, γ, finally adopted as explained p. 800, and it seems very desirable to obtain the expressions of l, g, h, in terms of these finally adopted constants: I have accordingly effected this transformation (which I found less laborious than I had anticipated). It will be convenient to imagine the a, n, e, γ of pp. 237, 238 replaced by A, N, E, Γ respectively. This being so, and writing $\frac{n'}{n}$ for the $\frac{n'}{n}$ of p. 800 we have, p. 800,

$$A = a\Big\{1 + [-\tfrac{2}{3}m^2 - \tfrac{223}{72}m^3]\tfrac{n'^2}{n^2} + (-1 + \tfrac{3}{2}\gamma^2 + \tfrac{3}{2}e^2 - \tfrac{3}{2}e'^2 - \tfrac{15}{4}\gamma^4 + \tfrac{15}{4}\gamma^2e^2 - \tfrac{15}{4}\gamma^2e'^2 + \tfrac{15}{4}e^4 - \tfrac{15}{4}e^2e'^2 - \tfrac{15}{4}e'^4)m^2 + \cdots\Big\}$$

[Long power-series expansion in γ, e, e', m for A, following pp. 237–238 and 800 of Delaunay; full coefficient string preserved in the displayed equation above; for full numerical detail consult the IA scan.]

[p. 529]

and hence calculating from the formula $N^2A^3 = n^2a^3$, we find

$$N = n\{1 + [\tfrac{1}{2}m^2 + \tfrac{61}{48}m^3]\tfrac{n'^2}{n^2} + \cdots\},$$

$$N = n + (-\tfrac{1}{2} + \tfrac{3}{4}\gamma^2 + \tfrac{3}{4}e^2 - \tfrac{3}{4}e'^2 - \tfrac{15}{8}\gamma^4 + \tfrac{15}{8}\gamma^2e^2 - \tfrac{15}{8}\gamma^2e'^2 + \tfrac{15}{8}e^4 - \tfrac{15}{8}e^2e'^2 - \tfrac{15}{8}e'^4)m^2 + \cdots$$
$$= n(1+Q) \text{ suppose.}$$

The values of E, Γ are given p. 800, but for the present purpose we only require E^2 and Γ^2 to the fifth order, viz. the values of these are at once found to be

$$E^2 = e^2(1 + \cdots), \quad \Gamma^2 = \gamma^2(1 + \cdots),$$

whence also $E^4 = e^4$ and $\Gamma^4 = \gamma^4$.

The formulae of pp. 237–238 now give

$$\begin{aligned}\frac{dl}{dt} = n &+ \big(\cdots\big)(1+Q)^{-1}\\ &+ \big(\cdots\big)m^2(1+Q)^{-2}\\ &+ \big(\cdots\big)m^3(1+Q)^{-3}\\ &+ \big(\cdots\big)m^4(1+Q)^{-4}\\ &+ \big(\cdots\big)m^5(1+Q)^{-5}.\end{aligned}$$

(Observe that writing herein $Q = 0$, and omitting the terms in m^4 and m^5 in the coefficient of $(1+Q)^{-1}$, and the term in m^5 in the coefficient of $(1+Q)^{-2}$, we have the original formula of p. 237.)

$$\frac{dg}{dt} = nt\{[\tfrac{1}{4}m^2 + \tfrac{19}{16}m^3]\tfrac{n'^2}{n^2} + \cdots\}(1+Q)^{-1}$$

[p. 530]

[The expansion continues with the analogous expressions for $\dfrac{dg}{dt}$ *and* $\dfrac{dh}{dt}$*, expanded in powers of* m *in the form* $(1+Q)^{-1}, (1+Q)^{-2}, \ldots, (1+Q)^{-5}$ *together with the "observe" parenthetical remarks that, on writing* $Q = 0$ *and omitting the highest-order coefficients of* m*, one recovers the original formulæ of Delaunay (pp. 237 and 238).]*

We hence have

$$l = nt\{A + (1+B)Q + CQ^2\}, \text{ i.e. } l = nt\{A - \tfrac{1}{3}Q + BQ + CQ^2\},$$

$$g = nt\{A' + B'Q + C'Q^2\},$$

$$h = nt\{A'' + B''Q + C''Q^2\},$$

where (omitting the terms in $\dfrac{n'^2}{n^2}$)

$$A = 1 + (-\tfrac{1}{4} + \tfrac{3}{4}\gamma^2 - \tfrac{1}{4}e^2 - \tfrac{1}{4}e'^2 - \tfrac{15}{16}\gamma^4 + \tfrac{15}{8}\gamma^2e^2 - \tfrac{15}{8}\gamma^2e'^2 + \tfrac{15}{8}e^4 - \tfrac{15}{8}e^2e'^2 - \tfrac{15}{8}e'^4)m^2$$
$$+ (-\tfrac{215}{16} + \tfrac{75}{8}\gamma^2 - \tfrac{15}{8}e^2 - \tfrac{15}{8}e'^2 - \tfrac{105}{16}\gamma^4 + \tfrac{35}{8}\gamma^2e^2 + \tfrac{35}{8}\gamma^2e'^2 + \tfrac{75}{8}e^4 - \tfrac{35}{8}e^2e'^2 - \tfrac{75}{8}e'^4)m^3 + \cdots$$

[p. 531]

$$B = (\cdots)m^2 + (\cdots)m^3 + \cdots$$

$$C = -\tfrac{1}{4}m^2 - \tfrac{105}{16}m^3 - \cdots$$

$$A' = (\tfrac{3}{4} - \tfrac{3}{4}\gamma^2 + \tfrac{1}{2}e^2 + \tfrac{1}{2}e'^2 - \tfrac{15}{8}\gamma^4 + \tfrac{15}{8}\gamma^2e^2 + \tfrac{15}{8}\gamma^2e'^2 - \tfrac{15}{8}\gamma^2e'^2 + \tfrac{15}{8}e^4 + \tfrac{15}{8}e^2e'^2 - \tfrac{15}{8}e'^4)m^2$$
$$+ (\cdots)m^3 + (\cdots)m^4$$

$$B' = (-\tfrac{15}{16} + \tfrac{15}{8}\gamma^2 - \tfrac{3}{8}e^2 + \tfrac{3}{8}e'^2)m^2 + \cdots$$

$$C' = \tfrac{3}{16}m^3 + \cdots$$

$$A'' = (-\tfrac{1}{4} + \tfrac{3}{4}\gamma^2 - \tfrac{1}{4}e^2 - \tfrac{1}{4}e'^2 - \tfrac{15}{16}\gamma^4 + \tfrac{15}{8}\gamma^2e^2 + \tfrac{15}{8}\gamma^2e'^2 + \tfrac{15}{8}e^4 - \tfrac{15}{8}e^2e'^2 - \tfrac{15}{8}e'^4)m^2 + \cdots$$

$$B'' = (\tfrac{1}{8} - \tfrac{3}{8}\gamma^2 + \tfrac{1}{8}e^2 + \tfrac{3}{8}e'^2)m^2 + \cdots$$

$$C'' = -\tfrac{3}{16}m^3 + \cdots$$

[p. 532]

And in terms BQ, $B'Q$, $B''Q$, we have

$$Q = (1 - \tfrac{1}{2}\gamma^2 + \tfrac{3}{4}e^2 + \tfrac{3}{4}e'^2)m^2 + \cdots$$

and in the terms CQ^2, $C'Q^2$, $C''Q^2$, simply $= Q^2 = m^4$. Hence finally the required values of l, g, h, are

$$l = nt\{1 + [-\tfrac{1}{4}m^2 - \tfrac{215}{48}m^3]\tfrac{n'^2}{n^2}\}$$
$$+ (-\tfrac{1}{4} + \tfrac{3}{4}\gamma^2 + \tfrac{1}{2}e^2 - \tfrac{1}{4}e'^2 - \tfrac{15}{16}\gamma^4 - \tfrac{15}{8}\gamma^2e^2 + \tfrac{15}{8}\gamma^2e'^2 + \tfrac{15}{8}e^4 - \tfrac{15}{8}e^2e'^2 - \tfrac{15}{8}e'^4)m^2 + \cdots$$

$$g = nt\big\{[\tfrac{3}{4}m^2 + \tfrac{61}{16}m^3]\tfrac{n'^2}{n^2} + \cdots\big\}$$
$$+ (\tfrac{3}{4} - \tfrac{3}{4}\gamma^2 + \tfrac{1}{2}e^2 + \tfrac{1}{2}e'^2 + \tfrac{15}{8}\gamma^4 - \tfrac{15}{8}\gamma^2e^2 - \tfrac{15}{8}\gamma^2e'^2 + \tfrac{15}{8}e^4 - \tfrac{15}{8}e^2e'^2 + \tfrac{15}{8}e'^4)m^2 + \cdots$$

$$h = nt\{[-\tfrac{3}{4}m^2 - \tfrac{61}{16}m^3]\tfrac{n'^2}{n^2}\}$$
$$+ (-\tfrac{1}{4} + \tfrac{3}{4}\gamma^2 - \tfrac{3}{4}e^2 - \tfrac{3}{4}e'^2 - \tfrac{15}{16}\gamma^4 + \tfrac{15}{8}\gamma^2e^2 + \tfrac{15}{8}\gamma^2e'^2 - \tfrac{15}{8}e^4 - \tfrac{15}{8}e^2e'^2 - \tfrac{15}{8}e'^4)m^2 + \cdots$$

[p. 533]

which values satisfy, as they should do, the equation $l+g+h = nt$. I recall that the precise signification of the constants is as follows: n is the coefficient of t in the expression of the Moon's longitude in terms of the time, a the corresponding elliptic value of the mean distance ($n^2a^3 =$ sum of masses), e the eccentricity, such that in the expression of the longitude the coefficient of the leading term of the equation of the centre has its elliptic value

$$= 2e - \tfrac{1}{4}e^3 + \tfrac{5}{96}e^5,$$

and γ the sine of the half-inclination, such that in the expression of the latitude the coefficient of the leading term has its elliptic value

$$= 2\gamma - 2\gamma e^2 - \tfrac{1}{2}\gamma^4 + \tfrac{1}{6}\gamma^2 e^2 + \tfrac{1}{2}\gamma^3 e^2 - \tfrac{1}{6}\gamma e^4.$$

n', a' are the mean motion and mean distance of the Sun, $m = \dfrac{n'}{n}$, and e' is the eccentricity of the Sun's orbit, considered as constant.

[p. 534]

481.

ON THE EXPRESSION OF M. DELAUNAY'S $h+g$ IN TERMS OF HIS FINALLY ADOPTED CONSTANTS.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXXII. *(1871–72), p. 74.]*

I HAD the pleasure of receiving from M. Delaunay a letter dated Paris, 17th Dec. 1871, in which he informs me that, on referring to his papers, he had found there expressions for l, g, h, identical with those given by me in the November Number of the *Monthly Notices*, with only a single typographical error, $-\frac{15}{8}e'^2m^3$ instead of $-\frac{15}{8}e'^4m^3$ [*ante* p. 532, corrected] in my expression of A.

M. Delaunay mentions also that he had obtained four additional terms in the expression for $h+g$ (longitude of the Moon's perigee), and that the complete expression in terms of the finally adopted constants is

$$\begin{aligned} h+g &= nt\Big\{(\tfrac{1}{2} - Gy^2 - \tfrac{1}{2}e^2 + \tfrac{1}{4}e'^2 - \tfrac{15}{16}\gamma^4 + \tfrac{15}{8}\gamma^2e^2 + \tfrac{15}{8}\gamma^2e'^2 - 9\gamma^2e^4 - \tfrac{15}{8}e^4 - \tfrac{15}{8}e'^4 + \tfrac{15}{8}e^2e'^2)m^2 \\ &\quad + (\cdots)m^3 + (\cdots)m^4 + (\cdots)m^5 \\ &\quad + [\tfrac{1}{2}m^2 + \tfrac{61}{16}m^3]\tfrac{n'^2}{n^2}\Big\}. \end{aligned}$$

[Observe that $h+g$ is $= nt - l$, and compare with the expression for l, *ante* p. 532.]

[p. 535]

482.

NOTE ON A PAIR OF DIFFERENTIAL EQUATIONS IN THE LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXXII. *(1871–72), pp. 31–32.]*

THE equations

$$\frac{d}{dt}\frac{d\rho}{dt} - \rho\left(\frac{dv}{dt}\right)^2 + \frac{1}{\rho^2} = km^2\rho\{\tfrac{1}{2} + \tfrac{3}{2}\cos(2v - 2mt)\},$$

$$\frac{d}{dt}\rho^2\frac{dv}{dt} = jm^2\rho^2\{-\tfrac{3}{2}\sin(2v - 2mt)\},$$

taking therein $j = k = 1$ in effect present themselves in the Lunar Theory, and particular integrals in series have been obtained, the development being carried to a great extent; but I give the results only as far as m^4, viz., writing

$$t - mt = D,$$

we have

$$v = t + (\tfrac{11}{8}m^2 + \tfrac{59}{12}m^3)\sin 2D + \tfrac{281}{96}m^4\sin 4D,$$

$$\tfrac{1}{\rho} = 1 + \tfrac{1}{6}m^2 - \tfrac{1}{12}m^4 + (\tfrac{2}{3}m^2 + \tfrac{5}{6}m^3 + \tfrac{17}{32}m^4)\cos 2D + \tfrac{1}{3}m^4\cos 4D.$$

In the Lunar Theory j and k are properly each $= \dfrac{1}{1 + \dfrac{E}{m'}}$ (E the mass of the Earth, m' that of the Sun), but they are taken to be $= 1$; the numerical difference is inappreciable; but there would be a considerable theoretical advantage in retaining

[p. 536]

in the equations the coefficients j, k [regarded as each of them $= k$]: in fact, the developments could then be arranged according to the powers of k, that is according to the powers of the disturbing force; whereas, when k is taken $= 1$, we have only a development in powers of m, and since m also presents itself through the coefficient $-2m$ of t in $2v - 2mt$, terms which are really of different orders in regard to the disturbing force, are united together into a single term: so that, instead of a term of the form $(Ak + Bk^2 + \text{&c.})\,m^p$, where A, B, are numerical, we have the term $(A + B + \cdots)\,m^p$, where of course $A + B + \cdots$ is given as a single numerical coefficient.

There is no equal advantage in retaining the two coefficients k, j, as this only serves to show how a term arises from the central and tangential forces respectively; thus retaining these coefficients, the integrals as far as m^2 are

$$v = t + (\tfrac{3}{8}k + j)\,m^2\sin 2D,$$

$$\tfrac{1}{\rho} = 1 + \tfrac{1}{6}m^2k + (\tfrac{1}{6}k + \tfrac{1}{2}j)\,m^2\cos 2D,$$

agreeing with the former result when $k = j = 1$; but there is, nevertheless, some interest in retaining the two coefficients. I hope to develope the results somewhat further, and to communicate them to the Society.

[p. 537]

483.

ON A PAIR OF DIFFERENTIAL EQUATIONS IN THE LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXXII. *(1871–72), pp. 201–206.]*

I CONSIDER the differential equations

$$\frac{d}{dt}\frac{d\rho}{dt} - \rho\left(\frac{dv}{dt}\right)^2 + \frac{1}{\rho^2} = km^2\rho\{\tfrac{1}{2} + \tfrac{3}{2}\cos(2v - 2mt)\},$$

$$\frac{d}{dt}\left(\rho^2\frac{dv}{dt}\right) = jm^2\rho^2\{-\tfrac{3}{2}\sin(2v - 2mt)\},$$

which when $j = k = 1$ give the following equations in the lunar theory ($D = t - mt$):

$$\begin{aligned}\frac{1}{\rho} = 1 + \tfrac{1}{6}m^2 - \tfrac{1}{12}m^4 - \tfrac{11}{63}m^5 - \tfrac{1721}{432}m^6 - \tfrac{74129}{3456}m^7 - \tfrac{145}{1296}m^8 \\ + \cos 2D\,[m^2 + \tfrac{5}{6}m^3 + \tfrac{17}{32}m^4 + \tfrac{89}{72}m^5 + (\cdots)m^6 + (\cdots)m^7 + (\cdots)m^8] \\ + \cos 4D\,[\tfrac{1}{3}m^4 + \tfrac{43}{60}m^5 + (\cdots)m^6 + (\cdots)m^7 + (\cdots)m^8] \\ + \cos 6D\,[(\cdots)m^6 + (\cdots)m^7 + (\cdots)m^8]\end{aligned}$$

or as far as m^5,

$$\begin{aligned}\frac{1}{\rho} = 1 + \tfrac{1}{6}m^2 - \tfrac{1}{12}m^4 - \tfrac{11}{63}m^5 + \cos 2D\,[m^2 + \tfrac{5}{6}m^3 + \tfrac{17}{32}m^4 + \tfrac{89}{72}m^5] \\ + \cos 4D\,[\tfrac{1}{3}m^4 + \tfrac{43}{60}m^5] \\ + \cos 6D\,[(\cdots)m^6].\end{aligned}$$

[p. 538]

($\frac{1}{\rho}$ is given by M. Delaunay only as far as m^5, the additional terms of $\frac{1}{\rho}$ and expression for ρ were kindly communicated to me by Prof. Adams); and

$$\begin{aligned}v = t \\ + \sin 2D\,[\tfrac{11}{8}m^2 + \tfrac{59}{12}m^3 + \tfrac{1471}{48}m^4 + (\cdots)m^5 + (\cdots)m^6 + (\cdots)m^7 + (\cdots)m^8] \\ + \sin 4D\,[\tfrac{281}{96}m^4 + \tfrac{1199}{144}m^5 + (\cdots)m^6 + (\cdots)m^7 + (\cdots)m^8] \\ + \sin 6D\,[(\cdots)m^6 + (\cdots)m^7 + (\cdots)m^8]\end{aligned}$$

(Delaunay, t. II. pp. 815, 836, 845).

To integrate the original equations write

$$\rho = 1 + \rho_1 + \rho_2 + \rho_3 + \cdots,$$

$$v = t + v_1 + v_2 + v_3 + \cdots,$$

where the suffixes indicate the degrees in the coefficients k, j conjointly: the equations for ρ_n, v_n take the form

$$\frac{d}{dt}\frac{d\rho_n}{dt} - 3\rho_n - 2\frac{dv_n}{dt} + V_n = Q_n,$$

$$\frac{d}{dt}\left(\frac{dv_n}{dt} + 2\rho_n + U_n\right) = P_n,$$

where V_n, U_n, P_n, Q_n do not contain ρ_n or v_n. From the second equation we have

$$\frac{dv_n}{dt} + 2\rho_n + U_n = \Omega_n + \int P_n\, dt,$$

where Ω_n is a constant of integration, the integral $\int P_n\, dt$ containing only periodic terms; and then adding twice this to the first equation we have

$$\frac{d}{dt}\frac{d\rho_n}{dt} + \rho_n + V_n + 2U_n = 2\Omega_n + Q_n + 2\int P_n\, dt,$$

which determines ρ_n; and substituting its value in the other equation we have $\frac{dv_n}{dt}$ and thence v_n; the constant Ω_n is determined so that $\frac{dv_n}{dt}$ may contain no constant term. We have

$$V_2 = -\left(\frac{dv_1}{dt}\right)^2 - 2\rho_1\frac{dv_1}{dt} + 3\rho_1^2,\quad V_3 = 2\rho_1\frac{dv_1}{dt} + (2\rho_2 + \rho_1^2)\frac{dv_1}{dt} - 2\rho_1\rho_2,$$

$$U_2 = \rho_1\frac{dv_1}{dt} - \rho_2\frac{dv_1}{dt} - \rho_1\left(\frac{dv_1}{dt}\right)^2,\ \ \&\text{c.}$$

$$Q_2 = -2\rho_1\frac{d\rho_1}{dt} + 6\rho_1\rho_2 - 4\rho_1^3,\quad \&\text{c.}$$

[p. 539]

$$Q_1 = km^2(\tfrac{1}{2} + \tfrac{3}{2}\cos 2D), \quad P_1 = jm^2(-\tfrac{3}{2}\sin 2D),$$
$$Q_2 = km^2\{3\rho_1 \sin 2D + \rho_1(\tfrac{1}{2} + \tfrac{3}{2}\cos 2D)\},$$
$$Q_3 = km^2\{-3v_1 \sin 2D - 3v_1^2 \cos 2D + \rho_2(\tfrac{1}{2} + \tfrac{3}{2}\cos 2D)\}, \quad P_2 = jm^2(-3v_1 \cos 2D - 3\rho_1 \sin 2D),$$
$$P_3 = jm^2\{-3v_2 \cos 2D + 3v_1^2 \sin 2D - 6\rho_1 v_1 \cos 2D + (2\rho_2 + \rho_1^2)(-\tfrac{3}{2}\sin 2D)\},$$
$$\&\text{c.}$$

In particular attending to the values of P_1, Q_1 the equations for ρ_1, v_1 are in their original form

$$\frac{d}{dt}\frac{d\rho_1}{dt} - 3\rho_1 + 2\frac{dv_1}{dt} = km^2(\tfrac{1}{2} + \tfrac{3}{2}\cos 2D),$$
$$\frac{d}{dt}\left(\frac{dv_1}{dt} + 2\rho_1\right) = jm^2(-\tfrac{3}{2}\sin 2D),$$

whence in the transformed form they are

$$\frac{dv_1}{dt} + 2\rho_1 = \Omega_1 + \tfrac{3}{4}\frac{jm^2}{1-m}\cos 2D,$$

and

$$\frac{d}{dt}\frac{d\rho_1}{dt} + \rho_1 = 2\Omega_1 + \tfrac{1}{2}km^2 + (\tfrac{3}{2}km^2 + \tfrac{3}{2}\tfrac{jm^2}{1-m})\cos 2D.$$

Thus the constant term of ρ_1 is $2\Omega_1 + \frac{1}{2}km^2$, giving in $\dfrac{dv_1}{dt}$ a constant term $-3\Omega_1 - km^2$; this must vanish, or we have $\Omega_1 = -\frac{1}{3}km^2$; and the equations thus become

$$\frac{d}{dt}\frac{d\rho_1}{dt} + \rho_1 = -\tfrac{1}{6}km^2 + \left(\tfrac{3}{2}km^2 + \tfrac{3}{2}\tfrac{jm^2}{1-m}\right)\cos 2D,$$
$$\frac{dv_1}{dt} + 2\rho_1 = -\tfrac{1}{3}km^2 + \left(\tfrac{3}{4}\tfrac{jm^2}{1-m}\right)\cos 2D,$$

and then completing the integration

$$\rho_1 = -\tfrac{1}{6}km^2 + \left\{\frac{-\frac{3}{2}km^2}{(3-8m+4m^2)} + \frac{-\frac{3}{4}jm^2}{(1-m)(3-8m+4m^2)}\right\}\cos 2D,$$
$$v_1 = \left\{\frac{\frac{3}{8}km^2}{(1-m)(3-8m+4m^2)} + \frac{\frac{3}{8}jm^2(7-8m+4m^2)}{(1-m)^2(3-8m+4m^2)}\right\}\sin 2D,$$

which are the accurate values of ρ_1 and v_1.

Expanding as far as m^5 we have

$$\rho_1 = k(-\tfrac{1}{6}m^2) + \cos 2D\left[k(-\tfrac{1}{2}m^2 - \tfrac{4}{3}m^3 - \tfrac{37}{8}m^4 - \tfrac{43}{3}m^5) + j(-\tfrac{1}{4}m^2 - \tfrac{5}{6}m^3 - \tfrac{61}{32}m^4 - \tfrac{49}{6}m^5)\right]$$
$$= k(-\tfrac{1}{6}m^2 - \tfrac{3}{4}m^3 - \tfrac{19}{32}m^4 - \tfrac{55}{12}m^5 - \tfrac{135}{16}m^5),\ \text{which for } j = k \text{ is}$$

[p. 540]

and

$$v_1 = \sin 2D\left\{k(\tfrac{1}{8}m^2 + \tfrac{4}{6}m^3 + \tfrac{17}{16}m^4 + \tfrac{73}{16}m^5) + j(\tfrac{3}{8}m^2 + \tfrac{2}{3}m^3 + \tfrac{47}{32}m^4 + \tfrac{31}{6}m^5 + \tfrac{47}{16}m^5)\right\}$$
$$= k(\tfrac{11}{8}m^2 + \tfrac{29}{12}m^3 + \tfrac{47}{32}m^4 + \tfrac{49}{16}m^5 + \tfrac{47}{16}m^5),\ \text{which for } j = k \text{ is}$$

I have, not in general, but for the value $j = k$, calculated ρ_2 and v_2 as far as m^5. I have not made the calculation for ρ_3 and v_3, but their values may be deduced from the foregoing values of ρ, v; the final expressions (when $j = k$) of $\rho = 1 + \rho_1 + \rho_2 + \rho_3 + \cdots$ and $v = t + v_1 + v_2 + v_3 + \cdots$ are

$$\rho = 1$$

$$+k(-\tfrac{1}{6}m^2)$$

$$+k^2(-\tfrac{1}{12}m^4)$$

$$+\cos 2D\left\{k(-\tfrac{1}{2}m^2 - \tfrac{4}{3}m^3 - \tfrac{37}{8}m^4 - \tfrac{43}{3}m^5 - \tfrac{135}{16}m^5) + k^2(\tfrac{1}{32}m^4 + \tfrac{4}{3}m^5 + \tfrac{61}{16}m^5) + j(-\tfrac{5}{12}m^5)\right\}$$

$$+\cos 4D\left\{k^2(\tfrac{1}{3}m^4 - \tfrac{4}{3}m^4 - \tfrac{8}{3}m^5 - \tfrac{37}{12}m^6) + k^3(\tfrac{8}{3}m^5 + \tfrac{61}{16}m^5)\right\}$$

$$+\cos 6D\left\{k^3(-\tfrac{1}{6}m^6)\right\},$$

and

$$v = t - \tfrac{1}{4}m^4 + \sin 2D\left\{k(\tfrac{11}{8}m^2 + \tfrac{29}{12}m^3 + \tfrac{47}{32}m^4 + \tfrac{49}{16}m^5 + \tfrac{47}{16}m^5) + k^2(\tfrac{1}{8}m^4 - \tfrac{4}{3}m^4 - \tfrac{43}{8}m^5 - \tfrac{27}{16}m^5)\right\}$$

$$+\sin 4D\left\{k^2(\tfrac{281}{96}m^4 + \tfrac{1199}{144}m^5 + \tfrac{47}{32}m^6)\right\} + \sin 6D\left\{k^3(\tfrac{47}{96}m^6 + \tfrac{47}{32}m^7)\right\},$$

which for $k = 1$ agree with the foregoing formulae (verifying them as far as m^4); the present formulae exhibit the manner in which the expressions depend on the several powers of the disturbing force.

[p. 541]

484.

ON THE VARIATIONS OF THE POSITION OF THE ORBIT IN THE PLANETARY THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXXII. *(1871–72), pp. 206–211.]*

IT has always appeared to me that in the Planetary Theory, more especially when the method of the variation of the elements is made use of, there is a difficulty as to the proper mode of dealing with the inclinations and longitudes of the nodes, hindering the ulterior development of the theory. Considering the case of two planets m, m', and referring their orbits to any fixed plane and fixed origin of longitudes therein, let θ, θ' be the longitudes of the nodes, ϕ, ϕ' the inclinations ($p = \tan\phi \sin\theta$, $q = \tan\phi\cos\theta$, &c., as usual); then the disturbing functions for m, m' respectively are developed, not explicitly in terms of ϕ, ϕ', θ, θ', but in terms of Φ, the mutual inclination of the two orbits, and of Θ, Θ' the longitudes in the two orbits respectively of the mutual node of the two orbits; Φ and Θ, Θ' being functions (and complicated ones) of ϕ, ϕ', θ, θ'. Moreover, although in the general theory of the secular variations of the orbits of the planetary system, θ, ϕ, &c., are, as above, referred to one fixed plane (the ecliptic of a certain date), yet in the theory of each particular planet it is the practice, and obviously the convenient one, to refer for such planet the θ, ϕ to its own fixed plane (the orbit of the planet at a certain date), the effect of course being that ϕ, and consequently p, q, instead of being of the order of the inclinations to the ecliptic, are only of the order of the disturbing forces. It has occurred to me that the last-mentioned plan should be adhered to throughout; viz., that for each planet m, the position of its variable orbit should be determined by θ, the longitude of its node, and ϕ, the inclination in reference to the appropriate fixed plane (orbit of the planet at a certain date) and origin of longitude therein. The disturbing functions for the planets m and m' will of course depend not only on θ, θ', ϕ, ϕ', but on the quantities Φ, Θ, Θ' which determine the mutual positions of the two fixed

[p. 542]

planes of reference and origins of longitude therein, these last being however absolute constants *not affected by any variation of the elements*; so that as regards the variation of the elements the disturbing functions are in fact given as explicit functions of the variable elements θ, ϕ, θ', ϕ'; and where ϕ, ϕ' and therefore also p, q, p', q' are only of the order of the disturbing forces.

I proceed to work out this idea, for the present considering the development of the Disturbing Function only as far as the first powers of p, q, &c. For comparison with the ordinary theory, observe that in this theory the disturbing function contains only the second powers of the p, q, &c., made use of therein; these are in fact of a form such as $P+p$, $Q+q$, $\ldots$ where P, Q are absolute constants and p, q, $\ldots$ are the p, q, $\ldots$ of the present theory; the ordinary theory gives therefore in the disturbing function a series of terms involving $(P+p)^2$, $(P+p)(Q+q),\ldots$ which I now take account of only as far as the first powers of p, $q,\ldots$ viz., they are in effect reduced to P^2+2Pp, $PQ+Pq+Qp$, &c. $\ldots$ The present theory is thus not now developed to the extent of giving the p, $q,\ldots$ of the ordinary theory in the more complete form as the solutions of a system of simultaneous linear differential equations, but only to the extent of obtaining for these p, $q,\ldots$ respectively the terms which are proportional to the time.

I commence with the following subsidiary problem. Consider a spherical triangle ABC (sides a, b, c, angles A, B, C, as usual), and taking the side c as constant, but the angles A and B as variable, let it be required to find the variations of C, a, b in terms of variations dA, dB and the variable elements C, a, b themselves. Although the geometrical proof would be more simple, I give the analytical one, as it may be useful.

We have

$$\cos C = -\cos A\cos B + \sin A\sin B\cos c,$$

and thence

$$-\sin C\,dC = (\sin A\cos B + \cos A\sin B\cos c)\,dA + (\sin A\cos B + \sin A\cos B\cos c)\,dB$$

$$= dA\ \sin b\frac{\sin c}{\tan b} + \sin A\,\frac{\sin c}{\tan a}\,dB,$$

that is

$$\frac{\sin C\,dC}{\sin c} = -dA\,\frac{\sin b\cos B}{\sin b} + \frac{\sin A\cos a}{\sin a}\,dB,$$

or finally

$$-dC = \cos b\,dA + \cos a\,dB.$$

Next

$$\sin a = \frac{\sin c}{\sin C}\sin A,$$

or, differentiating,

$$\cos a\,da = \frac{\sin c}{\sin^2 C}(\sin C\cos A\,dA - \cos C\sin A\,dC),$$

[p. 543]

or, substituting for dC its value,

$$= \frac{\sin c}{\sin^2 C}\,\{dA\,(\sin C\cos A + \cos C\sin A\cos b) + dB\ \cos C\sin A\cos a\},$$

$$= \frac{\sin c}{\sin^2 C}\left\{dA\,\frac{\sin C\sin b\cos a}{\sin a} + dB\ \cos C\sin A\cos a\right\},$$

that is

$$\frac{\sin c}{\sin C}\,da = \left\{dA\,\frac{\sin b}{\sin a} + dB\,\frac{\cos C\sin A}{\sin C}\right\},$$

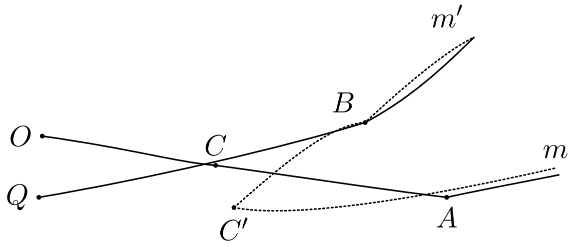
or, on the right-hand writing $\dfrac{\sin A}{\sin a}$ instead of $\dfrac{\sin C}{\sin c}$, this is

$$da = \frac{1}{\sin C}\,(dA\ \sin b + dB\ \cos C\sin a);$$

and similarly

$$db = \frac{1}{\sin C}\,(dB\ \sin a + dA\ \cos C \sin b).$$

Now let the continuous lines represent the orbits of m, m' at certain dates, O, Q the origins of longitude therein; and the dotted lines the variable orbits of the planets respectively.



Write

$$OQ = \Theta, \quad CA = \theta, \quad \angle CAC' = \phi,$$
$$QC = \Theta', \quad CB = \theta', \quad \angle CBC' = \phi',$$
$$\angle C = \Phi.$$

Then, answering to the notation of the lemma, we have

$$a = \theta', \quad b = \theta, \quad C = \Phi, \quad dA = d\phi, \quad dB = -d\phi',$$

or say $= \tan\phi$, $= -\tan\phi'$, whence

$$C'B = a + da,$$
$$= \theta' + \frac{1}{\sin\Phi}(\tan\phi\ \sin\theta - \tan\phi'\ \cos\Phi \sin\theta'),$$
$$= \theta' + \frac{1}{\sin\Phi}(p - p'\cos\Phi),$$

[p. 544]

$$\begin{gathered}
C'A = b + db,\\
= \theta + \frac{1}{\sin\Phi}(-\tan\phi'\sin\theta' + \tan\phi\cos\Phi\sin\theta),\\
= \theta - \frac{1}{\sin\Phi}(p' - p\cos\Phi),\\
\angle C' = \Phi - q + q' \ = \Phi + \tan\phi\cos\theta - \tan\phi'\cos\theta',\ = \Phi - q + q'.
\end{gathered}$$

Suppose v, v' are the longitudes of the planets in their two orbits respectively; that is

$$v = OA + Am = \Theta + \theta + Am,$$

$$v' = QB + Bm' = \Theta' + \theta' + Bm',$$

whence

$$\begin{gathered}
C'm = C'A + Am,\ = v - \Theta - \frac{1}{\sin\Phi}(p' - p\cos\Phi),\\
C'm' = C'B + Bm',\ = v' - \Theta' + \frac{1}{\sin\Phi}(p - p'\cos\Phi),\\
\angle C = \Phi - q + q'.
\end{gathered}$$

Say these values are $v - \Theta + x$, $v' - \Theta' + x'$, $\Phi + y$. Then if H is the angular distance mm' of the two planets,

$$\begin{aligned}
\cos H \ &= \ \cos(v-\Theta+x)\cos(v'-\Theta'+x') + \sin(v-\Theta+x)\sin(v'-\Theta'+x')\cos(\Phi+y),\\
&= \ \cos(v-\Theta)\cos(v'-\Theta') + \sin(v-\Theta)\sin(v'-\Theta')\cos\Phi\\
&\quad\ + x\,[-\sin(v-\Theta)\cos(v'-\Theta') + \cos(v-\Theta)\sin(v'-\Theta')\cos\Phi]\\
&\quad\ + x'[-\cos(v-\Theta)\sin(v'-\Theta') + \sin(v-\Theta)\cos(v'-\Theta')\cos\Phi]\\
&\quad\ + y\,[-\sin(v-\Theta)\sin(v'-\Theta')\sin\Phi],\\
&= \ \cos H + \nabla,\ \text{suppose.}
\end{aligned}$$

The disturbing function for the planet m disturbed by m' is

$$R = m'\Big\{\frac{1}{\sqrt{r^2 + r'^2 - 2rr'\cos H}} - \frac{r\cos H}{r'^2}\Big\}$$

(R, if $\Omega = -R$, is the disturbing function of the *Mécanique Céleste*); and the term hereof which involves ∇ is

$$= \nabla\frac{d\cos H}{d\cos H}\Big\{\frac{rr'}{(r^2 + r'^2 - 2rr'\cos H)^{\frac{3}{2}}} + \frac{r}{r'^2}\Big\}\nabla,$$

where after the differentiation $\cos H$ is replaced by $\cos H$,

[p. 545]

viz., this is a linear function of x, x', y, that is of p, q, p', q', with coefficients which of course involve the other variable elements and the time; but it will be remembered that Θ, Θ', Φ are not variable elements, but are absolute constants. The variations of p depend upon $\frac{d\Omega}{dq}$ and those of q on $\frac{d\Omega}{dp}$, and the quantities p, q, p', q',… disappear from these differential coefficients $\frac{d\Omega}{dq}$, $\frac{d\Omega}{dp}$; that is, disregarding periodic terms, and the variations of the elements, we obtain $\frac{dp}{dt}$, $\frac{dq}{dt}$ as absolute constants, or reckoning the time from the epoch belonging to the fixed orbit of m, we have p, q as mere multiples of the time ($p = At$, $q = Bt$, where A and B are constants); agreeing with the statement preceding the investigation.

Observe that the p, q, as used above, have reference not only to the fixed orbit of m, but also to the node thereon of the fixed orbit of m': we may, if we please, write $p = \tan\phi\sin(\theta+\Theta)$, $q = \tan\phi\cos(\theta+\Theta)$, that is, $p = \mathfrak{p}\cos\Theta + \mathfrak{q}\sin\Theta$, $q = -\mathfrak{p}\sin\Theta + \mathfrak{q}\cos\Theta$ (or $\mathfrak{p} = p\cos\Theta - q\sin\Theta$, $\mathfrak{q} = p\sin\Theta + q\cos\Theta$), and in place of p, q introduce into the formulae $\mathfrak{p}$ and $\mathfrak{q}$, which have reference only to the fixed orbit of m, and similarly writing $p' = \tan\phi'\sin(\theta' + \Theta')$, $q' = \tan\phi'\cos(\theta' + \Theta')$, instead of p', q' introduce $\mathfrak{p}'$, $\mathfrak{q}'$ which have reference only to the fixed orbit of m'.

I remark that a table for the relative positions of the orbits of the eight Planets for the Epoch 1st January, 1850, is given in Leverrier's *Annales de l'Observ. de Paris*, t. II. (1856), pp. 64–66.

[p. 546]

485.

PROBLEMS AND SOLUTIONS.

[From the Mathematical Questions with their Solutions from the Educational Times, vols. v. to xii. (1866–1869).]

[Vol. v., January to July, 1866, p. 17.]

1791. (*Proposed by Professor Cayley.*) —Given a quartic curve $U = 0$, to find three cubic curves $P = 0$, $Q = 0$, $R = 0$, each meeting the quartic in the same six points 1, 2, 3, 4, 5, 6, and such that $P = 0$, $R = 0$ may besides meet the quartic in the same three points a, b, c, and that $Q = 0$, $R = 0$ may besides meet the quartic in the same three points α, β, γ.

[Vol. v., pp. 25, 26.]

Note on the Problems in regard to a Conic defined by five Conditions of Intersection.

I USE the word "intersection" rather than "contact" because it extends to the case of a 1-pointic intersection, which cannot be termed a contact. The conditions referred to are that the conic shall have with a given curve, at a point given or not given, a 1-pointic intersection, a 2-pointic intersection (= ordinary contact), a 3-pointic intersection, &c., as the case may be. It may be noticed that when the point on the curve is a given point, the condition of a k-pointic intersection is really only the condition that the conic shall pass through k given points; though from the circumstance that these are consecutive points on a conic, the formulae for a conic passing through k discrete points require material alteration; for instance, in the two questions to find the equation of a conic passing through five given points, and to find the equation of a conic having at a given point of a given curve 5-pointic intersection with the curve, the forms of the solutions are very different from each other.

The foregoing remark shows, however, that it is proper to detach the conditions which relate to intersections at given points; and consequently attending only to the